UI/UX Design Specification

UTeM Micro-Credential Management System (MCMS)

Document Control

Document Information	
Title	UI/UX Design Specification for the UTeM Micro-Credential Management System (MCMS)
Date	27 October 2026
Status	Approved design basis for the Release 1 prototype; supervisor review planned once the prototype is shown (D-23)
Version	1.1
Prepared for	Supervisor review; Centre for Open and Distance Learning (CODL), Universiti Teknikal Malaysia Melaka (assumed system owner)
Prepared by	Aliff, Intern
Reference	MCMS_UIUX_V1.1_Oct_2026
Written against	Glossary and Decision Log v0.5 (MCMS_FND_V0.5_Oct_2026); Software Requirements Specification v1.0 (MCMS_SRS_V1.0_Oct_2026); Product Requirements Document v1.0 (MCMS_PRD_V1.0_Oct_2026); DESIGN.md (ulearn-carbon)
Design source	Clickable prototype v2, `UIUX/v2/prototype`, approved 26 October 2026
Document set	SRS, PRD, UI/UX Design Specification (this document), Software Design Description (SDD)

Disclaimer:

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Revision History

Version	Date	Change
1.0	26 Oct 2026	First issue, written from the approved clickable prototype v2: design principles, visual system, interaction rules, 114 screens, 26 flows and 16 printable documents. Includes the corrections from the planner review of the prototype (Section 9).
1.1	27 Oct 2026	Added Appendix B (every screen with its tabs, wizard steps, pop-ups, warnings, side panels and menus), Appendix C (every page of every printable document and report template) and Appendix D (a storyboard of every step of every flow), all captured from the approved prototype. Fixed in the prototype: the Import CSV and Add buttons of the CODL reference data screen were not shown.

Glossary

All terms of the Glossary and Decision Log v0.5 apply with exactly the same meaning. The table below adds only the design terms used here.

Term	Definition
Confirmation dialog	A pop-up that states the effect of an action and asks the user to confirm it before anything is saved.
Drawer	A panel that slides in from the right to show a record without leaving the current list.
Flow (FL-XX)	A guided walk-through of a process across screens and roles, with a step bar at the bottom of the prototype.
Hotspot	An outline the prototype draws around every clickable element when the viewer presses H.
Persona	A fictional representative user (Section 2.3). Personas and their data are illustrative.
Route	The address of a screen in the prototype, for example `#/dd/verify/DS-2026-2`. Used as the screen's identifier.
Screen ID	Short identifier of a screen in this document, for example DD-12 (Section 6).
Status tag	A small label with a pale tint and dark text showing the status of a record, always with its word.
Toast	A short message at the top right that confirms a finished action and names the next step.

1. Introduction

1.1 Purpose

This UI/UX Design Specification describes how MCMS looks and behaves for each of its users: the visual system, the interaction rules, every screen, every flow between screens and roles, and every printable document. It turns the requirements of the SRS and the product decisions of the PRD into a design that developers can build without guessing.

The specification was written from the approved clickable prototype v2. The prototype is part of the specification: where a screen is described here, the prototype shows it with sample data, and where this text and the prototype differ, this text prevails and the prototype is corrected. Where this document and the SRS differ on a requirement, the SRS prevails.

1.2 Document Conventions

Convention	Meaning
Screen ID	Role prefix and number, for example DD-12 or ST-04. The full list with routes is in Section 6.4.
Route	Prototype address of a screen, written as `#/dd/verify/:id`. A word after a colon is a record identifier.
FL-XX	A flow of Section 7.
Release 1 / 2	Follows the SRS: **SHALL** is Release 1, **SHOULD** is Release 2. Release 2 controls appear disabled and labelled "Release 2".
FR-XX, BR-XX, NFR-XX	Requirement IDs of the SRS v1.0, as published there.
D-XX, A-XX, OI-XX	Decisions, assumptions and open items of the Glossary and Decision Log v0.5.
Dates and money	Written in this document as "26 Oct 2026"; shown to users as DD/MM/YYYY and RM 0.00 (Section 4.9).

> **Note:** Names, records and figures in the screenshots are sample data of the prototype. They are consistent with each other but are not real people or decisions.

1.3 Intended Audience

Reader	Use of this document	Read first
Supervisor and CODL	Check that every user can do their work and that the design keeps the Guide's stages separate.	Sections 2, 6, 7
Developers and AI development agents	Build screens, components and behaviour exactly as specified.	Sections 3, 4, 5, 6
Testers	Derive UI tests from the interaction rules, flows and screen states.	Sections 4, 7, 9
Content and translation	Write and translate interface text consistently in BM and EN.	Sections 4.9, 8

1.4 Scope

The specification covers the Release 1 web application for all roles: participants (students), lecturers and colleague reviewers, deputy deans, the CODL system administrator and the industry expert. It also covers the public website for visitors and employers, and the documents the system prints or exports.

The application is web first and responsive down to a phone width of 360 px (NFR responsiveness); there is no native mobile app (SRS 1.4).

Out of scope: the visual design of UCreds, the national ABC platform and other external systems; the SDD's technical design of components; and Release 2 features beyond showing them as disabled.

1.5 References

1. Glossary and Decision Log for the UTeM MCMS, version 0.5, 26 October 2026 (MCMS_FND_V0.5_Oct_2026).
2. Software Requirements Specification, UTeM MCMS, version 1.0, 26 October 2026 (MCMS_SRS_V1.0_Oct_2026).
3. Product Requirements Document, UTeM MCMS, version 1.0, 26 October 2026 (MCMS_PRD_V1.0_Oct_2026).
4. DESIGN.md, design system "ulearn-carbon" (IBM Carbon with uLearn Navy), held in the project folder.
5. MCMS UI design basis v2 (`UIUX/v2/DESIGN-BASIS.md`) and SRS screen inventory (`UIUX/v2/basis/srs-fr-inventory.md`).
6. MCMS clickable prototype v2 (`UIUX/v2/prototype`) and its sample PDFs (`UIUX/v2/samples`).
7. WCAG 2.1; Garis Panduan Micro-Credential UTeM, Versi 1.0 (the Guide).

2. Design Principles and Users

2.1 Design Principles

Seven principles drive every design decision. They come from the Guide's separation of recognition stages, the PRD design principles (PRD 8.2) and the user needs of the personas.

Principle	What it means on screen	Source
One stage, one record, one look	Approval, offering, result verification, completion, credential, QV, ACR, ABC deposit, transfer and redemption are separate decisions with separate screens. No button combines two stages. Verifying results creates module achievements; confirming completion is a later, separate action that makes the system issue credentials.	D-15, BR-16, G 12–14
Land on what needs me today	Staff open on a task list with age and overdue flags; participants open on their learning and credit.	PRD 8.2
Only the authorised role sees a decision	Decision buttons appear only for the role that may decide. CODL has no academic decision anywhere. The Senate never signs in; the deputy dean records its decisions with the meeting reference.	D-11, D-12, D-28, BR-42
Recognition text is generated	Credit, MQF level and QV claims come only from verified records and templates. A SAMC shows "Credit after QV" until QV is recorded; a lapsed QV removes the claim.	D-17, BR-14, BR-22, BR-23
The rule sits beside the field	Rules are checked as the user types; blocking errors stop submission, warnings let the user continue with the warning on record.	PRD 8.2, ir-forms
History is one click away	Every record has a history from the append-only audit trail. Nothing is deleted; a correction creates a new revision.	D-20, BR-38
Honest about credit	Wherever a participant meets credit, the screen says that deposit is storage, not transfer, and that each transfer is decided case by case.	BR-26, BR-31

2.2 Users and Permissions

The design follows the six kinds of user of decision D-28. Table 1 shows what each role can and cannot do in the interface; buttons for actions a role cannot take are not shown at all, rather than shown disabled.

Table 1: *Roles and their permissions in the interface*

Role (persona)	Signs in	Can	Cannot
Student P8 (Amir Hakimi)	SSO	Browse, register and pay (stub), withdraw, view own records, share credentials, give or decline consent to deposit, apply for transfer, view own stackable plan	See anyone else's records
Lecturer P1 (Dr. Aina Zulkifli)	SSO + MFA	Propose and edit, answer review comments, create offerings, complete the lecturer's readiness items, enter results, send for verification, write quality reviews, gather QV evidence, review colleagues' proposals after declaring conflict of interest	Review or decide own proposal; verify results they entered

Role (persona)	Signs in	Can	Cannot
Deputy dean P2 (Hakim Rahman, FTMK)	SSO + MFA	Assign reviewers, verify category, endorse or return, record Senate decisions, confirm readiness, verify results, confirm completion, correct, replace or revoke on a recorded decision, record QV results, confirm quality reviews, confirm ACR, record deposits, decide transfers into FTMK degrees, redeem or release credit, map pathways and plans, periodic review, suspension	Act outside FTMK; issue credentials by hand (the system issues them)
Senate P3	Never	Decides in its meeting	Use MCMS
CODL S1/P5 (Farid Hashim)	SSO + MFA	Users and scoped roles, reference data, settings with history, bilingual templates, UCreds files, adapter status, test mailbox, audit trail, reports, retention	Any academic decision: approve, verify, issue, ACR, deposit, transfer, module credit flag
Industry expert X2 (Ir. Siti Aminah Yusof)	SSO (limited)	Declare conflict of interest, comment on invited proposals	Rate, recommend or decide
Public X3 (Tan Mei Ling)	Never	Catalogue, pathways, how it works, FAQ, verify a credential	See results, identity numbers, email or matric numbers

A person may hold several roles, each with its own scope. Separation of duties is checked per record: a lecturer who is also a reviewer cannot review their own proposal, and a person who entered a result cannot be its only verifier.

2.3 Personas and Sample Data

The personas of PRD Section 3 appear in the prototype with consistent sample data. The world is set on 26 October 2026 at FTMK, with records at different stages so that every flow can be walked: the UMC Data Structures (BITS 1123, 3 credits, 120 hours) with cohorts from 2025/2 to 2027/1, the SAMC Cloud Native Development (2 credits, MQF level 4, Quality-Verified to 6 December 2026), the MCE Leadership Essentials (16 hours, no credit), and Amir's two deposited or pending credits. Figures add up across screens, for example the 28 learners of Data Structures 2026/2, of whom 27 pass.

3. Visual Design System

MCMS uses the design system of DESIGN.md, "ulearn-carbon": IBM Carbon geometry and type with uLearn Navy as the single brand colour. This section fixes how that system is applied to working screens. Figure 1 shows the result on the deputy dean's task queue.

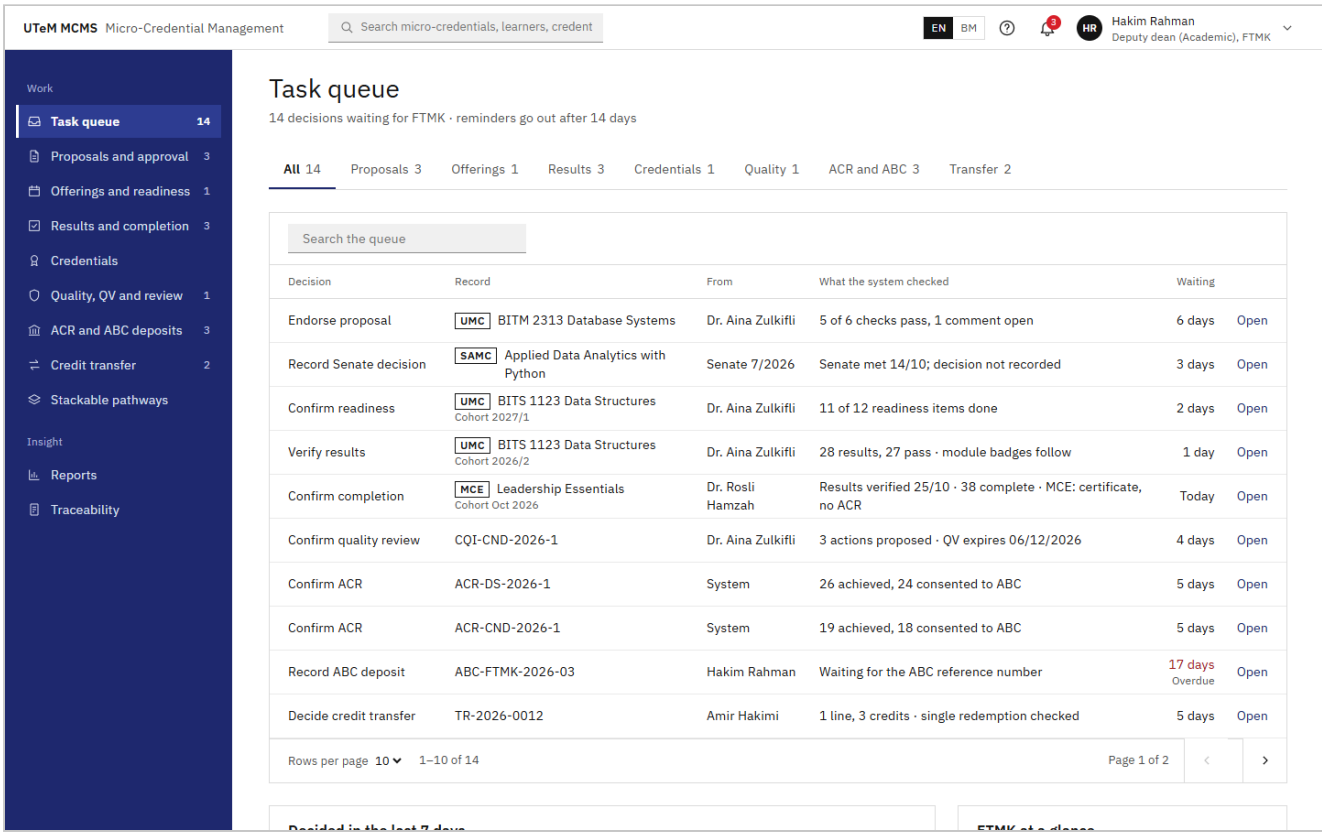


Figure 1: *Application shell: top bar, navy side navigation and white content area (deputy dean task queue, DD-01).*

3.1 Colour

Table 2: *Colour roles*

Role	Value	Use
Primary (uLearn Navy)	#1e2a6e	Side navigation background, the one filled primary button per screen, links, focus outline, selected tab underline
Navy hover / selected	#18225a / #2e3d94	Hover on navigation items; the current page in the side navigation (with a 3 px white left bar)
Ink	#161616	Text, secondary buttons, the dark flow bar
Muted	#525252	Labels, table headers, meta text (meets 4.5:1 on white)
Surface 1	#f4f4f4	Input fields, table bands, hover rows
Hairline	#e0e0e0	Panel borders and table header rule; row rules #ececec
Success / Warning / Error	#24a148 / #f1c21b / #da1e28	Dots, tag tints, notice bars and the danger button only. Never a full background. One red mark per object.

> **Note:** The navy side navigation is the owner's decision of 26 October 2026, overriding the neutral-sidebar guidance of the design craft rules, to give the uLearn brand colour more presence. Content areas stay white.

3.2 Typography

IBM Plex Sans is the only typeface; IBM Plex Mono is used only for identifiers such as setting keys and file names. Every number in a table, KPI or form is set in tabular figures and right-aligned.

Role	Size and weight	Use
Page title	28 / 400	One per screen, sentence case
KPI value	28 / 300	Figures in a KPI strip; light weight per DESIGN.md
Section title	16 / 600	Panel and section headings
Body, cells, navigation	14 / 400, tracking 0.16 px	Everything else
Labels, meta, table headers	12 / 400	Sentence case, grey; never uppercase tracked
Documents	11 pt body, 20 pt light title	A4 printable documents (Section 8)

3.3 Shape, Spacing and Elevation

Corners are square (0 px) for buttons, inputs, panels and modals; only status tags are pill-shaped. Spacing follows a 4 px grid: 24 px between sections, 16 px between fields, 20 px panel padding. Panels are separated by 1 px hairlines and surface change, never by shadows. Shadows are reserved for what floats: menus, toasts, dialogs and drawers.

3.4 Layout and Shell

Region	Specification
Top bar (48 px, white, sticky)	Product name (goes home) · global search (staff) · language switch EN/BM · help · notification bell with unread count · profile button with avatar, name and post
Side navigation (248 px, navy, sticky)	Grouped items with outline icons (16 px) and counts on the right that come from the live task data; the current page is highlighted
Content (white, max 1400 px)	Breadcrumb · title row (title, subtitle, status tag, actions on the right) · content blocks. One filled primary button per screen.
Public site	Top navigation (Home, Catalogue, Pathways to a degree, How it works, FAQ, Verify a credential), Sign in, and a dark footer with contact and policy links
Maintenance banner	A warning bar under the top bar, shown from 48 hours before planned maintenance

3.5 Components

Table 3: *Component catalogue*

Component	Specification
Buttons	40 px high, square. Primary (navy, one per screen), secondary (ink), tertiary (white with navy border), ghost (text), danger (red). A disabled button stays readable and states why, for example "Submit · 1 check failing" or "Confirm ready · 1 item open".
Fields	Label above (12 px), input on Surface 1 with a bottom rule; focus is a 2 px navy outline; error is a 2 px red outline with the message under the field in plain language. Required labels end with an asterisk.
Choice cards	Radio or checkbox inside a bordered card with a title and one-line consequence; the chosen card has a 2 px navy border.
Status tags	Pale tint with dark text of the same hue, always with the status word (Section 4.8). Never above a heading.

Component	Specification
Category mark	UMC, SAMC or MCE in a 1 px ink box beside the micro-credential name.
Tables	12 px grey sentence-case headers, rules between rows, no zebra striping, numbers right-aligned. Search bar above long lists; pager below (Section 4.5).
Tabs	Text tabs with counts; the selected tab has a 2 px navy underline.
KPI strip	One bordered container split by hairlines; label over value; no icons, no sparklines.
Recognition track	A horizontal track of stages (for example Earned, ACR, ABC, Transfer): done stages in ink, the current stage in navy, later stages grey.
Notices	Inline bar with a coloured left edge: info (navy), warning, error, success. Never used as a page background.
Checks list	Dot, statement and one-line evidence; used for system checks, readiness items and rules.
Dialogs, drawers, menus, toasts	See Section 4.

4. Interaction Design

4.1 Every Control Has a Defined Result

Each button, link, tab, table row and menu item does exactly one of the things in Table 4. A control that does nothing is a defect. A control that cannot be used yet is shown disabled with the reason in its label or beside it.

Table 4: *Possible results of a control*

Result	Used for
Go to a screen	Opening a record, the next step of a wizard, a list, a printable document
Confirmation dialog	Every action listed in Section 4.2
Drawer	Reading a record without leaving a list: history, evidence, an email, import problems
In-place change and toast	Saving a draft, marking notifications read, adding a row, accepting evidence
Menu	Notifications, profile, row actions
Download	A real file: signed credential (.json), report and export tables (.csv)

4.2 Confirmation Dialogs

The system asks for confirmation, with a summary of the effect, before: submit, endorse, return, record a decision, verify results, confirm completion (which issues credentials), correct, replace, revoke, confirm ACR, record a deposit, decide a transfer, redeem or release credit, change a setting, assign or remove a role, deactivate an account, import a file, withdraw an enrolment, give or withdraw consent, pay, and sign out with unsaved work (ir-confirm).

Part	Rule
Kicker	The record the dialog is about, for example the credential number
Title	A question naming the action and the count, for example "Verify 28 results?"
Body	What will happen, what will not happen, and who is told. Stage separation is stated where it matters: verifying results does not issue certificates; a deposit is not a transfer; refusing a transfer leaves the ACR and credentials unchanged.
Fields	Any input the decision needs, such as a meeting reference, a reason or an effective date. Required fields are marked and checked.
Footer	Cancel on the left; the action as a verb on the right
Destructive actions	Revoke, deactivate, reject, release, withdraw and terminate use the danger style with a red top edge. The record name appears in the body, a reason is required for the audit trail, and focus starts on a safe control.
Non-blocking warnings	The same dialog states the risk and asks the user to confirm, for example an unresolved non-blocking review comment.

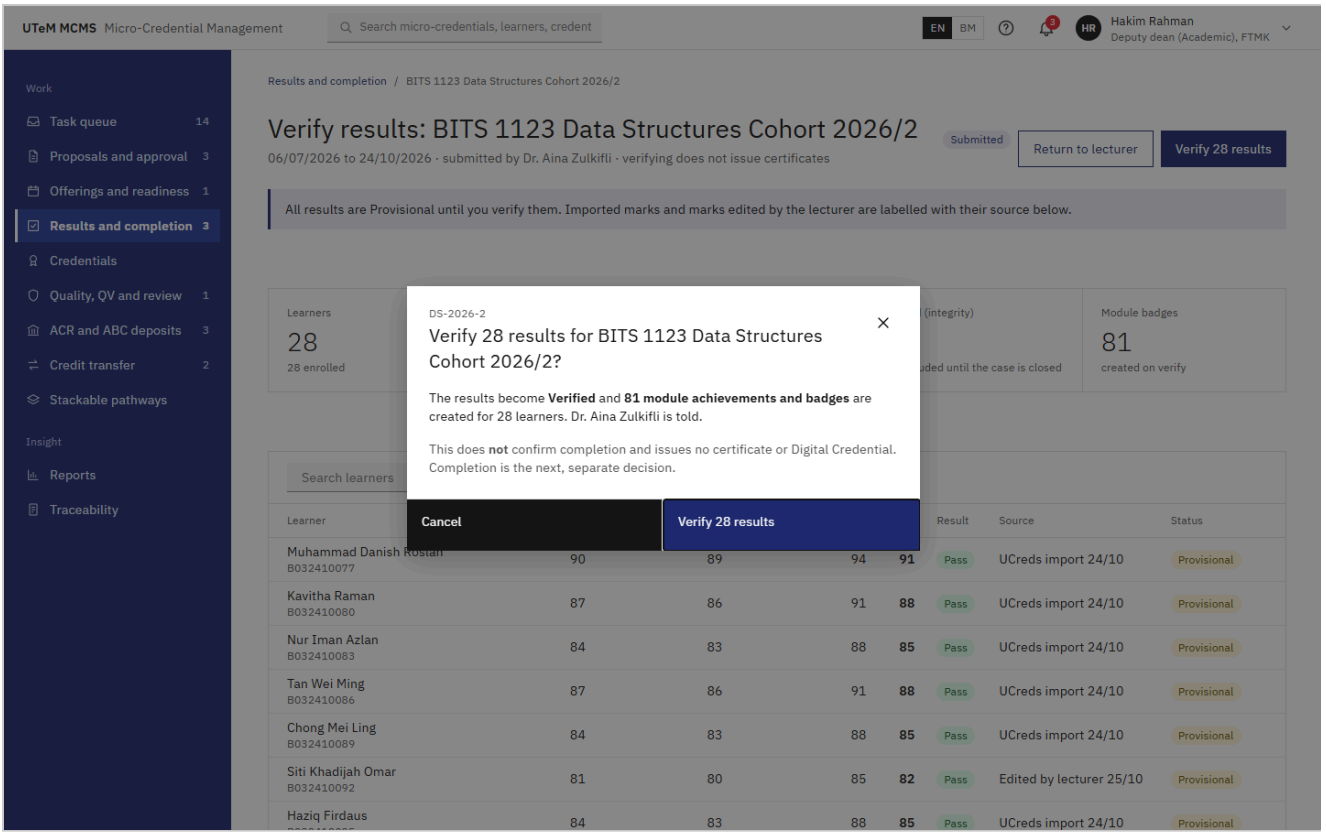


Figure 2: *Confirmation dialog: verifying results states that it creates module achievements only and that completion is confirmed separately.*

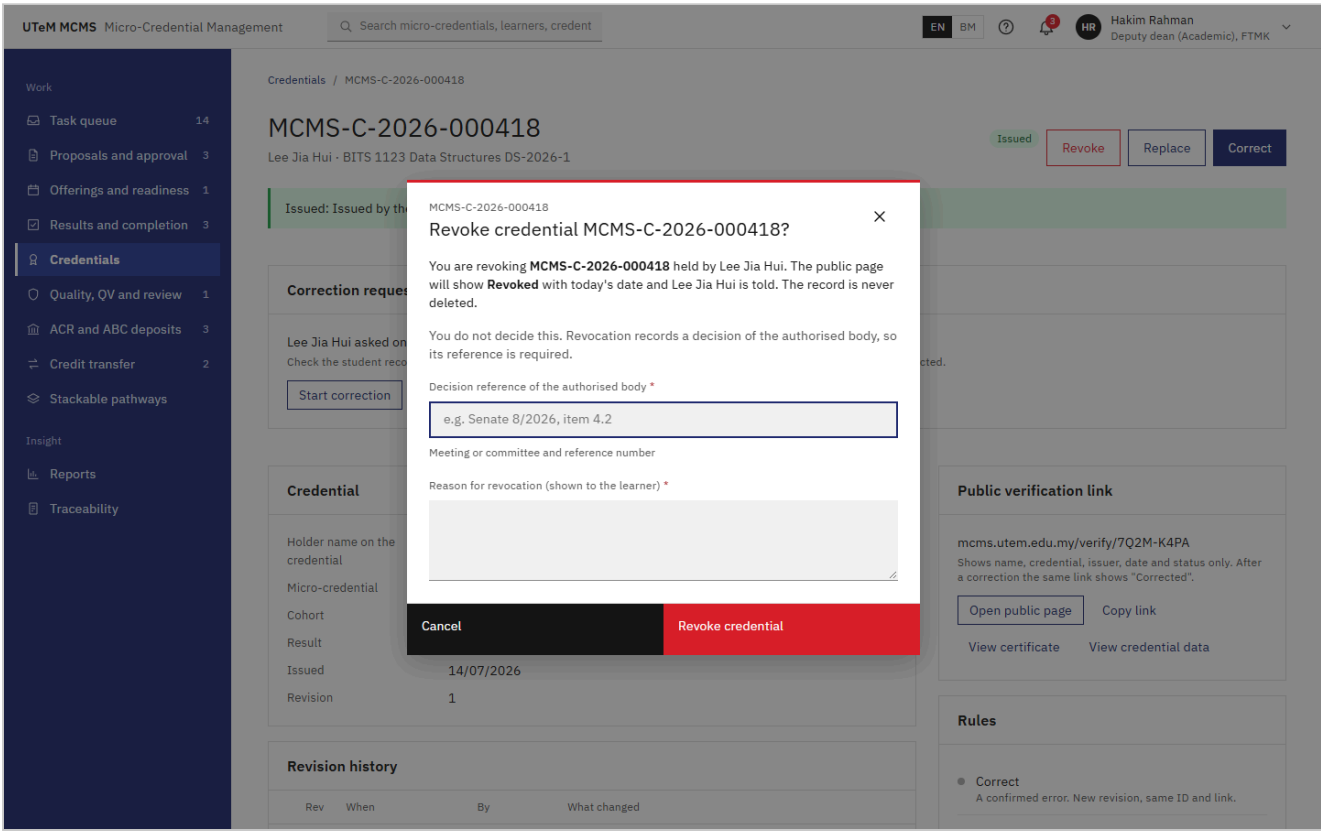


Figure 3: *Destructive dialog: revoking a credential needs the authorised body's decision reference and a reason; the record is never deleted.*

4.3 Feedback

Situation	Design
Action finished	A toast at the top right for 5 seconds: what happened and the next step, for example "Results verified. 27 module badges issued. Confirm completion next."
Field error	The message under the field in plain language; entered data is kept; a toast counts the errors (ir-forms).
System error	Inside the block that failed: what failed, a reference code such as ERR-..., and Retry. No technical detail (nfr-errors).
Long job	Queued, running with a count ("issuing 14 of 27"), done; the user is notified when it finishes (nfr-queue). Used for issuing credentials, imports and large reports.
Empty state	The frame and header stay; one sentence says why it is empty and offers the first action.
Blocked action	The button is disabled and says why; the rule is shown beside the field that causes it.

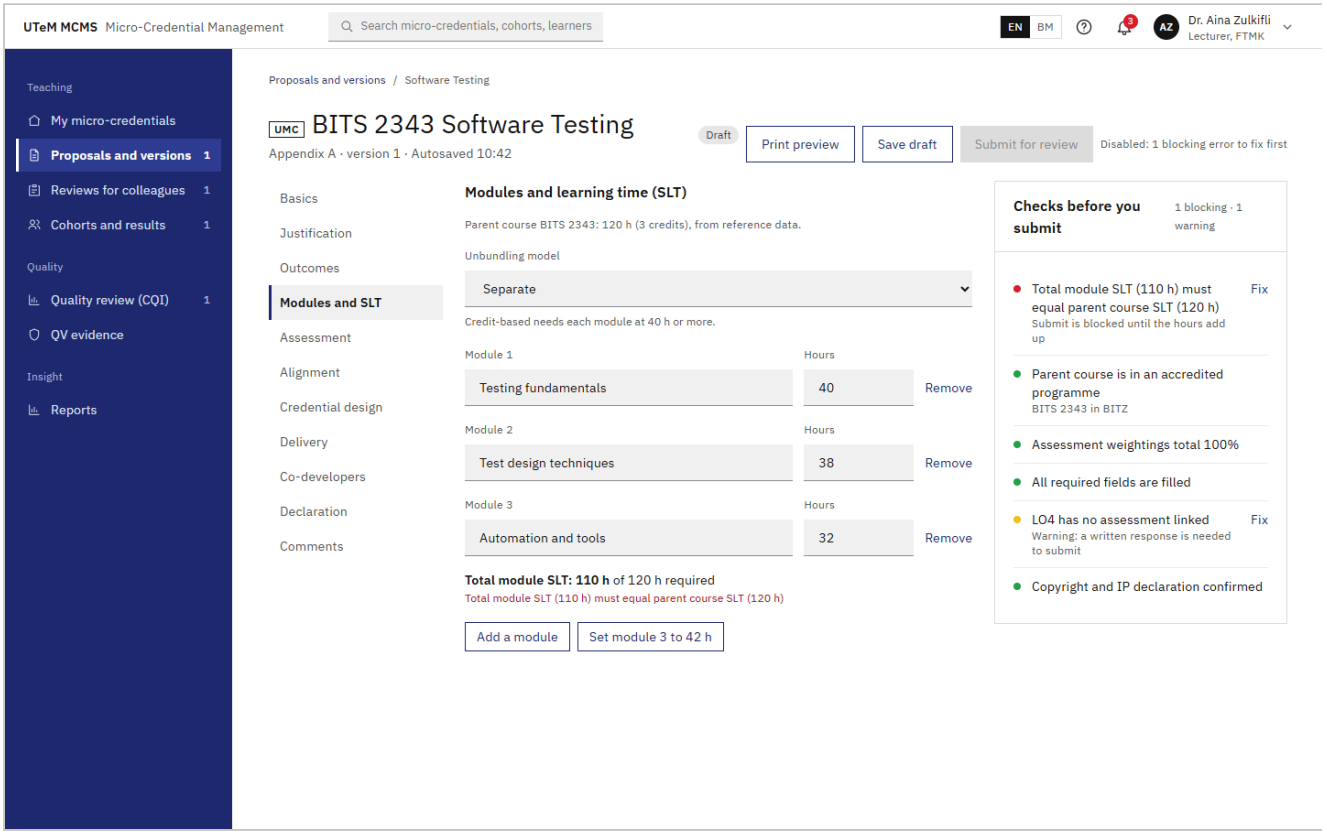


Figure 4: *Rule beside the field: a UMC whose modules total 110 hours cannot be submitted against a 120-hour parent course; the fix is offered in place.*

4.4 Notifications and Profile

The bell in the top bar shows the unread count and opens a panel with the five latest notifications, "Mark all read" and "View all". The notifications page filters by read state, marks a notification read when it is opened and takes the user to the record. Decisions on the user's records, credential actions, consent requests and enrolment changes are mandatory notices that cannot be switched off (fr-ntprefs). Lecturers and deputy deans may choose a daily digest instead of single emails.

The profile button opens a menu: My profile, Notification preferences, Sign-in and sessions, Help and guides, and Sign out (which asks for confirmation). The profile page has four tabs: Details (identity from

SSO, read-only; language, phone and name on certificates editable), Notifications, Sign-in and sessions (MFA, active sessions, idle timeout) and My personal data (view and request a correction, sr-access).

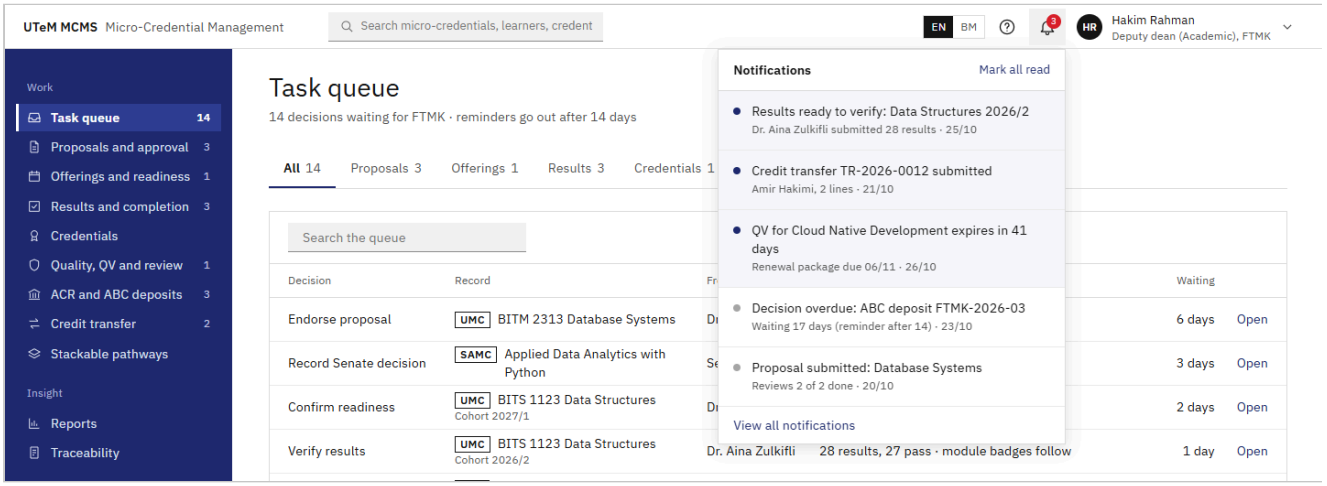


Figure 5: *Notification panel opened from the bell.*

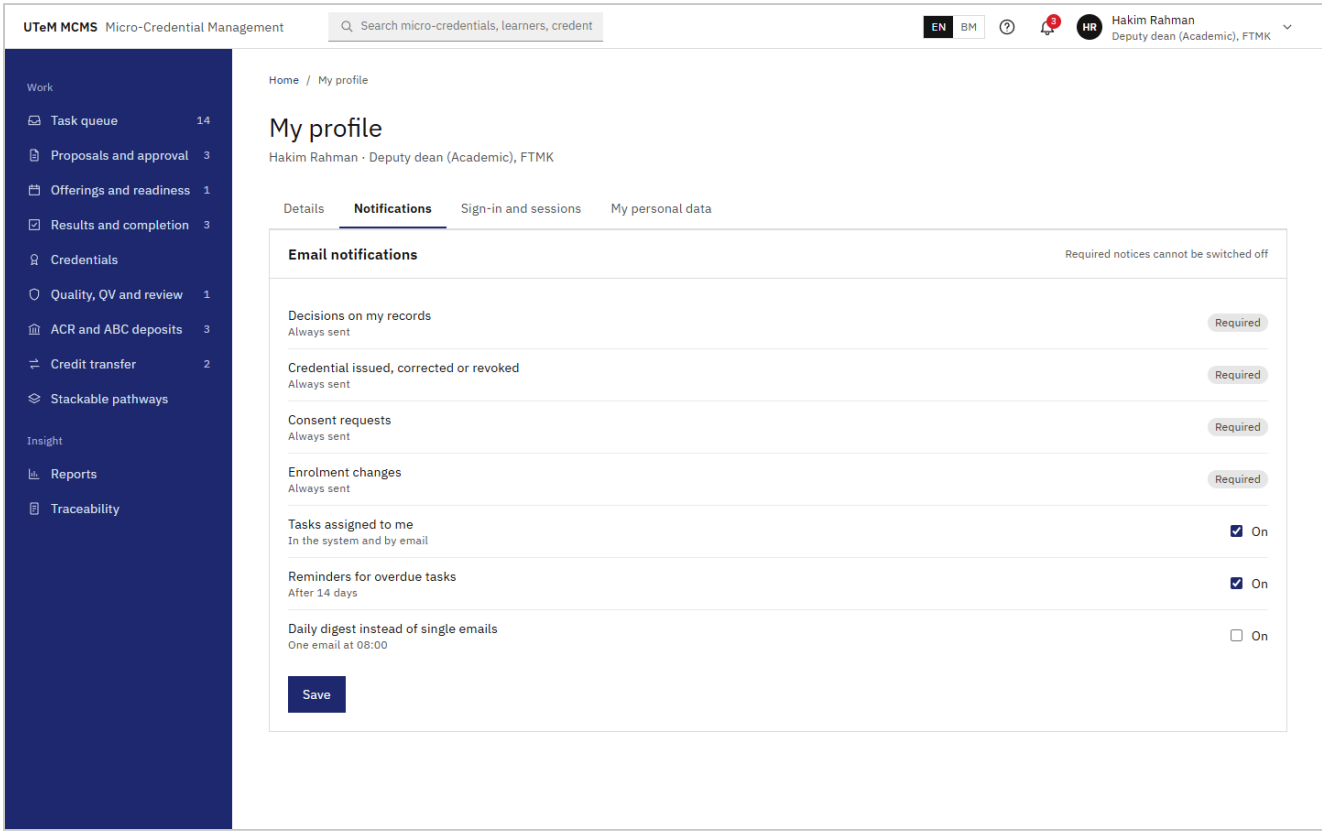


Figure 6: *Profile, notification preferences: required notices are marked and cannot be switched off.*

4.5 Tables and Pagination

Every list that can exceed one page is paginated: rows per page (5, 10, 25 or 50; default 10), a count "1–10 of 28", "Page 1 of 3" and previous and next buttons. Long lists have a search box above the table and filters as selects in the same bar; counts follow the filter. A row that opens a record is clickable across the whole row. A bulk action is one filled button above the table that names the count. Rows that need the user's attention, such as missing marks, are listed first.

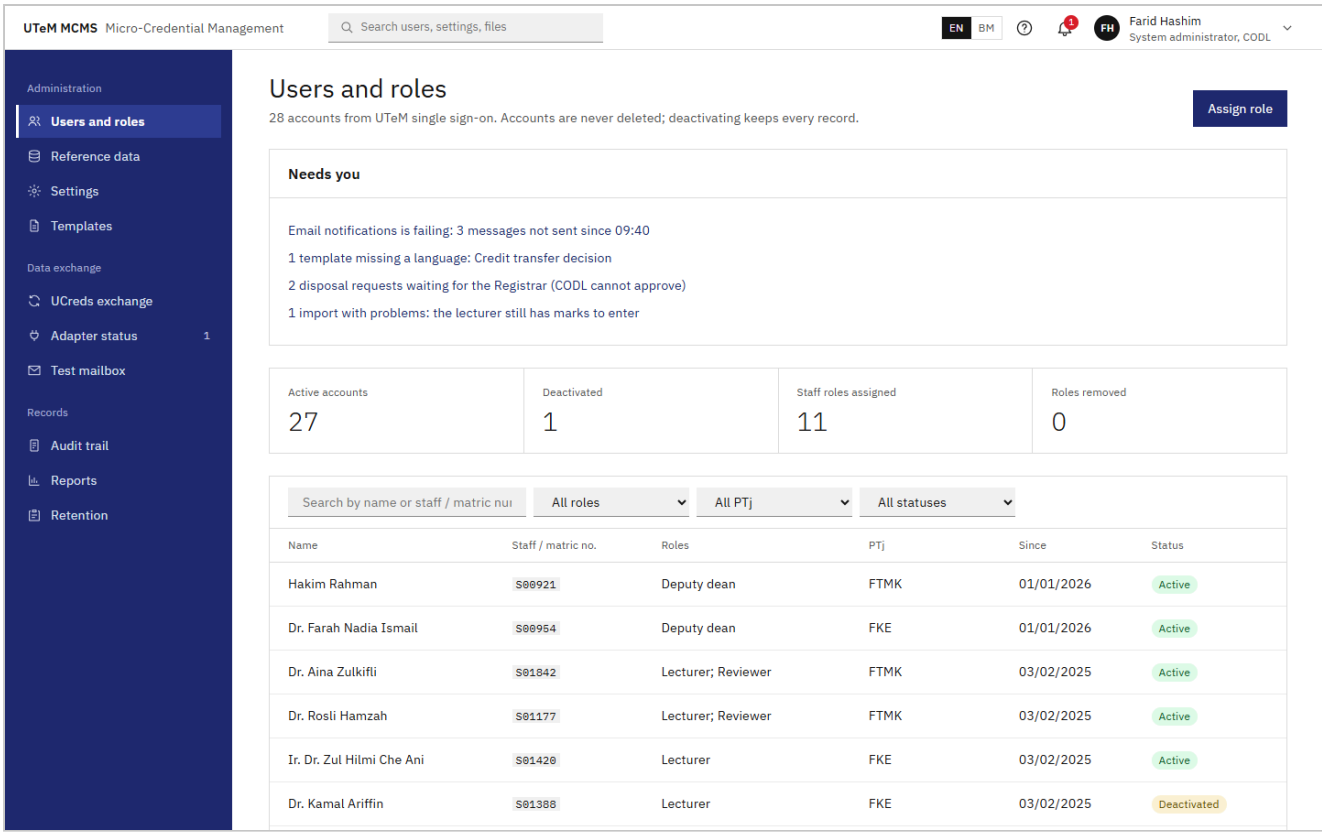


Figure 7: *Table with search, filters and pagination (CODL users and roles), with the "Needs you" panel above it.*

4.6 Session, Sign-in and Maintenance

Staff sign in with the UTeM Microsoft account and a second step (MFA); participants sign in without the second step. There is no MCMS password and no public sign-up; an account that is not recognised gets a page explaining how external learners register at Pejabat Pendaftar or HEPA (D-10, A-06). Two minutes before the 30-minute idle timeout a dialog offers "Stay signed in" or "Sign out now". Planned maintenance is announced in a banner 48 hours ahead.

4.7 Accessibility and Language

- Every control works by keyboard: links, tabs, table rows and menu items are focusable, and Enter or Space activates them. Escape closes dialogs, drawers and menus; focus stays inside an open dialog and returns to the control that opened it.
- Every field has a visible label; placeholders never replace labels. Status is never shown by colour alone; every tag carries its word.
- Text meets 4.5:1 contrast; the navy navigation uses white and light-blue text that meets it on #1e2a6e.
- Every screen, notification and credential exists in BM and EN. The language switch changes the language in place and keeps what the user has typed (nfr-langswitch). Bilingual content fields mark a missing language.
- The layout reflows to one column below 760 px; the side navigation becomes a scrollable strip, and tables scroll inside their frame (ir-responsive).

4.8 Status Vocabulary

Statuses use the exact words of PRD 8.3 and the SRS state diagrams. Each status has a short explanation where it first appears on a screen (nfr-plain). Negative statuses are checked before positive ones, so "Not approved" is never shown as an approval.

Table 5: *Status words by record type*

Record	Statuses
Proposal or version	Draft, Submitted, In review, Returned for amendment, Endorsed, Pending approval, Approved, Not approved, Superseded, Suspended, Terminated
Offering	Not ready, Ready, Open, Full, Running, Ended, Cancelled
Enrolment	Registered, Pending payment, Enrolled, Cancelled, Withdrawn, Completed, Not completed
Module or result	Not started, In progress, Provisional, Verified, Held (integrity), Achieved
Completion or credential	Complete, Issued, Corrected, Replaced, Revoked
QV	Not applied, Preparing, Ready, Submitted, Quality-Verified, Expired, Suspended, Withdrawn
ACR or deposit	ACR prepared, Verified, Consent given, Consent declined, Deposited, Correction needed
Credit	Available, Provisionally redeemed, Redeemed, Expired
Transfer	Submitted, Under evaluation, Approved, Partly approved, Not approved, More evidence needed
Stackable plan	Suitable, Needs review, Not yet suitable
CQI action	Open, In progress, Done (effectiveness pending), Closed

4.9 Formats

Dates are shown as DD/MM/YYYY and times as 24-hour Malaysia time (ir-dates, ir-time). Money is shown as "RM 450.00" (nfr-currency). Names are printed in full, including bin, binti, a/l and a/p (nfr-names). Setting keys and file names use the mono face.

5. Navigation

Each role has its own side navigation, grouped by the kind of work, with its home page first. Counts beside an item come from the live task data and fall as work is done. The profile menu, notifications, help and search are the same for every role; search results always stay inside the signed-in role's own screens, so a search never switches role.

Table 6: *Side navigation by role*

Role	Groups and items (home first)
Student	**Learning:** My learning · Find a micro-credential · Registrations and payments. **Achievements:** Achievement record · Credentials. **Credit:** Credit ledger and ABC (count: consents waiting) · Credit transfer · Stackable plan
Lecturer	**Teaching:** My micro-credentials · Proposals and versions · Reviews for colleagues · Cohorts and results. **Quality:** Quality review (CQI) · QV evidence. **Insight:** Reports
Deputy dean	**Work:** Task queue · Proposals and approval · Offerings and readiness · Results and completion · Credentials · Quality, QV and review · ACR and ABC deposits · Credit transfer · Stackable pathways. **Insight:** Reports · Traceability
CODL administrator	**Administration:** Users and roles · Reference data · Settings · Templates. **Data exchange:** UCreds exchange · Adapter status · Test mailbox. **Records:** Audit trail · Reports · Retention
Industry expert	**Invitations:** My review invitations
Public website	Home · Catalogue · Pathways to a degree · How it works · FAQ · Verify a credential · Sign in; footer: contact, Privacy notice (PDPA), Terms of use, Accessibility

Links that lead from one role's screen into another role's work are not offered: for example, CODL is told which marks are missing but cannot open the lecturer's marks screen, and the deputy dean's traceability view does not open CODL's audit screen. Breadcrumbs show the path inside a role.

6. Screens

This section shows the key screens of each role and what they must do. The complete list of 114 screens with their routes is in Section 6.6. Every screen, with every tab, wizard step, pop-up, warning and side panel, is pictured in Appendix B; every printable document page in Appendix C; and every step of every flow in Appendix D. Every screen is in the prototype with sample data; the behaviour of each is described by the SRS requirements listed in `UIUX/v2/basis/srs-fr-inventory.md`.

6.1 Deputy Dean

The deputy dean lands on the task queue (Figure 1): every decision waiting for FTMK, grouped by area, with the system's checks and how long each has waited. Overdue items (14 days or more) are marked in words. Key screens:

Screen	Must do
Proposal review and endorsement	Show system checks, reviews (Support, Amend, Do not support), comments with blocking flags; verify the category as a separate field; assign reviewers when Submitted; Endorse only when no blocking comment is open and the category is verified; Return for amendment with a reason; open the decision pack.
Record Senate decision	Meeting reference and date (not before endorsement), decision (Approved, Approved with conditions, Not approved, Returned for amendment, Deferred), conditions, minutes extract; state that no credit, QV or transfer status is set; for a new version, the transition rule for enrolled learners.
Confirm readiness	Appendix B checklist from the lecturer; accept uploaded evidence; Ready, Ready with conditions or Not ready; the offering opens only after the decision.
Verify results	Learners with provisional results and their source; held results excluded until the case is closed; Verify creates module achievements only.
Confirm completion	Pre-issue checks per learner; failures listed with reasons and not issued; progress of the issuing job; MCE gets a certificate and no ACR; SAMC without QV covering the cohort gets the credential without the QV claim.
Credential	Revision history; Correct (new revision, same link), Replace (new credential, old shows Replaced) and Revoke (decision reference of the authorised body and a reason).
ACR and deposits	Confirm ACR per batch; deposit queue of consented, verified ACR only; export for ABC; record the ABC reference, giving credit valid for 10 years.
Credit transfer	Outcome-by-outcome comparison per line; Approve blocked for a line with no outcome match or no comparison; decisions Approved, Partly approved, Not approved, More evidence needed; redemption and release of credit.
Stackable pathways and plans	Map micro-credentials to courses (MCE refused); individual plans with the 30% UTeM credit and residency checks, stating that 30% does not mean 70% transfers.
Quality, QV and review	QV records with validity and history; QV result (no more than 3 years); quality reviews with action effectiveness; periodic review outcomes; suspension or termination with a transition plan.

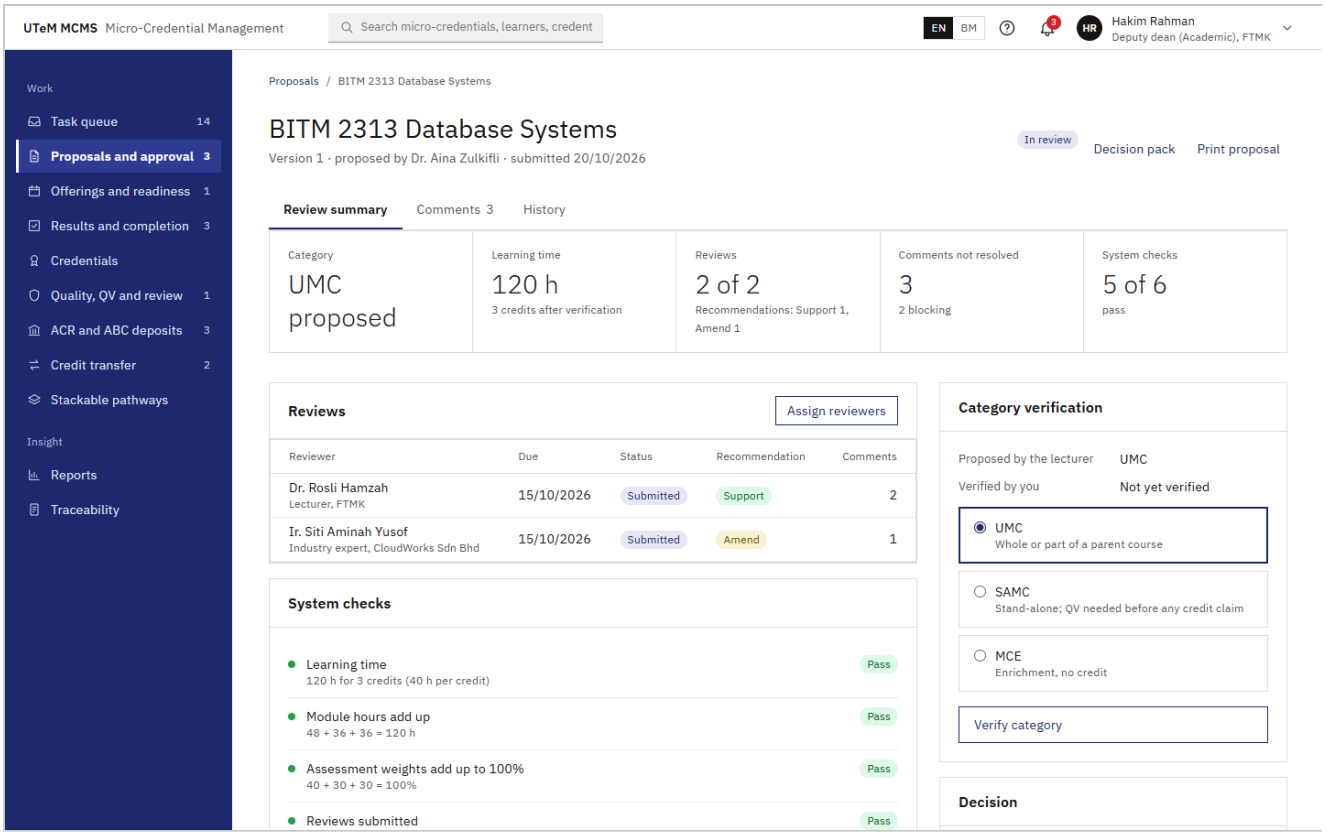


Figure 8: *Proposal review and endorsement (BITM 2313 Database Systems).*

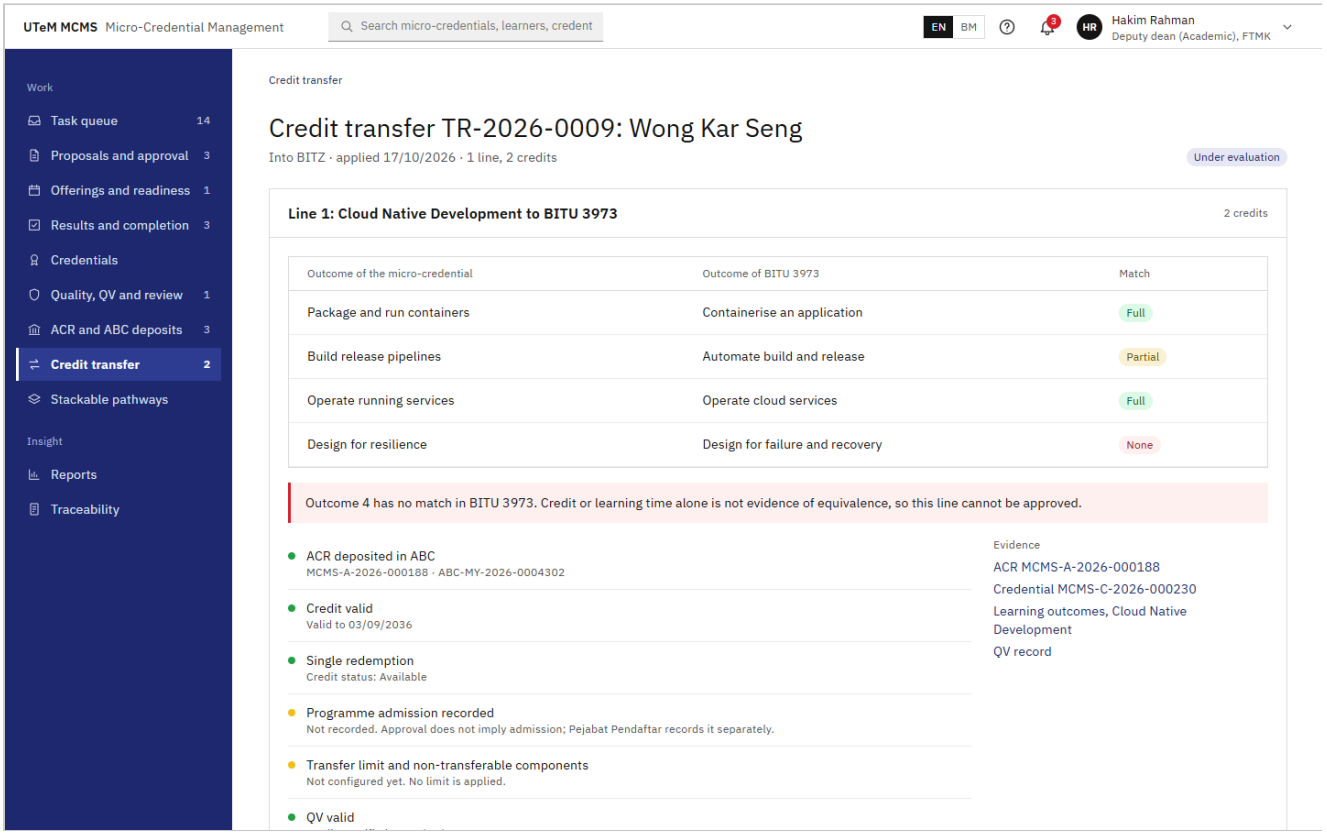


Figure 9: *Credit transfer: a line whose fourth outcome has no match cannot be approved.*

6.2 Lecturer and Reviewer

Screen	Must do
My micro-credentials	"Needs you" list (draft to finish, results to send, quality review due, review invitation, readiness item, QV expiry) and all owned micro-credentials with stage.
New proposal	Three questions decide UMC, SAMC or MCE and explain what the category means for the proposal.
Proposal workspace	Appendix A sections with live checks; SLT must equal the parent course for a UMC; weightings must total 100%; recognition fields follow the category (an MCE has no credit); comments tab to answer reviewers; Submit only when checks pass.
Change to an approved version	Minor or material; a material change becomes a new version submitted for review while learners stay on their version.
Review a colleague's proposal	Conflict of interest first, then the proposal; rate items; recommend Support, Amend or Do not support.
Cohort results	Results from UCreds marked Provisional; missing marks listed first and required; integrity flag; Send for verification; the lecturer cannot verify their own entries.
Readiness, quality review, QV evidence	Lecturer's readiness items with evidence upload; quality review with indicators and actions closed only after an effectiveness check; QV evidence checklist and "Ready to submit" gated by a completed cohort.

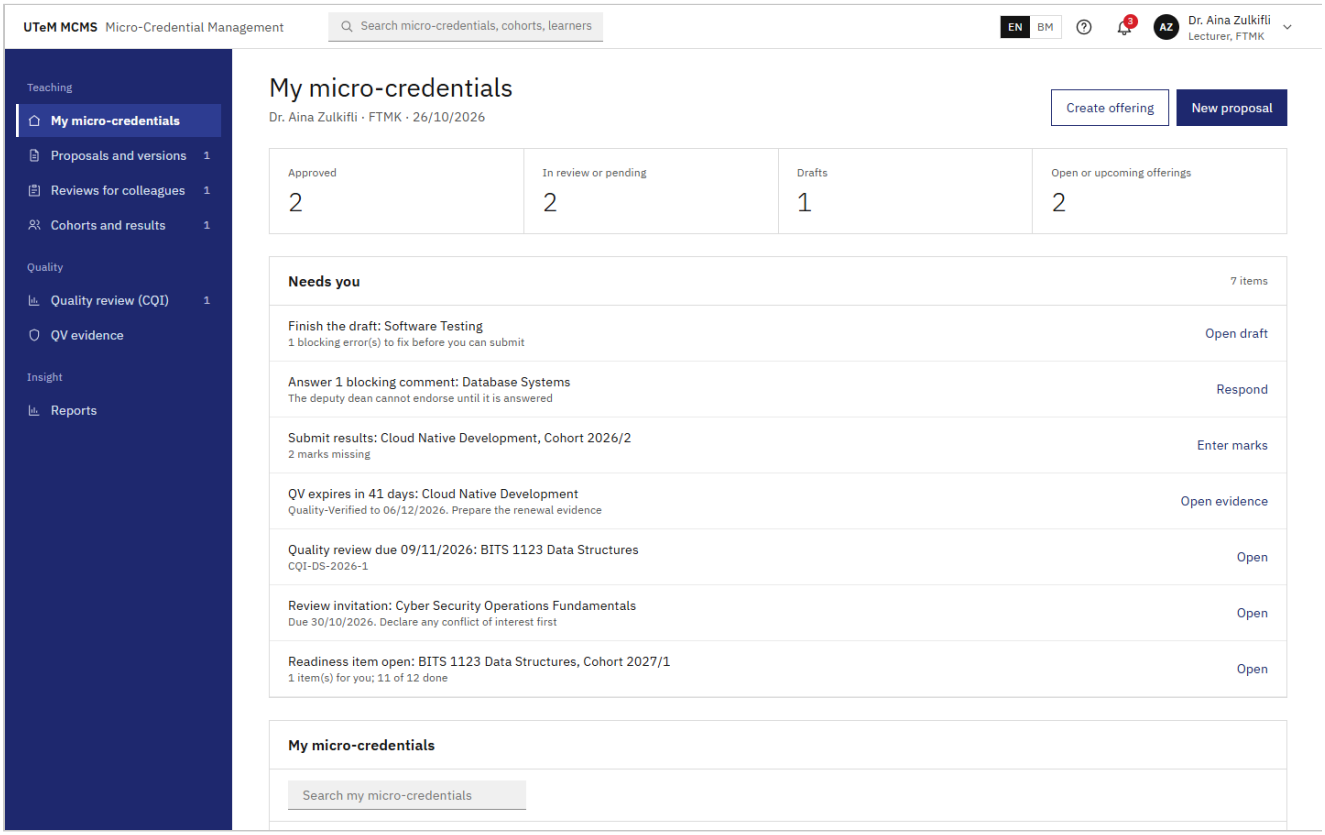


Figure 10: *Lecturer home: what needs Dr. Aina today, then her micro-credentials.*

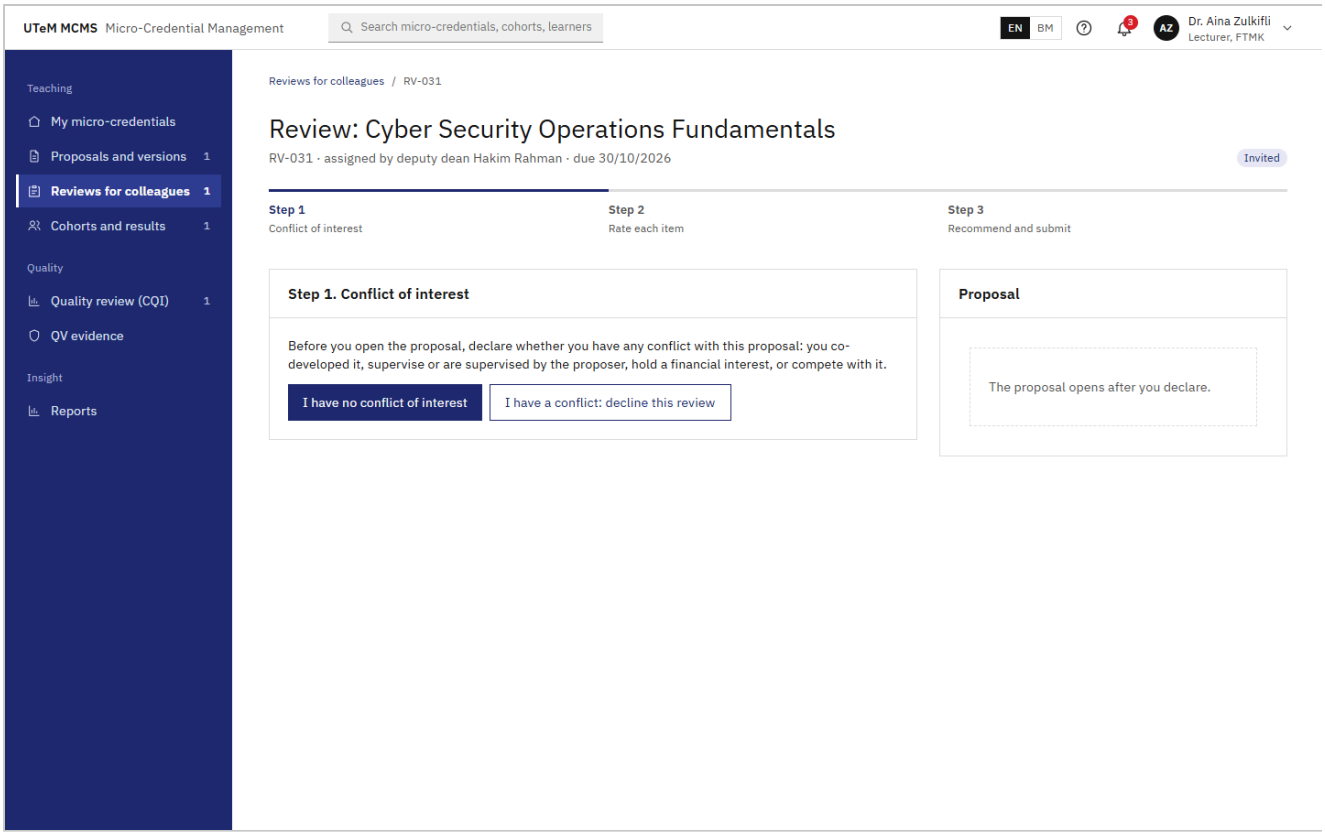


Figure 11: *Colleague review: the proposal opens only after the conflict-of-interest declaration.*

6.3 Student

Student screens are web screens in the same shell as staff, and reflow to a phone.

Screen	Must do
My learning	In progress with module progress, waiting for verification or completion (worded from the cohort's real state), achieved, and actions for the student.
Register	Four steps: eligibility (each rule shown with its result), acknowledgement of the recognition statement and PDPA notice (version and date recorded), dummy payment (failure and retry, never a double charge, capacity checked), confirmation with UCreds access and receipt.
Registrations and payments	Status, receipt, withdraw before the deadline, cancel a pending payment.
Achievement record and credentials	Achievement record labelled "Internal record, not an Academic Credit Record"; credentials with PDF certificate, signed credential, share link and revision history; Replaced and Revoked shown honestly.
Credit ledger and ACR	One ledger with Available, Provisionally redeemed, Redeemed and Expired; consent to deposit with Give, Decline or Postpone; withdrawing consent asks for a reason.
Credit transfer and plan	Apply with deposited, available credit only (MCE and used credit not selectable, with the reason); decision and reason shown; upload more evidence when asked; stackable plan from real records.

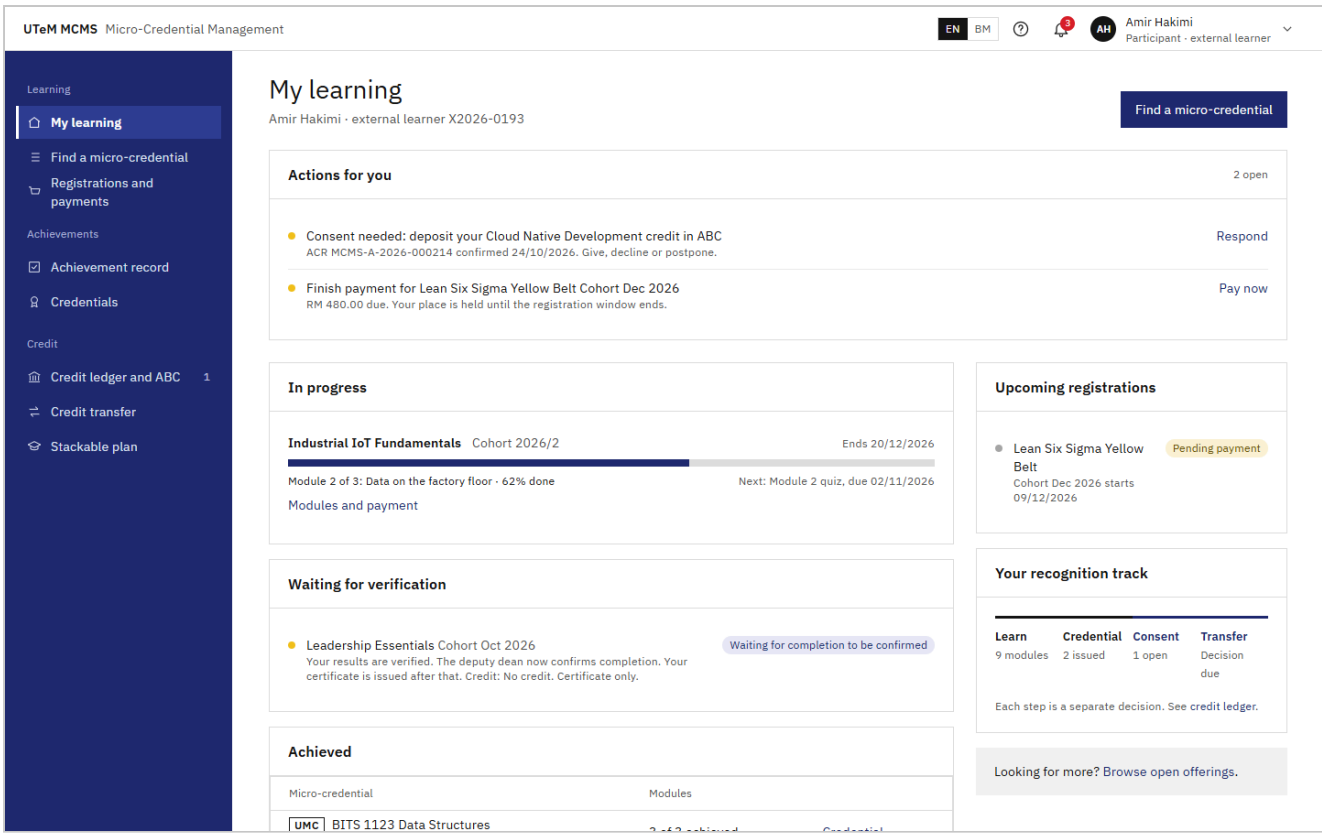


Figure 12: *Student home (Amir Hakimi).*

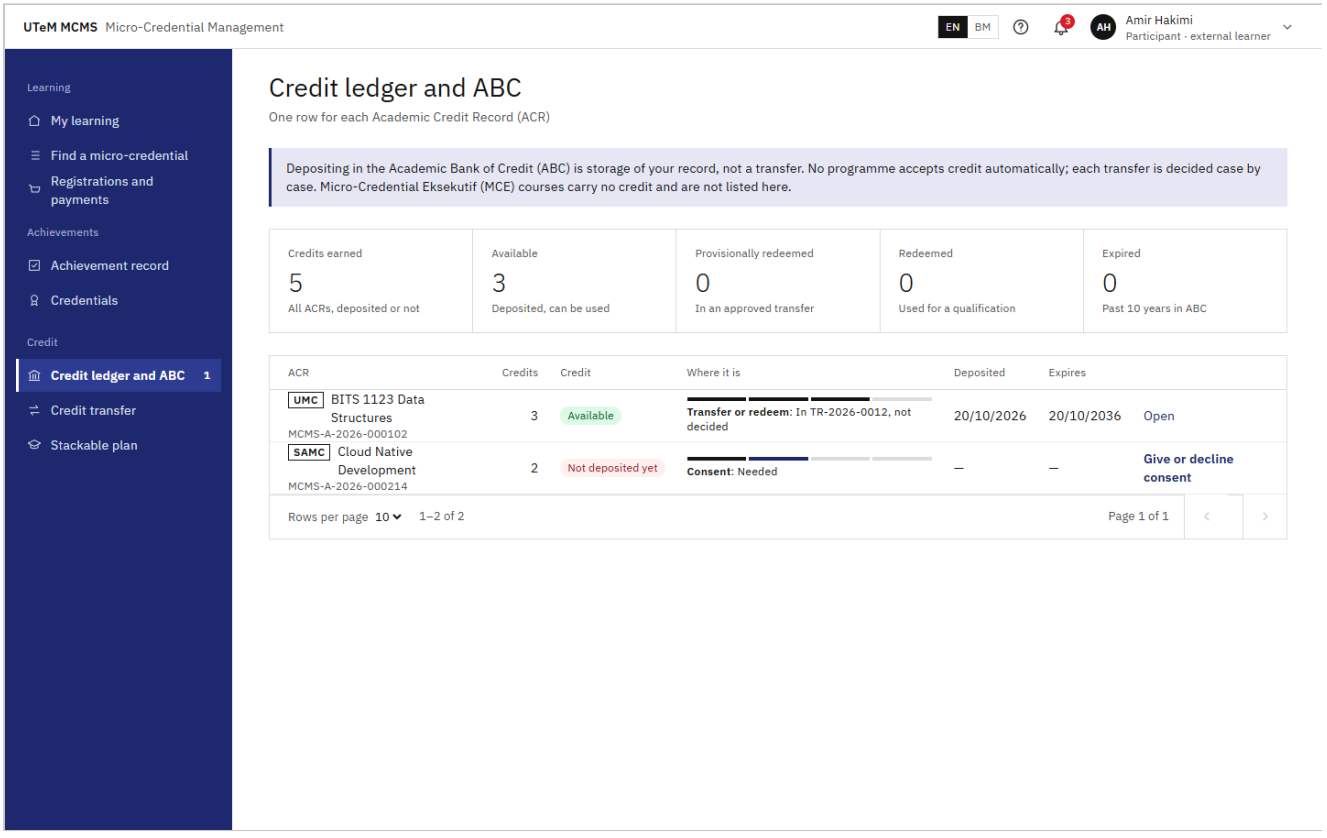


Figure 13: *Credit ledger: one row per ACR with its stage and the honest note that deposit is storage, not transfer.*

6.4 Public Website

The public site informs before it sells. The home page explains the three categories, features the next open offerings, explains how recognition works and links to verification. Offering pages follow the

structure of the UCreds course pages: a facts block (institution, format, effort per week, dates, duration, cost, credit, places) and tabs for Overview, Syllabus, Assessment, Prerequisites, Recognition and FAQ, with an Enrol button. Credit is shown only when it is verified for the cohort's dates; the separate-stages notice states that completion, ACR, deposit and transfer are separate with no guarantee.

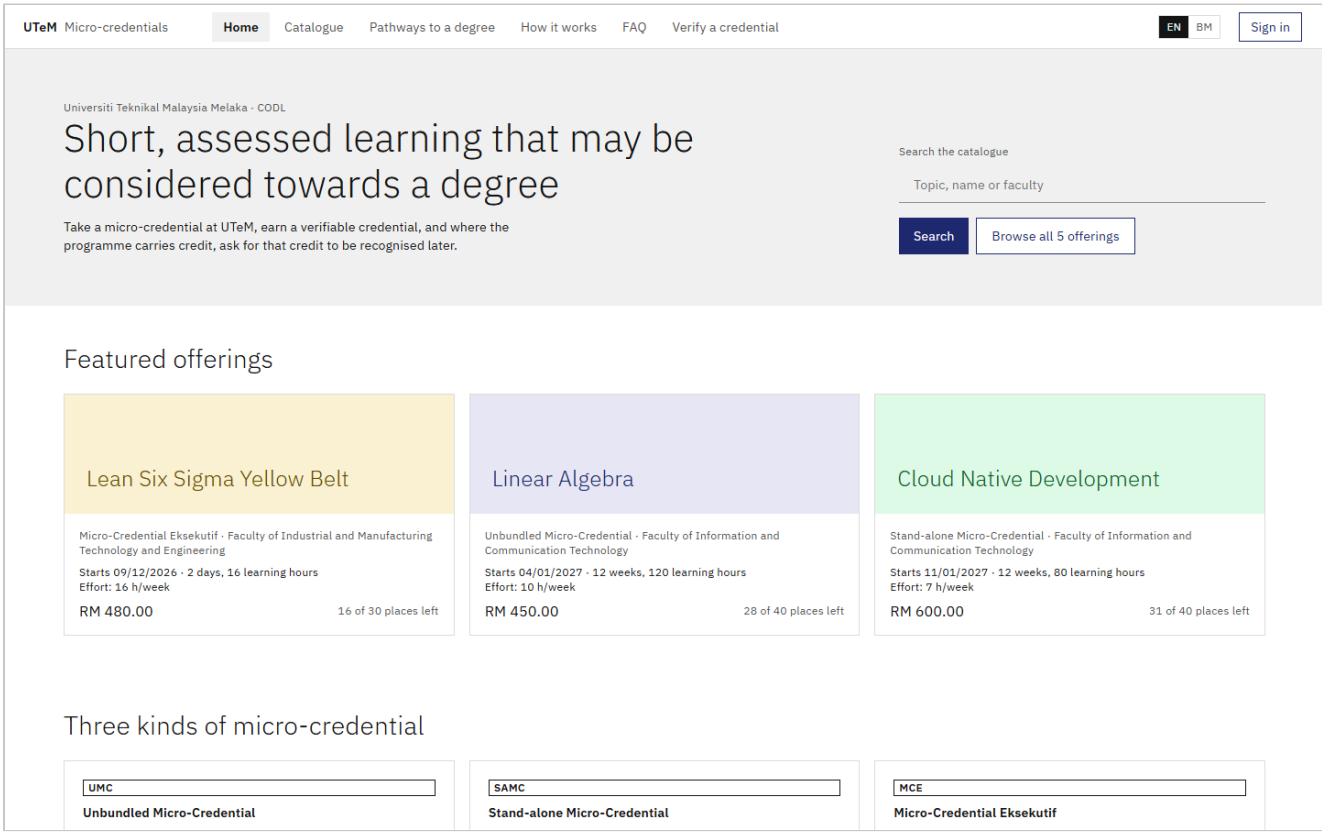


Figure 14: *Public home page.*

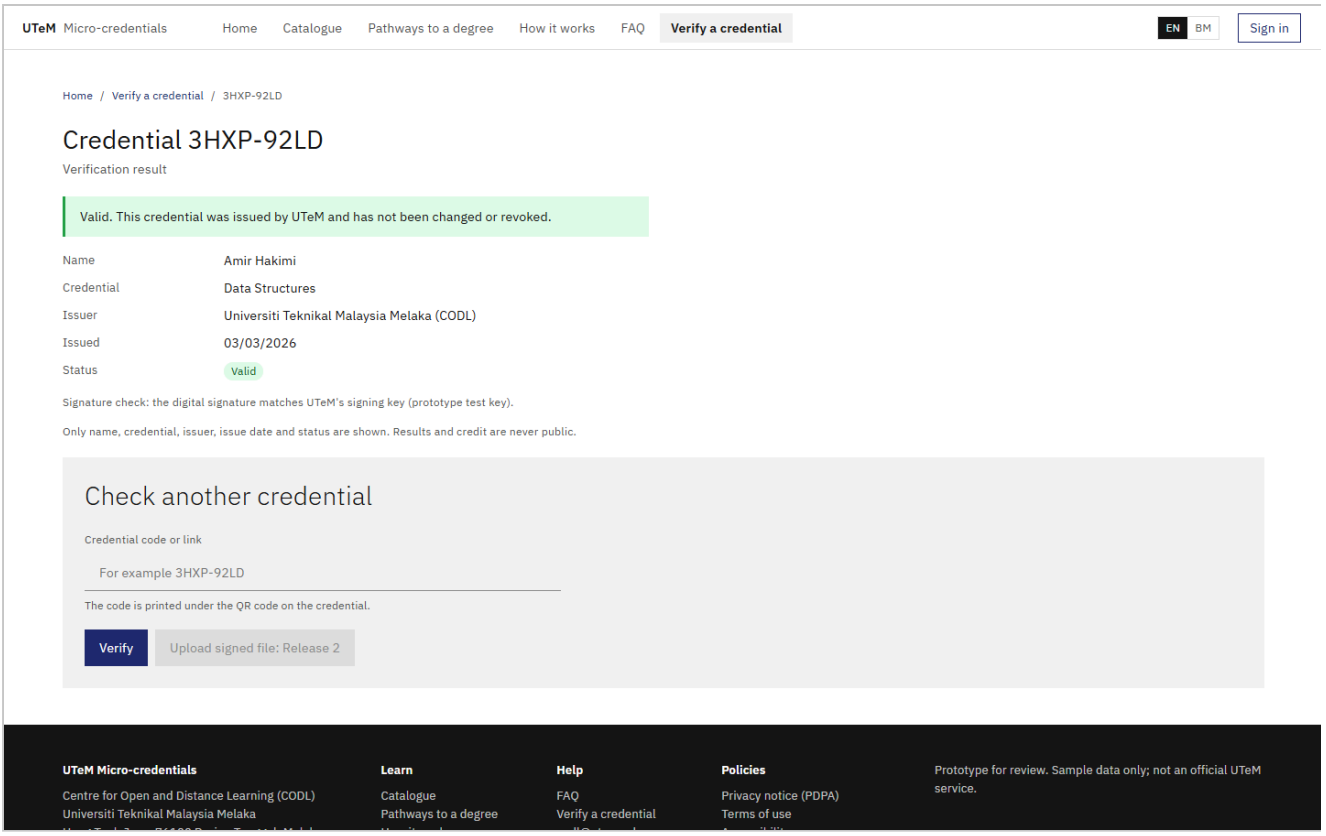


Figure 15: *Public verification: only name, credential, issuer, issue date and status, without signing in. Corrected, Replaced, Revoked and Not found each have their own result.*

6.5 CODL Administrator

CODL lands on users and roles with a "Needs you" panel (failing adapter, templates missing a language, disposals waiting for the Registrar, imports with problems). Role assignment requires a scope, a reason and an effective date; a future date is scheduled; an administrator cannot change their own roles or remove the only deputy dean of a faculty. Settings changes need a reason and a future effective date and keep a history; the public verification fields can only be chosen from the five allowed fields. Templates cannot be published when the BM and EN placeholders differ or when the text makes an unsupported claim such as an automatic transfer. UCreds imports stage rows as Provisional and never create achievements. No CODL screen has an academic decision; where one might be expected, a note names who decides.

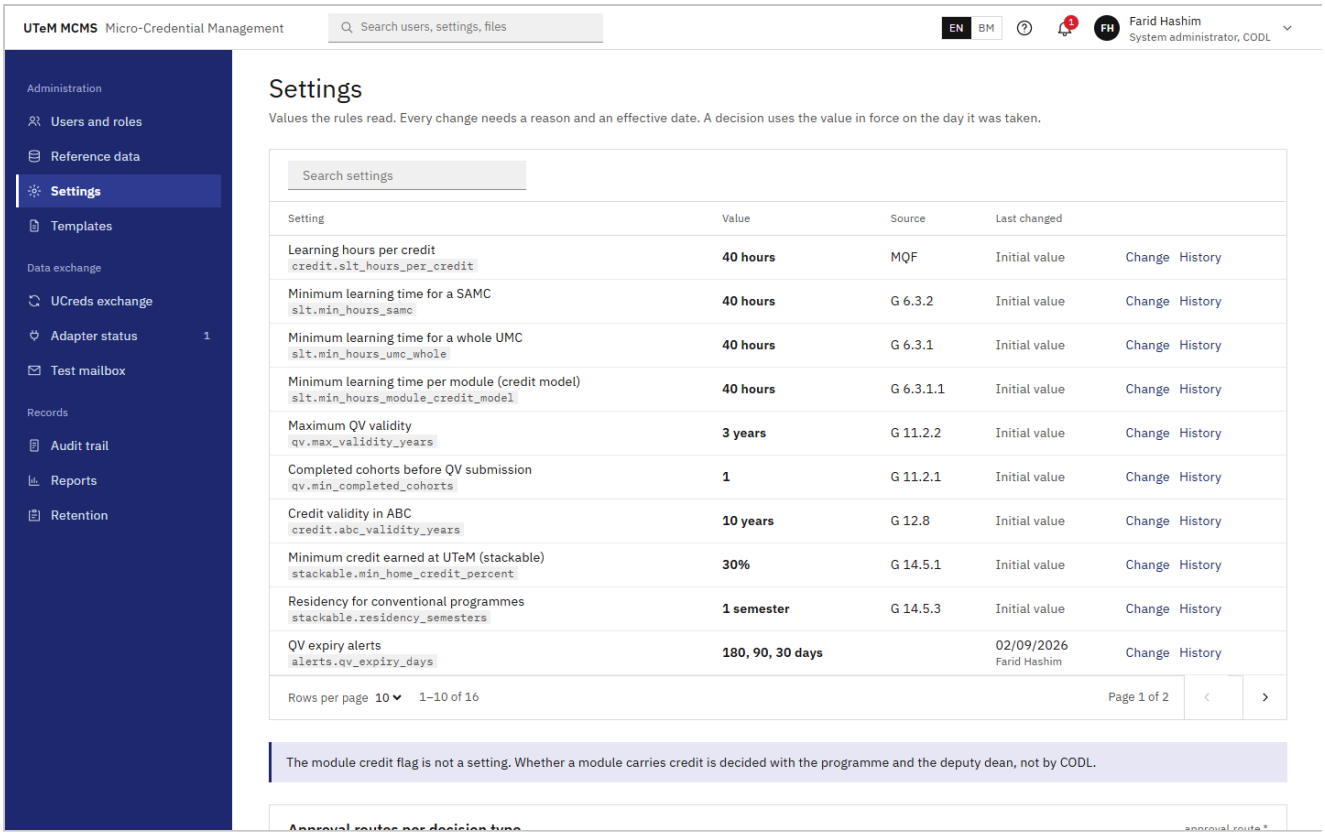


Figure 16: *Settings with source and history; changes take effect from a stated date.*

6.6 Screen Inventory

The 114 screens of the approved prototype, by role. The ID is used in tests and in the SDD; the route opens the screen in the prototype with sample data. Routes with a `:` take a record identifier.

Common and sign-in

Table 7: *Screens for Common and sign-in (10)*

ID	Screen	Route
CMN-01	Personas	`#/`
CMN-02	Sign in	`#/signin`
CMN-03	Verify it is you	`#/signin/mfa`
CMN-04	Account not recognised	`#/signin/noaccount`
CMN-05	Test sign-in	`#/signin/stub`
CMN-06	My profile	`#/profile`
CMN-07	Notifications	`#/notifications`
CMN-08	Help	`#/help`
CMN-09	Search	`#/search`
CMN-10	Something went wrong	`#/error`

Public website

Table 8: *Screens for Public website (11)*

ID	Screen	Route
PUB-01	Micro-credentials at UTeM	`#/public/home`

ID	Screen	Route
PUB-02	Catalogue	`#/public/catalogue`
PUB-03	Offering	`#/public/offering/:id`
PUB-04	Pathways to a degree	`#/public/pathways`
PUB-05	How recognition works	`#/public/how`
PUB-06	FAQ	`#/public/faq`
PUB-07	Verify a credential	`#/public/verify`
PUB-08	Verification result	`#/public/verify/:code`
PUB-09	Privacy notice	`#/public/privacy`
PUB-10	Terms of use	`#/public/terms`
PUB-11	Accessibility statement	`#/public/accessibility`

Student (P8)

Table 9: *Screens for Student (P8) (14)*

ID	Screen	Route
ST-01	My learning	`#/st/home`
ST-02	Find a micro-credential	`#/st/catalogue`
ST-03	Register	`#/st/register/:cohort`
ST-04	Registrations and payments	`#/st/enrolments`
ST-05	Registration	`#/st/enrolment/:id`
ST-06	Achievement record	`#/st/record`
ST-07	Credentials	`#/st/credentials`
ST-08	Credential	`#/st/credential/:id`
ST-09	Credit ledger and ABC	`#/st/ledger`
ST-10	Academic Credit Record	`#/st/acr/:id`
ST-11	Credit transfer	`#/st/transfer`
ST-12	New transfer application	`#/st/transfer/new`
ST-13	Transfer application	`#/st/transfer/:id`
ST-14	Stackable plan	`#/st/plan`

Lecturer (P1)

Table 10: *Screens for Lecturer (P1) (18)*

ID	Screen	Route
LEC-01	My micro-credentials	`#/lec/home`
LEC-02	Micro-credential	`#/lec/mc/:id`
LEC-03	Proposals and versions	`#/lec/proposals`
LEC-04	New proposal	`#/lec/new`
LEC-05	Proposal workspace	`#/lec/proposal/:id`
LEC-06	Request a change	`#/lec/change/:id`
LEC-07	Reviews for colleagues	`#/lec/reviews`
LEC-08	Review a proposal	`#/lec/review/:id`

ID	Screen	Route
LEC-09	Cohorts and results	`#/lec/cohorts`
LEC-10	Cohort results	`#/lec/cohort/:id`
LEC-11	Create an offering	`#/lec/offering/new`
LEC-12	Readiness checklist	`#/lec/readiness/:id`
LEC-13	Quality review (CQI)	`#/lec/cqi`
LEC-14	Quality review record	`#/lec/cqi/:id`
LEC-15	QV evidence	`#/lec/qv`
LEC-16	QV evidence	`#/lec/qv/:id`
LEC-17	Reports	`#/lec/reports`
LEC-18	Report	`#/lec/report/:id`

Industry expert (X2)

Table 11: *Screens for Industry expert (X2) (2)*

ID	Screen	Route
EXT-01	My review invitations	`#/ext/home`
EXT-02	Comment on a proposal	`#/ext/review/:id`

Deputy dean (P2)

Table 12: *Screens for Deputy dean (P2) (29)*

ID	Screen	Route
DD-01	Task queue	`#/dd/queue`
DD-02	Proposals and approval	`#/dd/proposals`
DD-03	Proposal review	`#/dd/proposal/:id`
DD-04	Record Senate decision	`#/dd/senate/:id`
DD-05	Decision pack	`#/dd/pack`
DD-06	Offerings and readiness	`#/dd/offerings`
DD-07	Confirm readiness	`#/dd/readiness/:id`
DD-08	Results and completion	`#/dd/results`
DD-09	Verify results	`#/dd/verify/:id`
DD-10	Confirm completion	`#/dd/completion/:id`
DD-11	Credentials	`#/dd/creds`
DD-12	Credential	`#/dd/cred/:id`
DD-13	Quality, QV and review	`#/dd/quality`
DD-14	QV record	`#/dd/qv/:id`
DD-15	Quality review	`#/dd/cqi/:id`
DD-16	Periodic review	`#/dd/review/:id`
DD-17	Suspension or termination	`#/dd/suspend/:id`
DD-18	ACR and ABC deposits	`#/dd/acr`
DD-19	Confirm ACR	`#/dd/acr/:id`
DD-20	Record ABC deposit	`#/dd/deposit/:id`

ID	Screen	Route
DD-21	Credit transfer	`#/dd/transfer`
DD-22	Transfer decision	`#/dd/transfer/:id`
DD-23	Stackable pathways	`#/dd/stack`
DD-24	Pathway mapping	`#/dd/pathway/:id`
DD-25	Individual stackable plan	`#/dd/plan/:id`
DD-26	Reports	`#/dd/reports`
DD-27	Report	`#/dd/report/:id`
DD-28	Traceability	`#/dd/trace`
DD-29	Trace a participant	`#/dd/trace/:who`

CODL system administrator (S1/P5)

Table 13: *Screens for CODL system administrator (S1/P5) (13)*

ID	Screen	Route
CODL-01	Users and roles	`#/codl/users`
CODL-02	User	`#/codl/user/:id`
CODL-03	Reference data	`#/codl/refdata`
CODL-04	Settings	`#/codl/settings`
CODL-05	Templates	`#/codl/templates`
CODL-06	Template	`#/codl/template/:id`
CODL-07	UCreds exchange	`#/codl/ucreds`
CODL-08	Adapter status	`#/codl/adapters`
CODL-09	Test mailbox	`#/codl/mailbox`
CODL-10	Audit trail	`#/codl/audit`
CODL-11	Reports	`#/codl/reports`
CODL-12	Report	`#/codl/report/:id`
CODL-13	Retention	`#/codl/retention`

Printable documents

Table 14: *Screens for Printable documents (17)*

ID	Screen	Route
DOC-01	Payment receipt	`#/doc/receipt/:id`
DOC-02	Enrolment confirmation	`#/doc/enrolment/:id`
DOC-03	Certificate	`#/doc/certificate/:id`
DOC-04	Digital credential	`#/doc/credential/:id`
DOC-05	Achievement record	`#/doc/record/:who`
DOC-06	Academic Credit Record	`#/doc/acr/:id`
DOC-07	Proposal print	`#/doc/proposal/:id`
DOC-08	Senate decision pack	`#/doc/pack/:id`
DOC-09	Report	`#/doc/report/:id`
DOC-10	ABC deposit export	`#/doc/abc/:id`

ID	Screen	Route
DOC-11	QV submission package	`#/doc/qvpack/:id`
DOC-12	Credit transfer decision letter	`#/doc/transfer/:id`
DOC-13	Suspension notice	`#/doc/suspend/:id`
DOC-14	Import validation report	`#/doc/import/:file`
DOC-15	UCreds enrolment export	`#/doc/export/:cohortId`
DOC-16	Audit trail export	`#/doc/audit`
DOC-17	Documents and PDFs	`#/doc/index`

7. User Flows

A flow is a process walked across screens and, often, across roles. Each flow of Table 16 is in the prototype's Flows page; starting one shows a step bar at the bottom of the screen that says what is happening and what to click next. Negative paths have flows of their own, for example a payment failure, a transfer line that cannot be approved, or a credential that was revoked.

7.1 End-to-End Journey

Figure 17 shows the journey of SRS use case A.5 as it runs in the prototype (flow FL-19). Each box is a separate decision by a separate role; the system issues credentials and generates the ACR only after the deputy dean's decision.

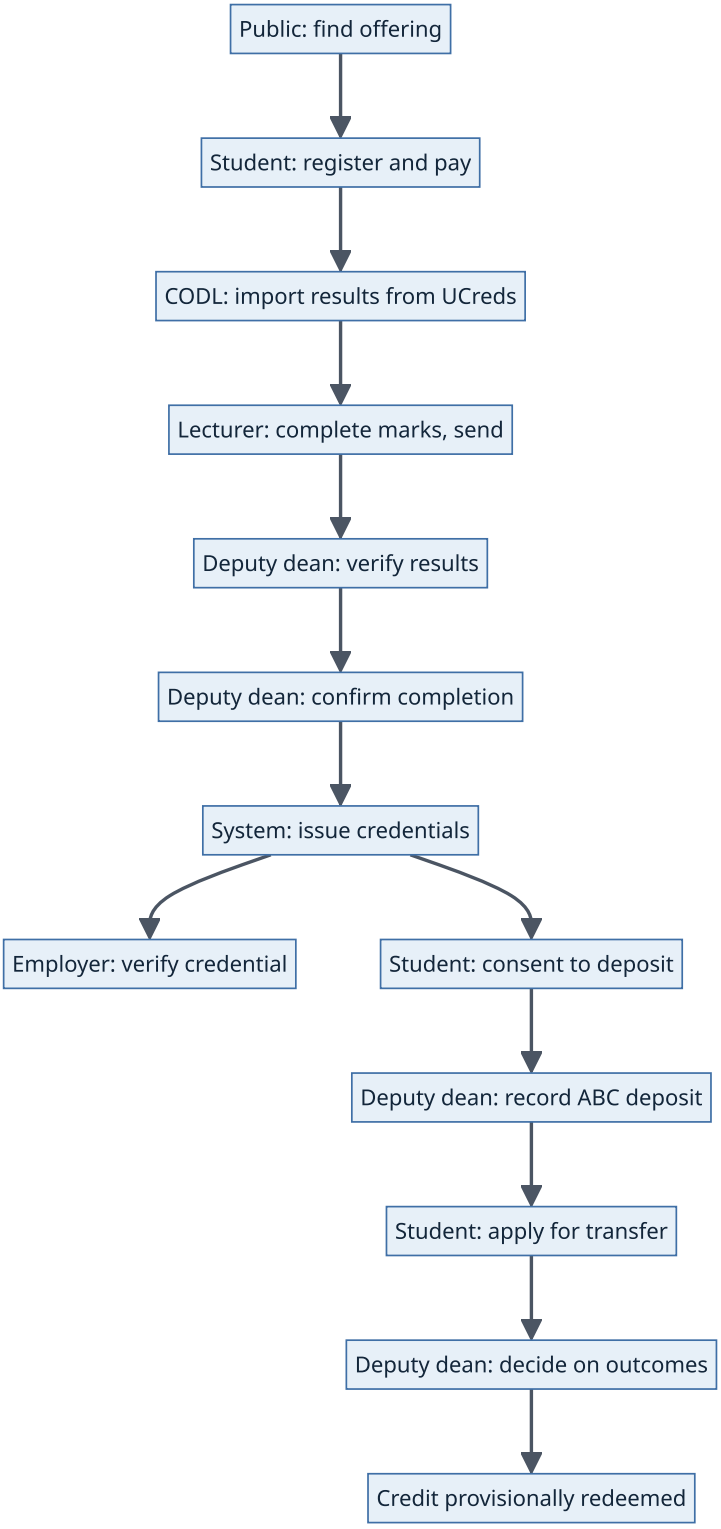


Figure 17: *End-to-end journey (FL-19): one stage, one decision, one role.*

7.2 Hand-offs Between Roles

When one role finishes a step, the next role sees it at once: a task appears in the deputy dean's queue, a notification reaches the person affected, and the record's status and history change. Table 15 lists the hand-offs the design guarantees; each is tested end to end in the prototype.

Table 15: *Hand-offs between roles*

From	Action	Lands with	Result seen
Lecturer	Sends cohort results for verification	Deputy dean	Task "Verify results"; every learner's provisional result on the verify screen
Lecturer	Sends the readiness checklist	Deputy dean	Task "Confirm readiness"; uploaded evidence ready to accept; after Ready the offering appears in the public catalogue
Lecturer	Answers a review comment	Deputy dean	Comment shows as Answered and can be resolved; endorsement unblocks when no blocking comment is open
Deputy dean	Confirms completion	Student	Credentials issued; enrolment Completed; notification
Student	Gives consent to deposit	Deputy dean	Task "Record ABC deposit"; ACR in the deposit queue
Deputy dean	Records the deposit	Student	Ledger shows the credit Available for 10 years
Student	Applies for transfer	Deputy dean	Task "Decide credit transfer" with the evidence attached
Deputy dean	Approves a transfer line	Student	Credit Provisionally redeemed; it can no longer be selected for another application
Deputy dean	Asks for more evidence	Student	Upload form on the application; the upload reopens the deputy dean's task
CODL	Imports UCreds results with problems	Lecturer	Notification naming the missing marks; rows needing marks listed first

7.3 Flow Catalogue

Table 16: *Walk-through flows in the prototype*

ID	Flow	Actors	Steps
FL-00	**Tour the deputy dean task queue**. How the queue, tabs, search and pagination behave	Deputy dean	2
FL-01	**Sign in with SSO and MFA, land on the task page**. UTeM Microsoft sign-in, second step, then the task queue	Any staff	3
FL-02	**Propose a UMC, fix the SLT, submit**. Category wizard, a blocked draft at 110 h, the fix to 120 h, and submit	Lecturer	5
FL-03	**Review a colleague's proposal**. Conflict of interest first, then rate, then recommend	Lecturer as reviewer	4
FL-04	**Endorse a proposal, then record the Senate decision**. Endorse is blocked until the category is verified and the blocking comment is resolved	Deputy dean	7
FL-05	**Create an offering and confirm readiness**. The cohort opens only after every Appendix B item is done and you confirm	Lecturer, deputy dean	4
FL-06	**Find, register and pay**. Public offering page to a paid registration for Cloud Native Development 2027/1	Student	6
FL-06b	**Payment failure and retry**. A declined card changes nothing; a second try works and charges once	Student	3
FL-07	**Import UCreds results with a bad row**. Import results-CND-2026-2.csv with 2 problems; the lecturer enters the missing marks and submits for verification	CODL, lecturer, deputy dean	3

ID	Flow	Actors	Steps
FL-08	**Verify results, then confirm completion**. Two separate decisions: verifying gives module badges, confirming completion makes the system issue credentials	Deputy dean	3
FL-09	**Share a credential and have it verified**. Share link, a genuine check, a revoked credential and a code that does not exist	Student, public	4
FL-10	**Correct a credential name**. A correction makes a new revision; the original is kept and the public link shows Corrected	Deputy dean	2
FL-11	**Consent to ABC deposit, confirm ACR, record deposit, see ledger**. Amir gives consent, Hakim finds the ACR in the deposit queue and records the deposit, and the ledger shows the credit	Student, deputy dean	4
FL-12	**Apply for credit transfer; decide by outcome; provisionally redeemed; second use blocked**. Amir applies, Hakim approves, the credit is held, and a second case with an unmatched outcome is blocked	Student, deputy dean	5
FL-13	**Stackable plan: 25% "Not yet suitable" and a suitable plan**. Two individual plans for BITZ, one below 30% UTeM credit and one that is suitable	Deputy dean	4
FL-14	**QV evidence, gate, record QV result; expiry removes claims**. Evidence on the lecturer side, then the deputy dean builds the package, records the submission and the result	Lecturer, deputy dean	4
FL-15	**Quality review: close an action only after the effectiveness check**. The lecturer view, then submit to the deputy dean	Lecturer, deputy dean	4
FL-16	**Suspend an offering with a transition plan**. Periodic review leads to a suspension with a transition plan; earned credentials stay valid	Deputy dean	4
FL-17	**Assign a scoped role and change a setting**. Give a role with scope, reason and date; change a setting with a reason; see its history and the audit trail	CODL administrator	4
FL-18	**Trace a participant from approval to credit use; audit trail**. Follow Amir through every stage, then read the audit rows for the same records	Deputy dean	3
FL-19	**End-to-end: Amir from catalogue to credit used in a degree**. SRS use case A.5 and PRD demo scenario: every recognition stage as its own decision	All roles	17
FL-20	**Industry expert comments on a proposal**. Conflict of interest, then comment only	Industry expert	3
FL-21	**Withdraw from an enrolment**. Withdraw with a reason; achieved modules stay on record	Student	3
FL-22	**A visitor explores the public site**. From the home page to a filtered catalogue, an offering, how recognition works and pathways	Public visitor	5
FL-23	**An employer verifies a credential**. Valid, corrected, revoked and not-found results	Tan Mei Ling, HR manager (public)	7
FL-24	**Documents and PDFs**. Open the printable documents: receipt, certificate, Digital Credential, full report, ABC export.	All roles	8

8. Printable Documents and Content

8.1 Printable Documents

Every download, print or export in MCMS opens a document laid out on A4 pages, which the user can print or save as PDF. Where a machine-readable file is needed, the document also offers a real download. Each page has a header with the UTeM name, the document title, reference and date, and a footer with the reference, page number, generation time and "Generated by MCMS". Documents for participants are bilingual, with BM beneath EN (D-02). No identity number, email or marks appear on documents that leave UTeM.

Table 17: *Printable documents*

Document	Opened from	Content	File
Payment receipt	Student registrations	Receipt number, payer, offering, fee breakdown, method, reference, total; "No payment received" when unpaid	PDF
Enrolment confirmation	Student registrations	Access to UCreds, dates, withdrawal deadline, recognition statement, separate-stages notice	PDF
Certificate	Student and deputy dean credential pages	Landscape, bilingual; holder, micro-credential, category, verified credit only, credential ID, verification code and link, QR, two signatures; Revoked band or revision line when relevant	PDF
Digital Credential	Credential pages	Signed credential shaped on Open Badges 3.0 / W3C VC with a summary and proof	JSON, PDF
Achievement record	Student, profile	Labelled "Internal record, not an Academic Credit Record"; enrolments, modules, badges, completions	PDF
Academic Credit Record	Ledger, deposit screens	Credits, MQF level, completion, verifier, consent, ABC reference and expiry; storage, not transfer	PDF
Proposal (Appendix A)	Lecturer workspace, deputy dean review	All Appendix A sections with SLT totals, weightings, alignment, version and status	PDF
Senate decision pack	Deputy dean proposal review	Cover, summary, category verification, reviews, comment resolution, endorsement, space for the minute reference	PDF
Report (10 kinds)	Reports	Cover, scope, period, filters, summary figures, tables that add up, method notes, page numbers	PDF, CSV
ABC deposit export	ACR and deposits	Batch header, one row per ACR, totals, format note	PDF, CSV
QV submission package	QV records	Cover, evidence index, cohort results, quality reviews, declaration	PDF
Transfer decision letter	Transfer case	Outcome per line, credits approved, conditions, statement that refusal leaves ACR and credentials unchanged	PDF
Suspension notice	Suspension	Notice to participants with the transition plan; earned credentials stay valid	PDF
UCreds import report	UCreds exchange	File, row counts, rows staged as Provisional, rejected rows with reasons	PDF, CSV
Enrolment export	UCreds exchange, lecturer cohort	Template version, columns, one row per enrolled learner	PDF, CSV
Audit export	Audit trail	Entries with an integrity line	PDF, CSV

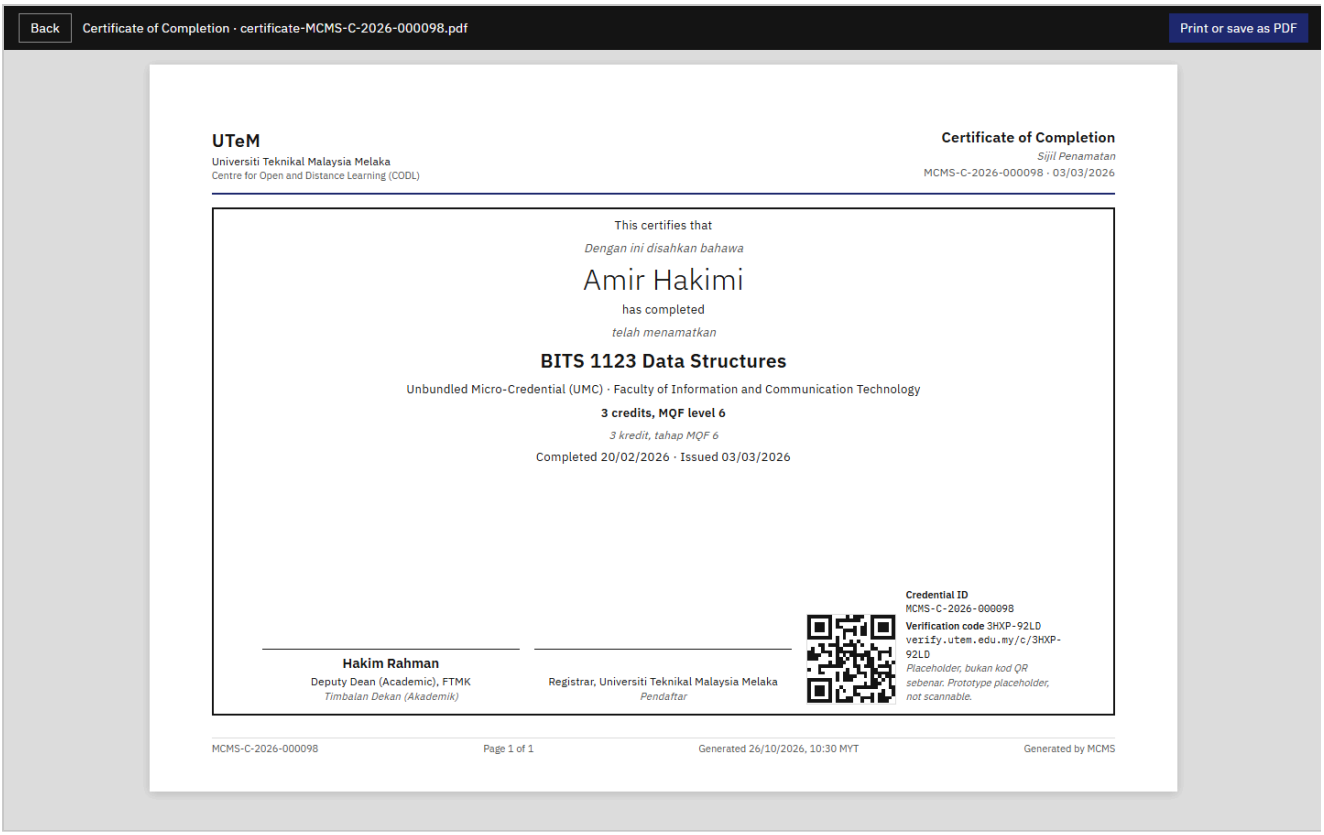


Figure 18: *Certificate document with the print bar (sample PDFs of every document are in* `UIUX/v2/samples`*).*

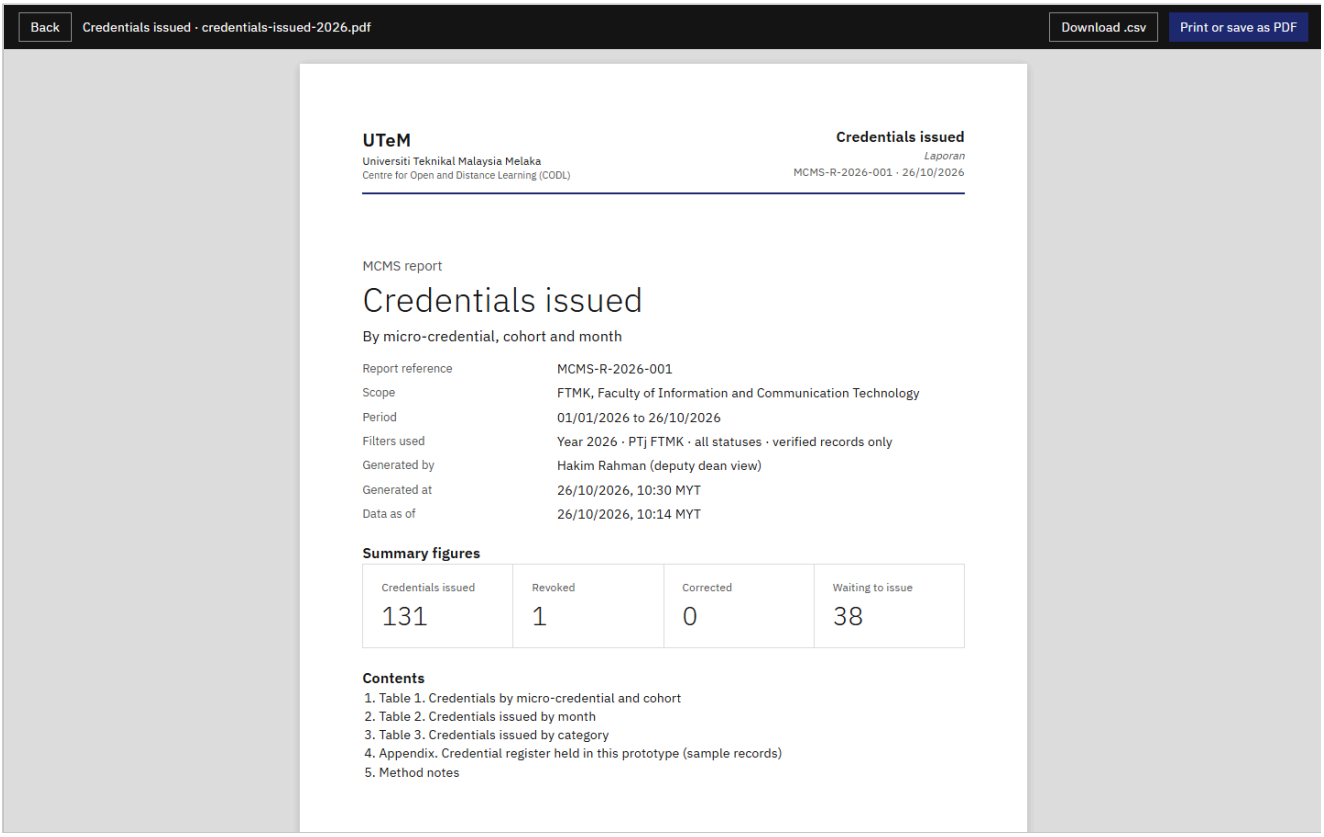


Figure 19: *Full report template (credentials issued), with PDF and CSV output.*

8.2 Content and Copy

Rule	Example
Plain, specific sentences with real figures	"Verifying issues module badges to 27 learners. Completion is confirmed separately."
Sentence case; no exclamation marks; no emoji	"Record Senate decision", not "RECORD SENATE DECISION!"
Never promise credit or transfer	"You may later apply to have it recognised; each transfer is decided case by case."
Name the category as in the Glossary	Unbundled Micro-Credential (UMC), Stand-alone Micro-Credential (SAMC), Micro-Credential Eksekutif (MCE)
No requirement codes in user text	Say the rule ("at least 1 completed cohort is required"), not its code
Say why a control is disabled	"Submit · 1 check failing", "Withdraw: the deadline 30/11/2026 has passed"
Status words exactly as in Section 4.8	"Provisionally redeemed", not "Used"
Both languages for every string	Missing BM or EN is marked on content fields and blocks template publishing

9. Traceability and Verification

9.1 Requirements to Screens

Table 18 maps each SRS functional module to the screens that deliver it. The per-requirement detail (actor, action, confirmation, result and blocking checks) is kept in `UIUX/v2/basis/srs-fr-inventory.md`.

Table 18: *SRS modules and the screens that deliver them*

SRS section	Module	Screens
3.1	Identity, roles and access	Sign-in, MFA, profile, CODL users and roles, settings, templates, reference data, adapter status
3.2	Category determination and proposal	Lecturer new proposal, proposal workspace, proposals and versions
3.3	Review, approval and versioning	Lecturer and expert review, deputy dean proposal review, Senate decision, decision pack, change to an approved version
3.4	Offering, catalogue and readiness	Lecturer create offering and readiness, deputy dean readiness, public catalogue and offering
3.5	Registration, enrolment and cohorts	Student register, registrations and payments, enrolment; lecturer cohorts
3.6	Learning platform connection	CODL UCreds exchange, import report, enrolment export
3.7	Assessment and achievement verification	Lecturer cohort results; deputy dean results and verify
3.8	Achievement Record	Student achievement record; deputy dean completion
3.9	Credential issuance and verification	Deputy dean completion and credentials; student credentials; public verify; certificate and credential documents
3.10	Quality Verification for SAMC	Lecturer QV evidence; deputy dean quality, QV record; QV package
3.11	ACR and ABC deposit	Student ledger and ACR consent; deputy dean ACR and deposit; ABC export
3.12–3.13	Credit transfer and single redemption	Student transfer; deputy dean transfer; transfer letter
3.14	Stackable degree pathway	Deputy dean pathways and plans; student plan; public pathways
3.15	Monitoring and CQI	Lecturer quality review; deputy dean quality review, periodic review, suspension
3.16	Records, audit trail and reporting	Reports, traceability, audit trail, retention, report documents
3.17	Notifications and integration hub	Notifications, preferences, adapter status, test mailbox

9.2 How the Design Was Verified

The prototype was checked in three ways before approval:

1. **Automated screen checks.** Every screen draws without script errors and every link, dialog, drawer and action it contains exists (the 114 screens plus the prototype's two index pages, 51 dialogs and panels, 0 problems).
2. **Hand-off test.** A scripted walk through the real screens checks the role hand-offs of Section 7.2 (14 checks, all passing).
3. **Design review.** Three independent reviews compared every screen with the design basis and the SRS. They found 124 issues: 23 that broke a rule or blocked a flow, 70 missing states or confusing

behaviours, and 31 polish items. All rule and flow issues were fixed.

9.3 Corrections to the First Design

The first design (v1, Figma) was corrected against the SRS before the prototype was built. The main corrections, kept in the design rules, are:

First design	Corrected rule
Verifying results issued credentials	Verify results gives module achievements; confirming completion is a separate step that triggers issue (D-15)
Public pages claimed SAMC credit and QV beyond their validity	Credit and QV shown only while verified and valid for the cohort dates (BR-14, BR-23)
The deputy dean could revoke alone; "superseded" wording	Revoke needs the authorised body's decision reference; statuses Corrected, Replaced, Revoked
Review labels "Ready / Not ready"	Support, Amend, Do not support; conflict of interest declared first
Endorse allowed with open comments	Endorse blocked by open blocking comments
Dates and money in mixed formats	DD/MM/YYYY and RM 0.00
Public verification showed credit	Minimum fields: name, credential, issuer, date, status
Transfer could approve a line with no outcome match	Approve disabled for that line, with the reason (BR-30)
Phone-first student screens	Web first, reflowing to a phone
Neutral side navigation	Navy side navigation (owner's decision)

10. Assumptions, Open Items and Prototype Limits

10.1 Design Assumptions

ID	Assumption
UA-01	The "advisor" of a stackable plan (PRD US-48) is the deputy dean of the degree's faculty.
UA-02	A lecturer's scope is their own micro-credentials plus the reviews they are invited to.
UA-03	DESIGN.md (Carbon with uLearn Navy) is accepted as the UTeM corporate identity for MCMS (ir-identity).
UA-04	Participants sign in without MFA; every staff role that approves, verifies, issues, deposits or administers uses MFA.
UA-05	The withdrawal deadline is 20 days before the cohort ends until the fee and refund rules are set.

10.2 Open Items Carried into Design

Item	Design treatment until it closes
OI-07 Retention periods	Retention categories shown with "TBD"; disposal requests wait for the Registrar.
OI-09 Transfer limits	Transfer case shows the limit check as "not configured yet; no limit is applied".
OI-11 ABC file format	ABC export marked "format not final"; columns shown can change.
OI-12 Fees and refunds	Receipts and withdrawals say refund rules are not decided yet.
OI-13 Completion window	Setting shown as TBD; no window reminder is sent.

10.3 Prototype Limits

- The prototype runs in a browser with sample data kept for the browser tab; "Reset data" restores the starting state. It has no server, accounts or real integrations.
- Screens show English text; the language switch works and is recorded, but BM strings are not written yet.
- The QR code on certificates is a drawn placeholder; signatures and hashes are test values.
- Excel export is marked Release 2; CSV and PDF are available.
- Screens were checked automatically at desktop width; the phone layout follows the responsive rules but has not been reviewed screen by screen.
- One sample learner appears in two Cloud Native Development cohorts; it does not affect any flow.

Appendix A. Using the Prototype

The prototype is the living part of this specification. It runs from the project folder without installation:

Task	How
Start it	Run `py UIUX/v2/tools/serve.py 8731` and open http://localhost:8731
Choose a person	The Personas page opens first: student, lecturer, deputy dean, CODL, industry expert, public visitor, or the real sign-in screens
Walk a process	Flows (bottom left) lists the 26 flows; Start shows a step bar with what to click next
See what is clickable	Hotspots, or the H key, outlines every clickable element
Start again	Reset data restores the sample records
Open a document	Any download or print button, or the document index at `#/doc/index`
Check it	`node UIUX/v2/tools/check.js` (screens and links), `clickcheck.js` (dialogs), `scenario.js` (hand-offs), `pdfs.js` (sample PDFs)

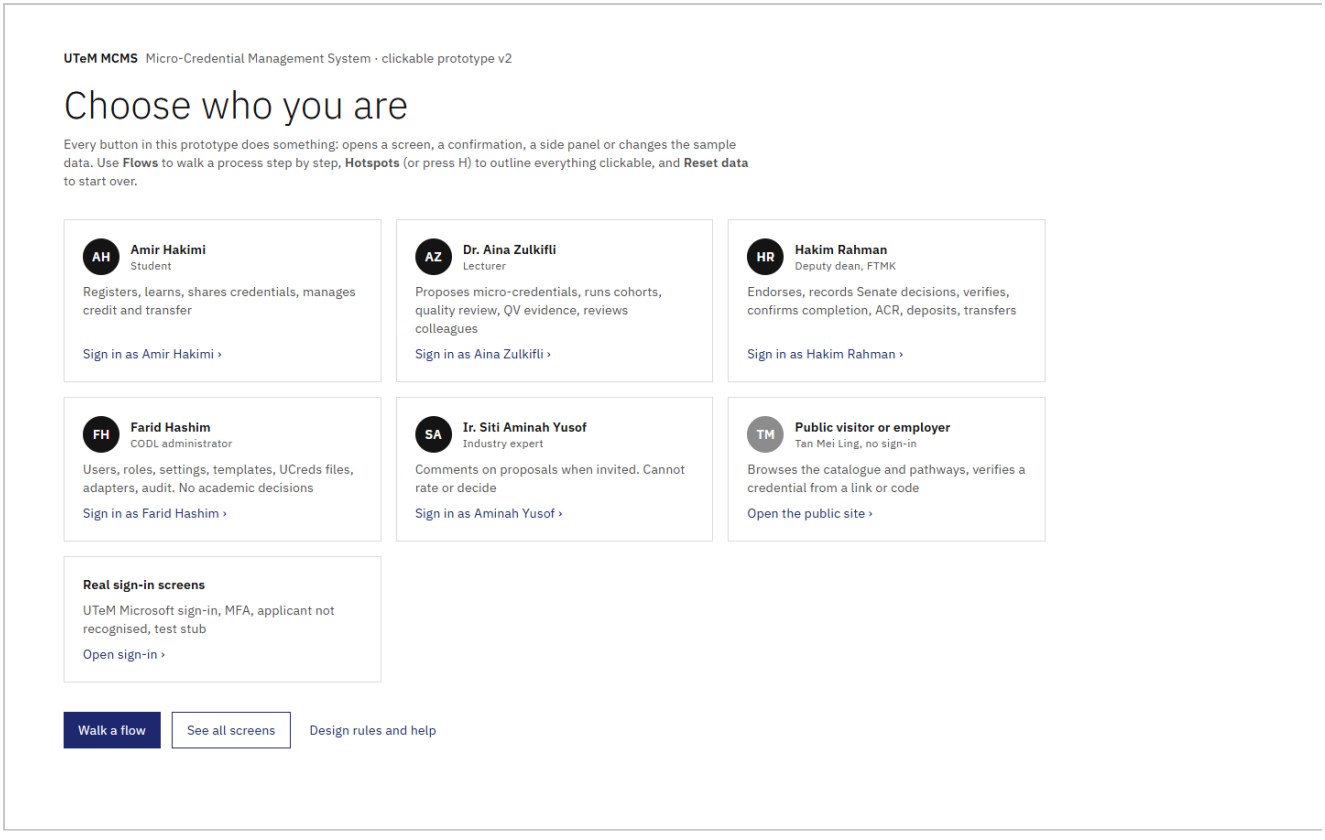


Figure 20: *Prototype start page: choose a persona.*

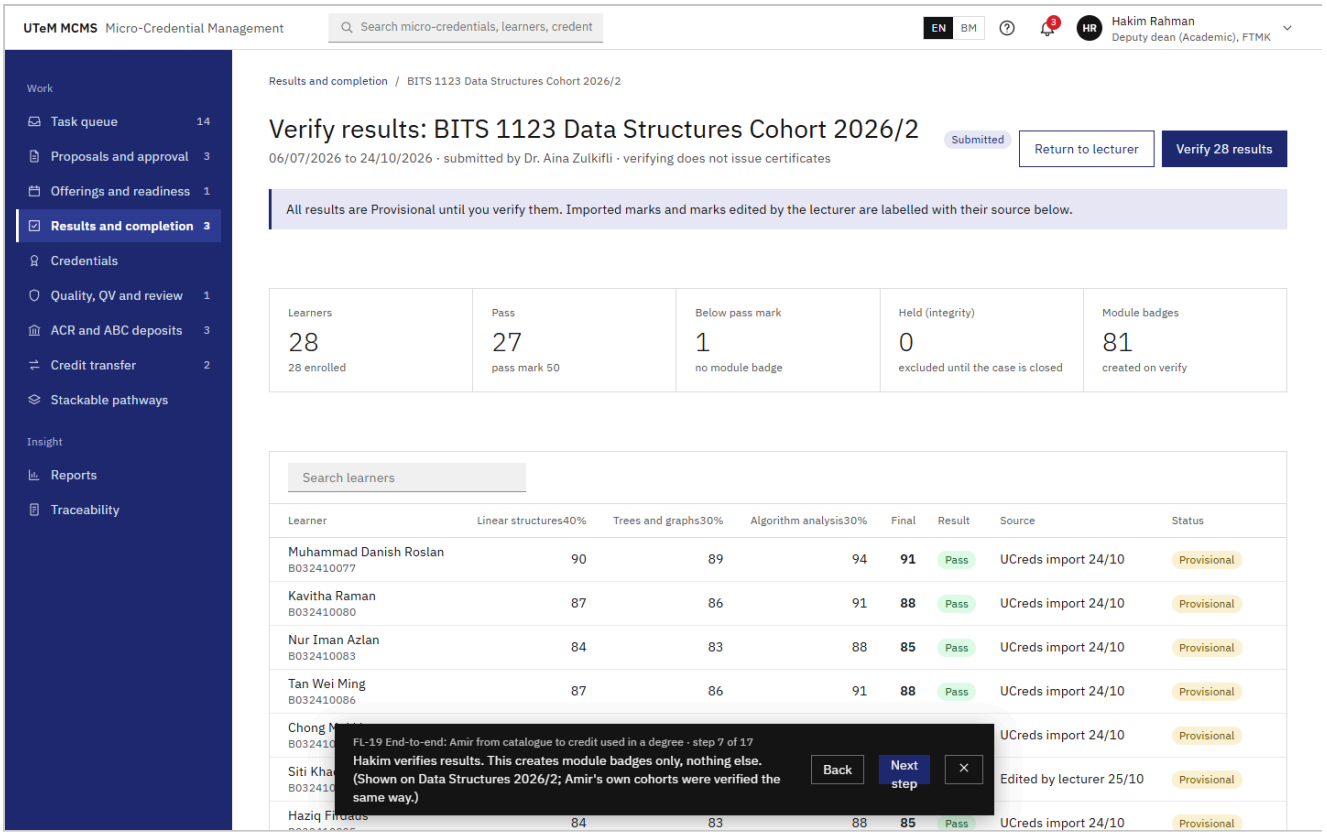


Figure 21: *A flow in progress: the step bar explains the step and offers Back, Next step and Exit.*

The pictures of Appendices B to D are captured from the prototype by `UIUX/v2/tools/gallery.js` and laid out by `UIUX/v2/tools/specgallery.js`, so they can be regenerated whenever the prototype changes.

Appendix B. Screen Gallery

Every screen of the approved prototype, captured from the prototype with its sample data: 98 screens, followed on the same page by each screen's other tabs and wizard steps (65 states) and every pop-up, warning and side panel it opens (67). The menus of each role (10) close each role's section. Screen IDs and routes match Section 6.6. Pop-ups that only appear for a record in a particular state were captured with that state set, for example Endorse after every blocking comment is resolved.

B.1 Common and sign-in

CMN-01 Personas

`#/`

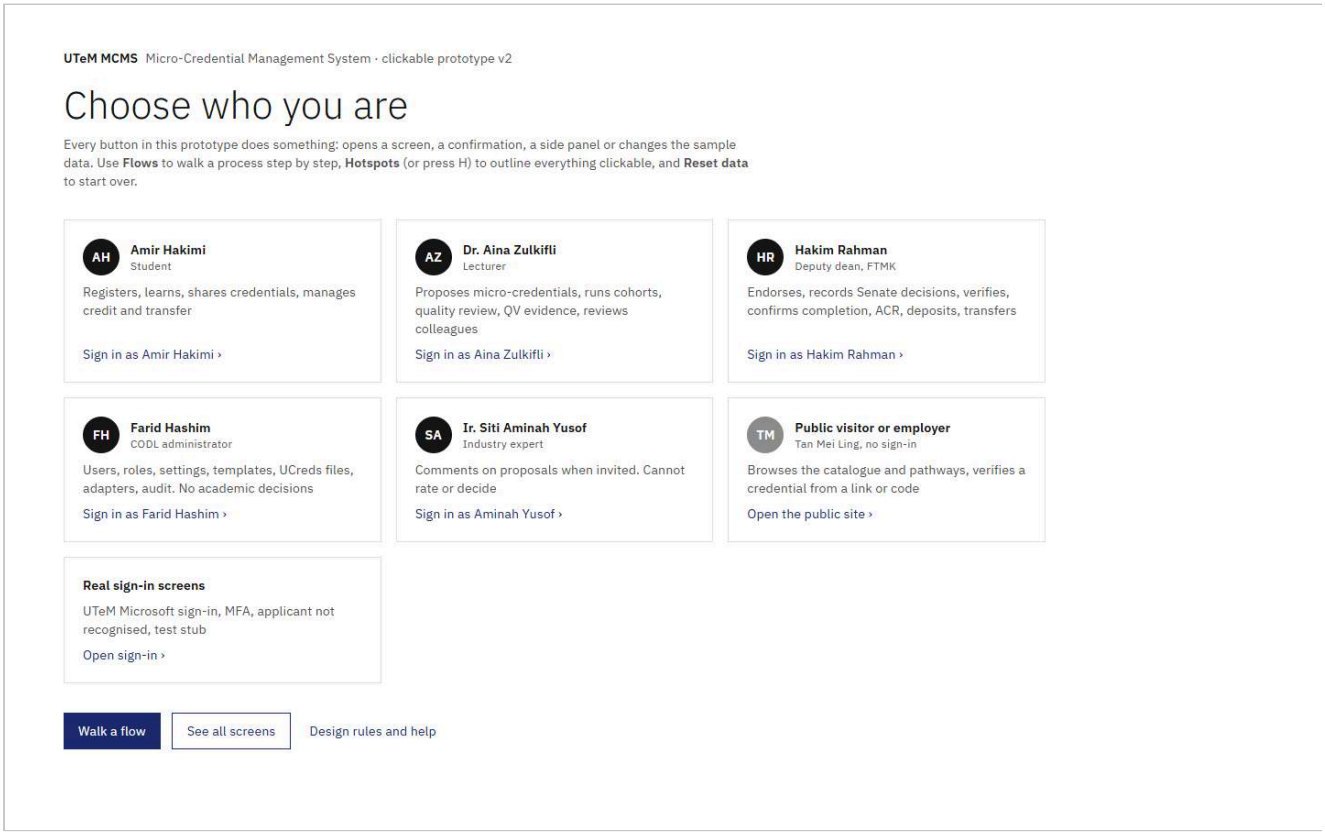


Figure 22: *CMN-01 Personas.*

CMN-02 Sign in

`#/signin`

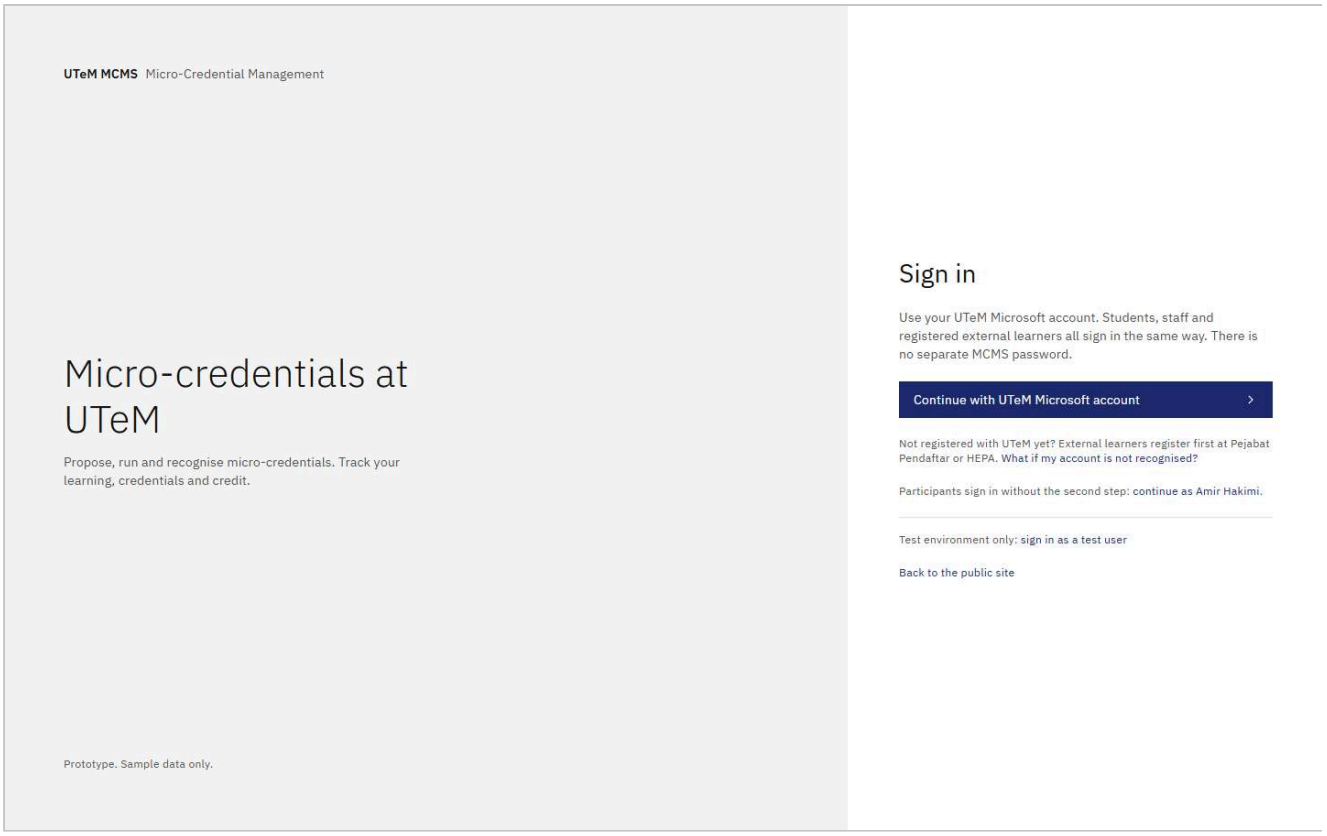


Figure 23: *CMN-02 Sign in.*

CMN-03 Verify it is you

`#/signin/mfa`

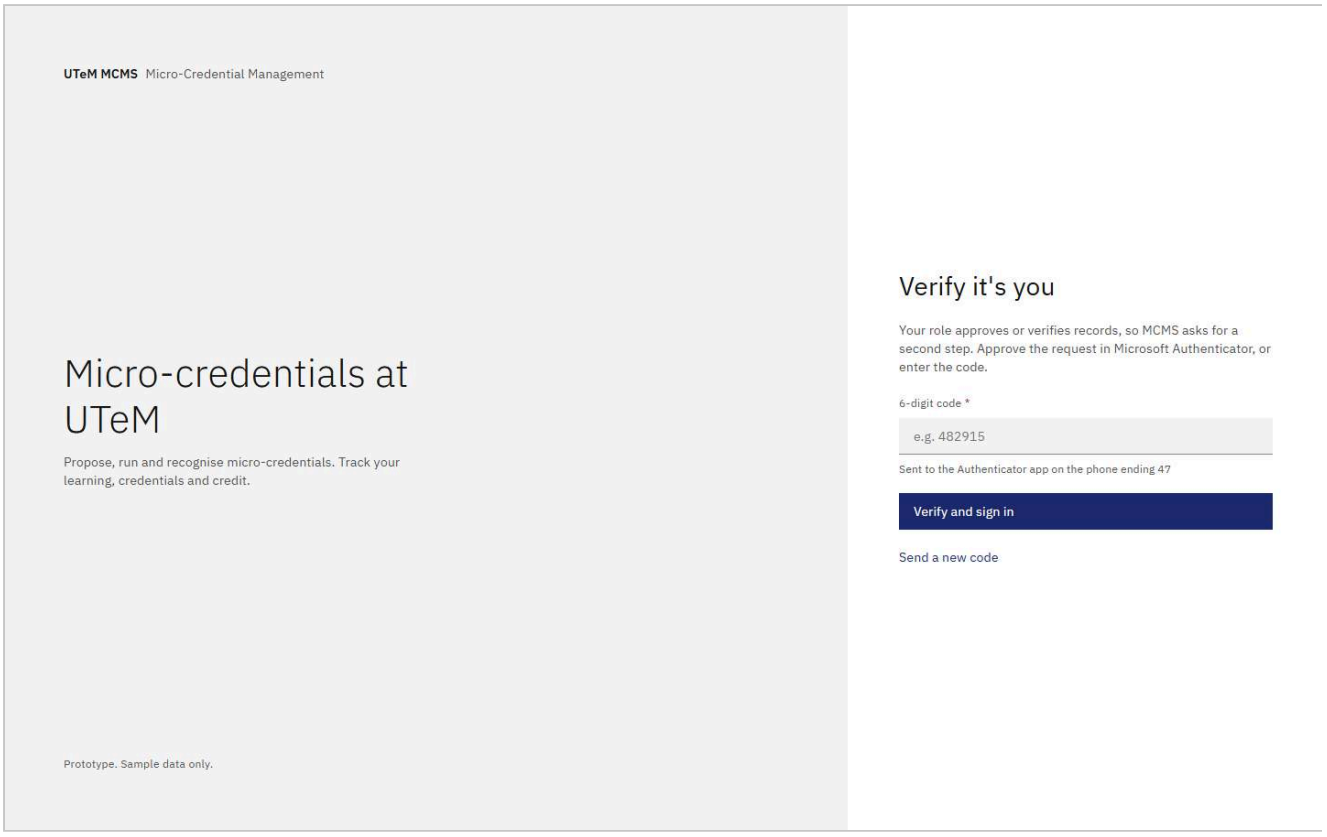


Figure 24: *CMN-03 Verify it is you.*

CMN-04 Account not recognised

`#/signin/noaccount`

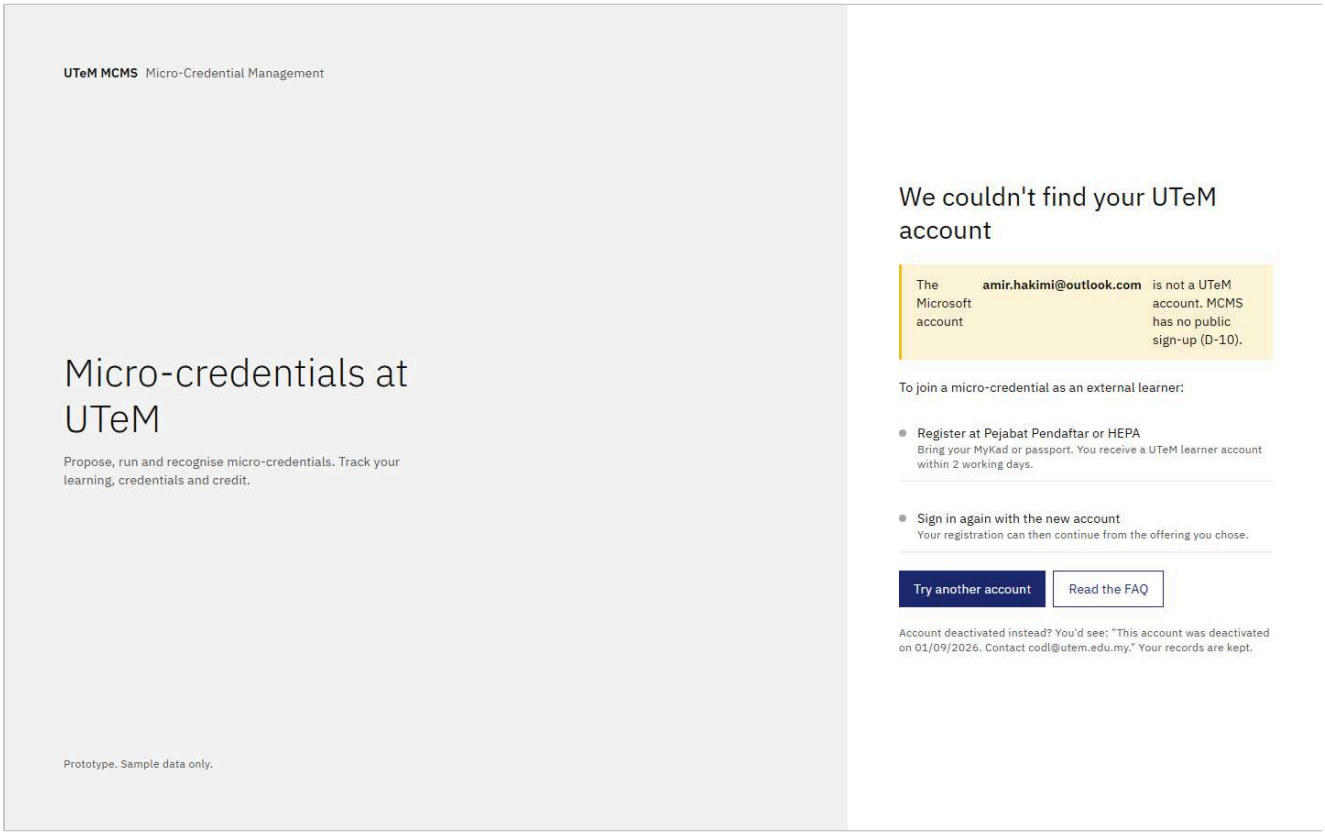


Figure 25: *CMN-04 Account not recognised.*

CMN-05 Test sign-in

`#/signin/stub`

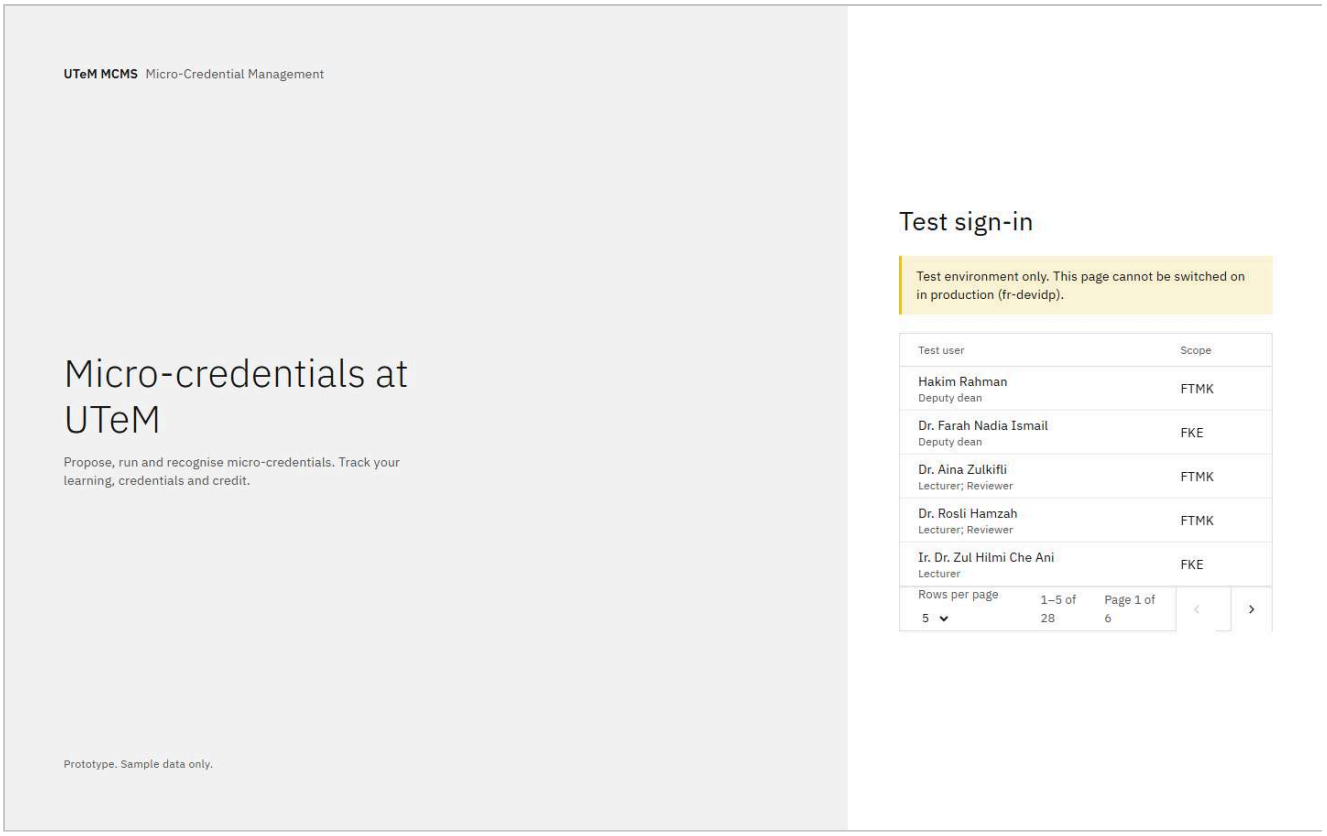


Figure 26: *CMN-05 Test sign-in.*

CMN-06 My profile

`#/profile`

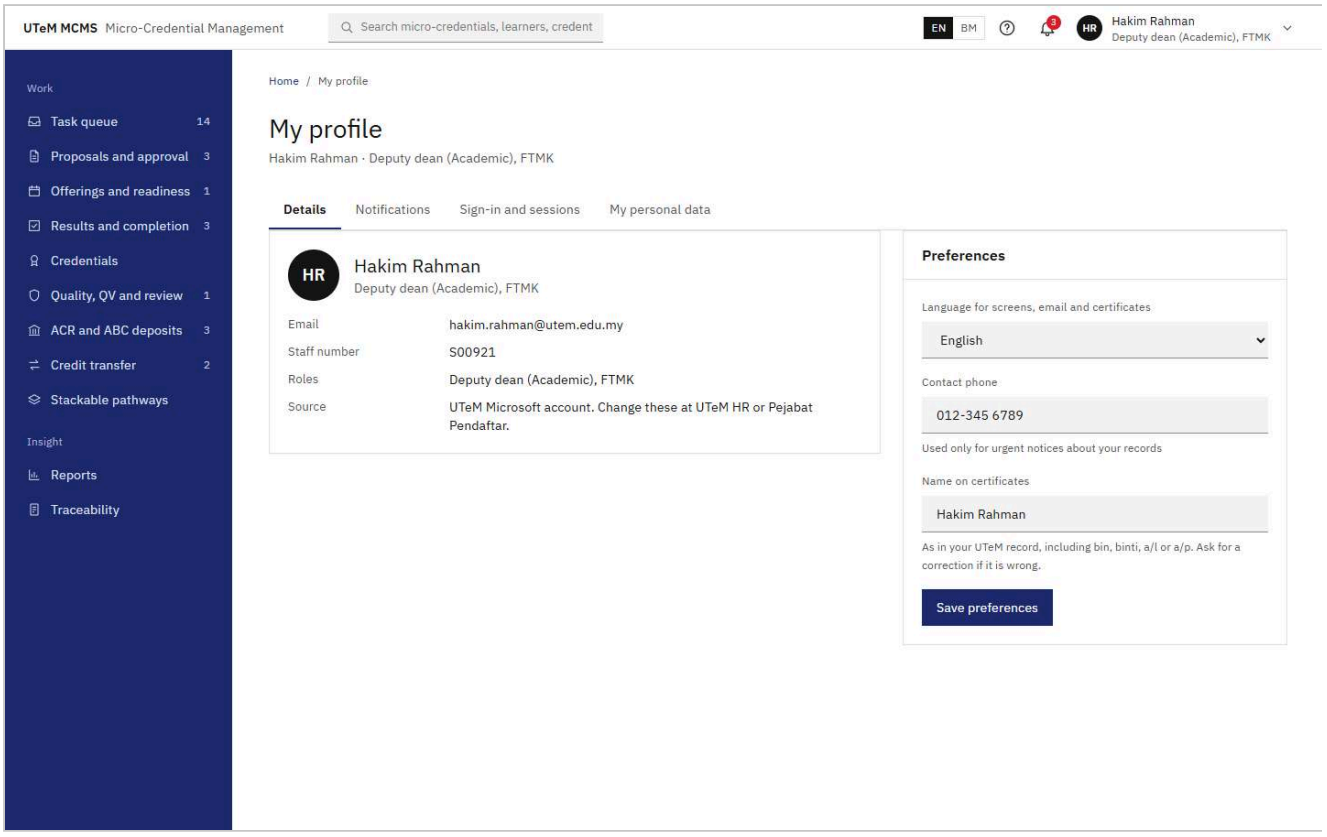


Figure 27: *CMN-06 My profile.*

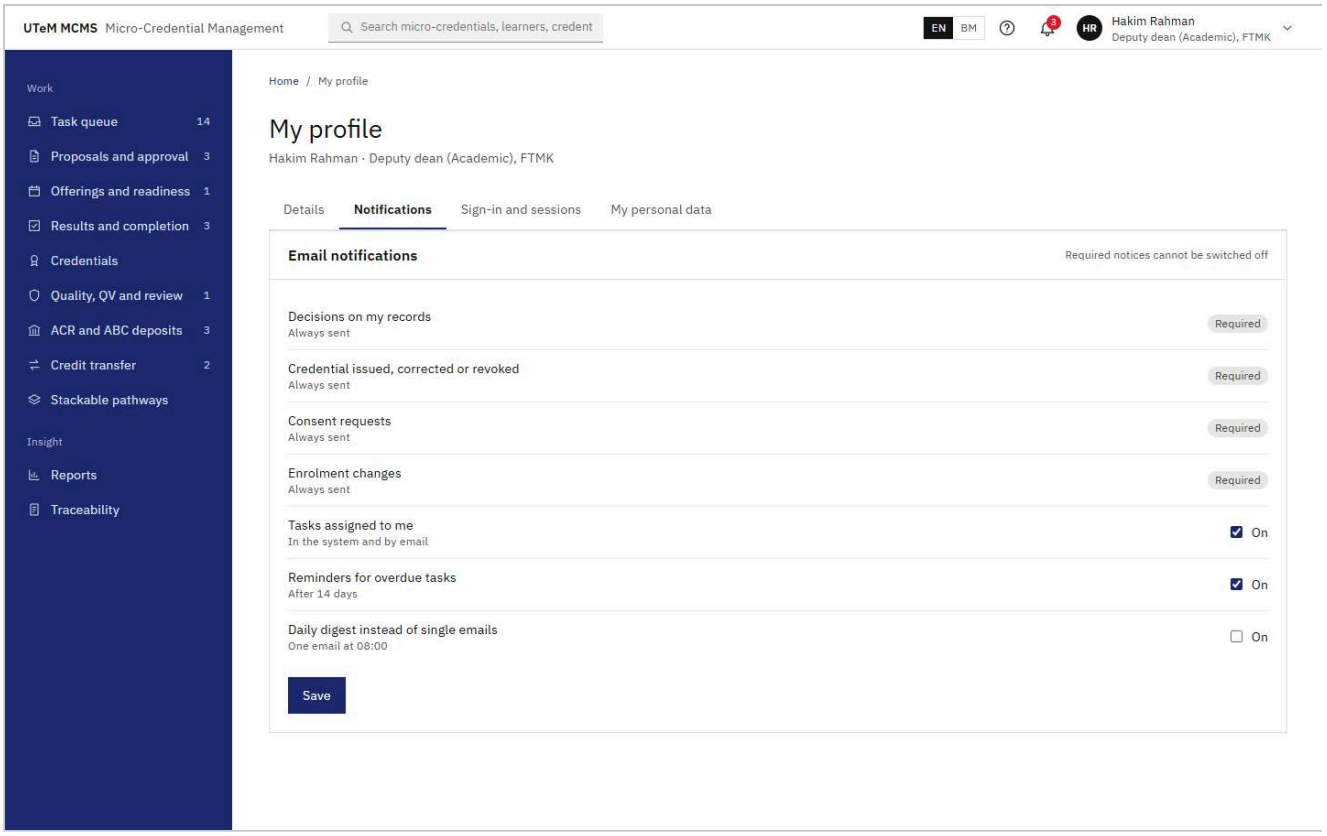


Figure 28: *CMN-06 My profile: Tab: Notifications.*

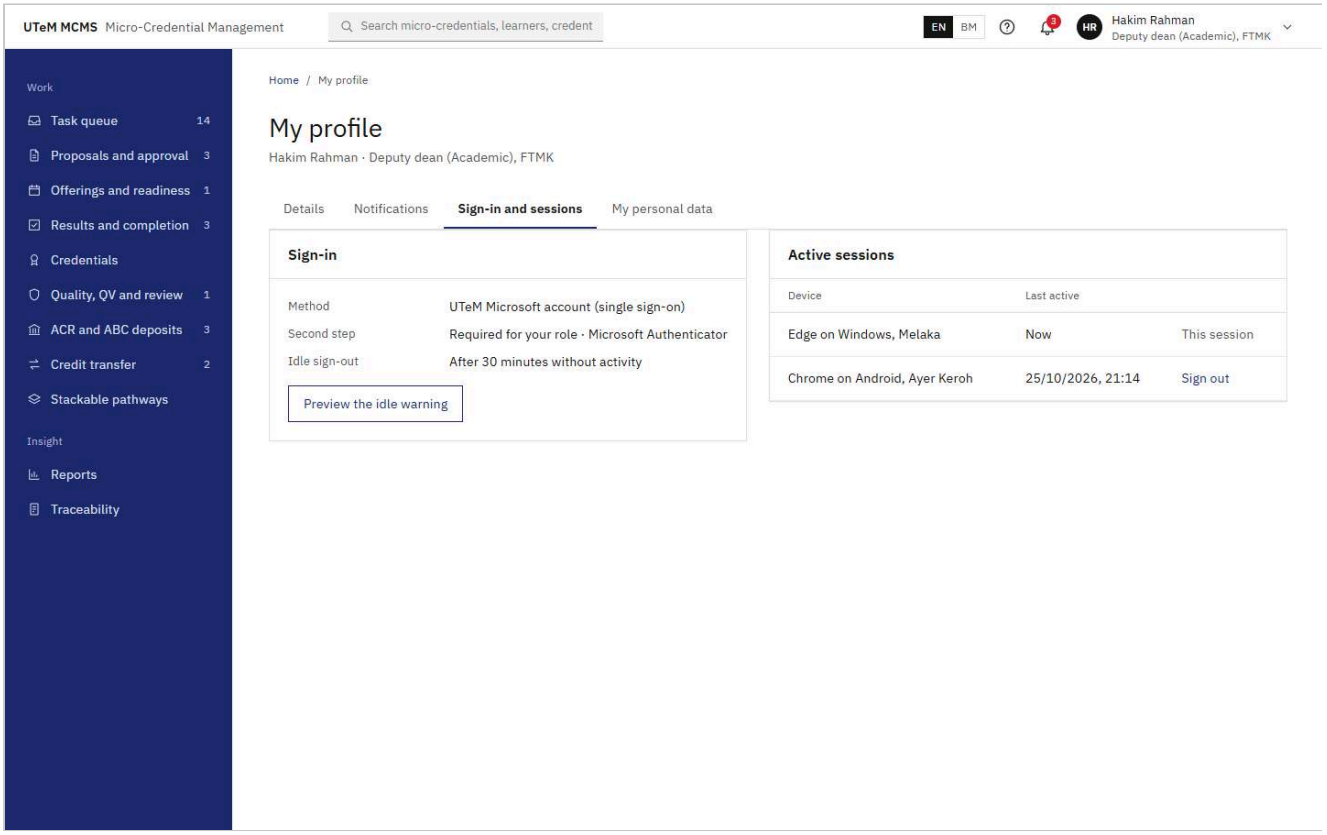


Figure 29: *CMN-06 My profile: Tab: Sign-in and sessions.*

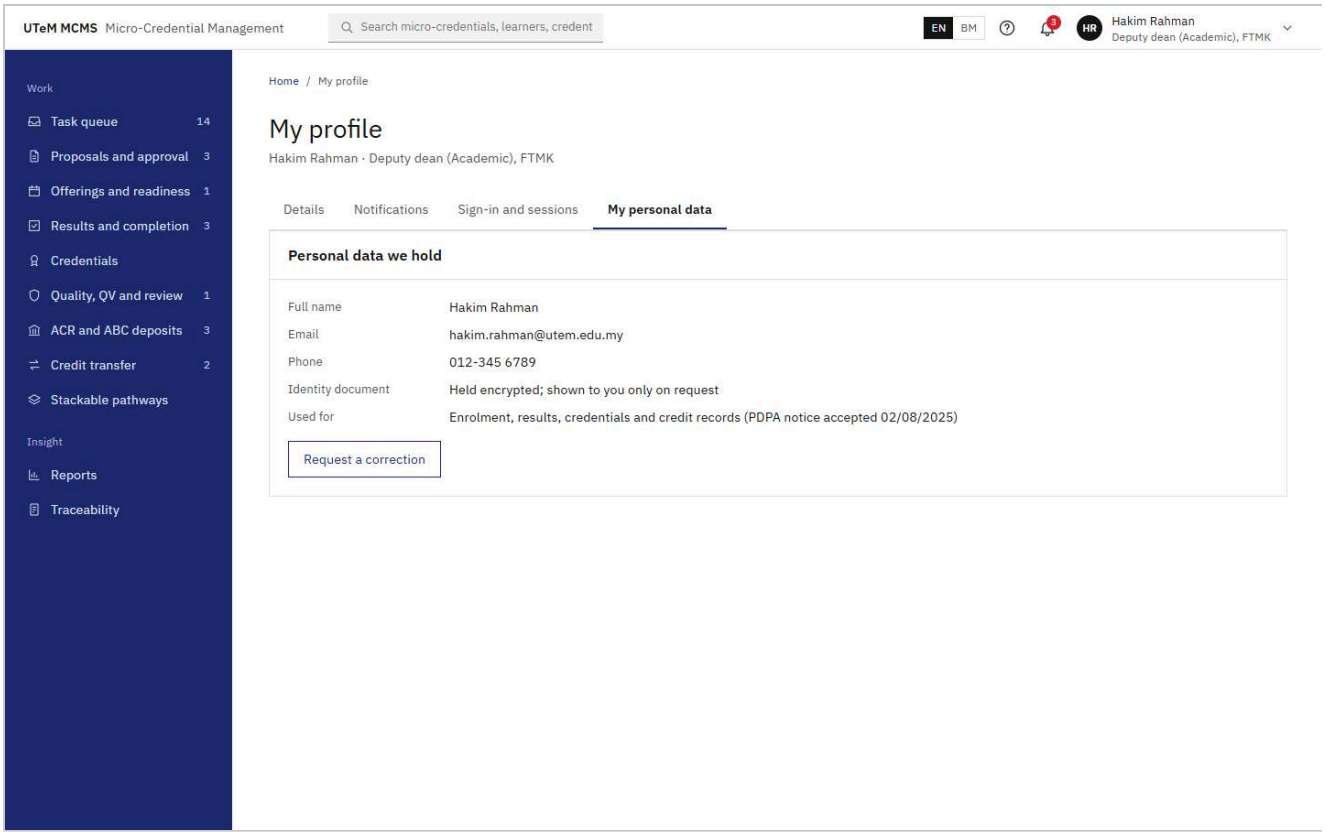


Figure 30: *CMN-06 My profile: Tab: My personal data.*

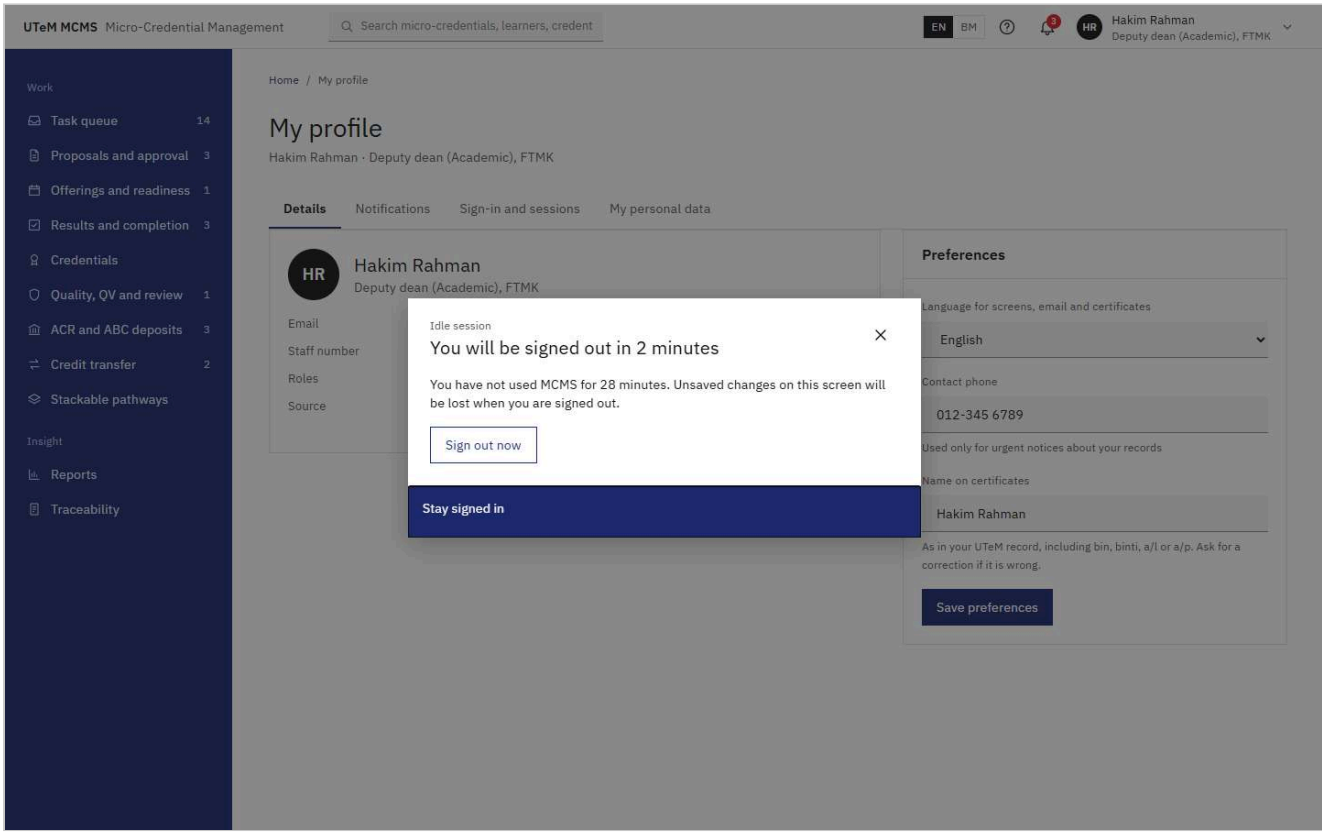


Figure 31: *CMN-06 Dialog: You will be signed out in 2 minutes.*

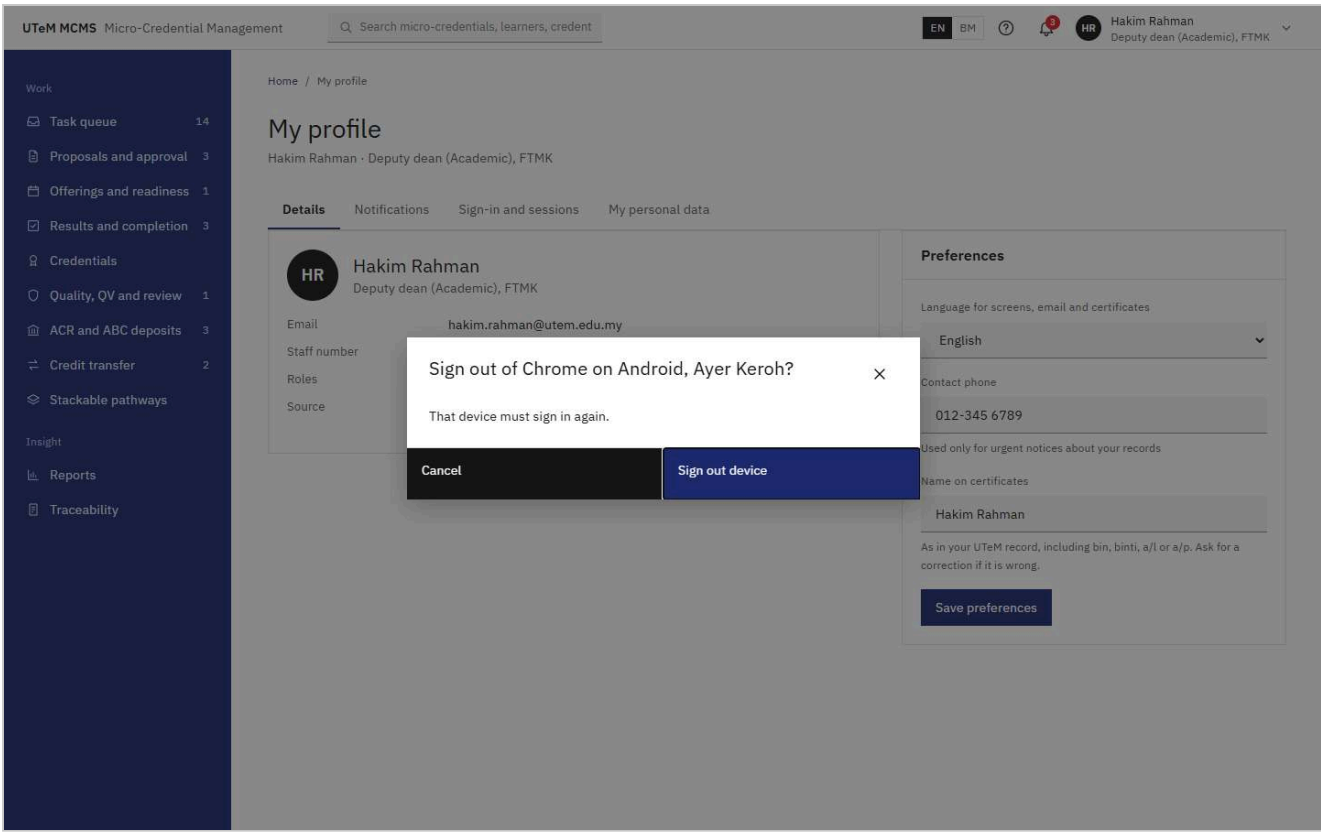


Figure 32: *CMN-06 Confirmation: Sign out of Chrome on Android, Ayer Keroh?*

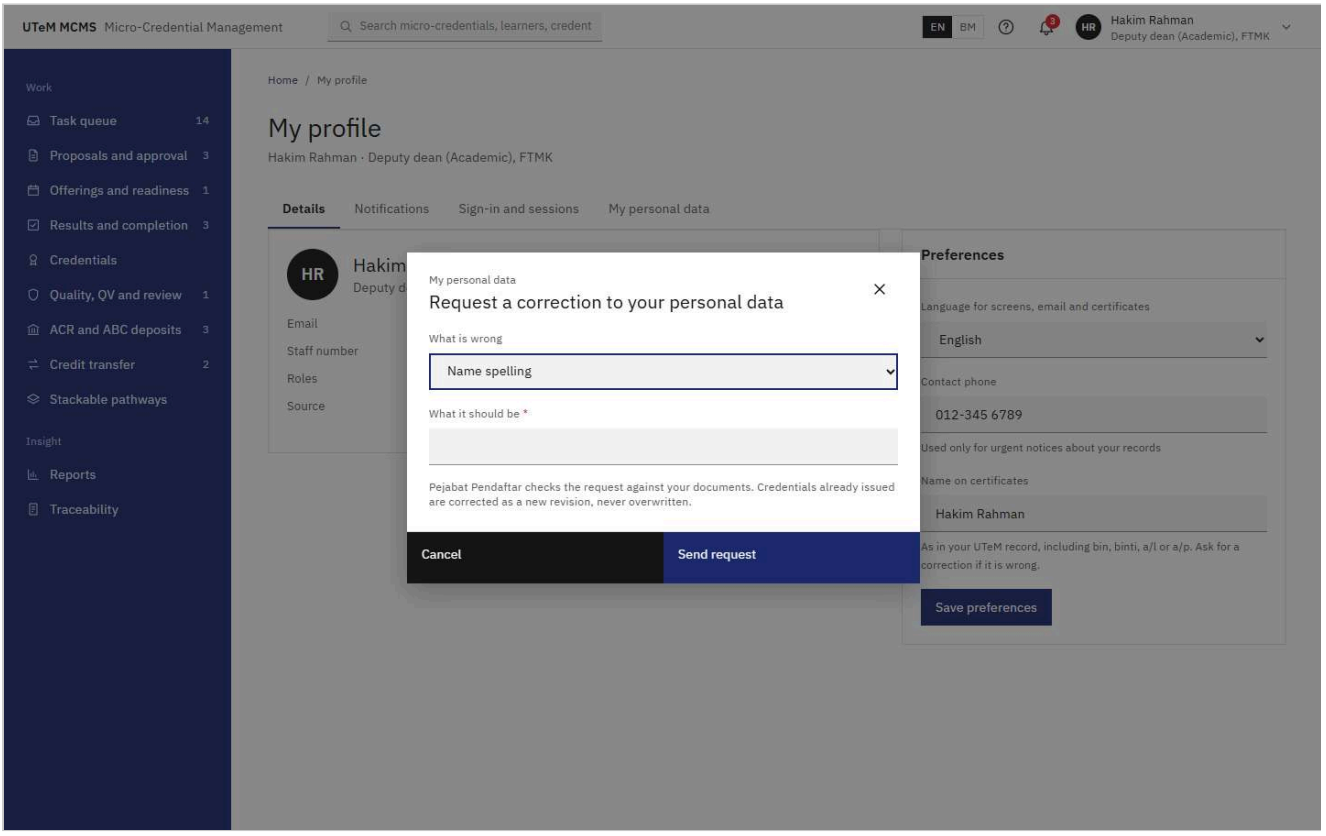


Figure 33: *CMN-06 Dialog: Request a correction to your personal data.*

CMN-07 Notifications

`#/notifications`

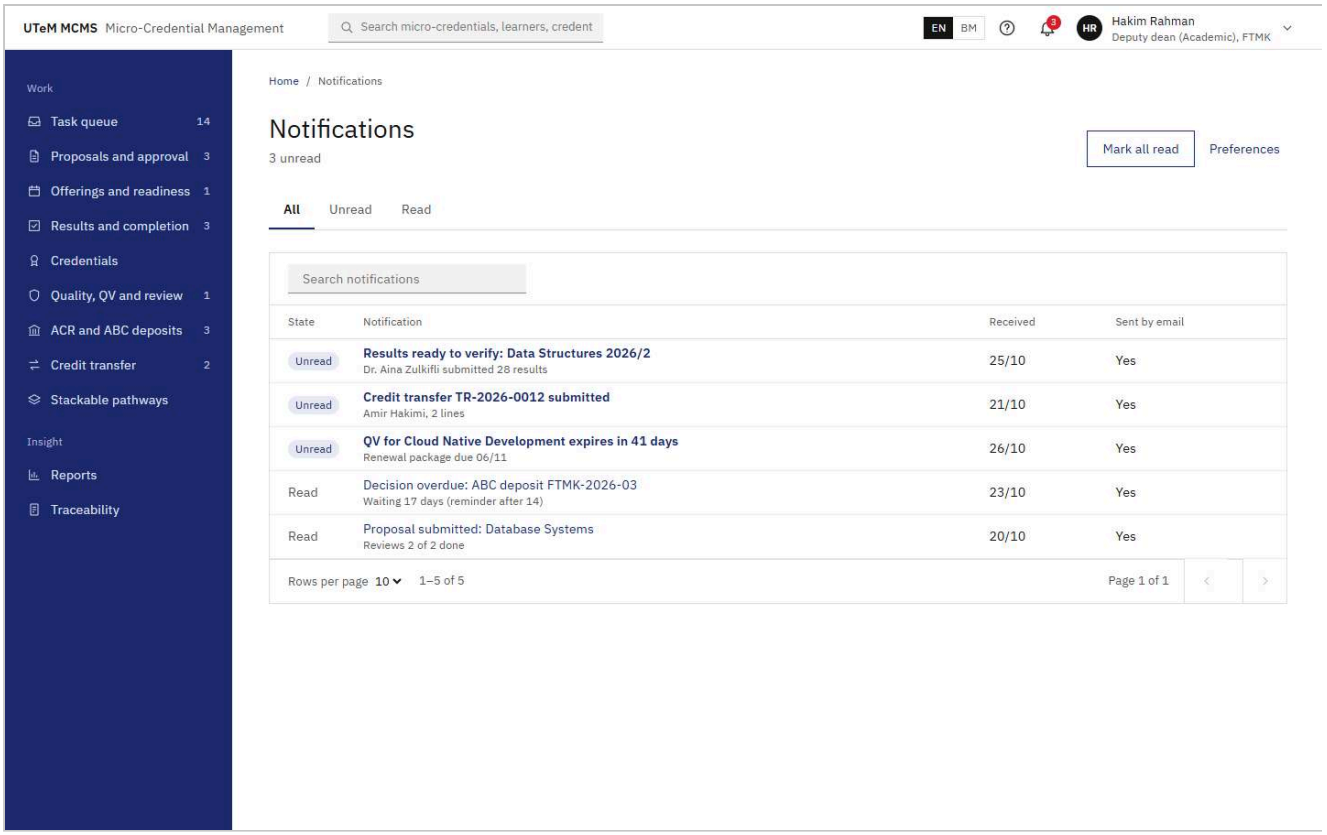


Figure 34: *CMN-07 Notifications.*

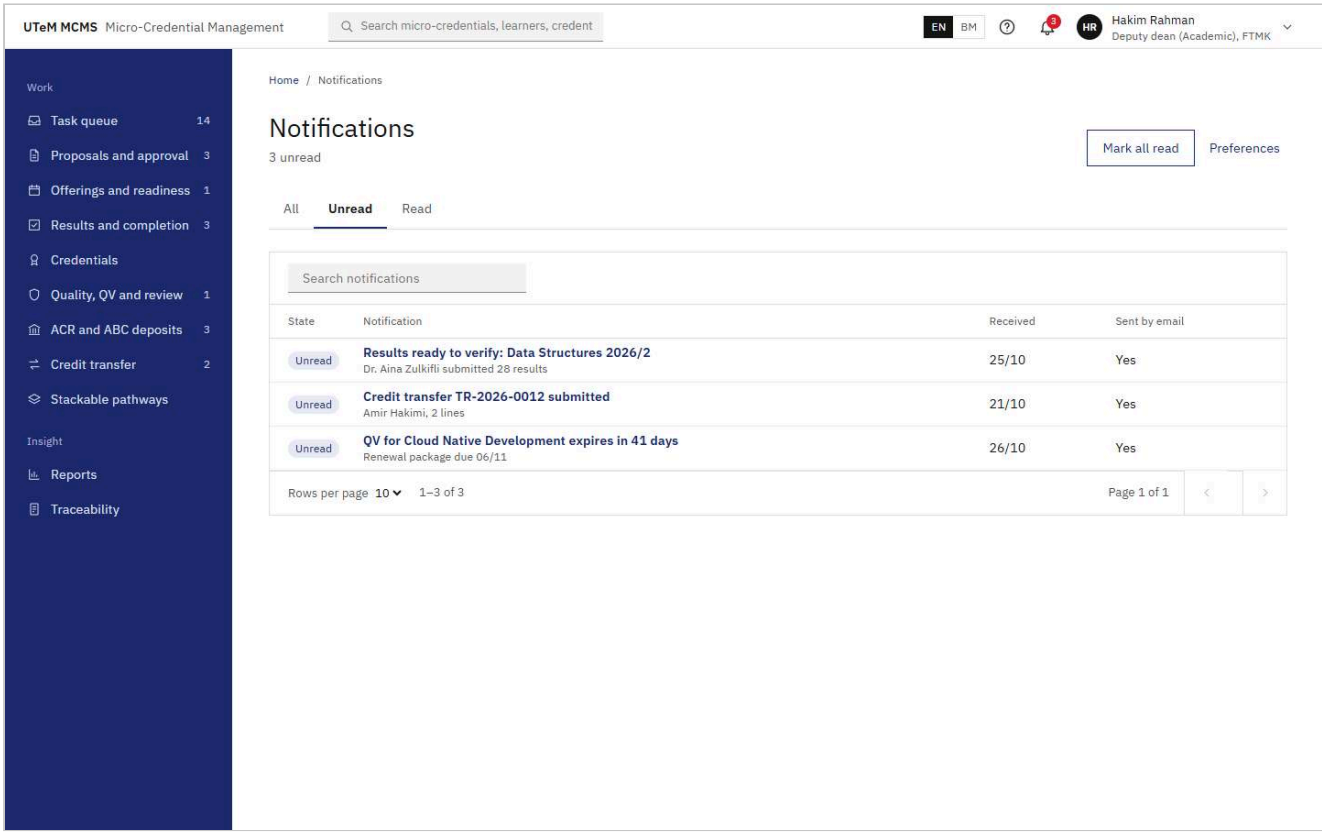


Figure 35: *CMN-07 Notifications: Tab: Unread.*

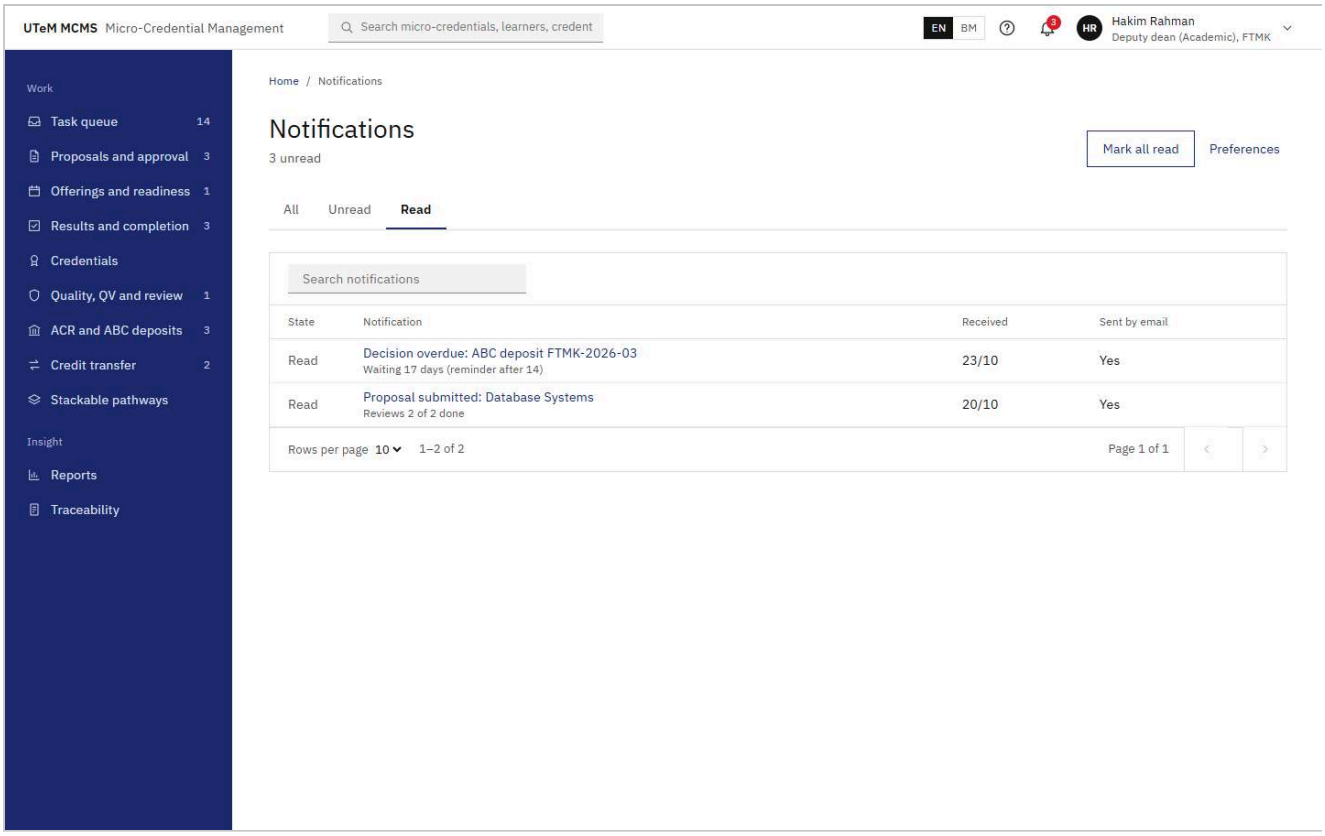


Figure 36: *CMN-07 Notifications: Tab: Read.*

CMN-08 Help

`#/help`

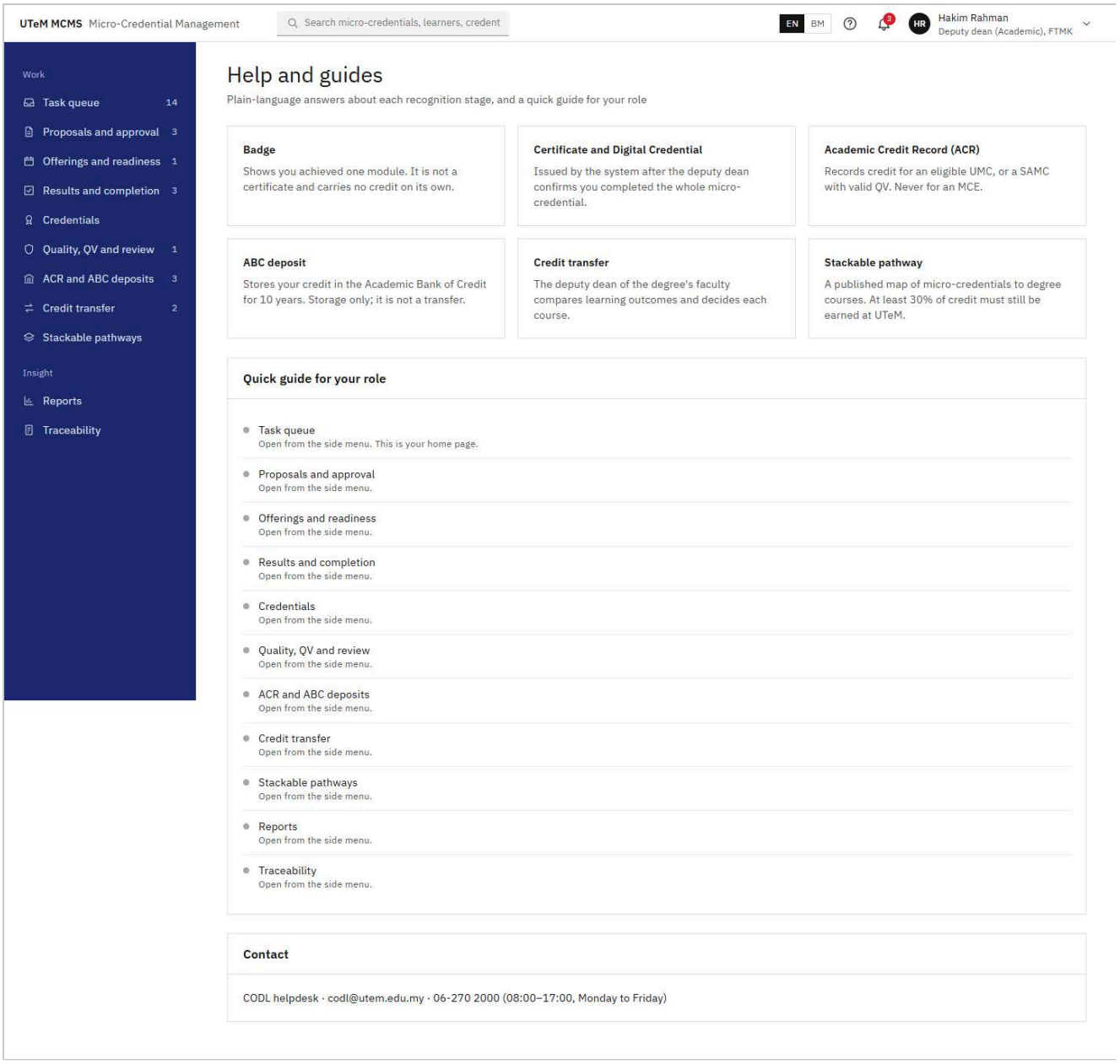


Figure 37: *CMN-08 Help.*

CMN-09 Search

`#/search`

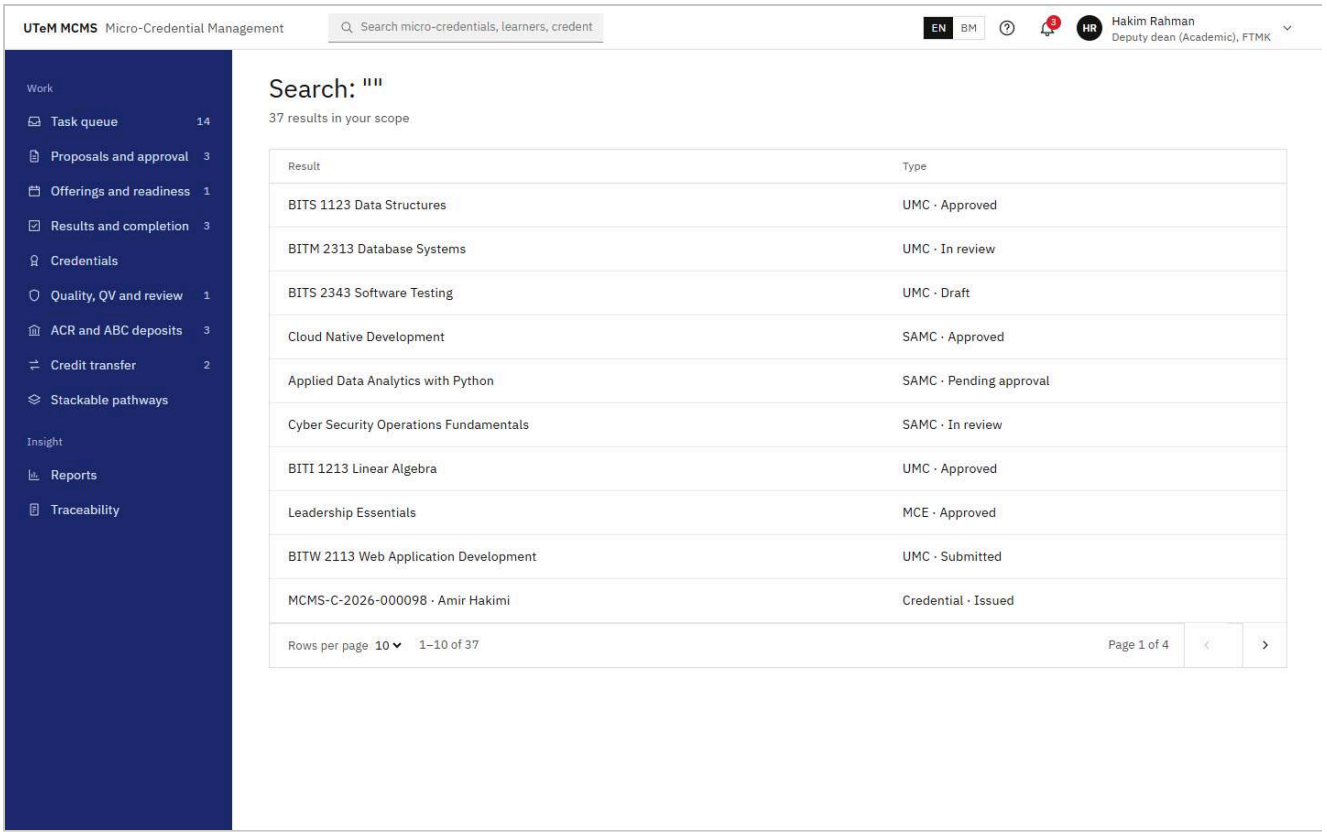


Figure 38: *CMN-09 Search.*

CMN-10 Something went wrong

`#/error`

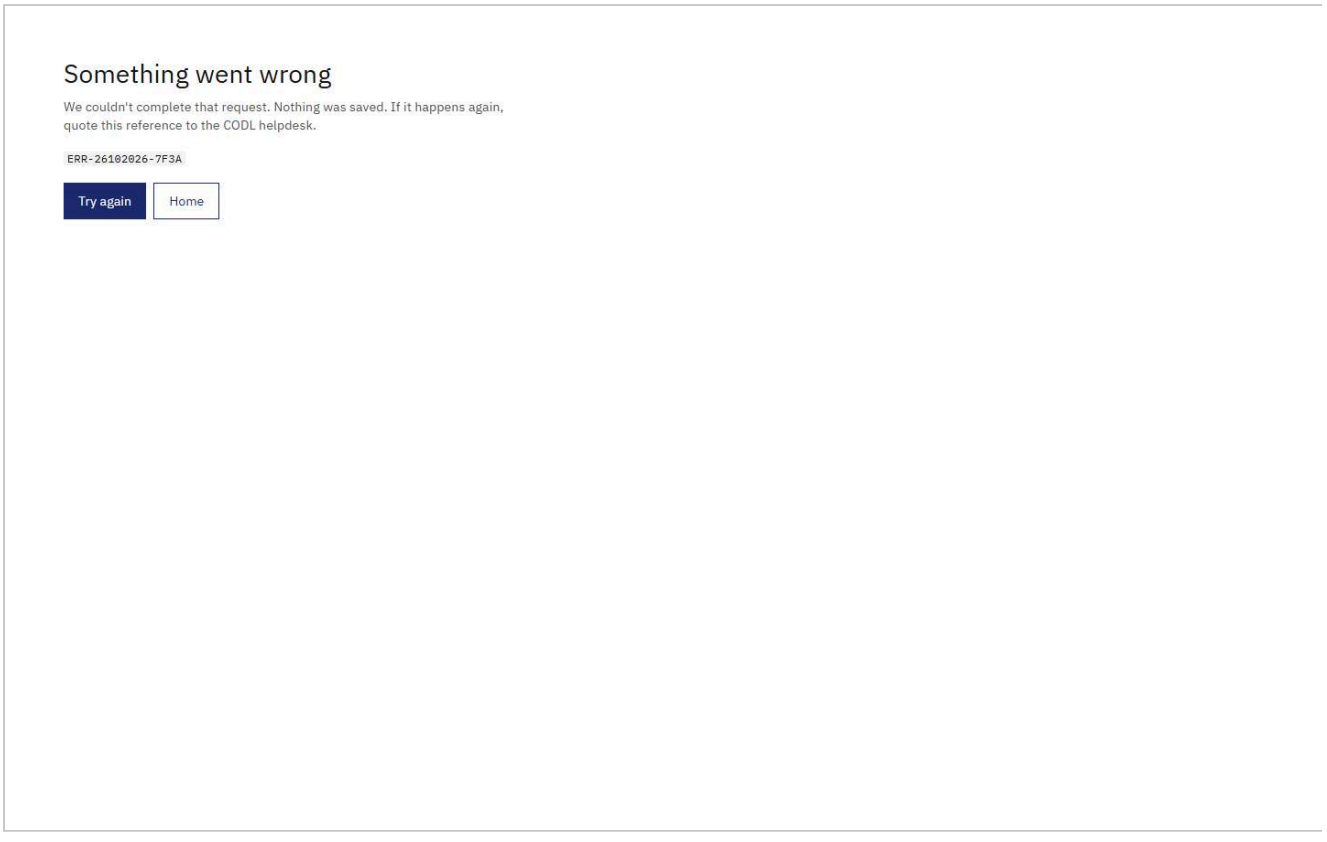


Figure 39: *CMN-10 Something went wrong.*

Any signed-in screen

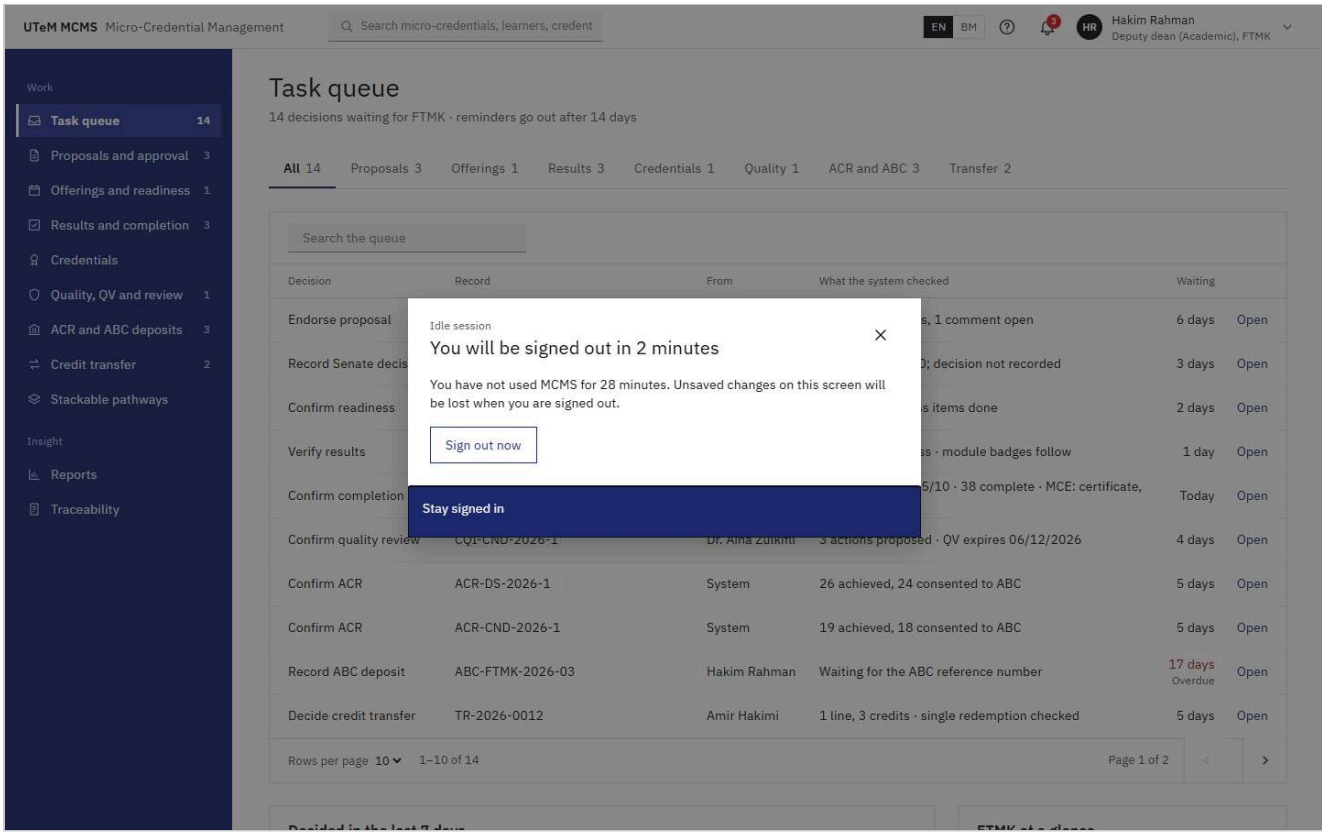


Figure 40: *Warning: idle sign-out (record state: Any signed-in screen).*

Profile menu

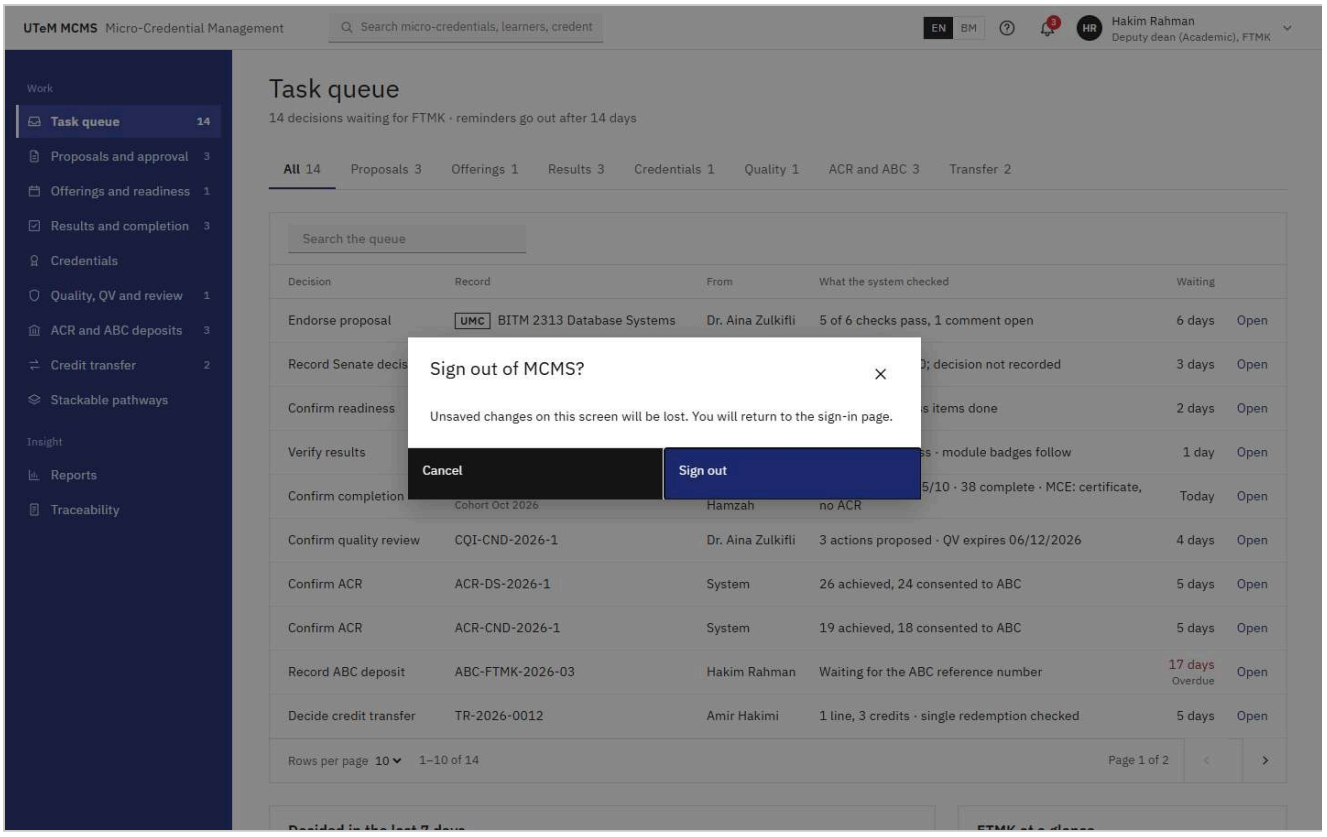


Figure 41: *Dialog: sign out (record state: Profile menu).*

B.2 Public website

PUB-01 Micro-credentials at UTeM

`#/public/home`

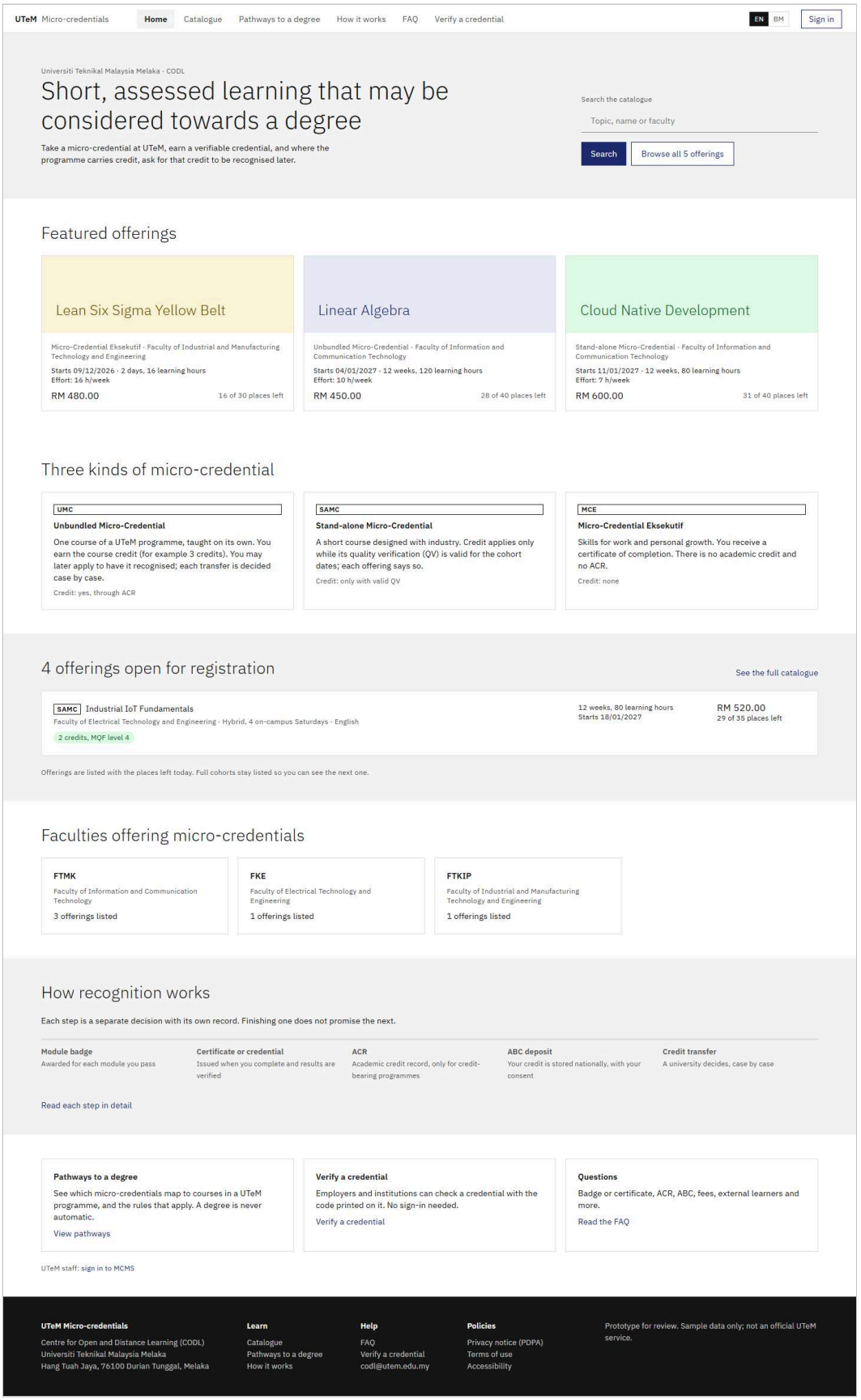


Figure 42: *PUB-01 Micro-credentials at UTeM.*

PUB-02 Catalogue

`#/public/catalogue`

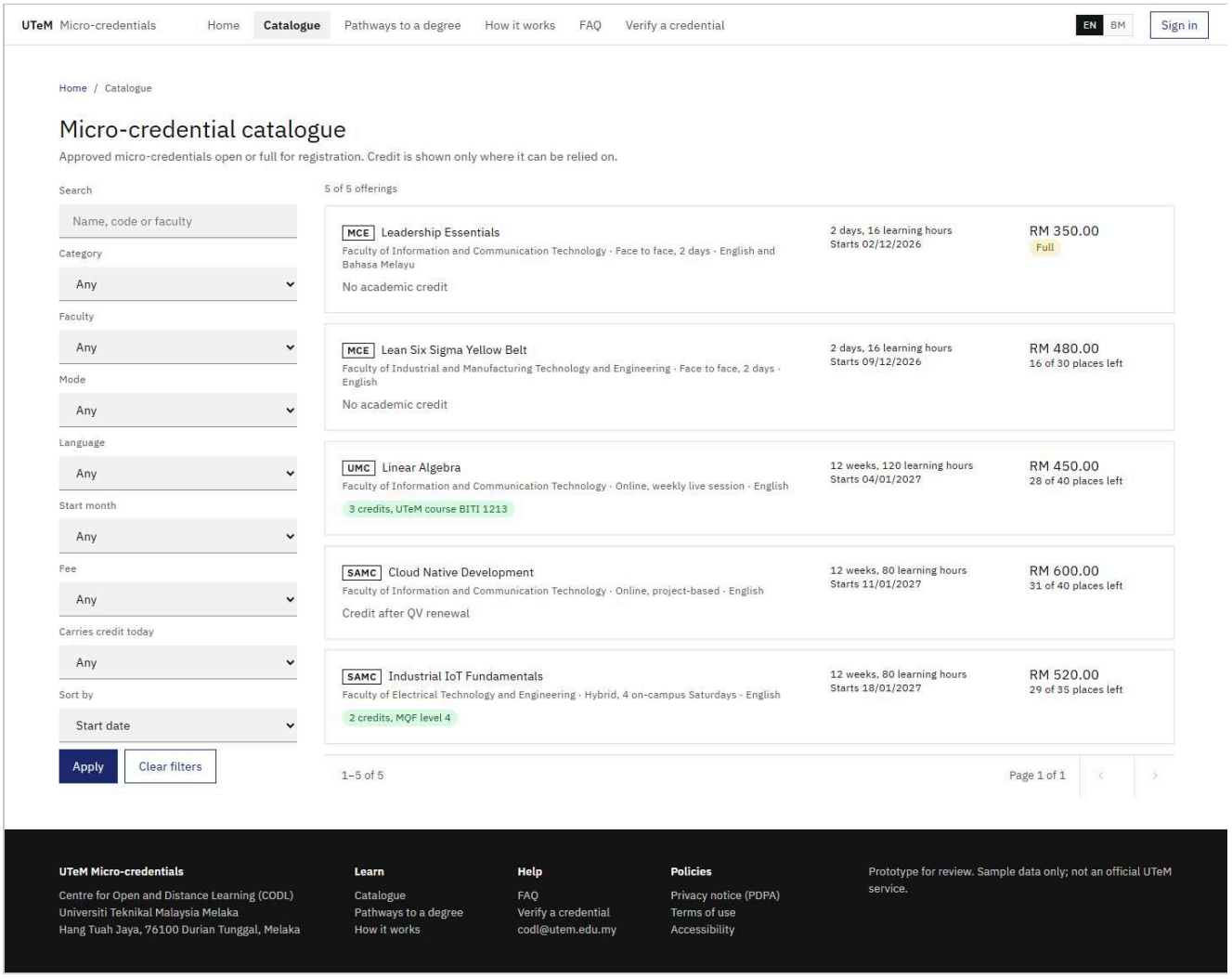


Figure 43: *PUB-02 Catalogue.*

PUB-03 Offering

`#/public/offering/CND-2027-1`

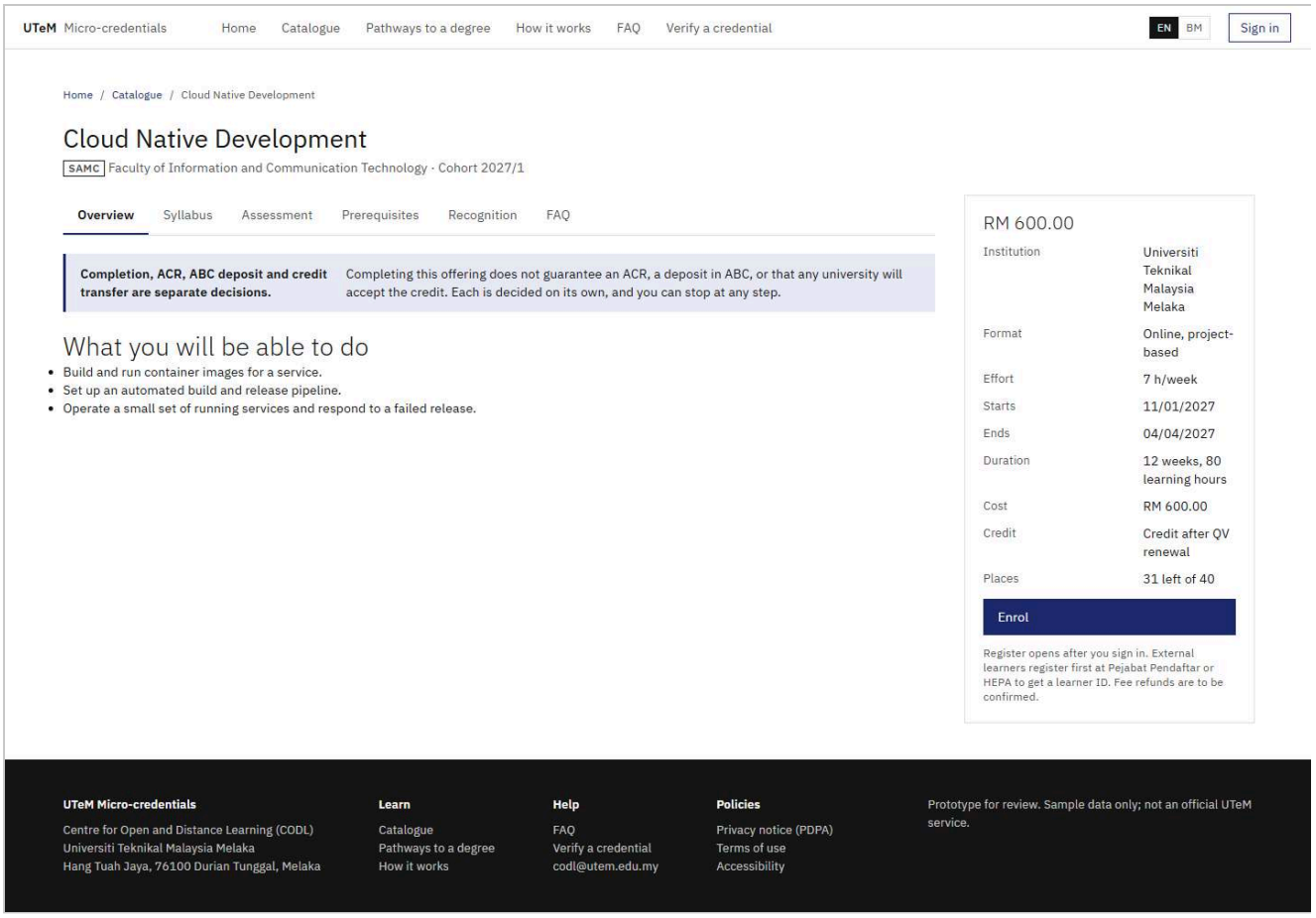


Figure 44: *PUB-03 Offering.*

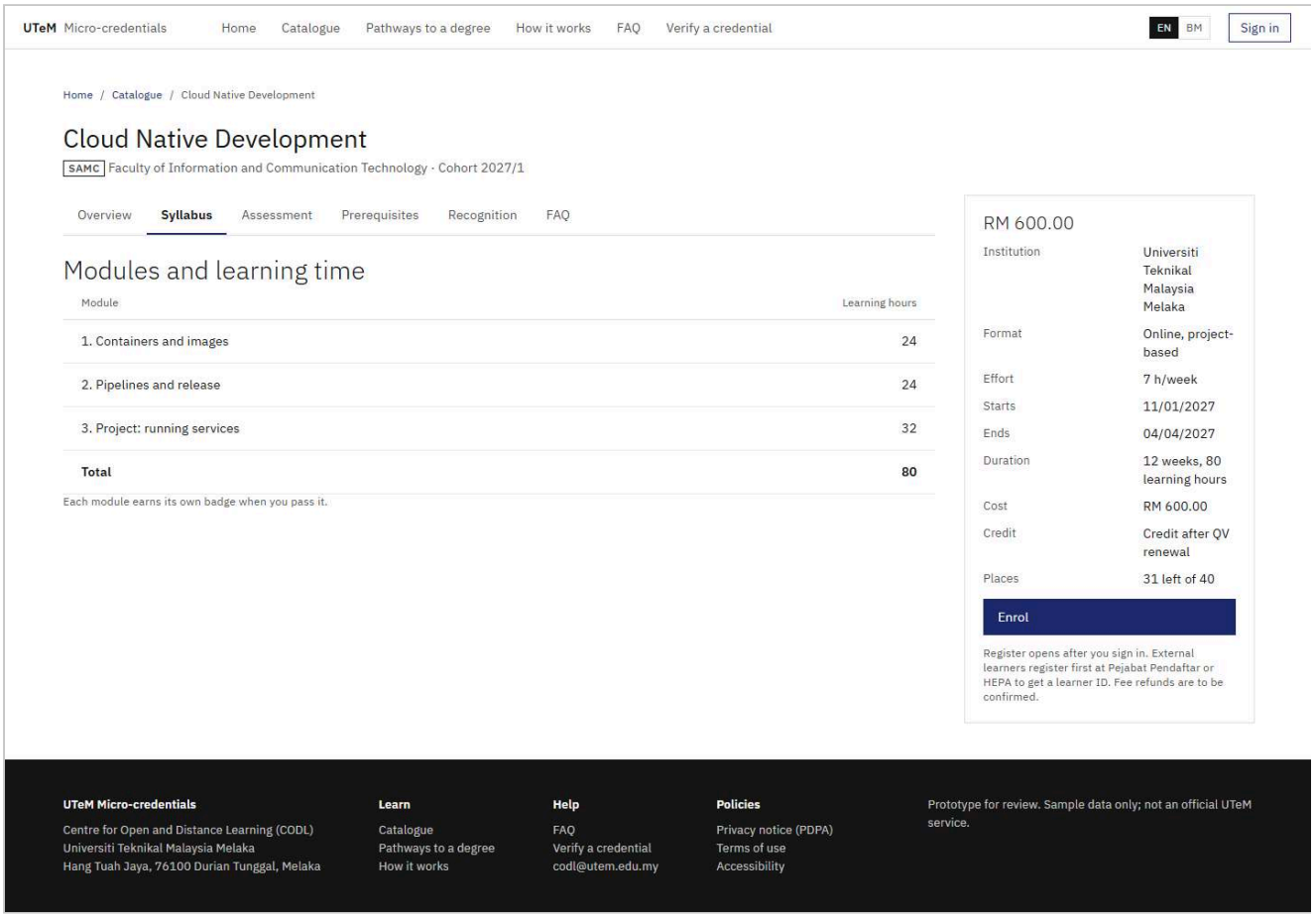


Figure 45: *PUB-03 Offering: Tab: Syllabus.*

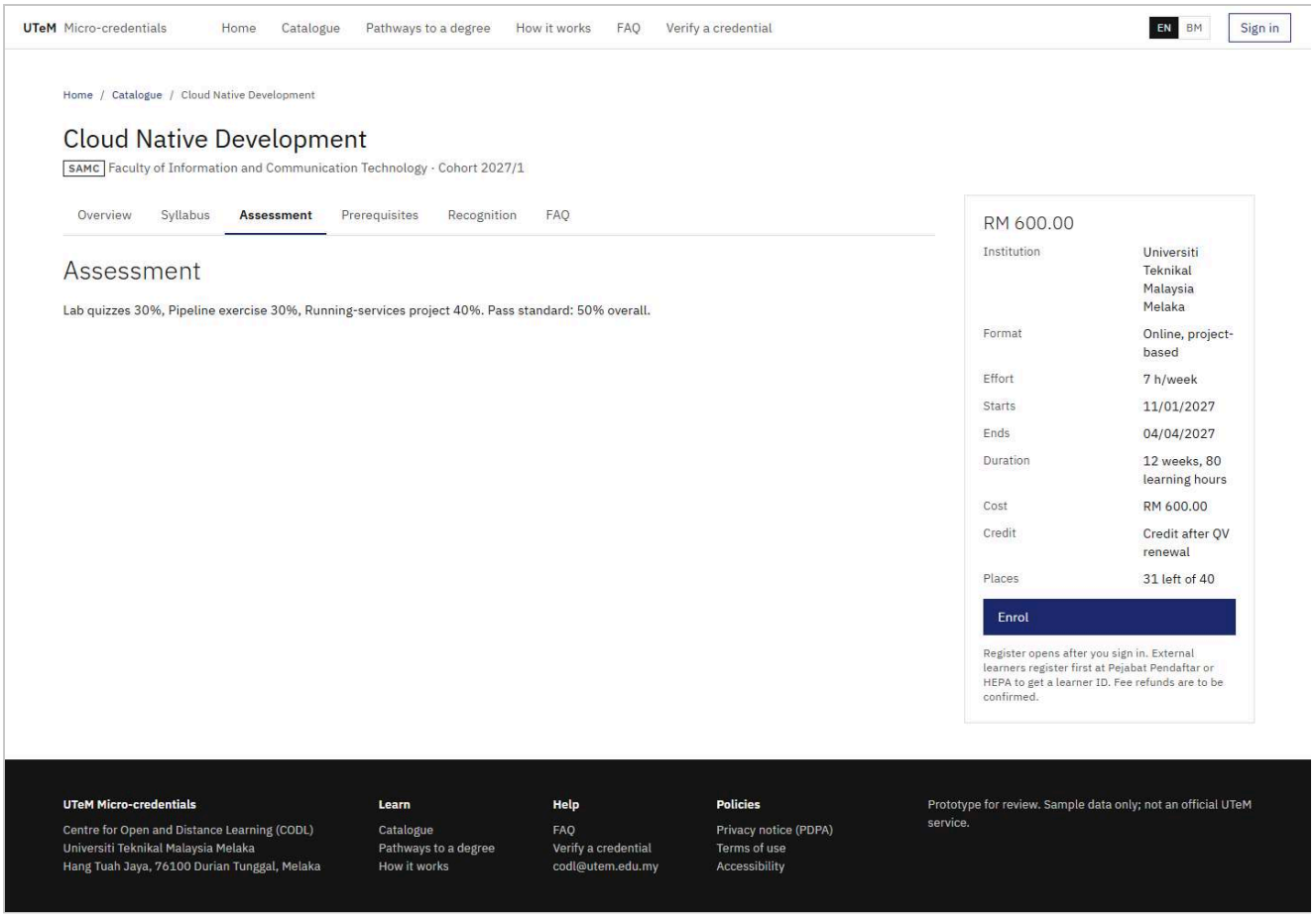


Figure 46: *PUB-03 Offering: Tab: Assessment.*

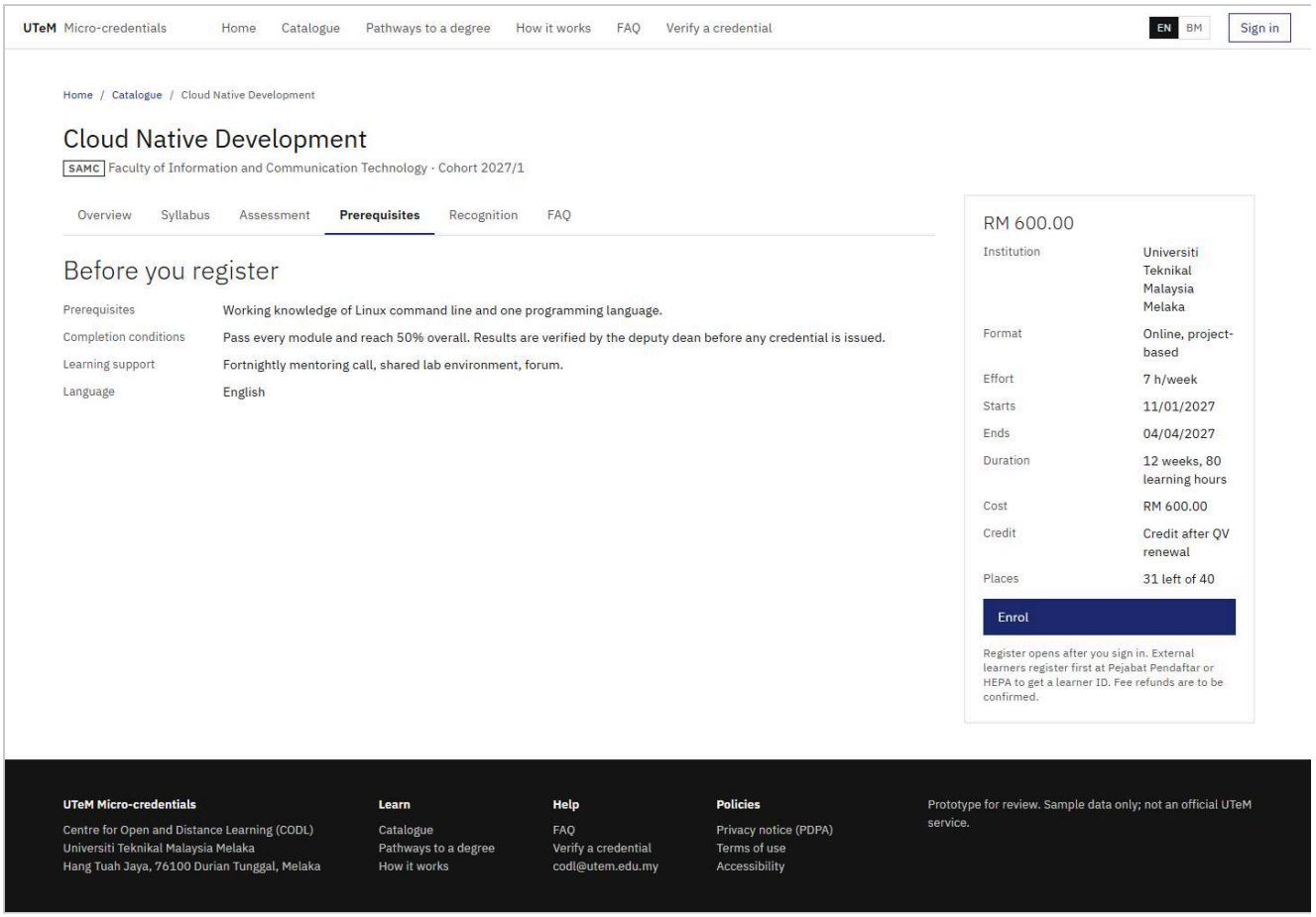


Figure 47: *PUB-03 Offering: Tab: Prerequisites.*

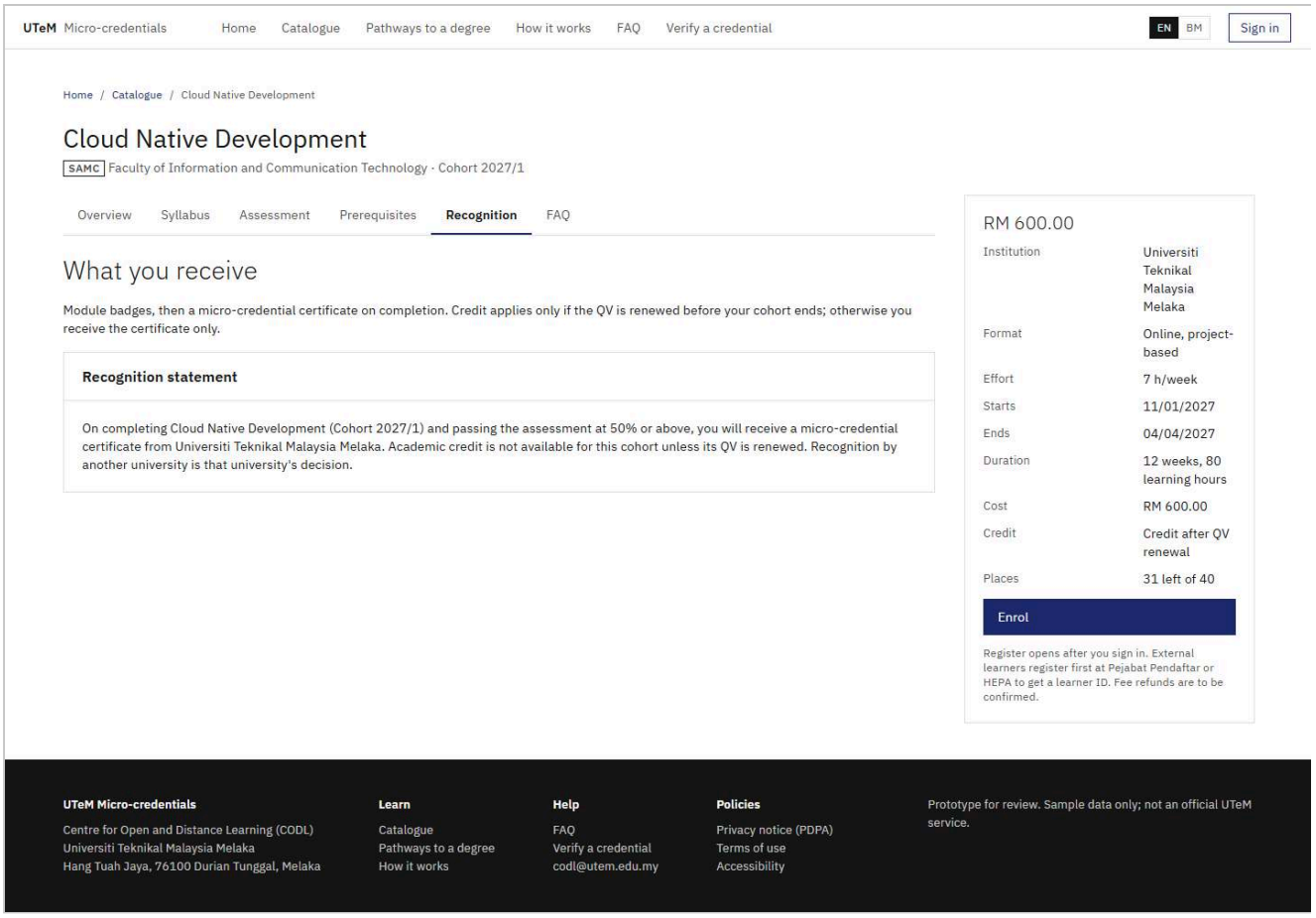


Figure 48: *PUB-03 Offering: Tab: Recognition.*

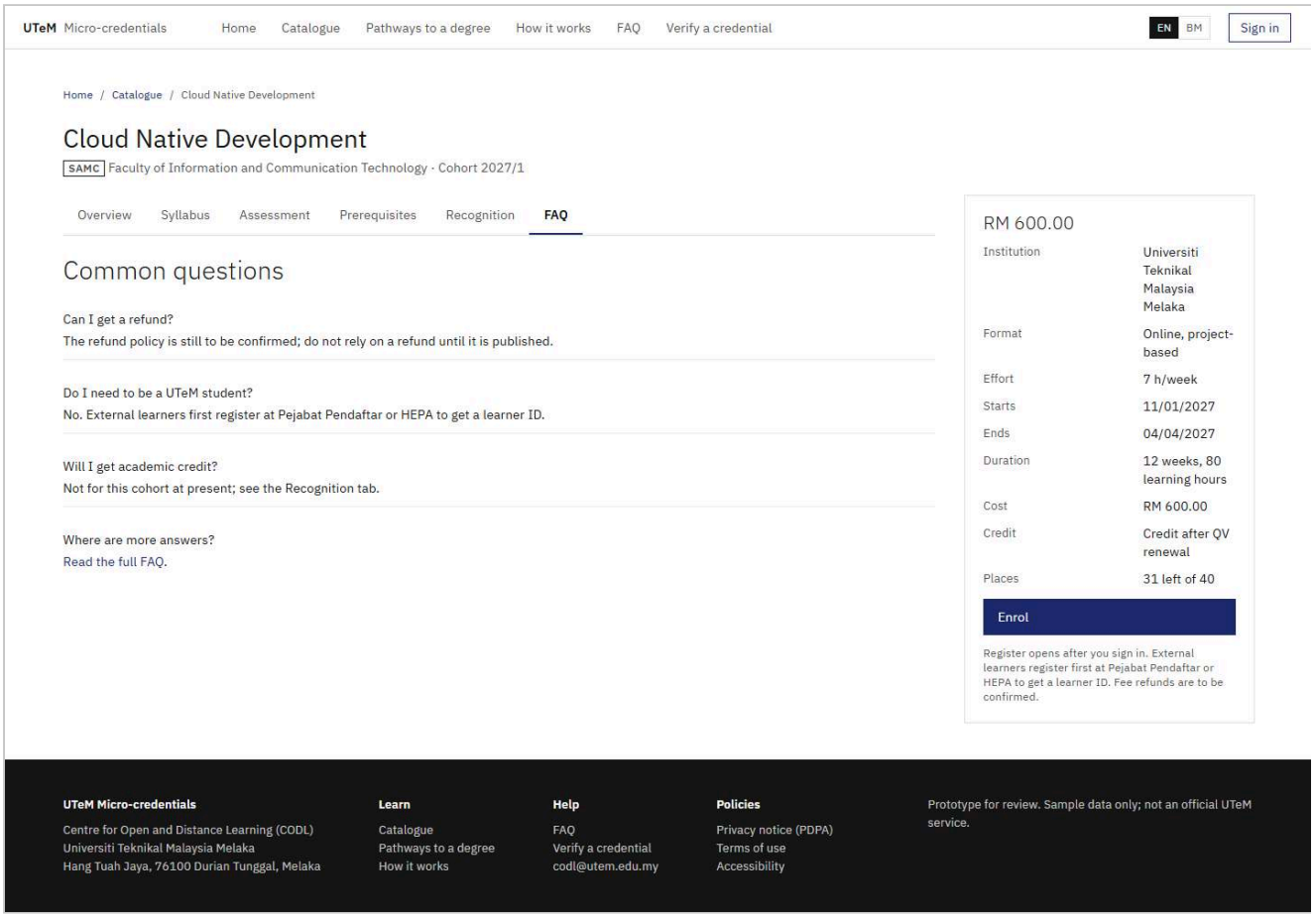


Figure 49: *PUB-03 Offering: Tab: FAQ.*

PUB-04 Pathways to a degree

`#/public/pathways`

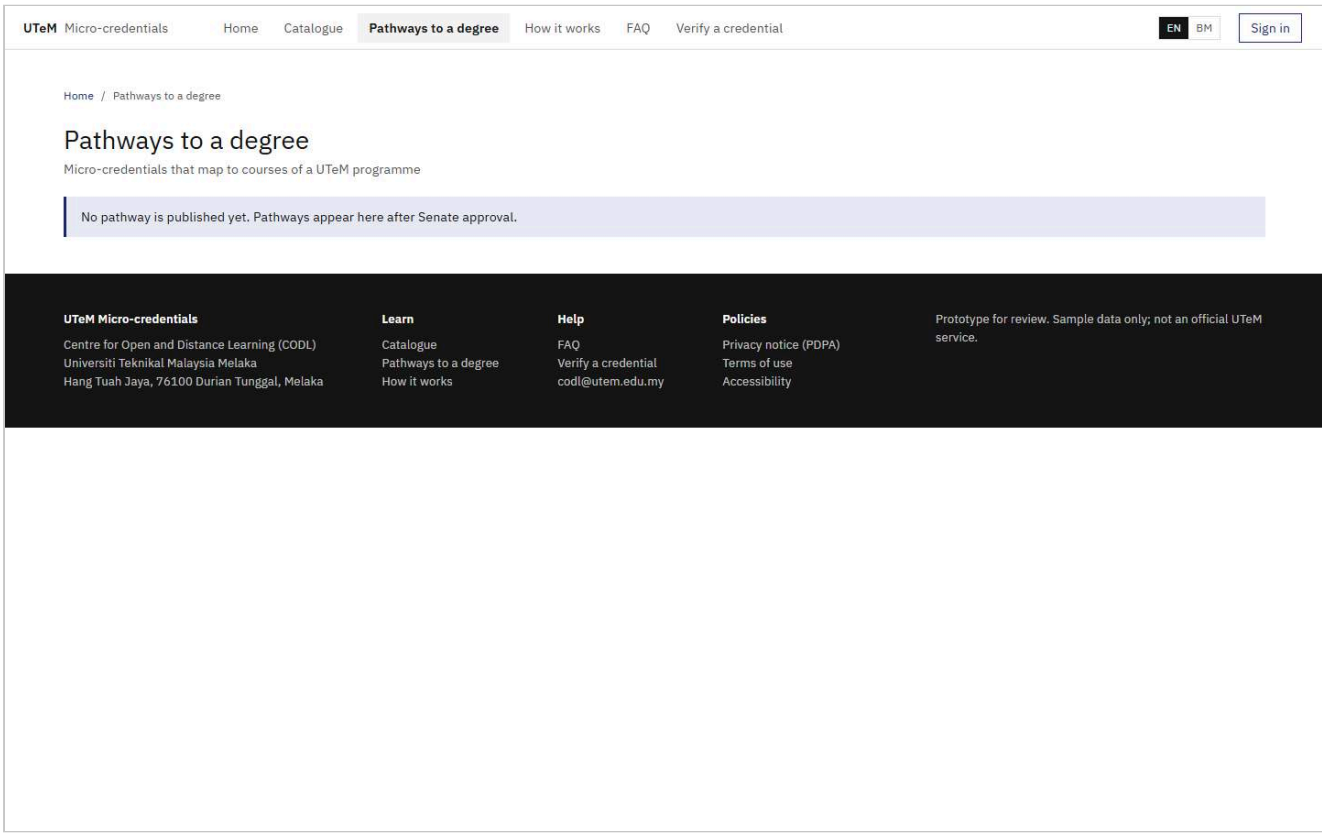


Figure 50: *PUB-04 Pathways to a degree.*

PUB-05 How recognition works

`#/public/how`

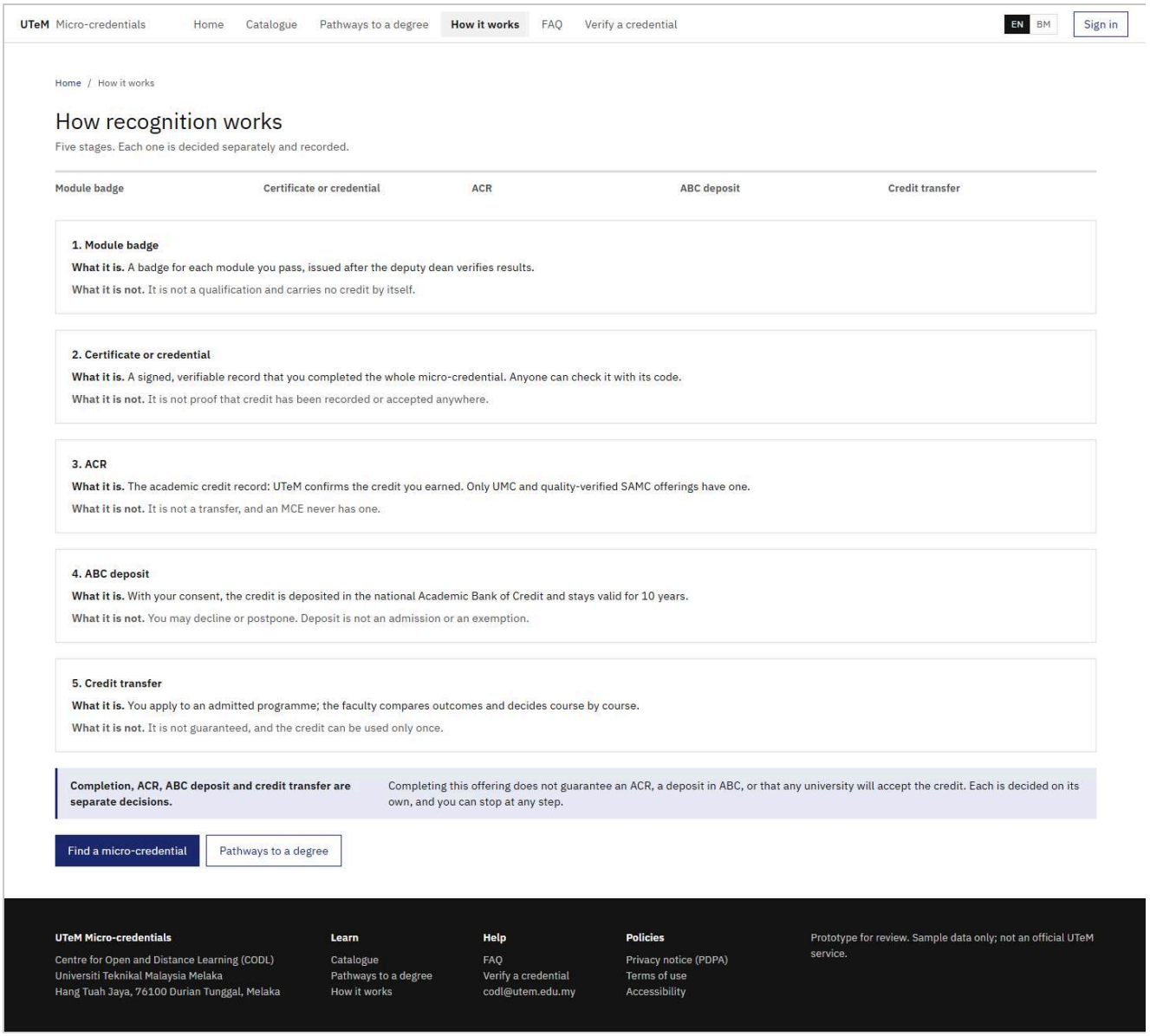


Figure 51: *PUB-05 How recognition works.*

PUB-06 FAQ

`#/public/faq`

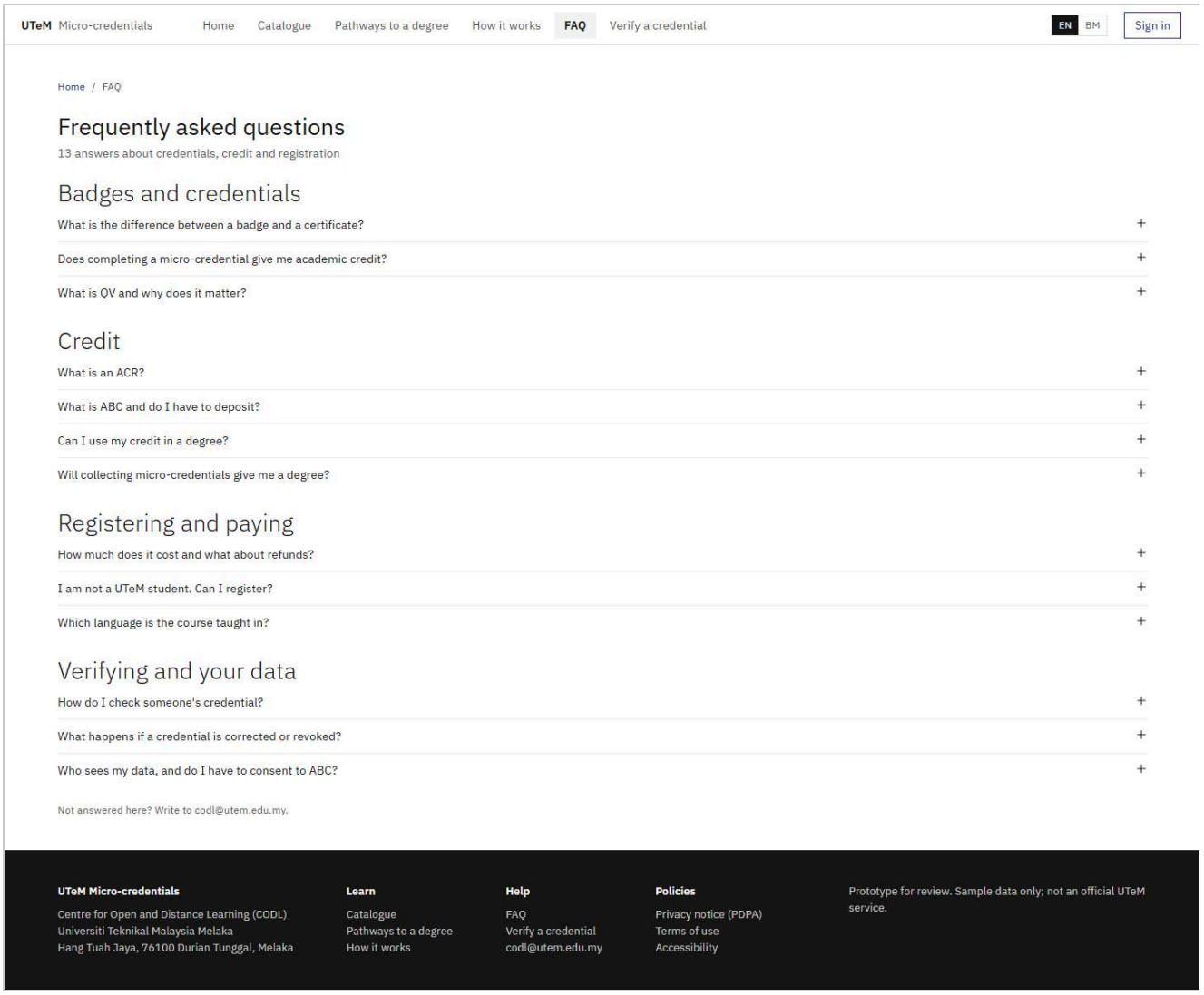


Figure 52: *PUB-06 FAQ.*

PUB-07 Verify a credential

`#/public/verify`

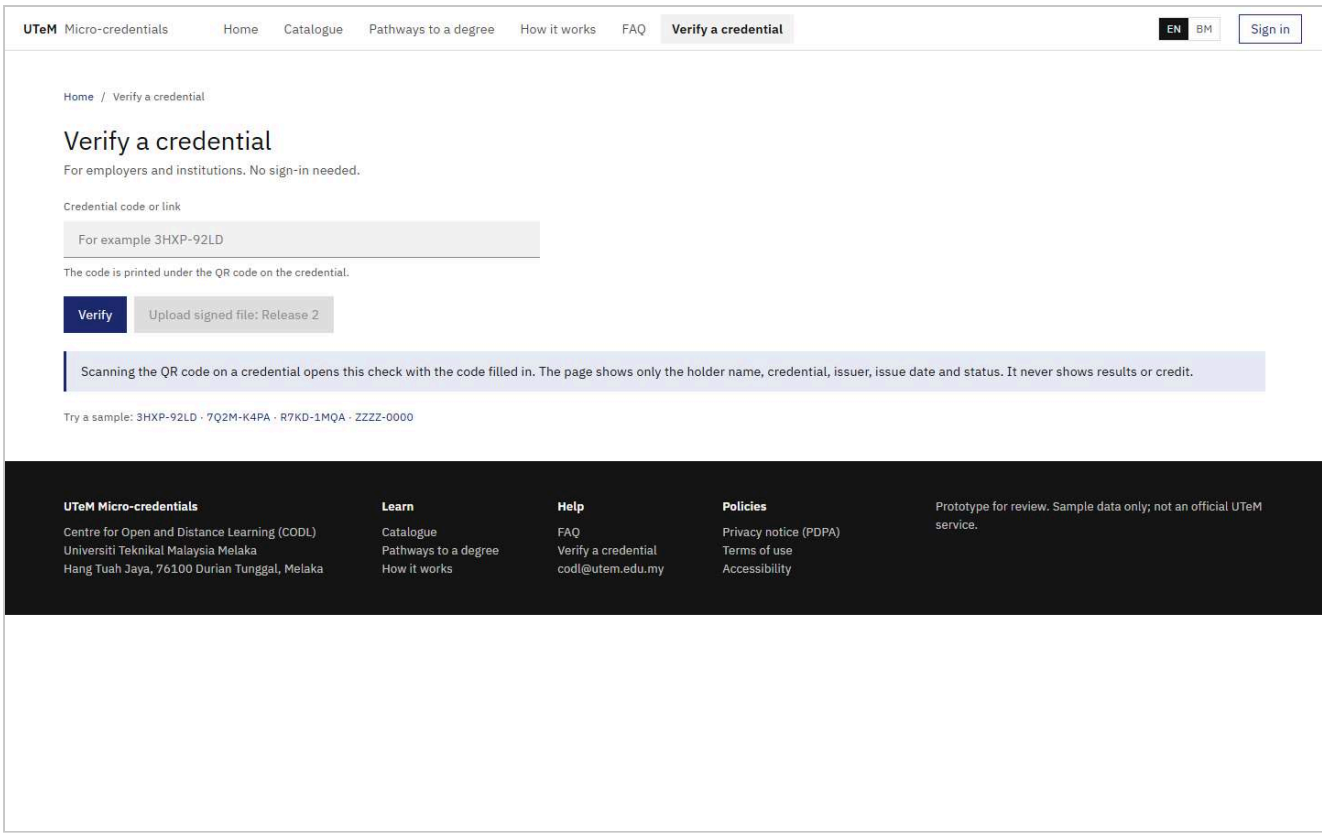


Figure 53: *PUB-07 Verify a credential.*

PUB-08 Verification result

`#/public/verify/3HXP-92LD`

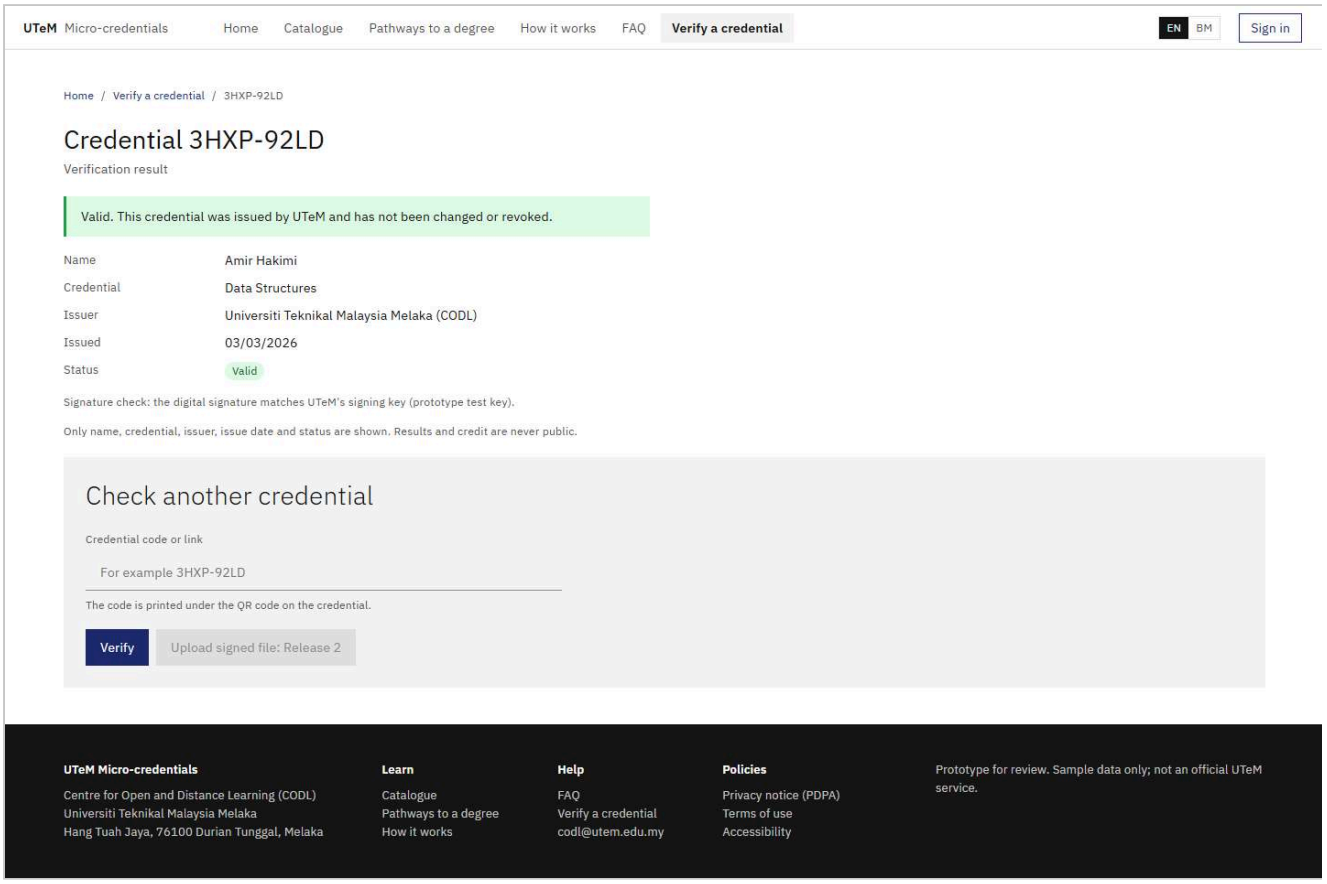


Figure 54: *PUB-08 Verification result.*

PUB-09 Privacy notice

`#/public/privacy`

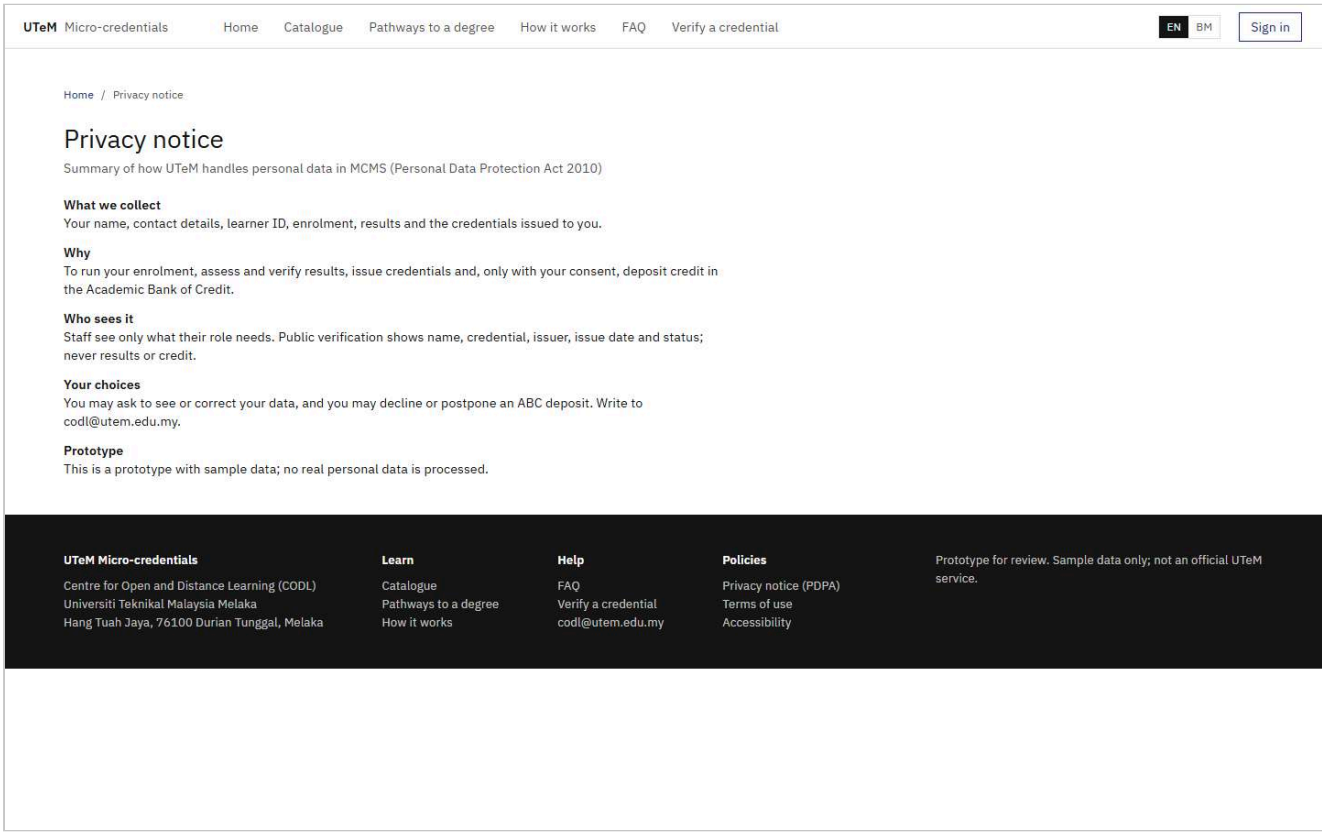


Figure 55: *PUB-09 Privacy notice.*

PUB-10 Terms of use

`#/public/terms`

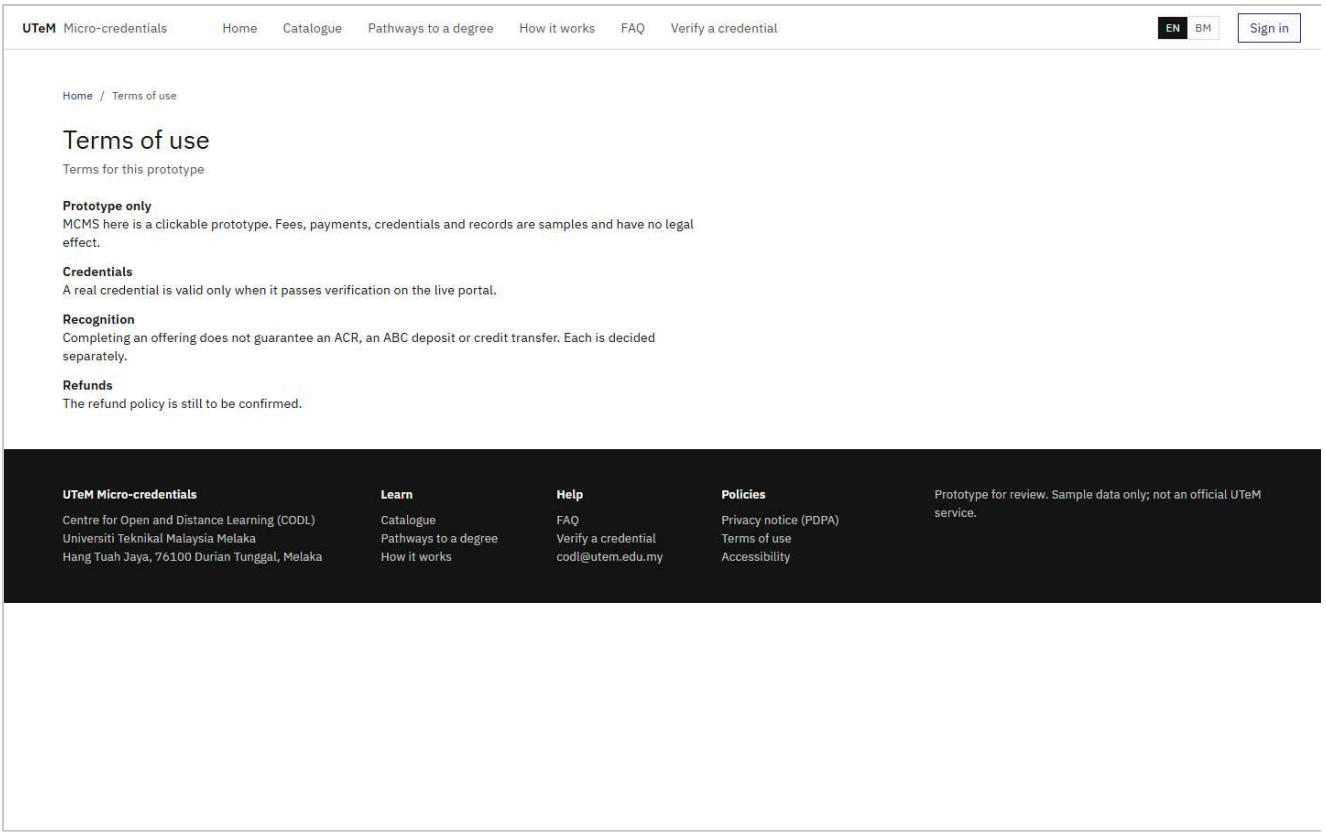


Figure 56: *PUB-10 Terms of use.*

PUB-11 Accessibility statement

`#/public/accessibility`

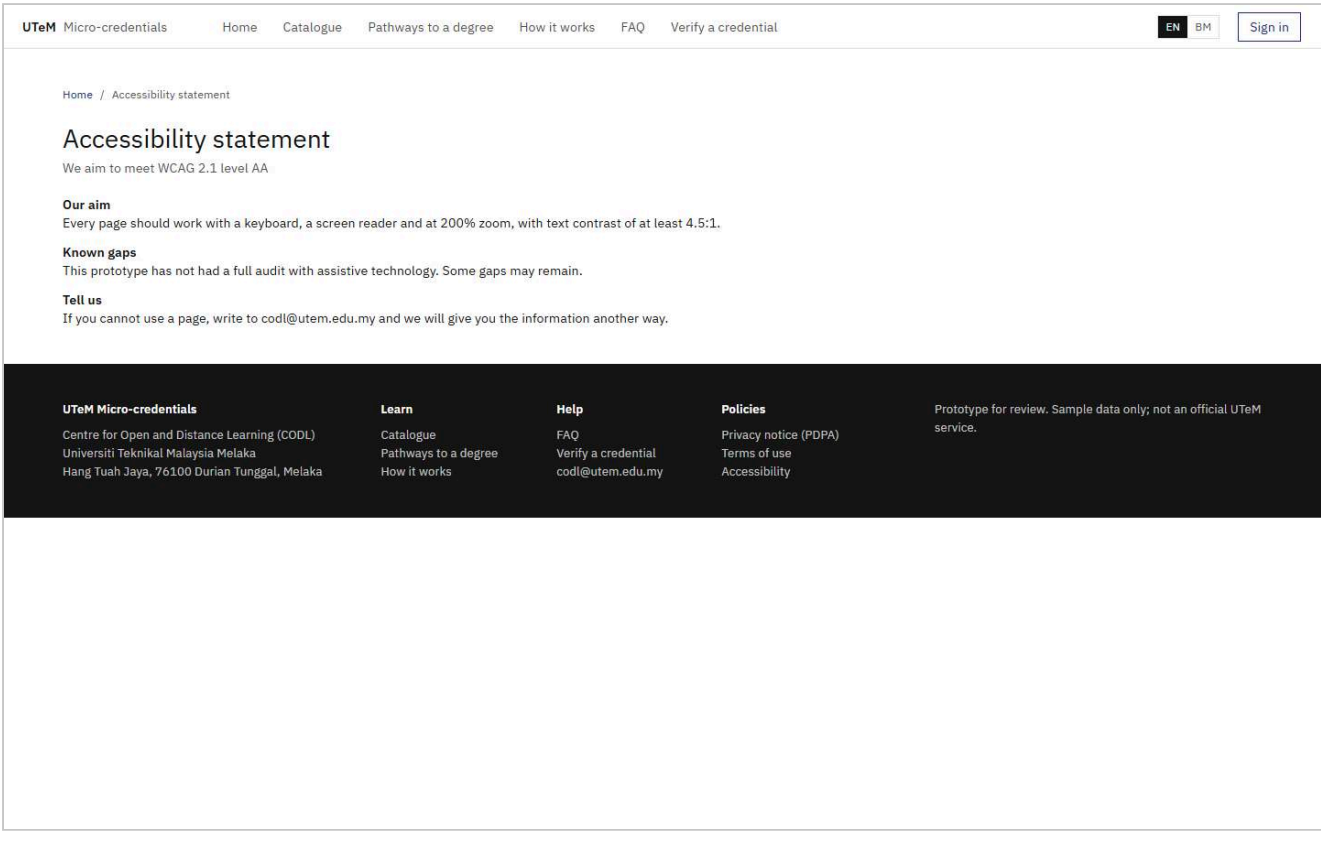


Figure 57: *PUB-11 Accessibility statement.*

B.3 Student

ST-01 My learning

`#/st/home`

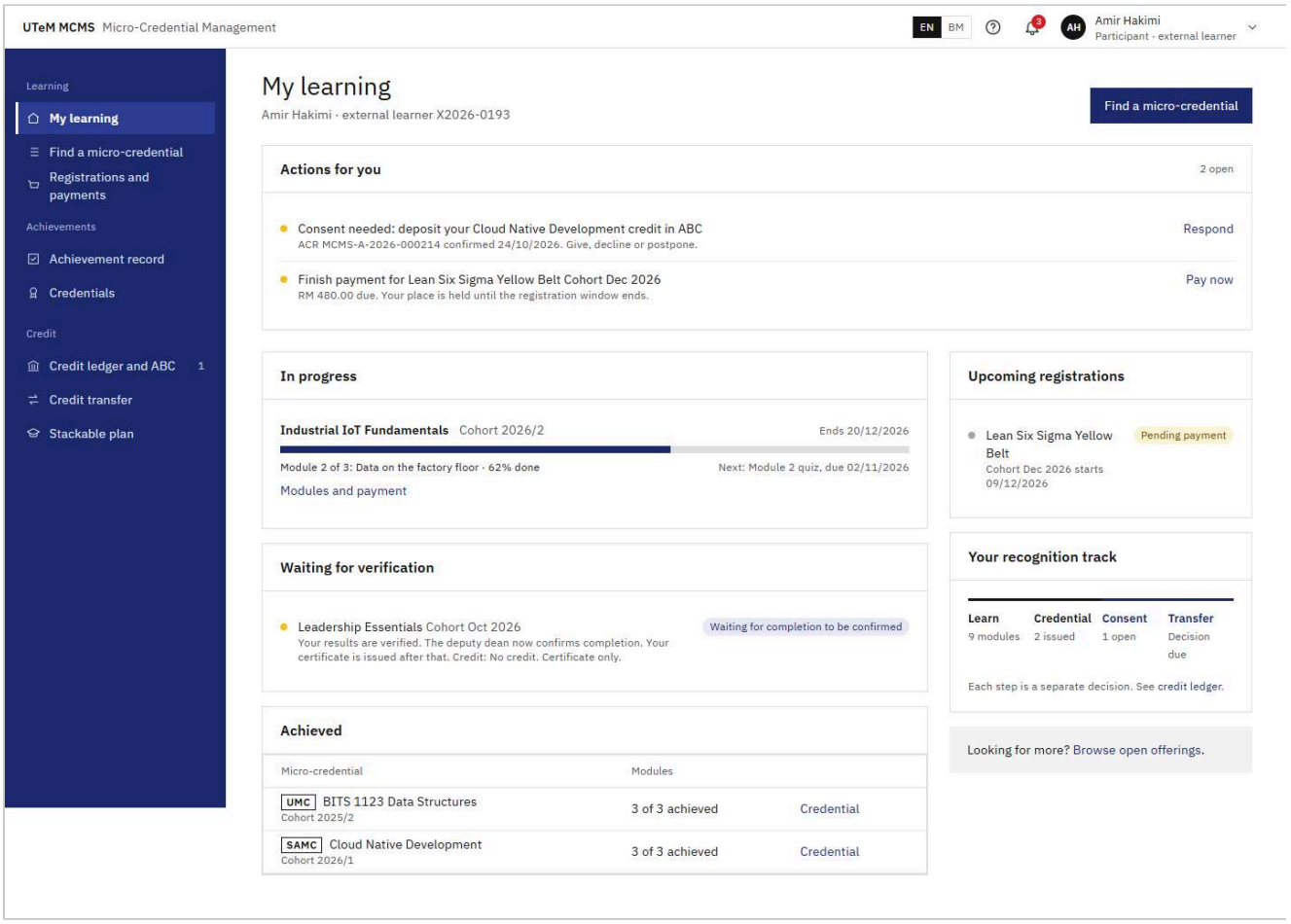


Figure 58: *ST-01 My learning.*

ST-02 Find a micro-credential

`#/st/catalogue`

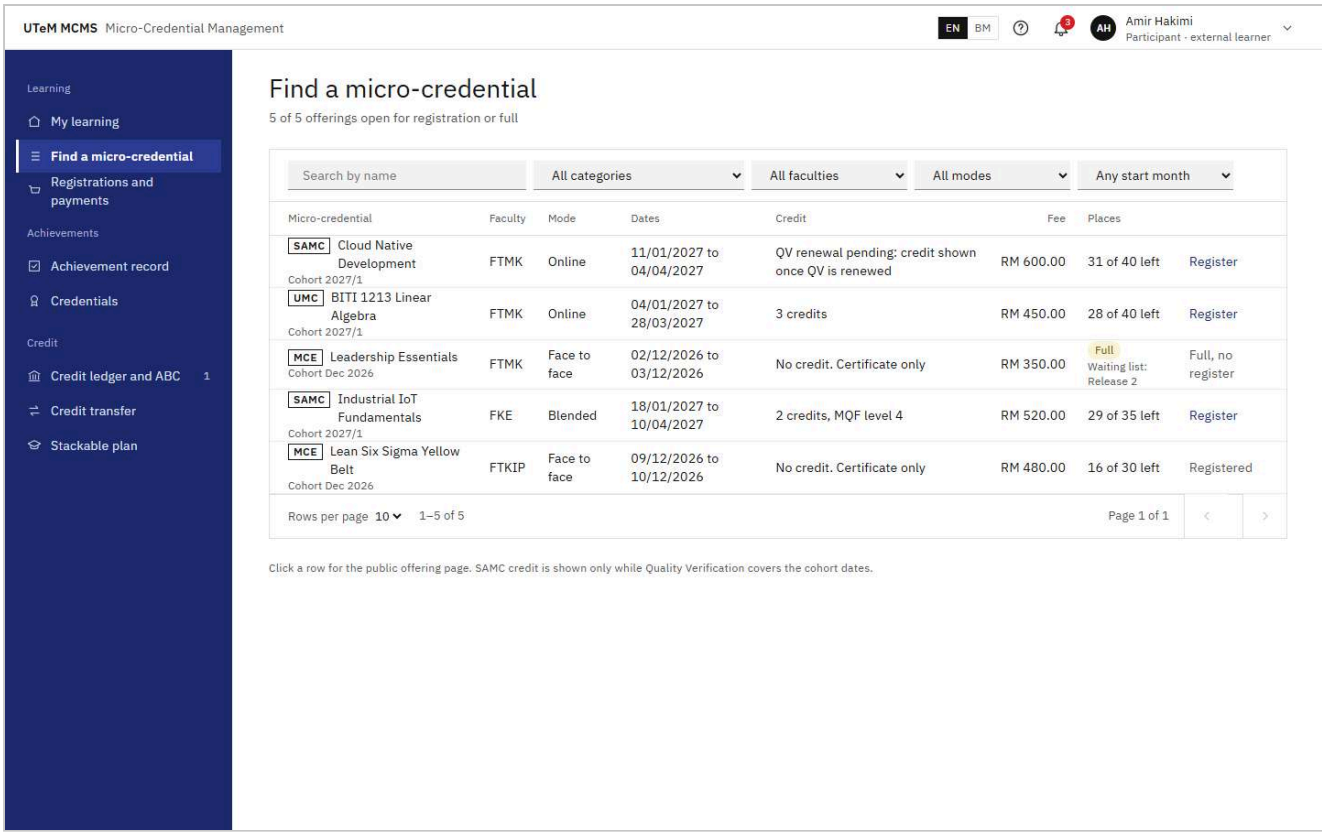


Figure 59: *ST-02 Find a micro-credential.*

ST-03 Register

`#/st/register/CND-2027-1`

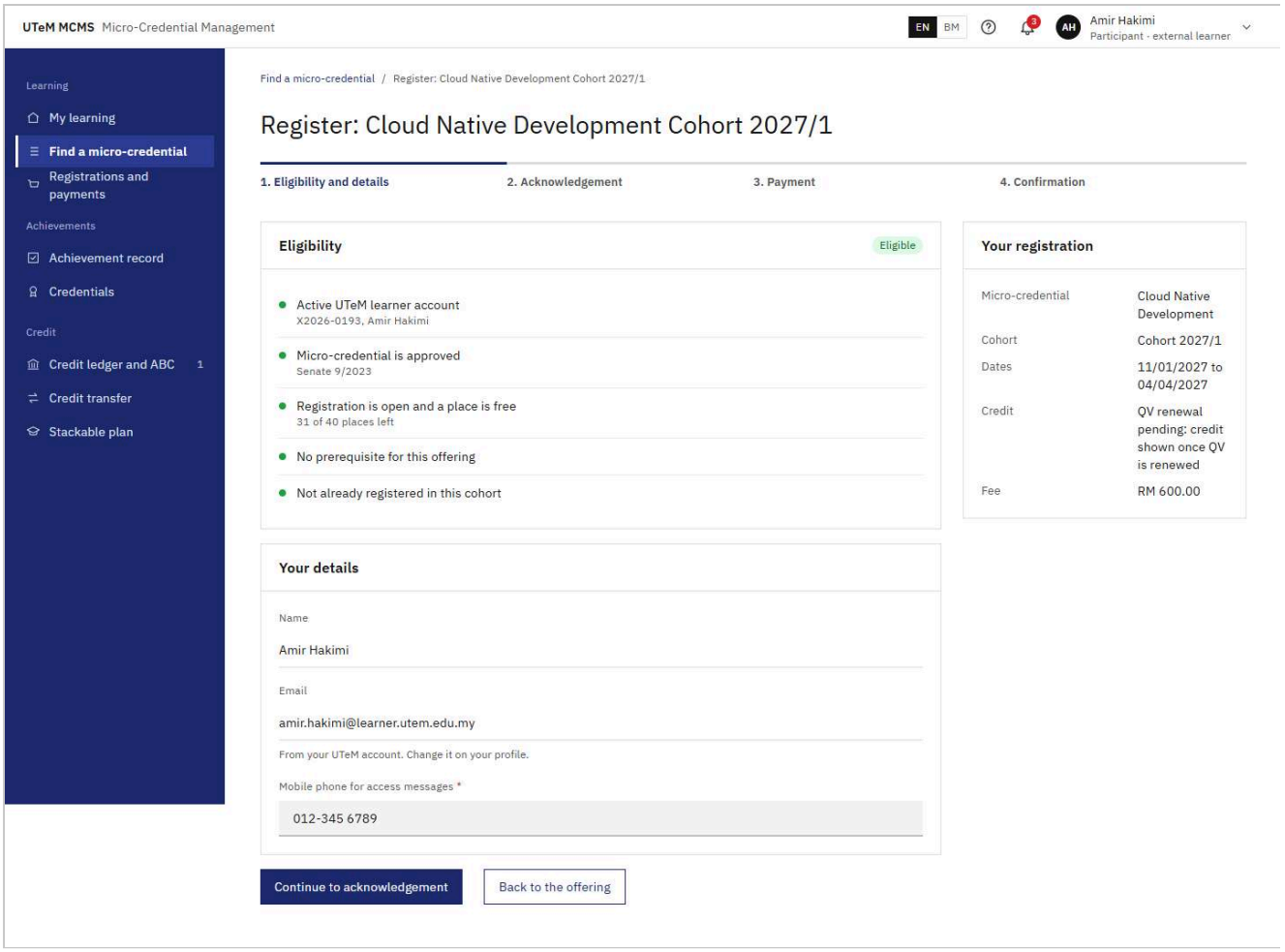


Figure 60: *ST-03 Register.*

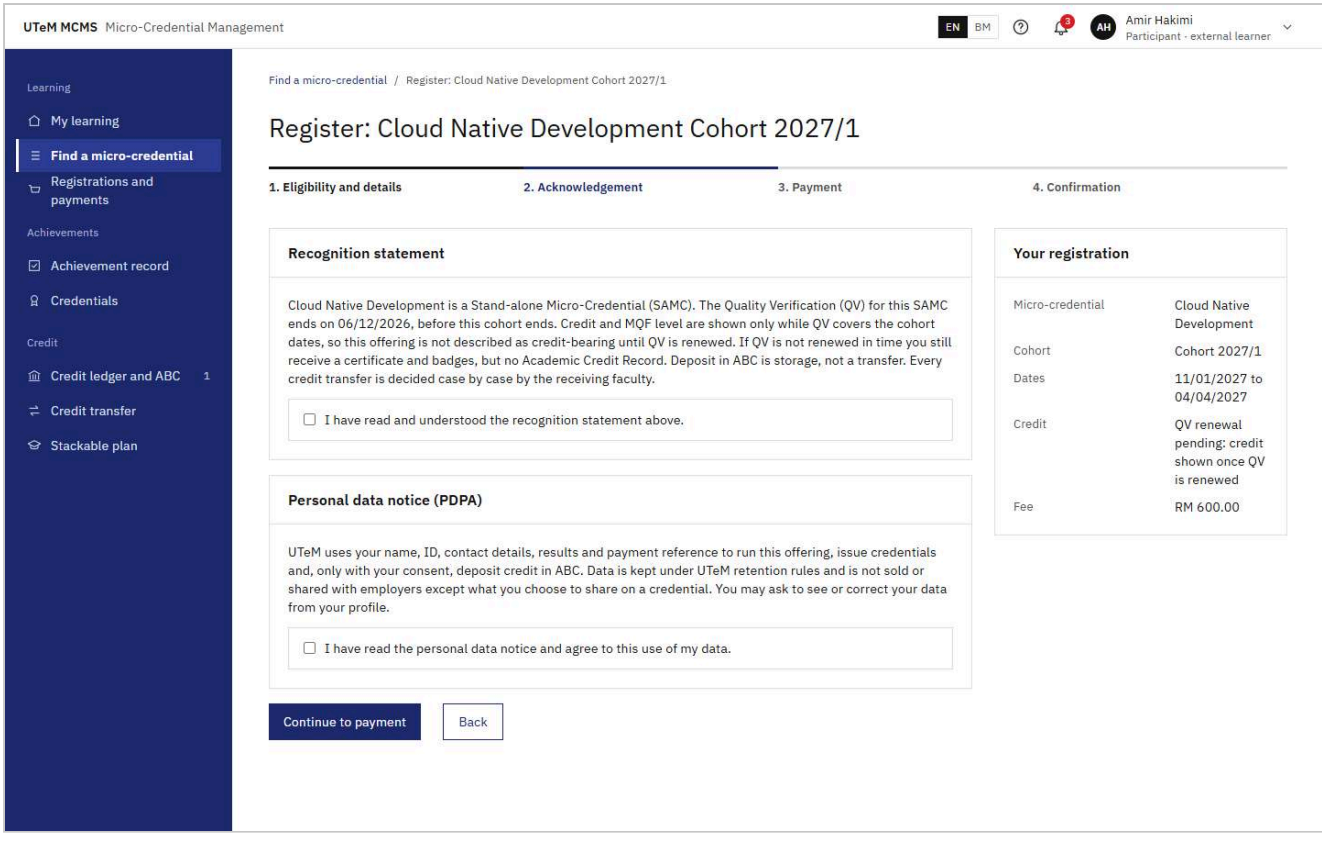


Figure 61: *ST-03 Register: Step 2.*

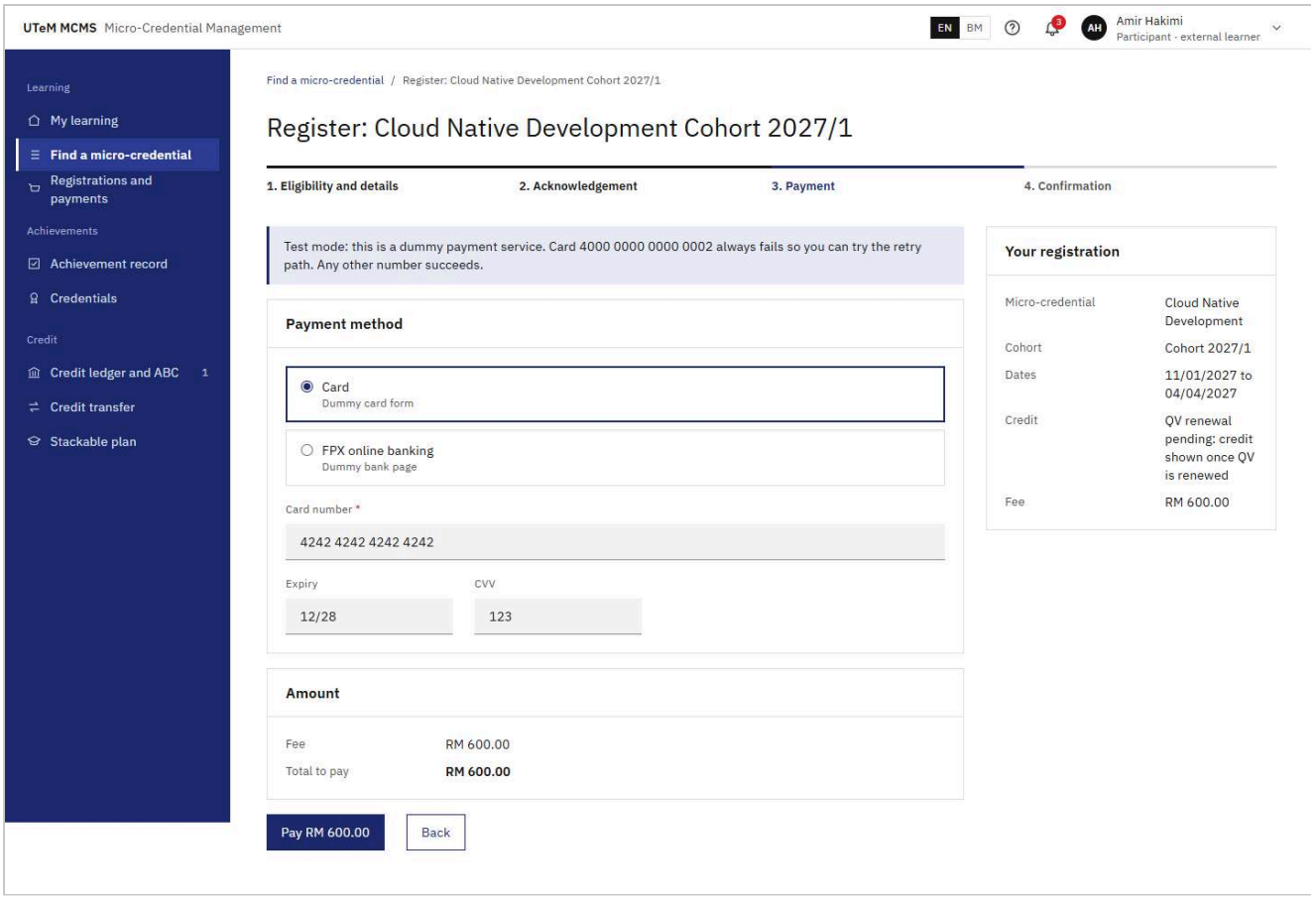


Figure 62: *ST-03 Register: Step 3.*

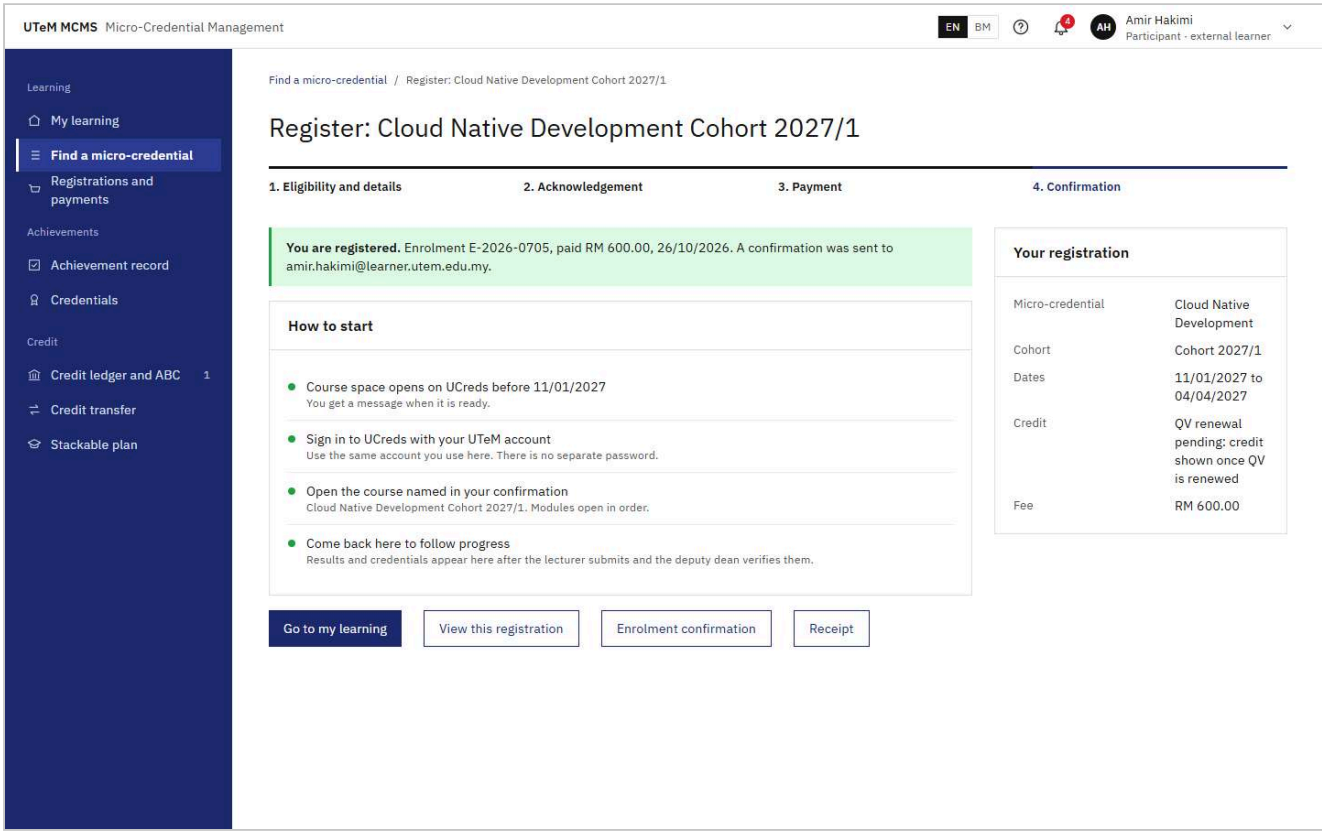


Figure 63: *ST-03 Register: Step 4: paid, registration confirmed.*

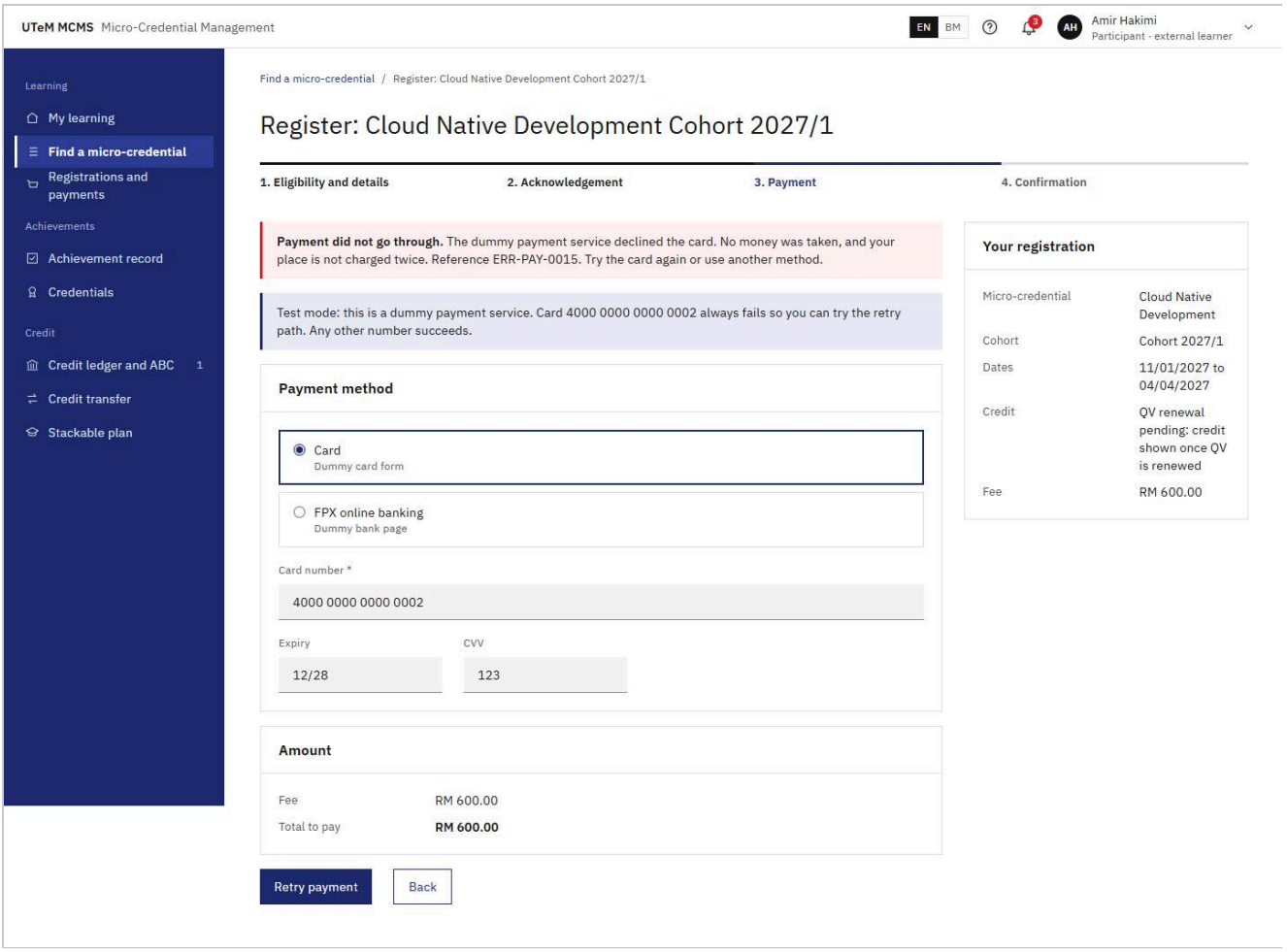


Figure 64: *ST-03 Register: Payment failed with the test card, ready to retry.*

ST-04 Registrations and payments

`#/st/enrolments`

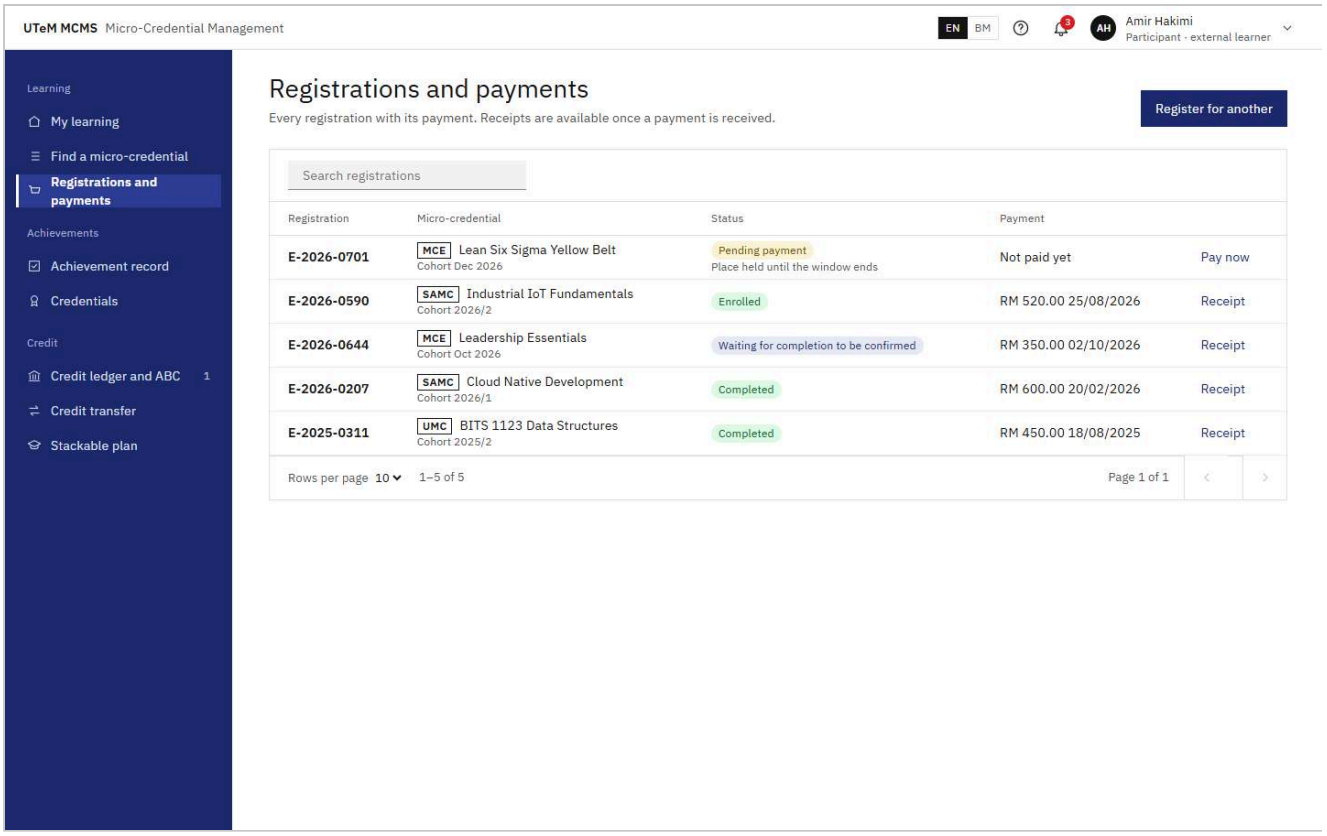


Figure 65: *ST-04 Registrations and payments.*

ST-05 Registration

`#/st/enrolment/E-2026-0590`

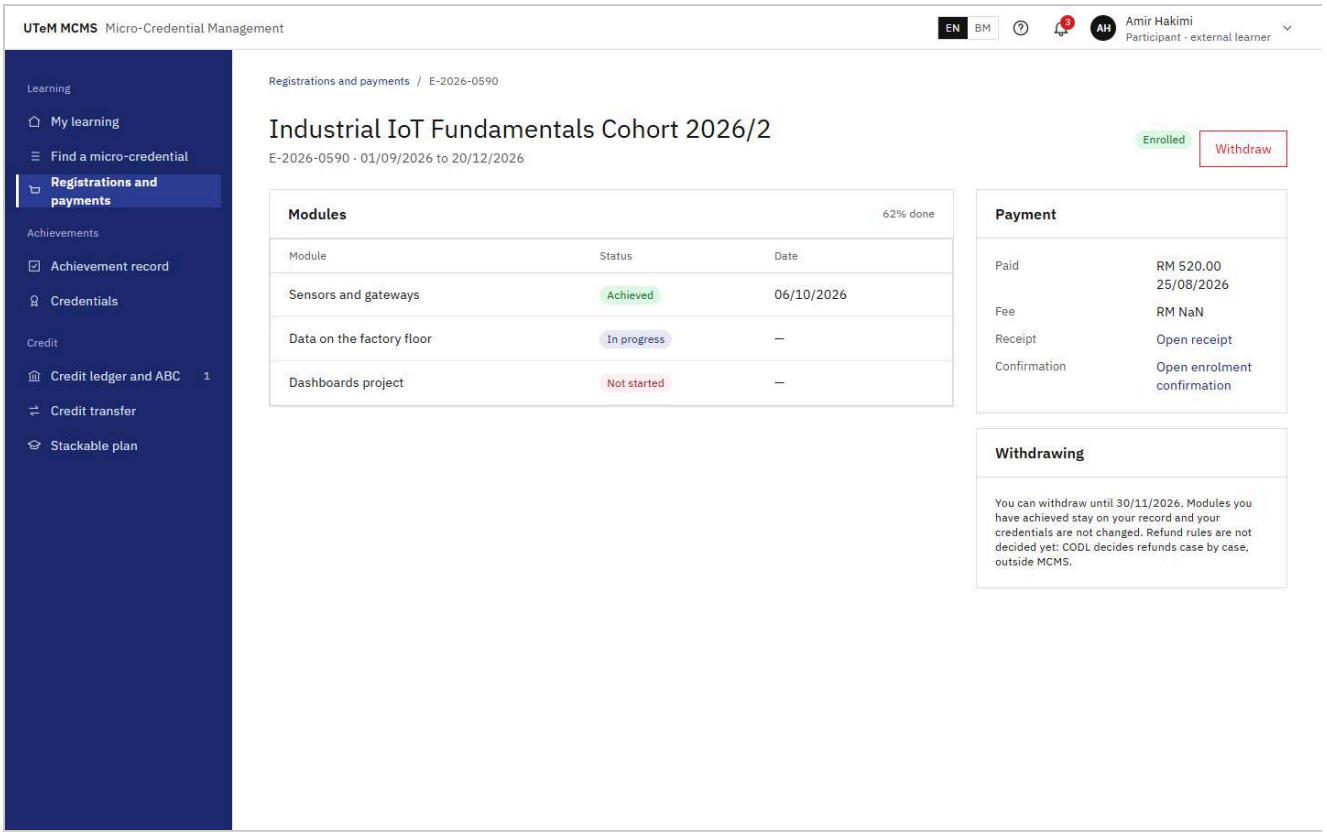


Figure 66: *ST-05 Registration.*

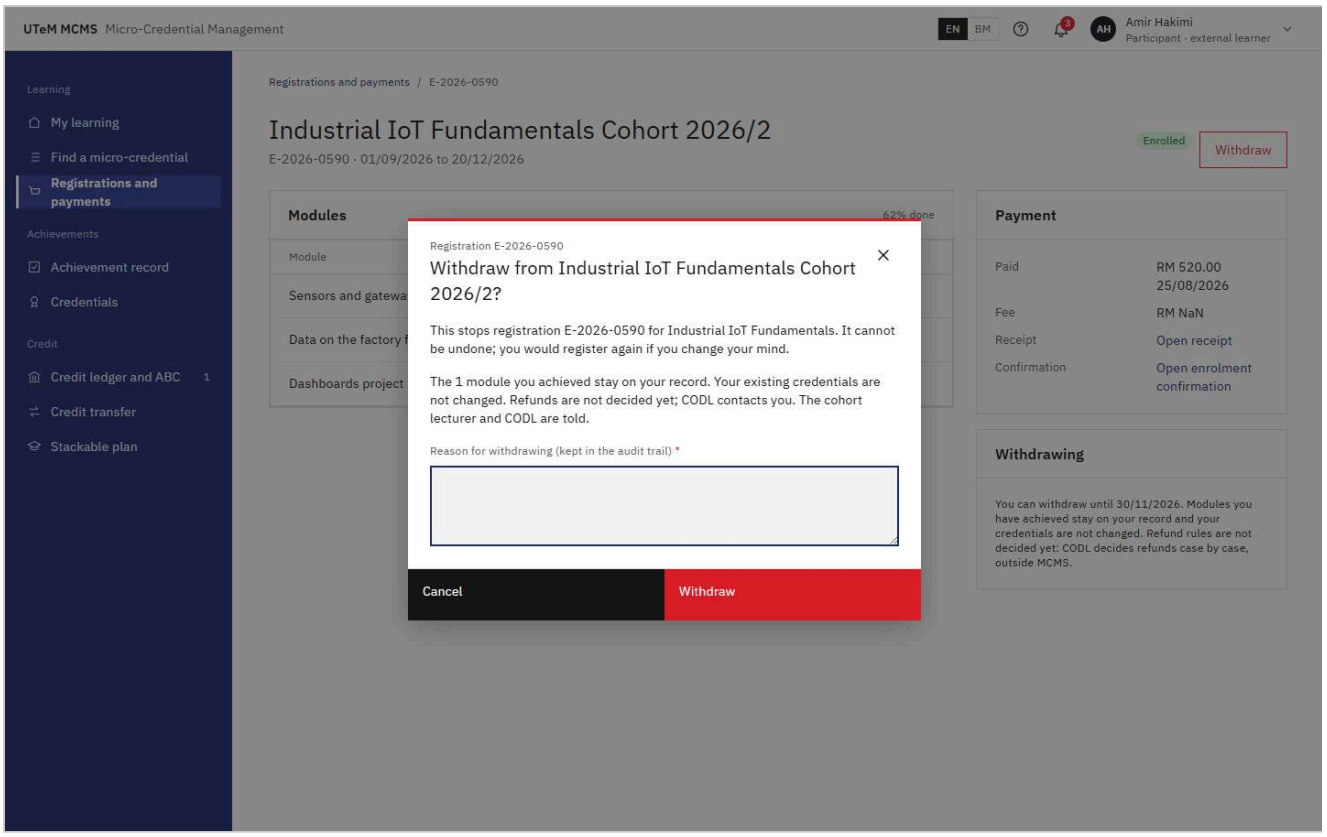


Figure 67: *ST-05 Warning (destructive): Withdraw from Industrial IoT Fundamentals Cohort 2026/2?*

ST-06 Achievement record

`#/st/record`

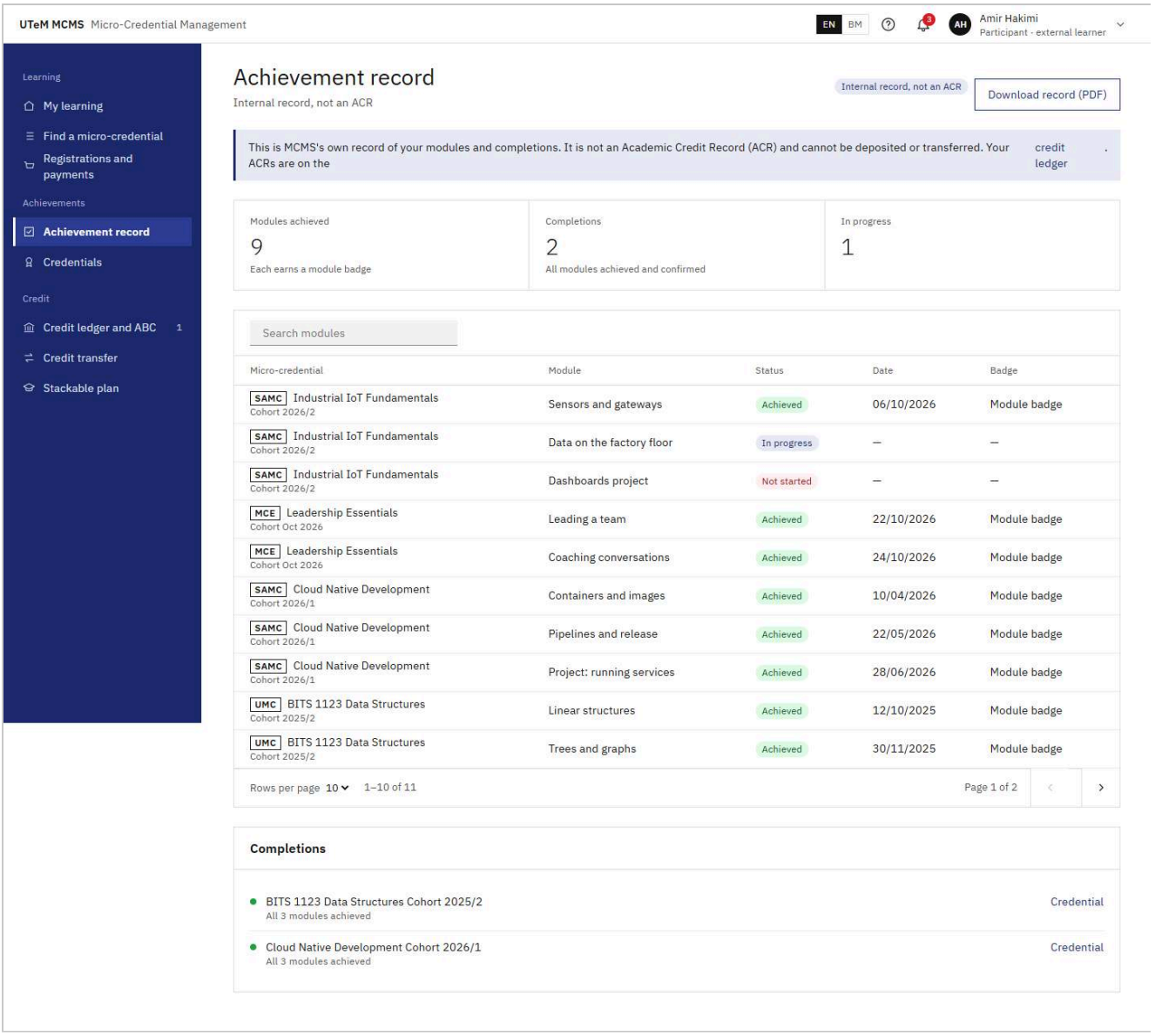


Figure 68: *ST-06 Achievement record.*

ST-07 Credentials

`#/st/credentials`

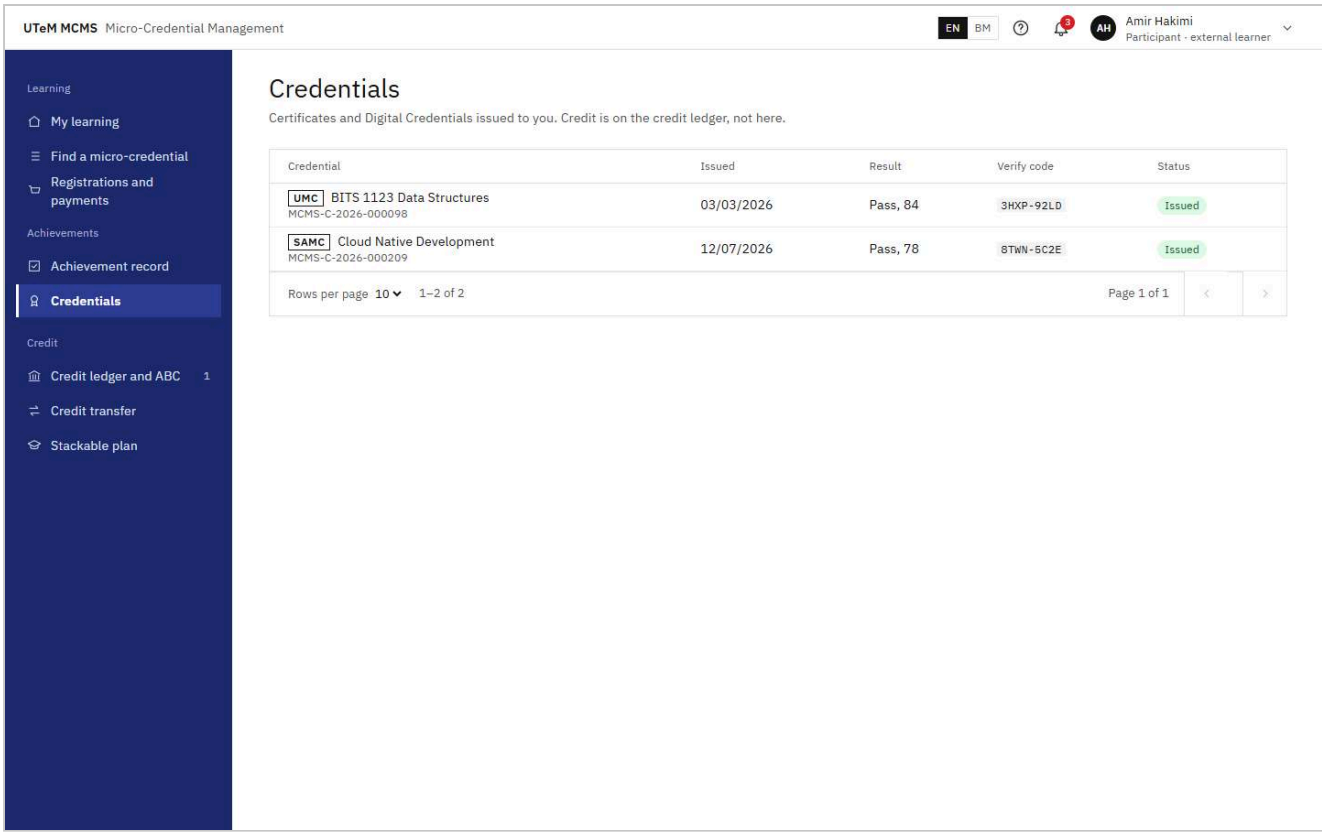


Figure 69: *ST-07 Credentials.*

ST-08 Credential

`#/st/credential/MCMS-C-2026-000098`

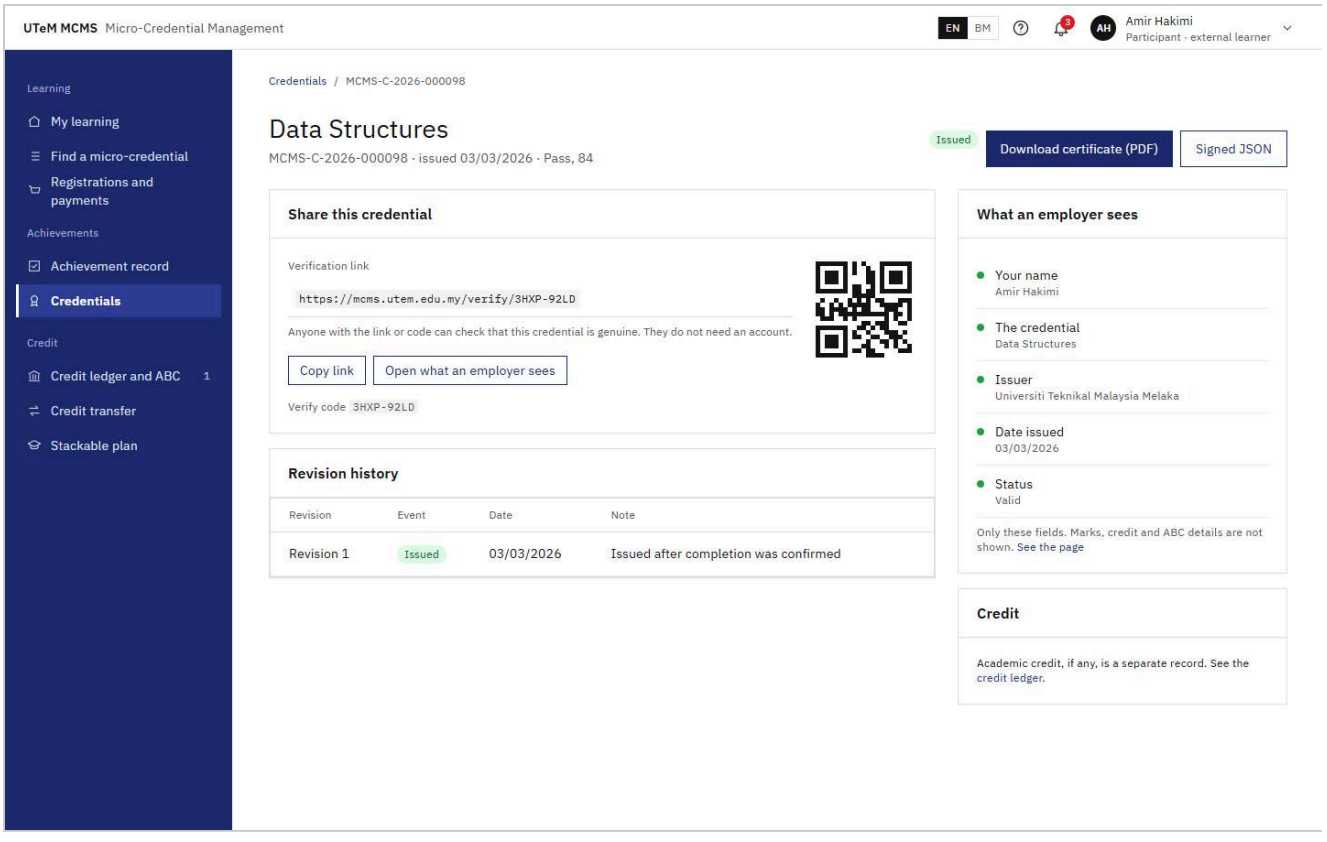


Figure 70: *ST-08 Credential.*

ST-09 Credit ledger and ABC

`#/st/ledger`

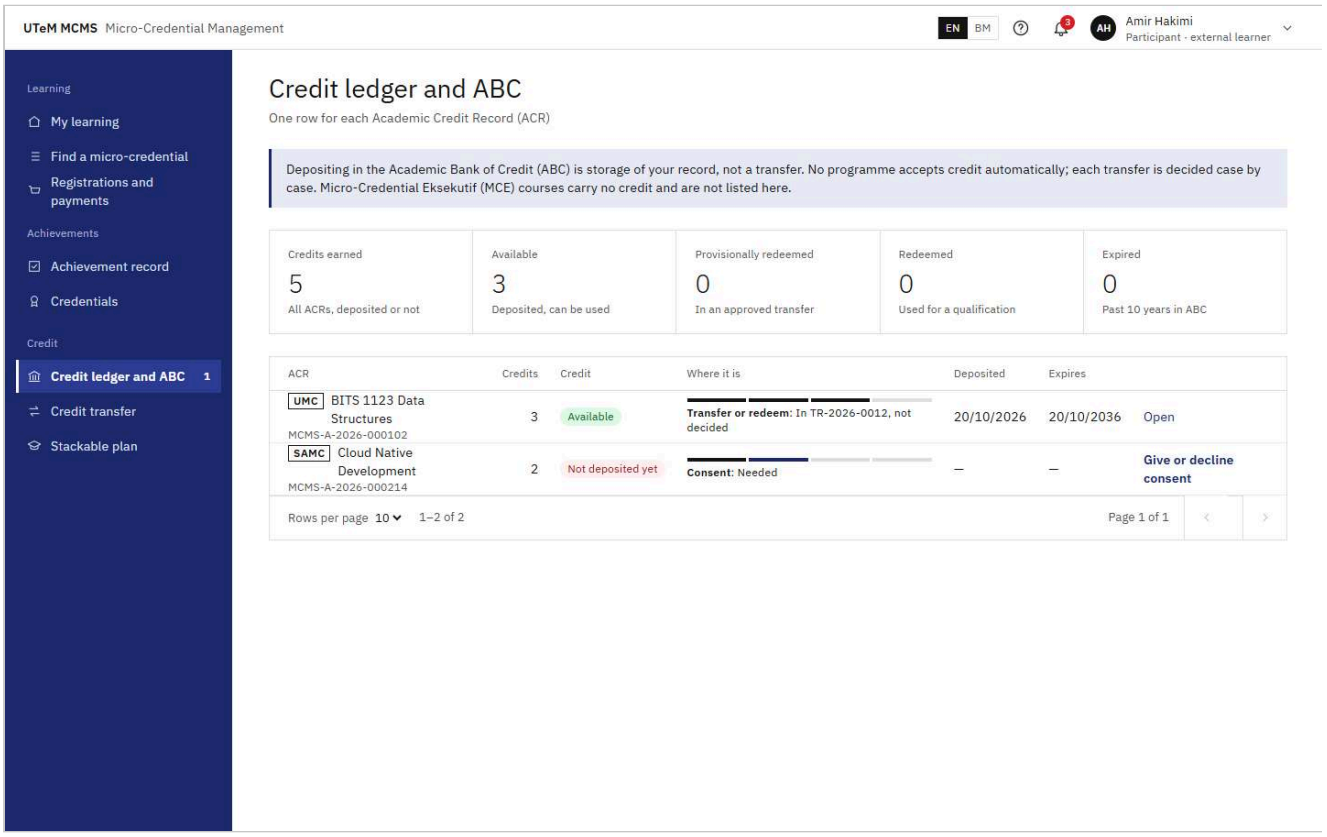


Figure 71: *ST-09 Credit ledger and ABC.*

ST-10 Academic Credit Record

`#/st/acr/MCMS-A-2026-000214`

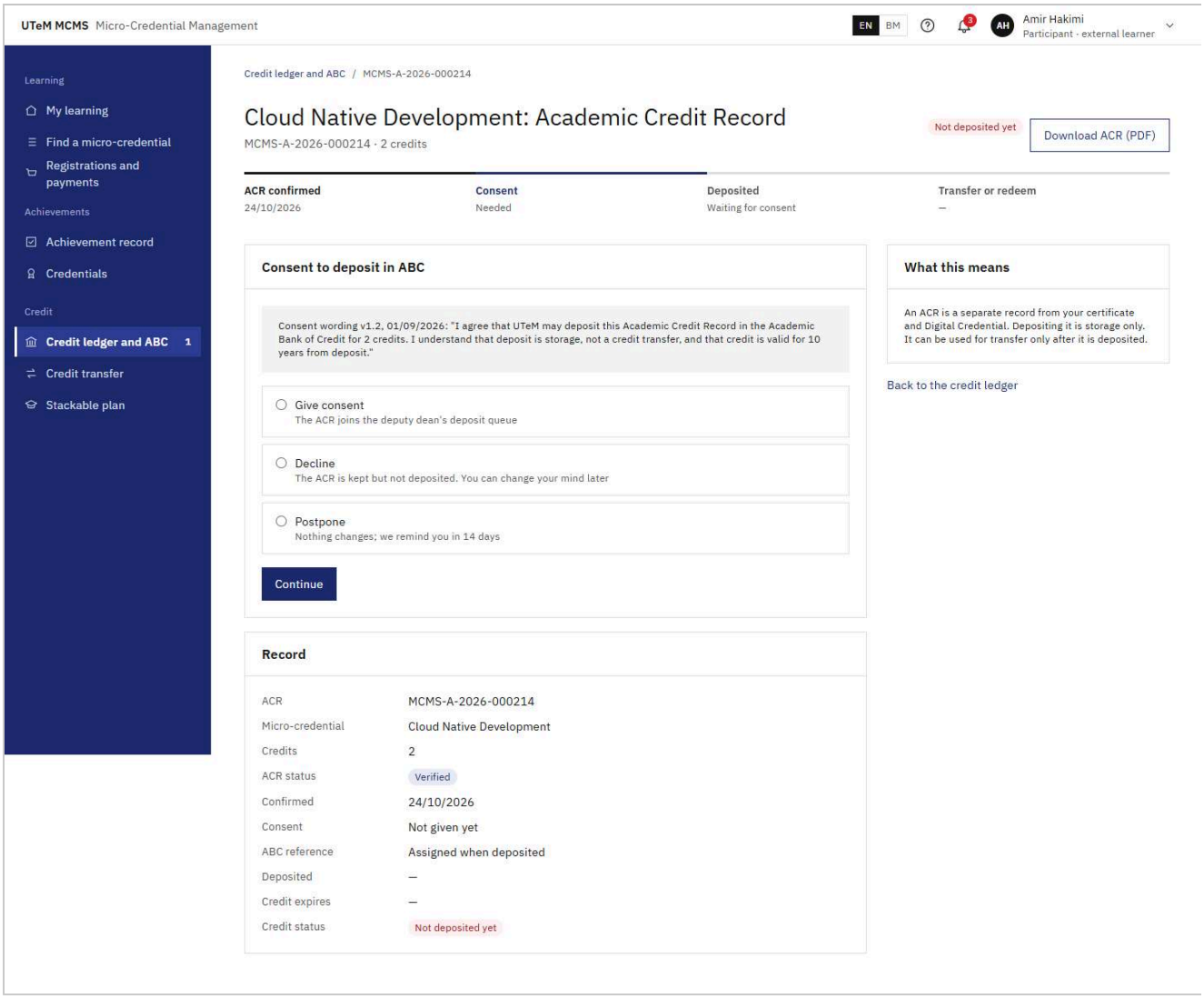


Figure 72: *ST-10 Academic Credit Record.*

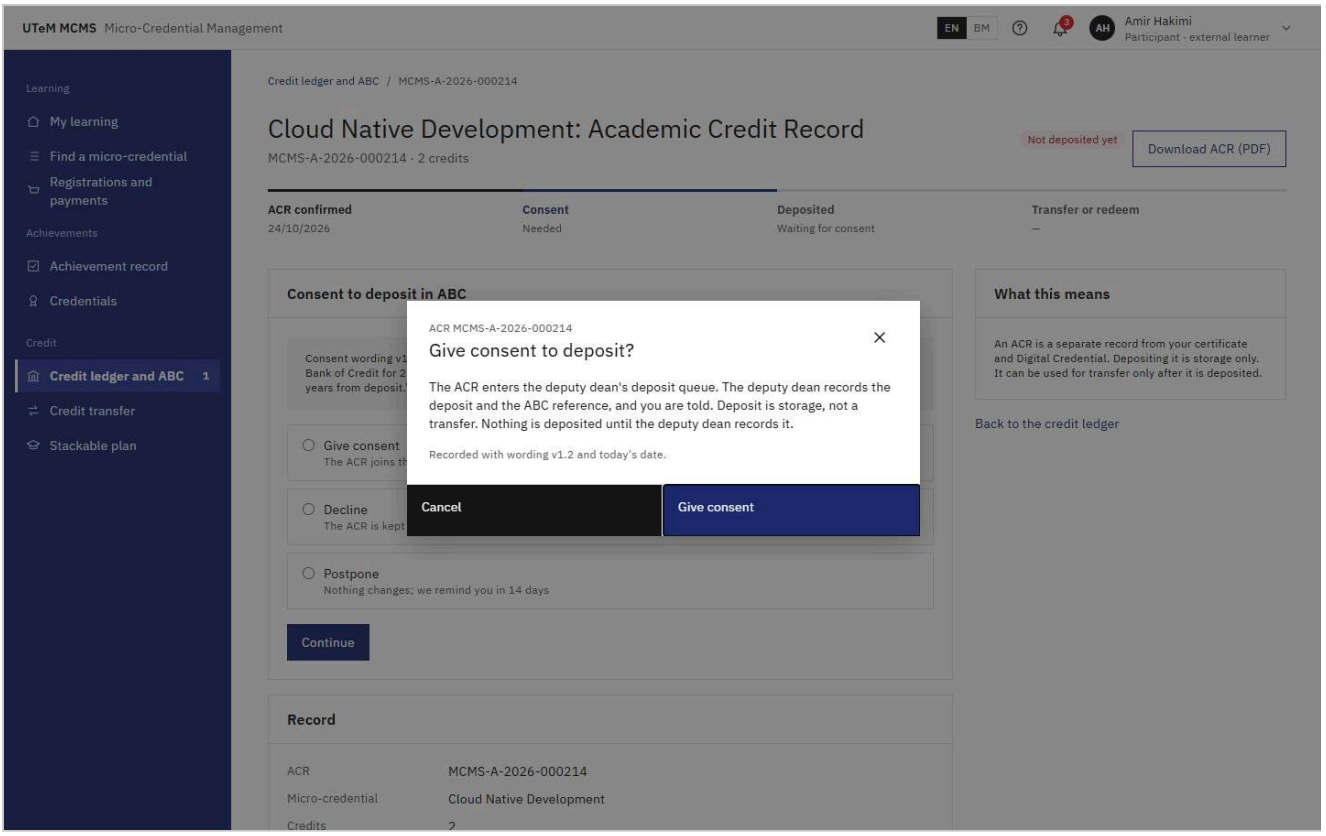


Figure 73: *ST-10 Confirmation: Give consent to deposit?*

ST-11 Credit transfer

`#/st/transfer`

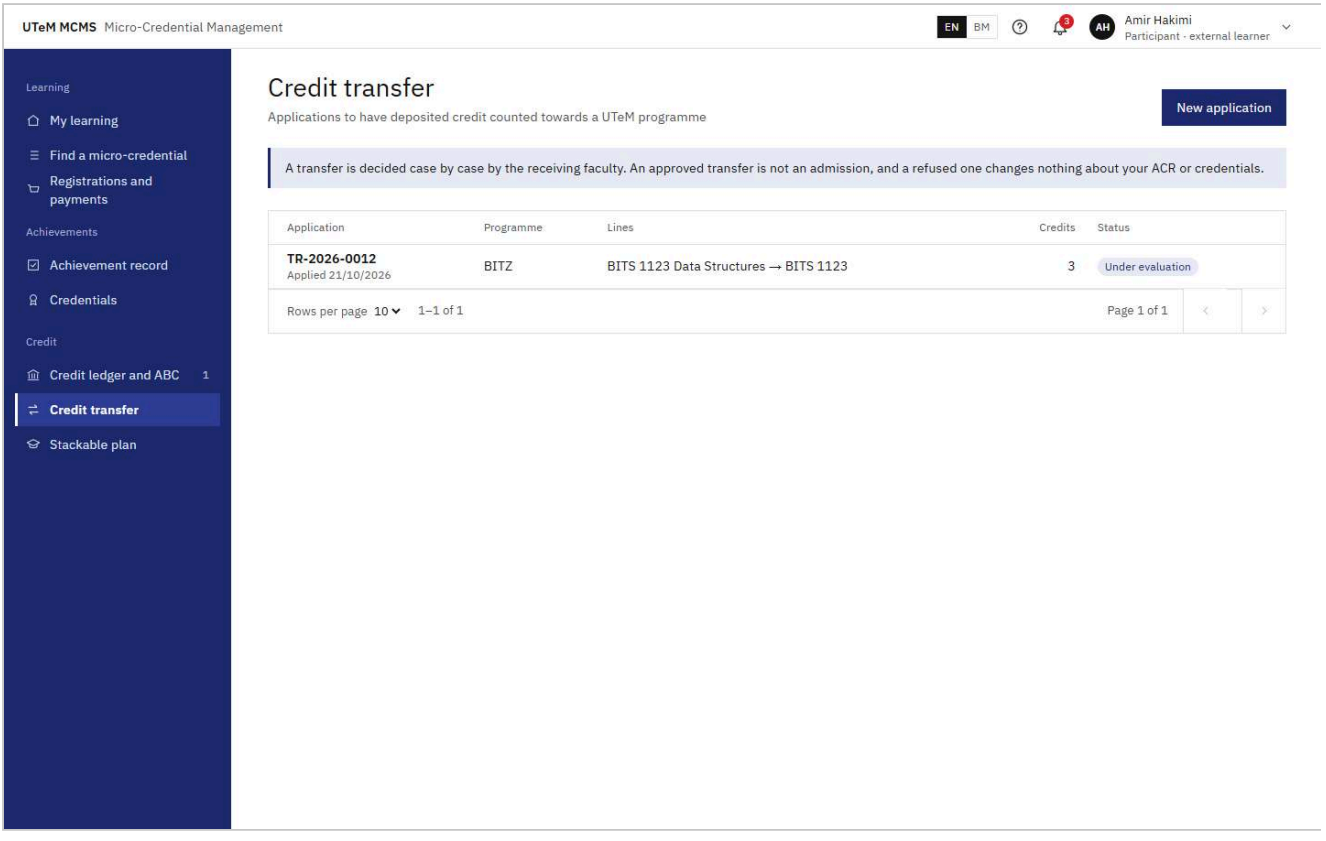


Figure 74: *ST-11 Credit transfer.*

ST-12 New transfer application

`#/st/transfer/new`

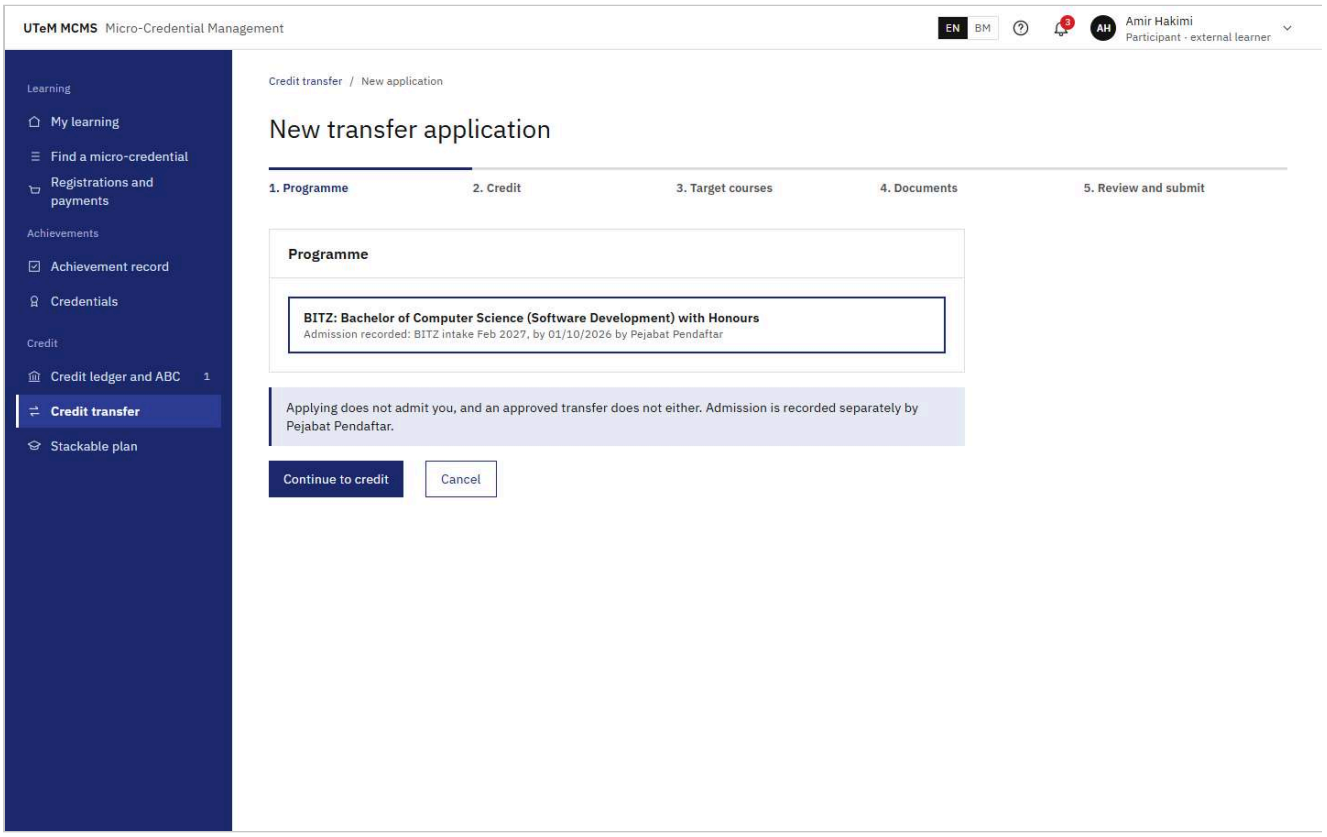


Figure 75: *ST-12 New transfer application.*

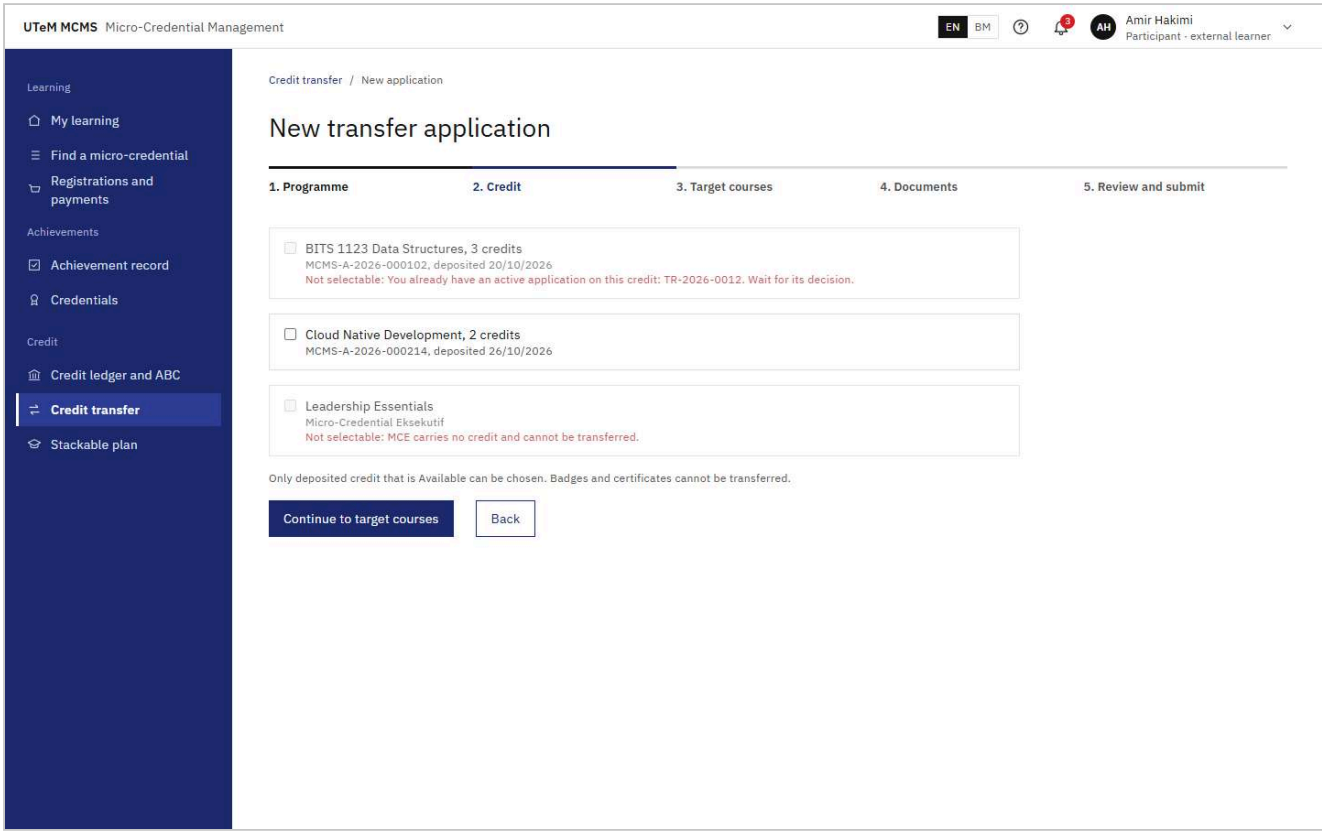


Figure 76: *ST-12 New transfer application: Step 2.*

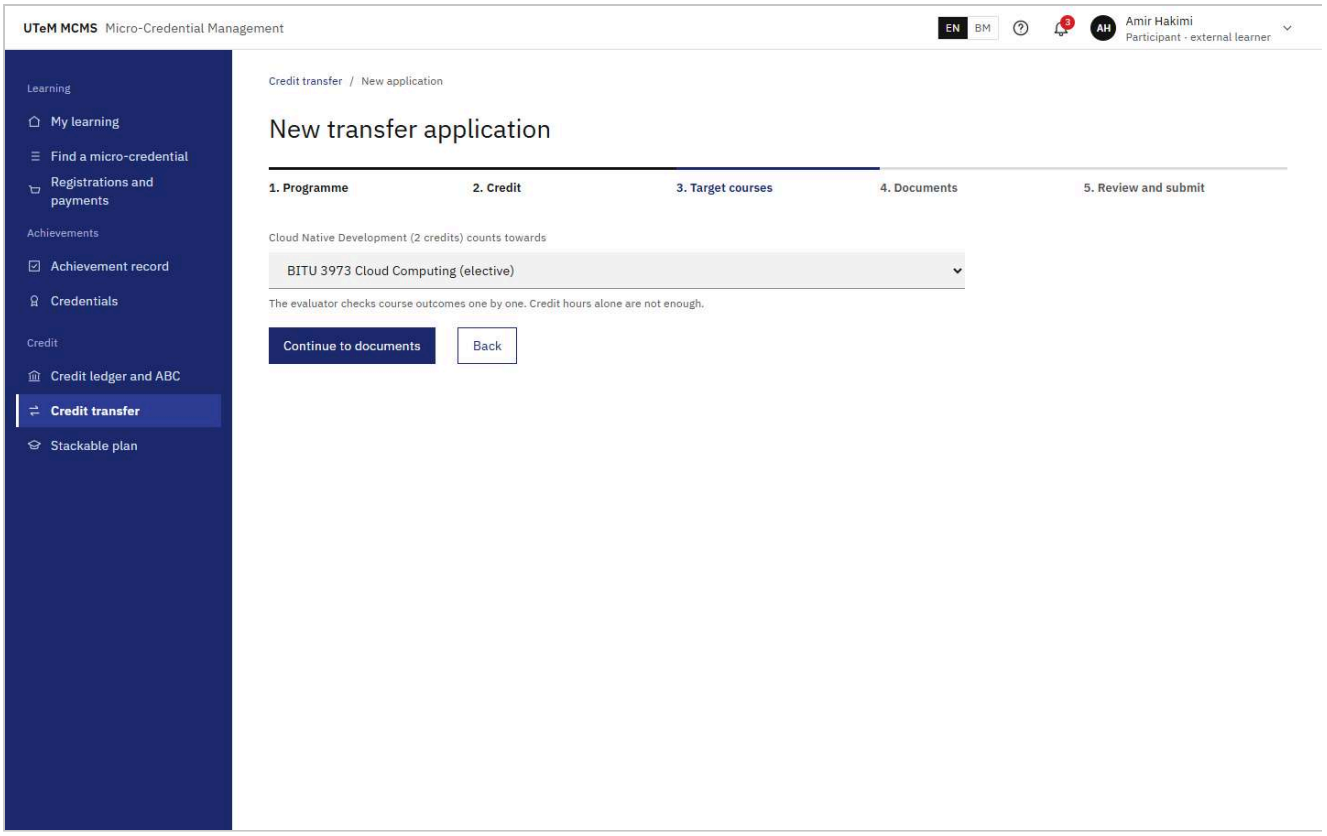


Figure 77: *ST-12 New transfer application: Step 3.*

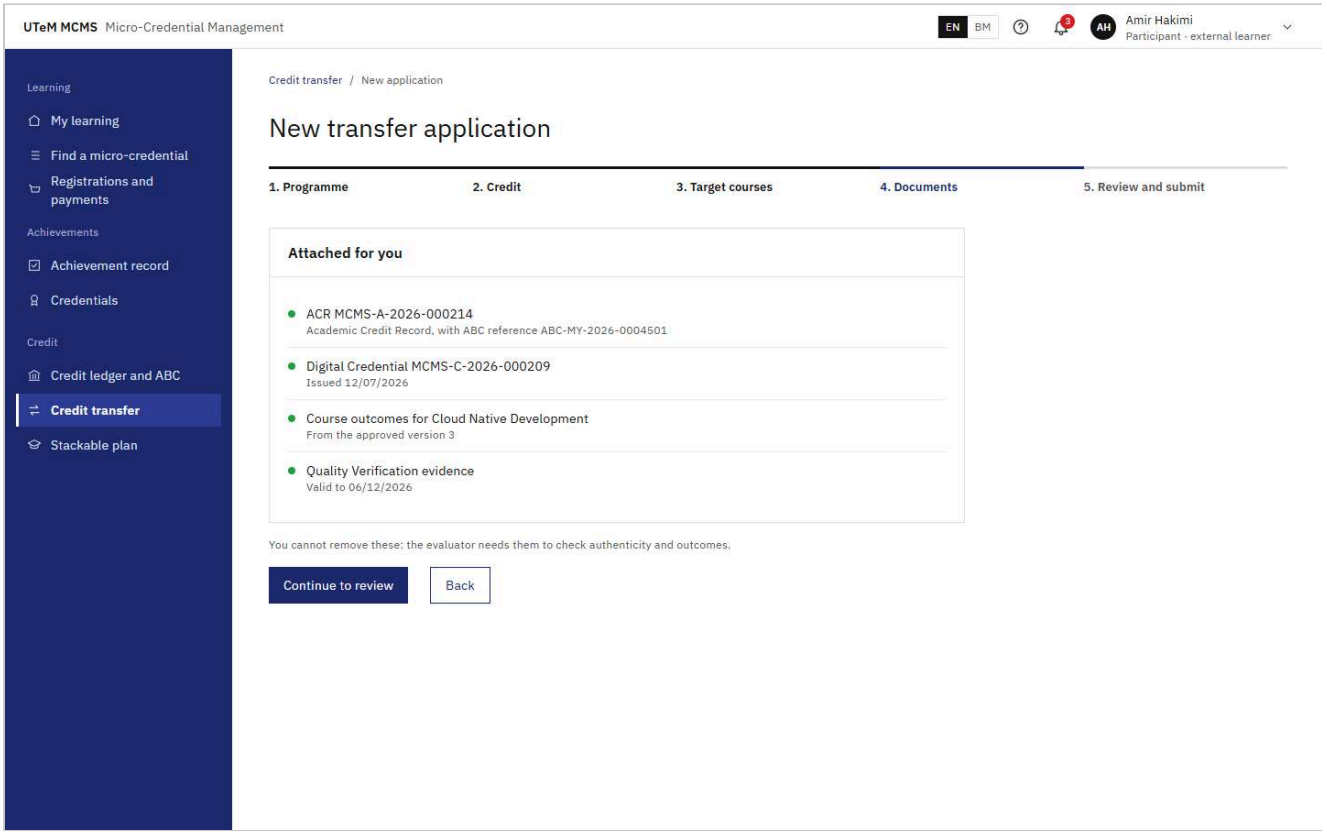


Figure 78: *ST-12 New transfer application: Step 4.*

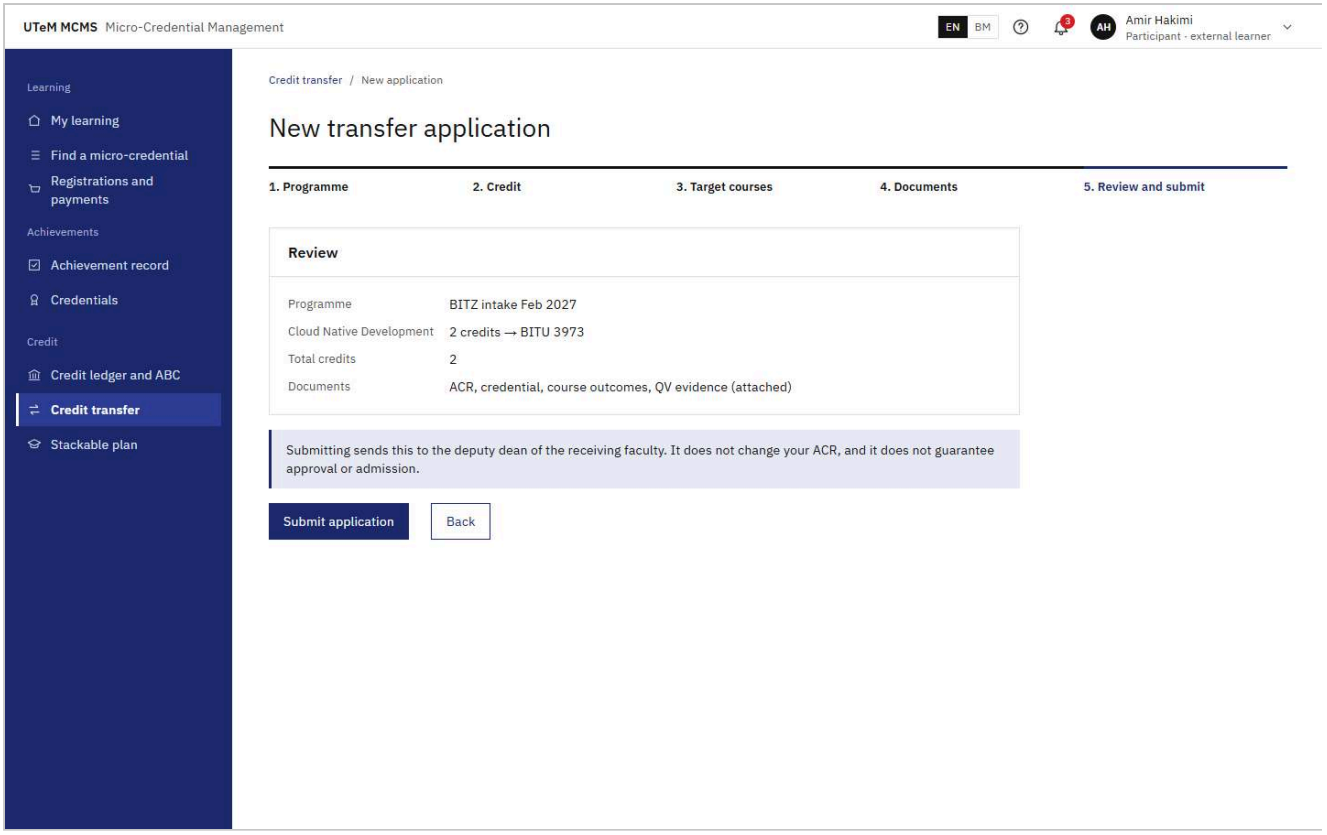


Figure 79: *ST-12 New transfer application: Step 5.*

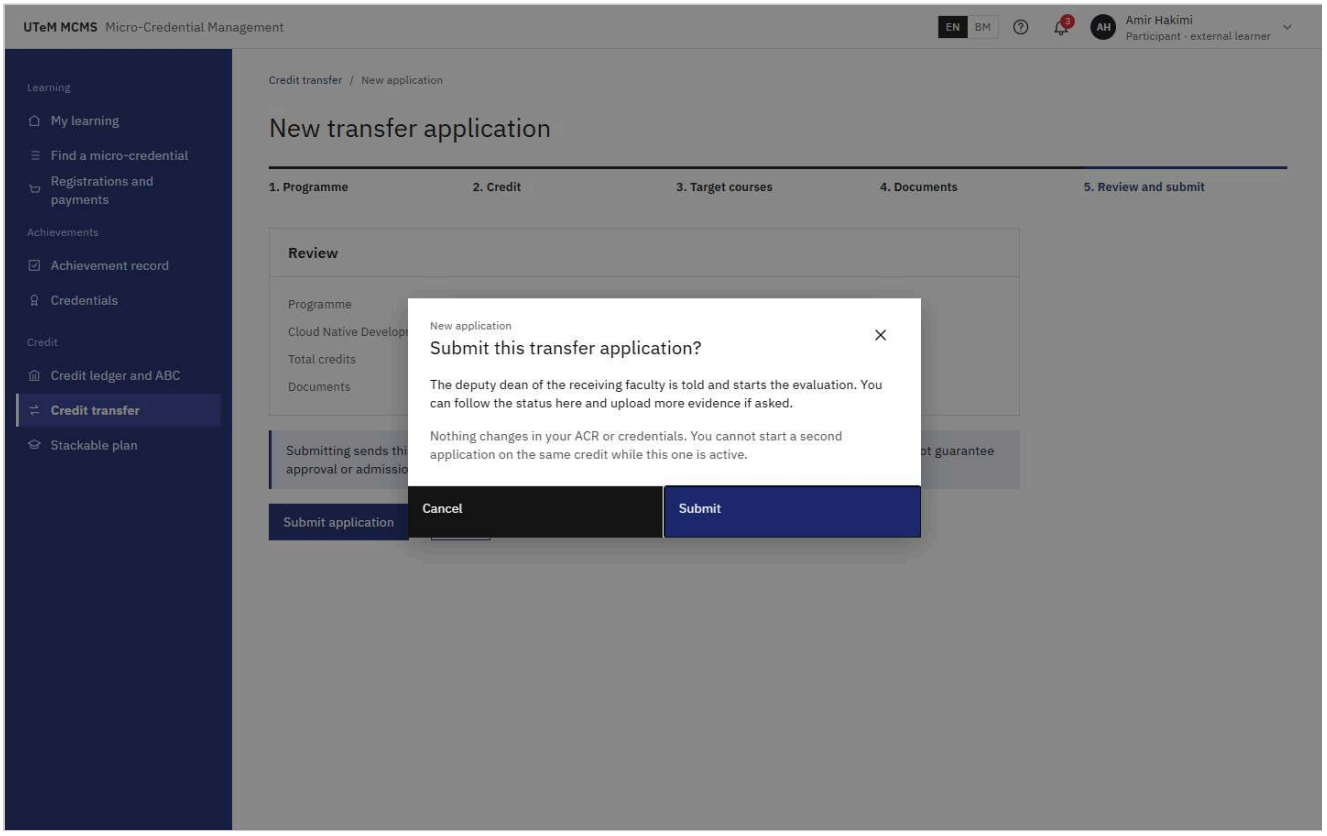


Figure 80: *ST-12 Confirmation: Submit this transfer application?*

ST-13 Transfer application

`#/st/transfer/TR-2026-0012`

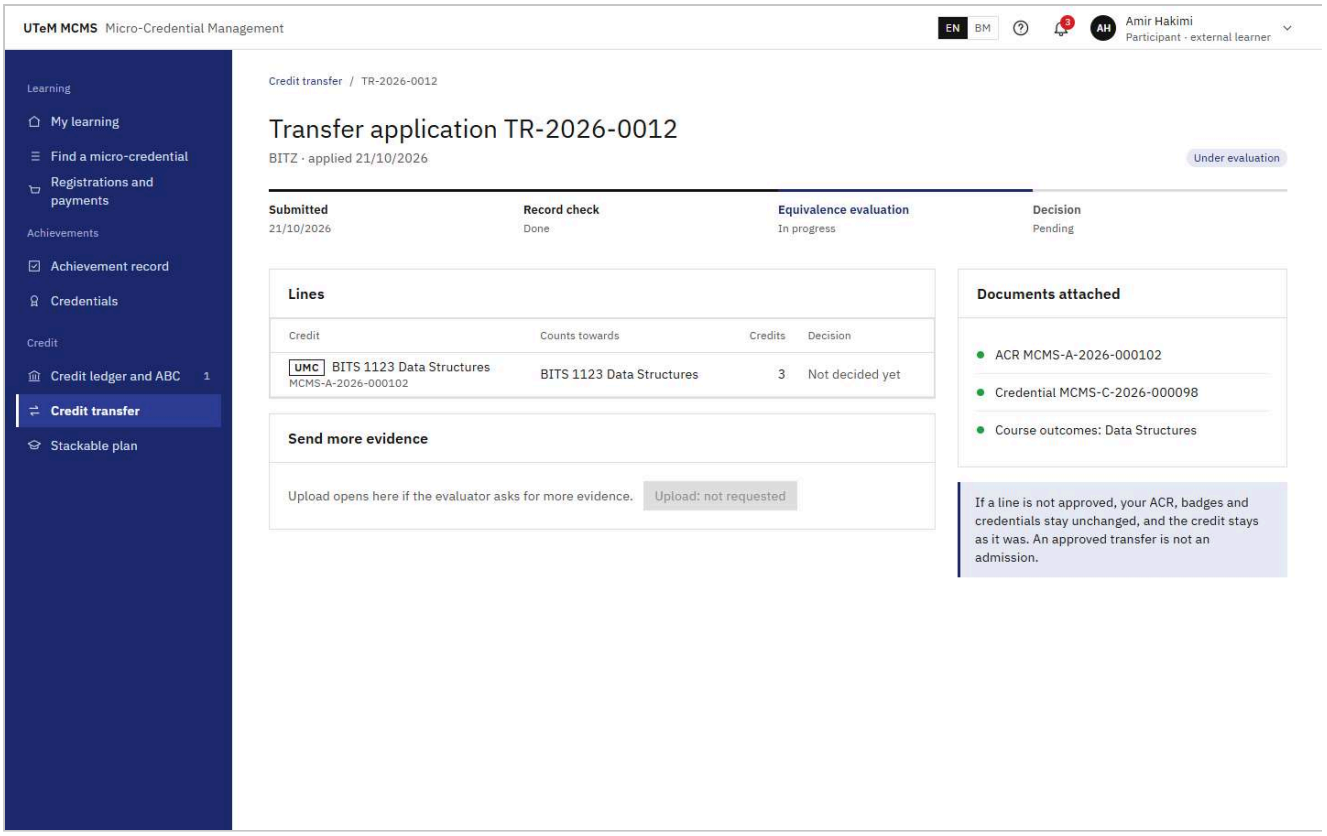


Figure 81: *ST-13 Transfer application.*

ST-14 Stackable plan

`#/st/plan`

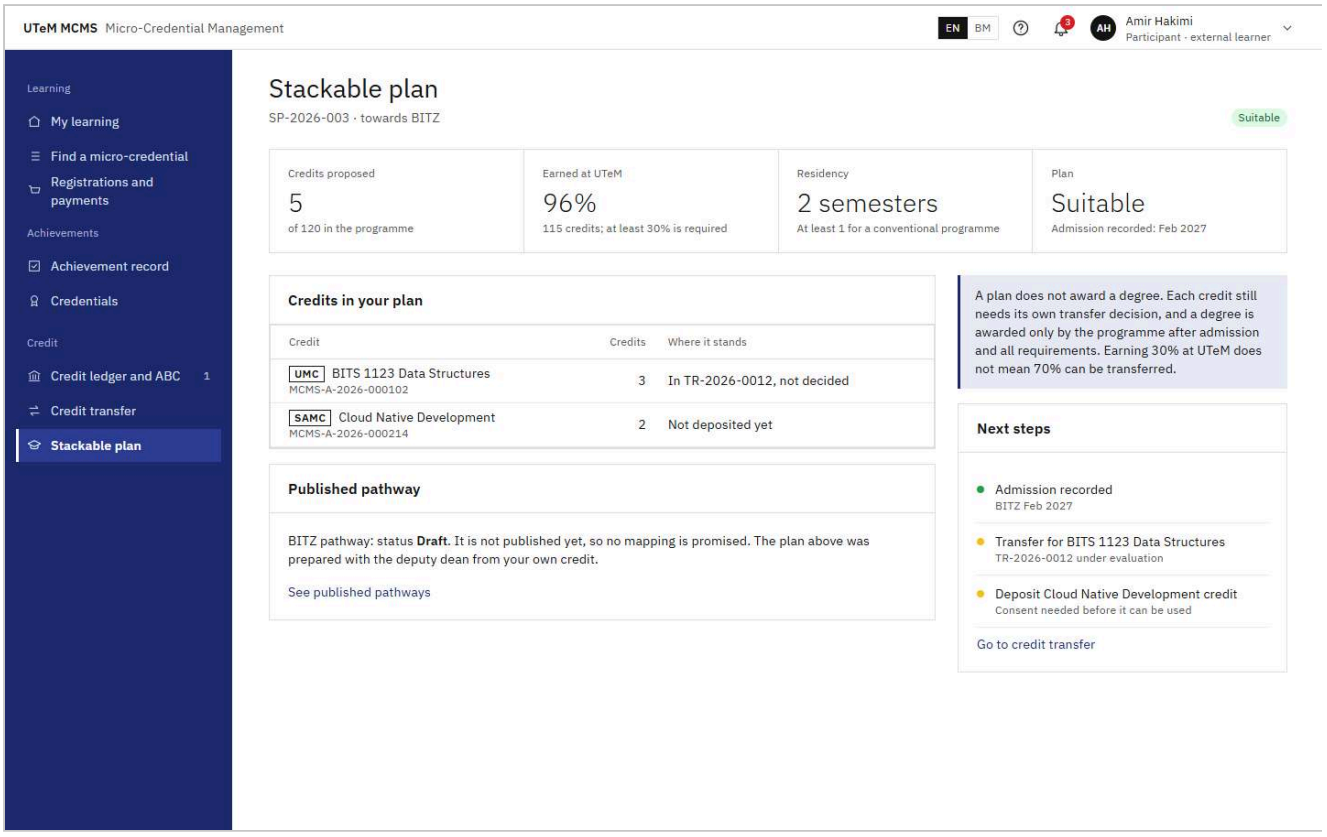


Figure 82: *ST-14 Stackable plan.*

Register, payment step

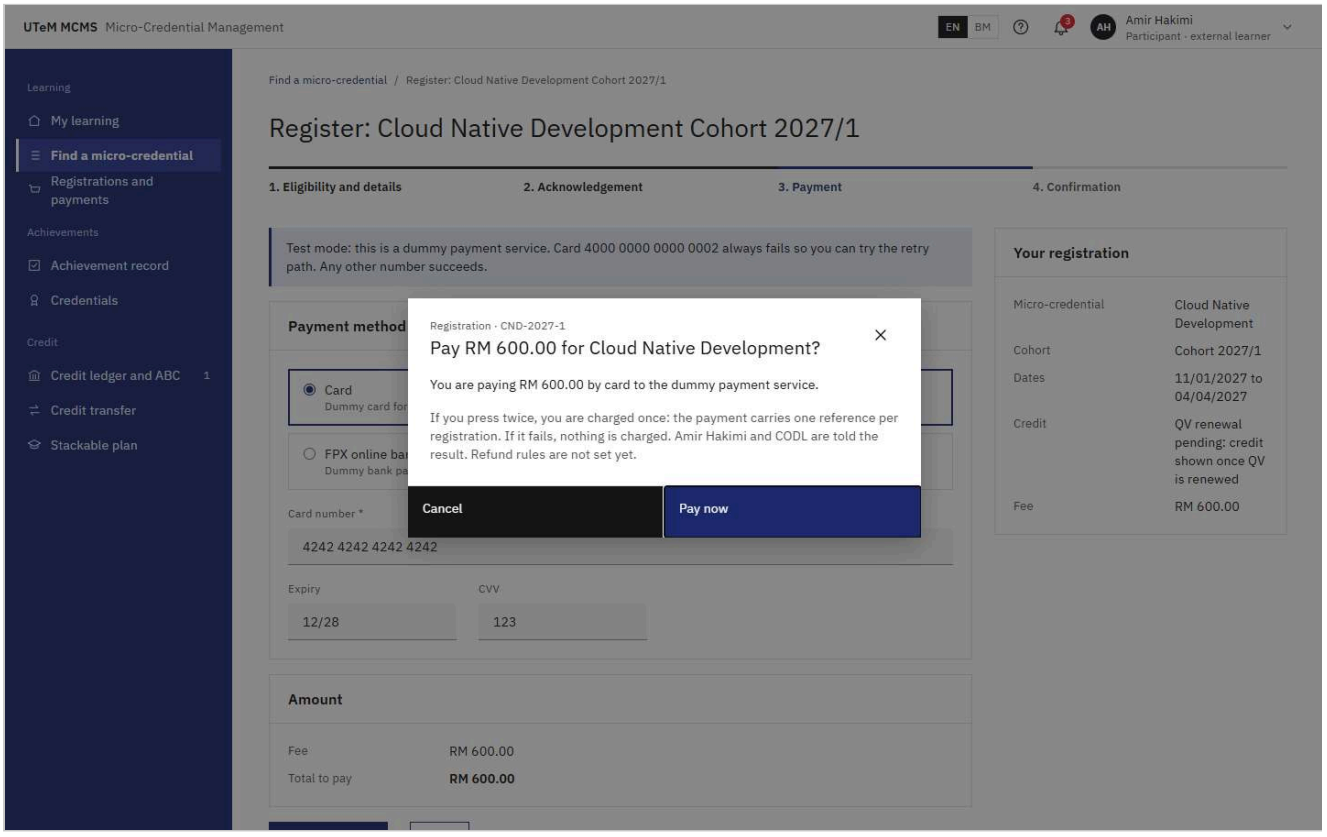


Figure 83: *Confirmation: Pay RM 600.00 for Cloud Native Development? (record state: Register, payment step).*

Registration pending payment

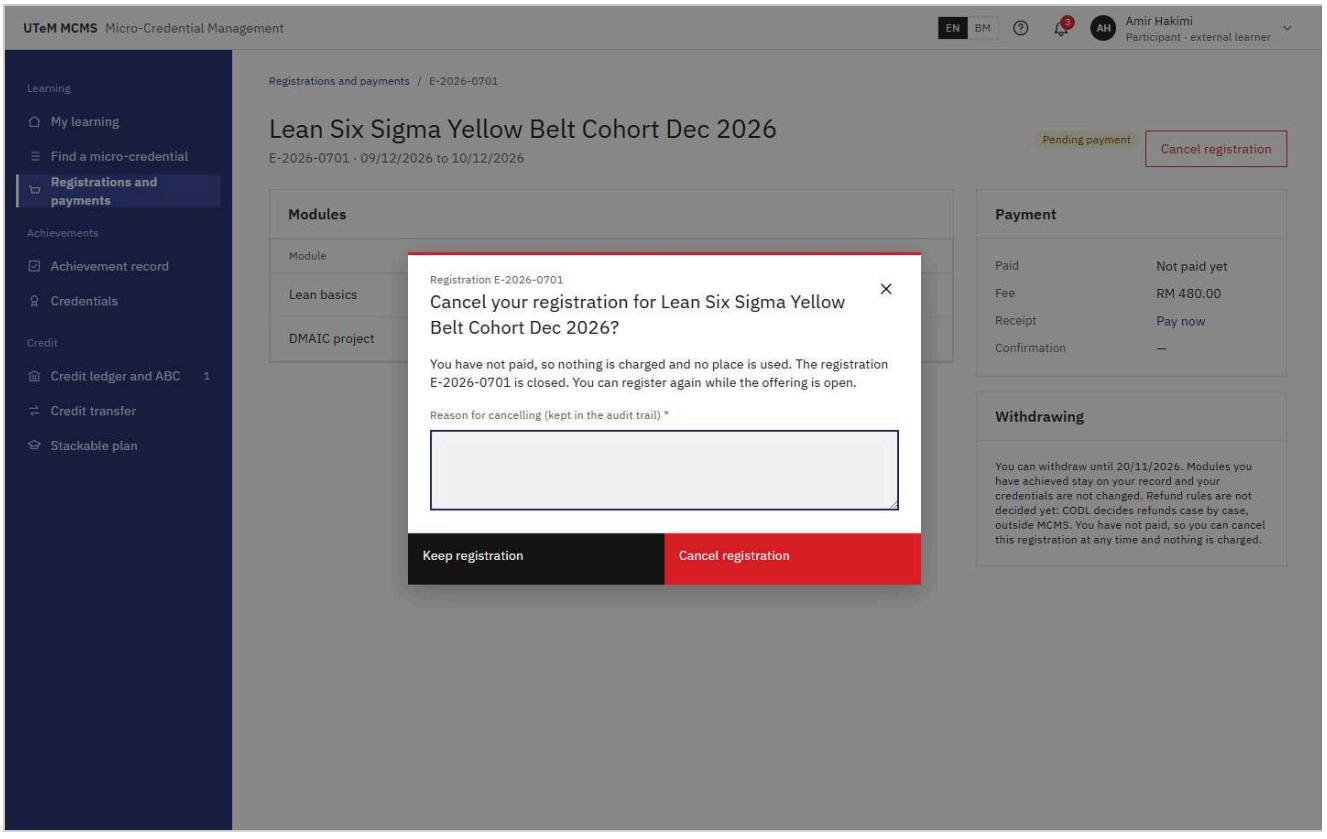


Figure 84: *Warning (destructive): Cancel your registration for Lean Six Sigma Yellow Belt Cohort Dec 2026? (record state: Registration pending payment).*

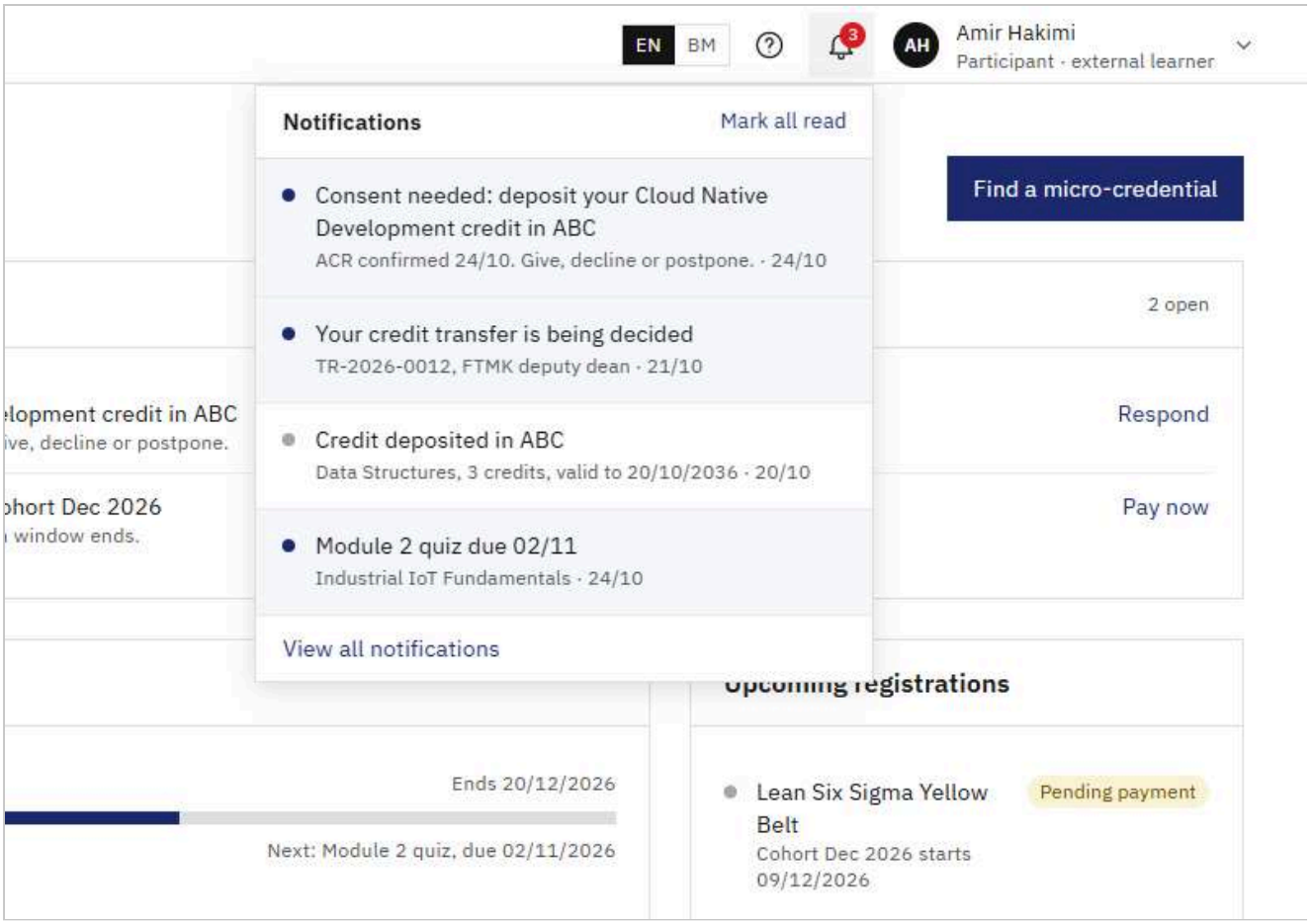


Figure 85: *Student: Notifications menu.*

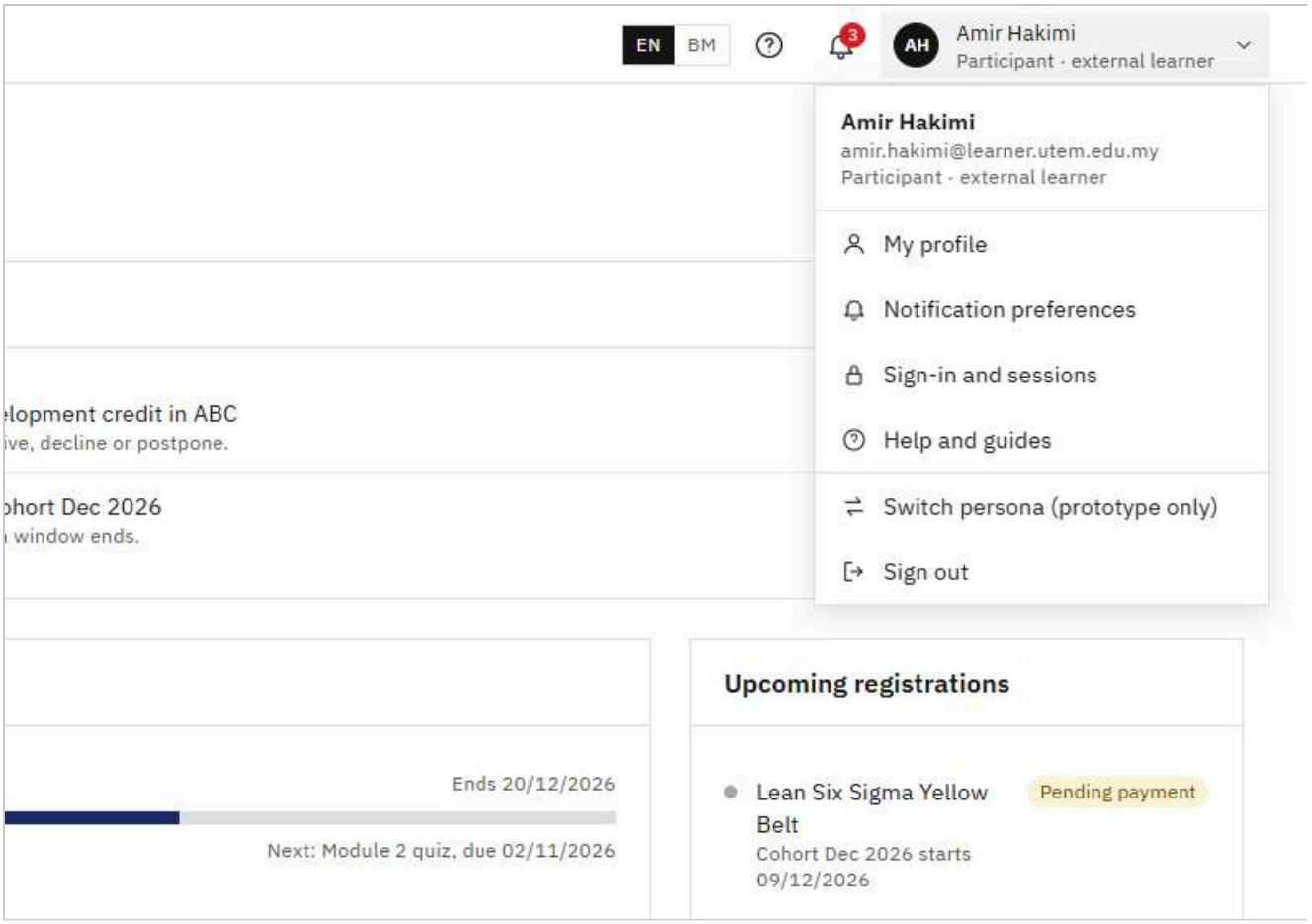


Figure 86: *Student: Profile menu.*

B.4 Lecturer and reviewer

LEC-01 My micro-credentials

`#/lec/home`

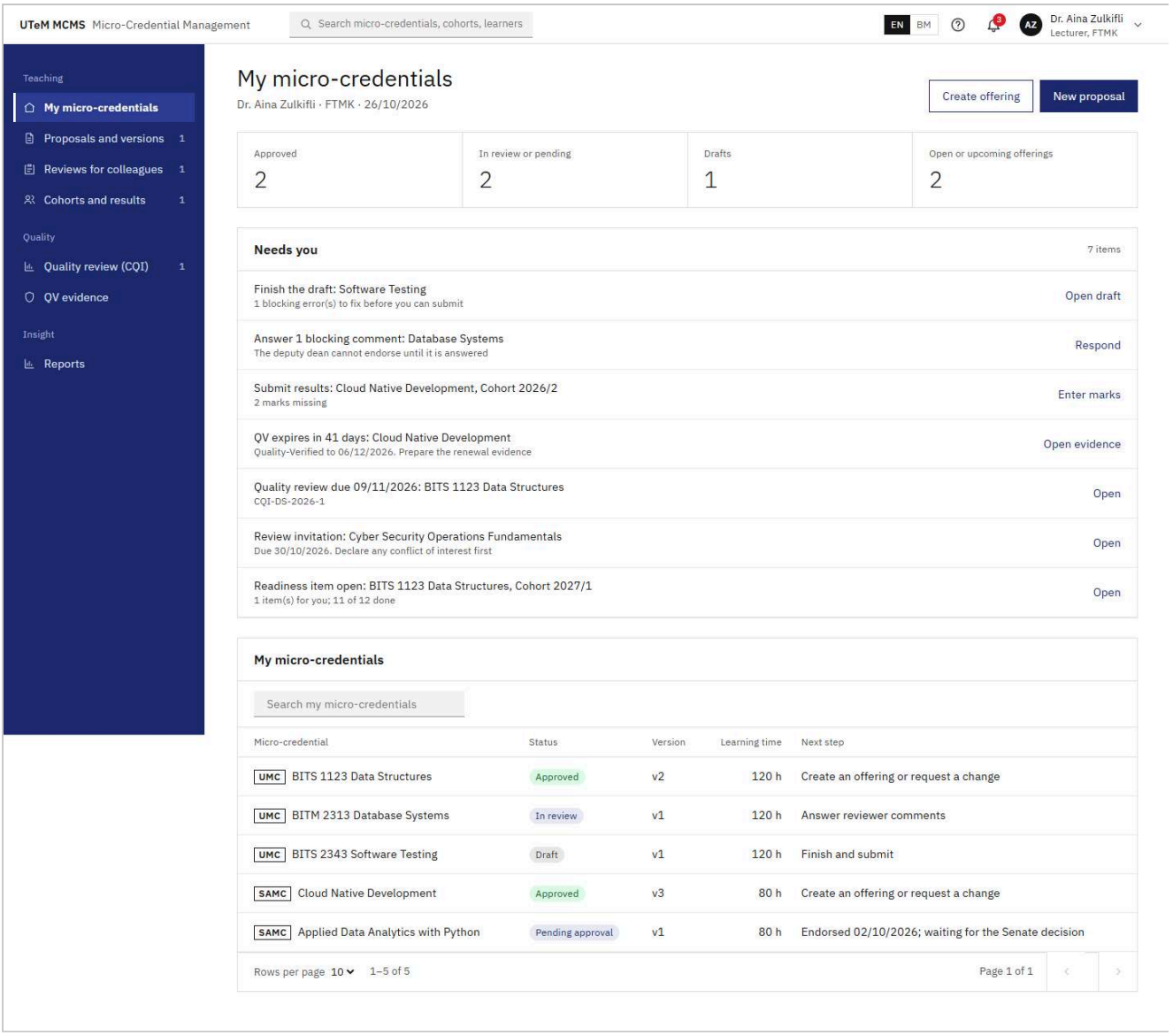


Figure 87: *LEC-01 My micro-credentials.*

LEC-02 Micro-credential

`#/lec/mc/DS`

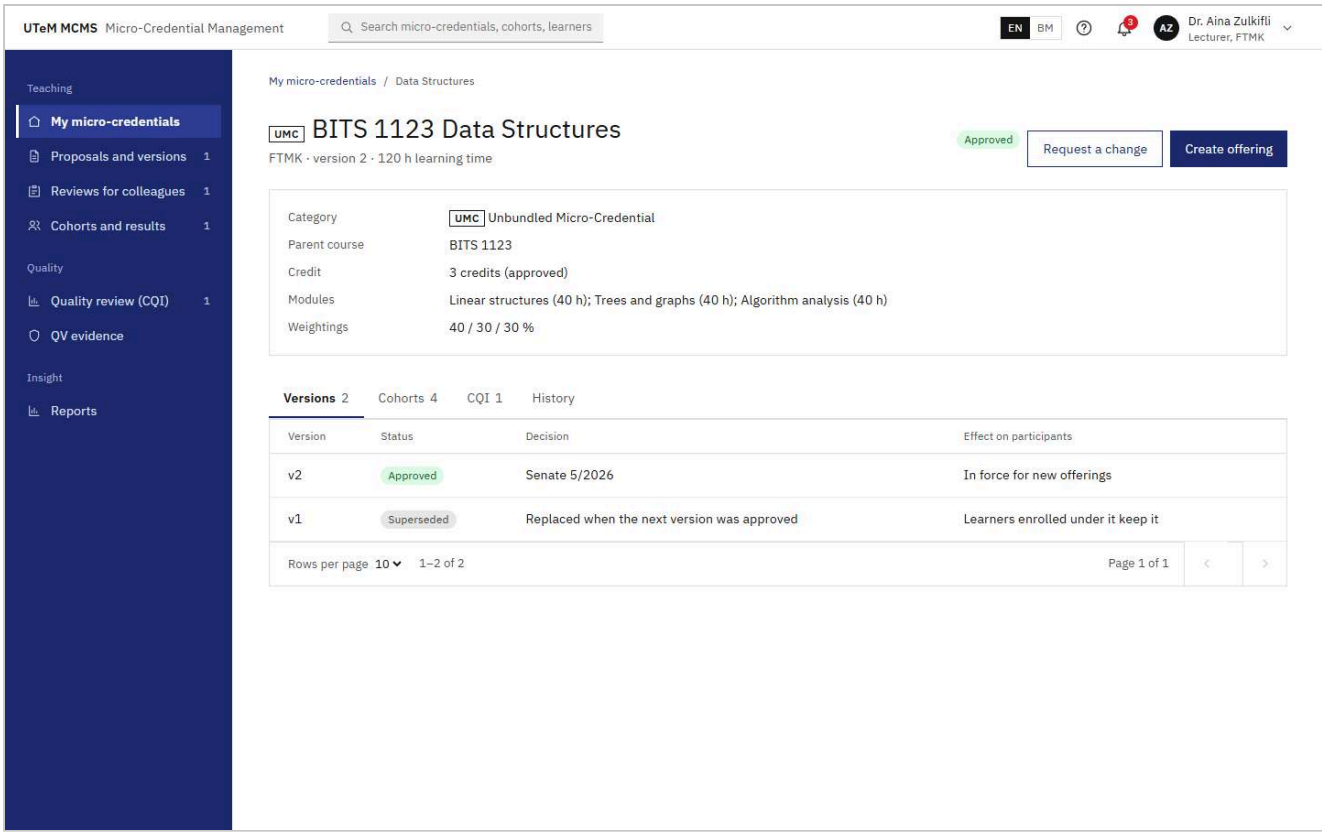


Figure 88: *LEC-02 Micro-credential.*

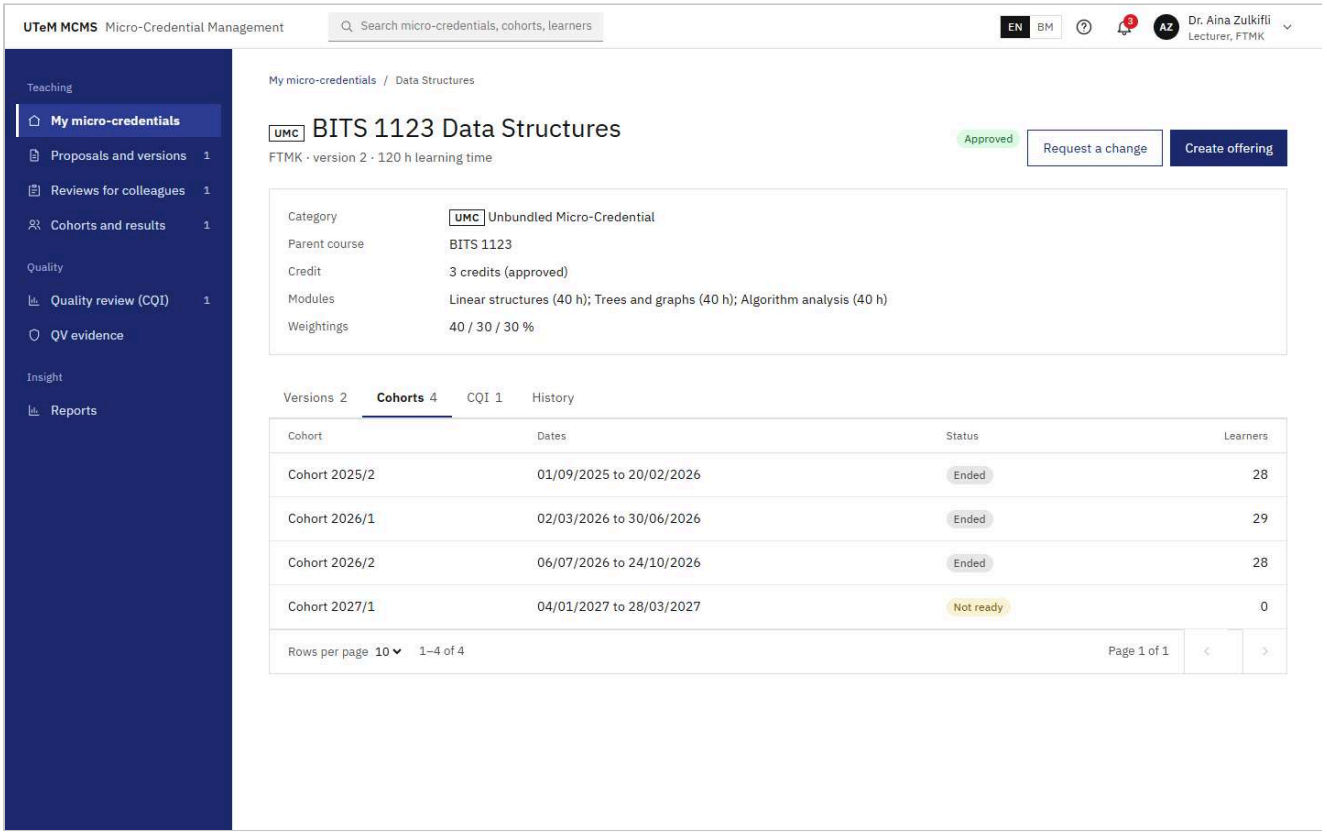


Figure 89: *LEC-02 Micro-credential: Tab: Cohorts.*

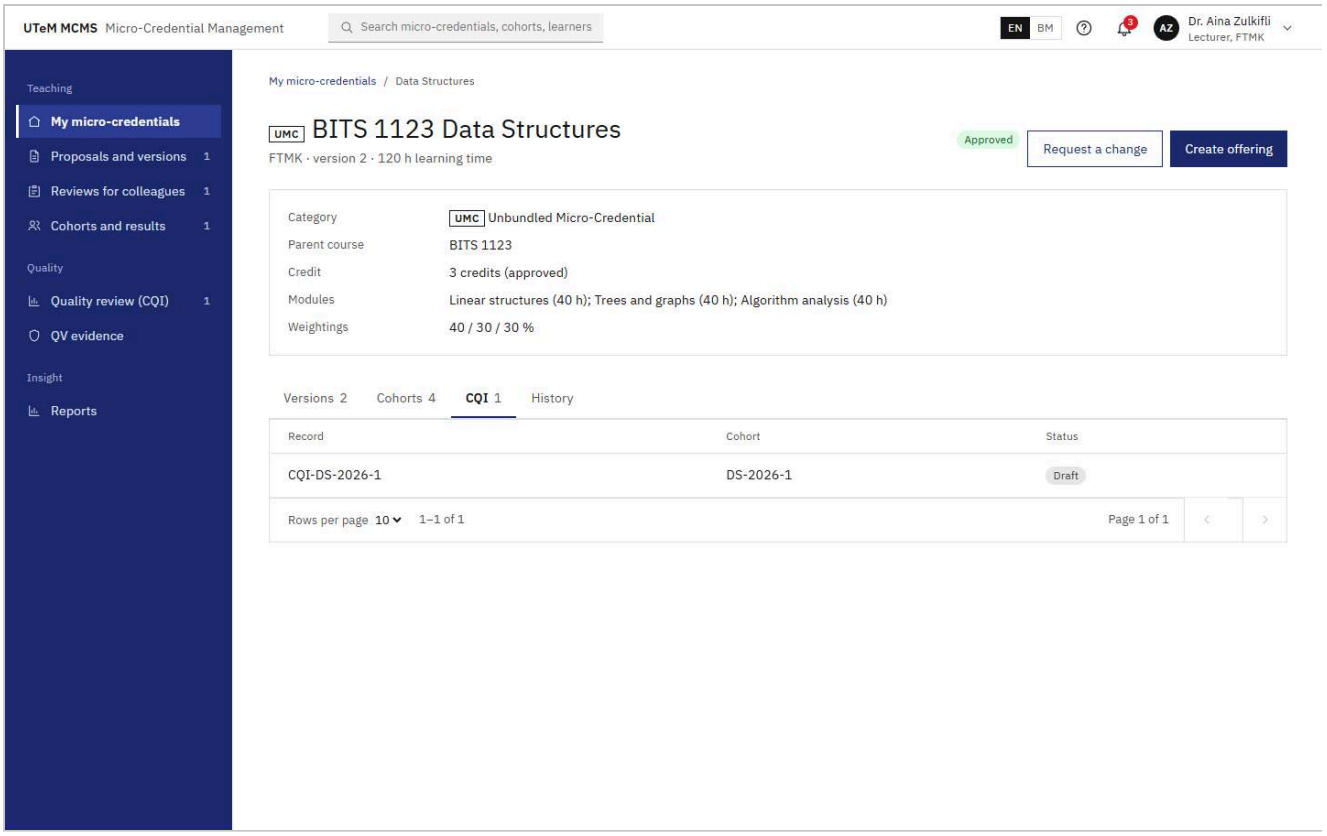


Figure 90: *LEC-02 Micro-credential: Tab: CQI.*

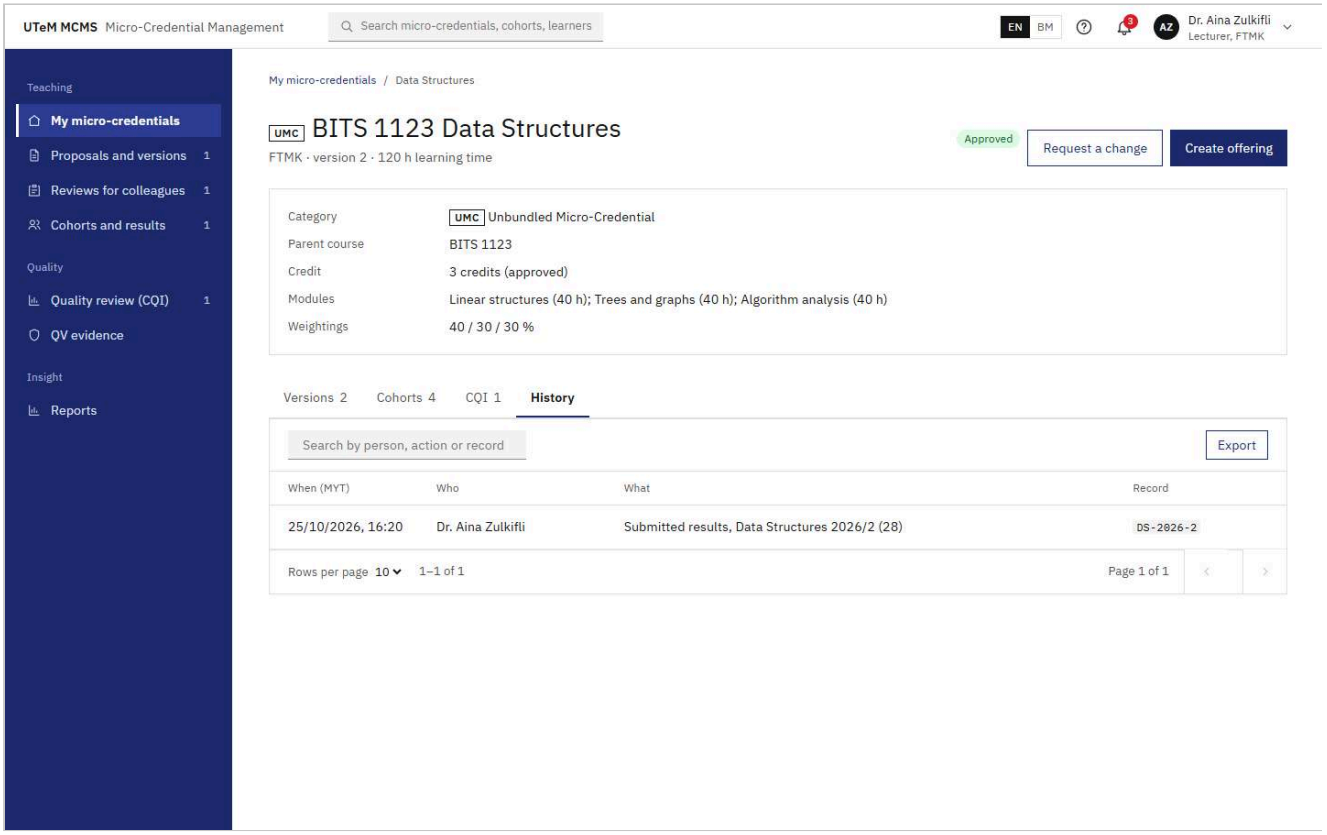


Figure 91: *LEC-02 Micro-credential: Tab: History.*

LEC-03 Proposals and versions

`#/lec/proposals`

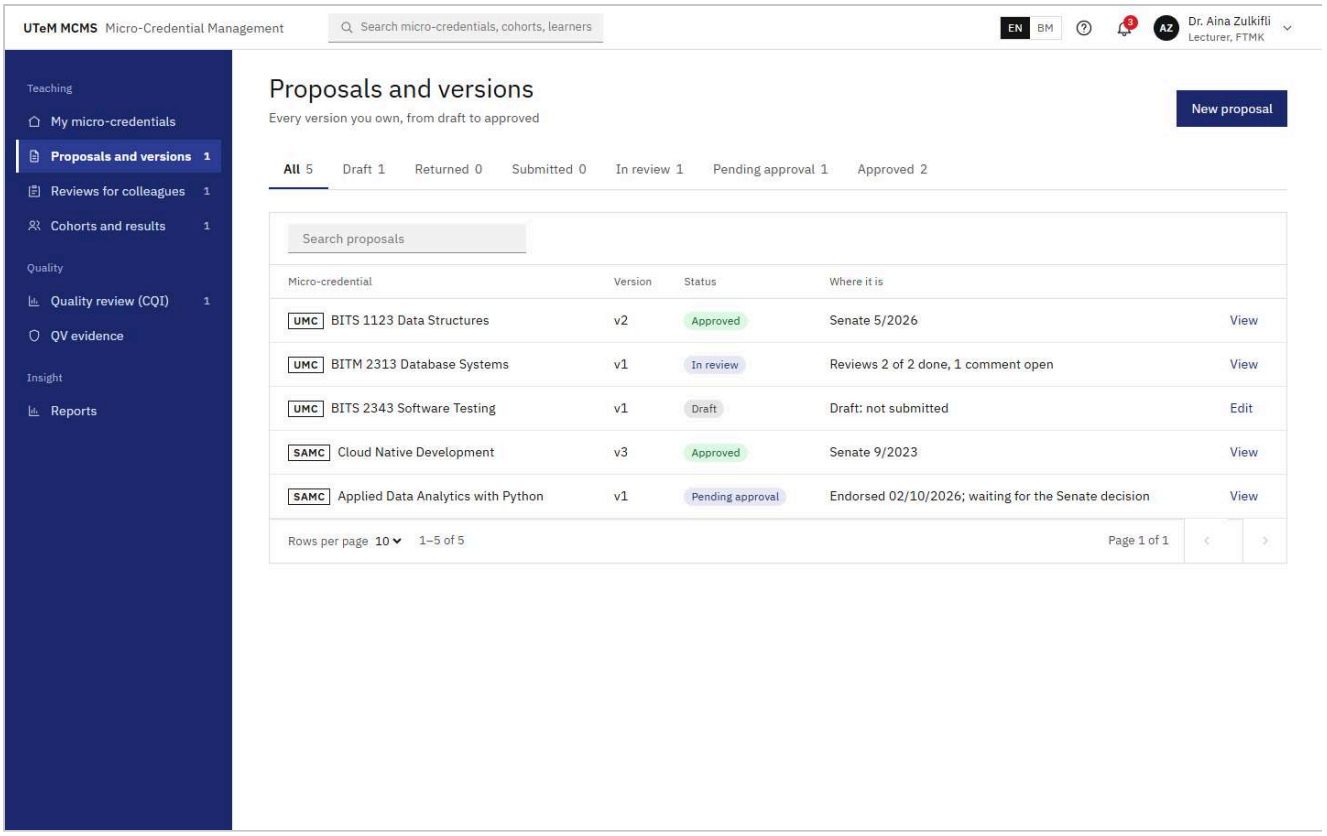


Figure 92: *LEC-03 Proposals and versions.*

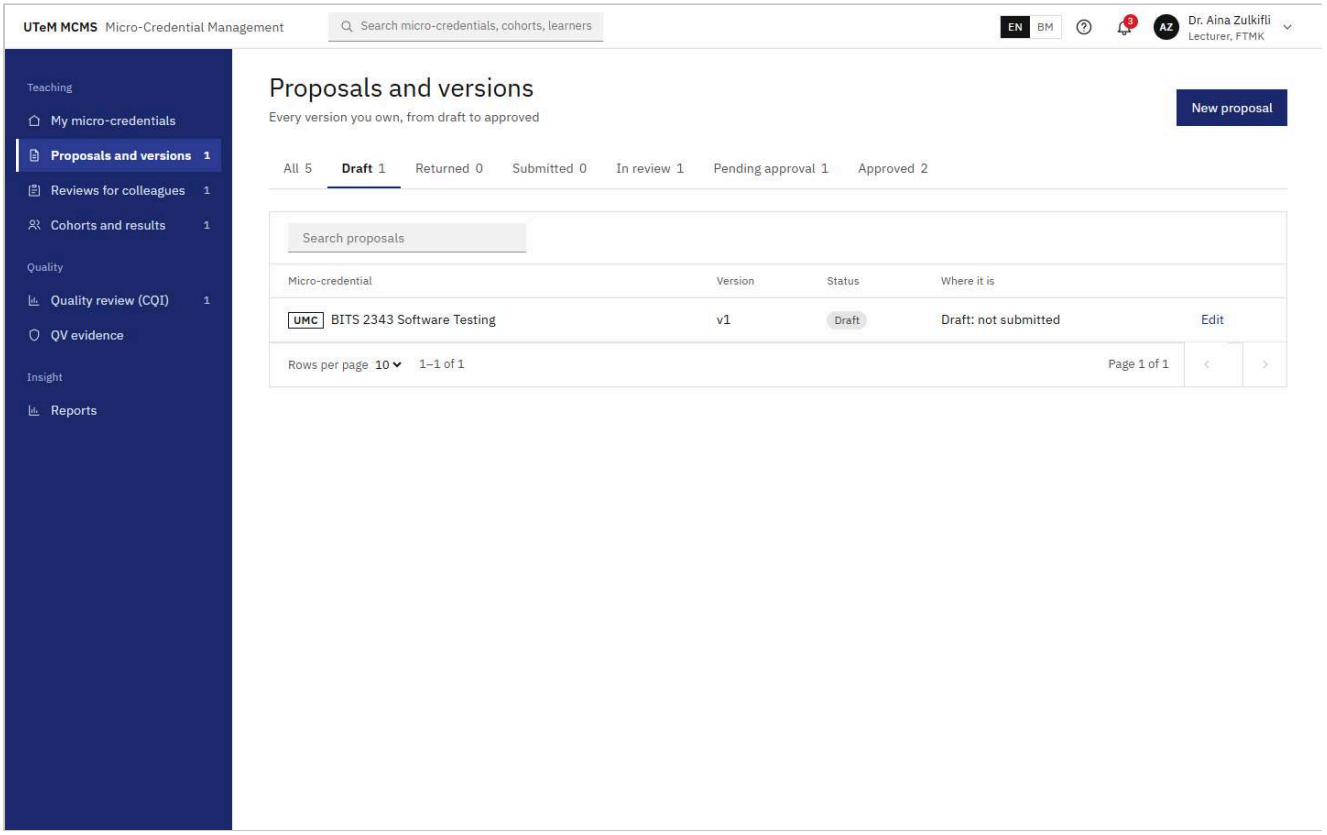


Figure 93: *LEC-03 Proposals and versions: Tab: Draft.*

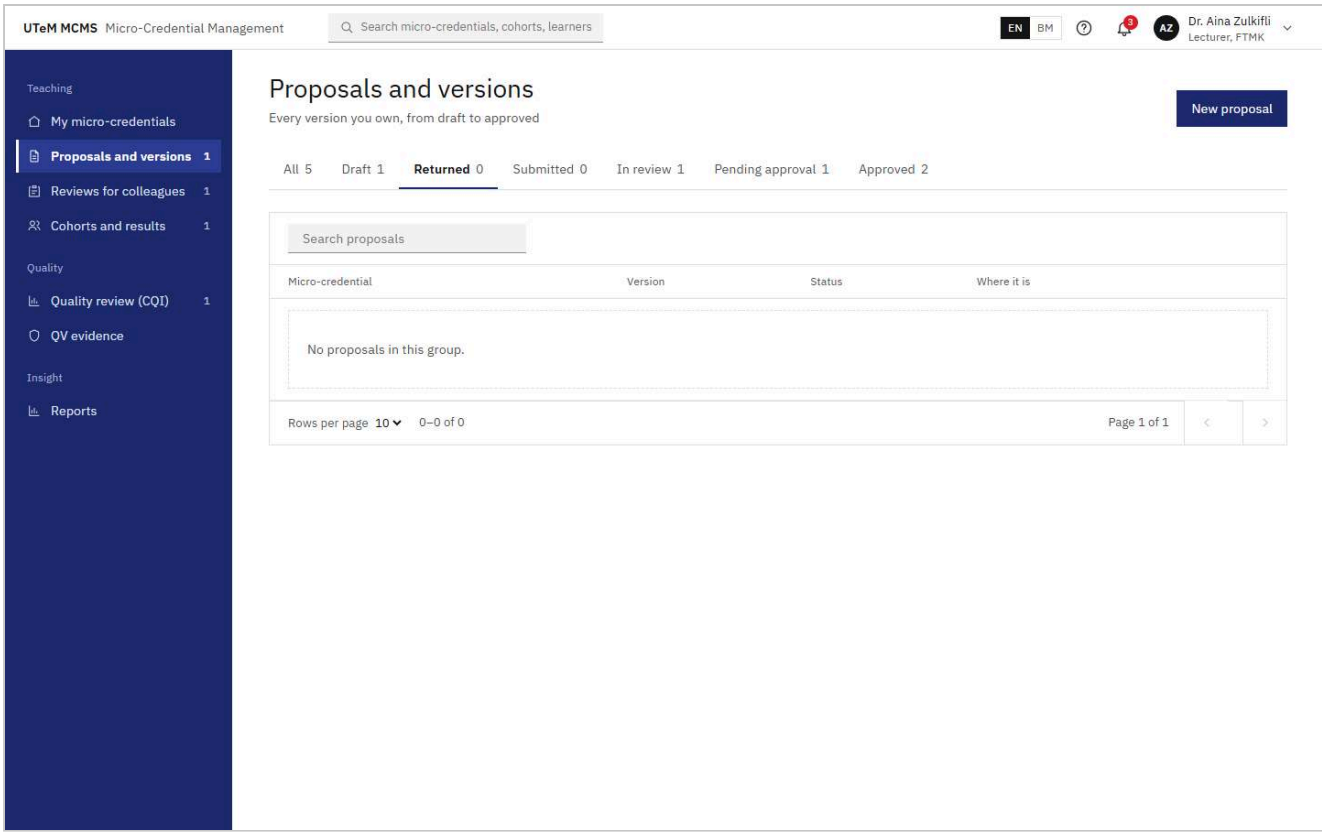


Figure 94: *LEC-03 Proposals and versions: Tab: Returned.*

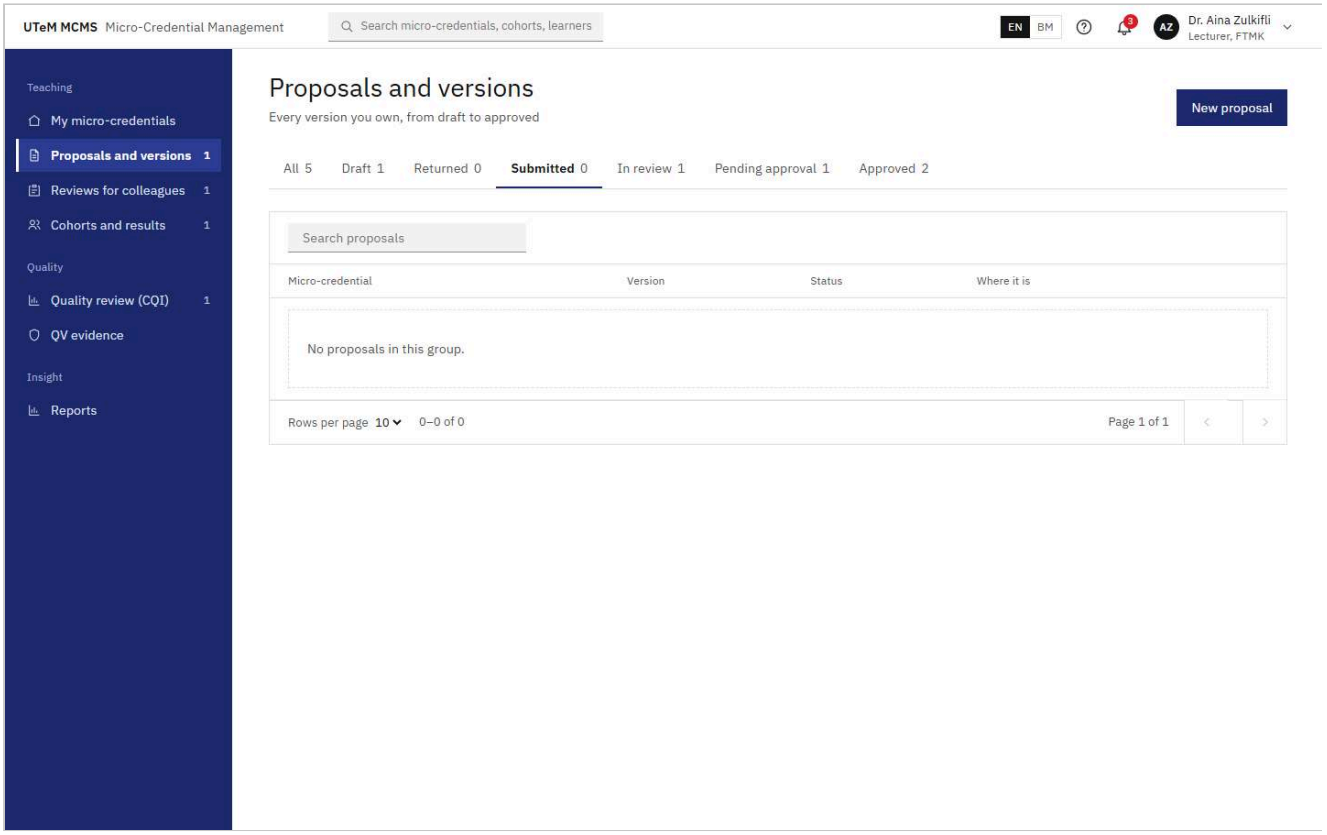


Figure 95: *LEC-03 Proposals and versions: Tab: Submitted.*

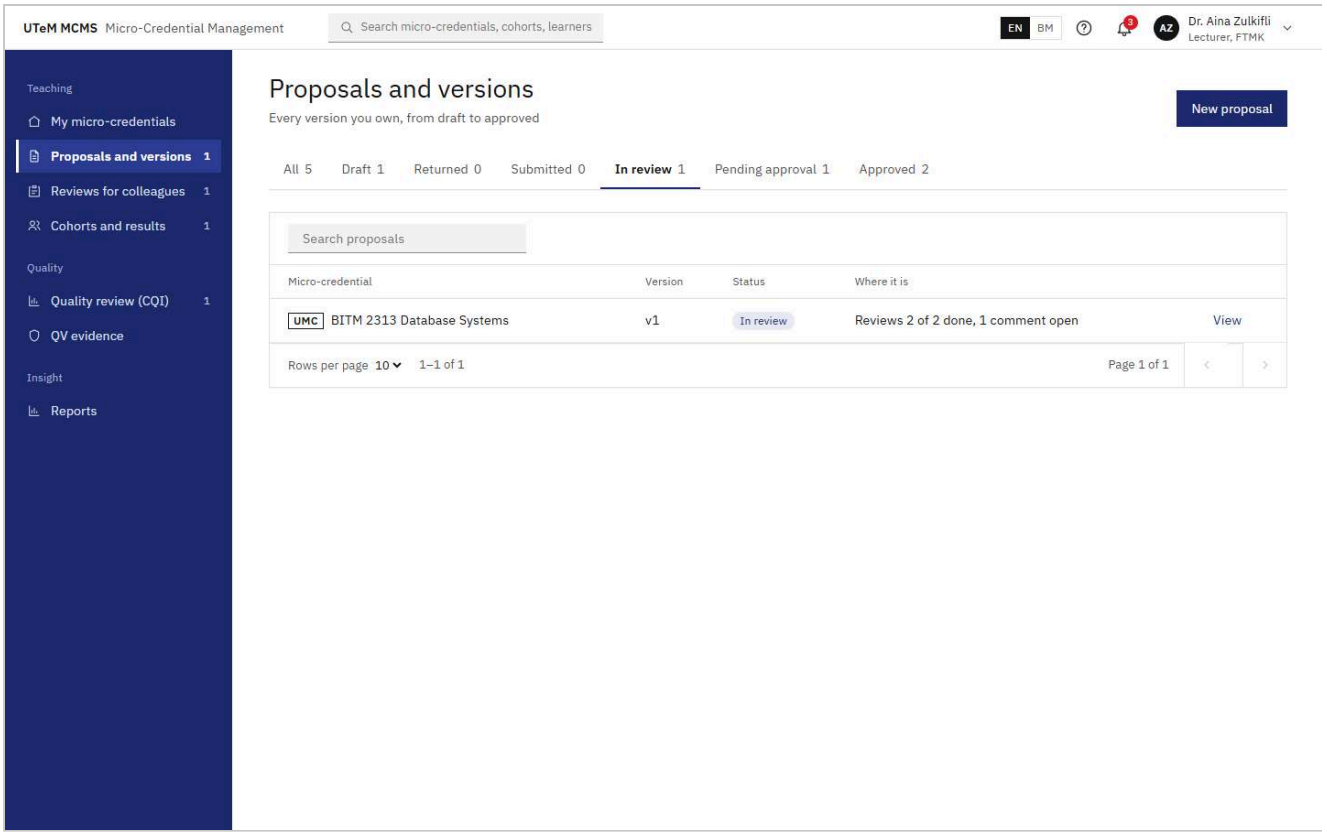


Figure 96: *LEC-03 Proposals and versions: Tab: In review.*

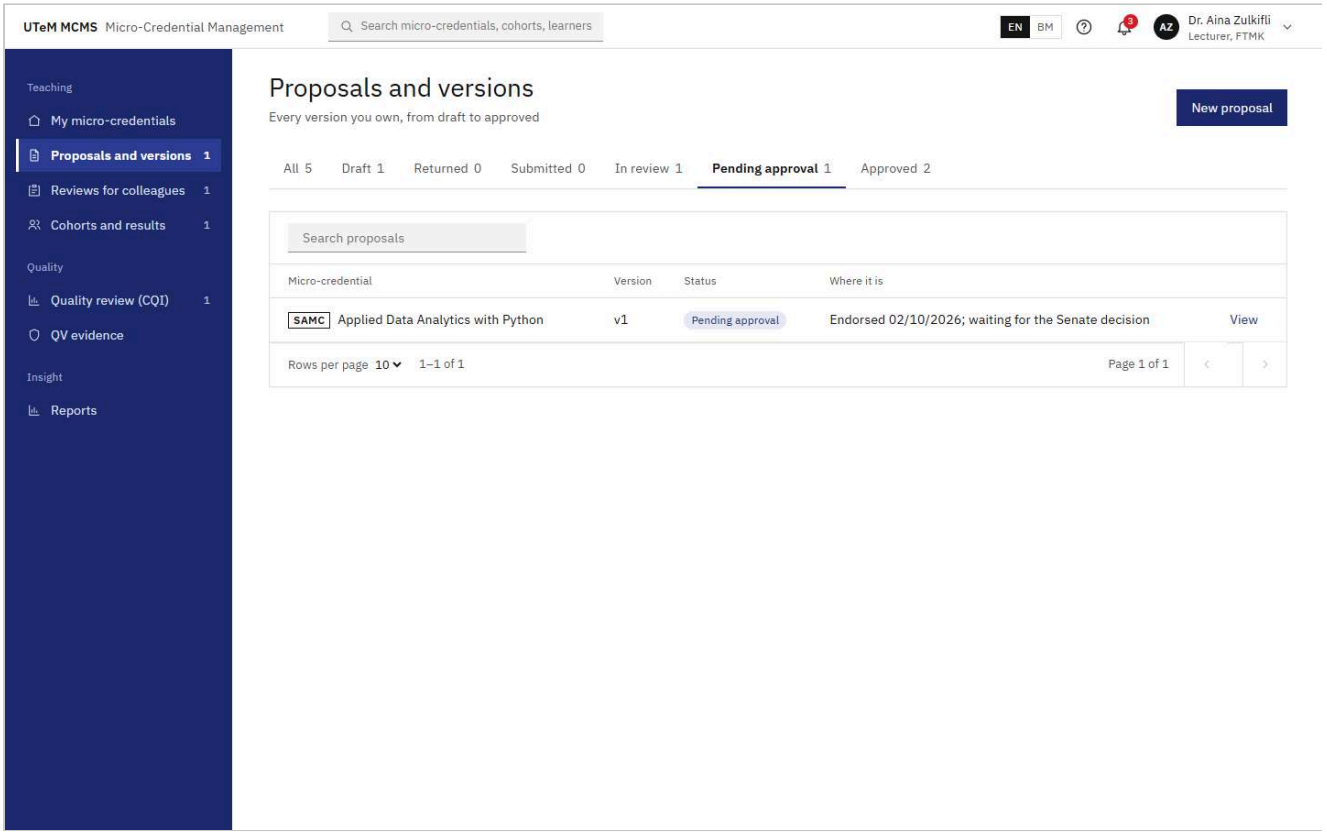


Figure 97: *LEC-03 Proposals and versions: Tab: Pending approval.*

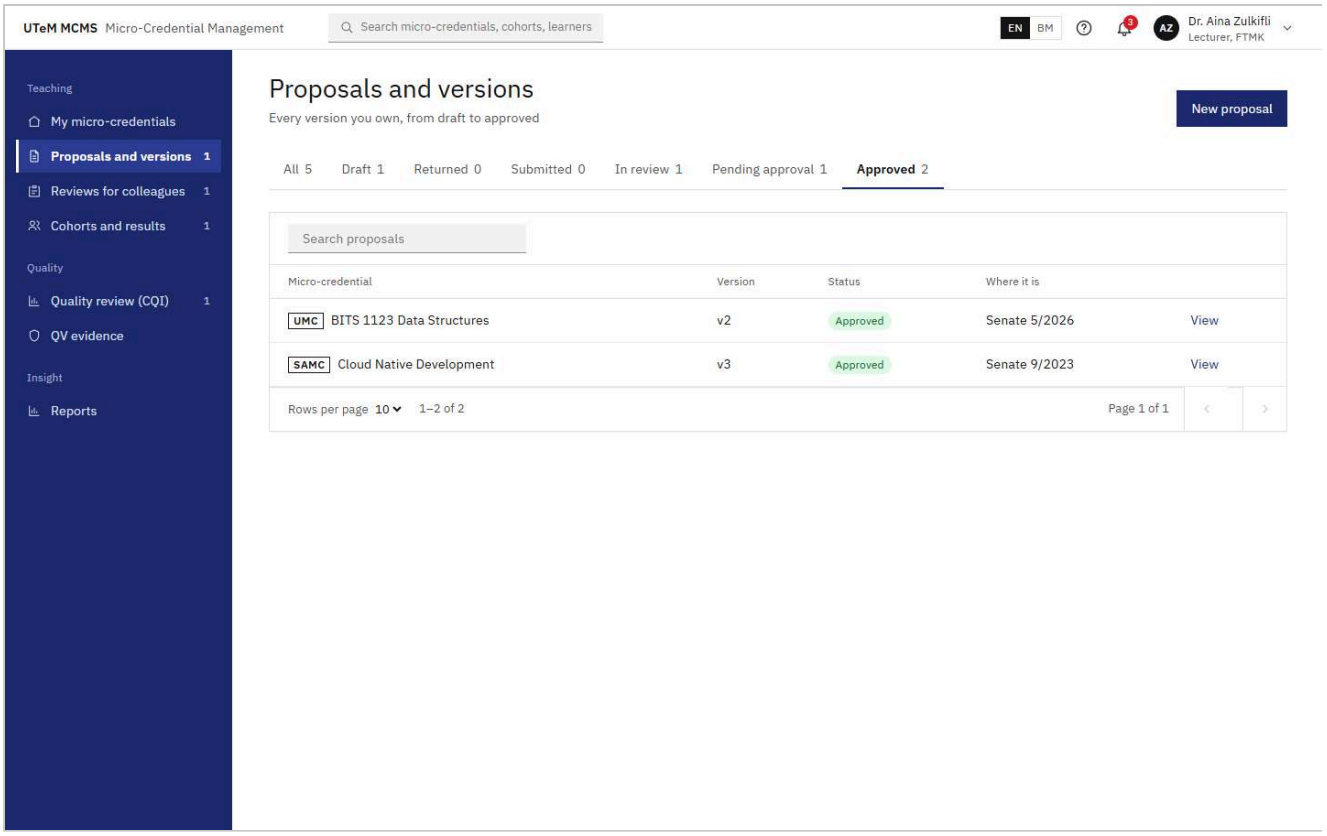


Figure 98: *LEC-03 Proposals and versions: Tab: Approved.*

LEC-04 New proposal

`#/lec/new`

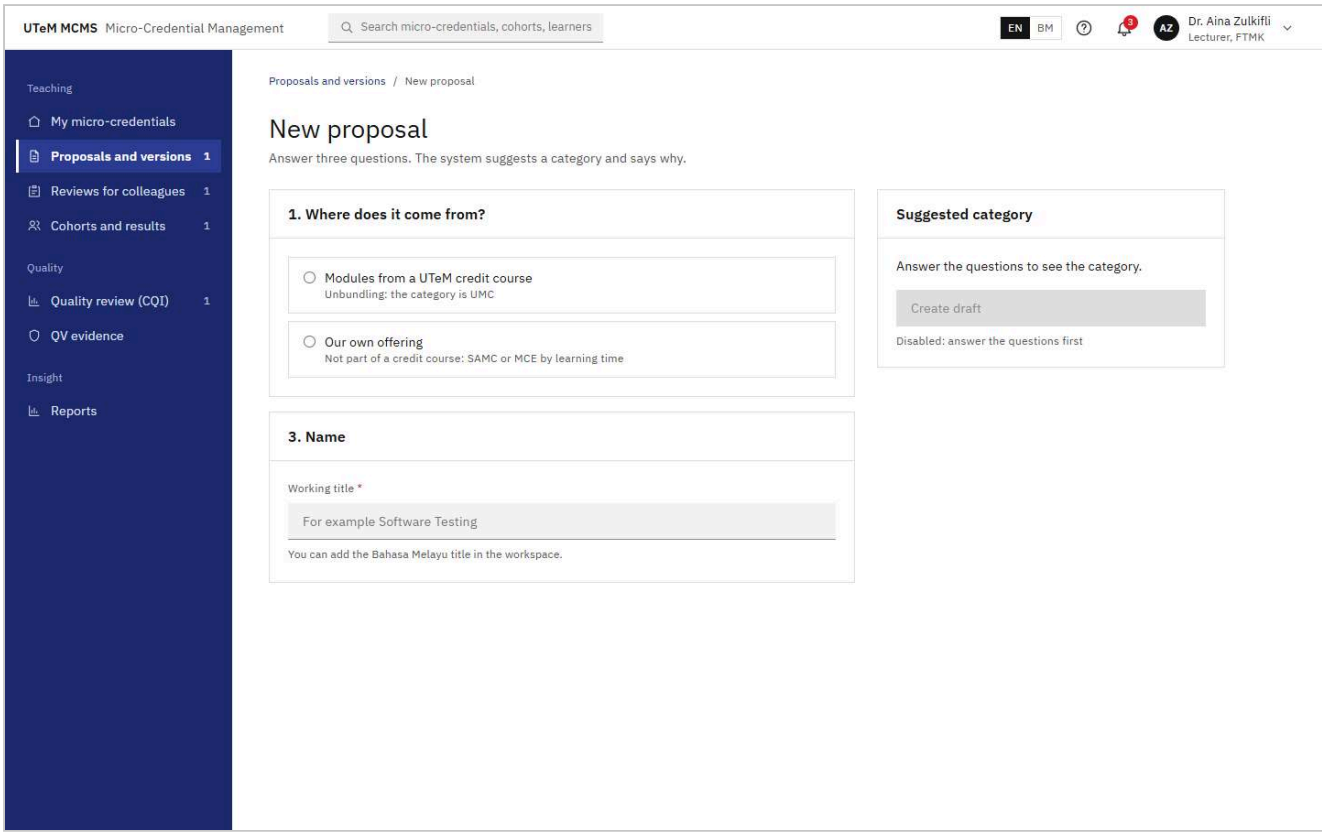


Figure 99: *LEC-04 New proposal.*

LEC-05 Proposal workspace

`#/lec/proposal/ST`

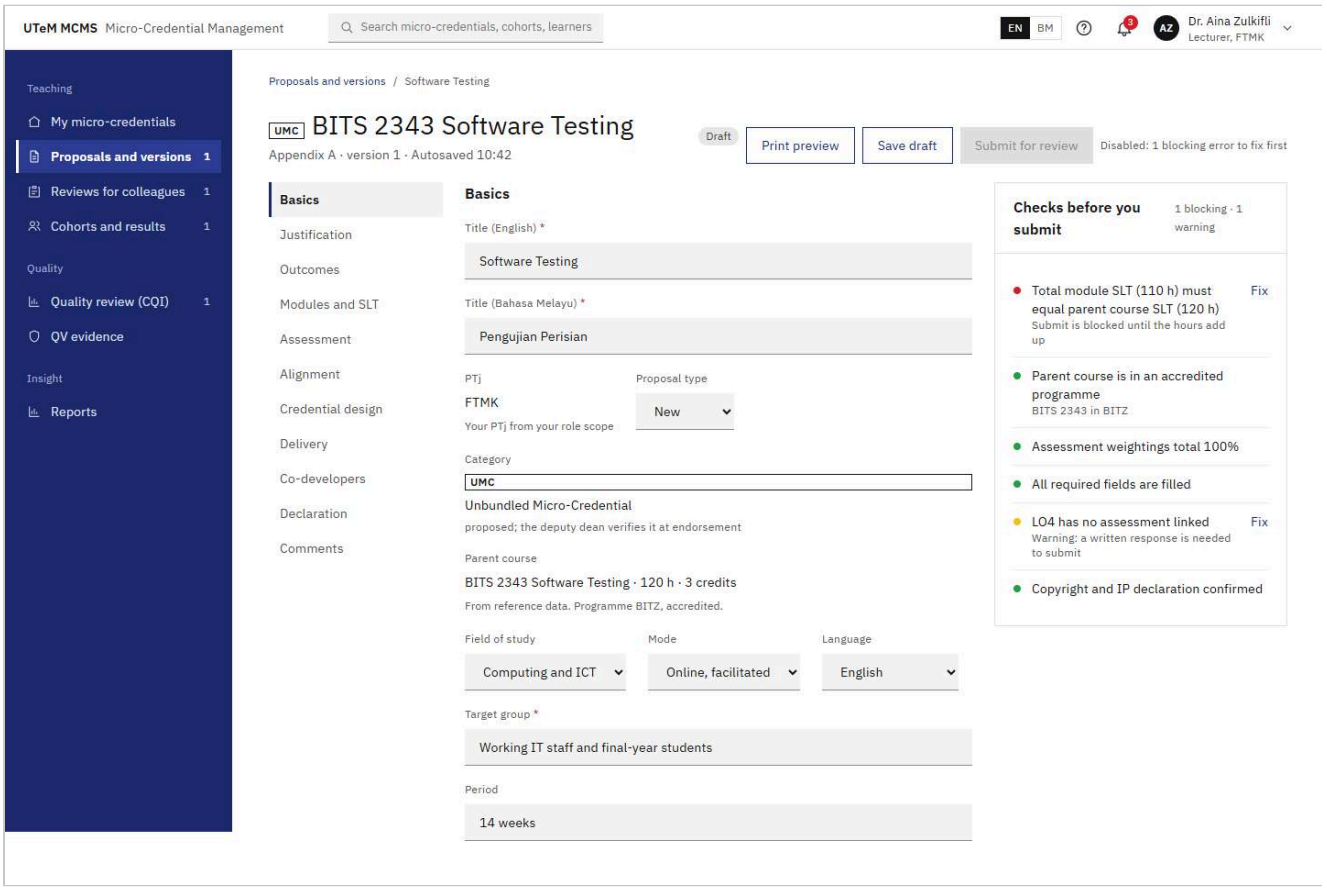


Figure 100: *LEC-05 Proposal workspace.*

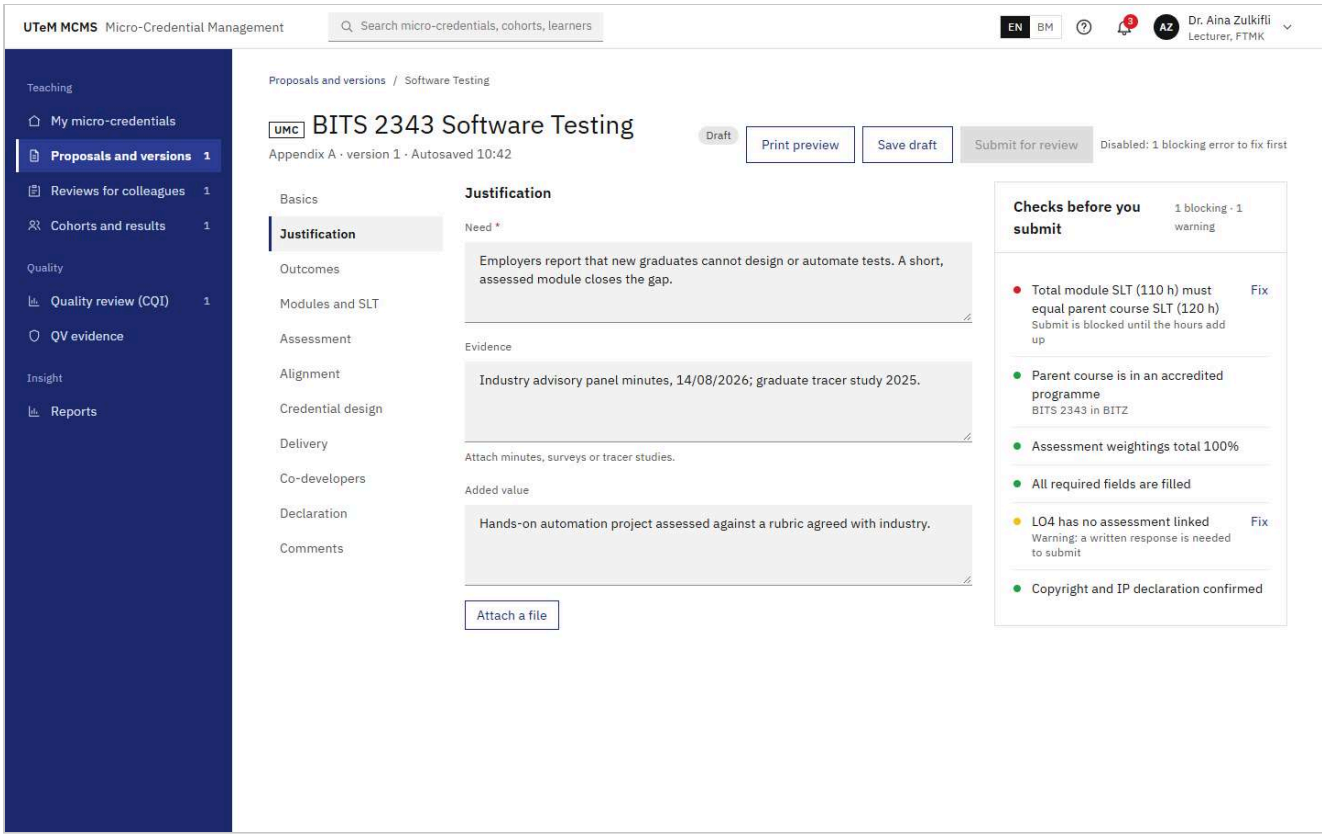


Figure 101: *LEC-05 Proposal workspace: Tab: Justification.*

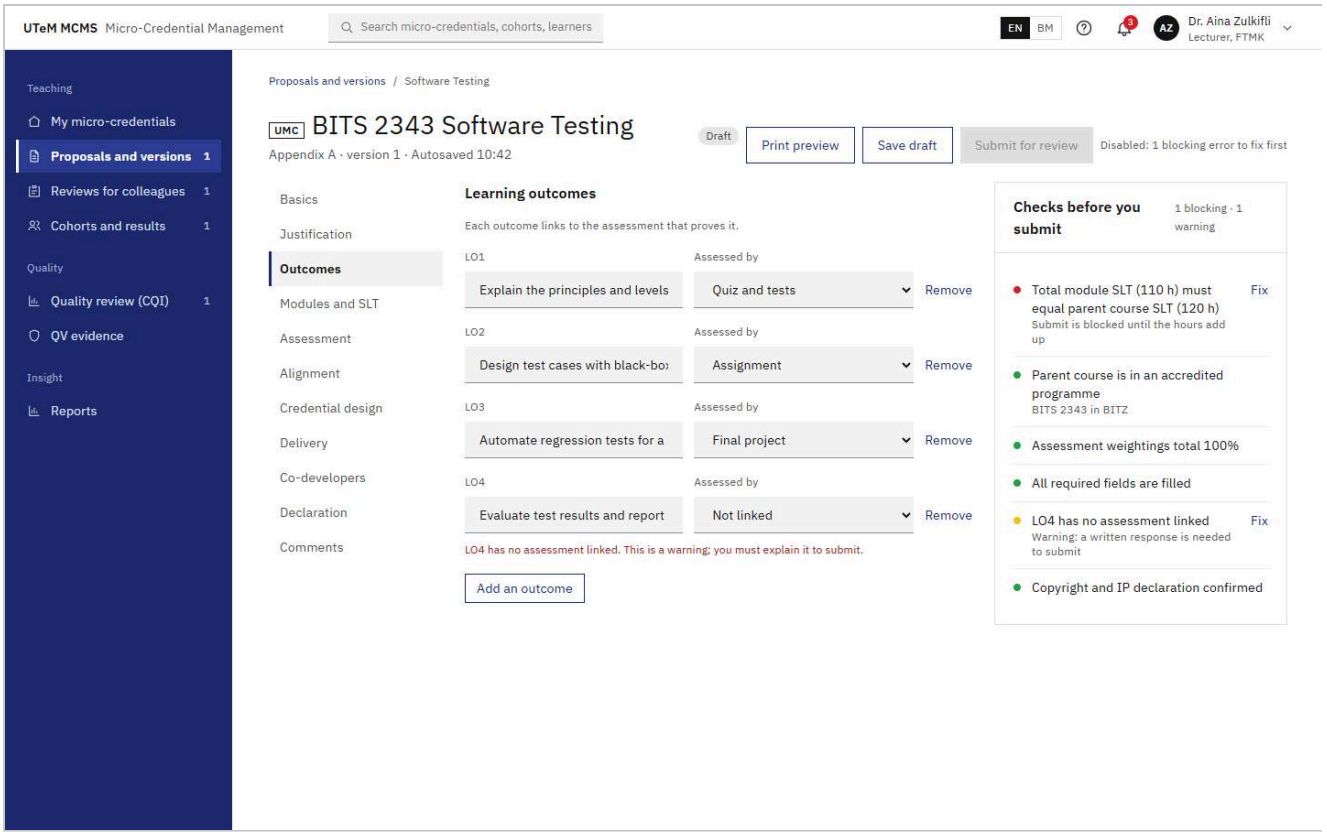


Figure 102: *LEC-05 Proposal workspace: Tab: Outcomes.*

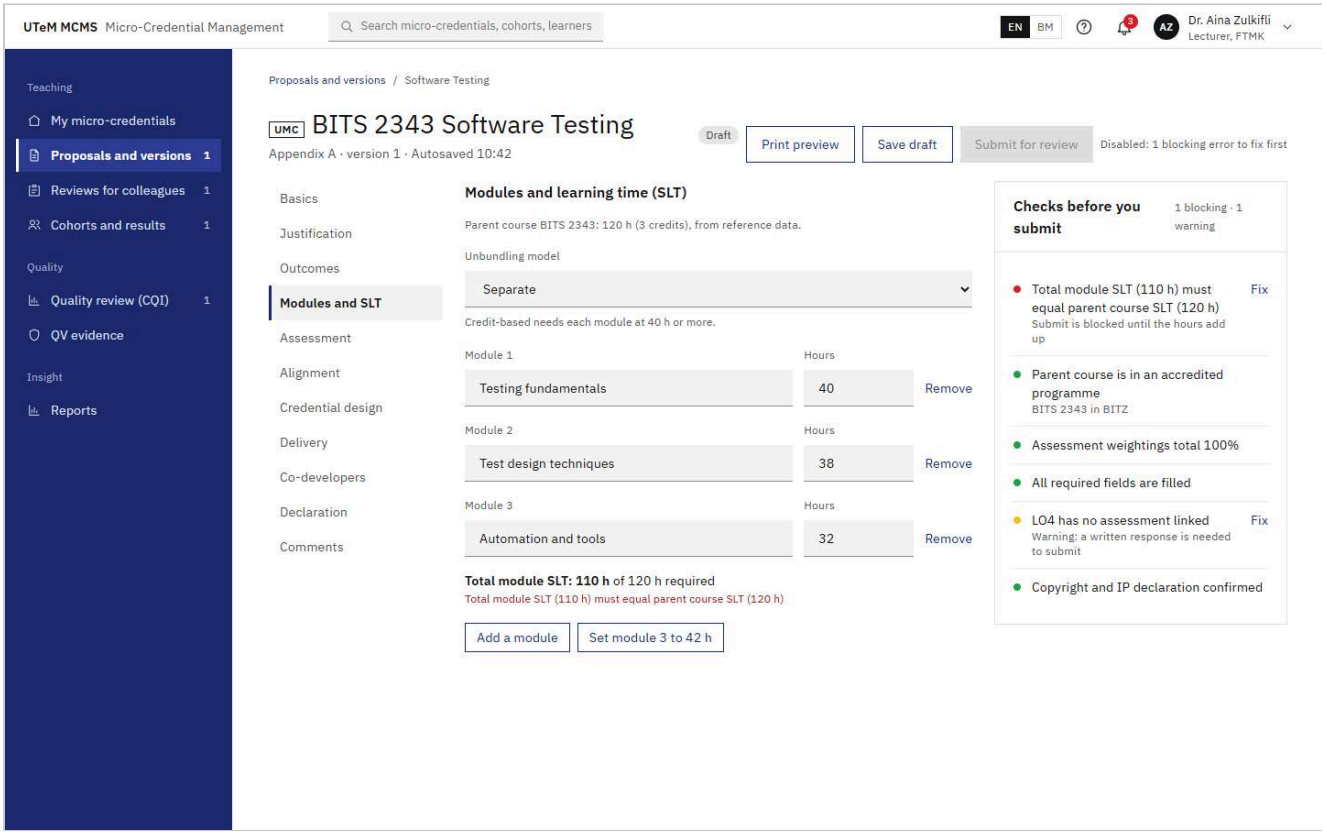


Figure 103: *LEC-05 Proposal workspace: Tab: Modules and SLT.*

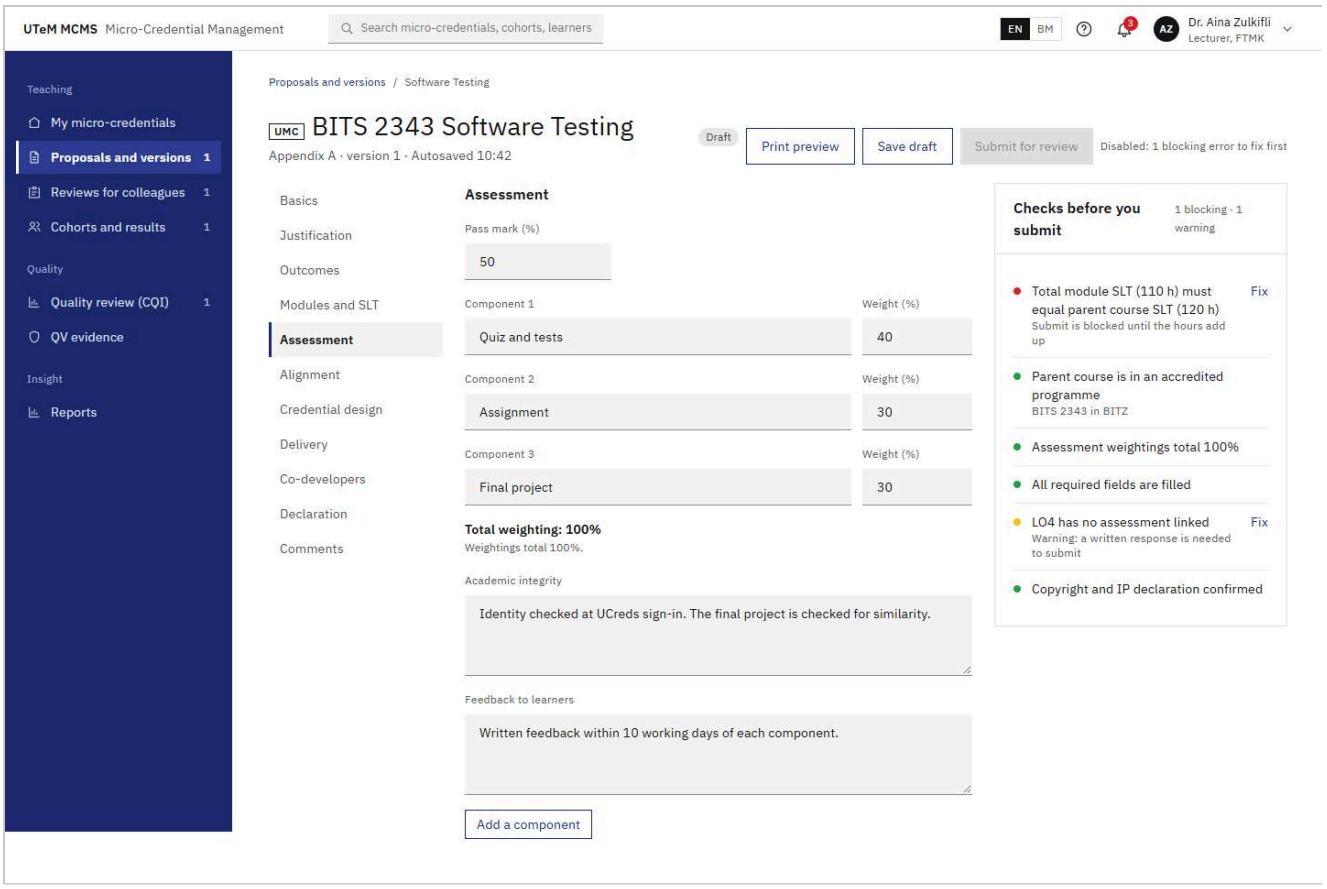


Figure 104: *LEC-05 Proposal workspace: Tab: Assessment.*

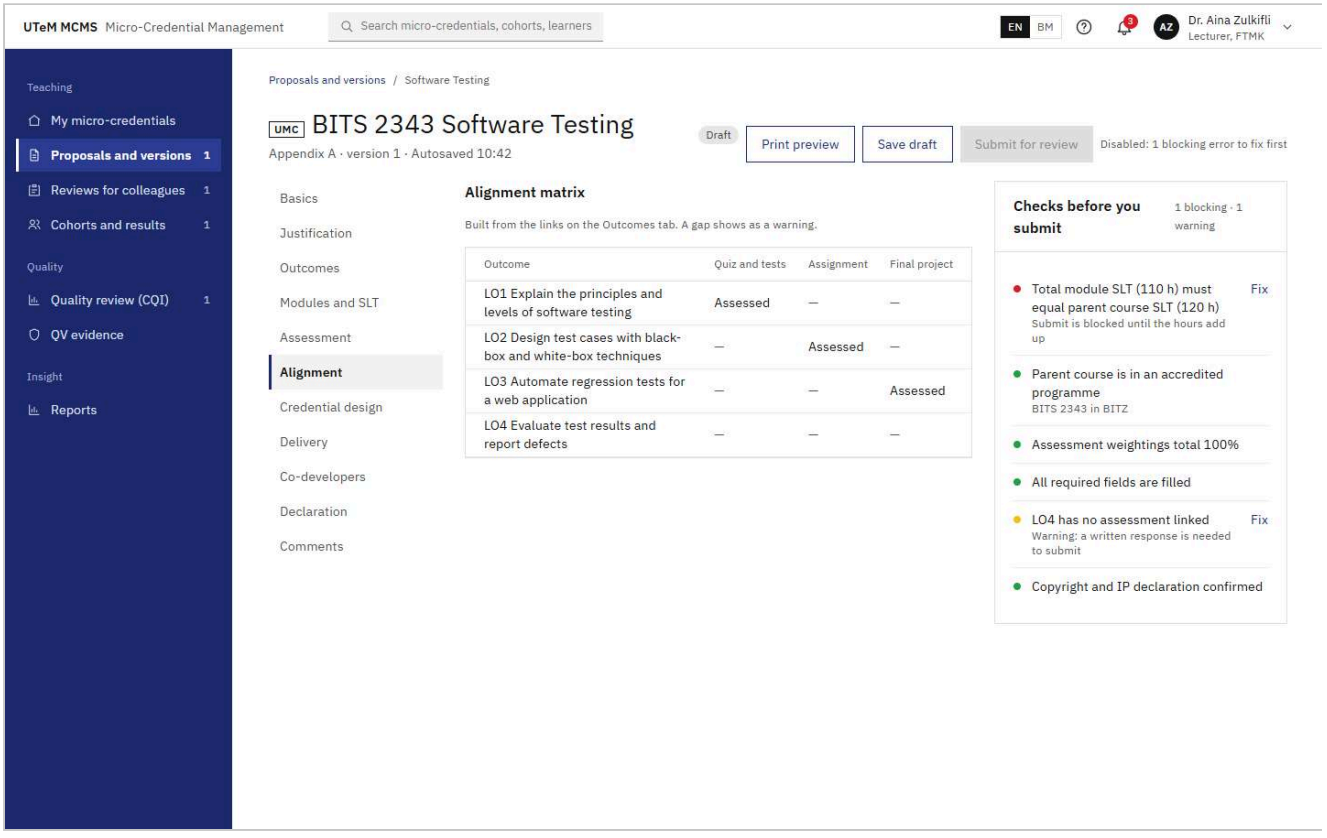


Figure 105: *LEC-05 Proposal workspace: Tab: Alignment.*

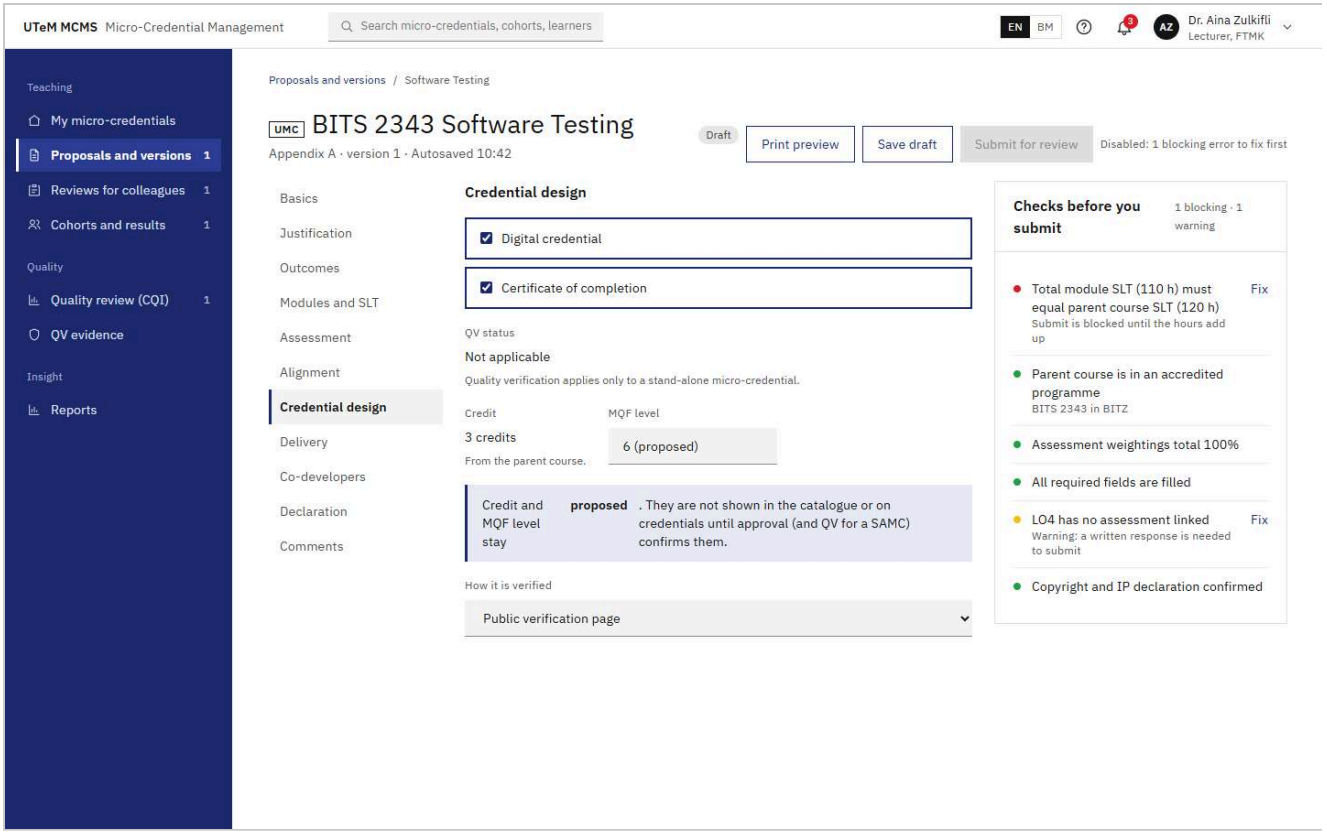


Figure 106: *LEC-05 Proposal workspace: Tab: Credential design.*

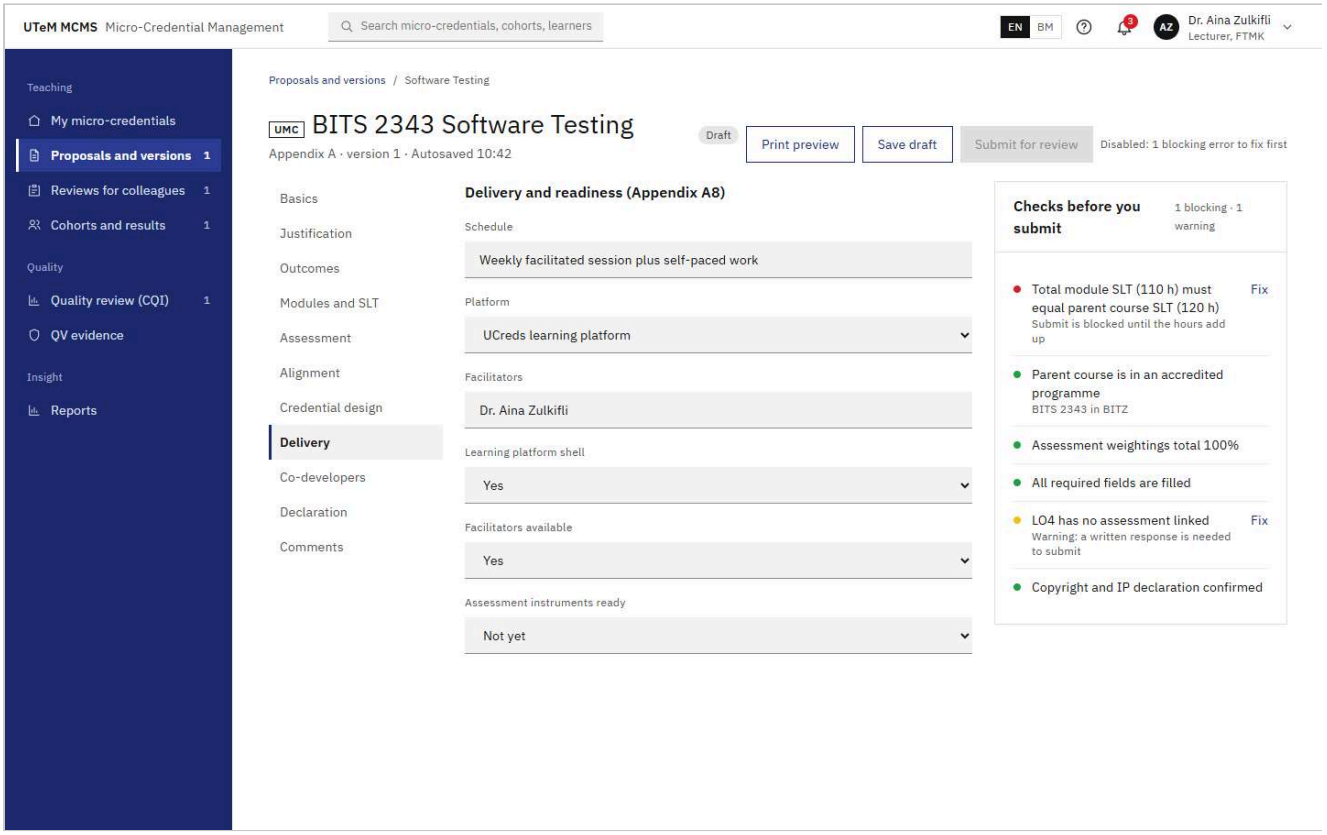


Figure 107: *LEC-05 Proposal workspace: Tab: Delivery.*

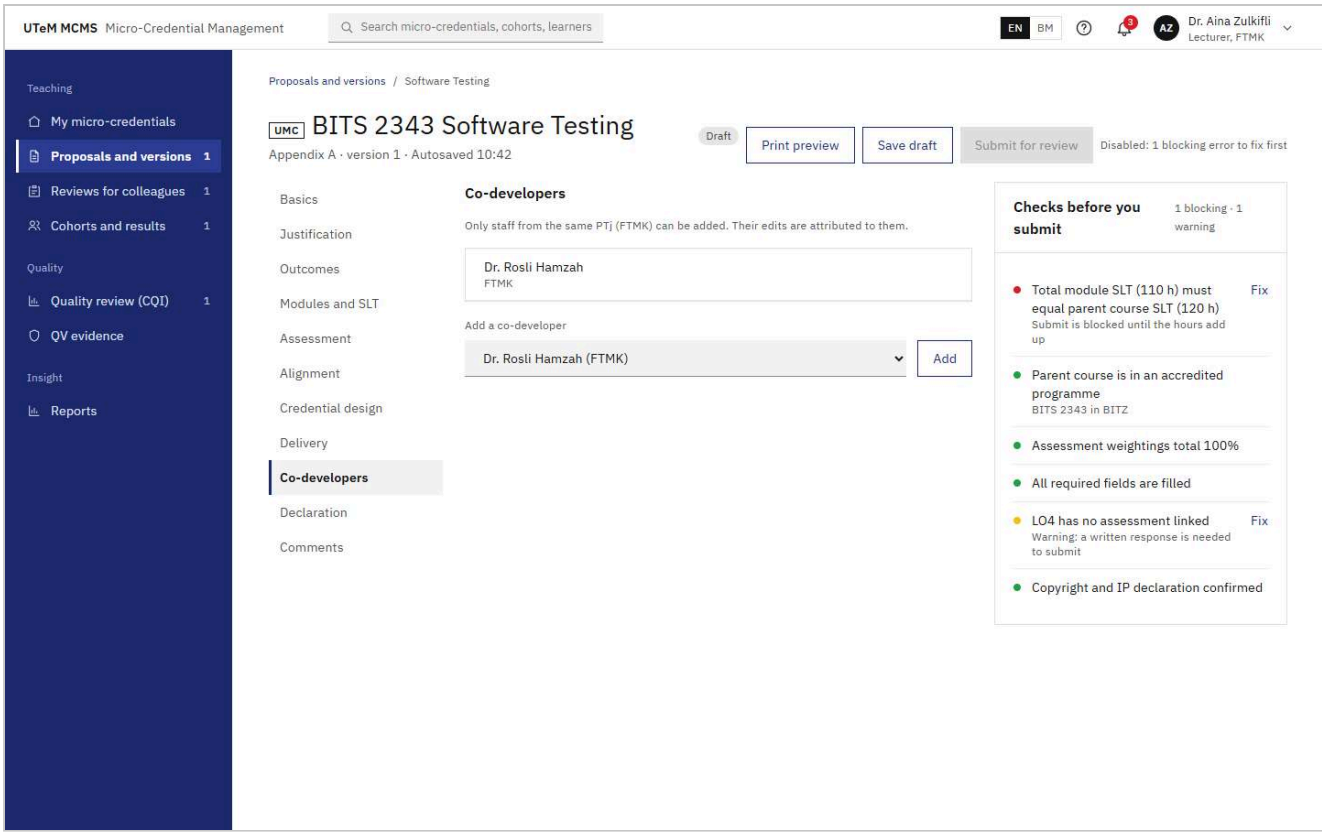


Figure 108: *LEC-05 Proposal workspace: Tab: Co-developers.*

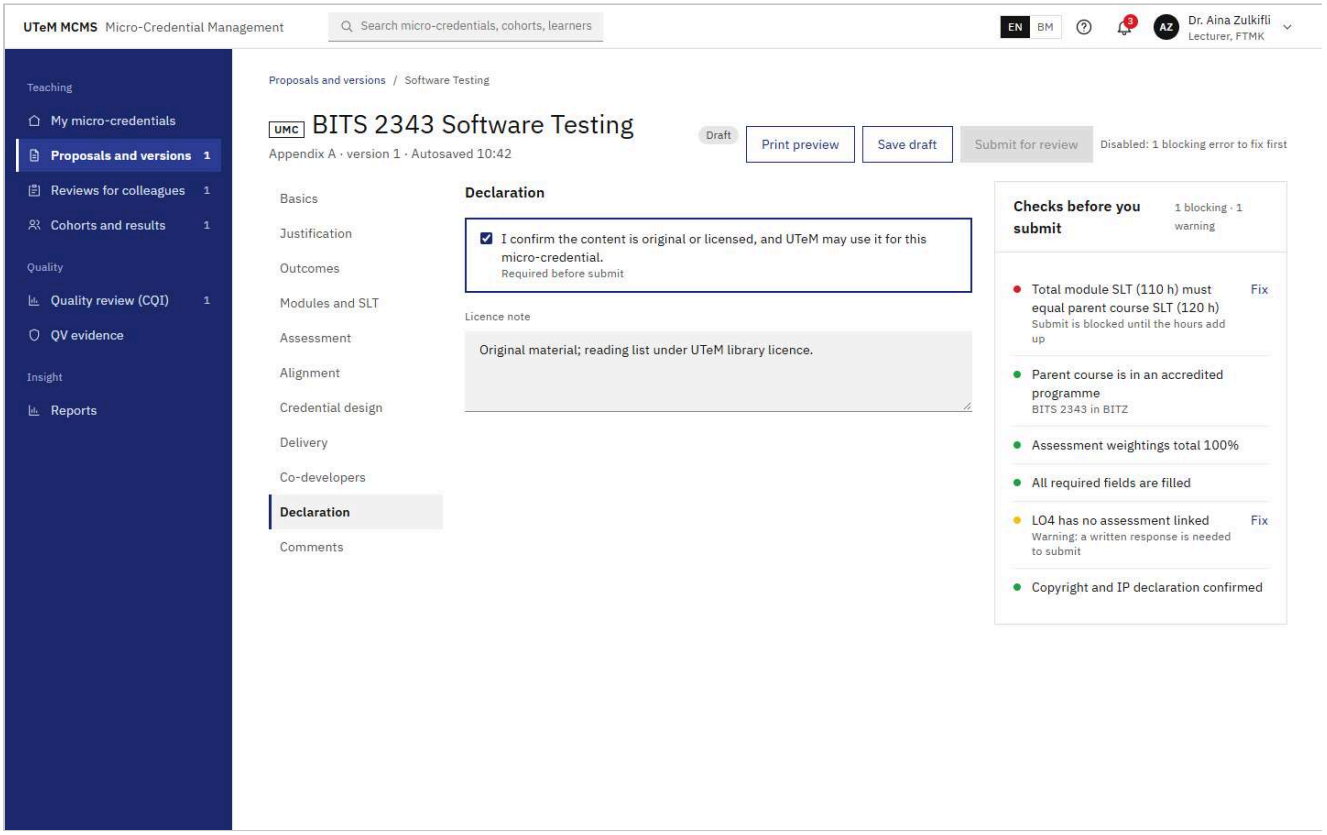


Figure 109: *LEC-05 Proposal workspace: Tab: Declaration.*

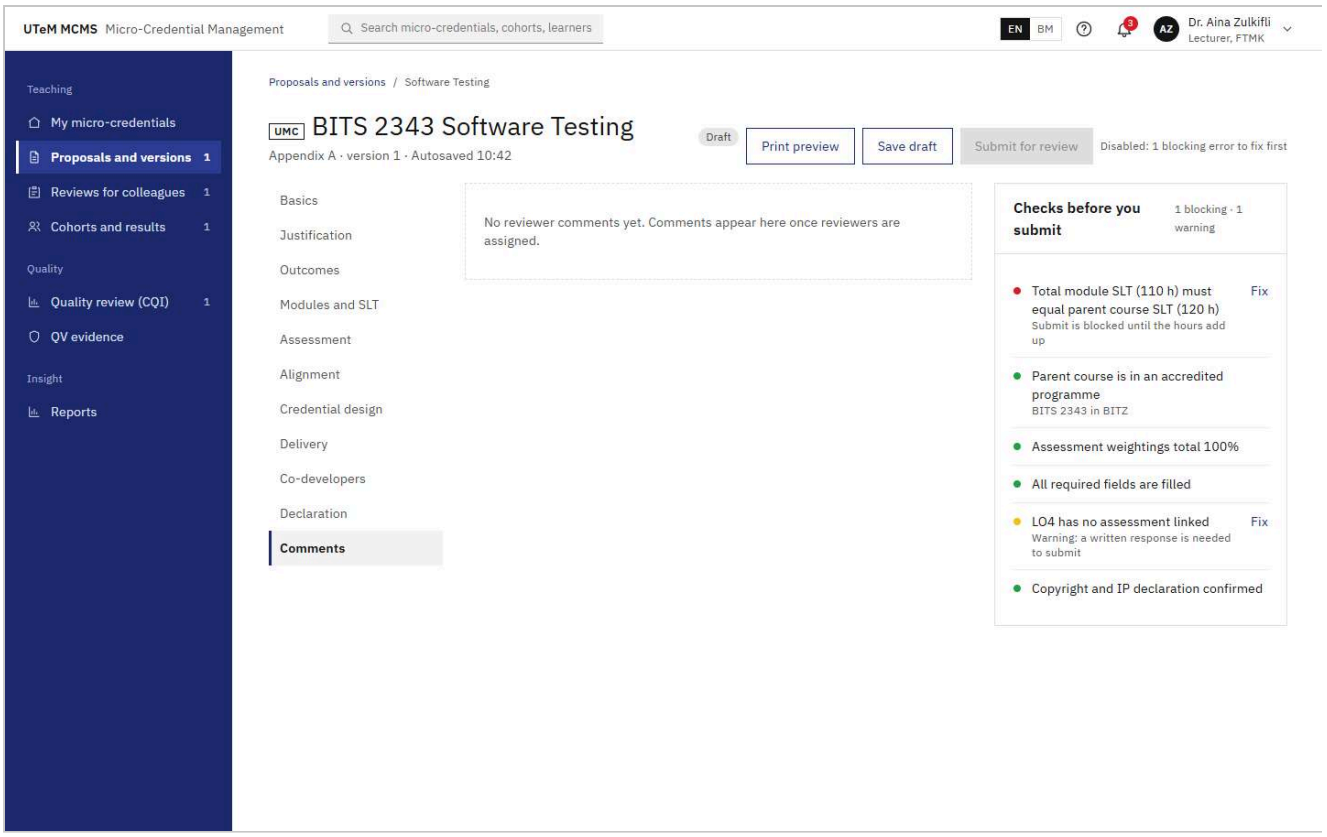


Figure 110: *LEC-05 Proposal workspace: Tab: Comments.*

LEC-06 Request a change

`#/lec/change/DS`

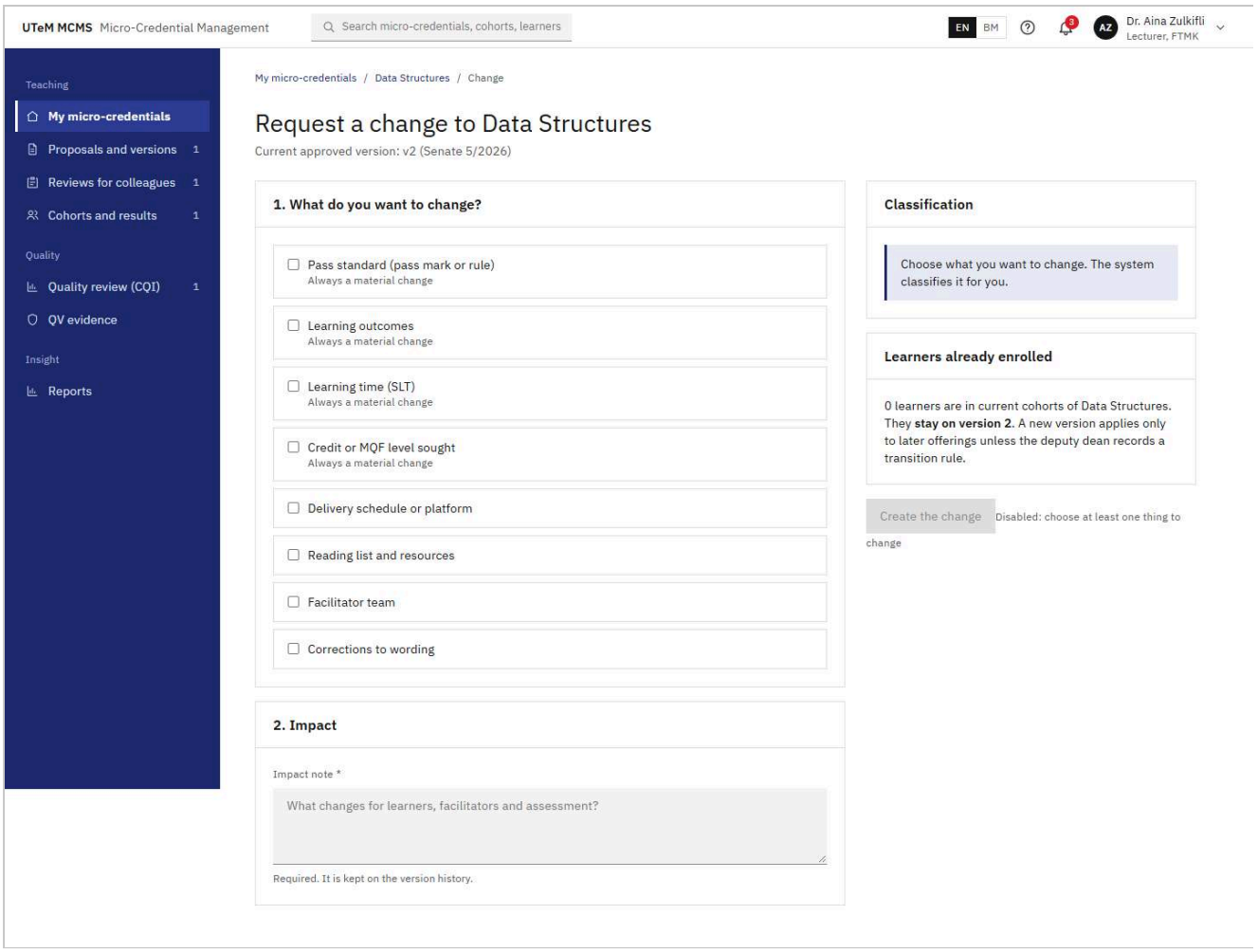


Figure 111: *LEC-06 Request a change.*

LEC-07 Reviews for colleagues

`#/lec/reviews`

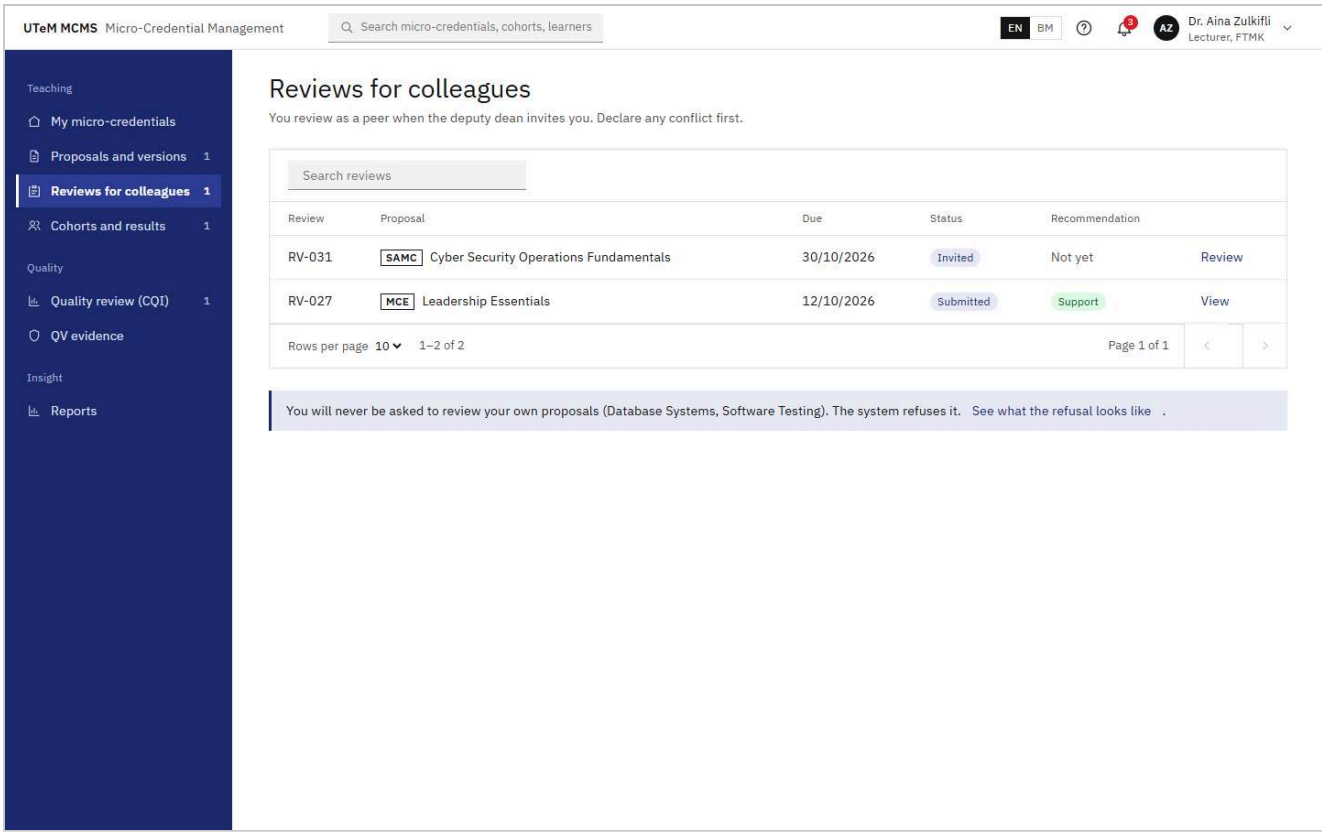


Figure 112: *LEC-07 Reviews for colleagues.*

LEC-08 Review a proposal

`#/lec/review/RV-031`

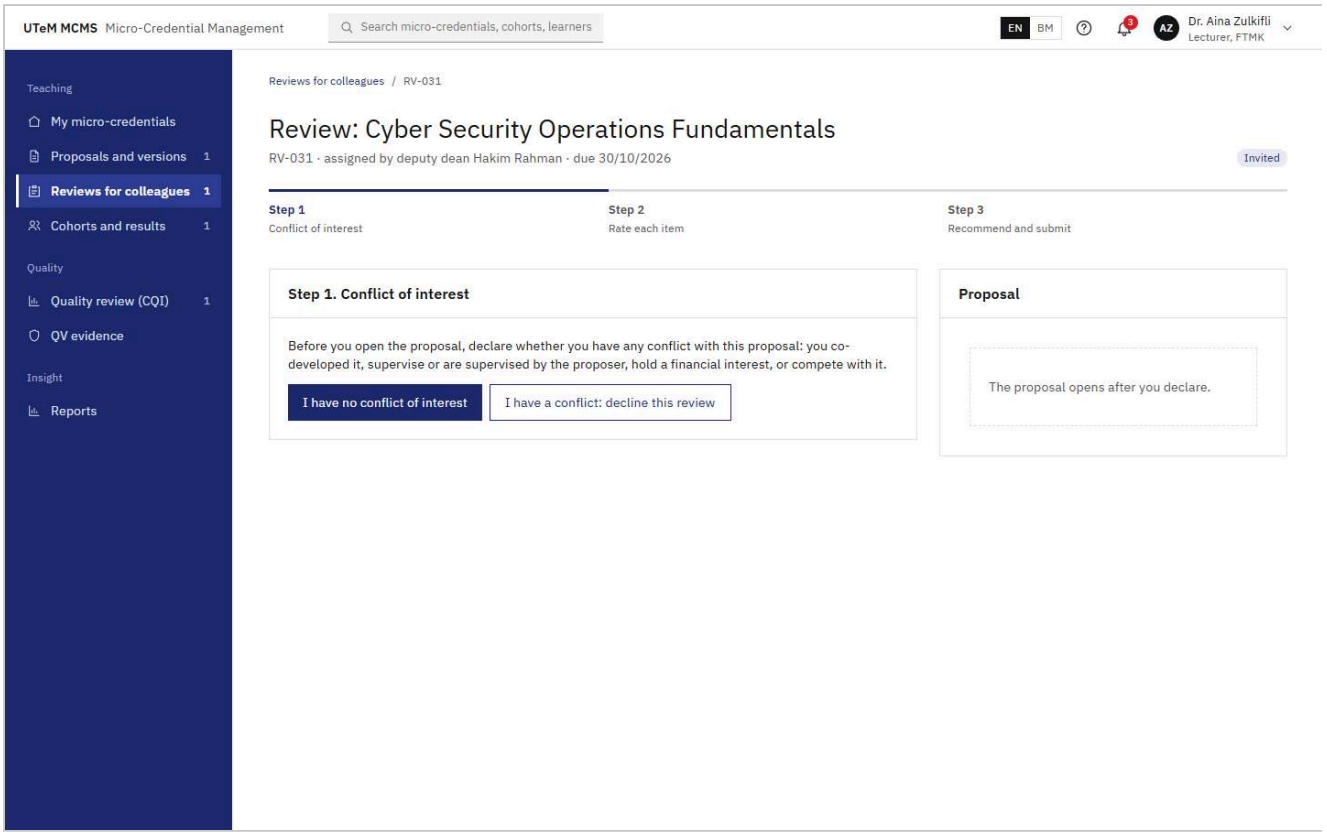


Figure 113: *LEC-08 Review a proposal.*

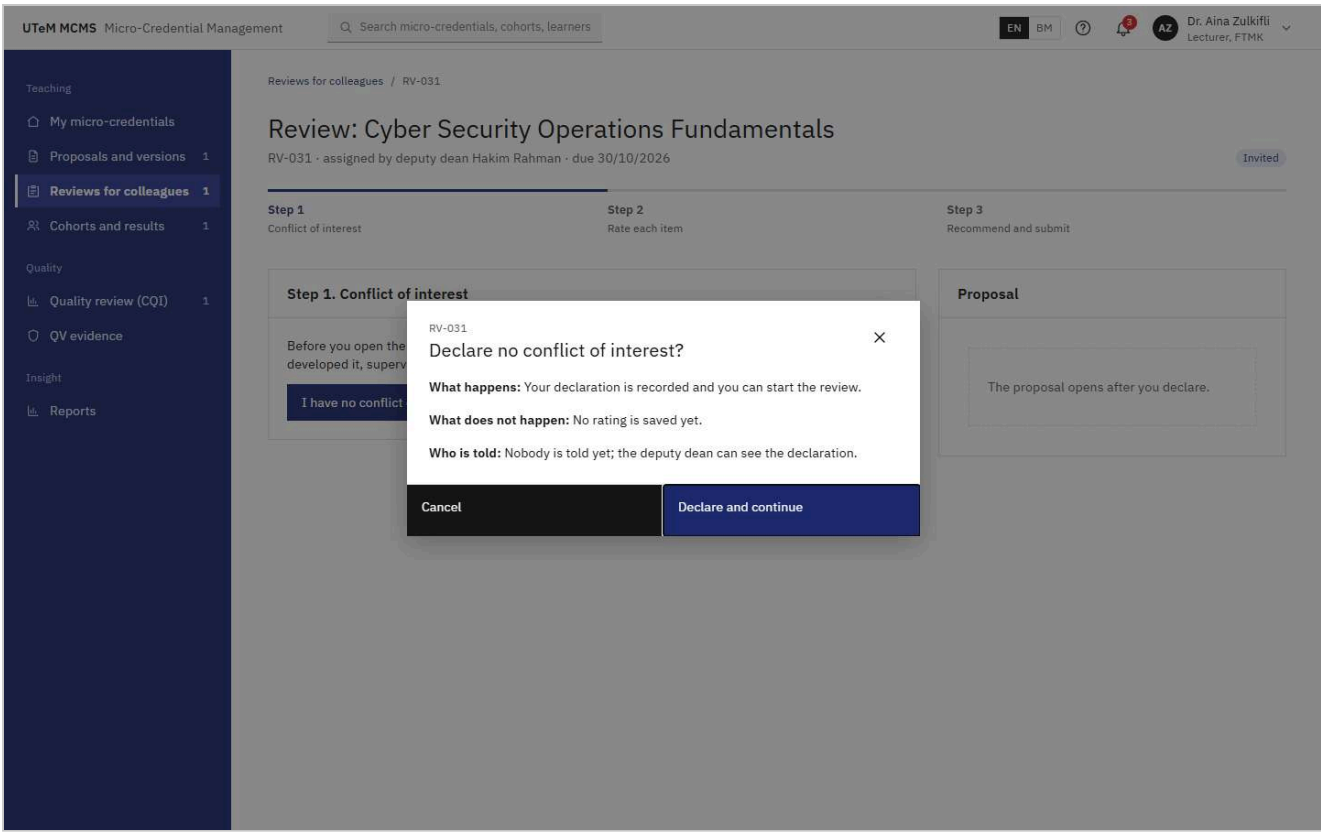


Figure 114: *LEC-08 Confirmation: Declare no conflict of interest?*

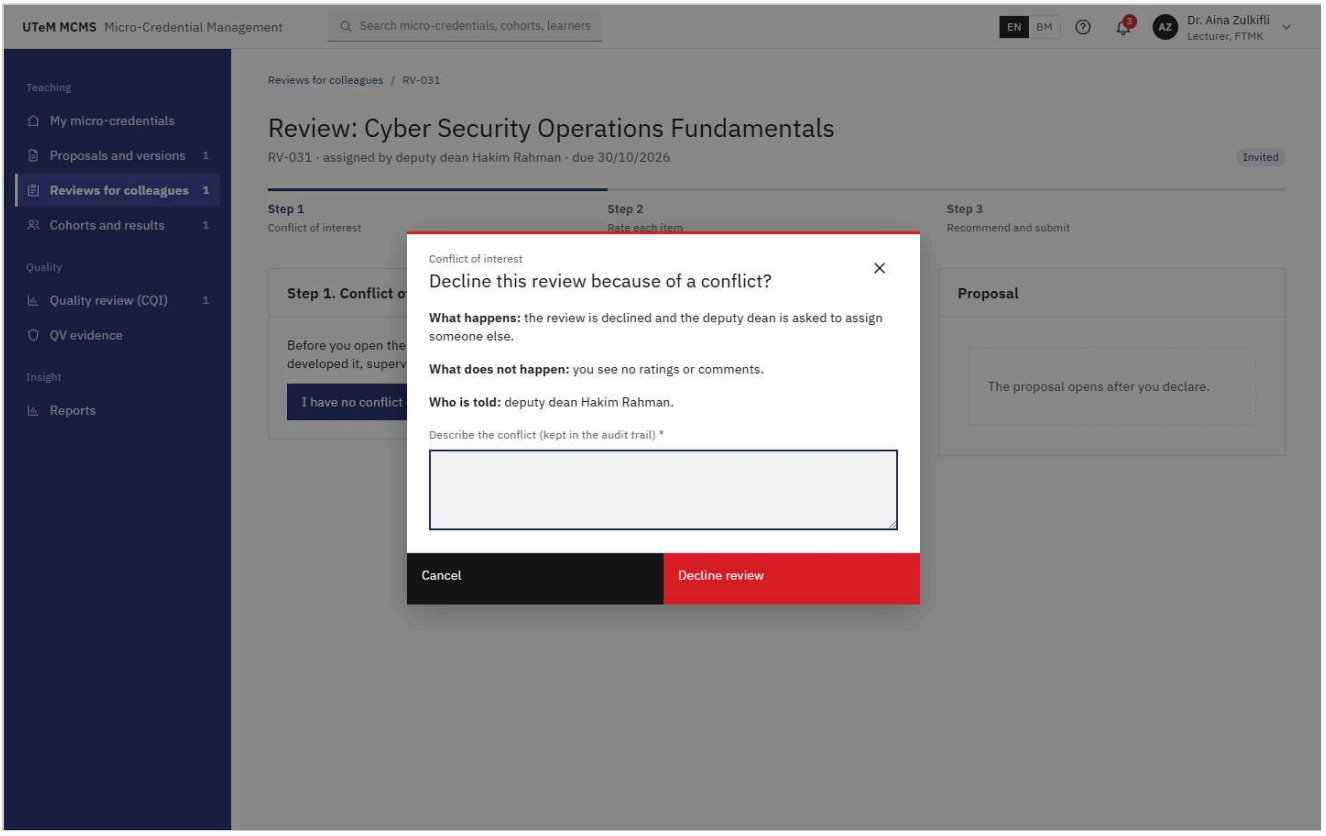


Figure 115: *LEC-08 Warning (destructive): Decline this review because of a conflict?*

LEC-09 Cohorts and results

`#/lec/cohorts`

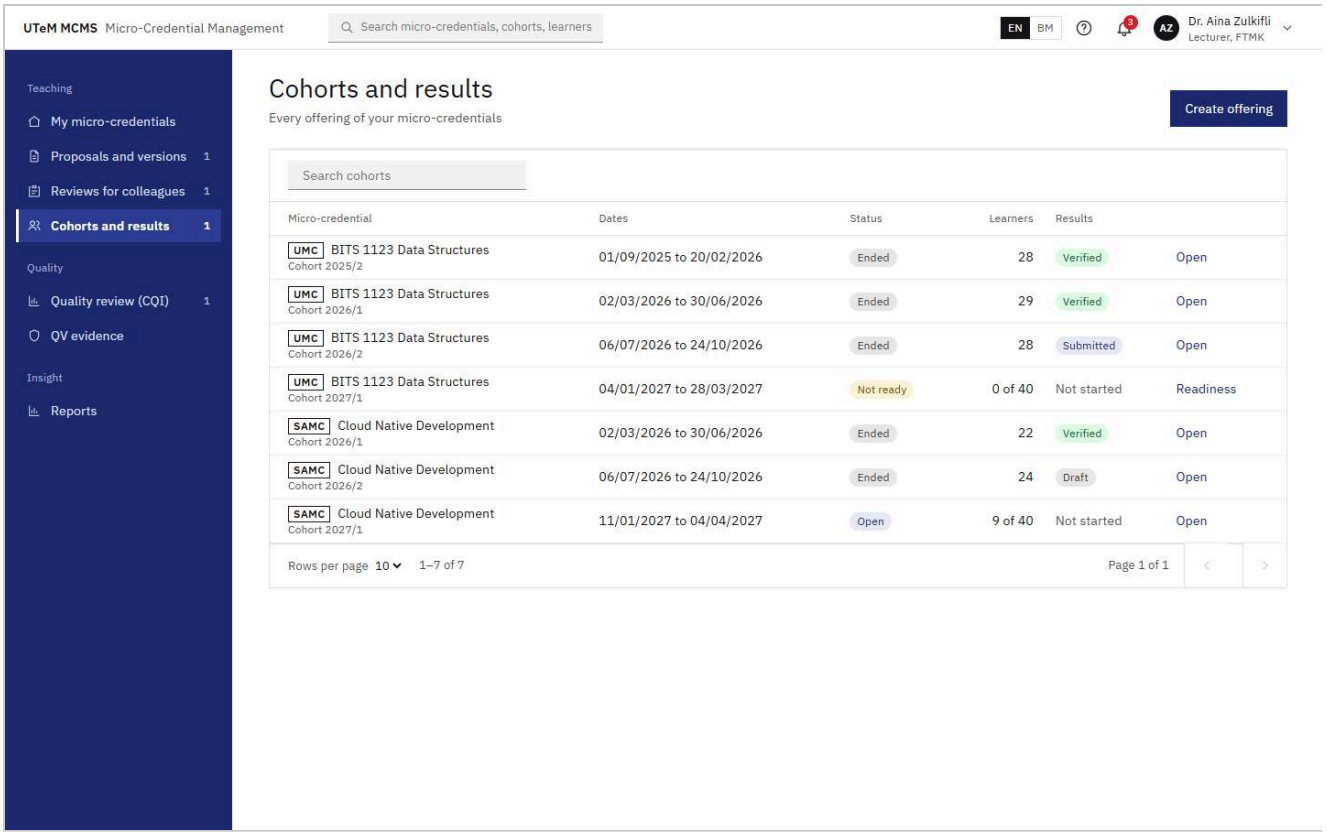


Figure 116: *LEC-09 Cohorts and results.*

LEC-10 Cohort results

`#/lec/cohort/CND-2026-2`

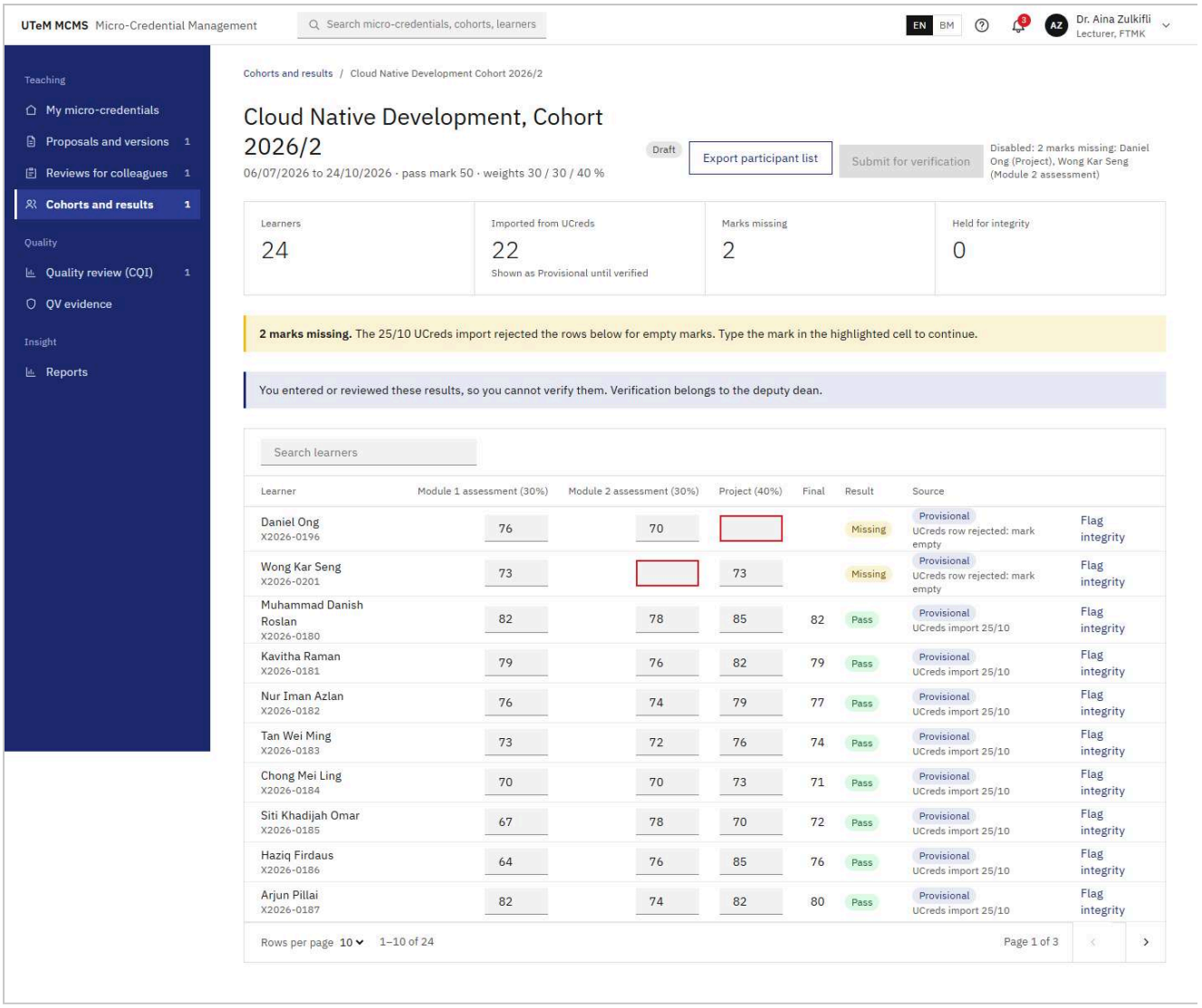


Figure 117: *LEC-10 Cohort results.*

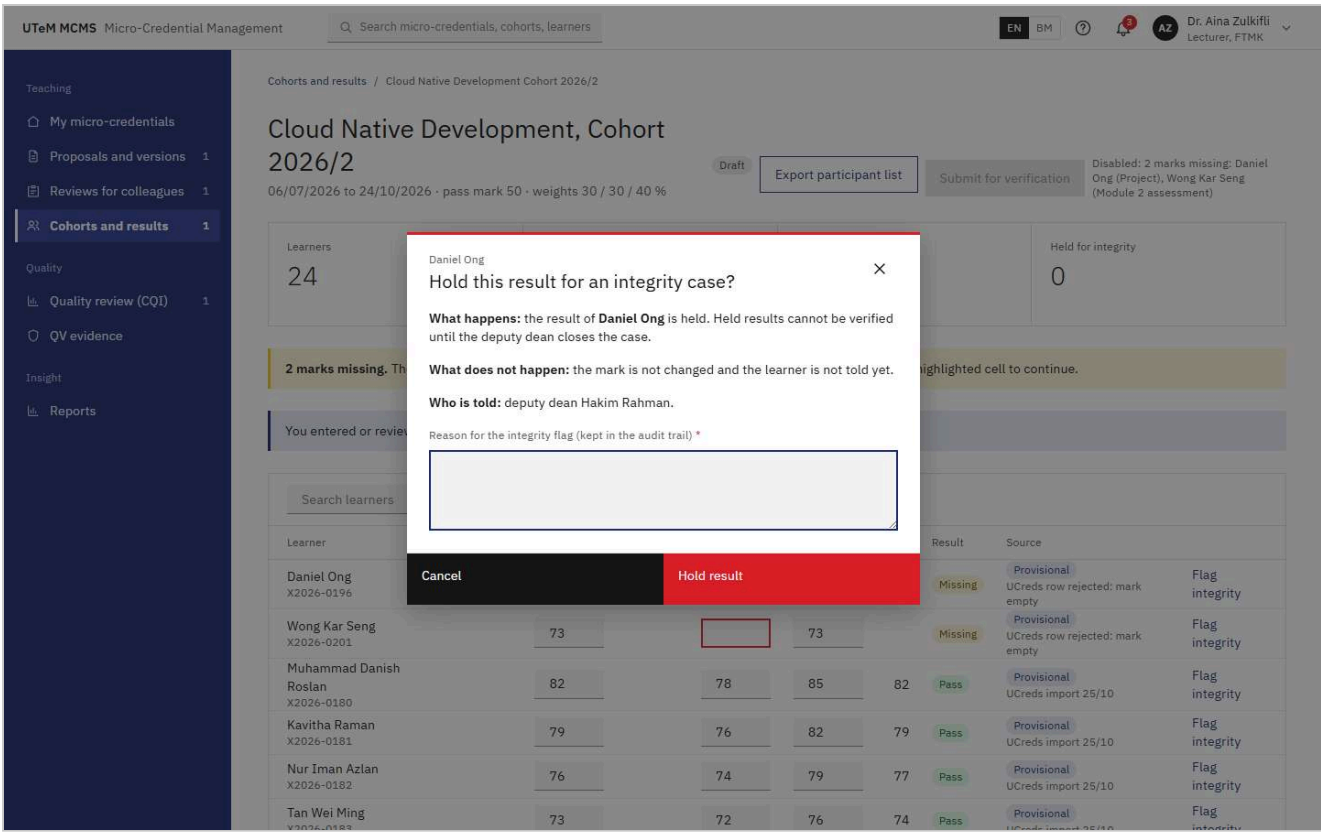


Figure 118: *LEC-10 Warning (destructive): Hold this result for an integrity case?*

LEC-11 Create an offering

`#/lec/offering/new`

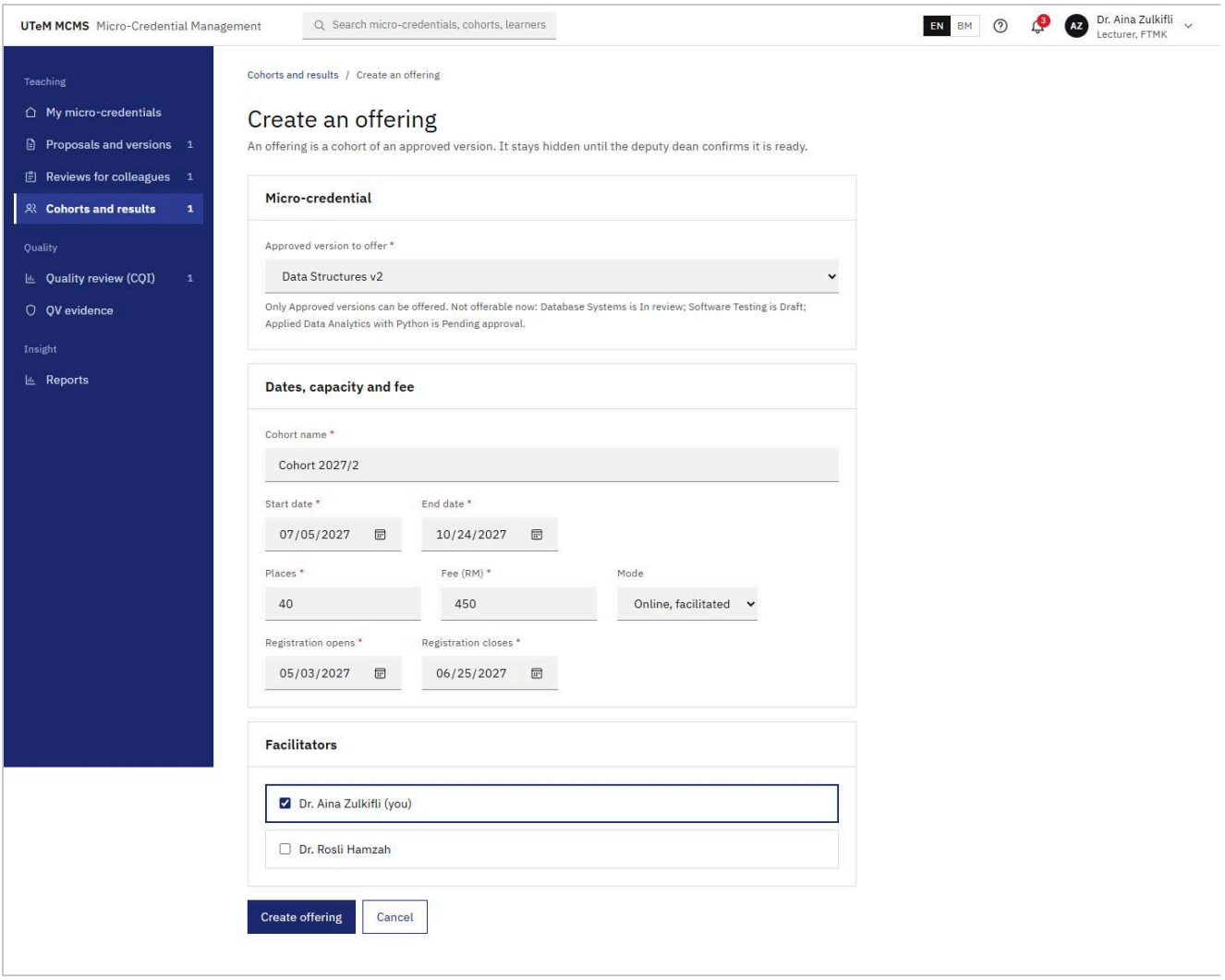


Figure 119: *LEC-11 Create an offering.*

LEC-12 Readiness checklist

`#/lec/readiness/DS-2027-1`

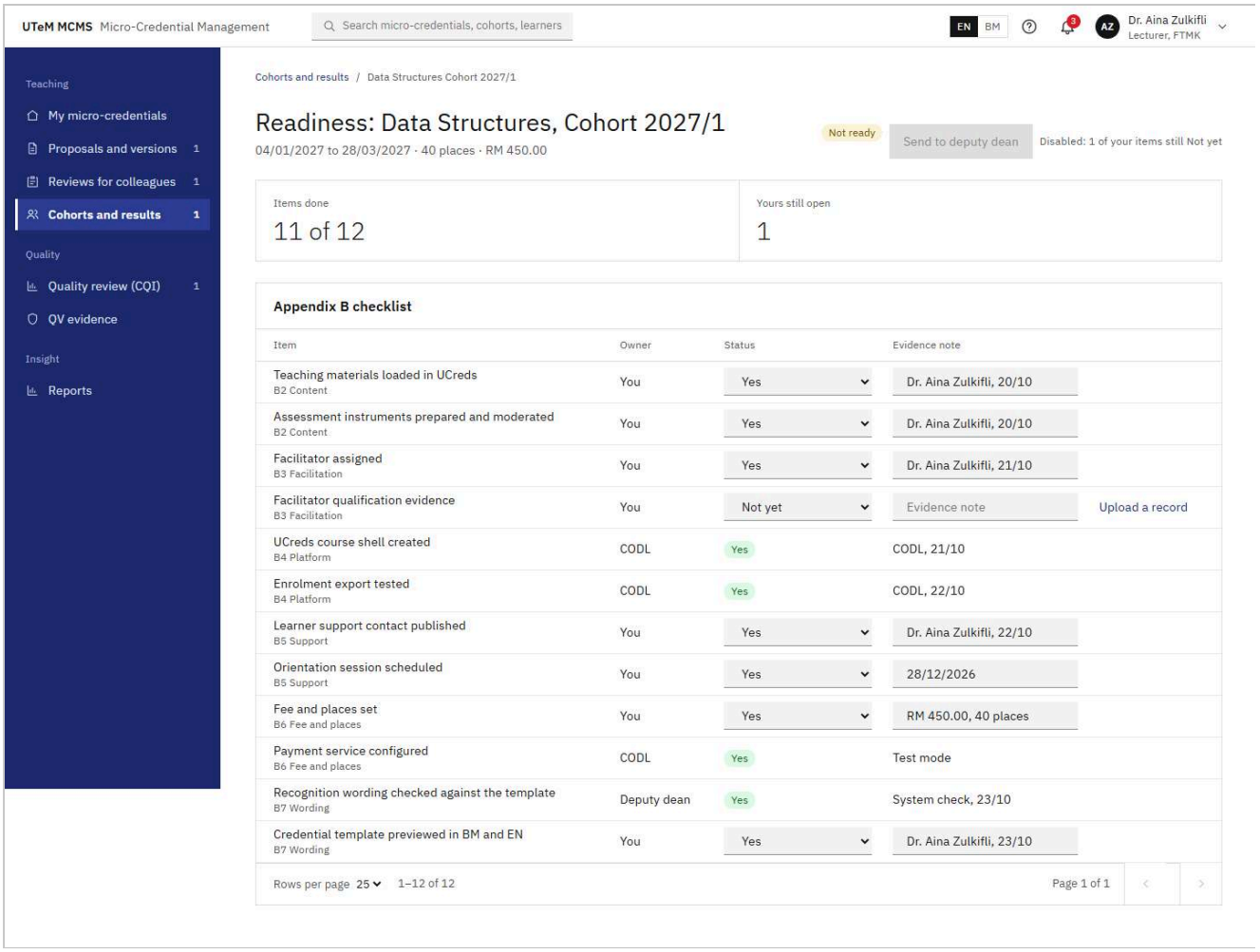


Figure 120: *LEC-12 Readiness checklist.*

LEC-13 Quality review (CQI)

`#/lec/cqi`

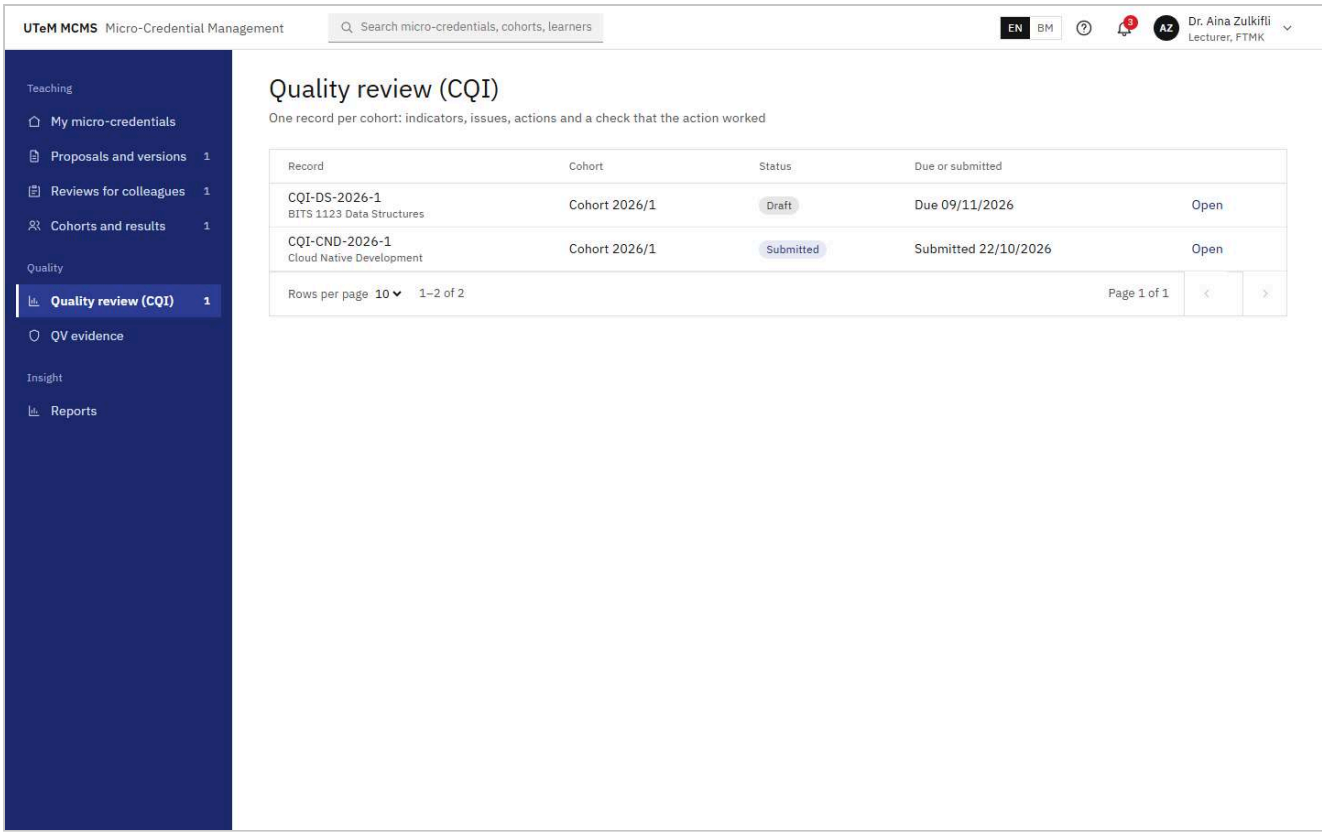


Figure 121: *LEC-13 Quality review (CQI).*

LEC-14 Quality review record

`#/lec/cqi/CQI-DS-2026-1`

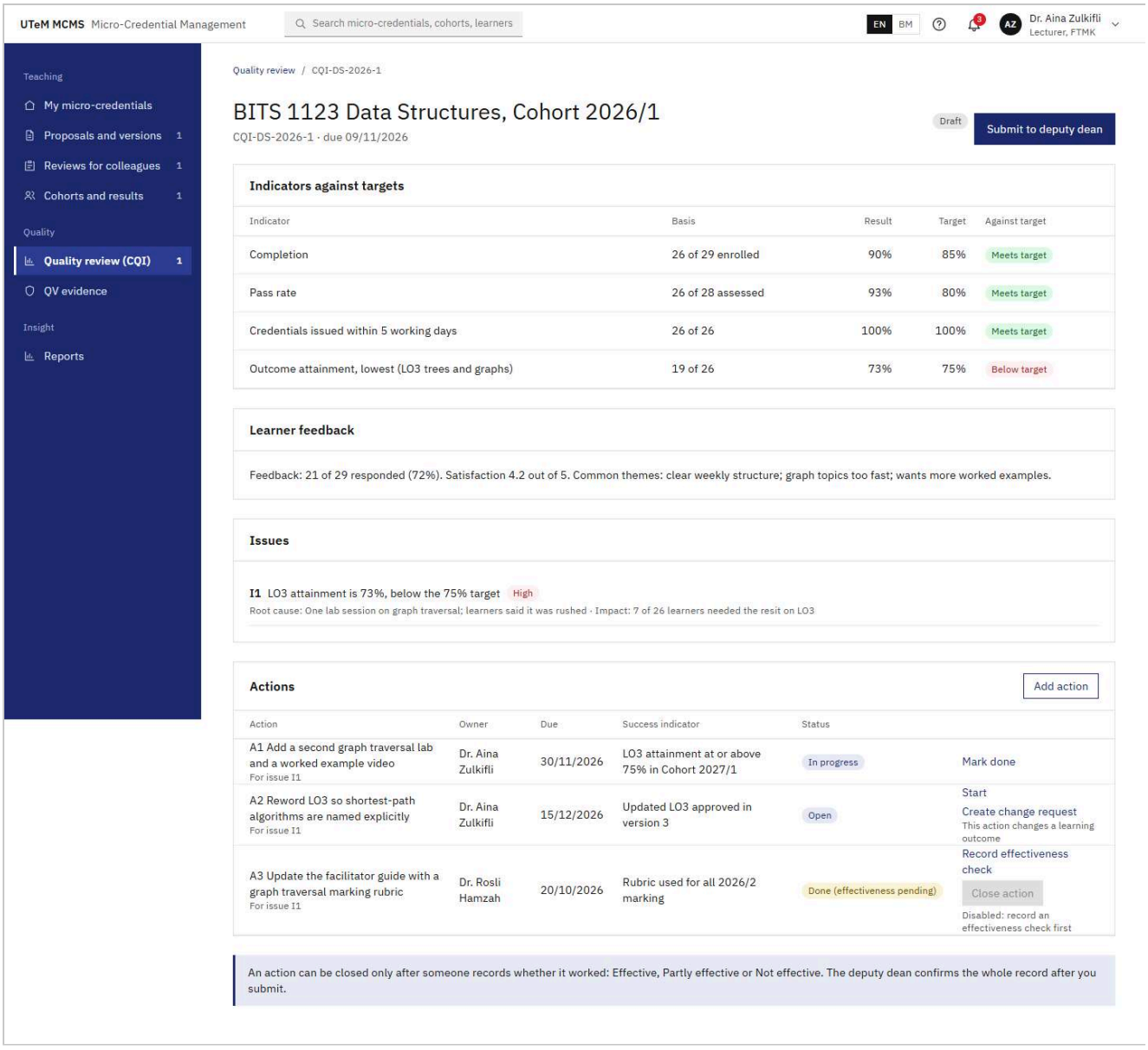


Figure 122: *LEC-14 Quality review record.*

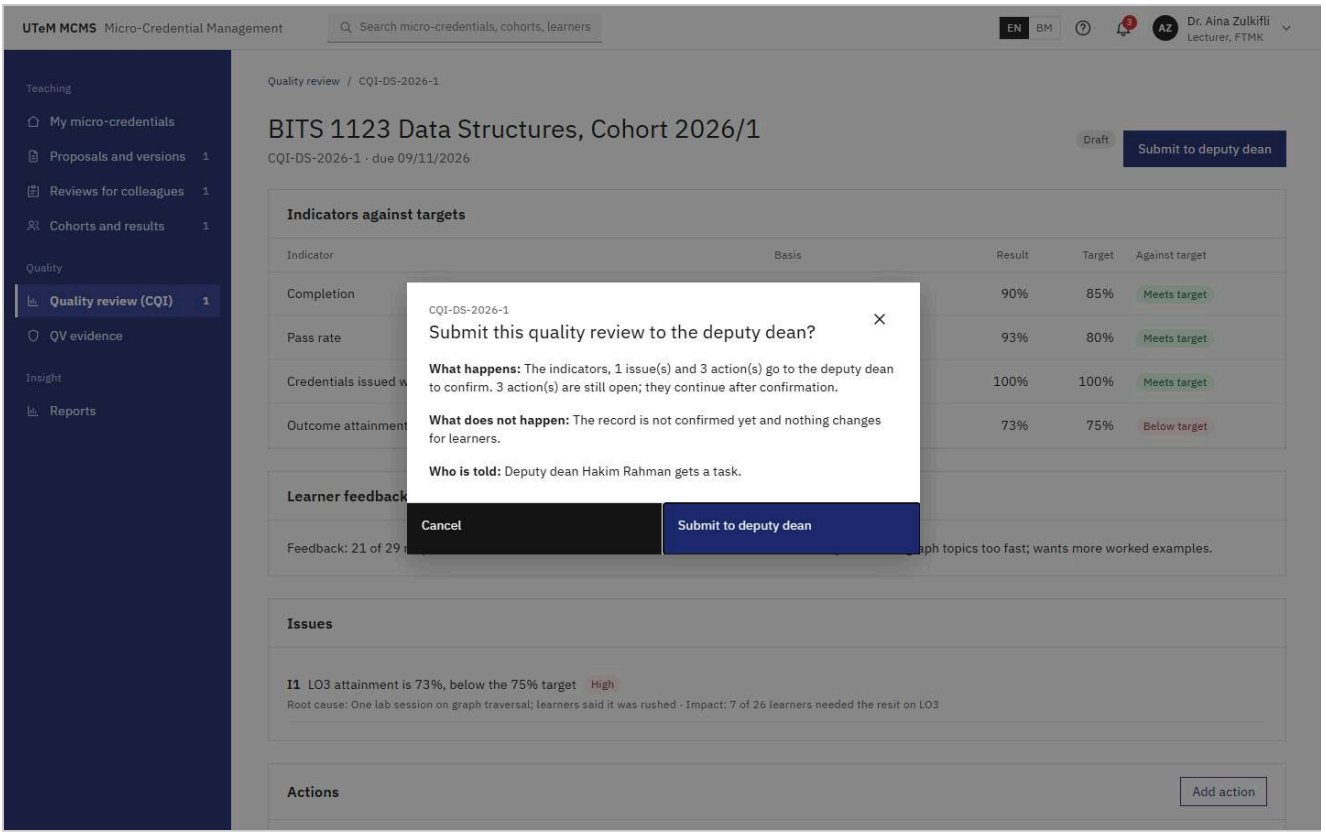


Figure 123: *LEC-14 Confirmation: Submit this quality review to the deputy dean?*

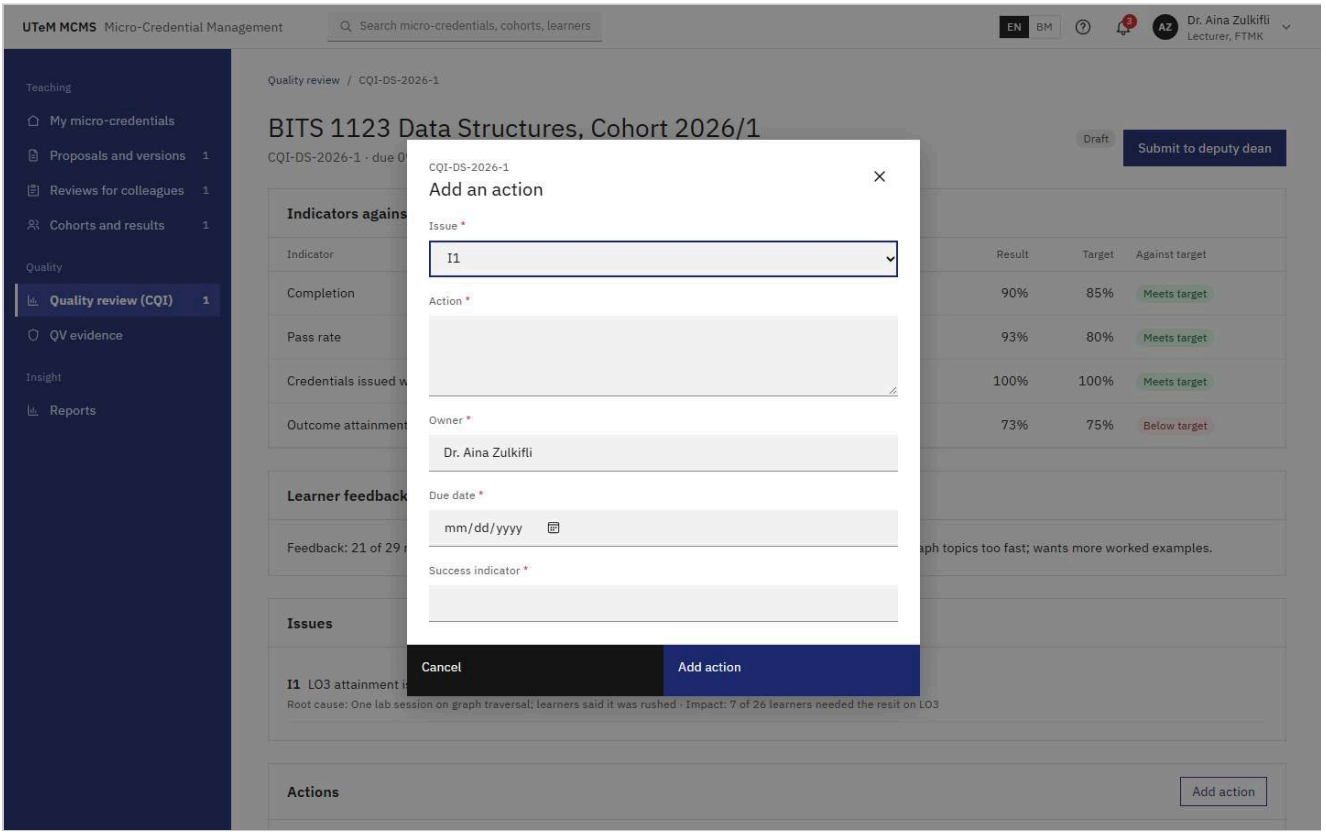


Figure 124: *LEC-14 Dialog: Add an action.*

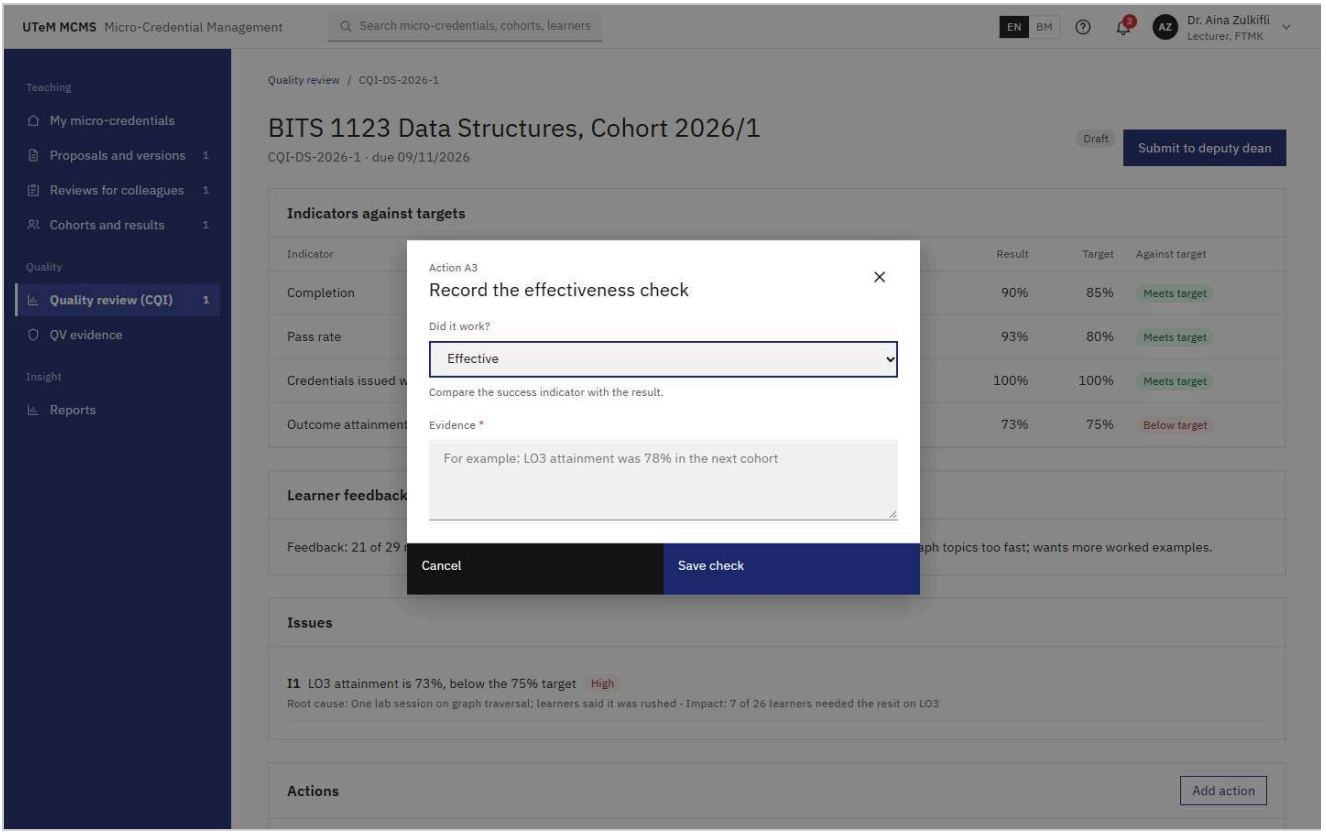


Figure 125: *LEC-14 Dialog: Record the effectiveness check.*

LEC-15 QV evidence

`#/lec/qv`

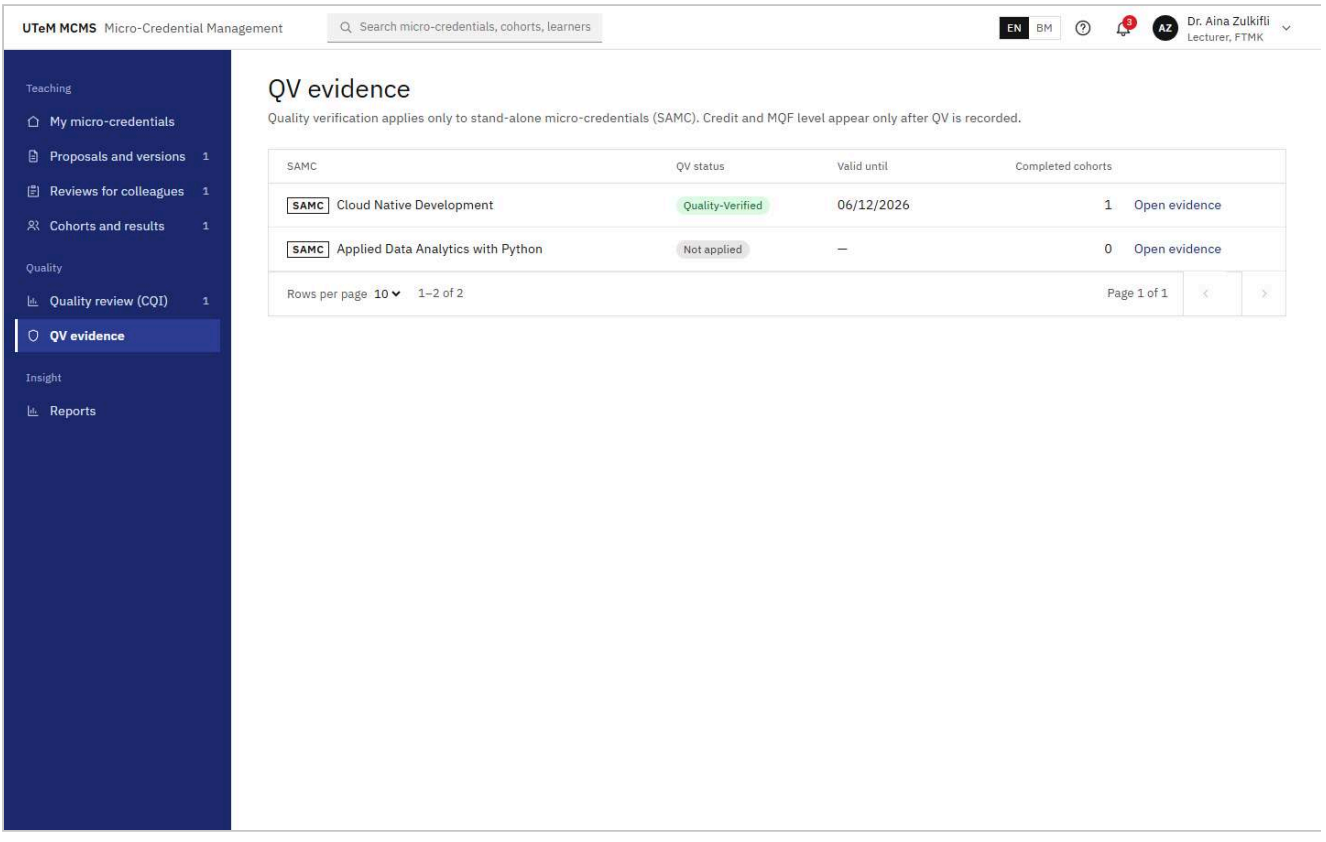


Figure 126: *LEC-15 QV evidence.*

LEC-16 QV evidence

`#/lec/qv/CND`

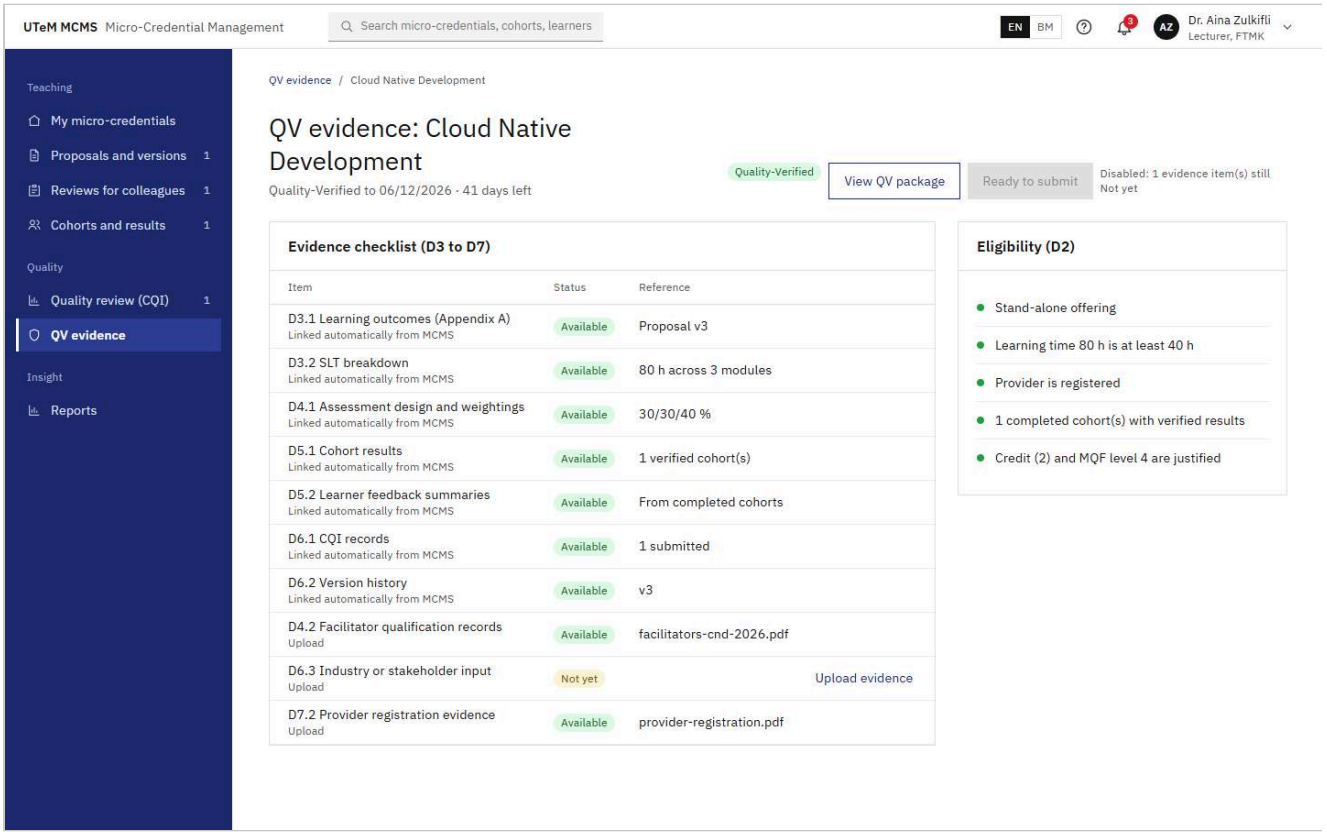


Figure 127: *LEC-16 QV evidence.*

LEC-17 Reports

`#/lec/reports`

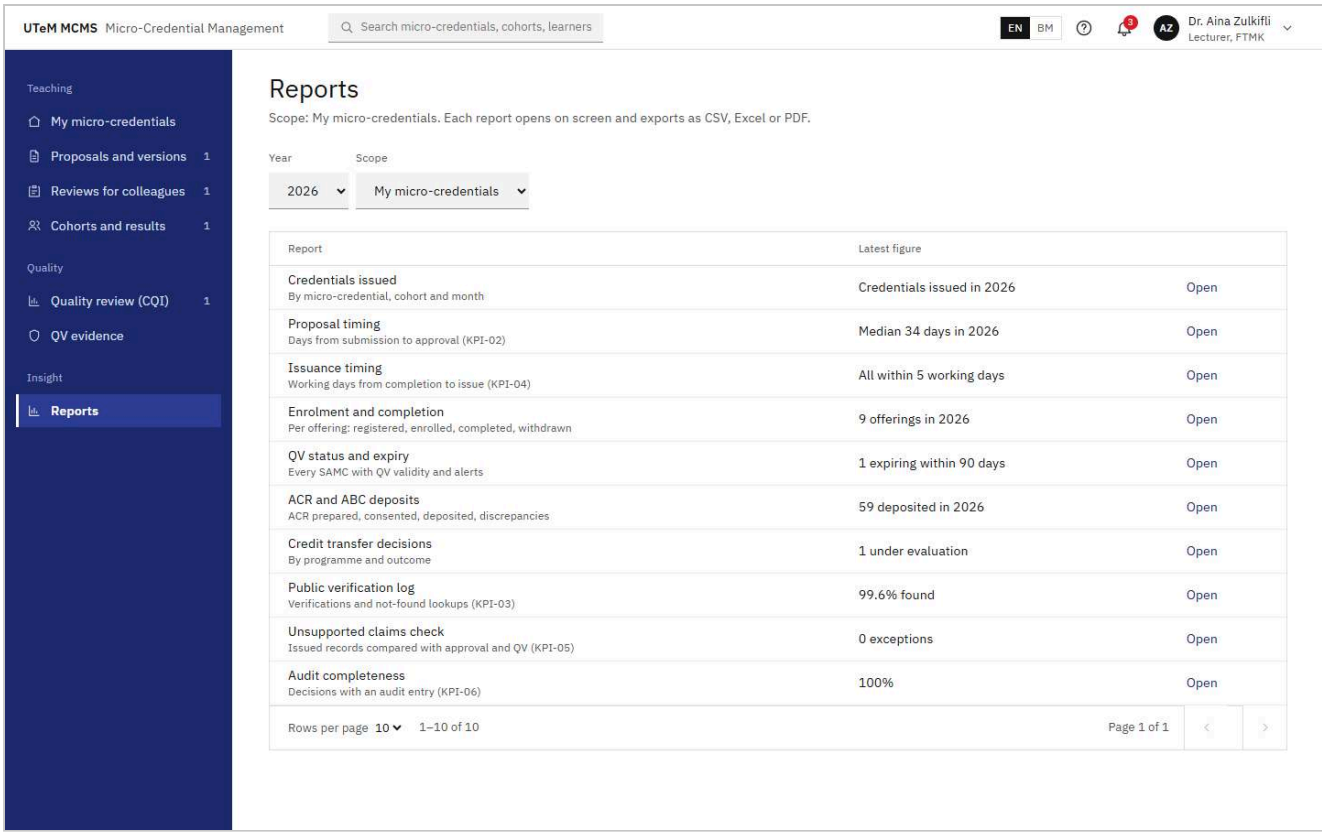


Figure 128: *LEC-17 Reports.*

LEC-18 Report

`#/lec/report/credentials-issued`

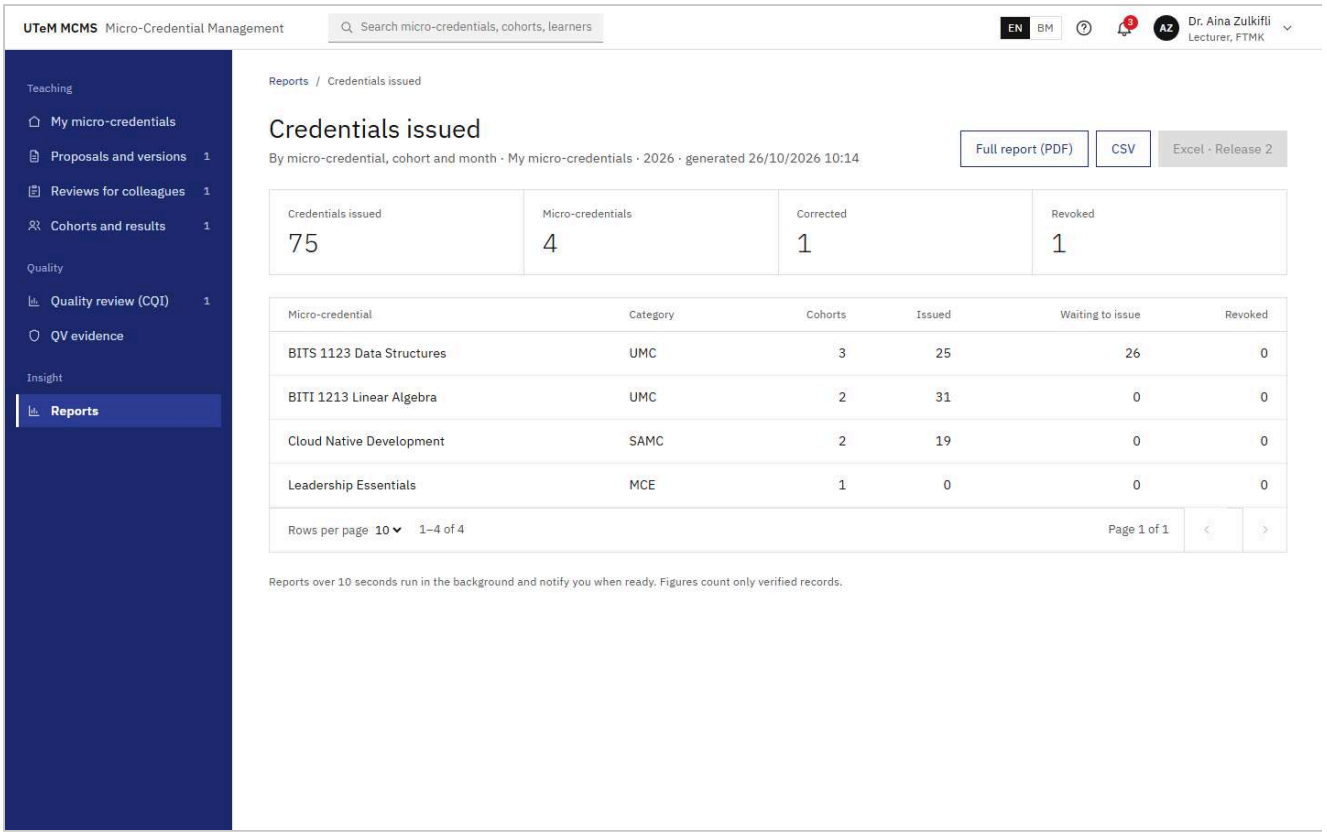


Figure 129: *LEC-18 Report.*

Proposal workspace, comments

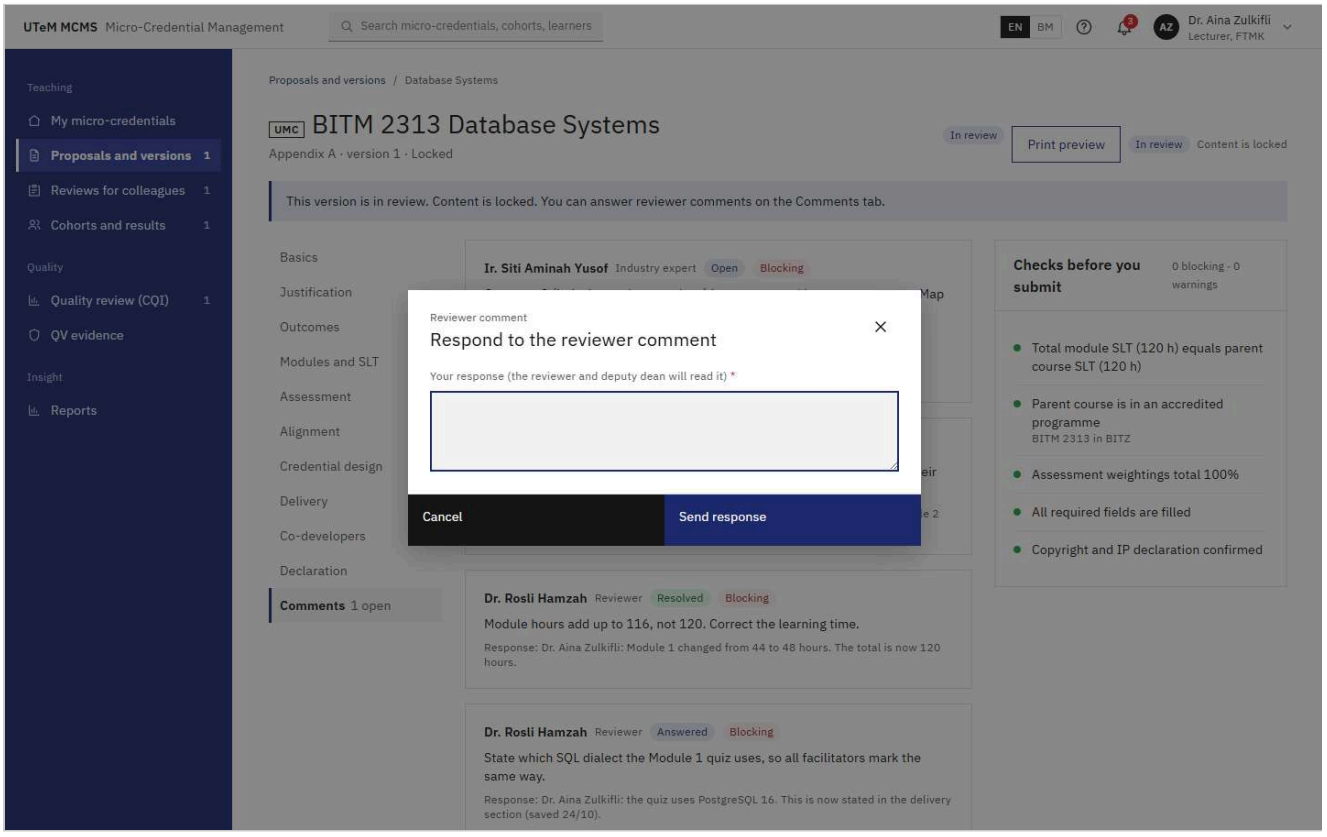


Figure 130: *Dialog: Respond to the reviewer comment (record state: Proposal workspace, comments).*

Change to an approved version (change chosen)

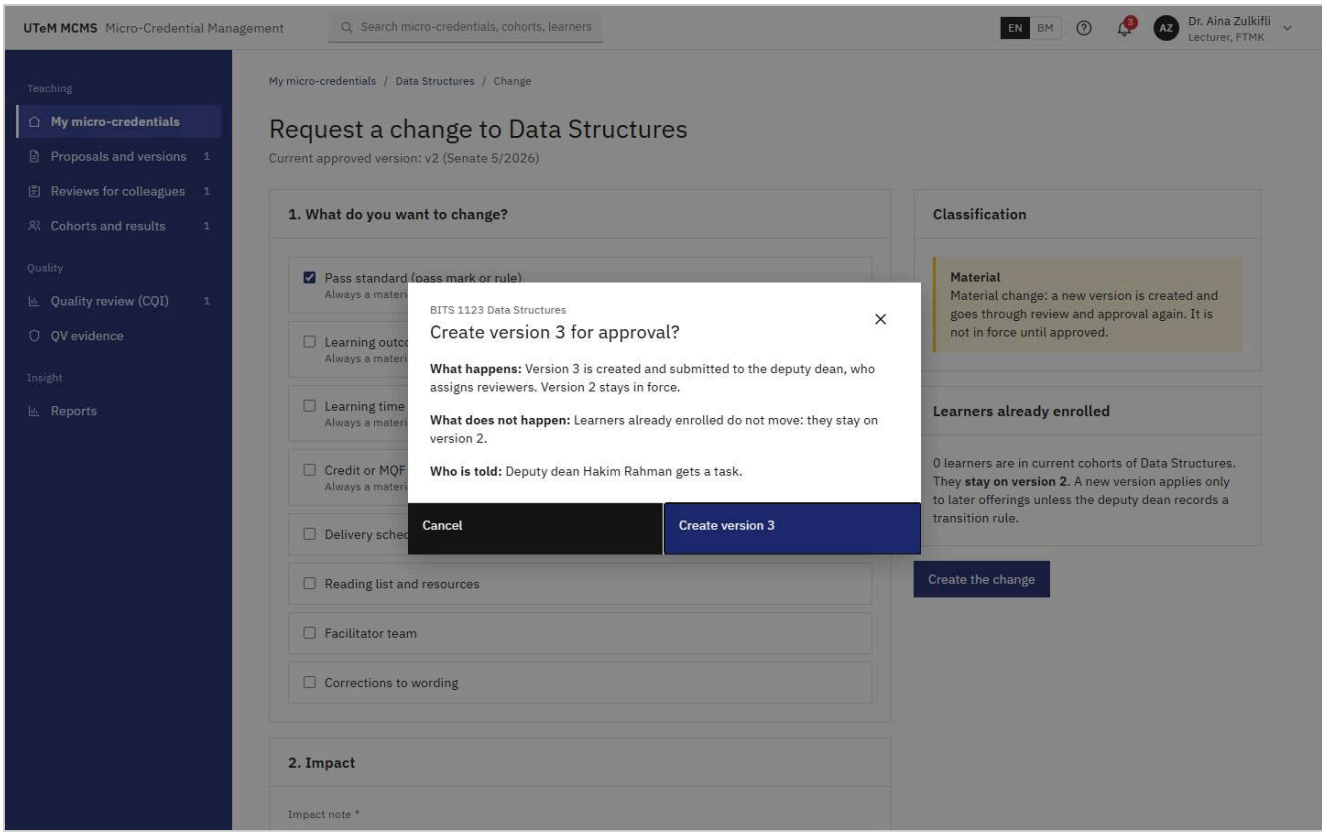


Figure 131: *Confirmation: Create version 3 for approval? (record state: Change to an approved version (change chosen)).*

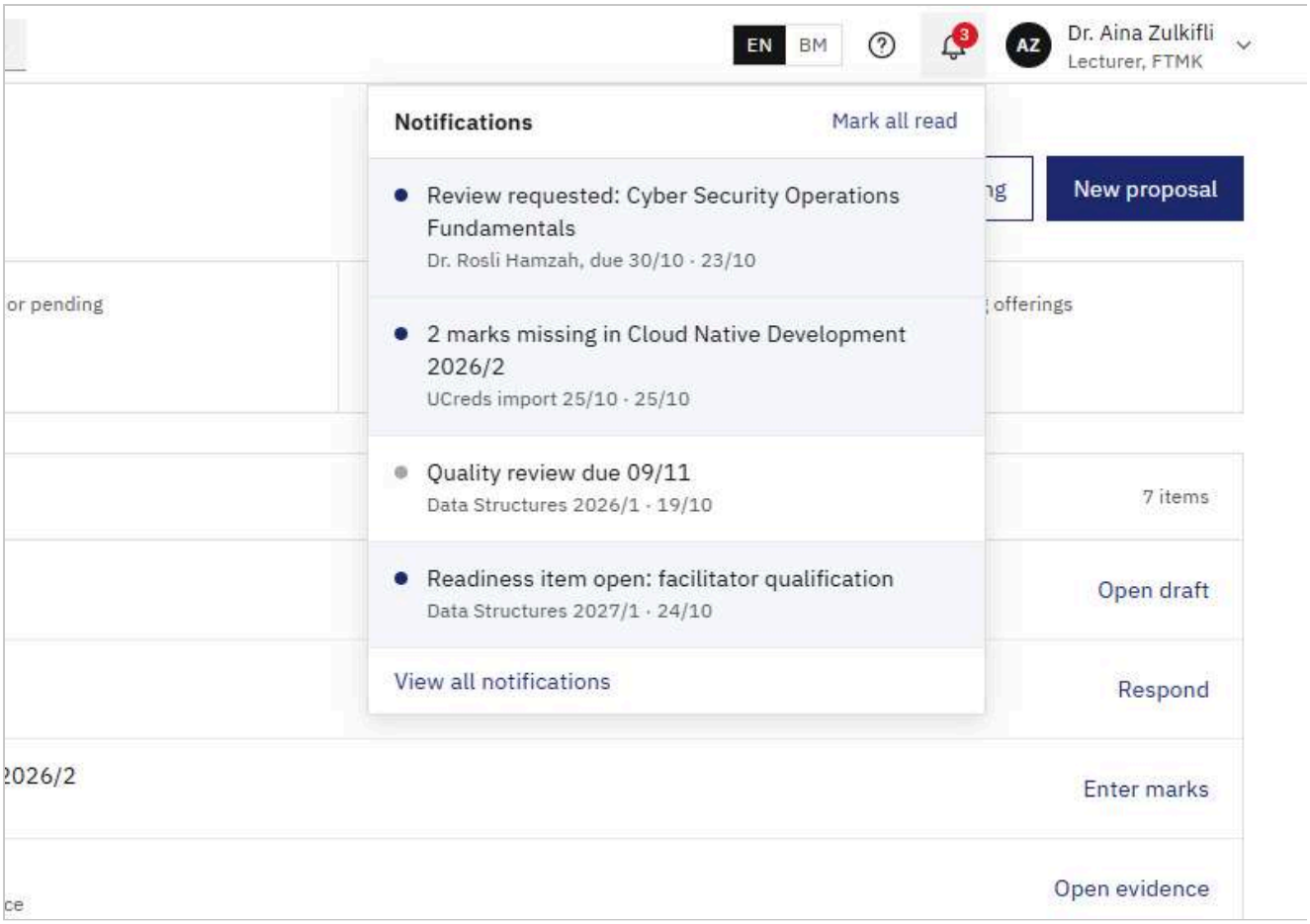


Figure 132: *Lecturer and reviewer: Notifications menu.*

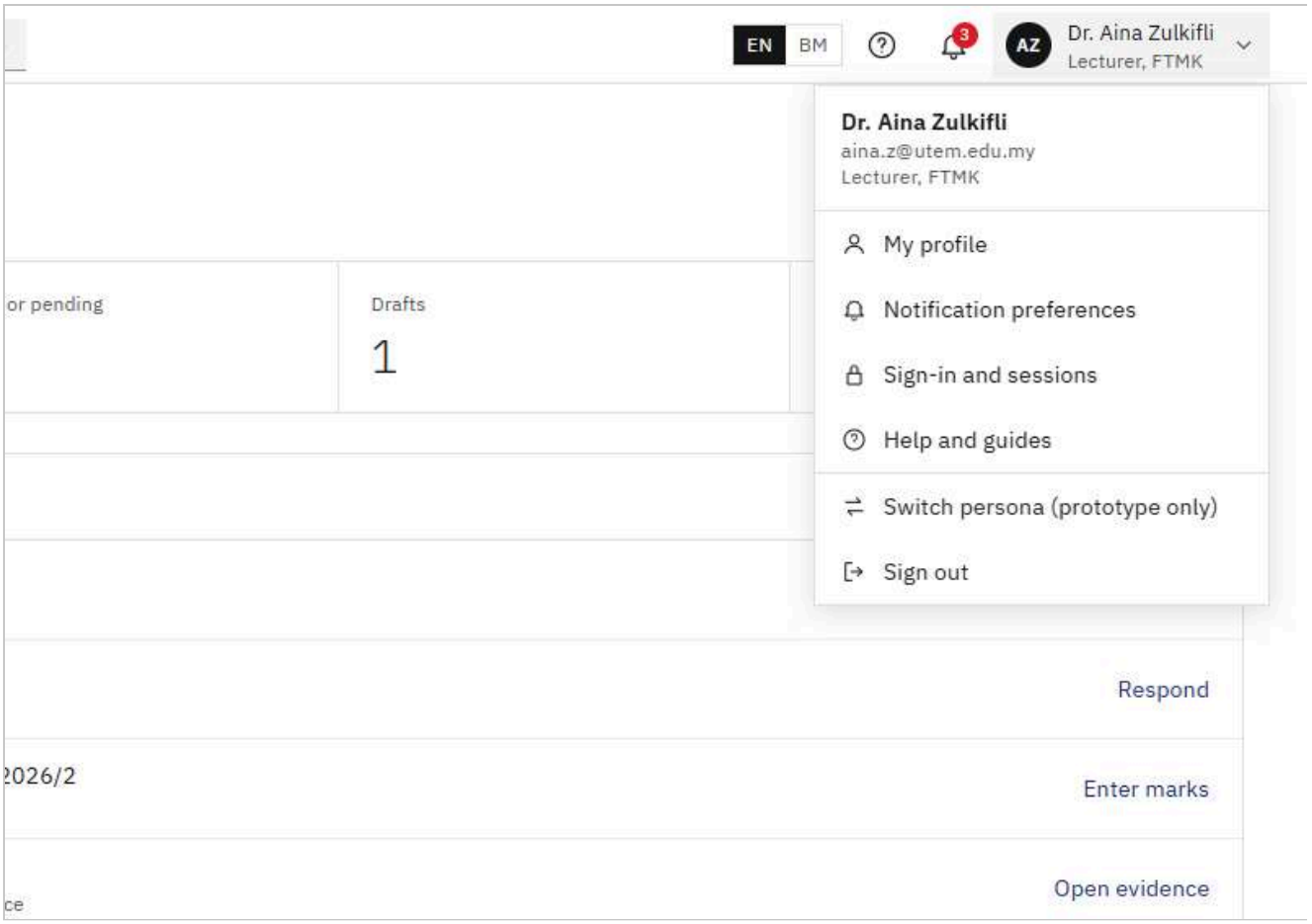


Figure 133: *Lecturer and reviewer: Profile menu.*

B.5 Industry expert

EXT-01 My review invitations

`#/ext/home`

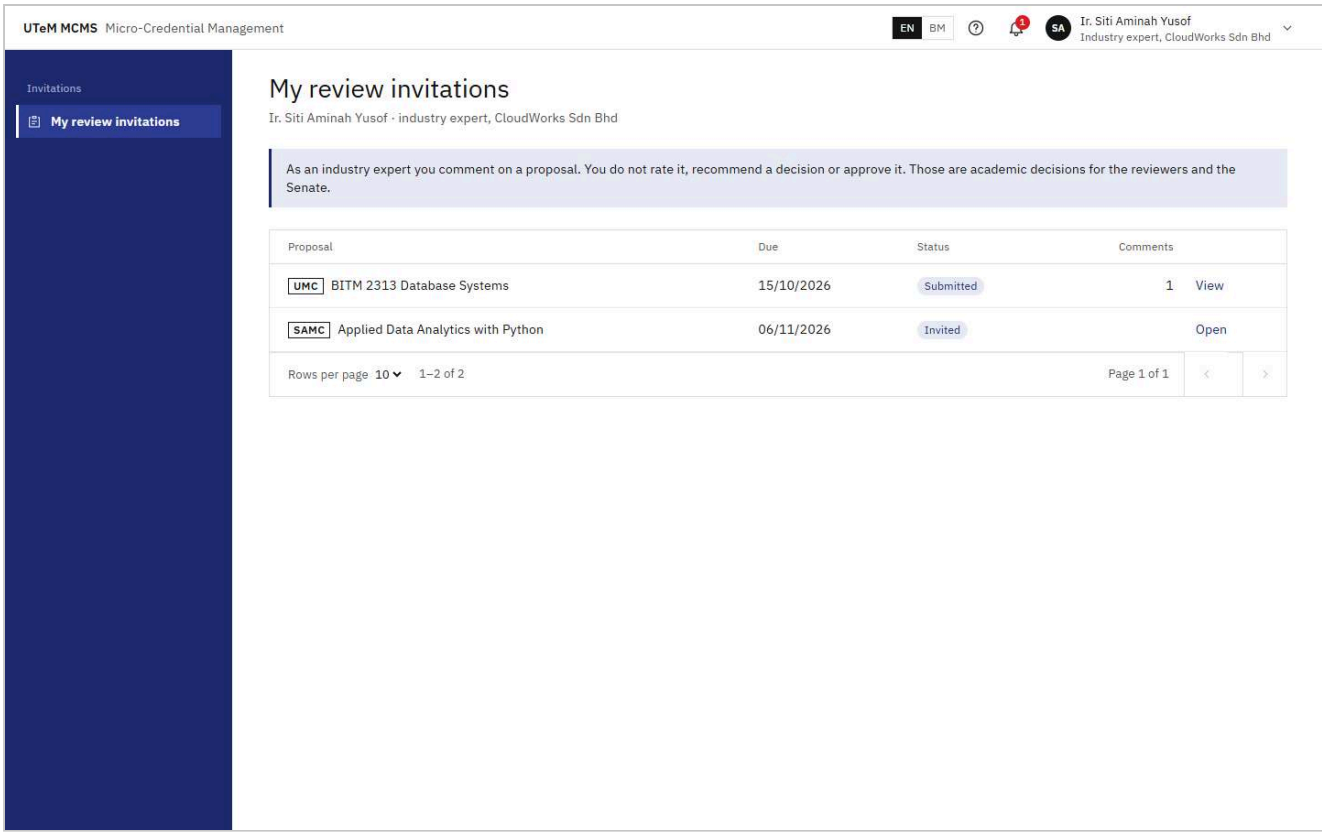


Figure 134: *EXT-01 My review invitations.*

EXT-02 Comment on a proposal

`#/ext/review/RV-033`

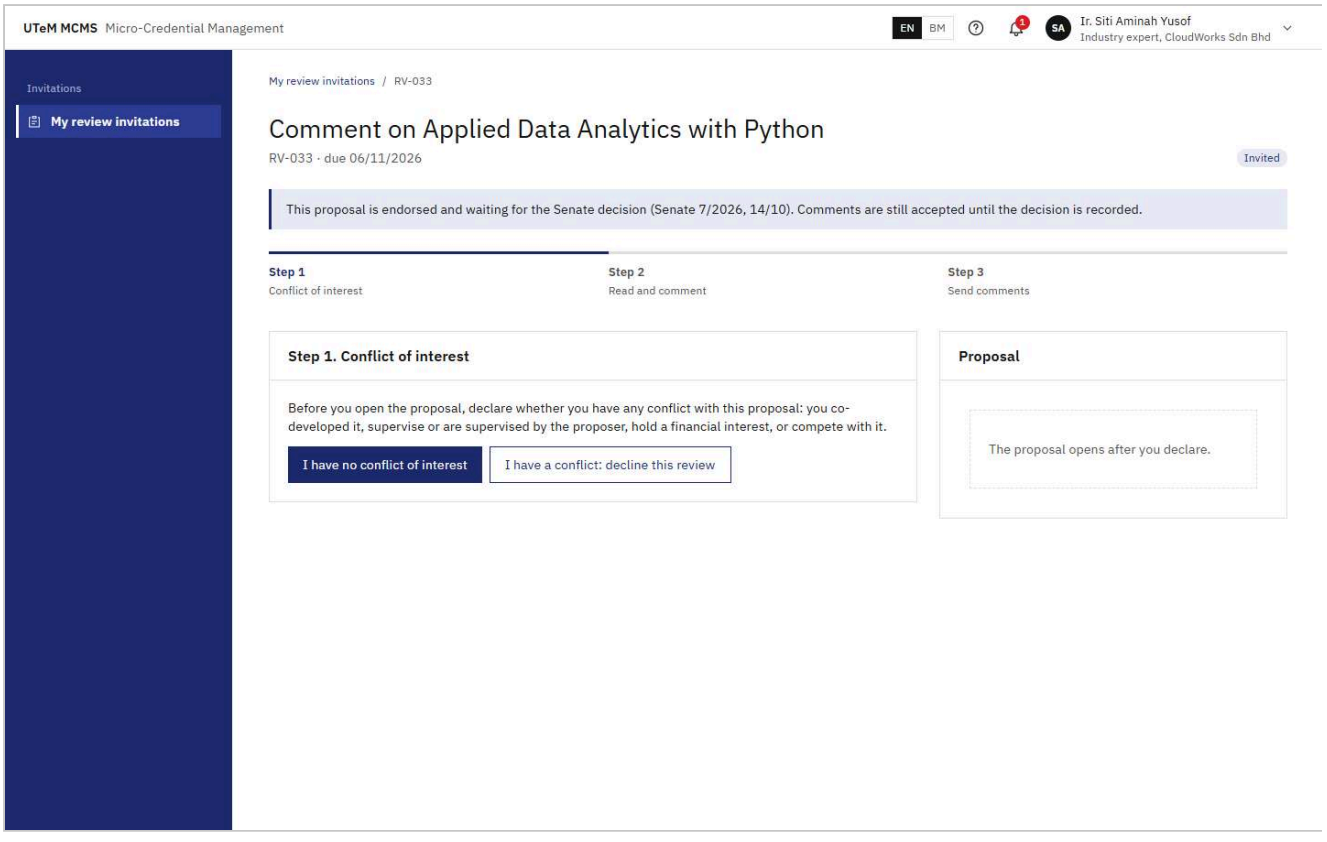


Figure 135: *EXT-02 Comment on a proposal.*

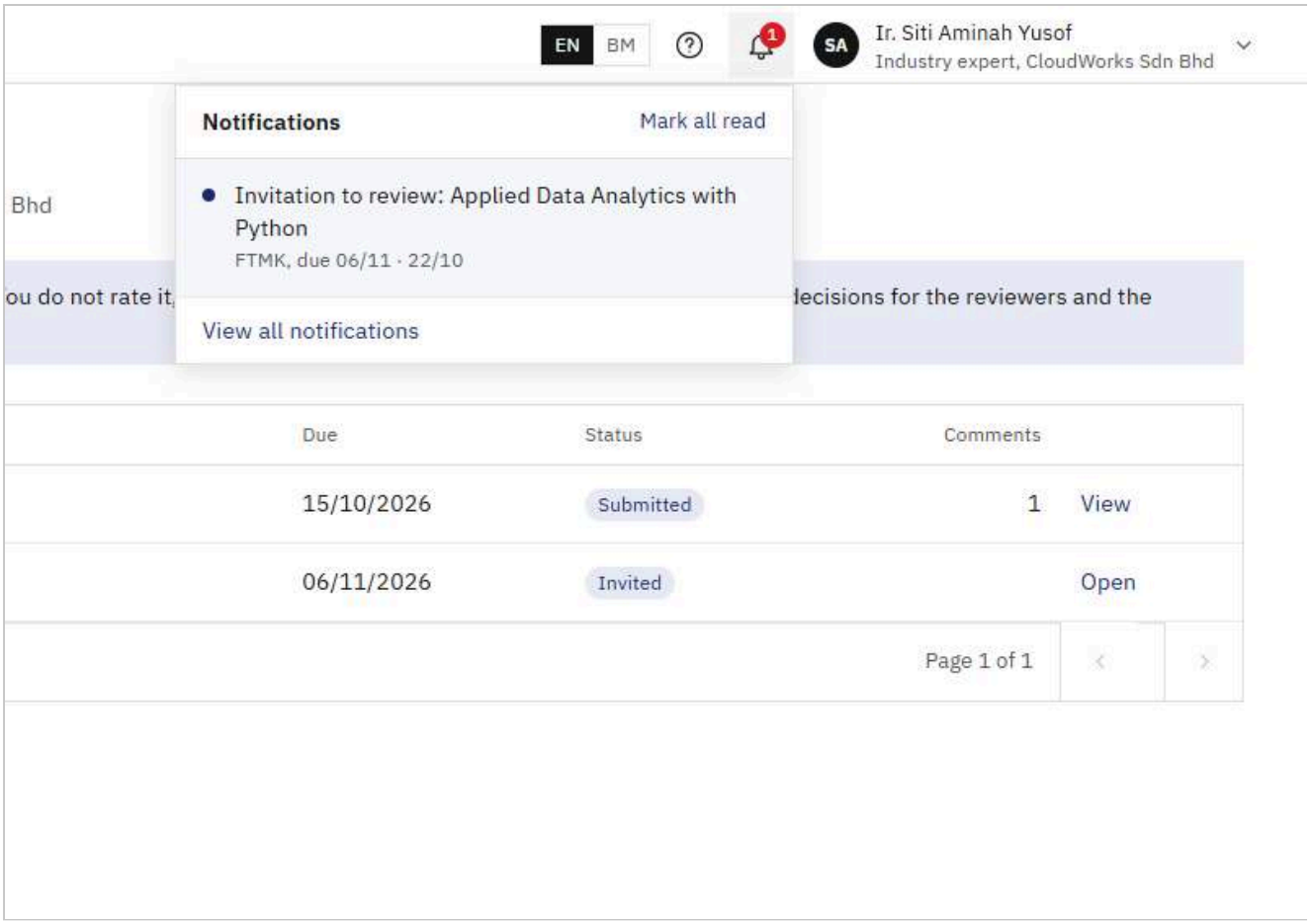


Figure 136: *Industry expert: Notifications menu.*

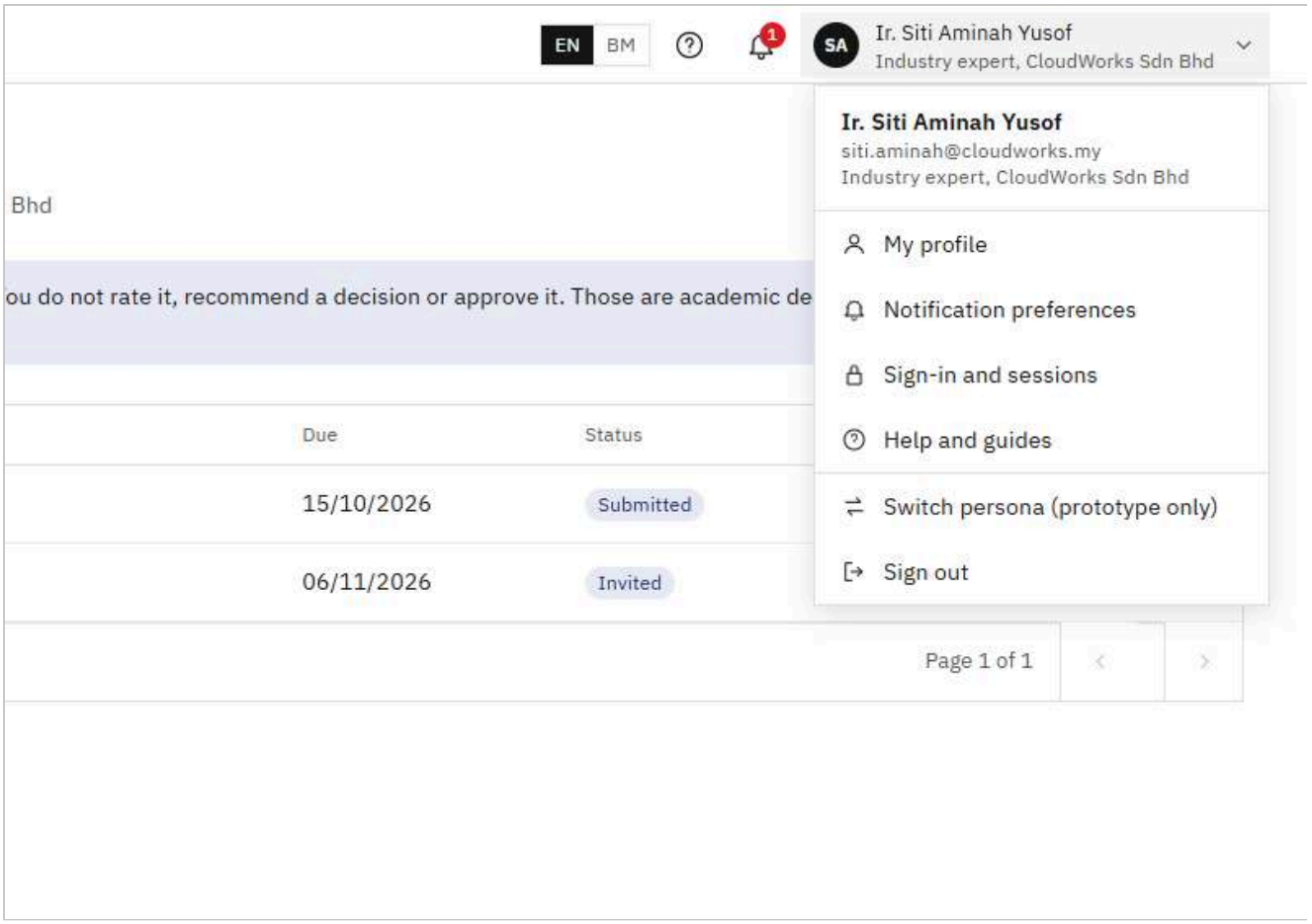


Figure 137: *Industry expert: Profile menu.*

B.6 Deputy dean

DD-01 Task queue

`#/dd/queue`

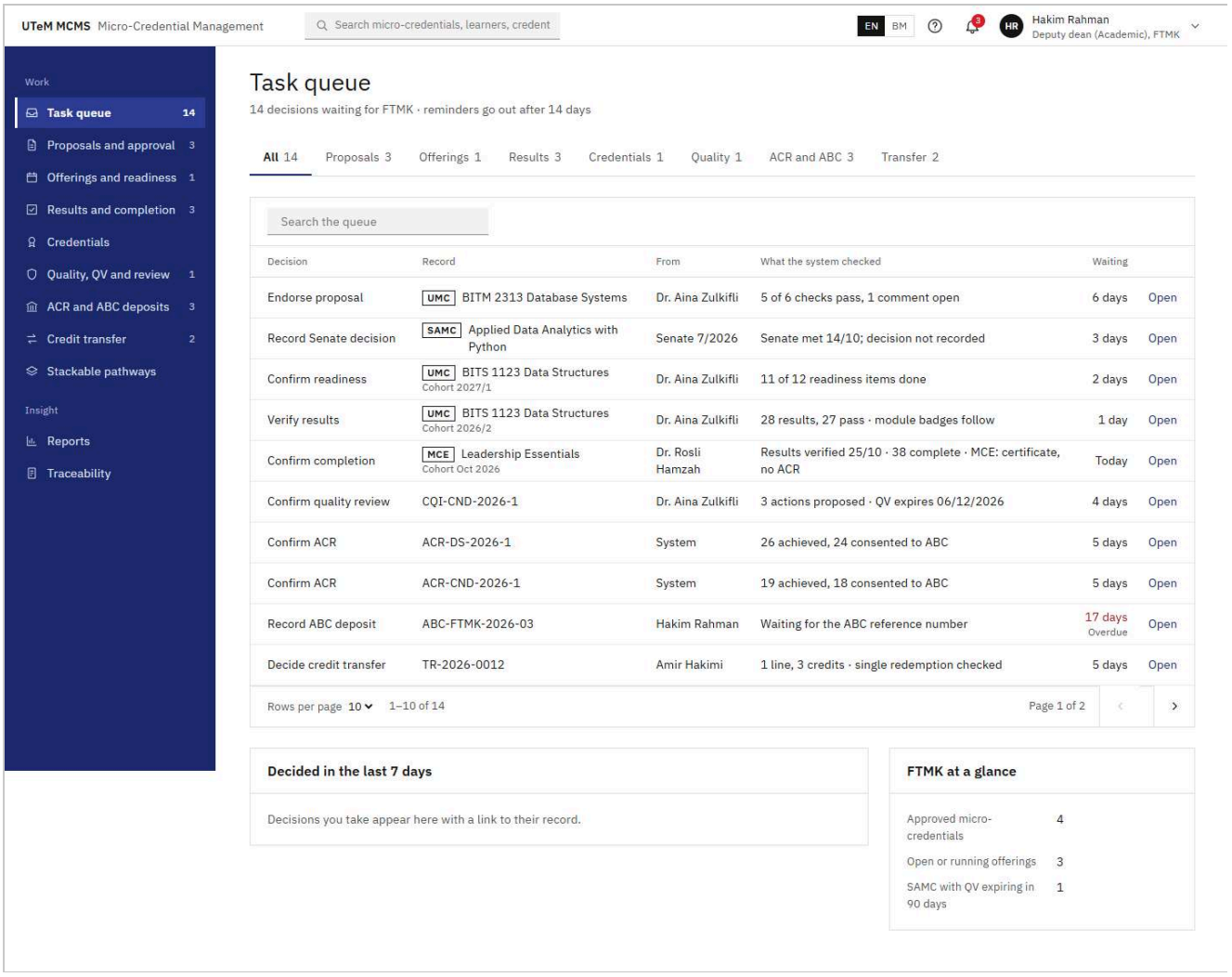


Figure 138: *DD-01 Task queue.*

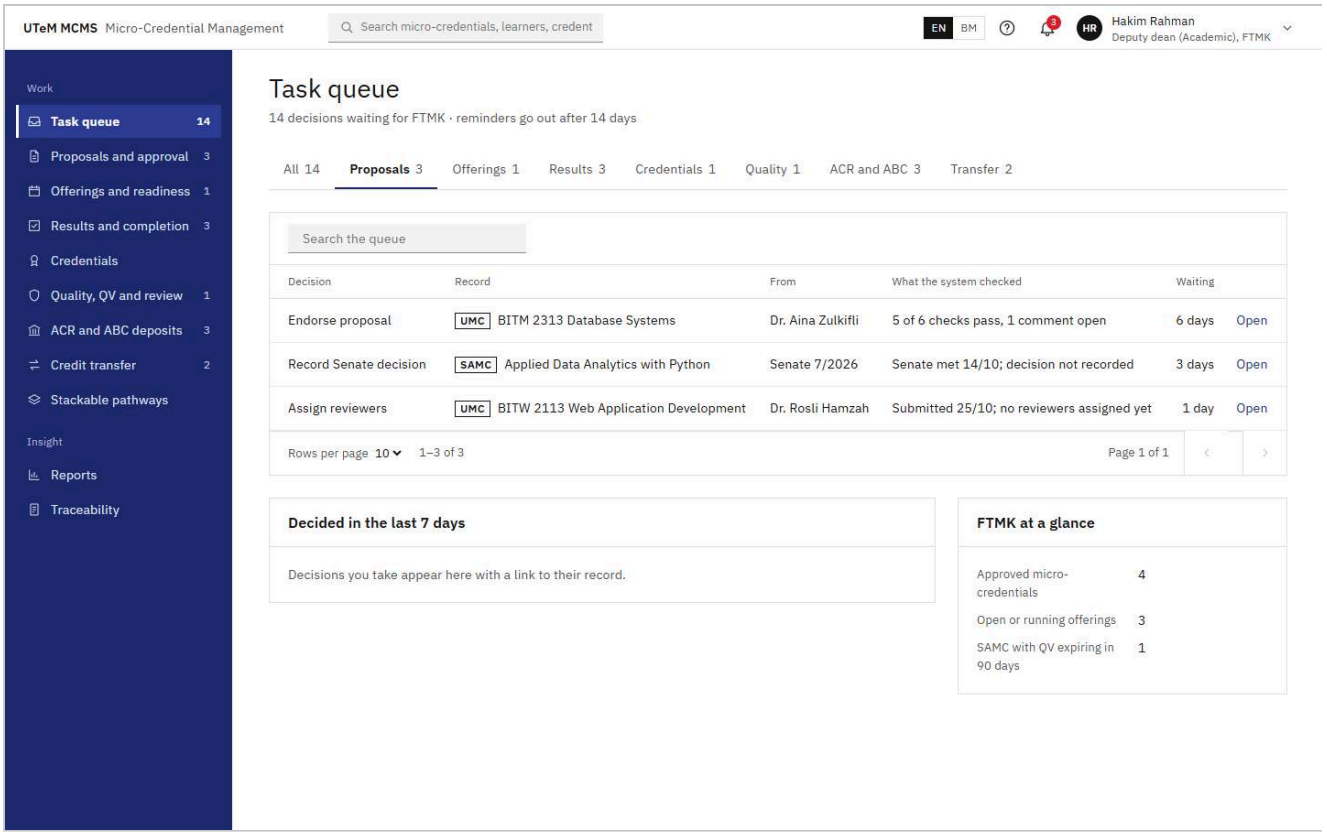


Figure 139: *DD-01 Task queue: Tab: Proposals.*

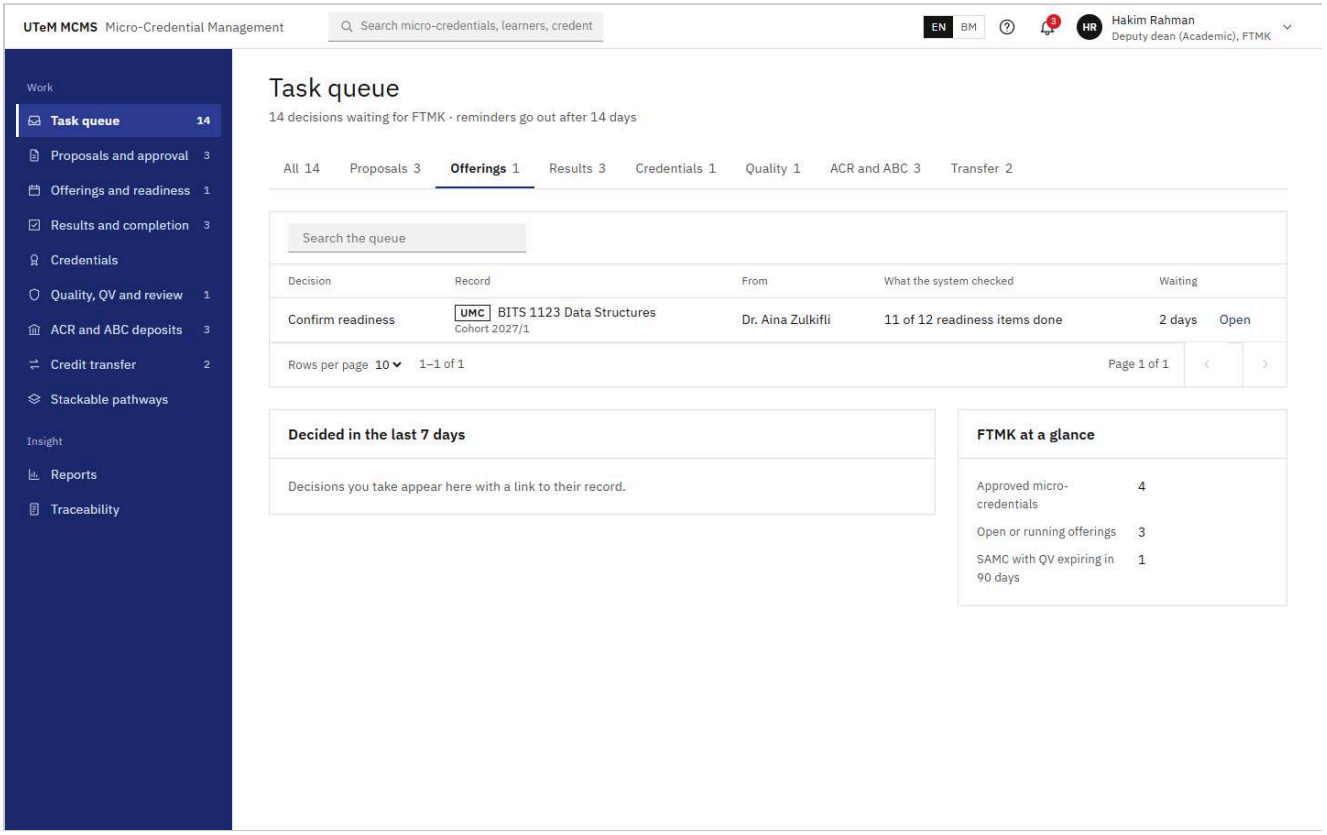


Figure 140: *DD-01 Task queue: Tab: Offerings.*

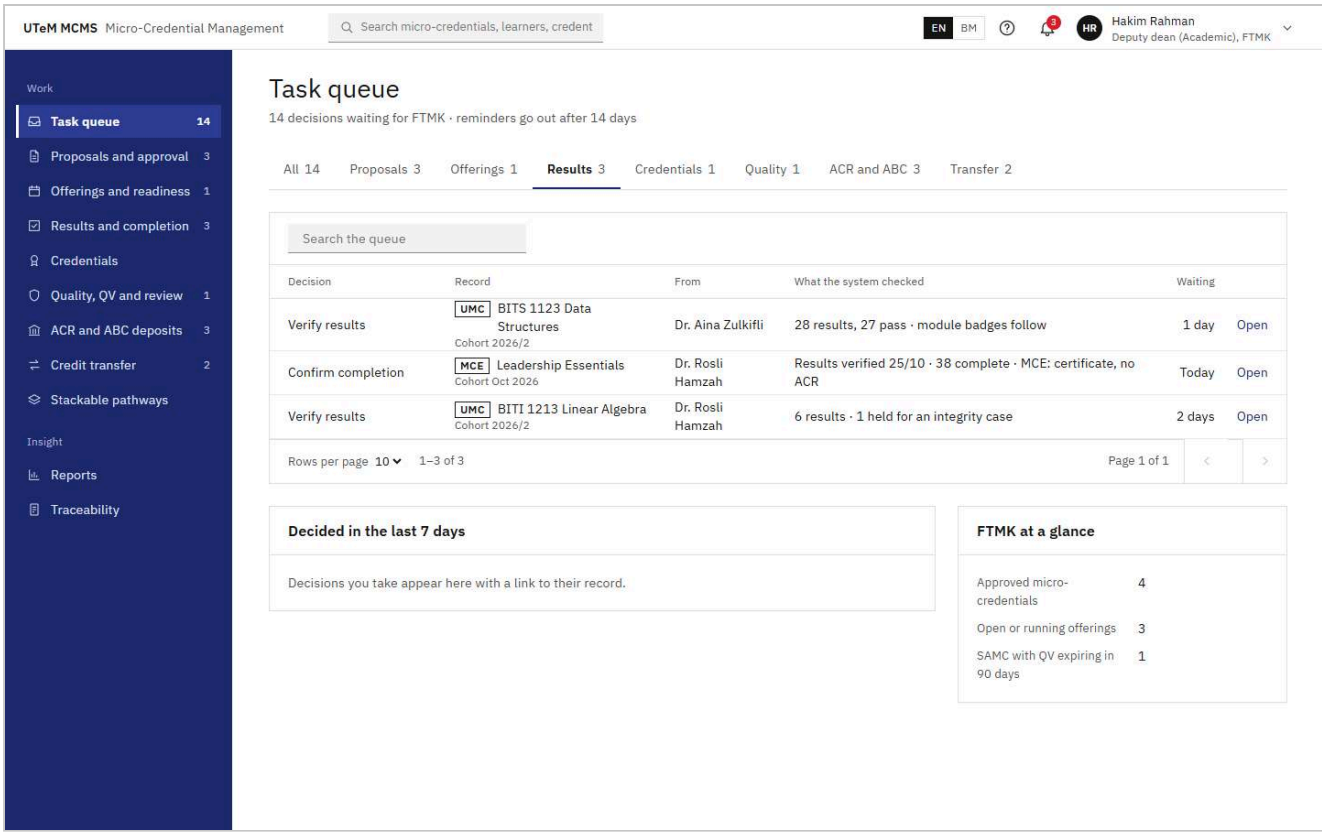


Figure 141: *DD-01 Task queue: Tab: Results.*

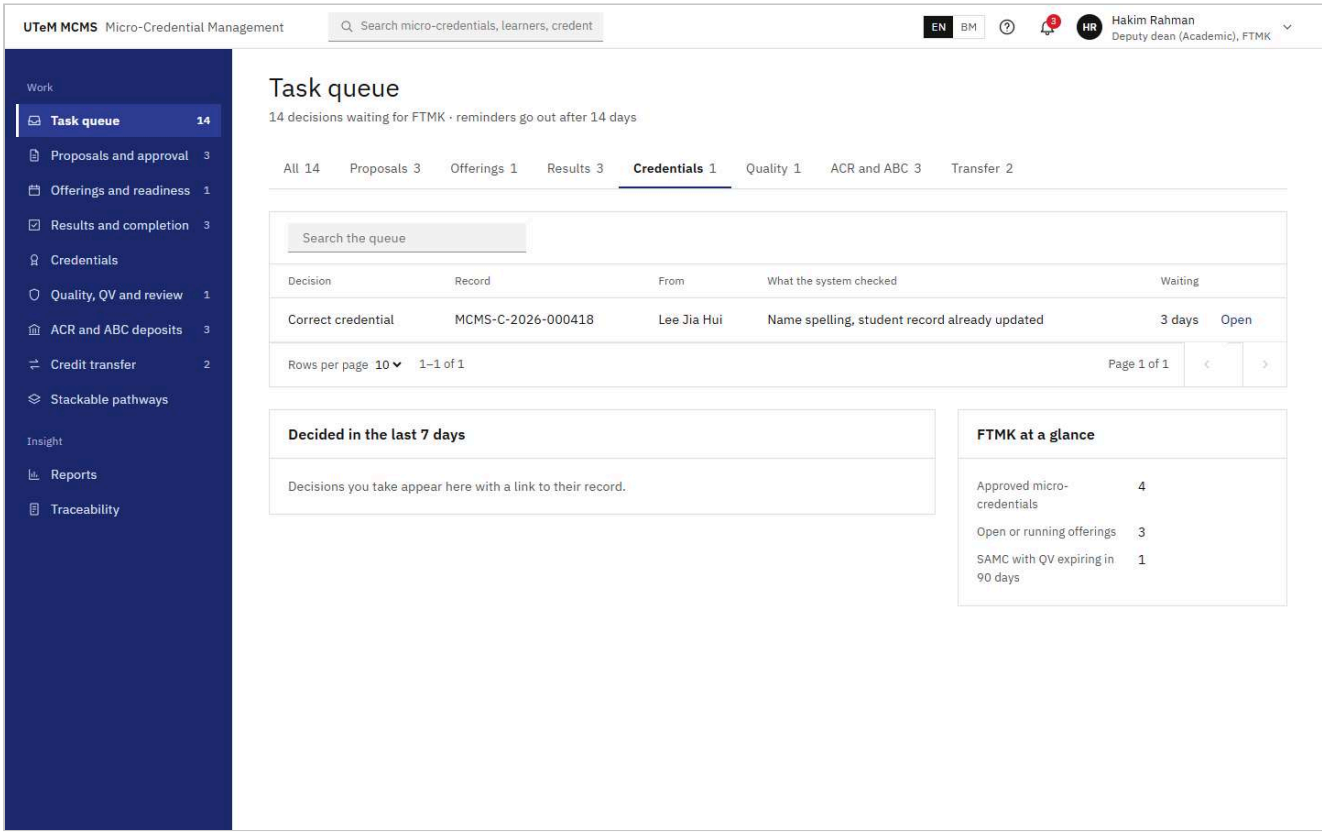


Figure 142: *DD-01 Task queue: Tab: Credentials.*

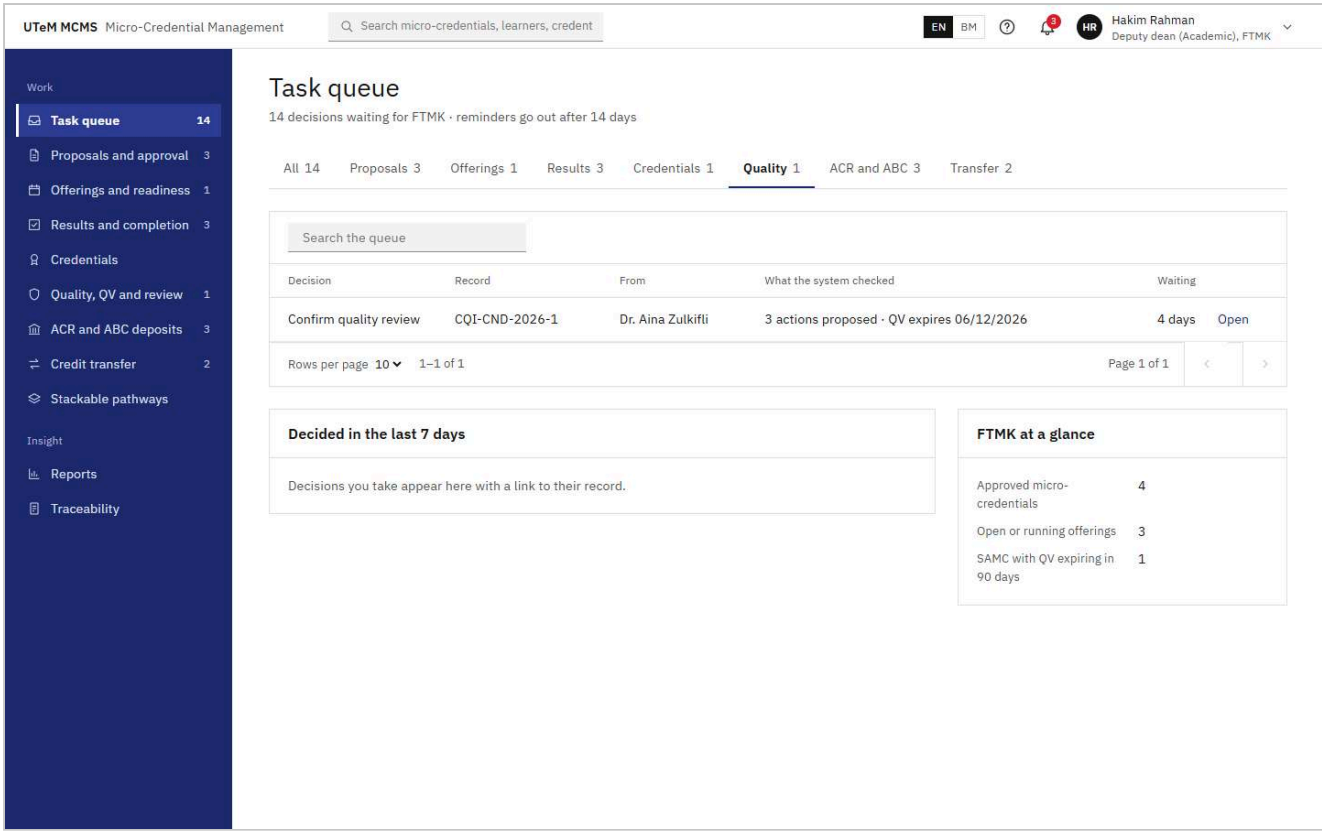


Figure 143: *DD-01 Task queue: Tab: Quality.*

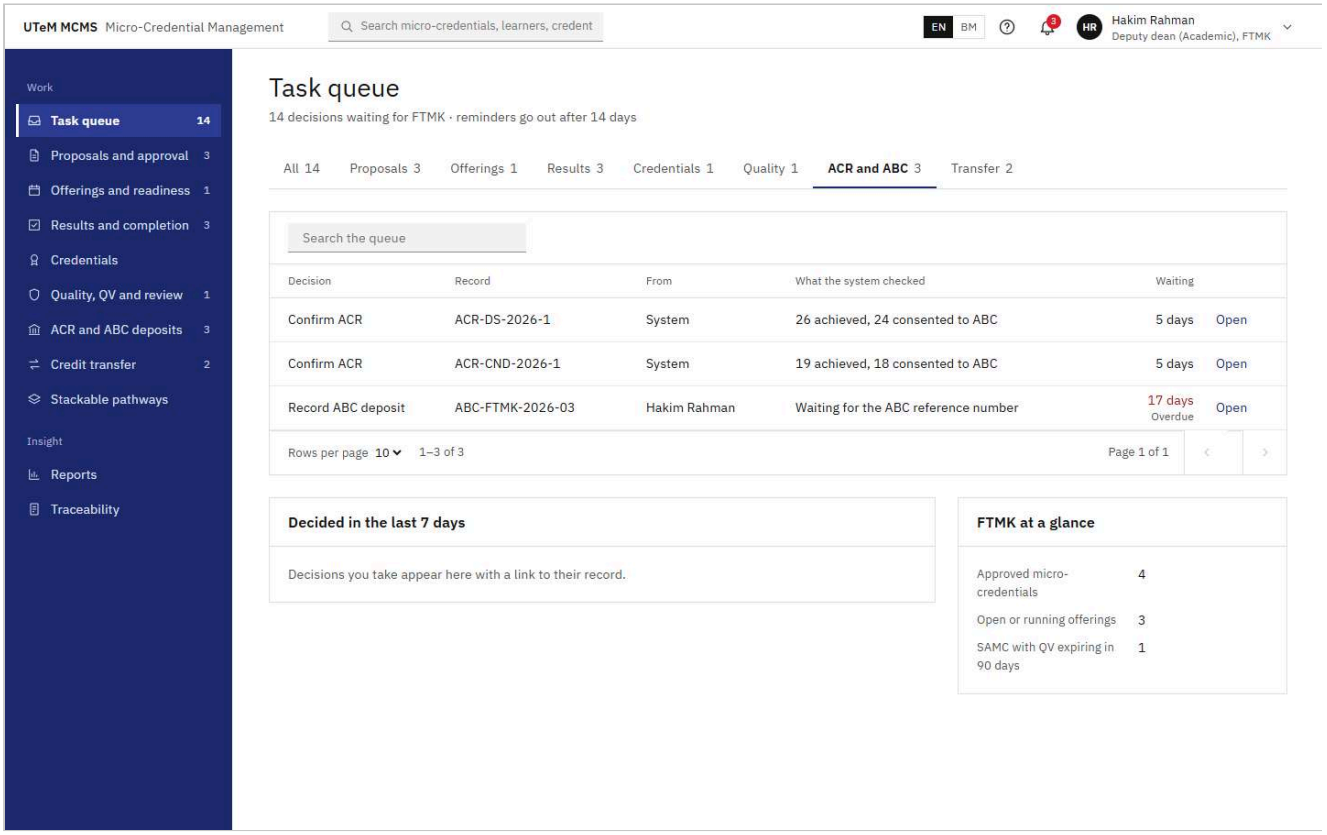


Figure 144: *DD-01 Task queue: Tab: ACR and ABC.*

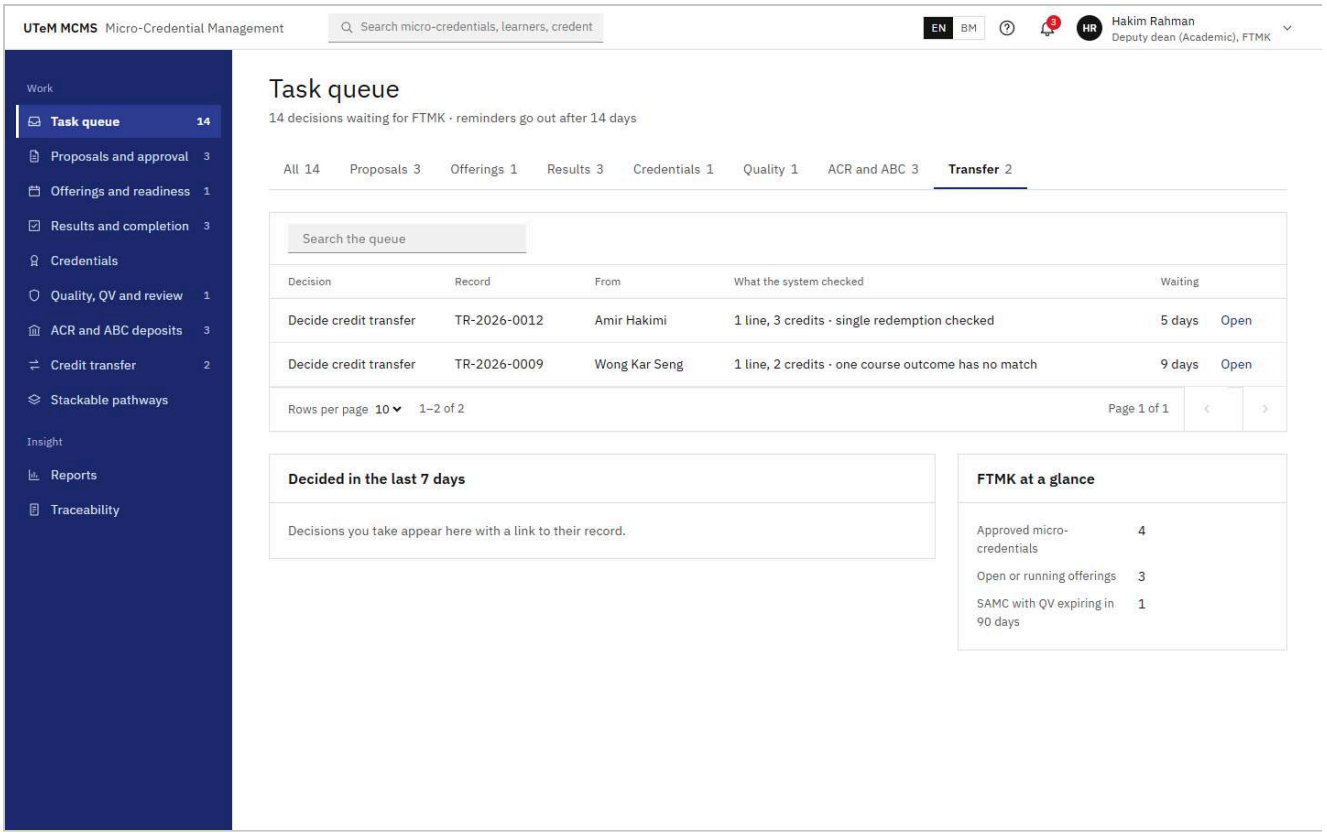


Figure 145: *DD-01 Task queue: Tab: Transfer.*

DD-02 Proposals and approval

`#/dd/proposals`

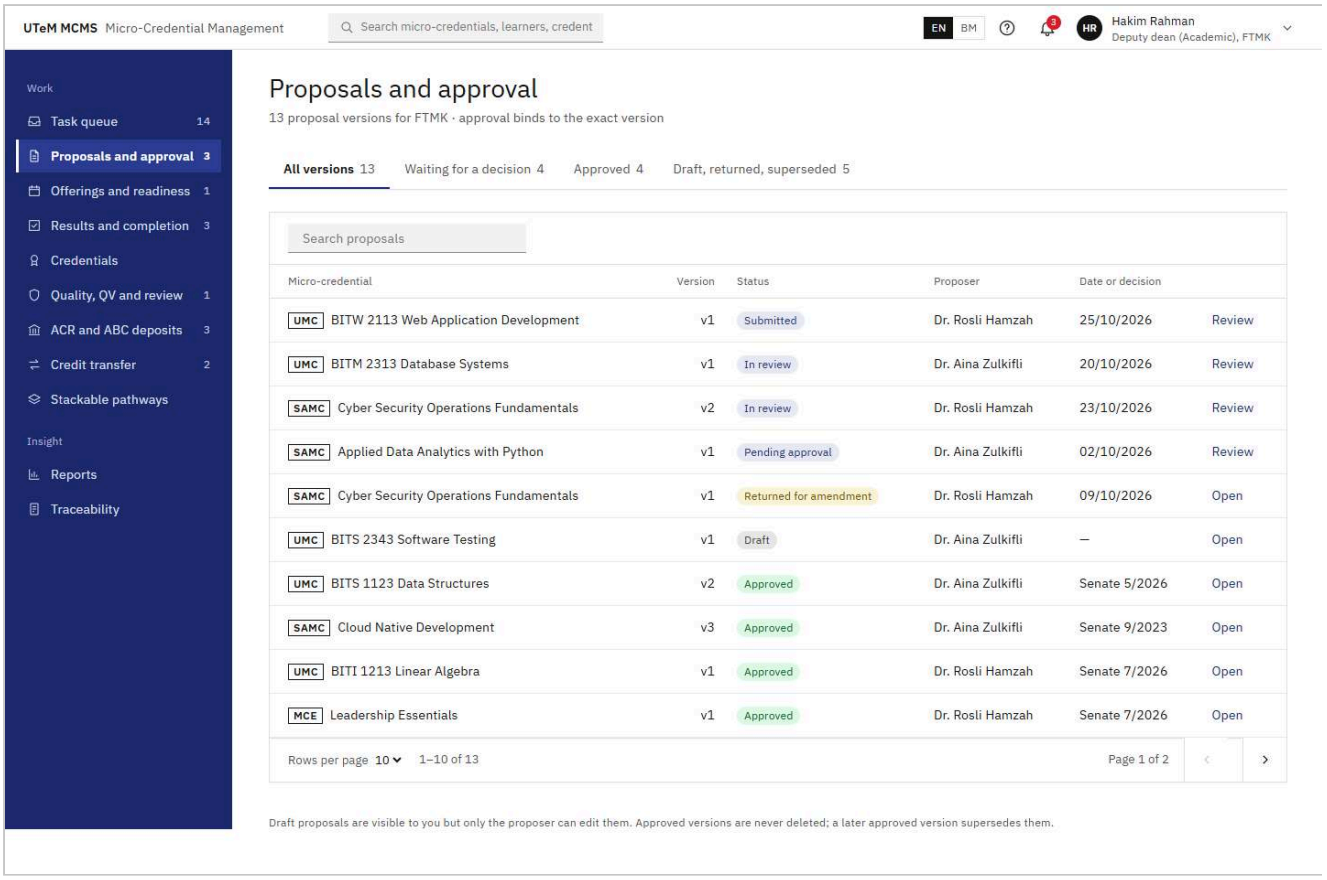


Figure 146: *DD-02 Proposals and approval.*

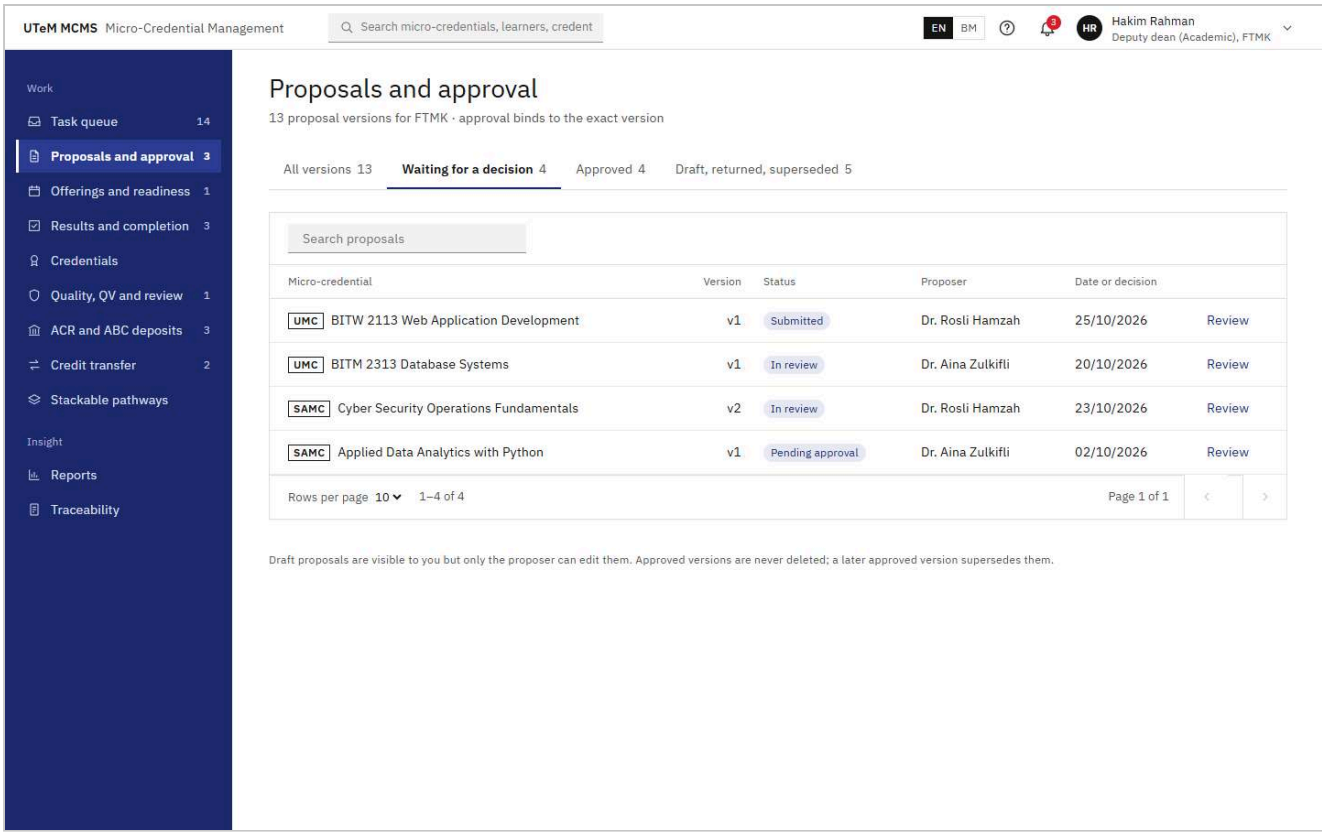


Figure 147: *DD-02 Proposals and approval: Tab: Waiting for a decision.*

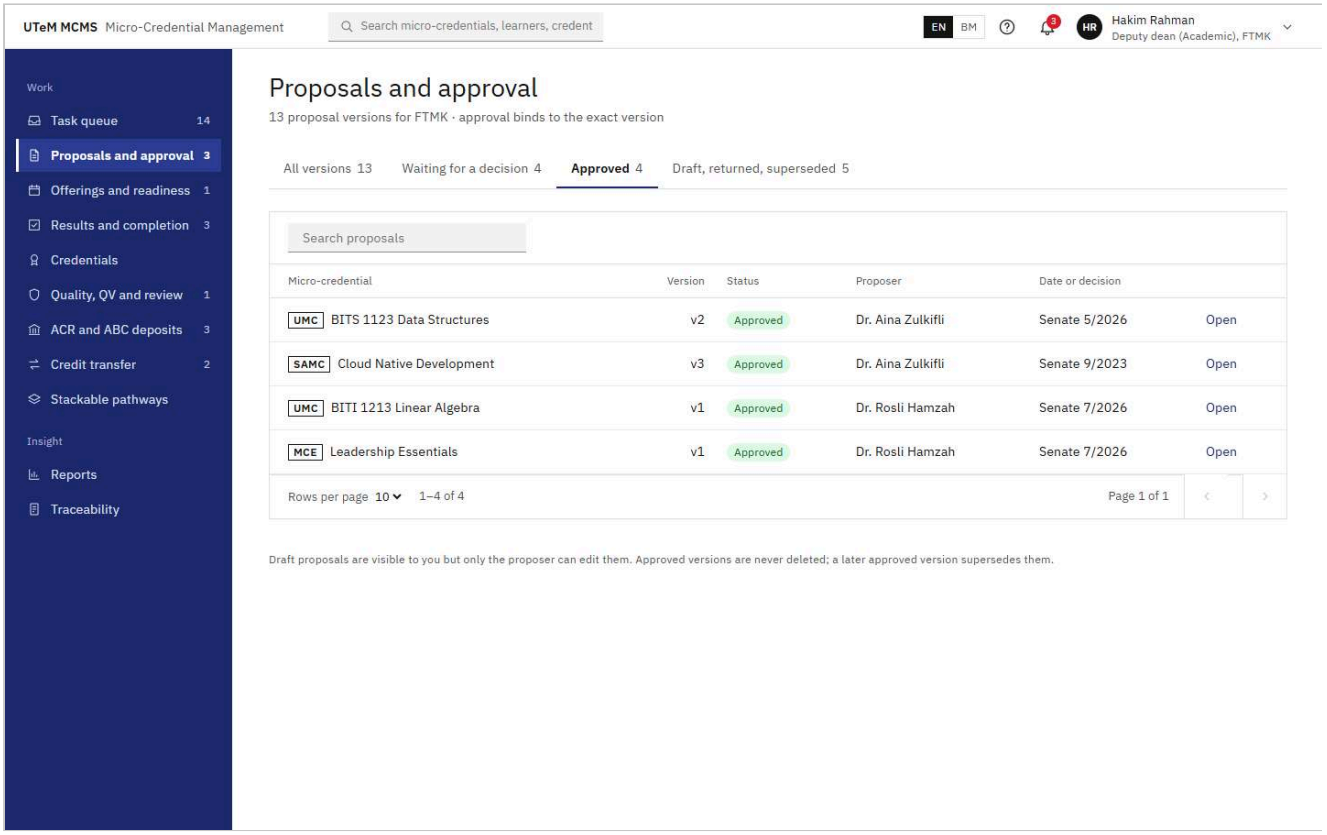


Figure 148: *DD-02 Proposals and approval: Tab: Approved.*

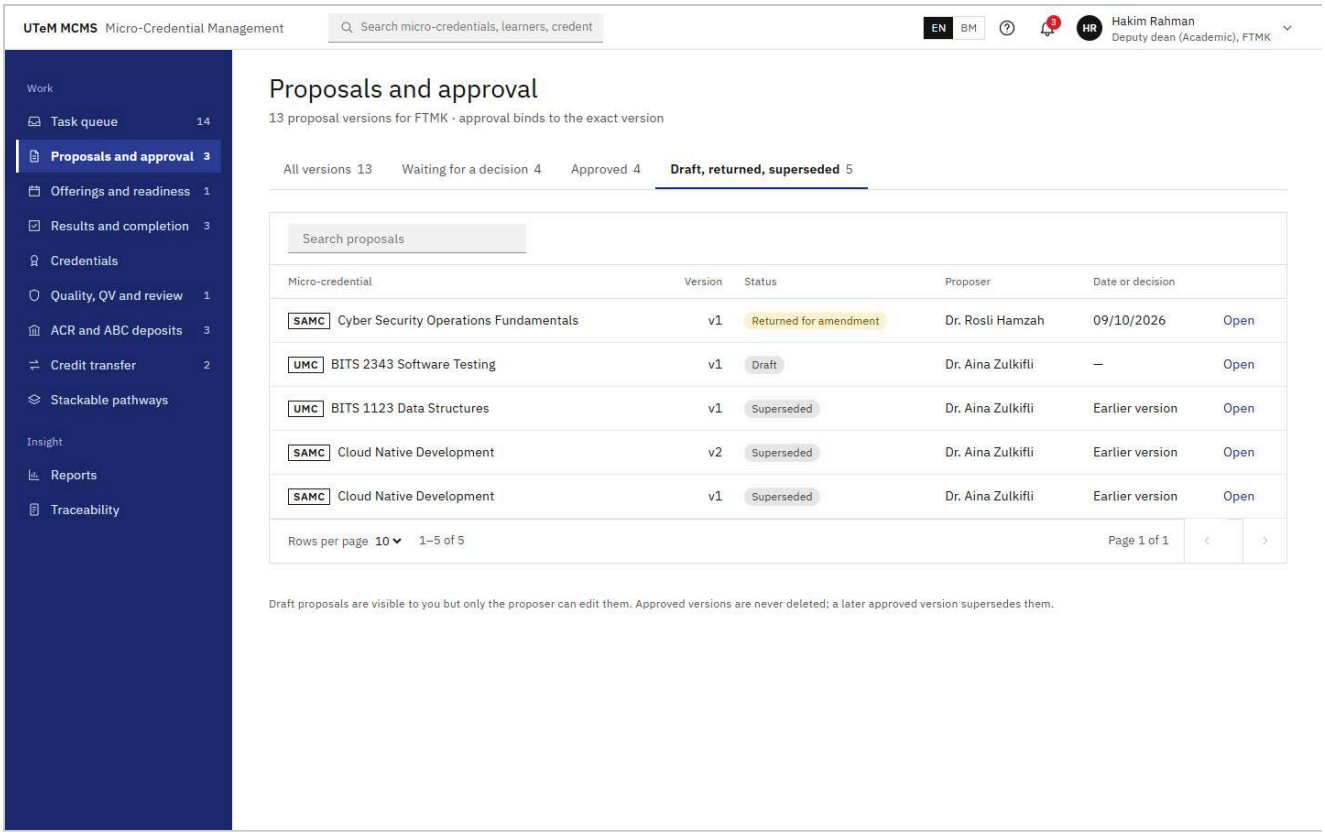


Figure 149: *DD-02 Proposals and approval: Tab: Draft, returned, superseded.*

DD-03 Proposal review

`#/dd/proposal/DB`

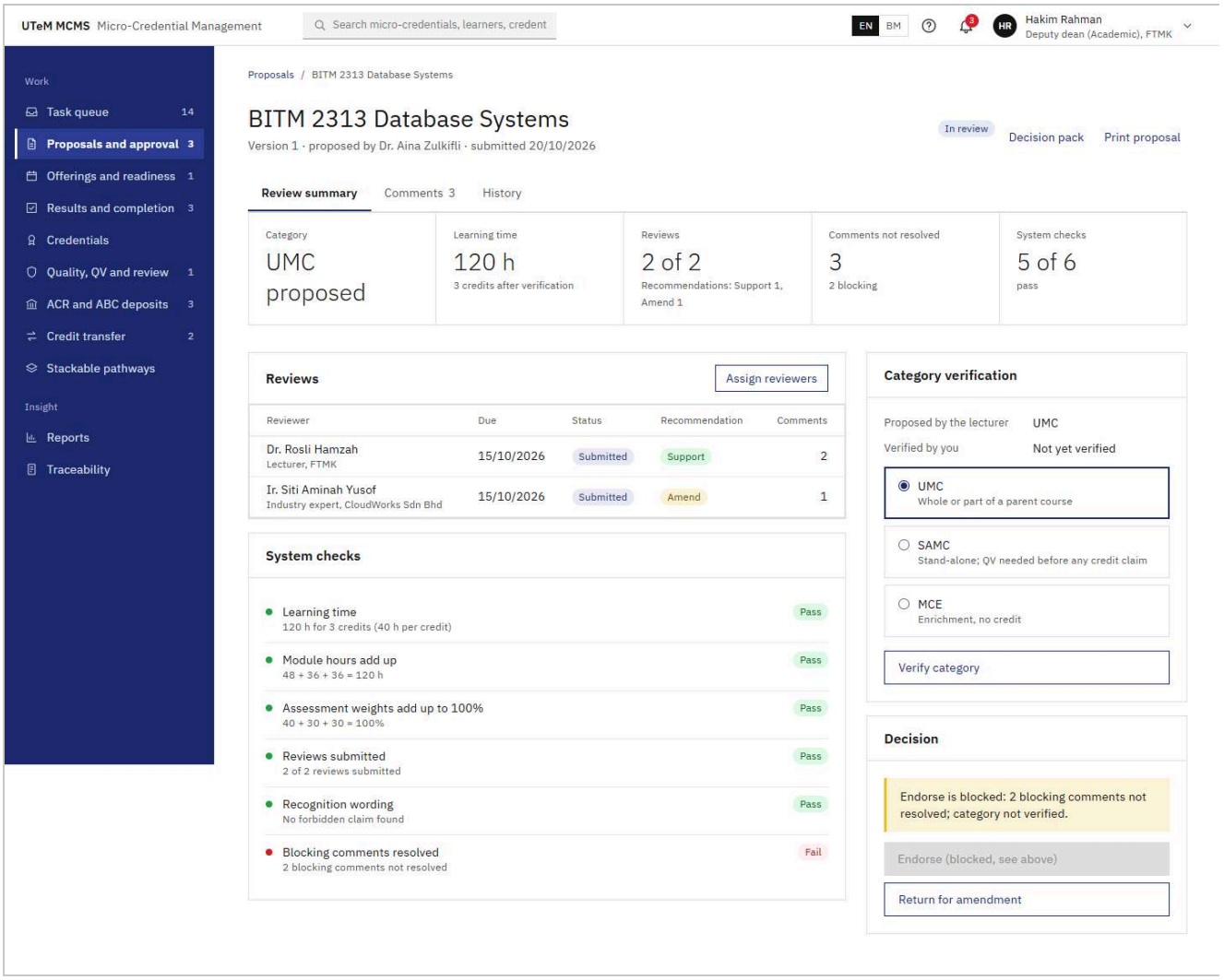


Figure 150: *DD-03 Proposal review.*

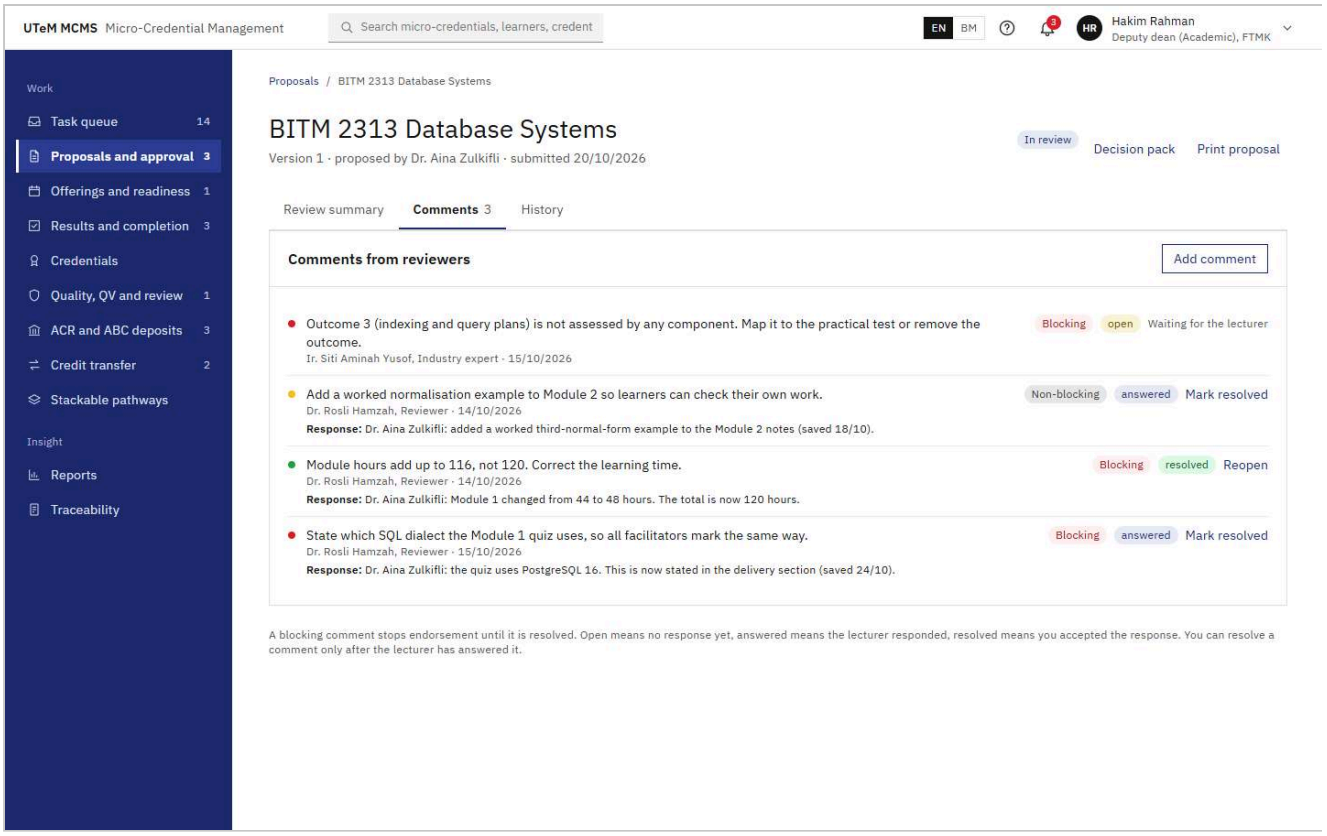


Figure 151: *DD-03 Proposal review: Tab: Comments.*

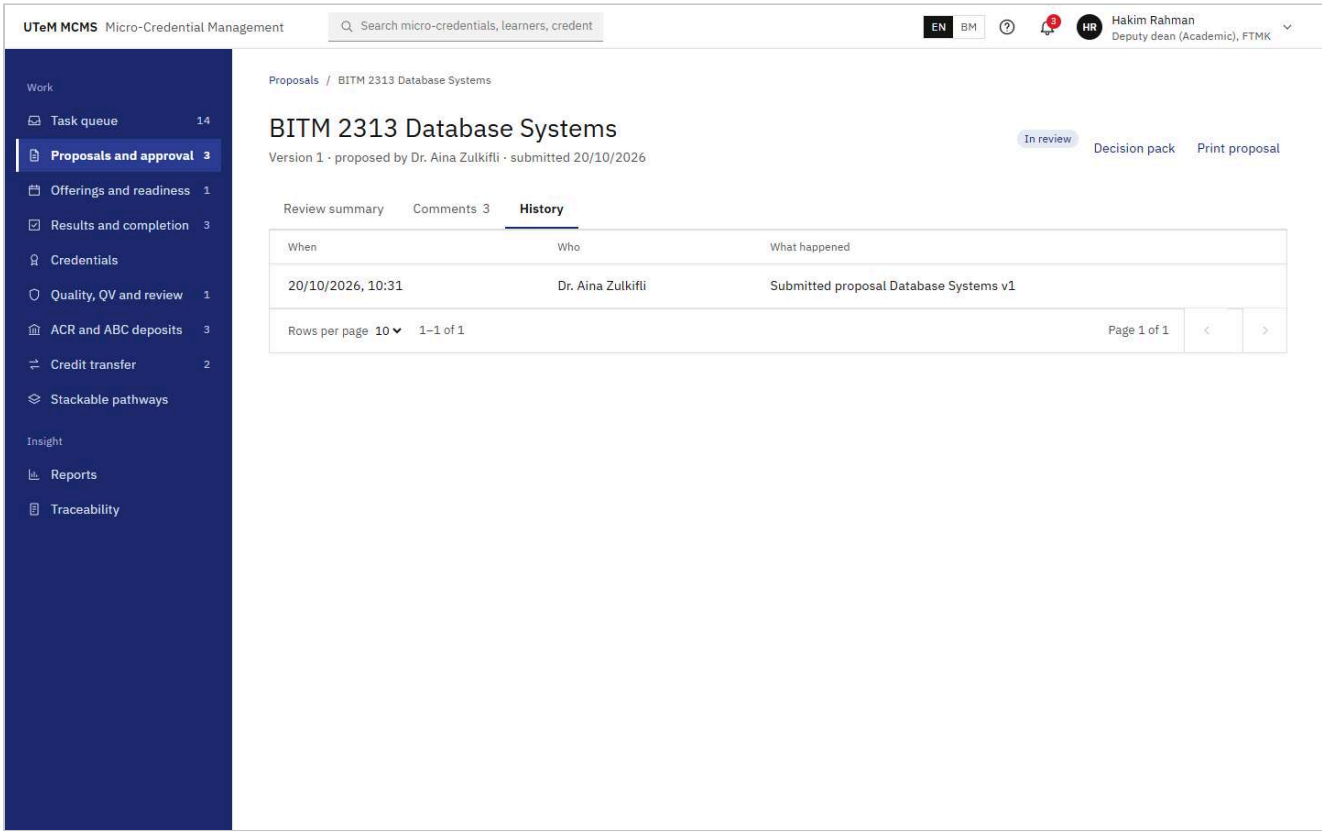


Figure 152: *DD-03 Proposal review: Tab: History.*

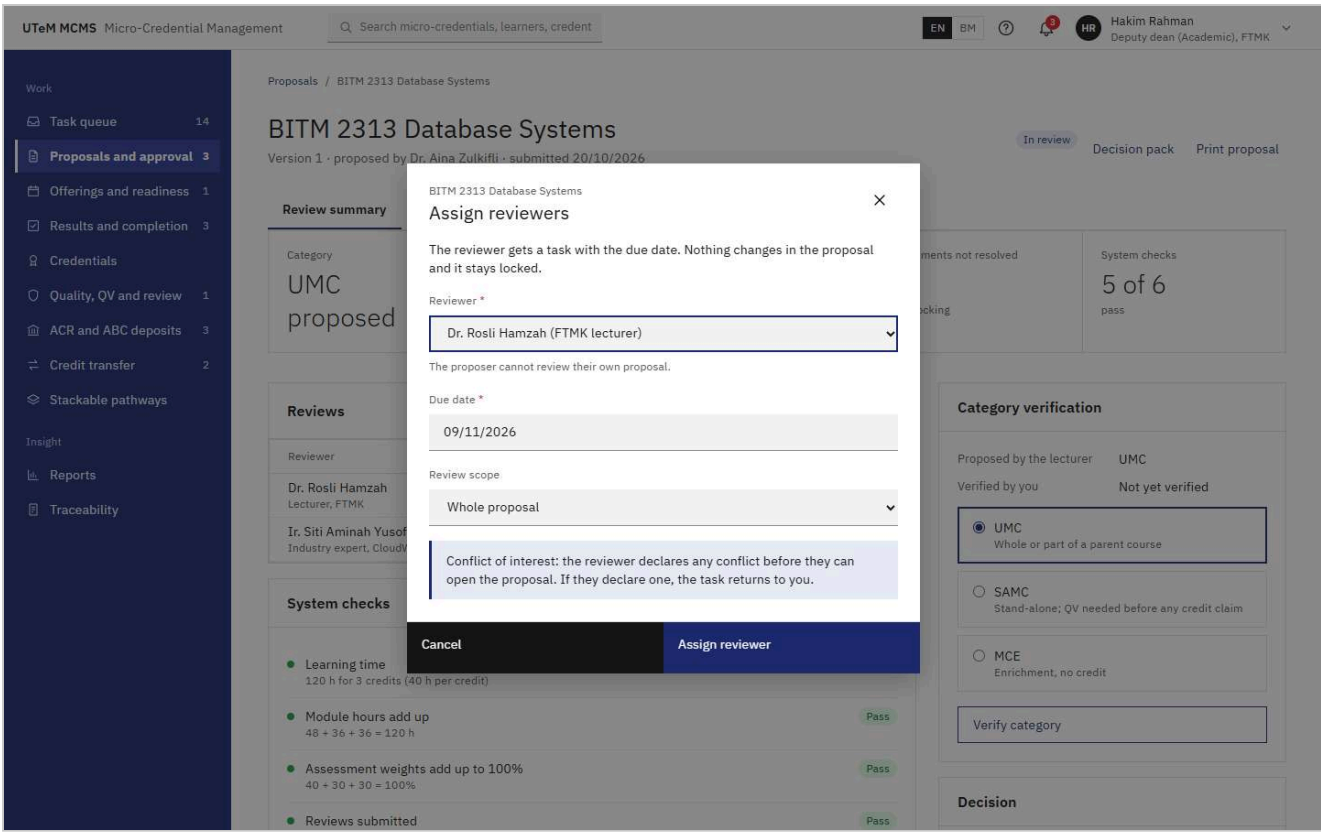


Figure 153: *DD-03 Dialog: Assign reviewers.*

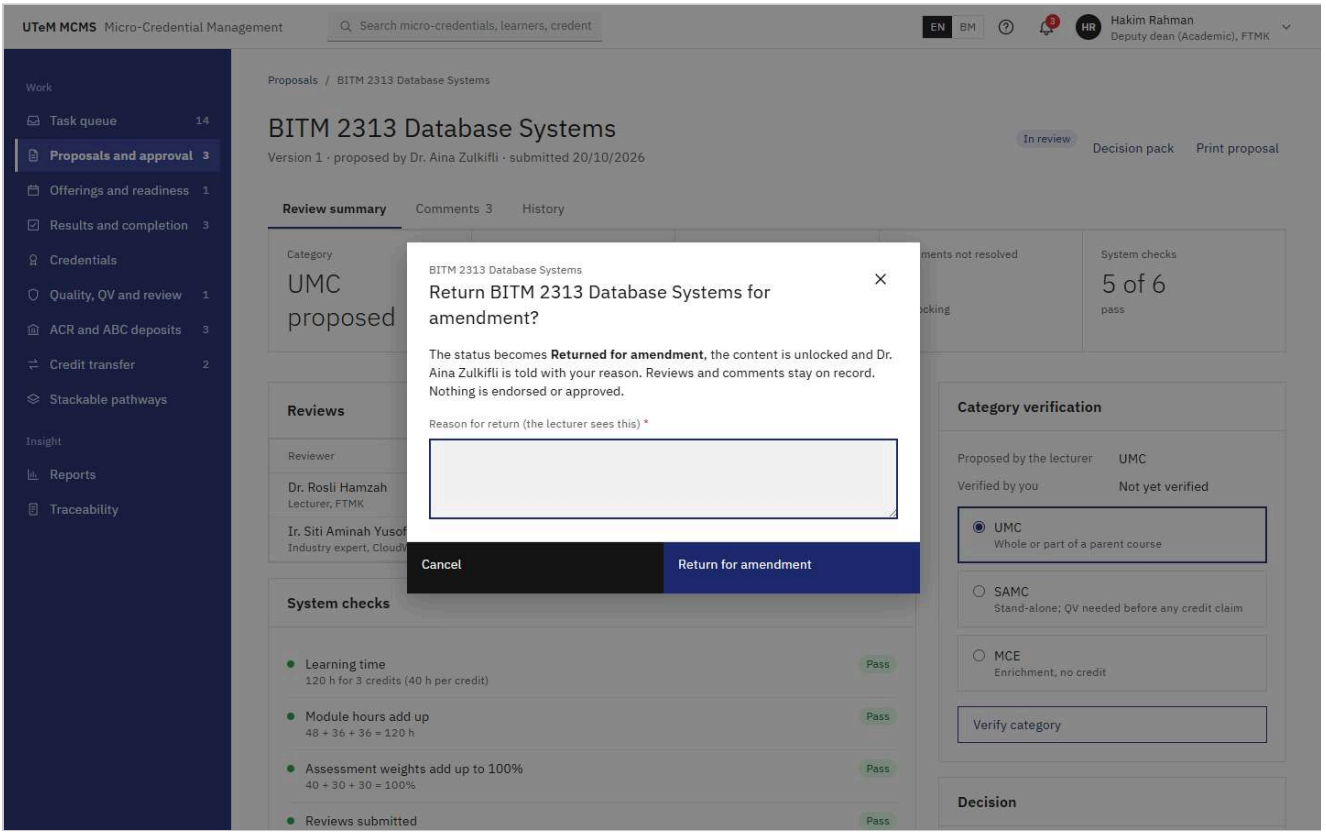


Figure 154: *DD-03 Confirmation: Return BITM 2313 Database Systems for amendment?*

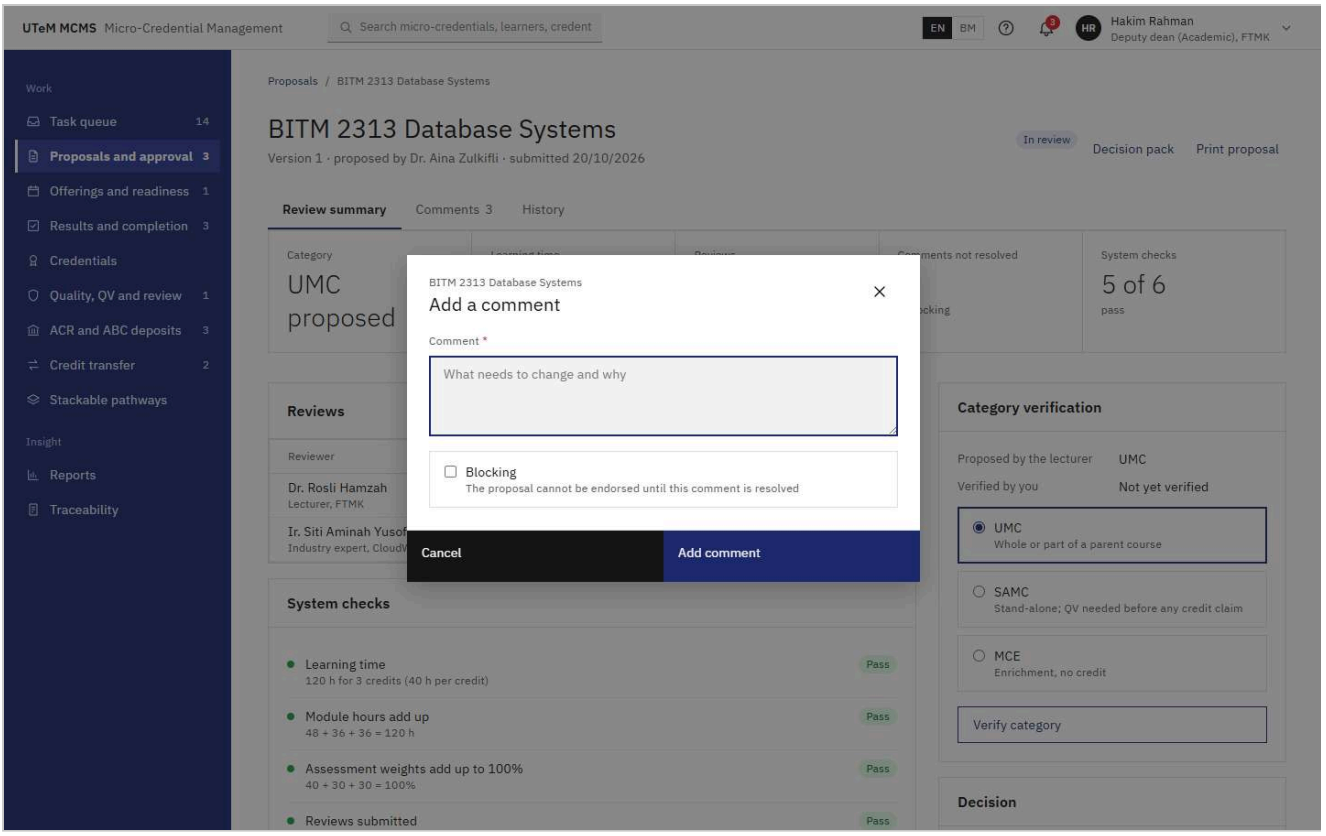


Figure 155: *DD-03 Dialog: Add a comment.*

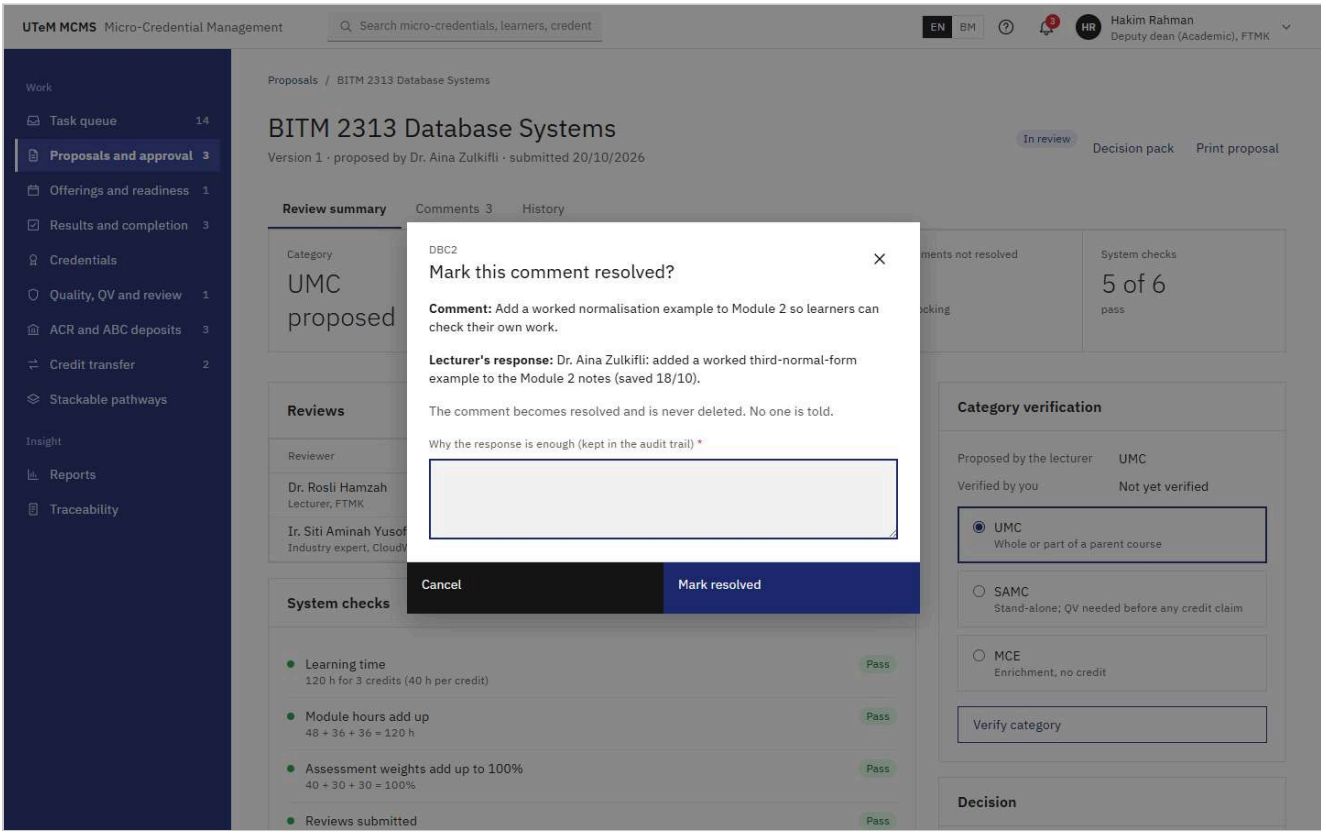


Figure 156: *DD-03 Confirmation: Mark this comment resolved?*

DD-04 Record Senate decision

`#/dd/senate/ADA`

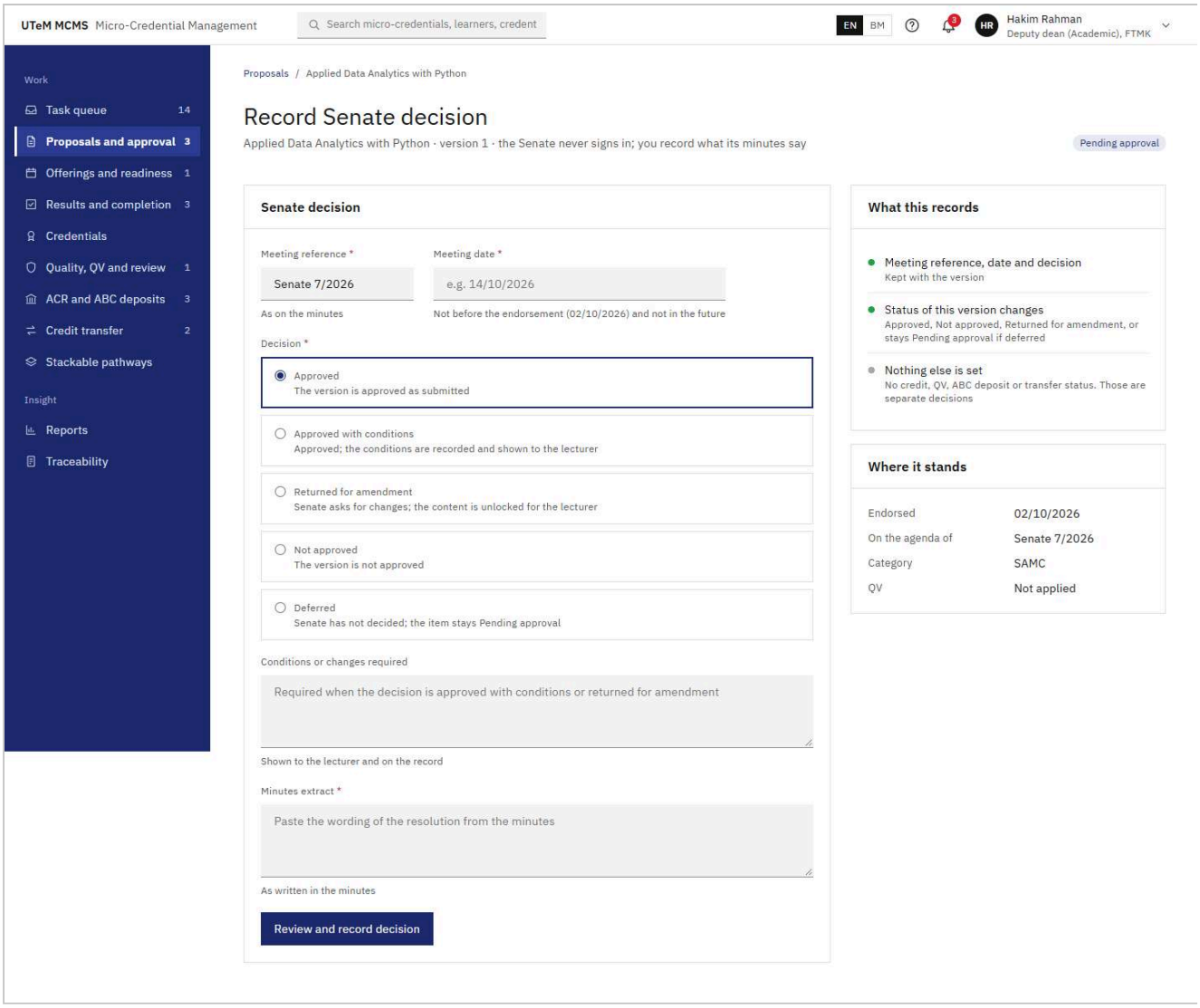


Figure 157: *DD-04 Record Senate decision.*

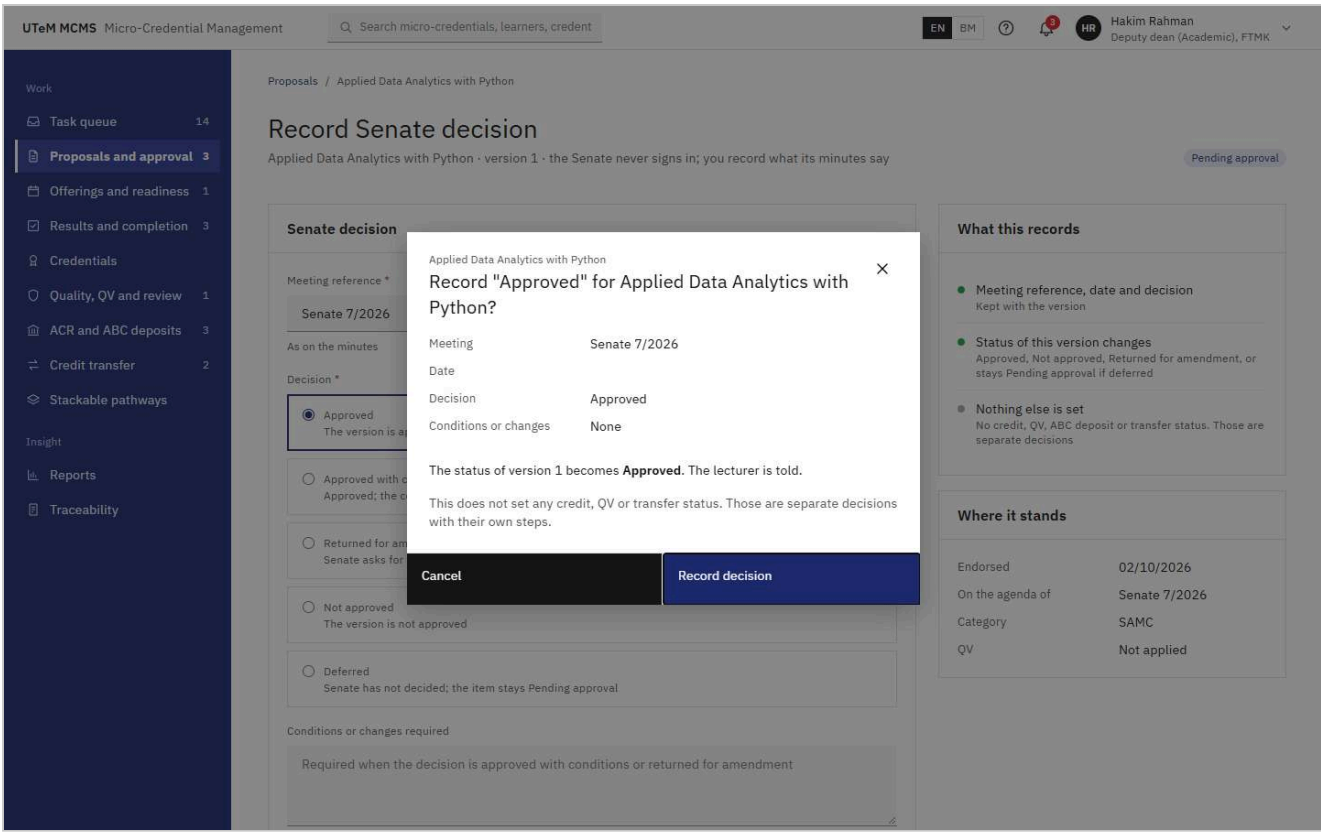


Figure 158: *DD-04 Confirmation: Record "Approved" for Applied Data Analytics with Python?*

DD-05 Decision pack

`#/dd/pack`

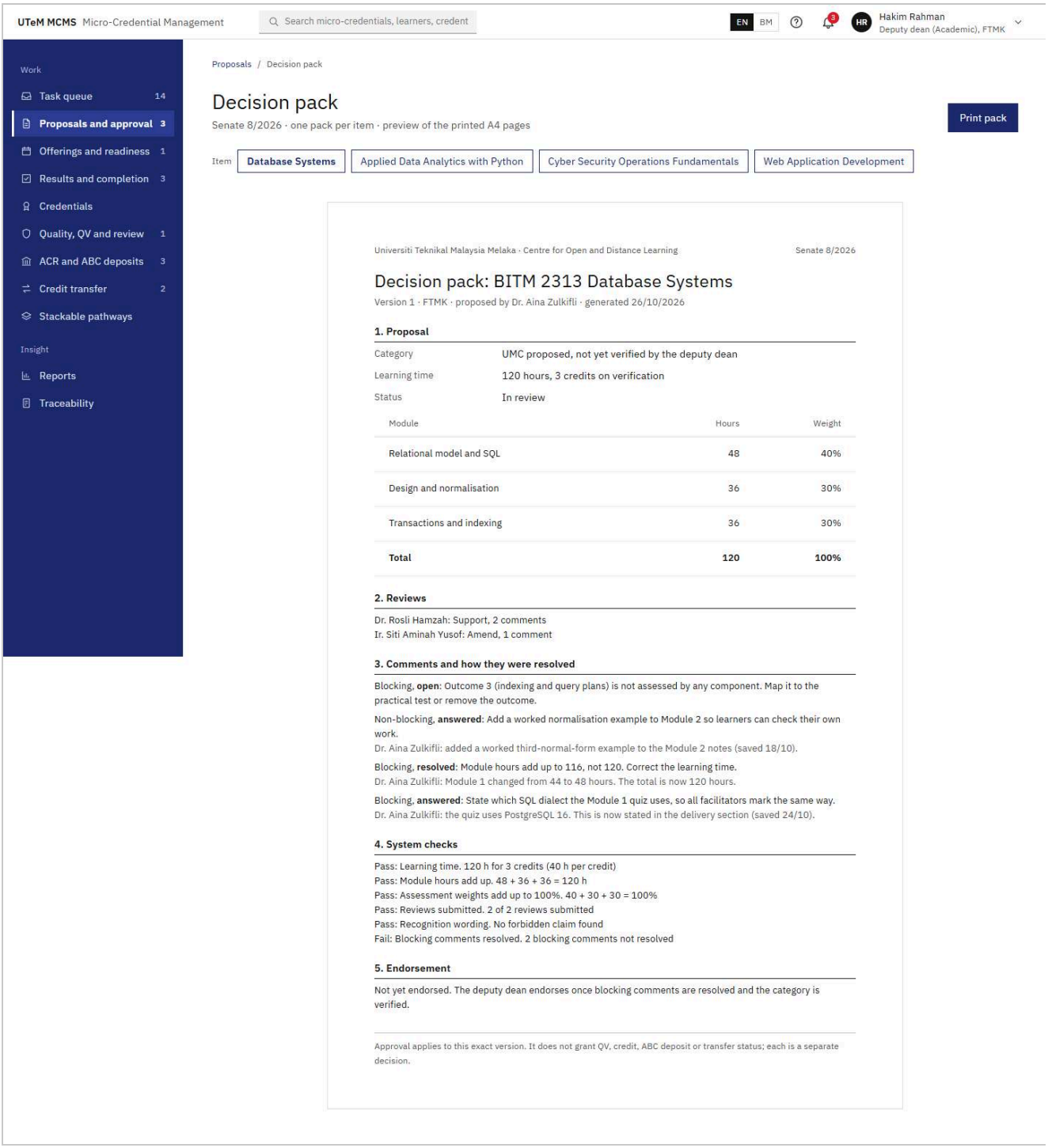


Figure 159: *DD-05 Decision pack.*

DD-06 Offerings and readiness

`#/dd/offerings`

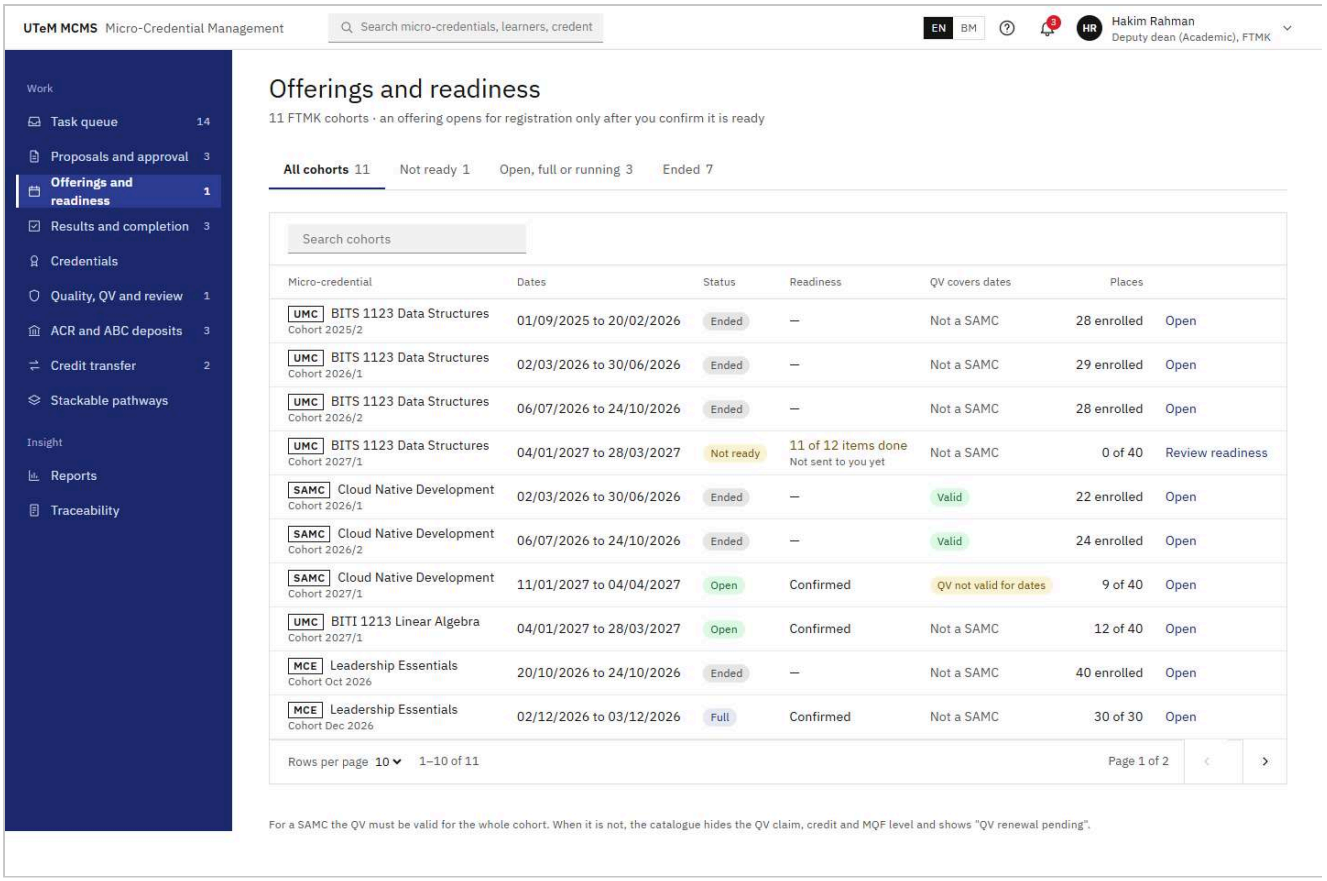


Figure 160: *DD-06 Offerings and readiness.*

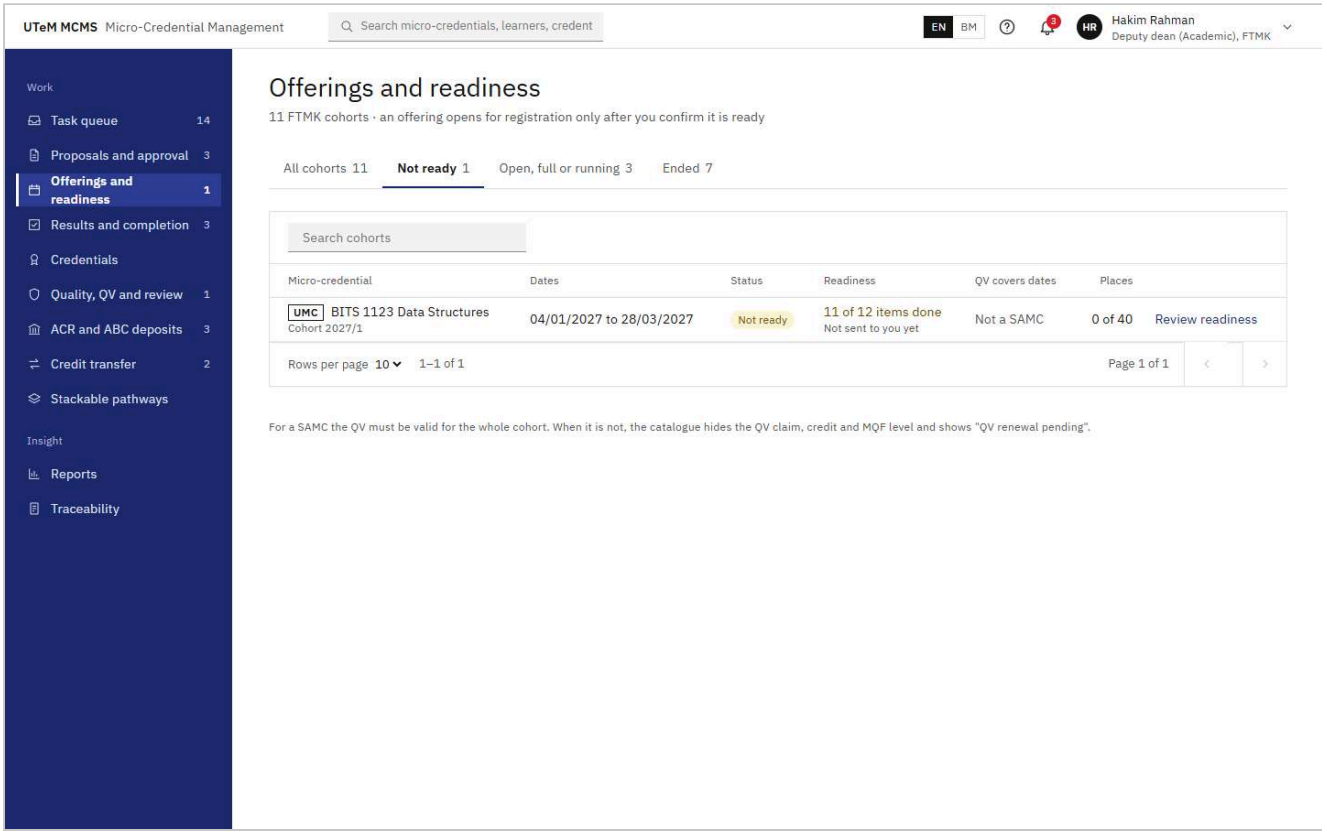


Figure 161: *DD-06 Offerings and readiness: Tab: Not ready.*

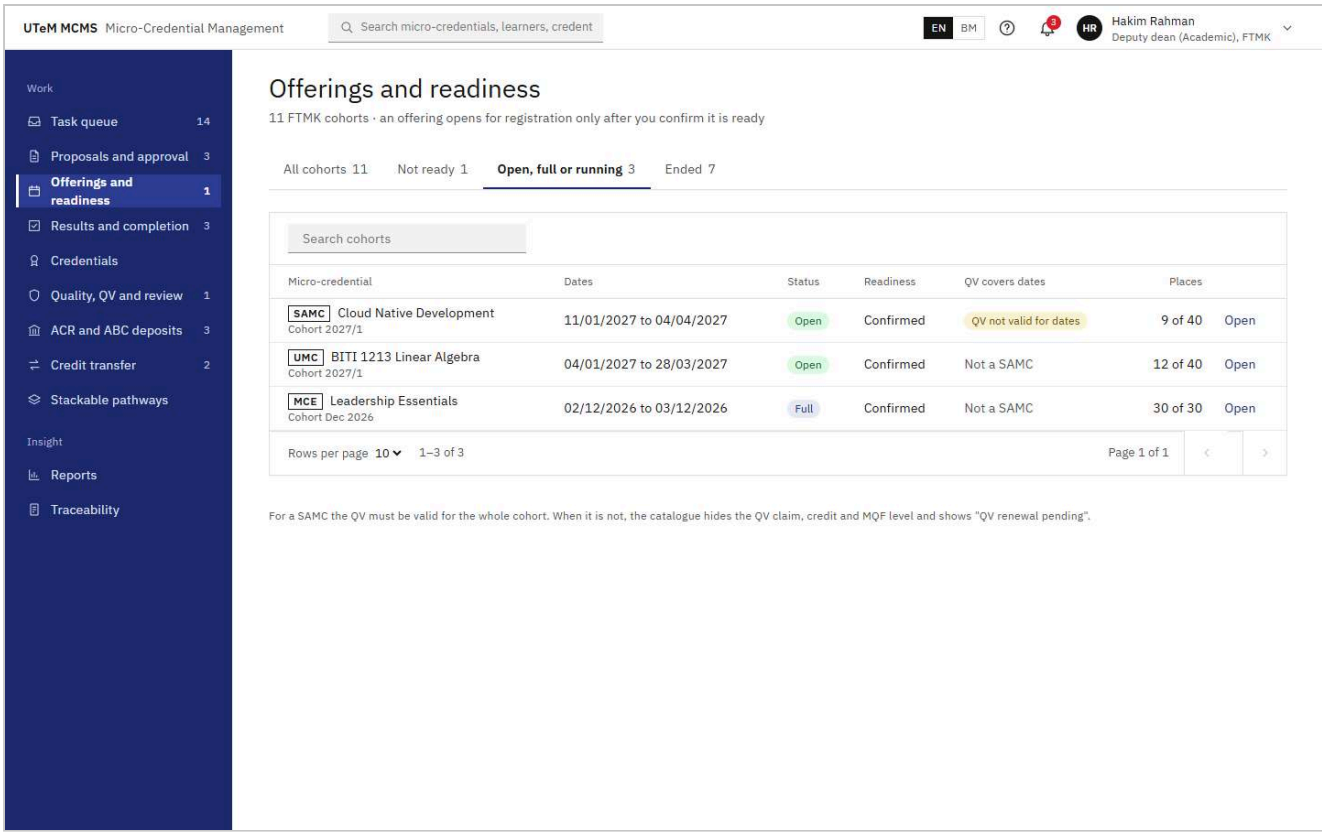


Figure 162: *DD-06 Offerings and readiness: Tab: Open, full or running.*

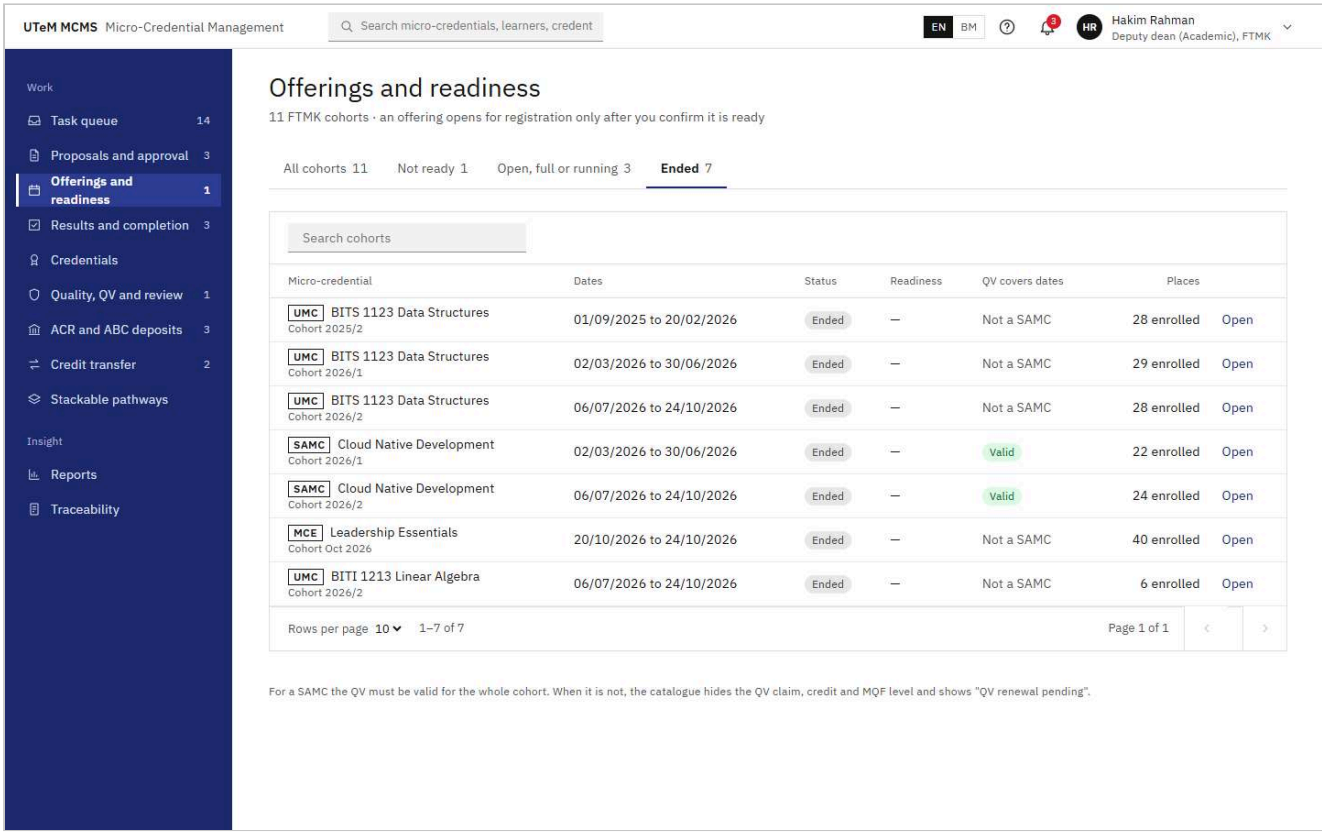


Figure 163: *DD-06 Offerings and readiness: Tab: Ended.*

DD-07 Confirm readiness

`#/dd/readiness/DS-2027-1`

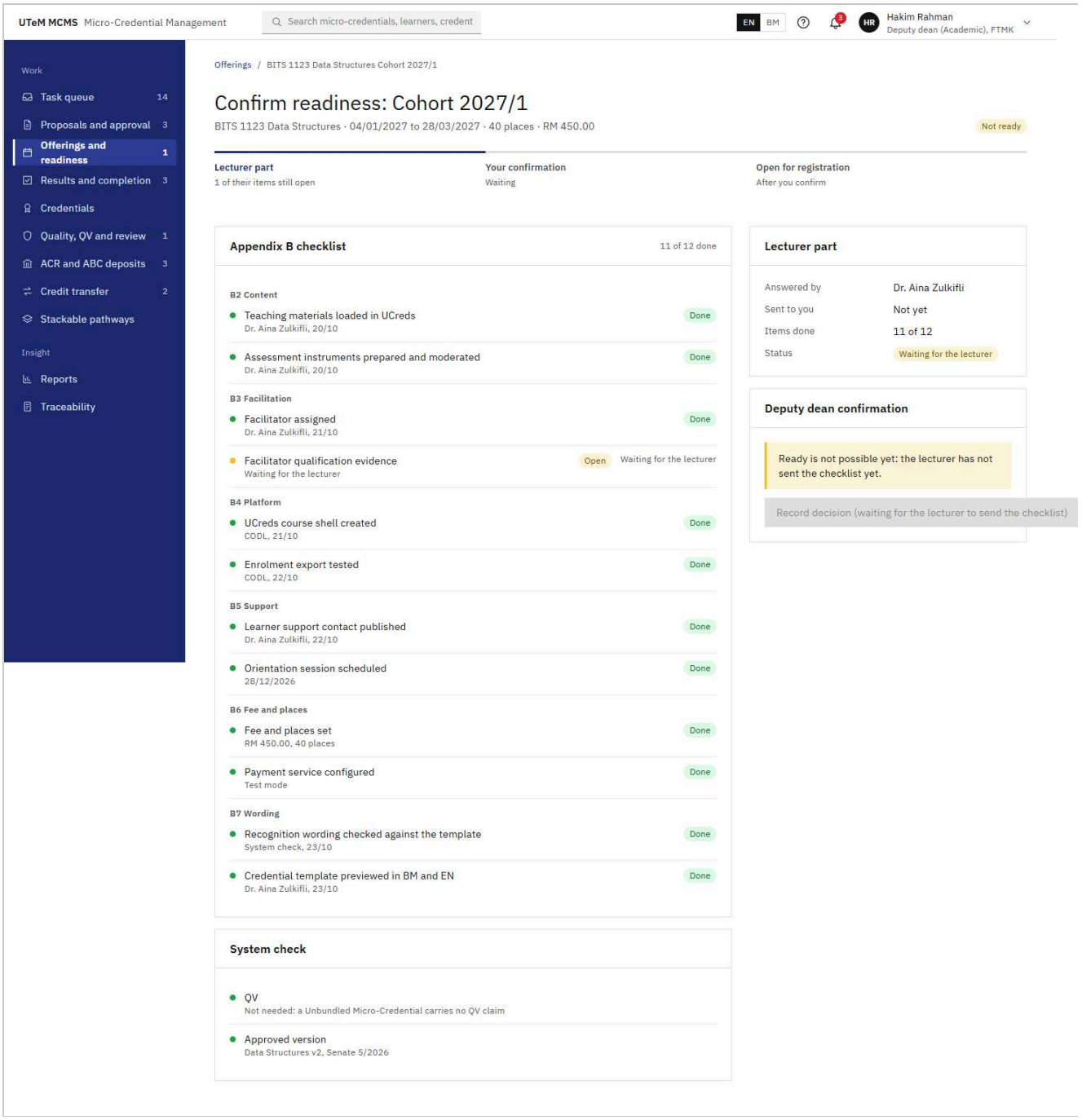


Figure 164: *DD-07 Confirm readiness.*

DD-08 Results and completion

`#/dd/results`

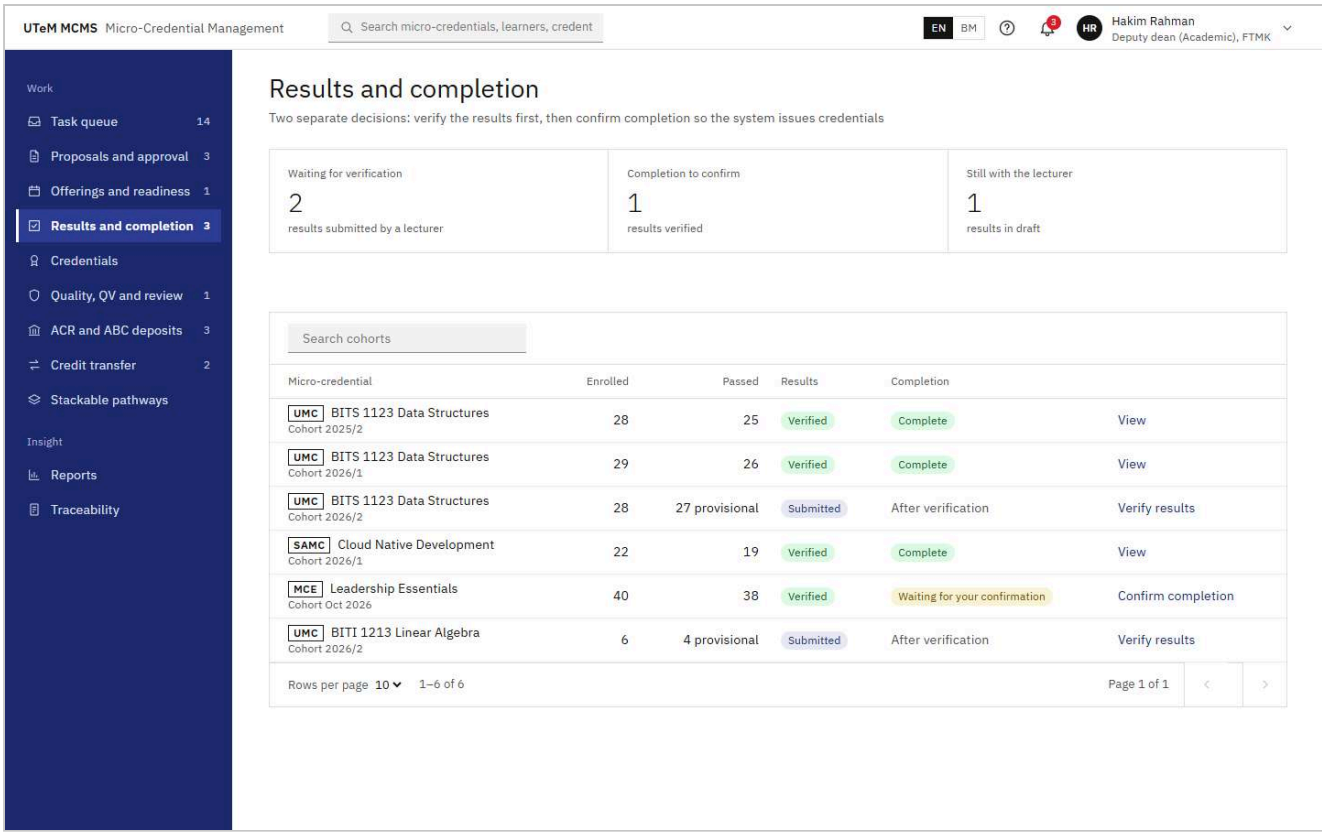


Figure 165: *DD-08 Results and completion.*

DD-09 Verify results

`#/dd/verify/DS-2026-2`

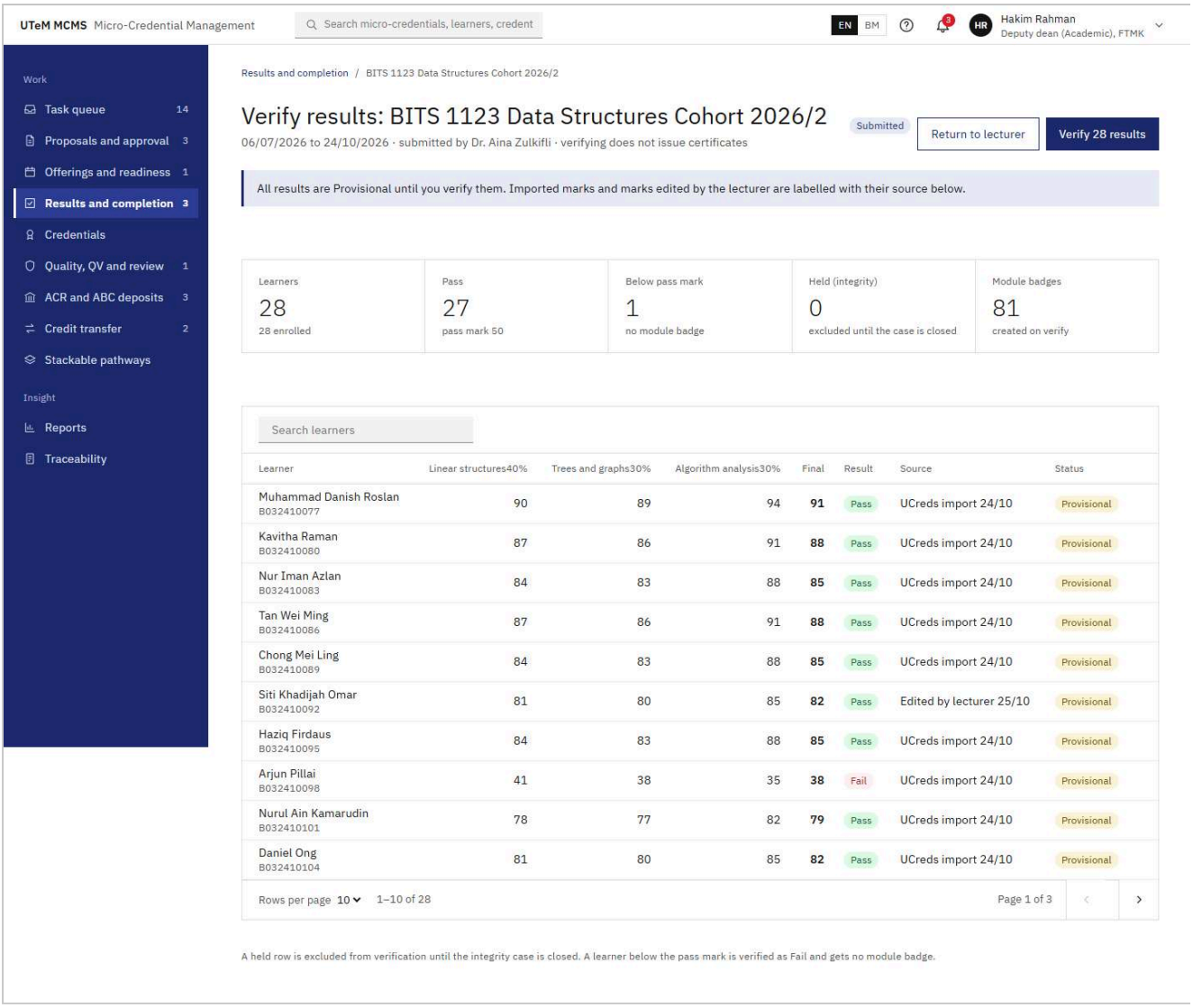


Figure 166: *DD-09 Verify results.*

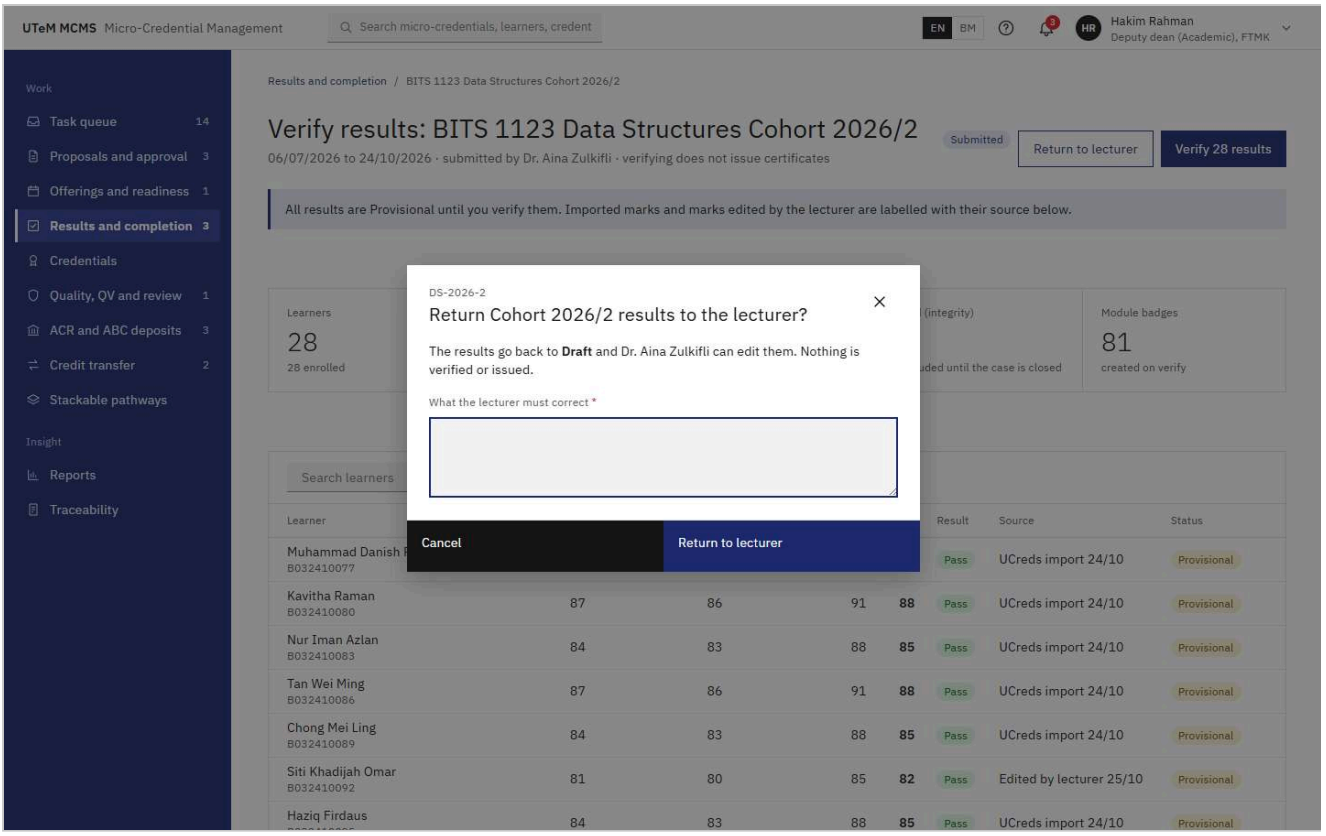


Figure 167: *DD-09 Confirmation: Return Cohort 2026/2 results to the lecturer?*

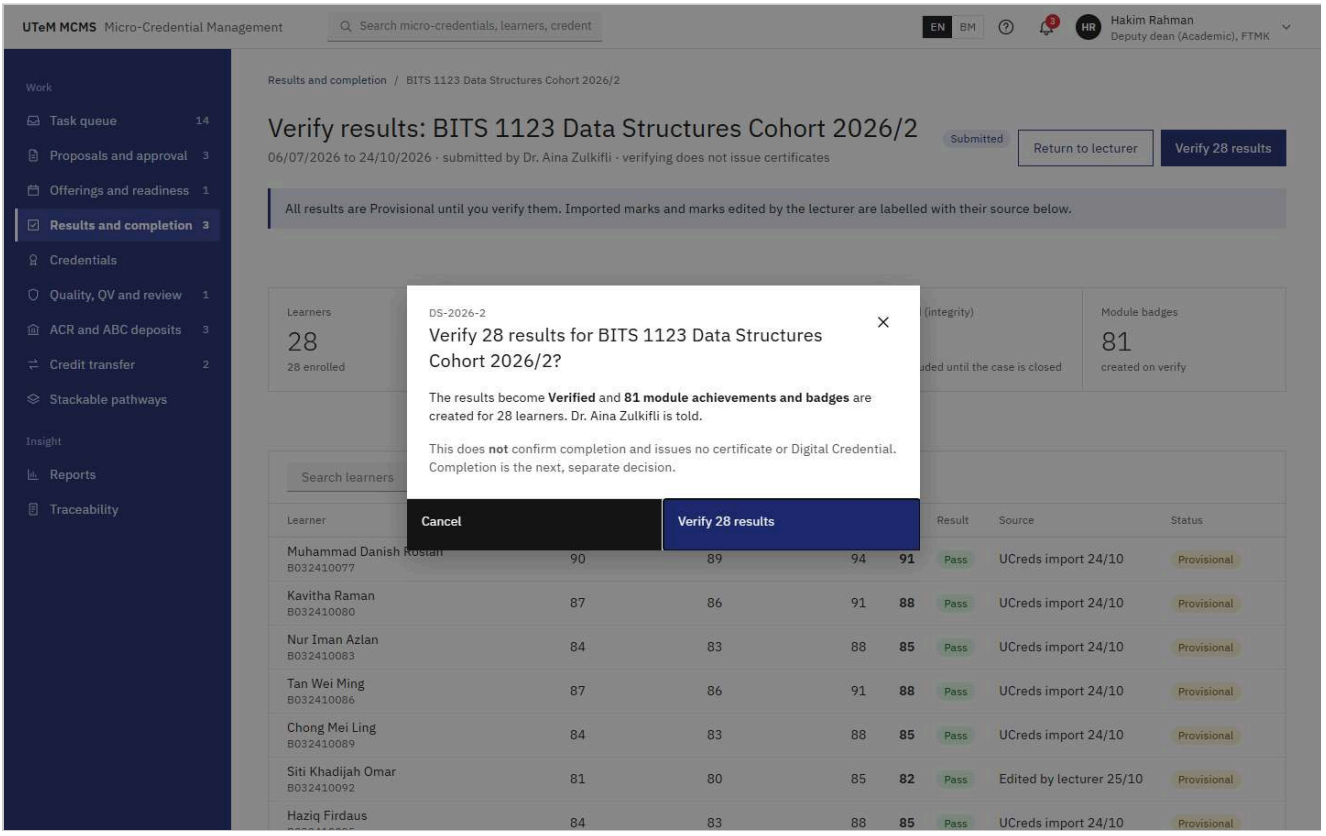


Figure 168: *DD-09 Confirmation: Verify 28 results for BITS 1123 Data Structures Cohort 2026/2?*

DD-10 Confirm completion

`#/dd/completion/LE-2026-10`

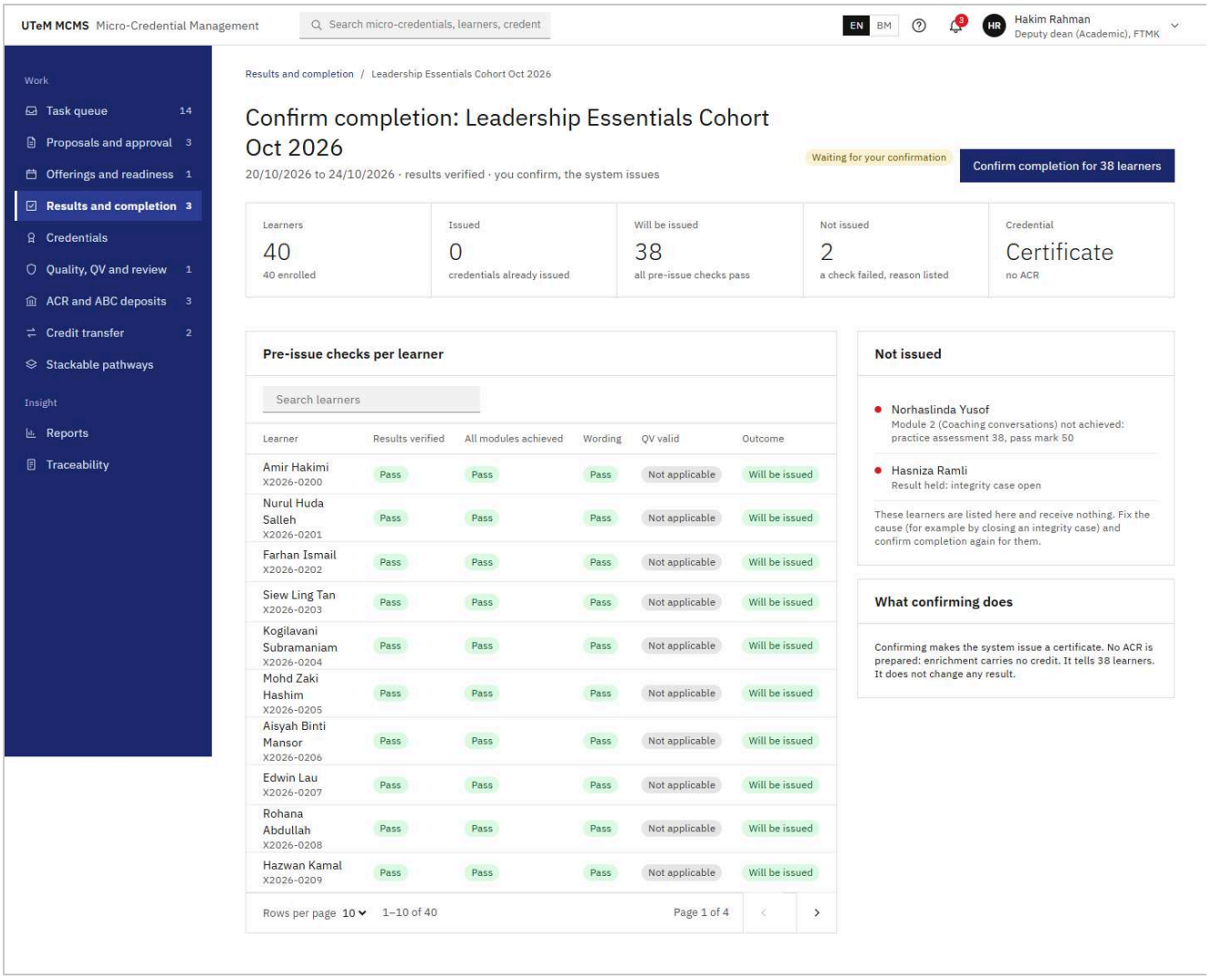


Figure 169: *DD-10 Confirm completion.*

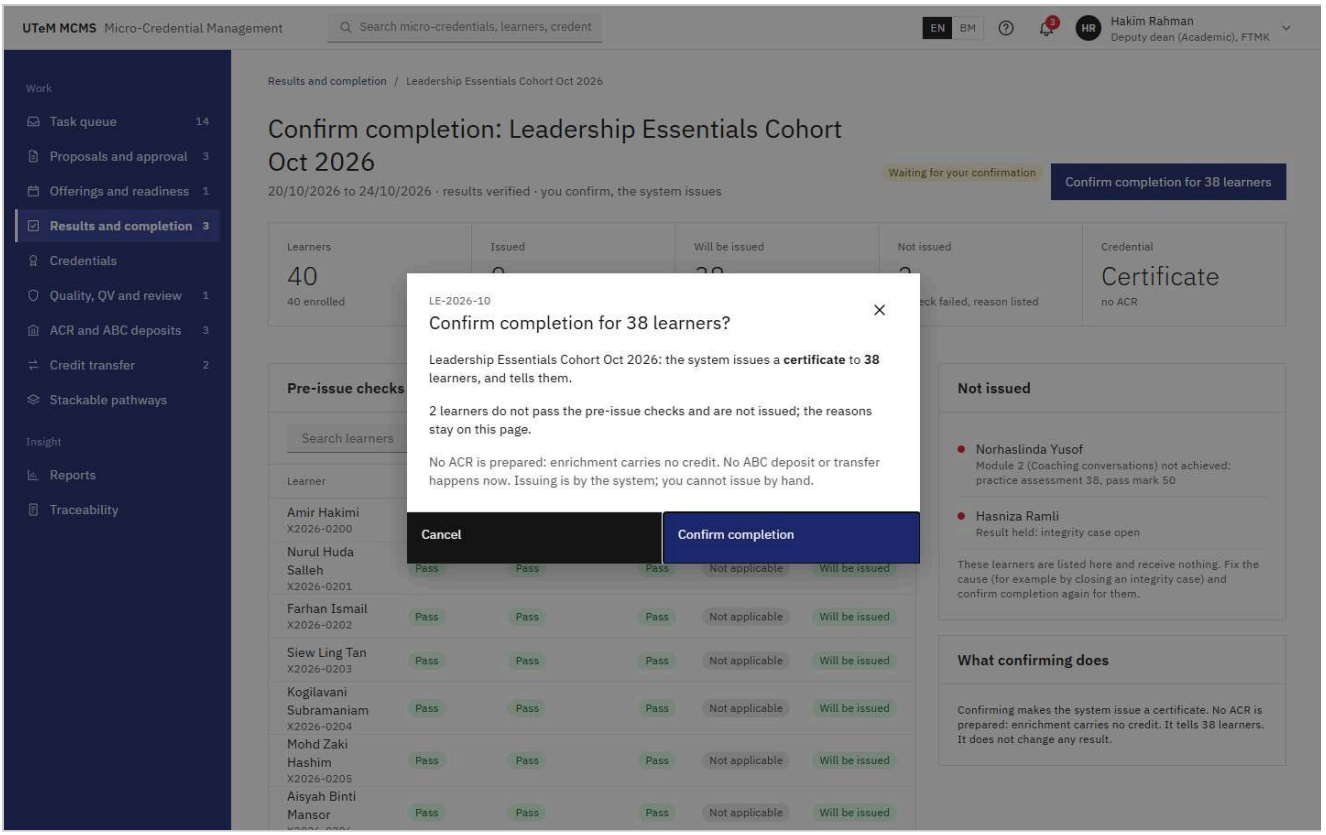


Figure 170: *DD-10 Confirmation: Confirm completion for 38 learners?*

DD-11 Credentials

`#/dd/creds`

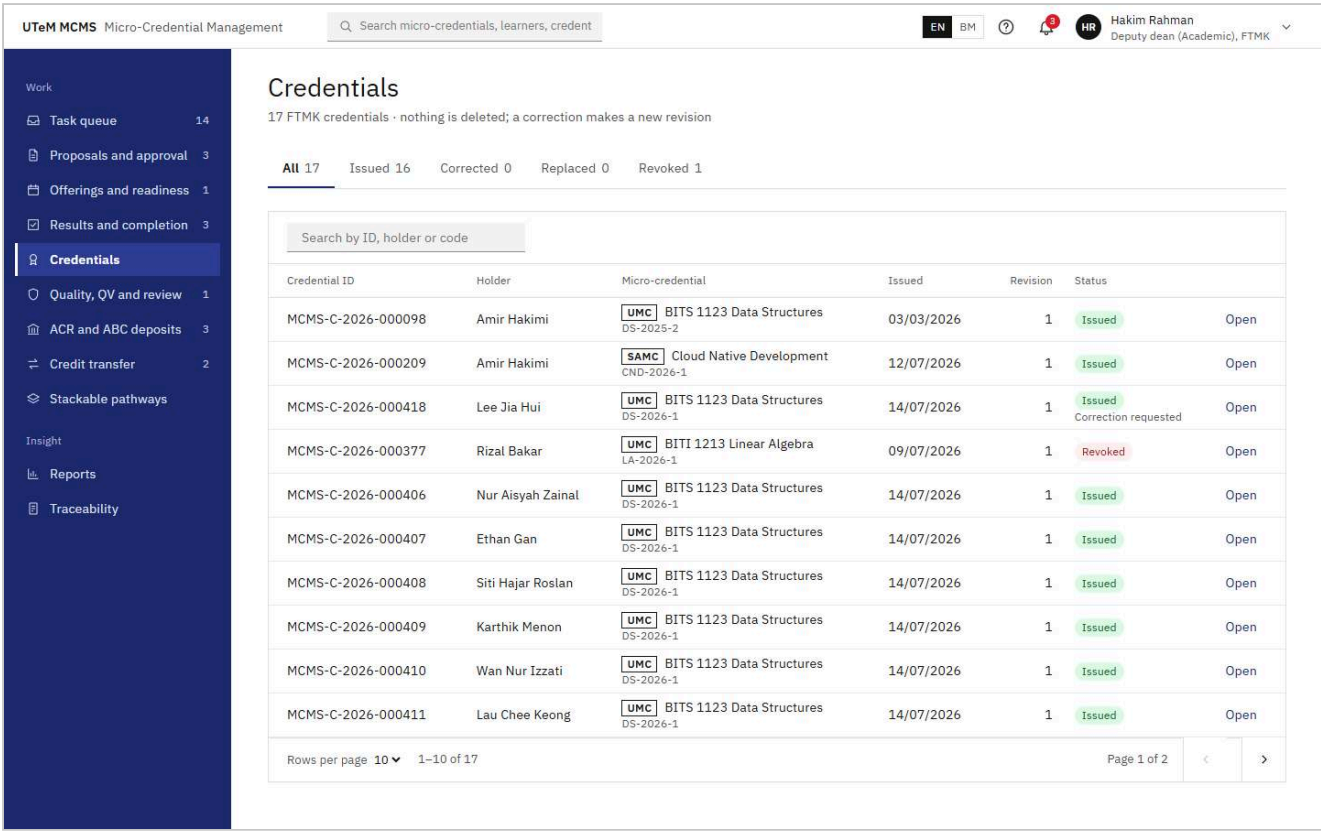


Figure 171: *DD-11 Credentials.*

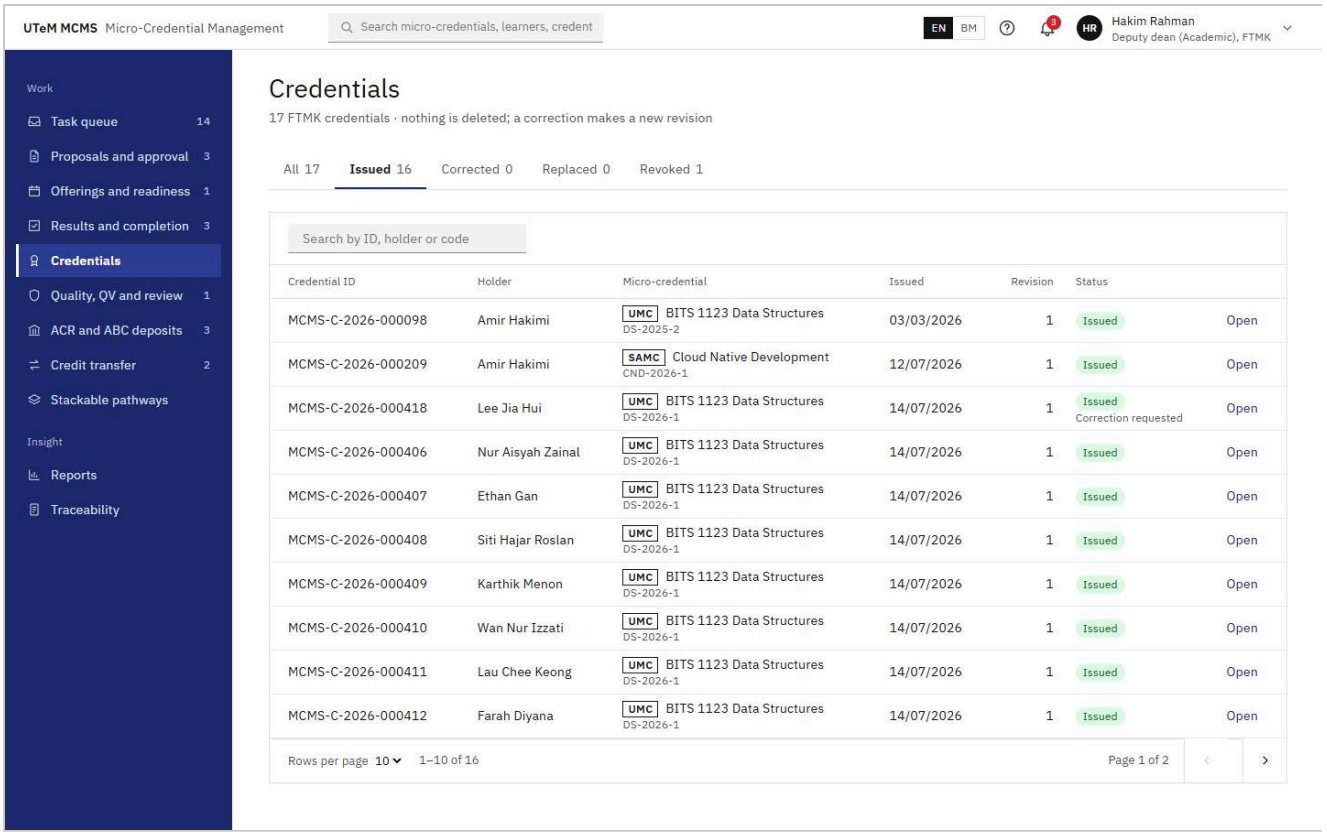


Figure 172: *DD-11 Credentials: Tab: Issued.*

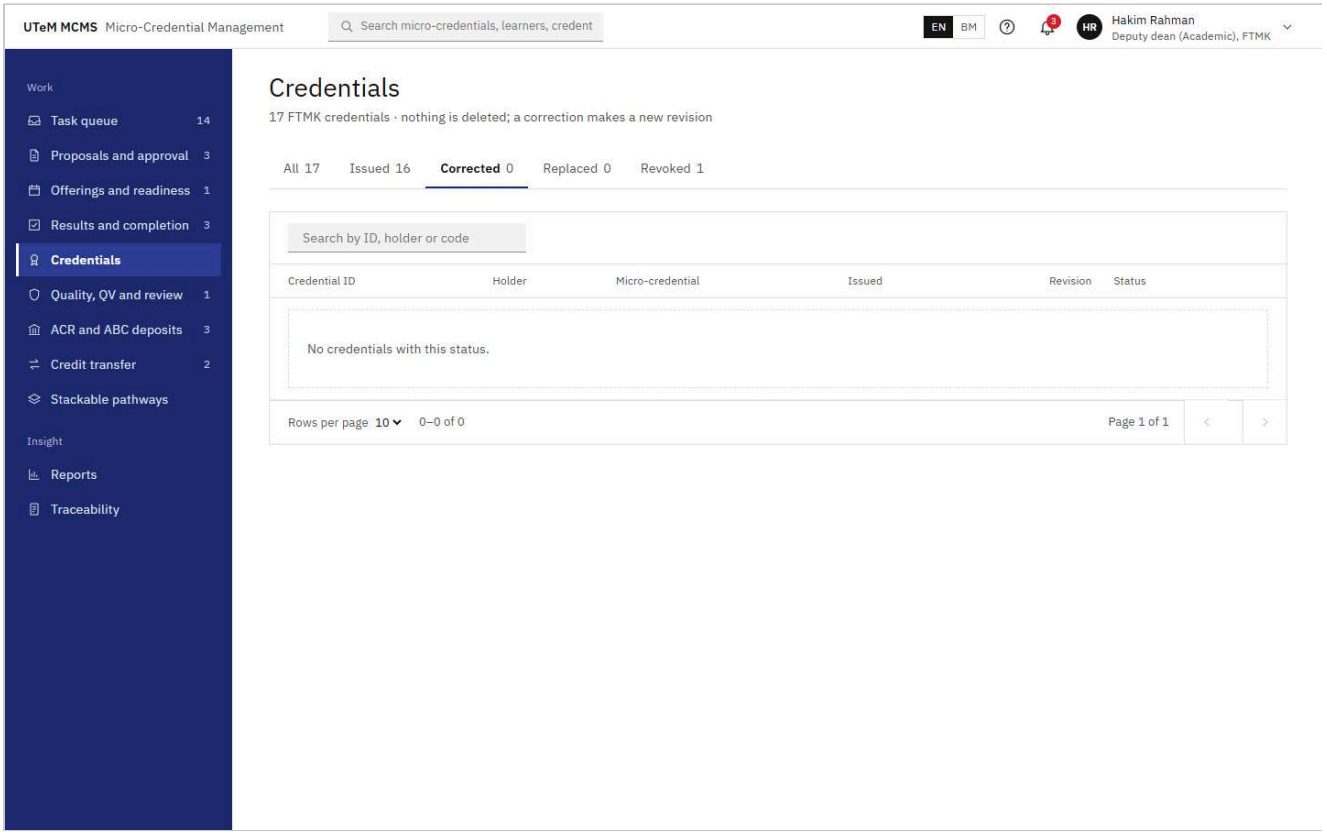


Figure 173: *DD-11 Credentials: Tab: Corrected.*

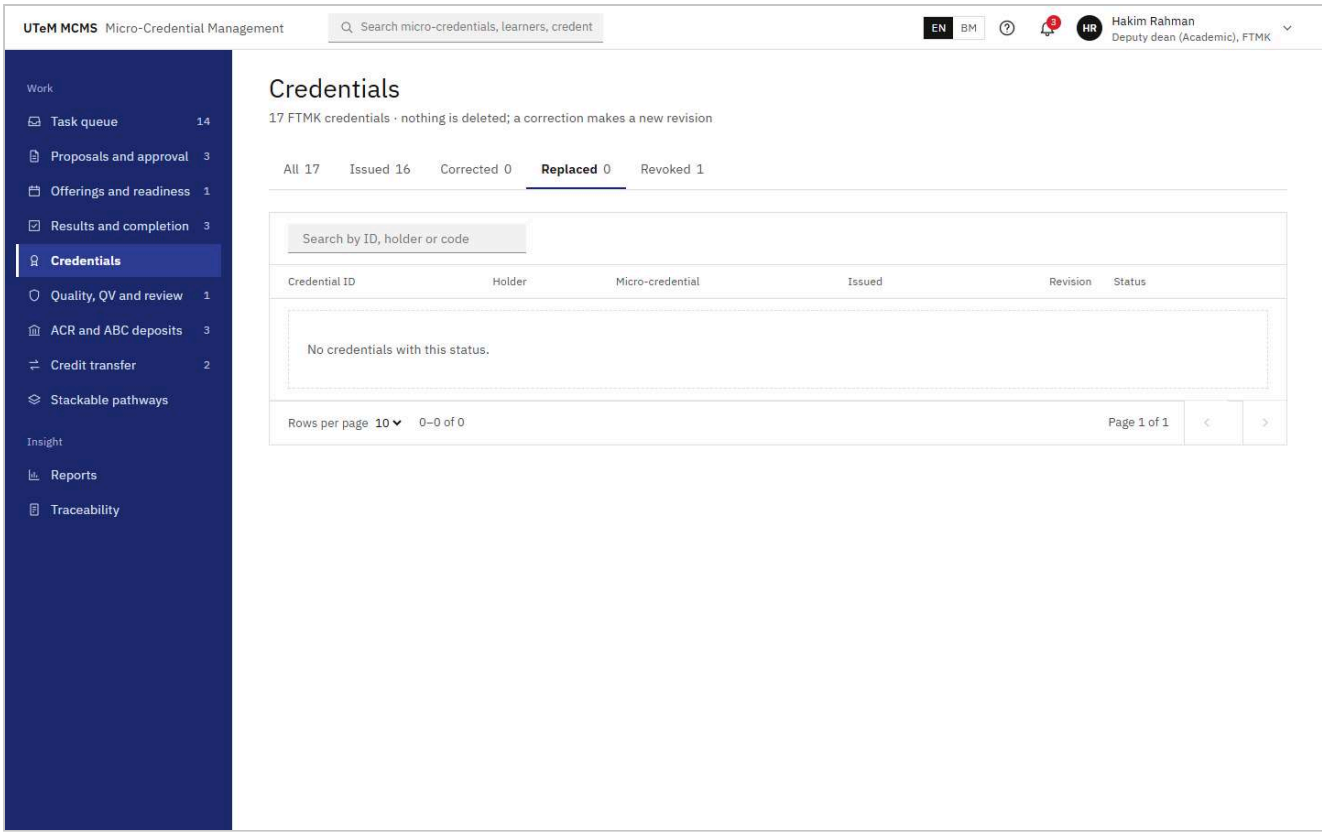


Figure 174: *DD-11 Credentials: Tab: Replaced.*

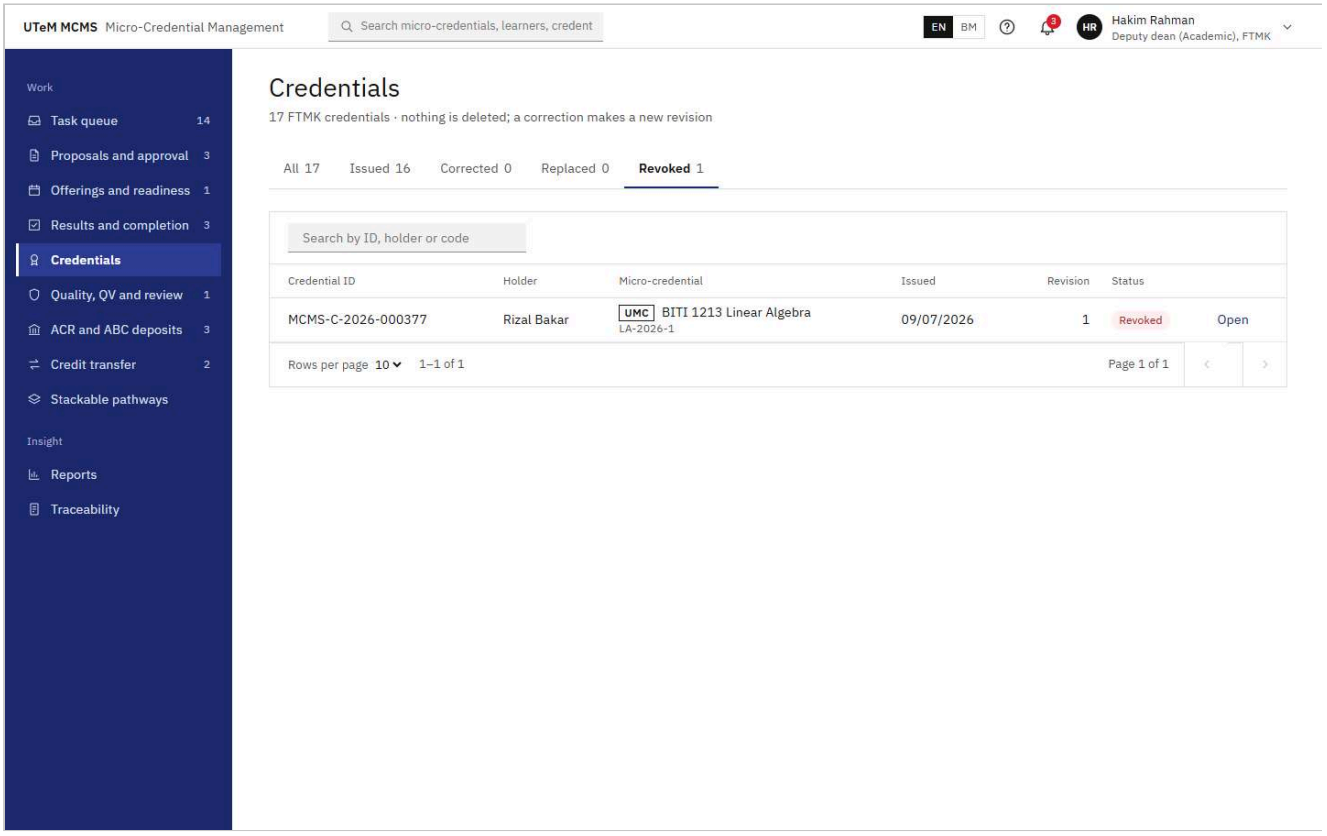


Figure 175: *DD-11 Credentials: Tab: Revoked.*

DD-12 Credential

`#/dd/cred/MCMS-C-2026-000418`

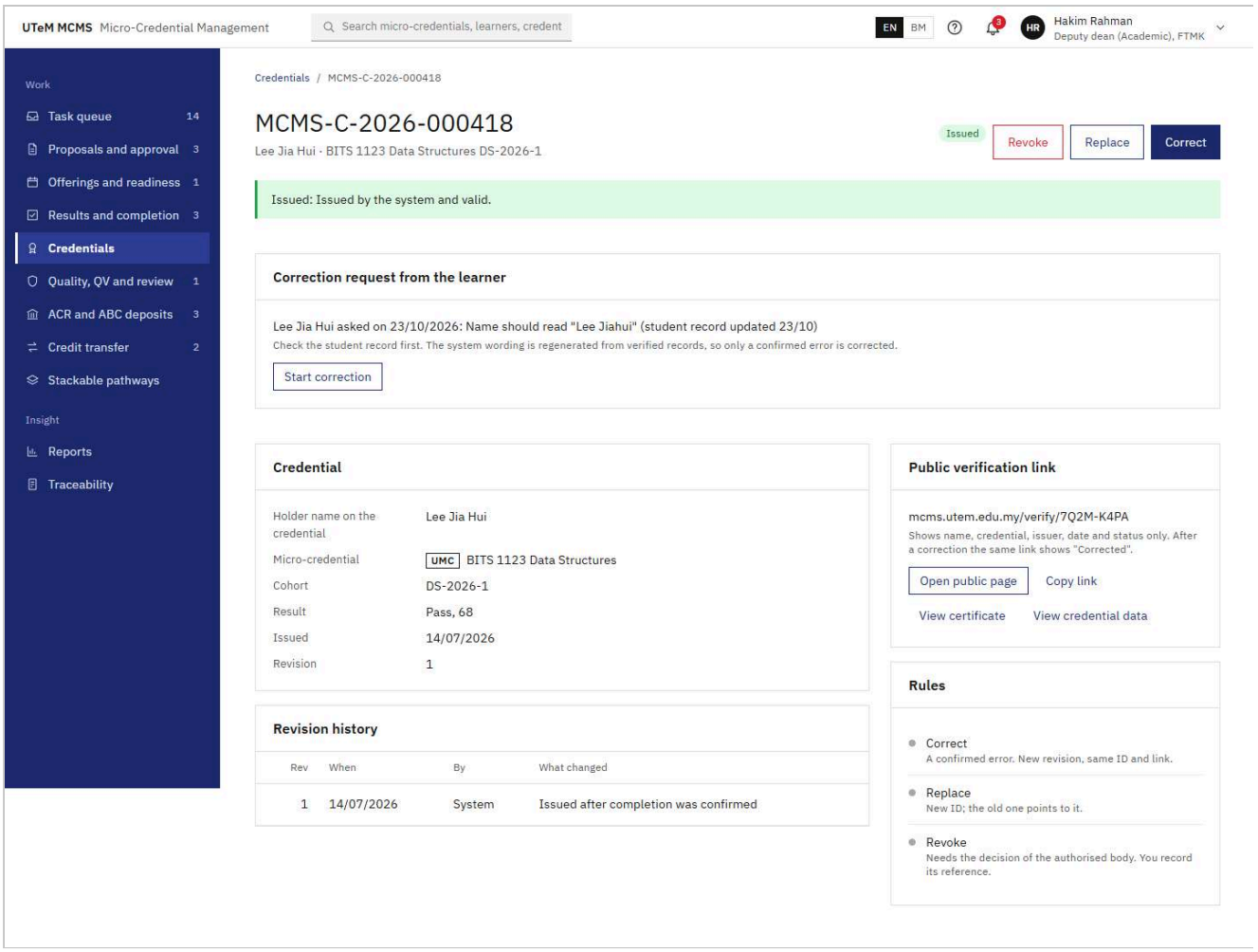


Figure 176: *DD-12 Credential.*

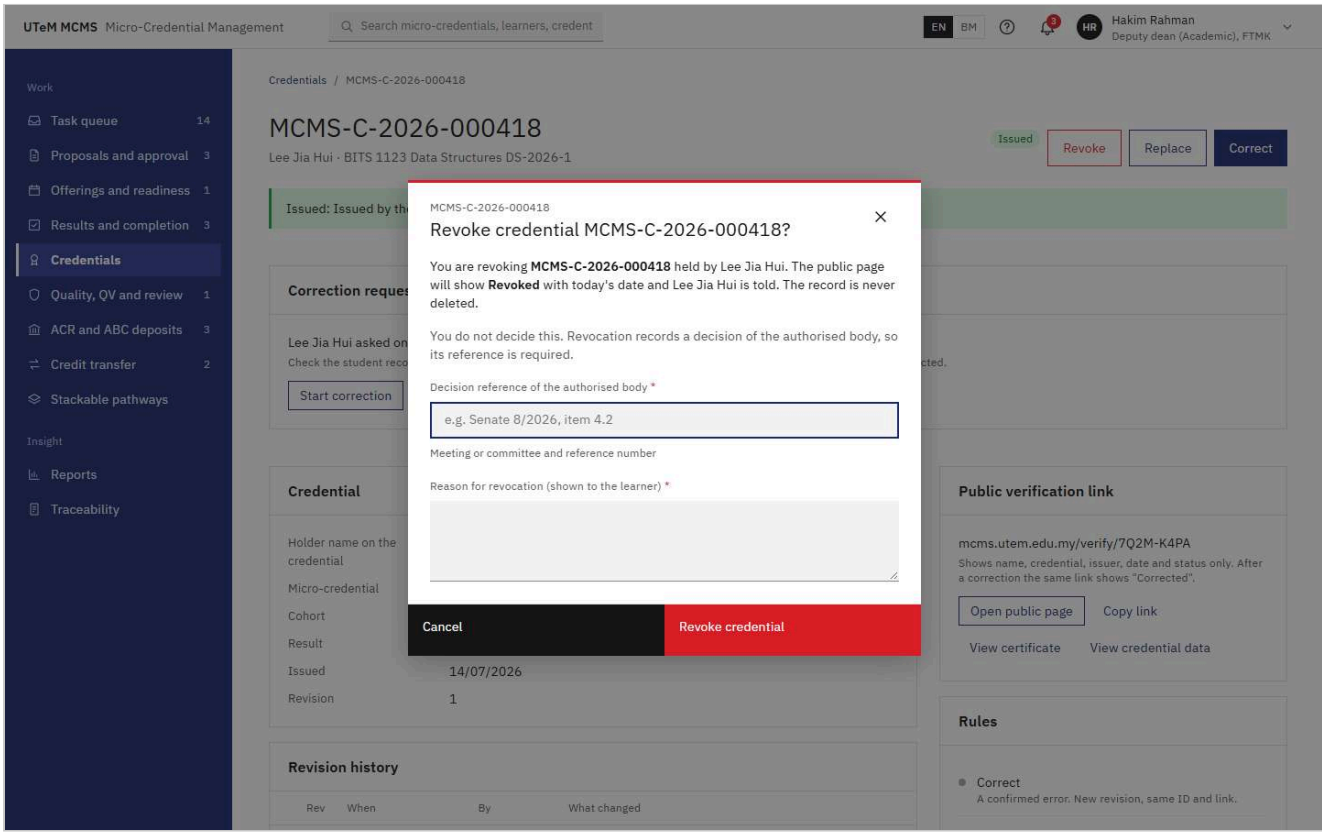


Figure 177: *DD-12 Warning (destructive): Revoke credential MCMS-C-2026-000418?*

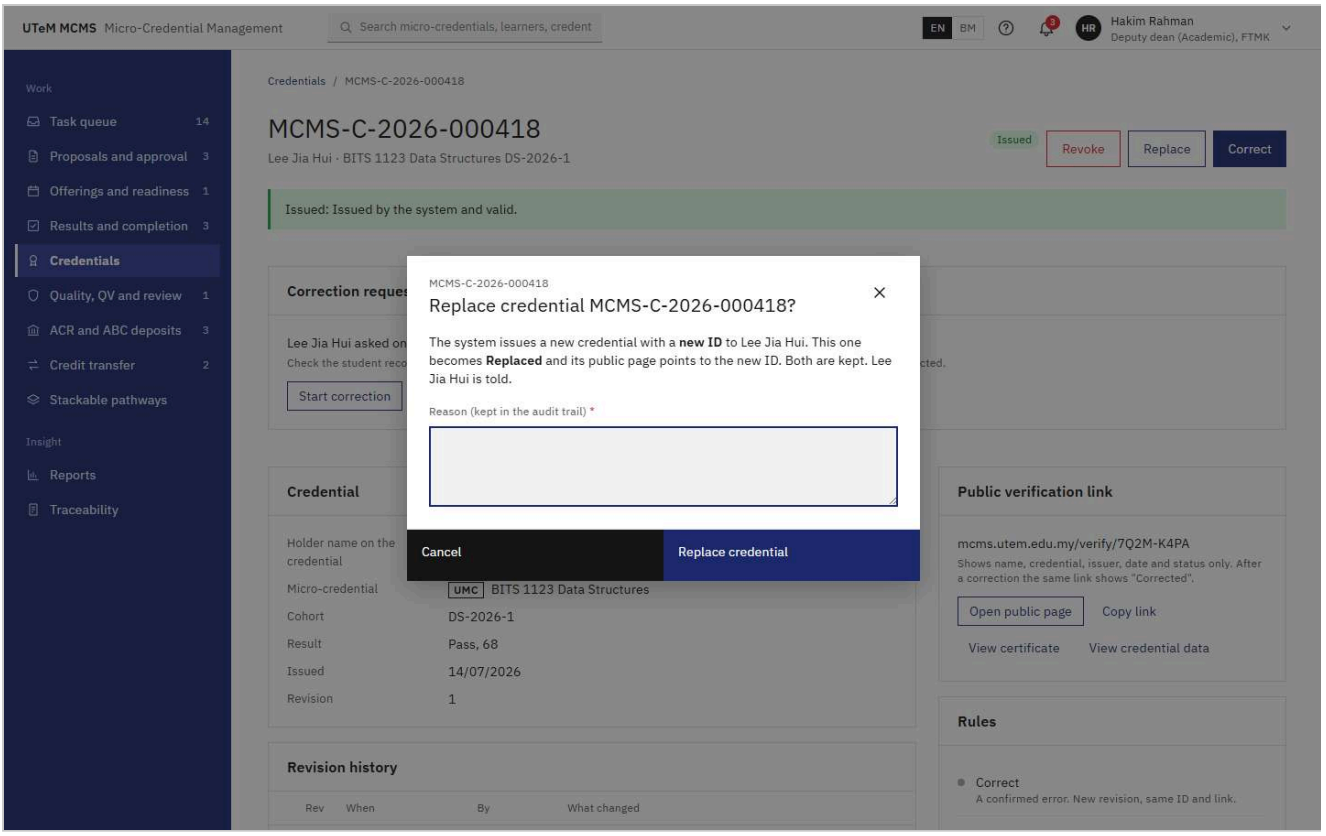


Figure 178: *DD-12 Confirmation: Replace credential MCMS-C-2026-000418?*

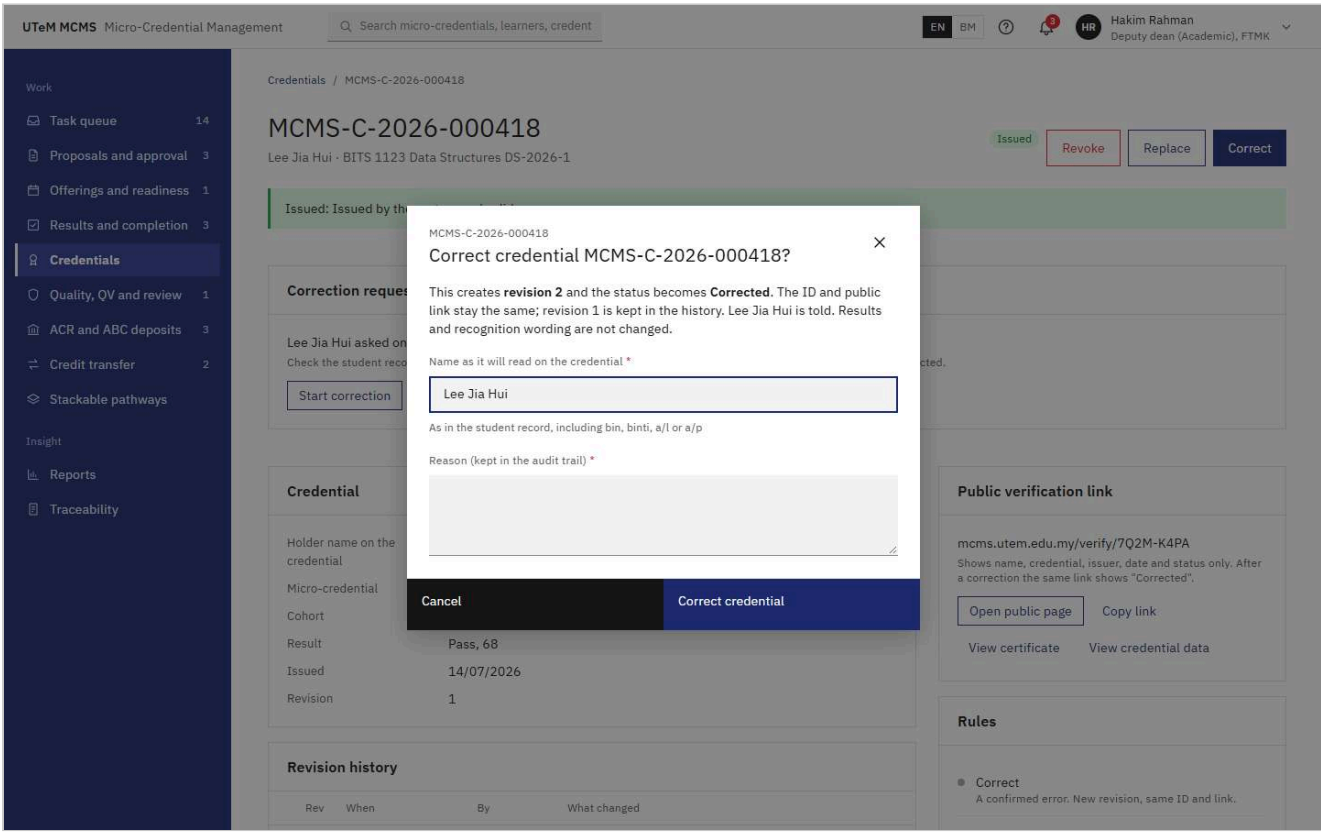


Figure 179: *DD-12 Confirmation: Correct credential MCMS-C-2026-000418?*

DD-13 Quality, QV and review

`#/dd/quality`

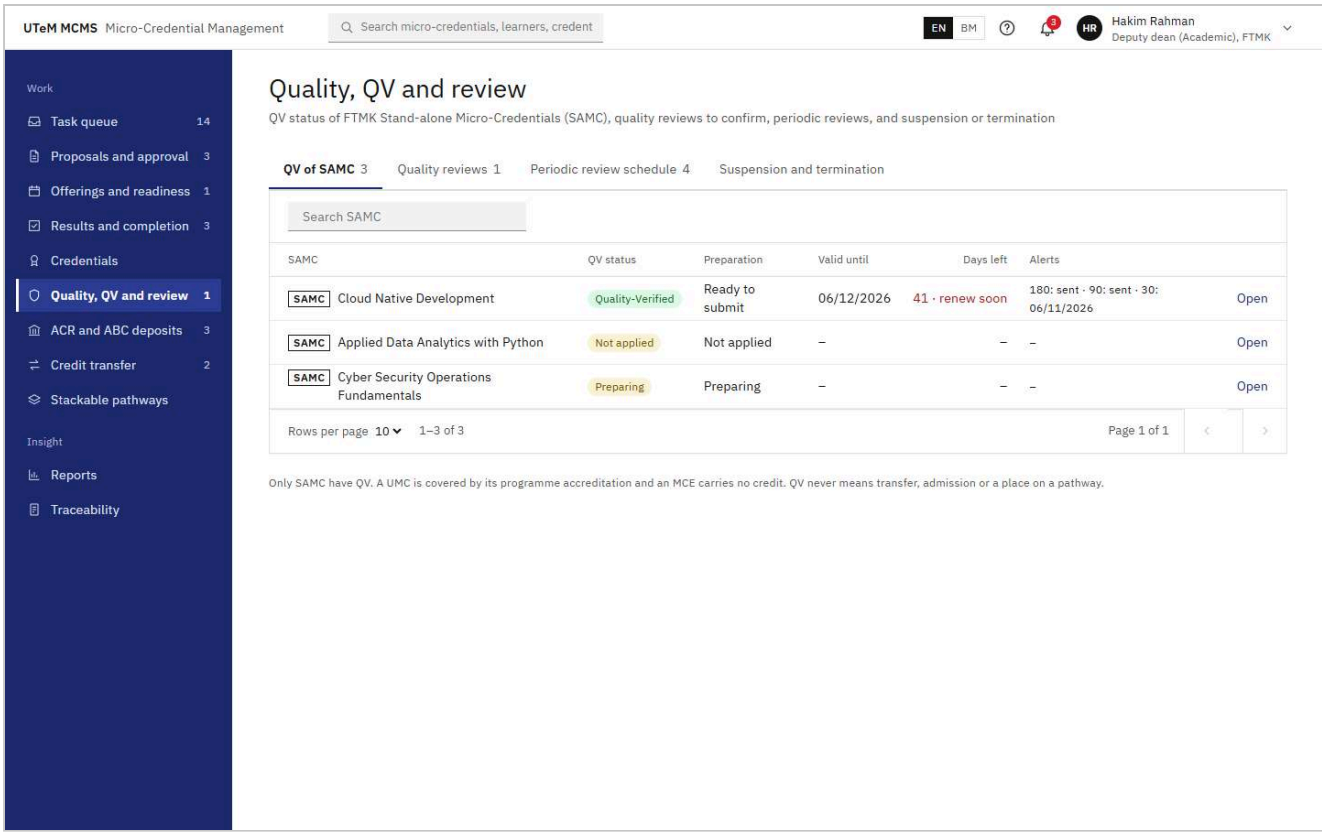


Figure 180: *DD-13 Quality, QV and review.*

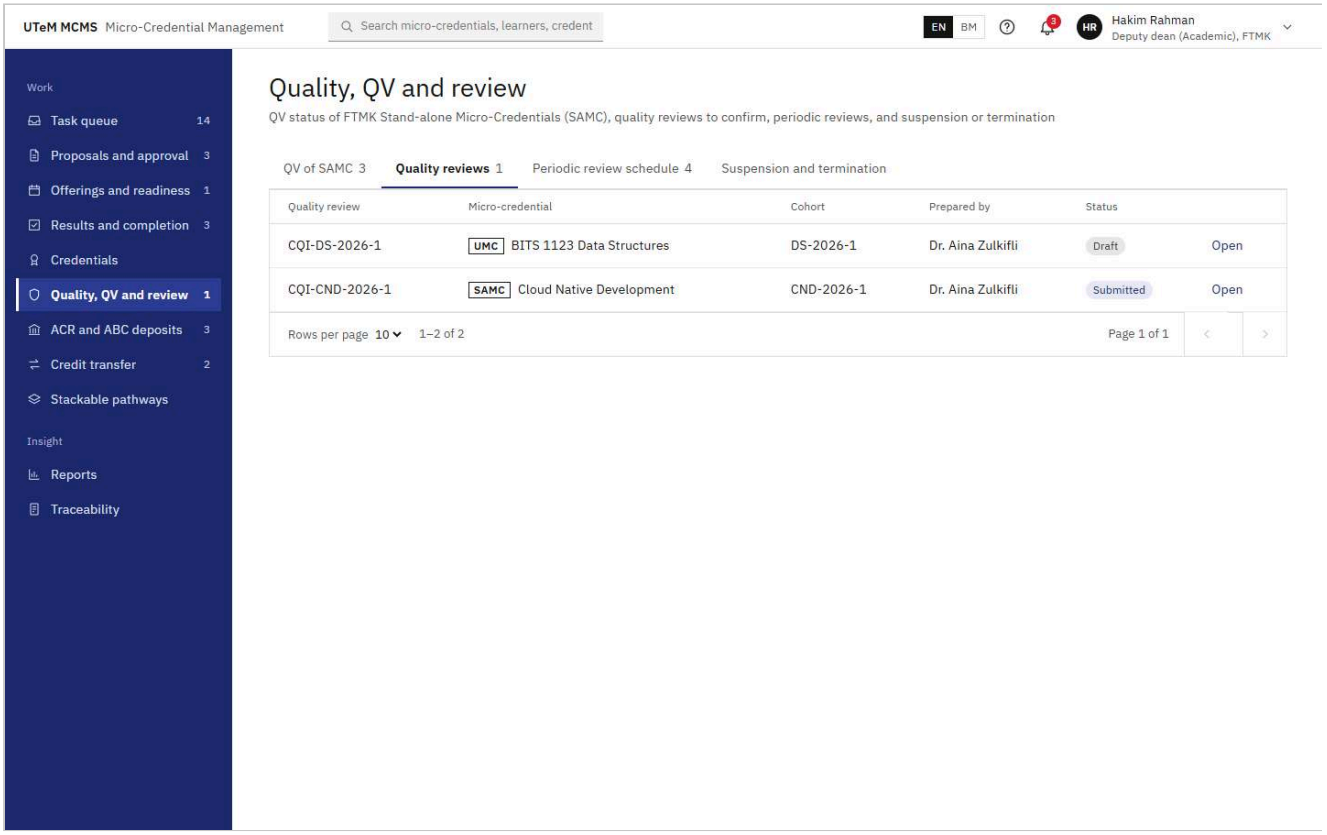


Figure 181: *DD-13 Quality, QV and review: Tab: Quality reviews.*

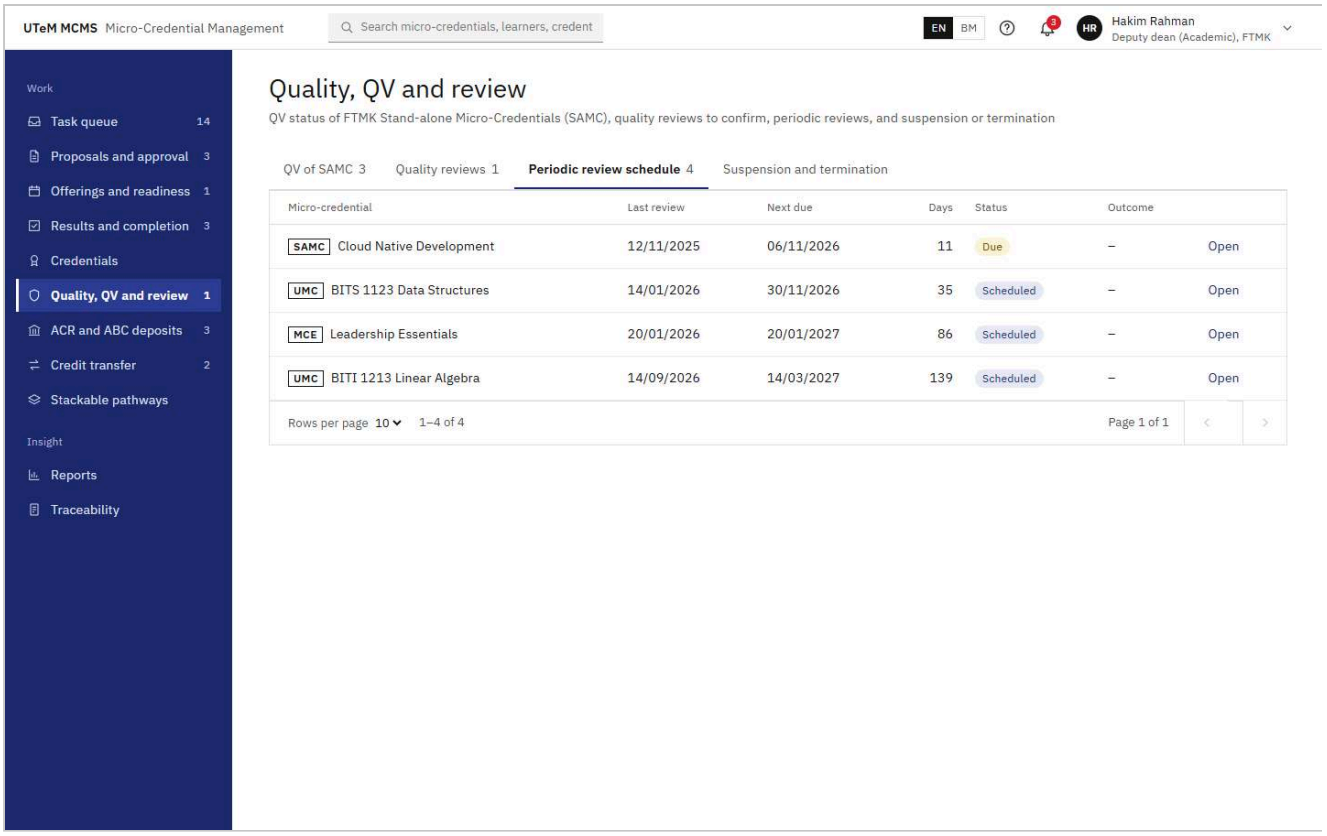


Figure 182: *DD-13 Quality, QV and review: Tab: Periodic review schedule.*

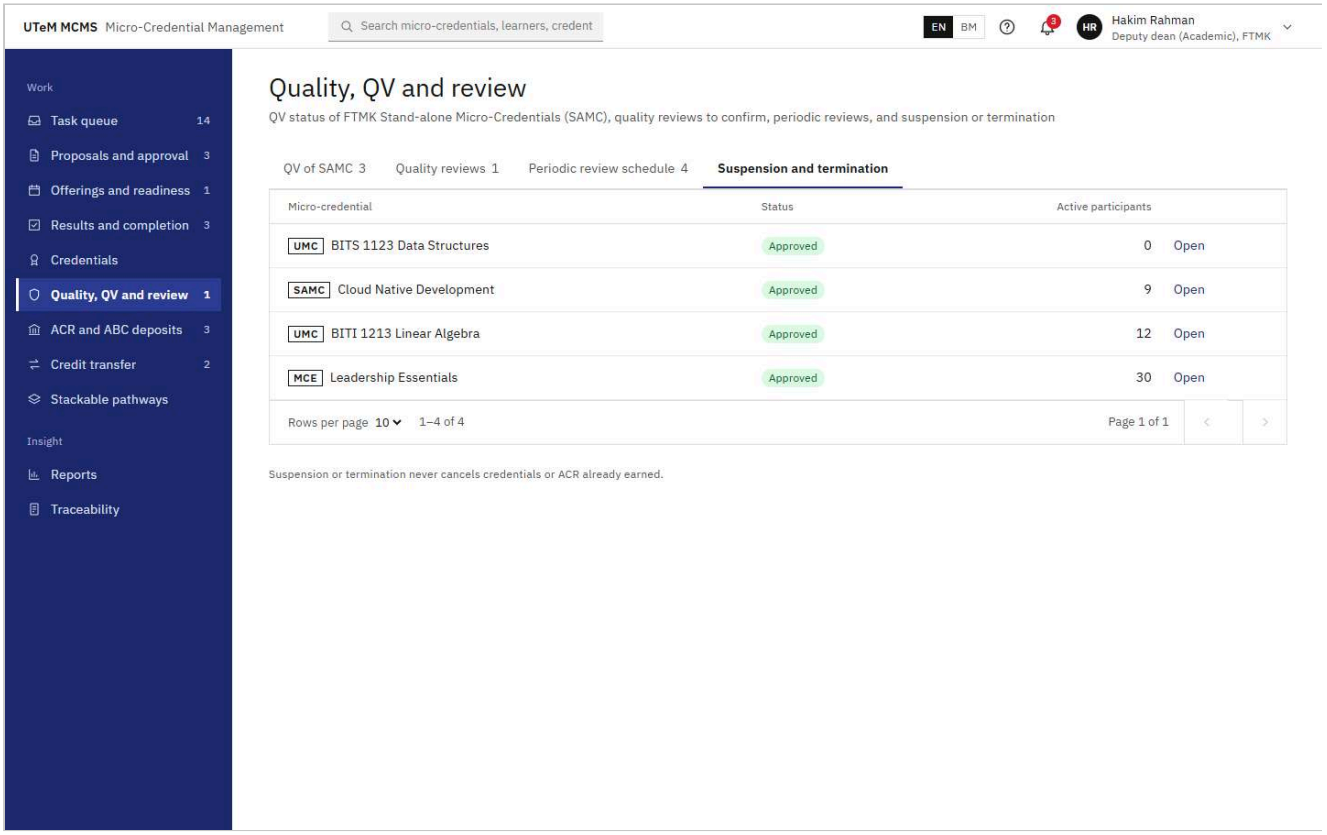


Figure 183: *DD-13 Quality, QV and review: Tab: Suspension and termination.*

DD-14 QV record

`#/dd/qv/CND`

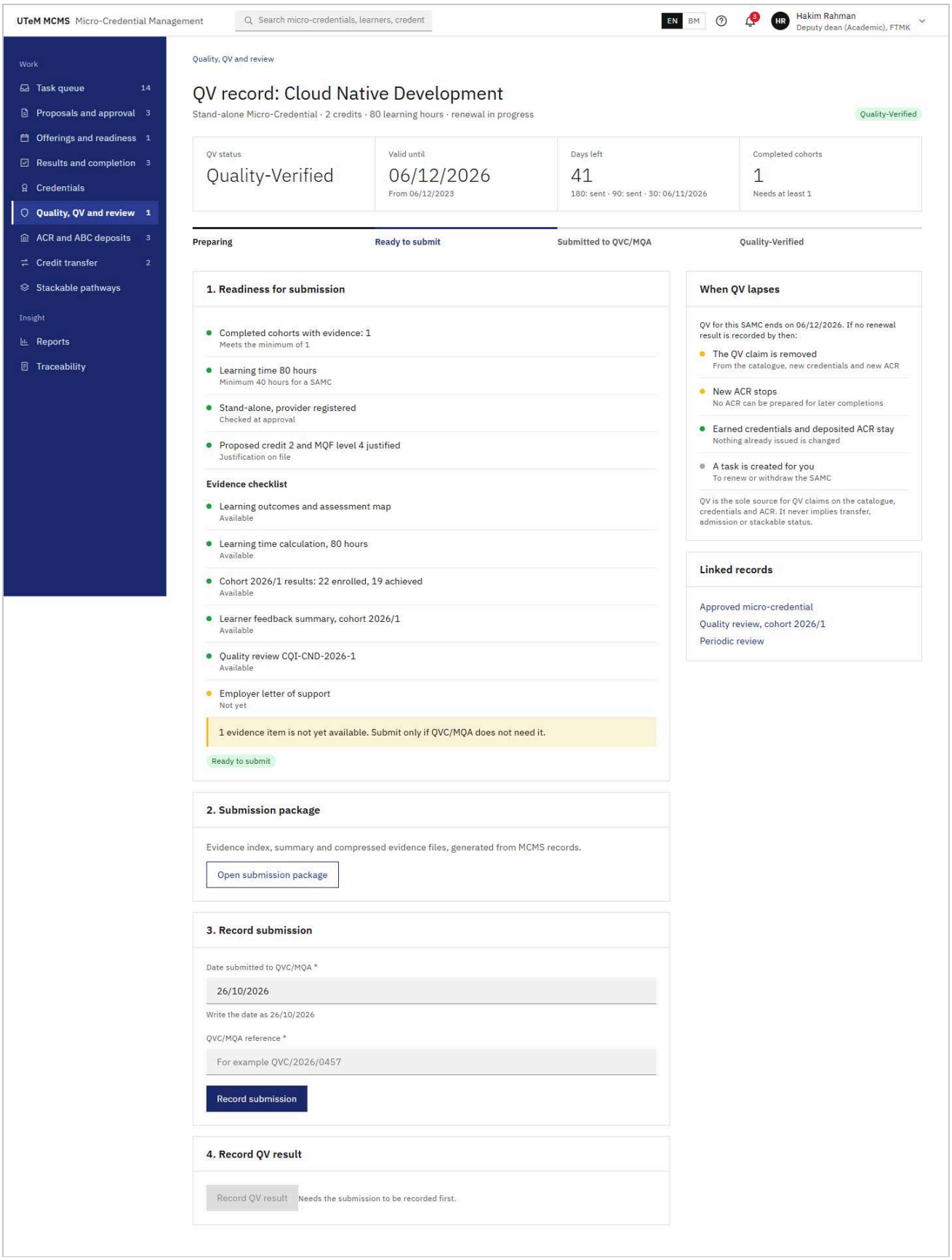


Figure 184: *DD-14 QV record.*

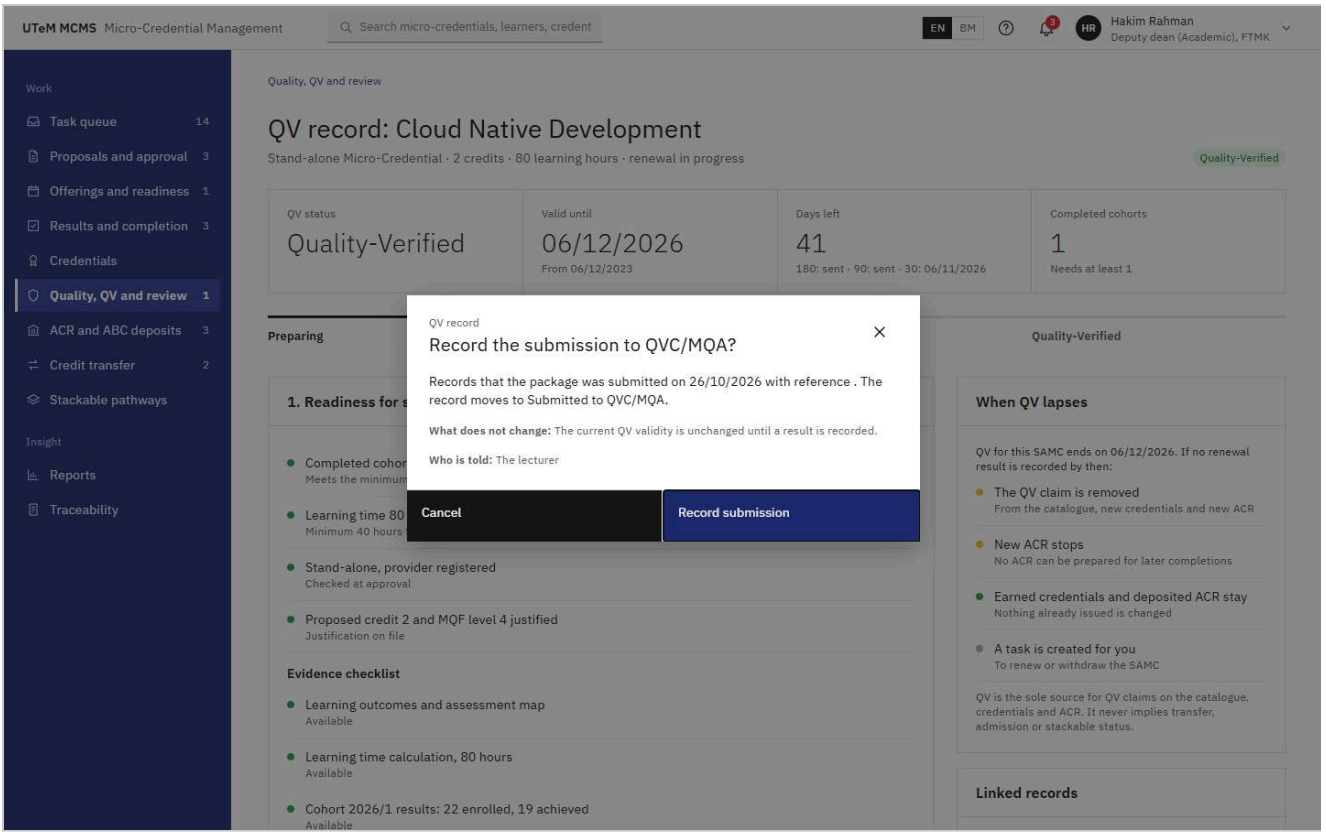


Figure 185: *DD-14 Confirmation: Record the submission to QVC/MQA?*

DD-15 Quality review

`#/dd/cqi/CQI-CND-2026-1`

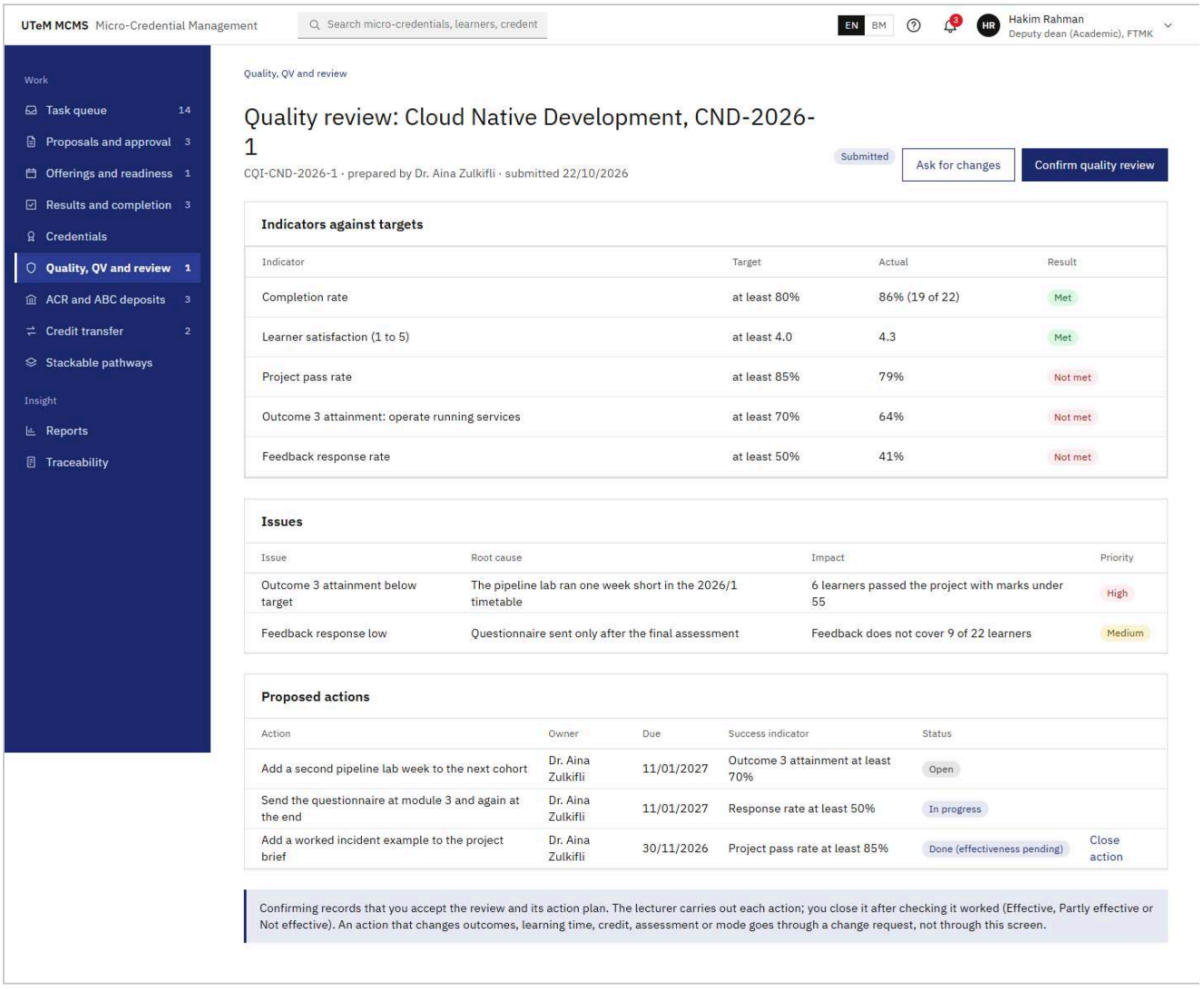


Figure 186: *DD-15 Quality review.*

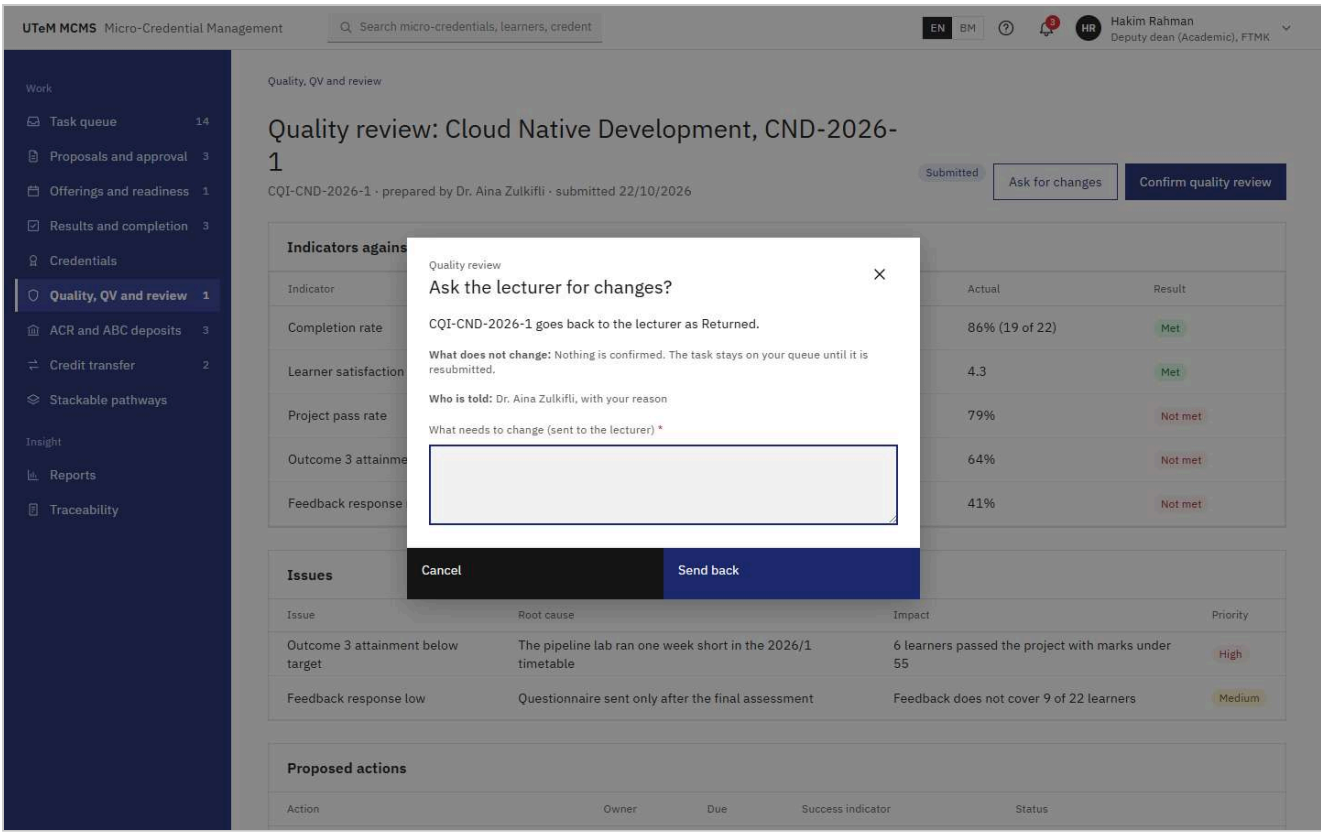


Figure 187: *DD-15 Confirmation: Ask the lecturer for changes?*

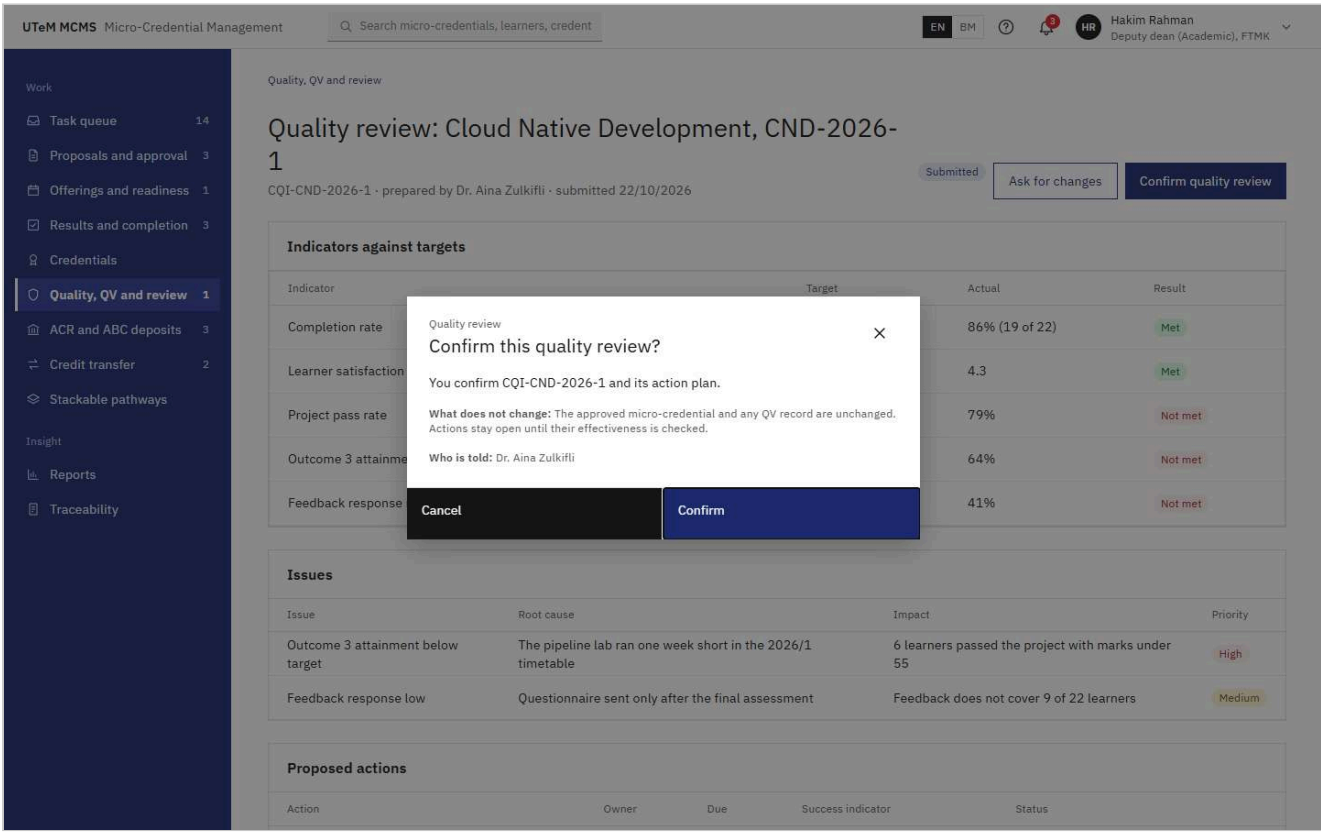


Figure 188: *DD-15 Confirmation: Confirm this quality review?*

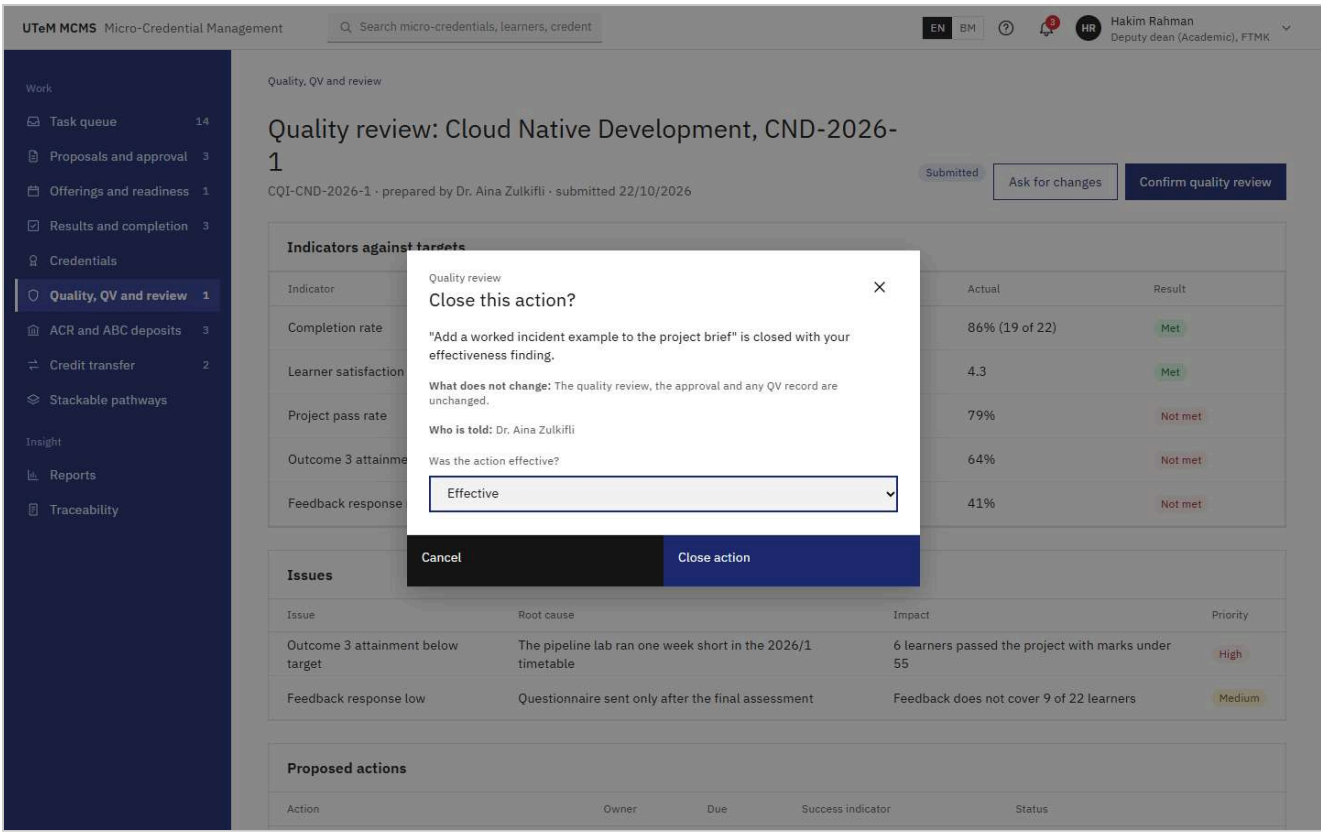


Figure 189: *DD-15 Confirmation: Close this action?*

DD-16 Periodic review

`#/dd/review/DS`

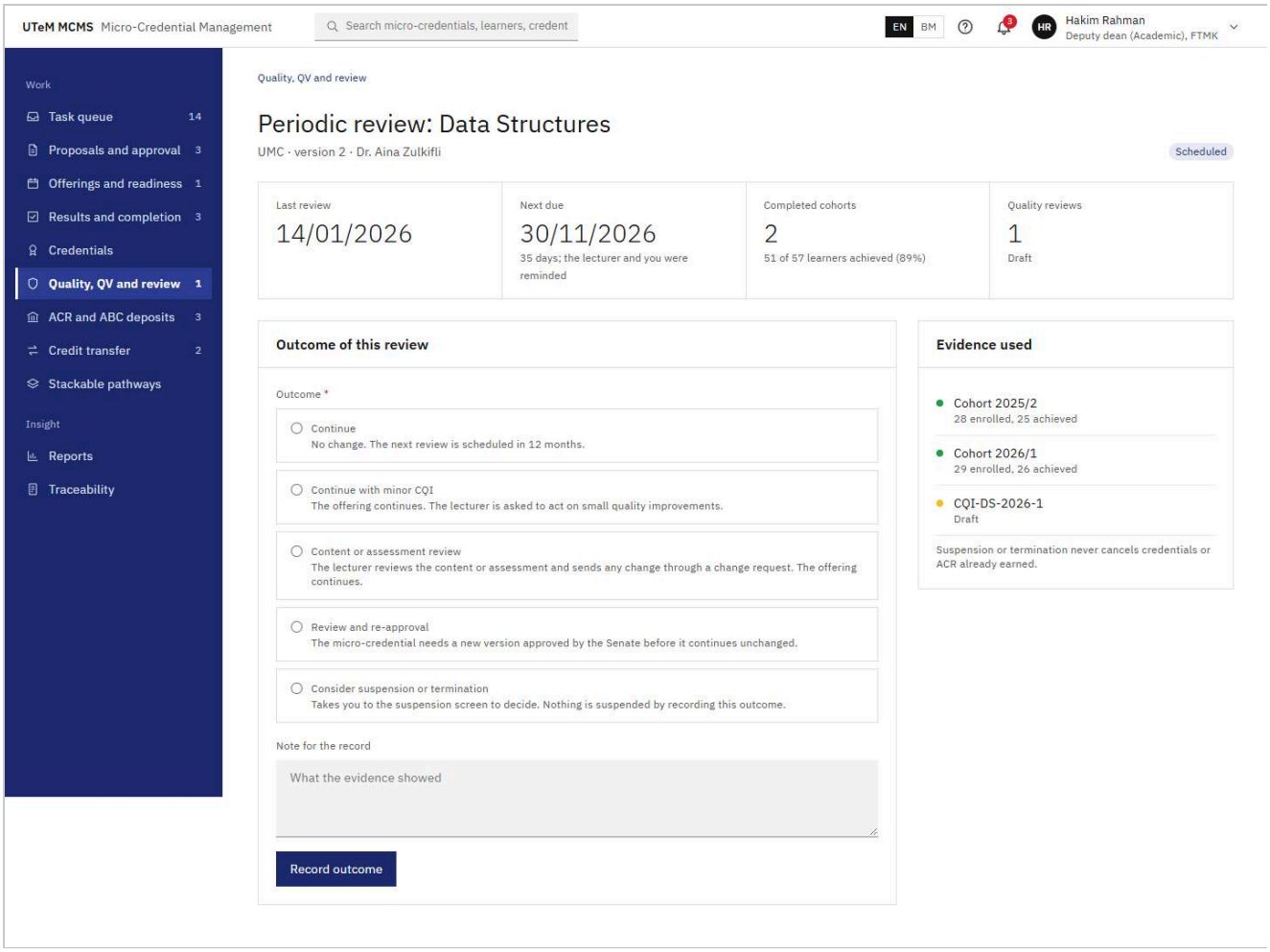


Figure 190: *DD-16 Periodic review.*

DD-17 Suspension or termination

`#/dd/suspend/CND`

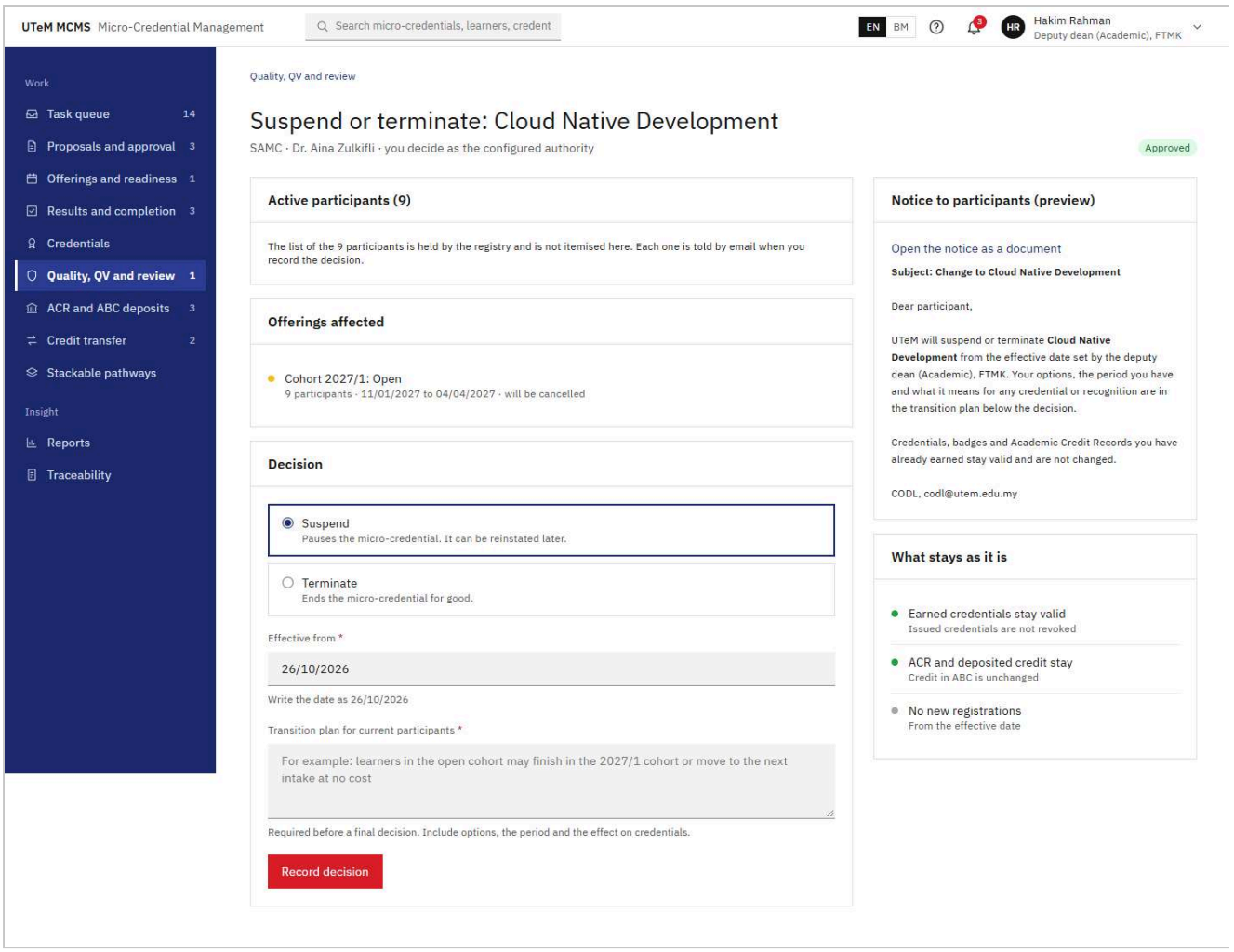


Figure 191: *DD-17 Suspension or termination.*

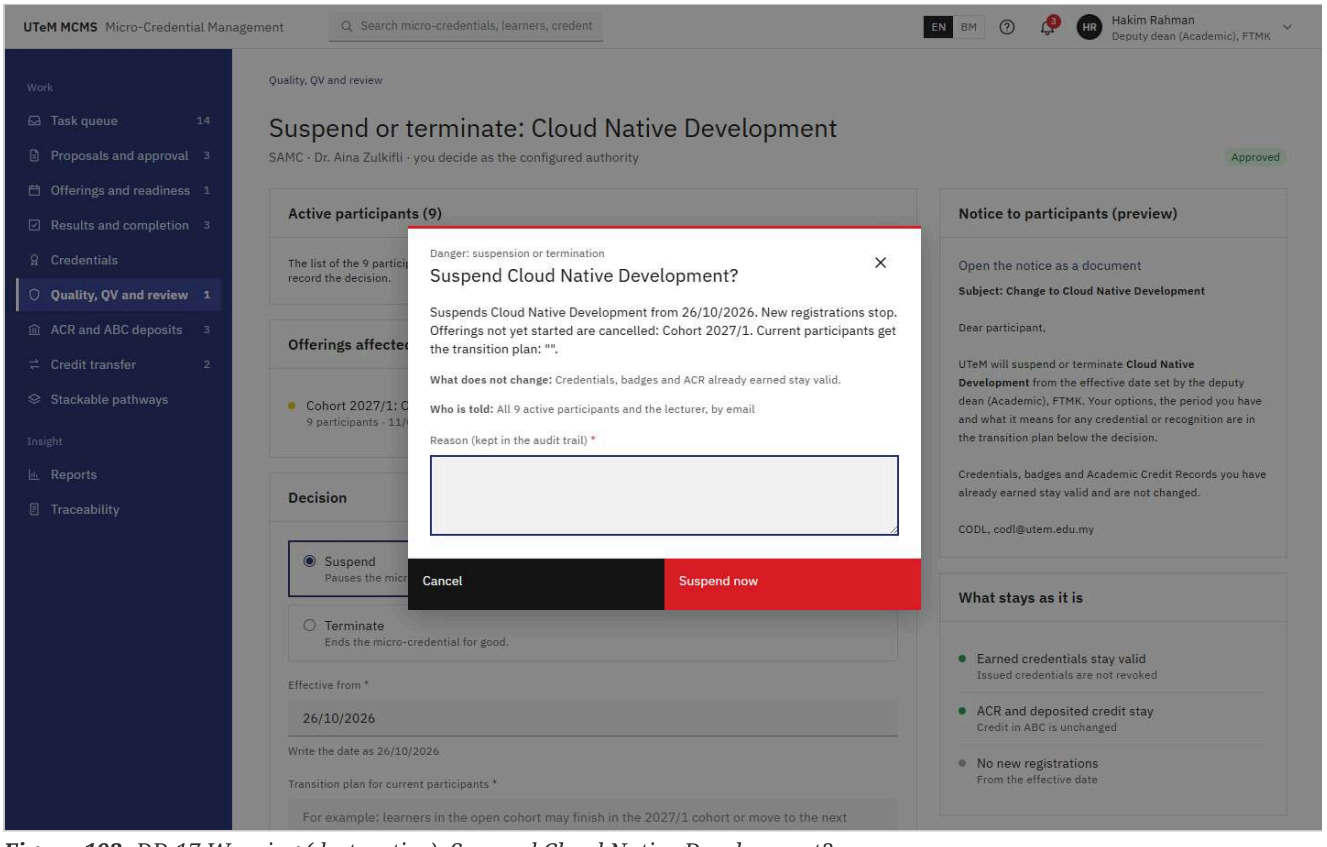


Figure 192: *DD-17 Warning (destructive): Suspend Cloud Native Development?*

DD-18 ACR and ABC deposits

`#/dd/acr`

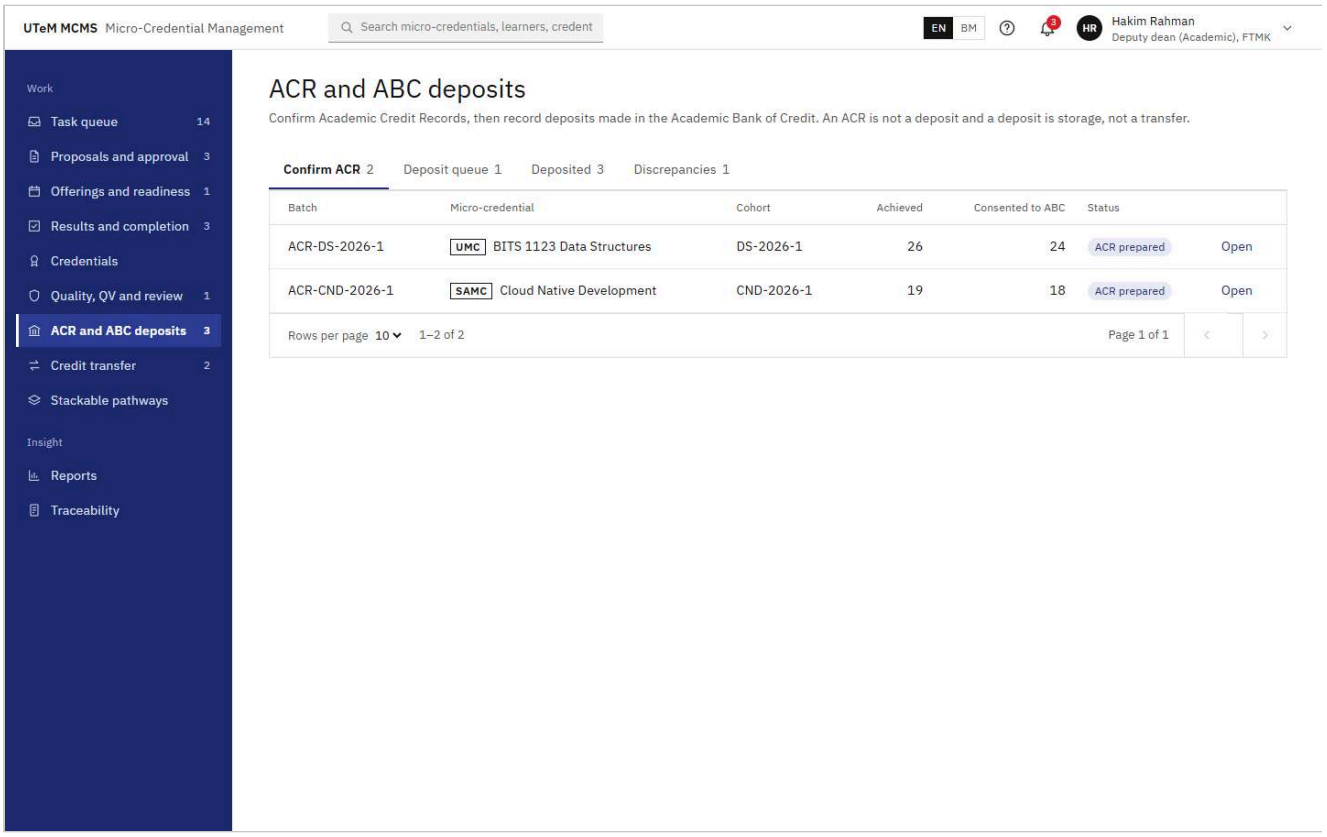


Figure 193: *DD-18 ACR and ABC deposits.*

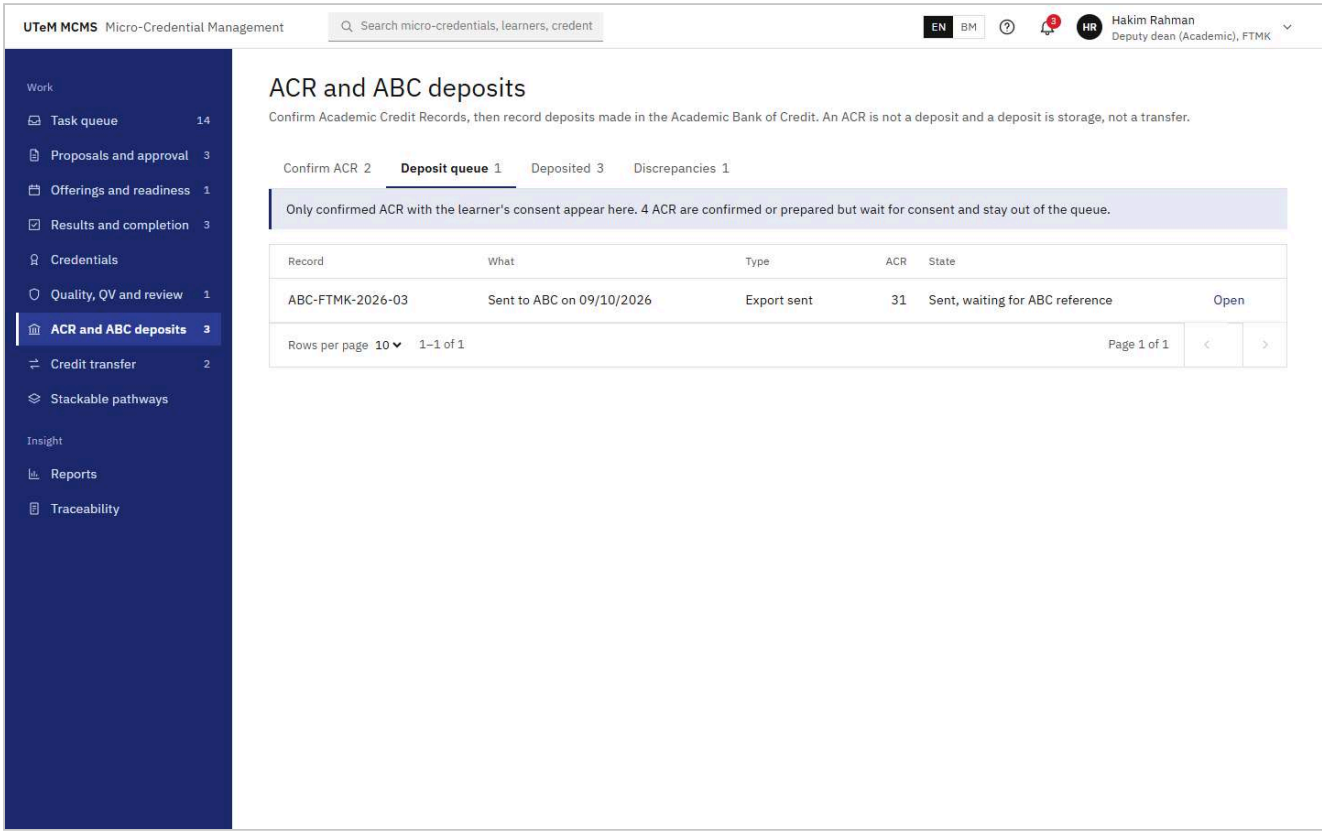


Figure 194: *DD-18 ACR and ABC deposits: Tab: Deposit queue.*

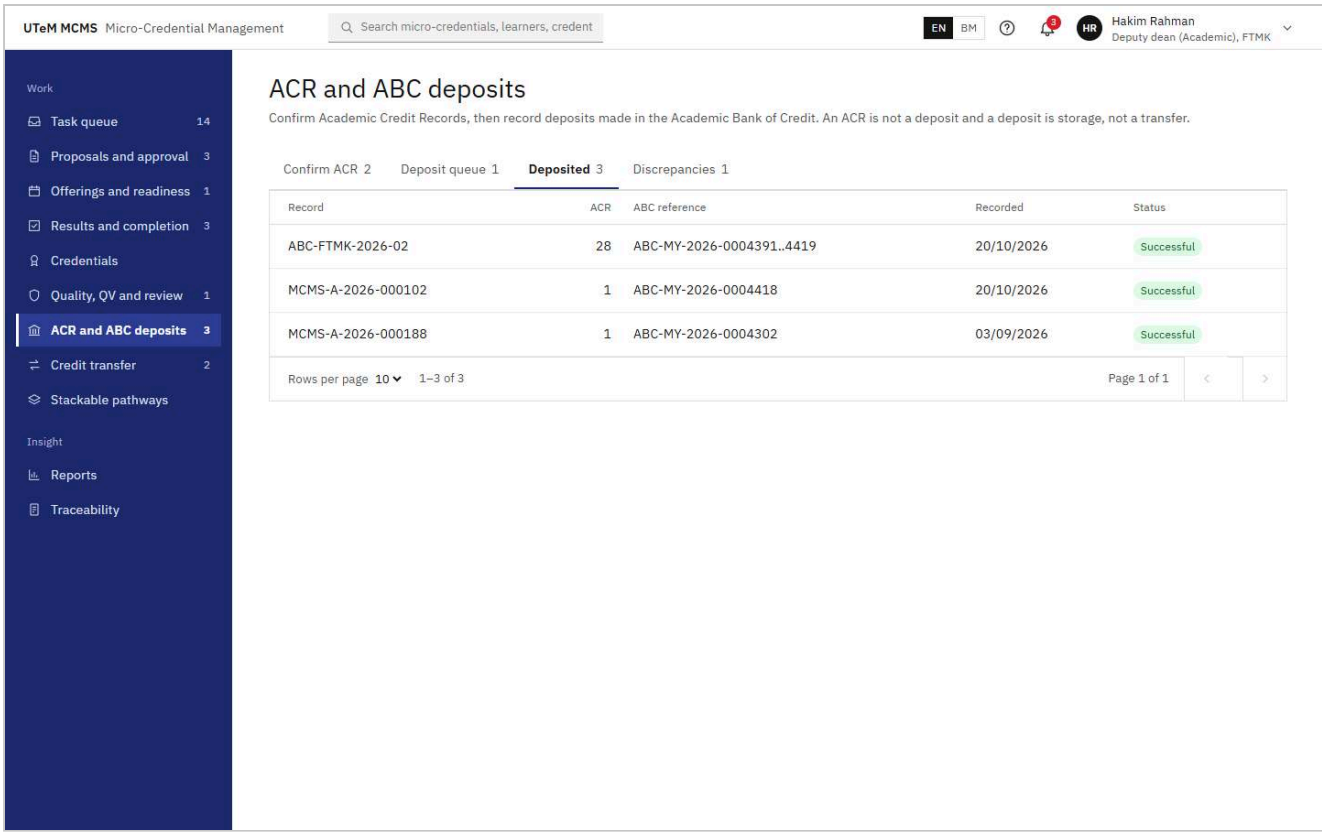


Figure 195: *DD-18 ACR and ABC deposits: Tab: Deposited.*

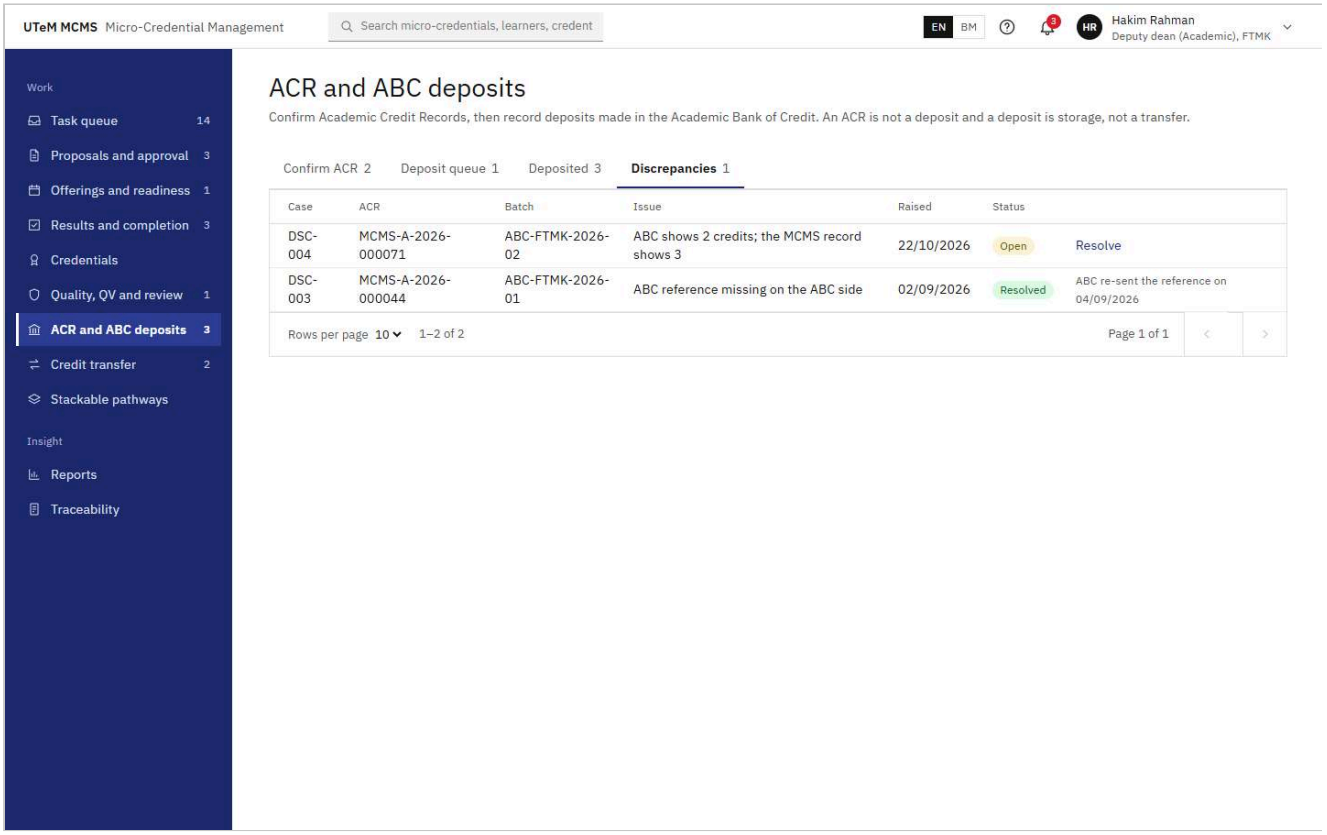


Figure 196: *DD-18 ACR and ABC deposits: Tab: Discrepancies.*

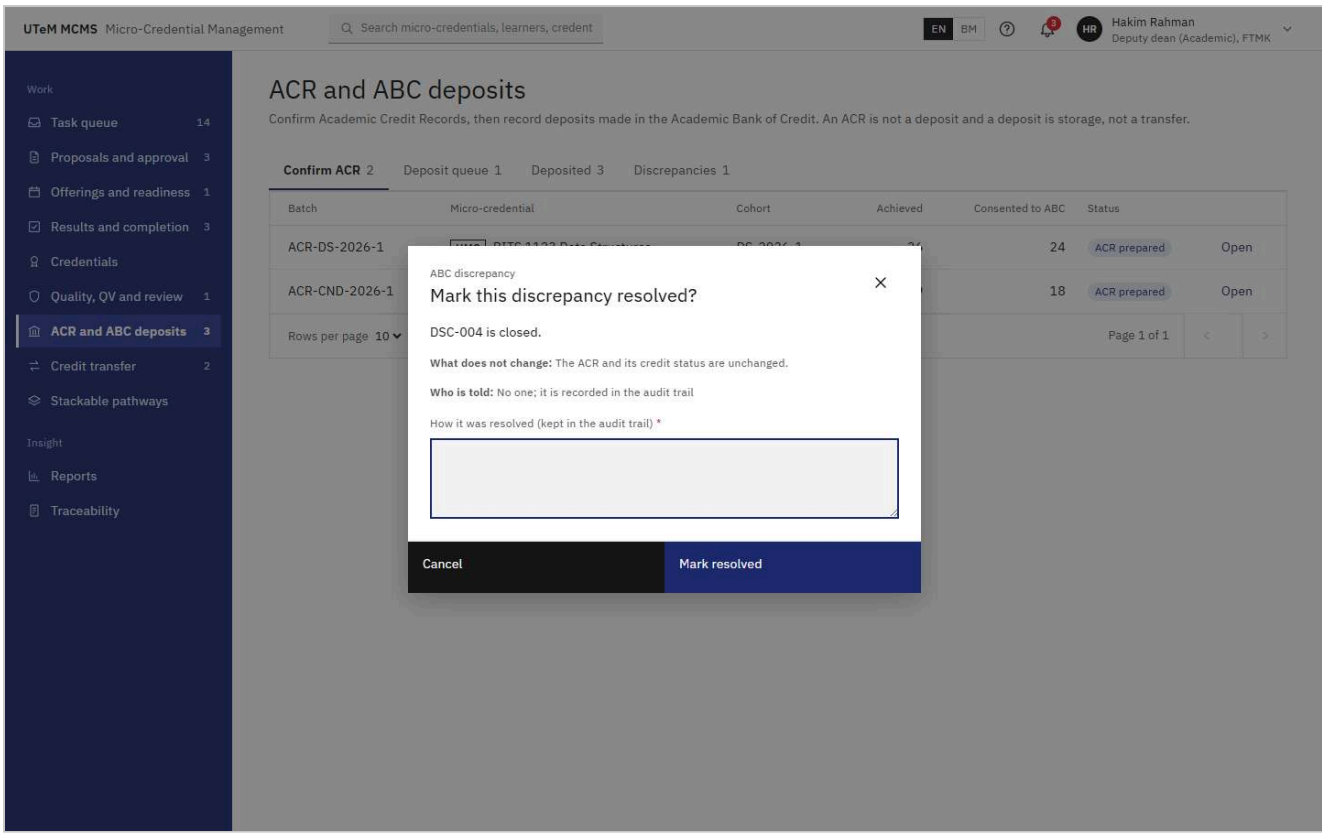


Figure 197: *DD-18 Confirmation: Mark this discrepancy resolved?*

DD-19 Confirm ACR

`#/dd/acr/ACR-DS-2026-1`

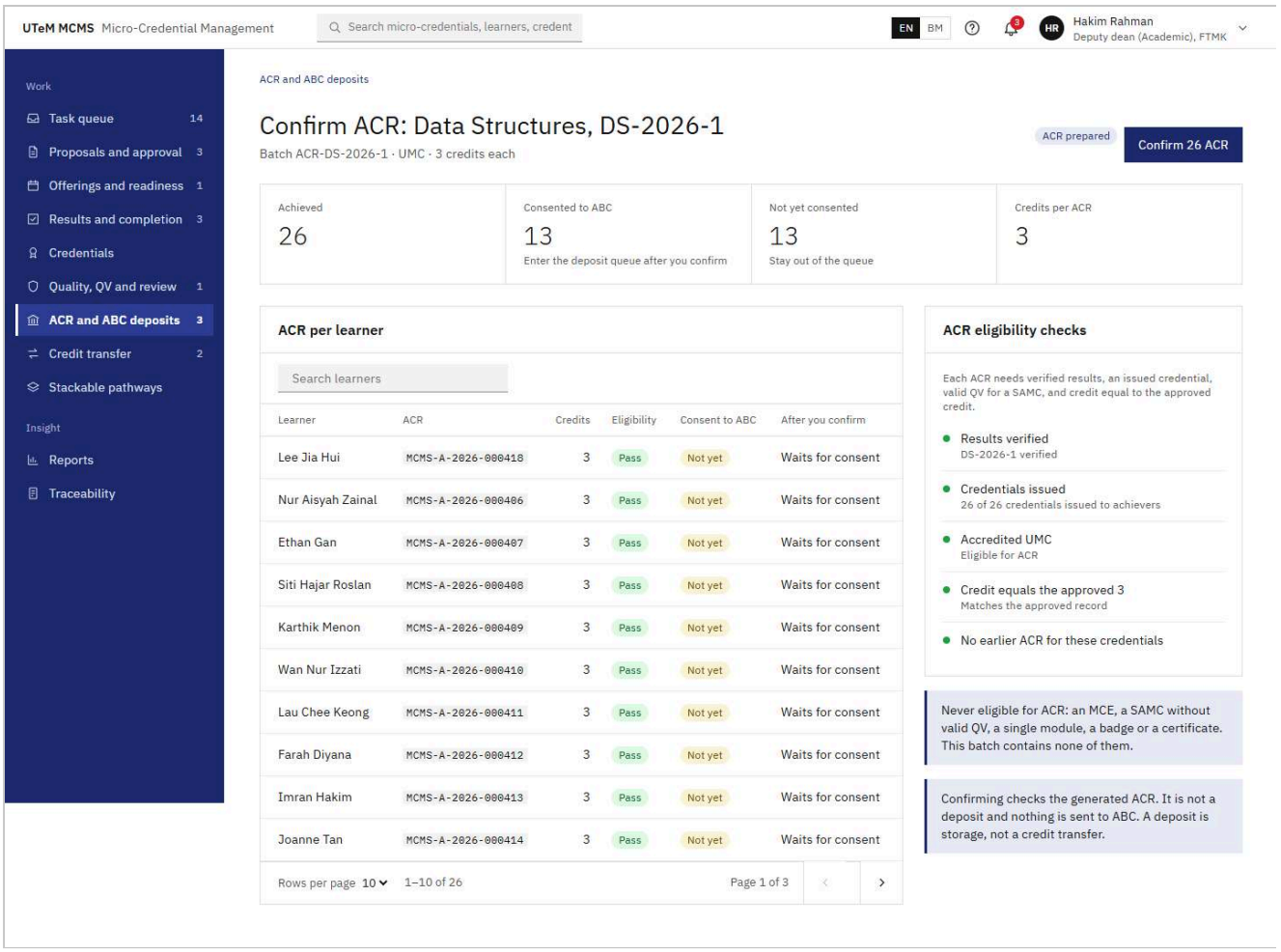


Figure 198: *DD-19 Confirm ACR.*

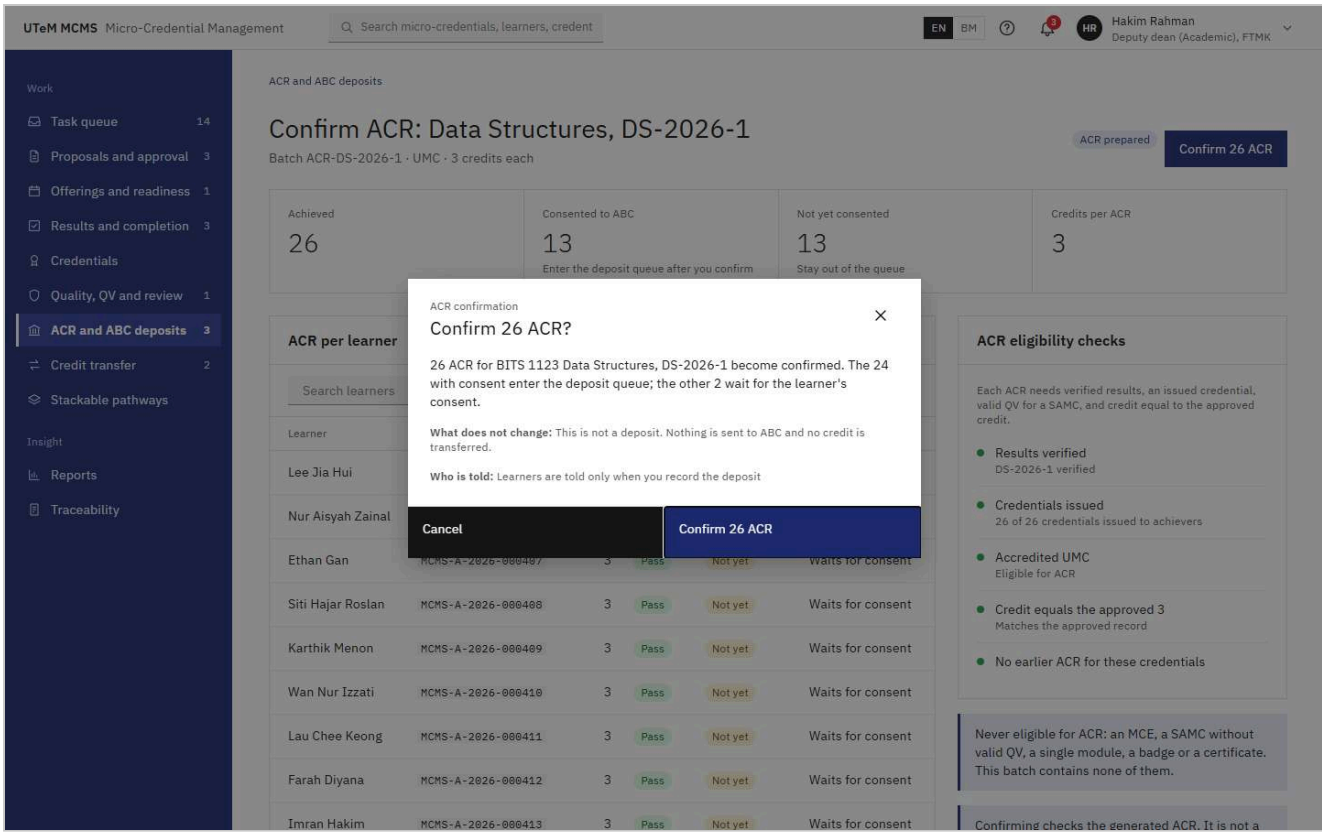


Figure 199: *DD-19 Confirmation: Confirm 26 ACR?*

DD-20 Record ABC deposit

`#/dd/deposit/ABC-FTMK-2026-03`

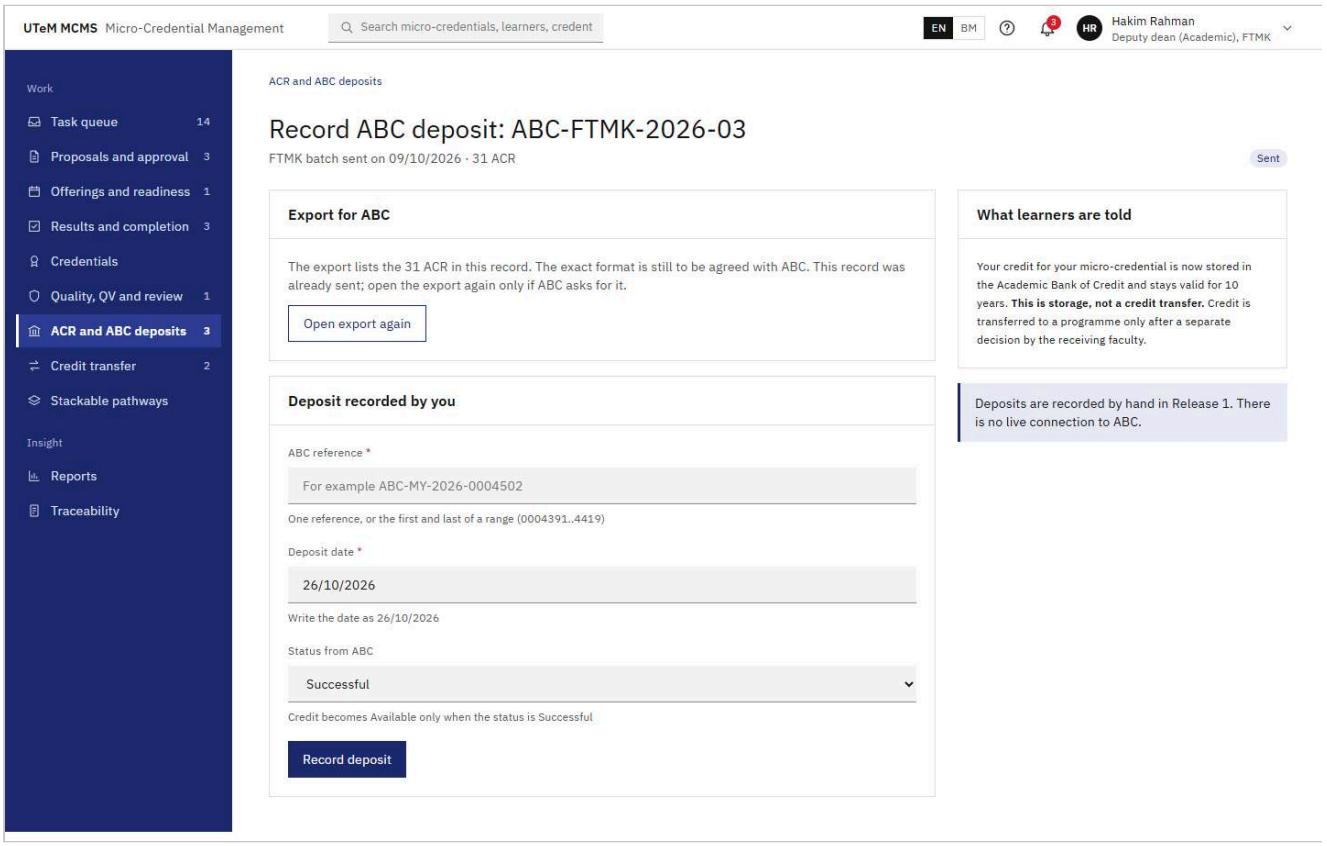


Figure 200: *DD-20 Record ABC deposit.*

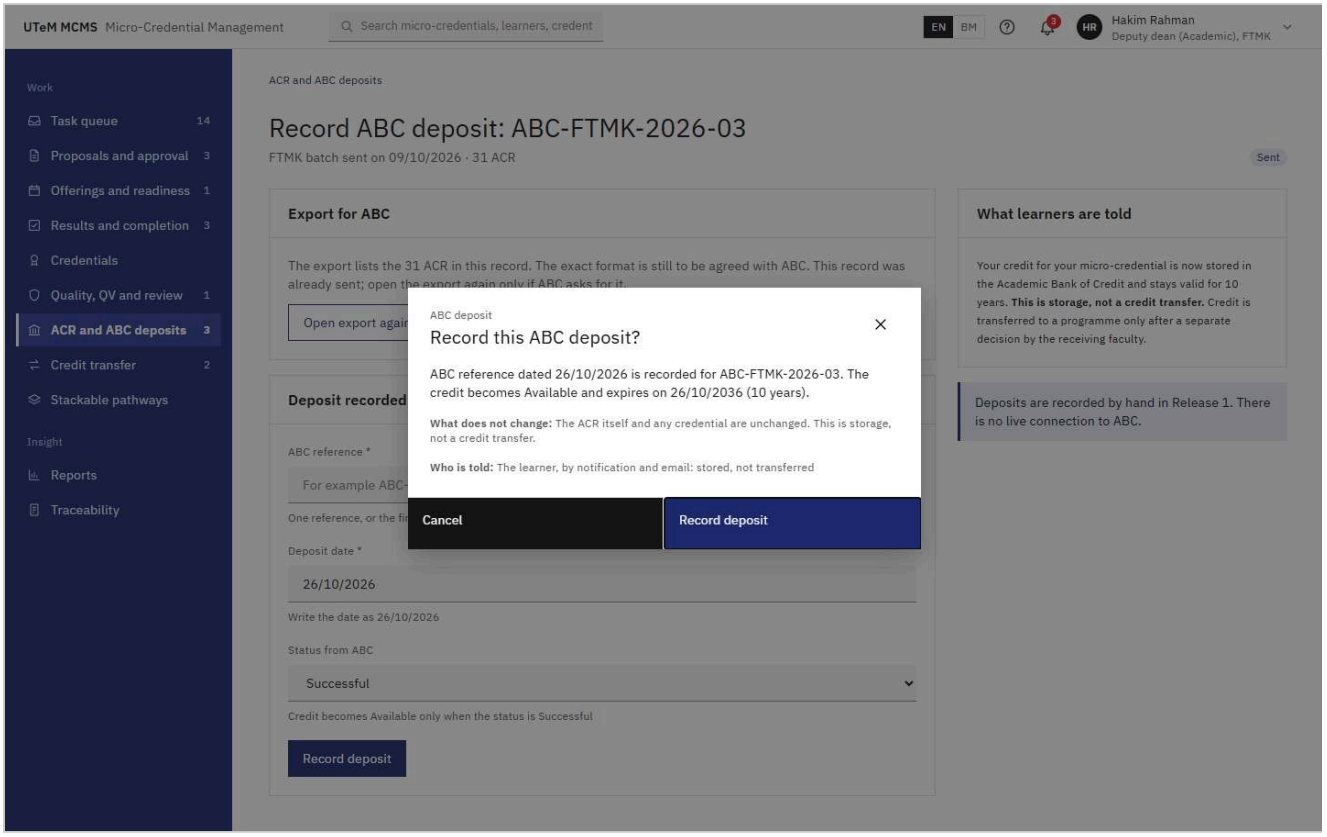


Figure 201: *DD-20 Confirmation: Record this ABC deposit?*

DD-21 Credit transfer

`#/dd/transfer`

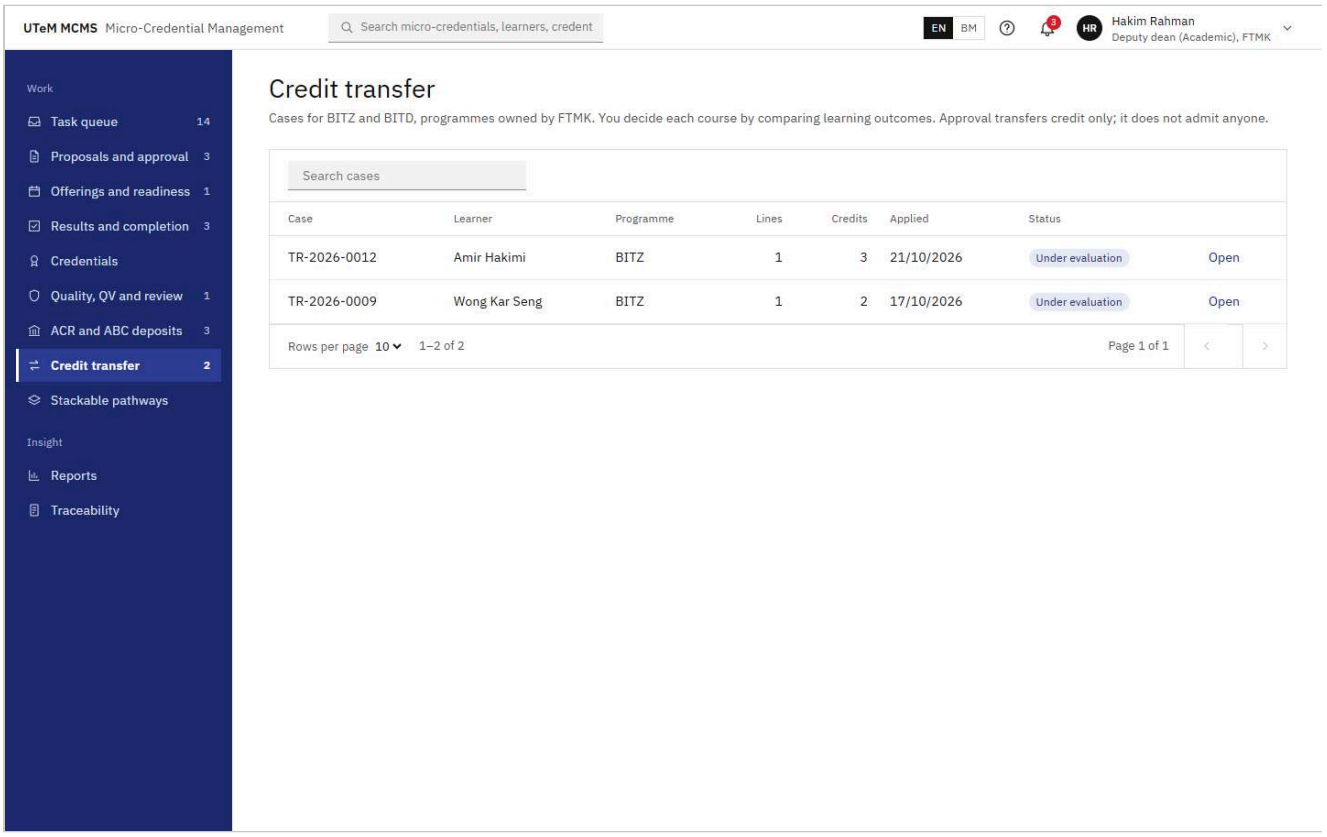


Figure 202: *DD-21 Credit transfer.*

DD-22 Transfer decision

`#/dd/transfer/TR-2026-0012`

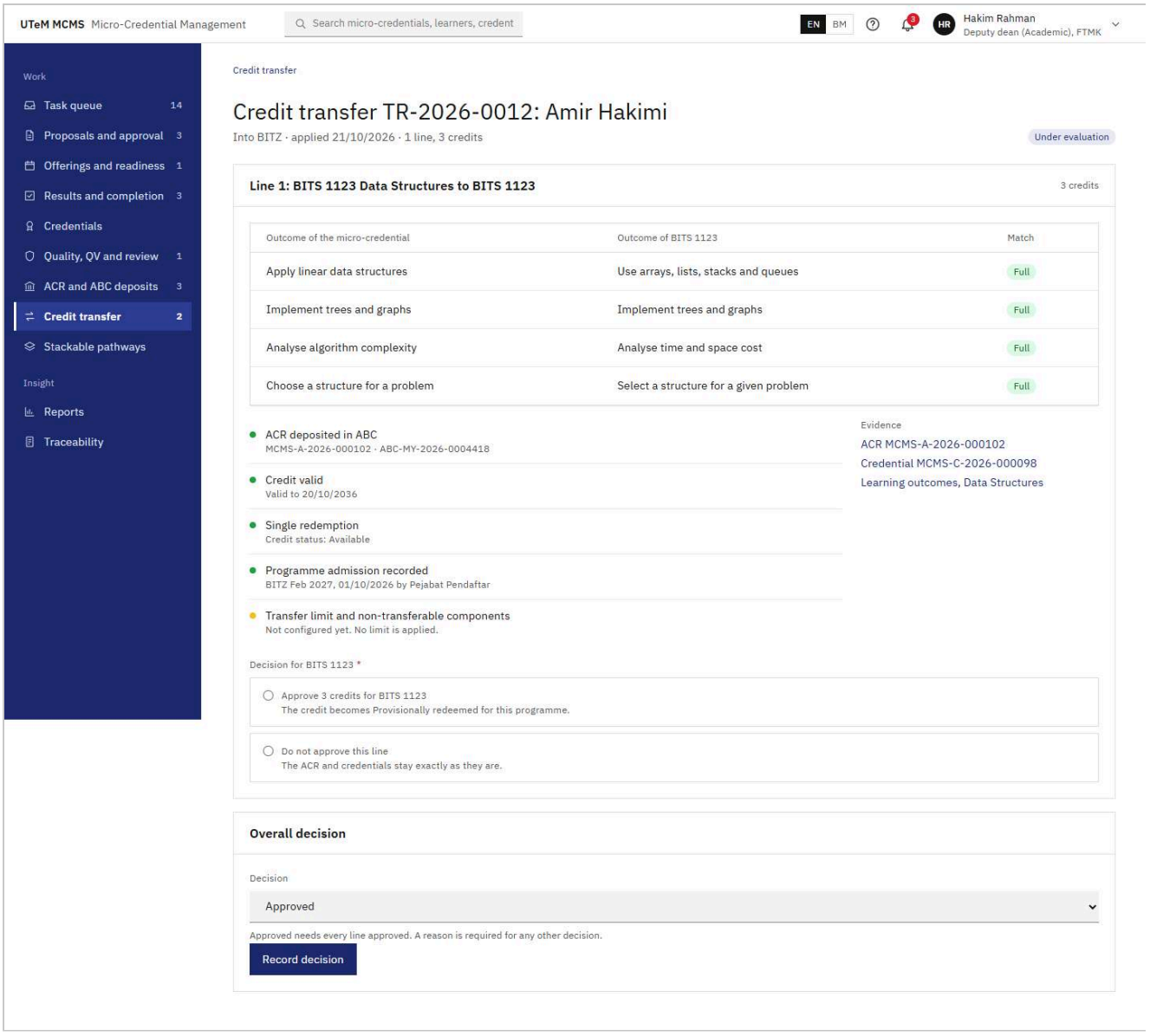


Figure 203: *DD-22 Transfer decision.*

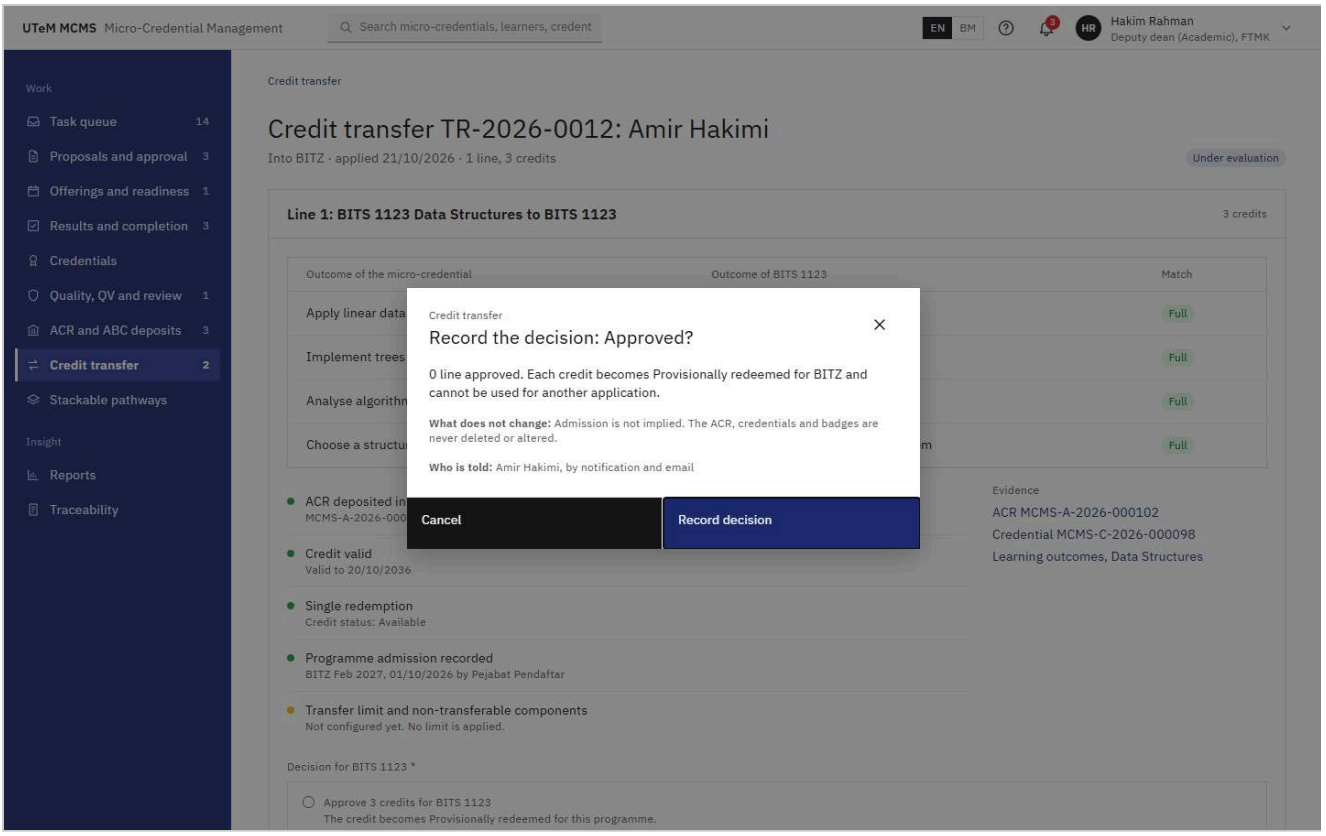


Figure 204: *DD-22 Confirmation: Record the decision: Approved?*

DD-23 Stackable pathways

`#/dd/stack`

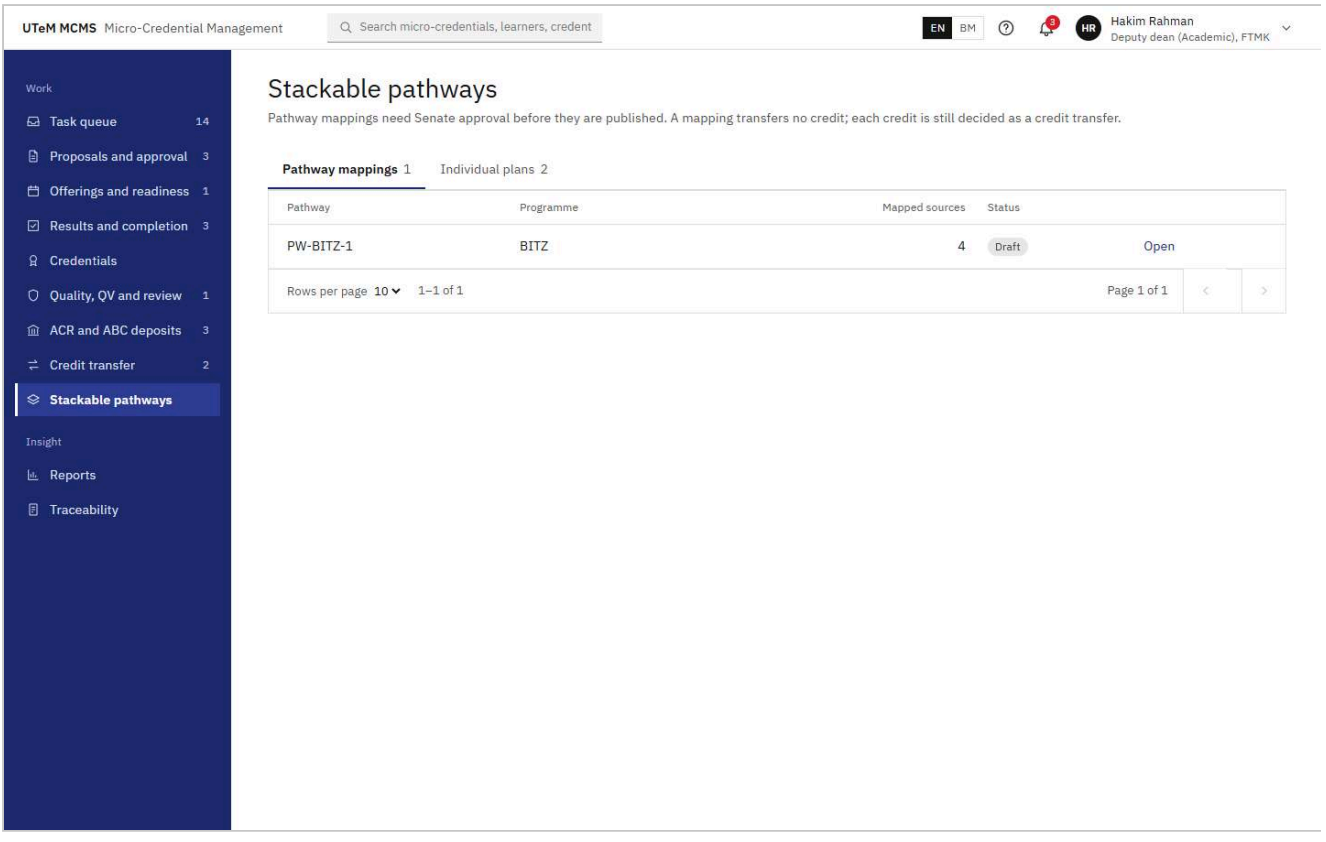


Figure 205: *DD-23 Stackable pathways.*

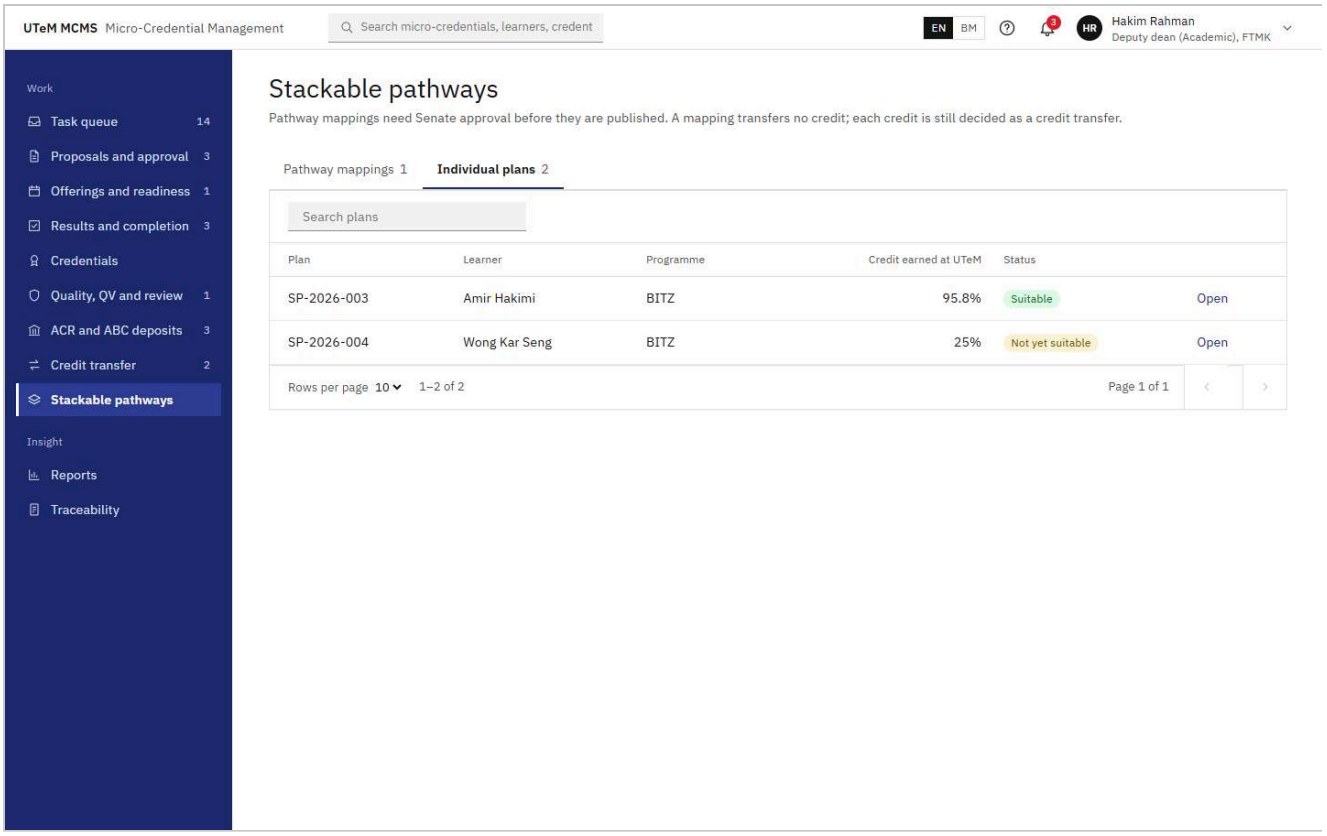


Figure 206: *DD-23 Stackable pathways: Tab: Individual plans.*

DD-24 Pathway mapping

`#/dd/pathway/PW-BITZ-1`

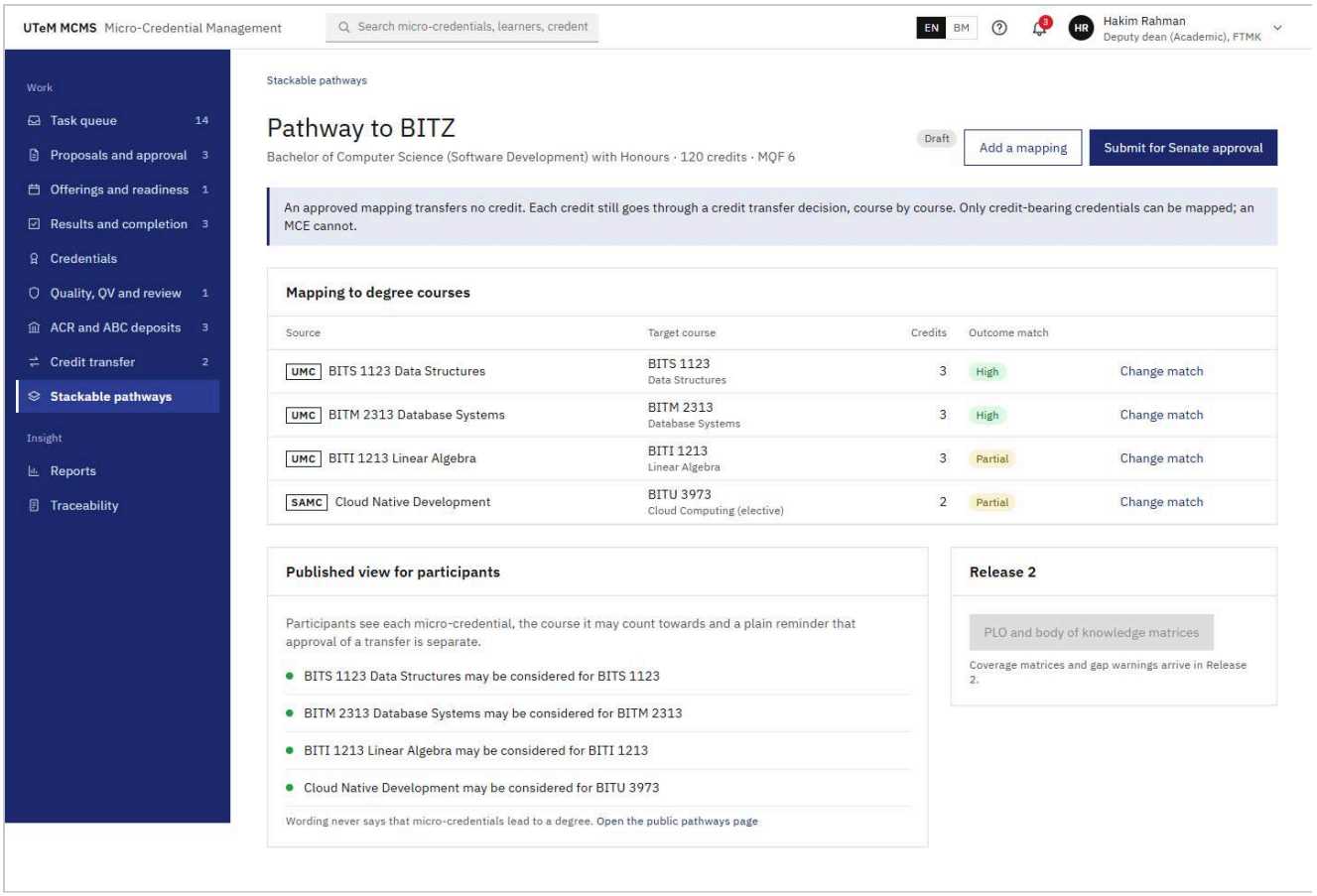


Figure 207: *DD-24 Pathway mapping.*

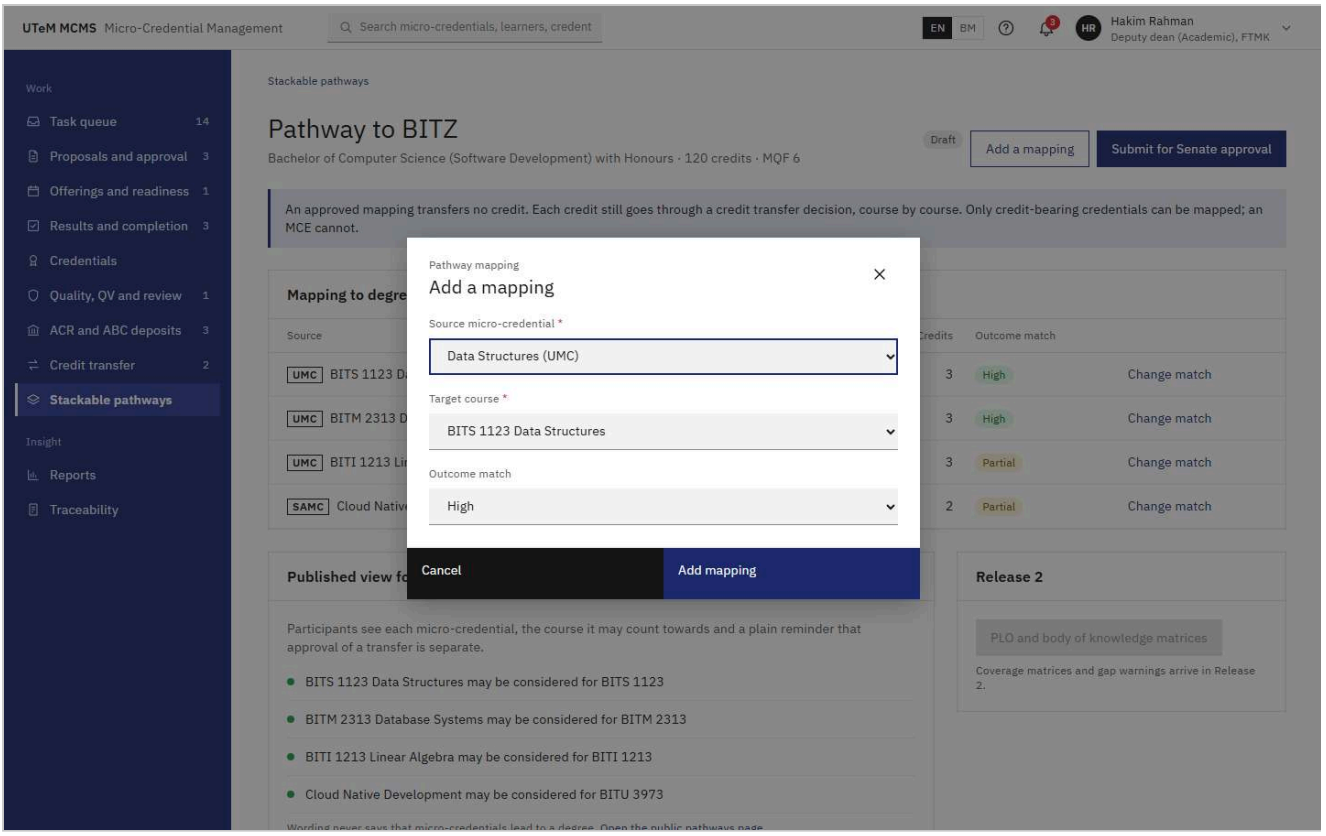


Figure 208: *DD-24 Dialog: Add a mapping.*

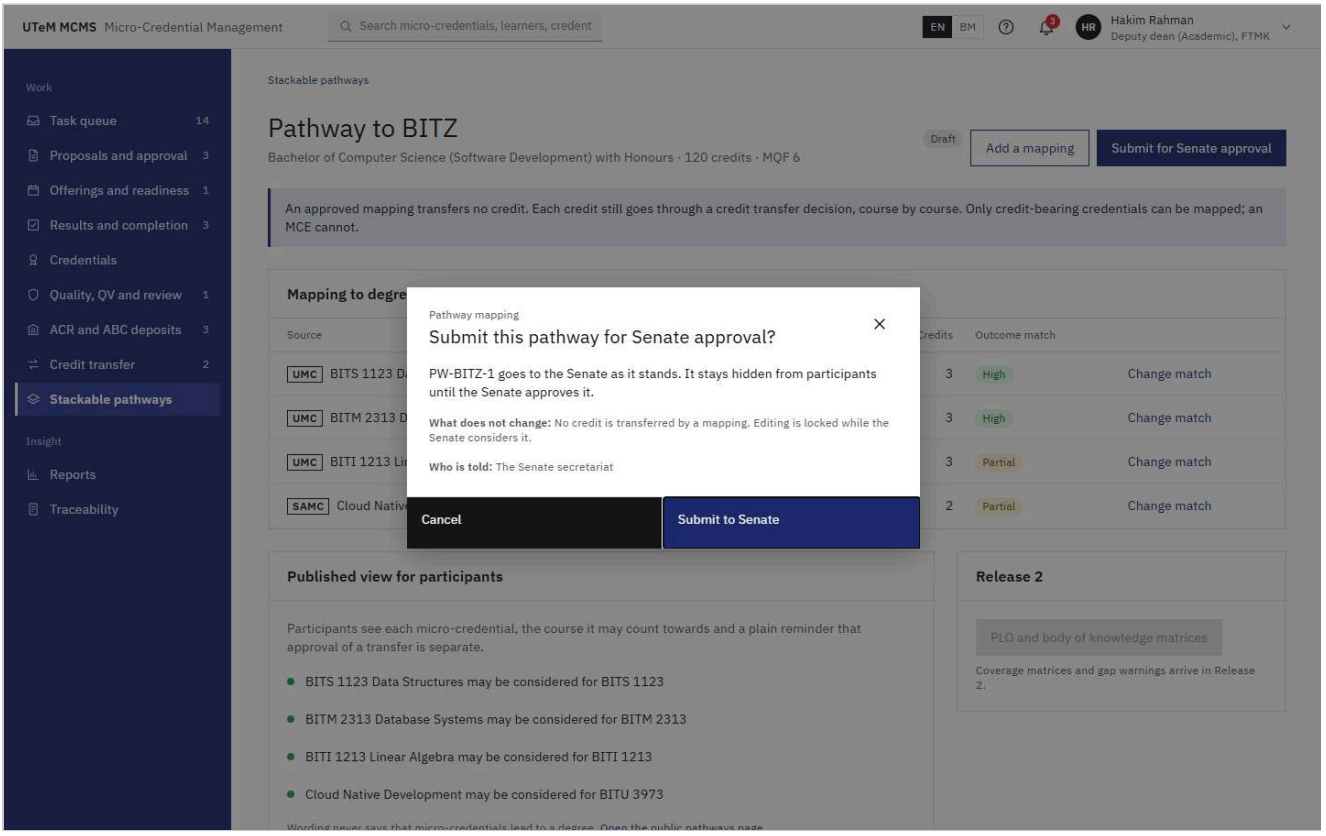


Figure 209: *DD-24 Confirmation: Submit this pathway for Senate approval?*

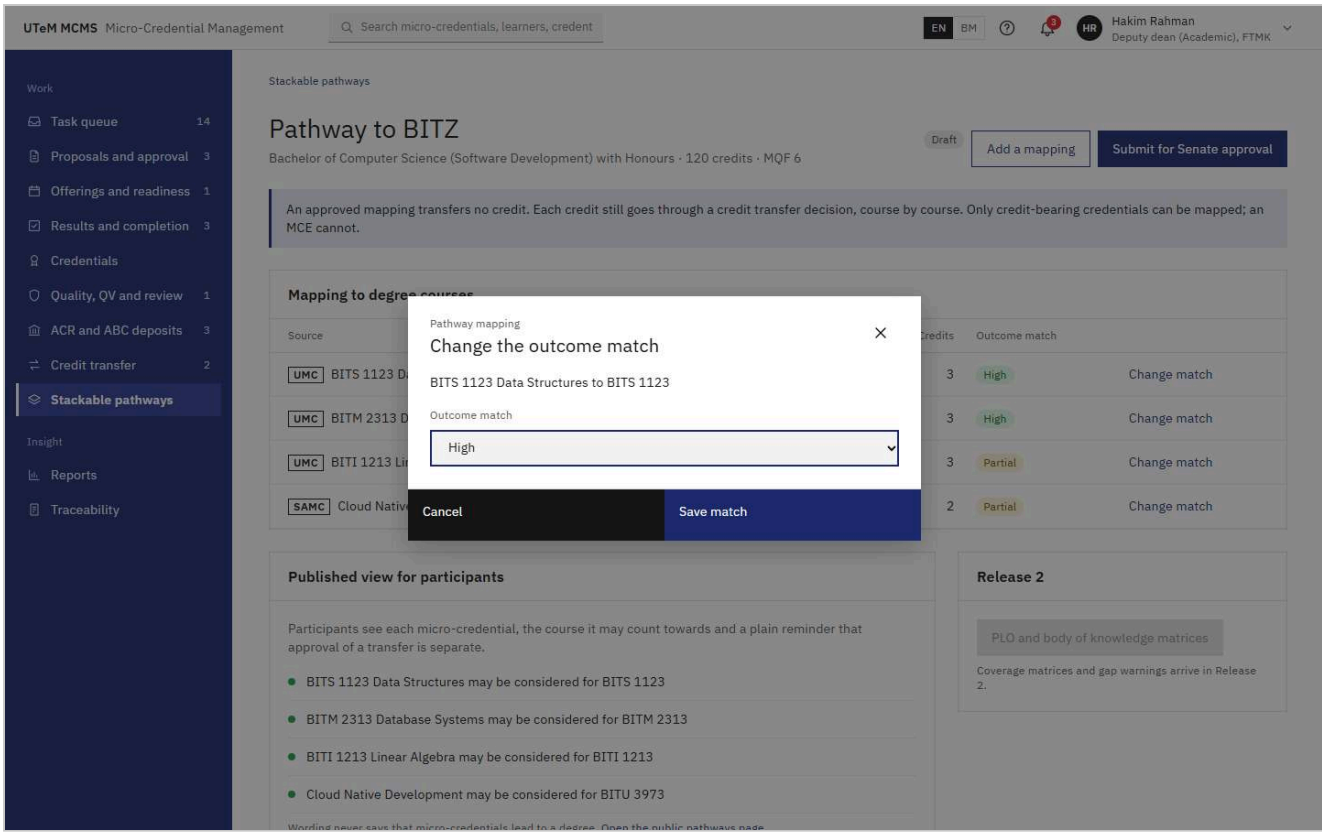


Figure 210: *DD-24 Dialog: Change the outcome match.*

DD-25 Individual stackable plan

`#/dd/plan/SP-2026-004`

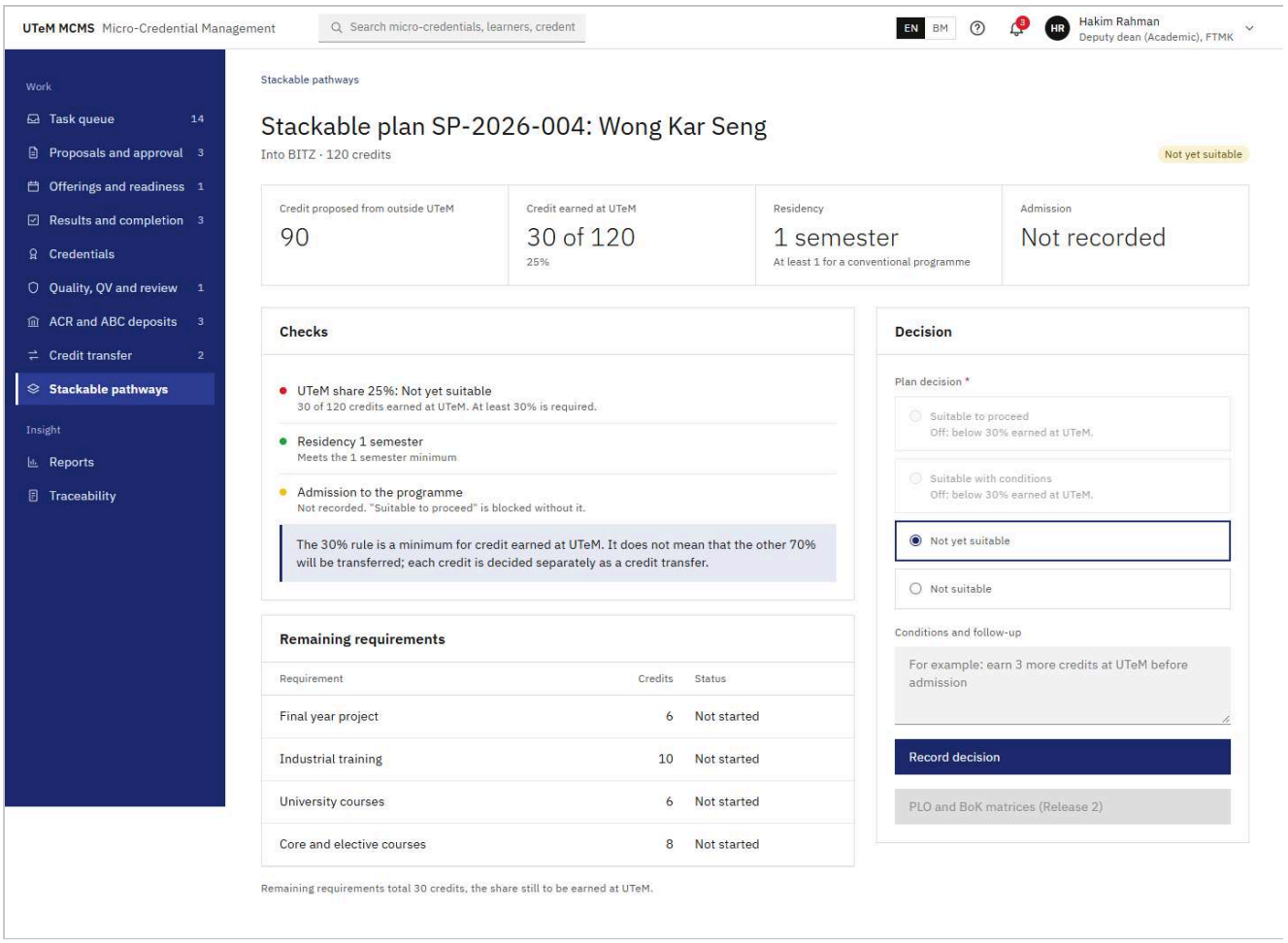


Figure 211: *DD-25 Individual stackable plan.*

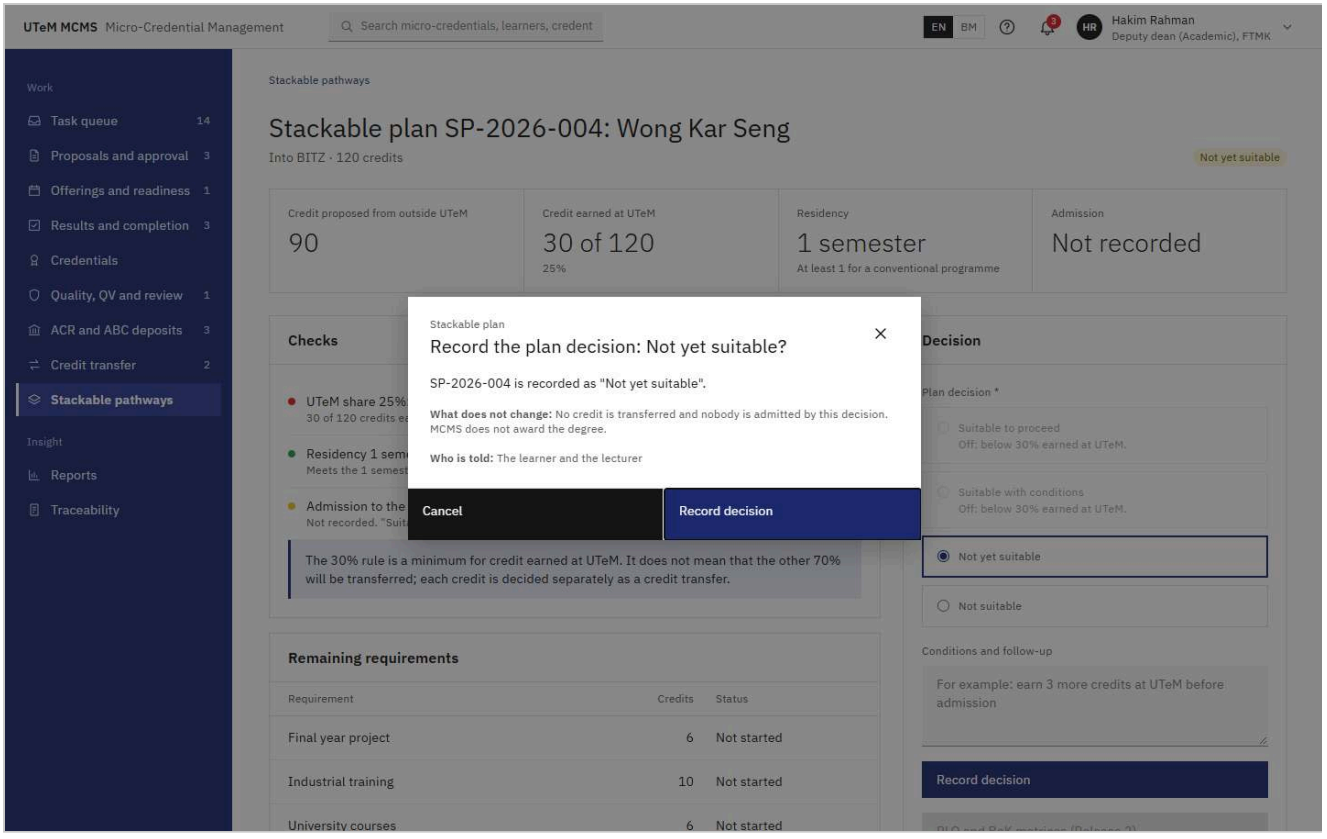


Figure 212: *DD-25 Confirmation: Record the plan decision: Not yet suitable?*

DD-26 Reports

`#/dd/reports`

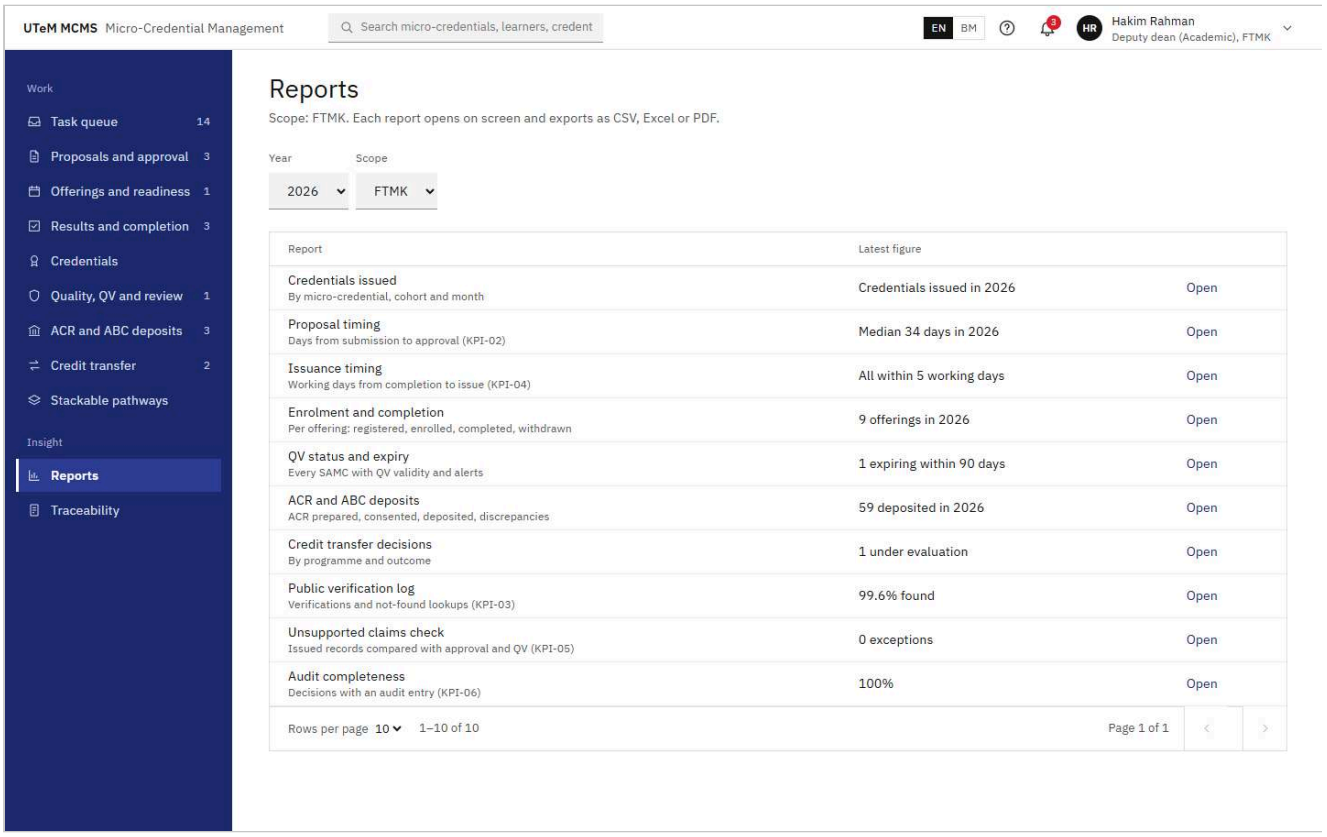


Figure 213: *DD-26 Reports.*

DD-27 Report

`#/dd/report/credentials-issued`

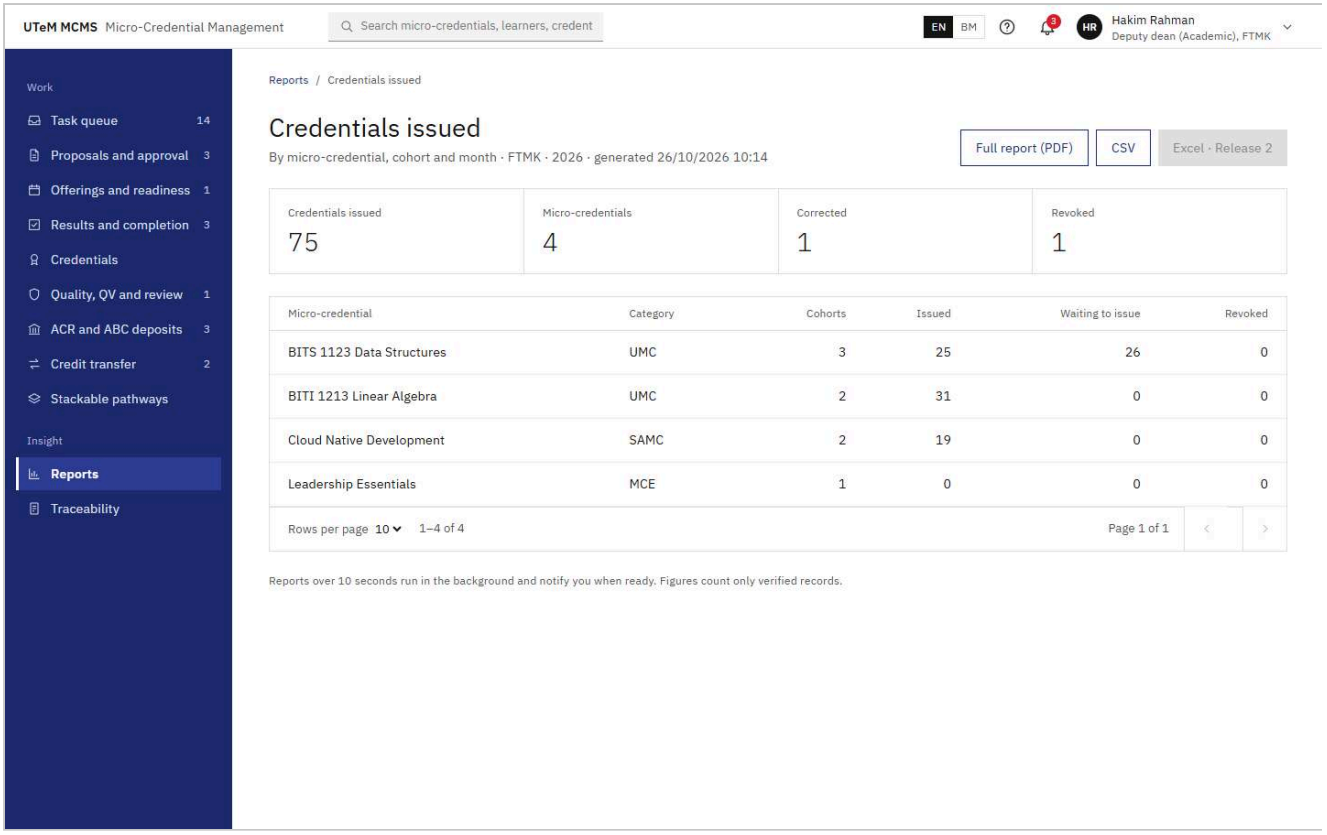


Figure 214: *DD-27 Report.*

DD-28 Traceability

`#/dd/trace`

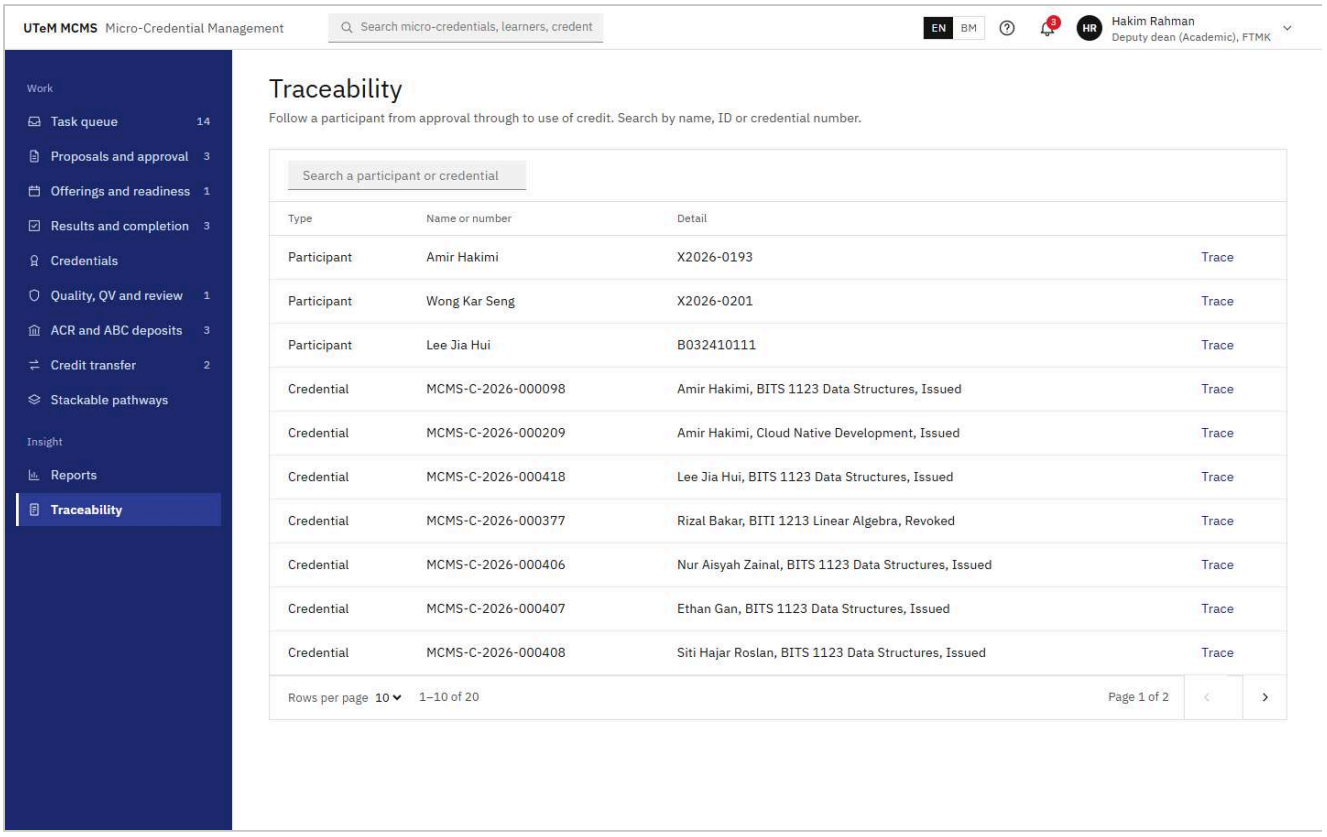


Figure 215: *DD-28 Traceability.*

DD-29 Trace a participant

`#/dd/trace/amir`

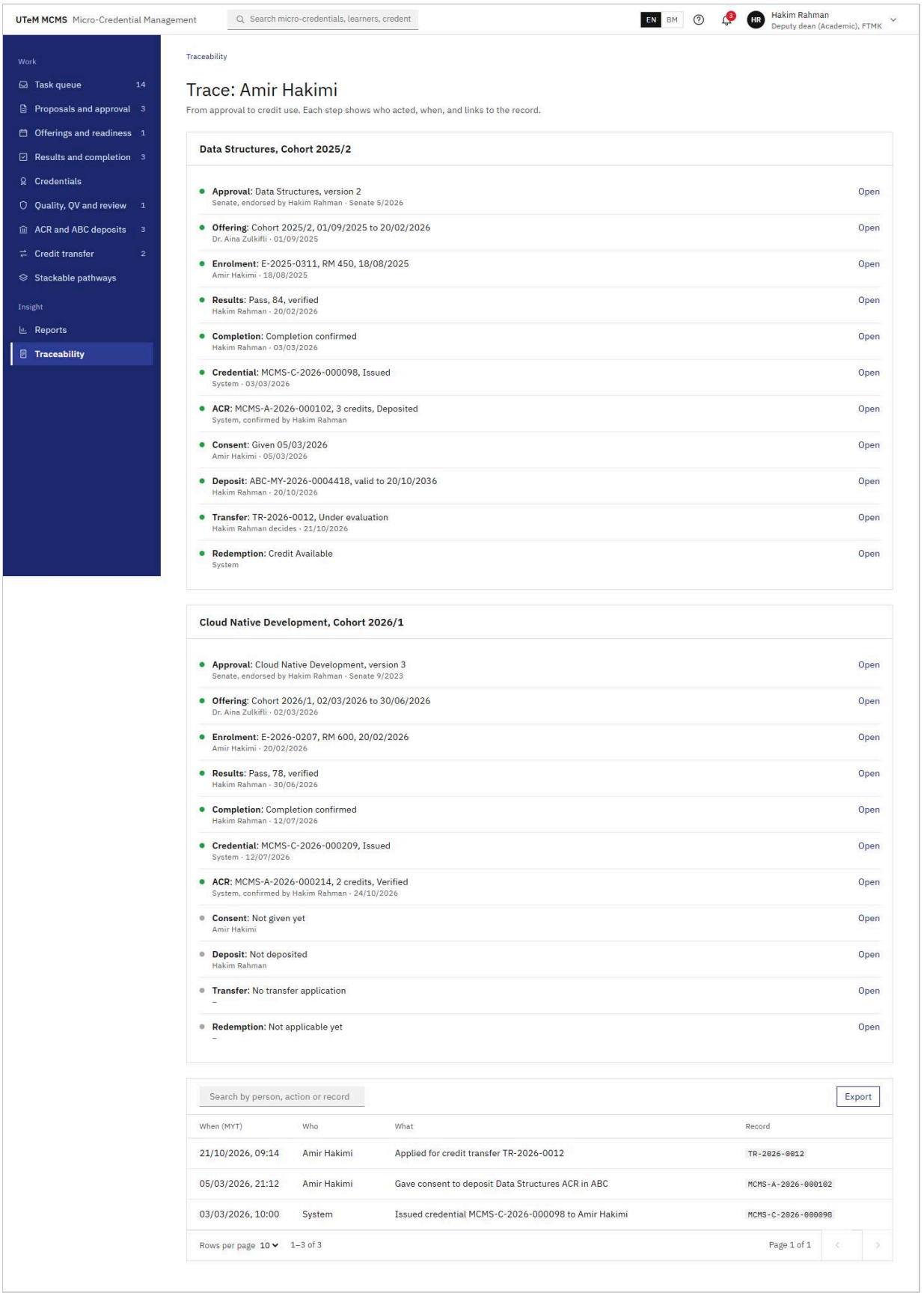


Figure 216: *DD-29 Trace a participant.*

Proposal review (all comments resolved, category verified)

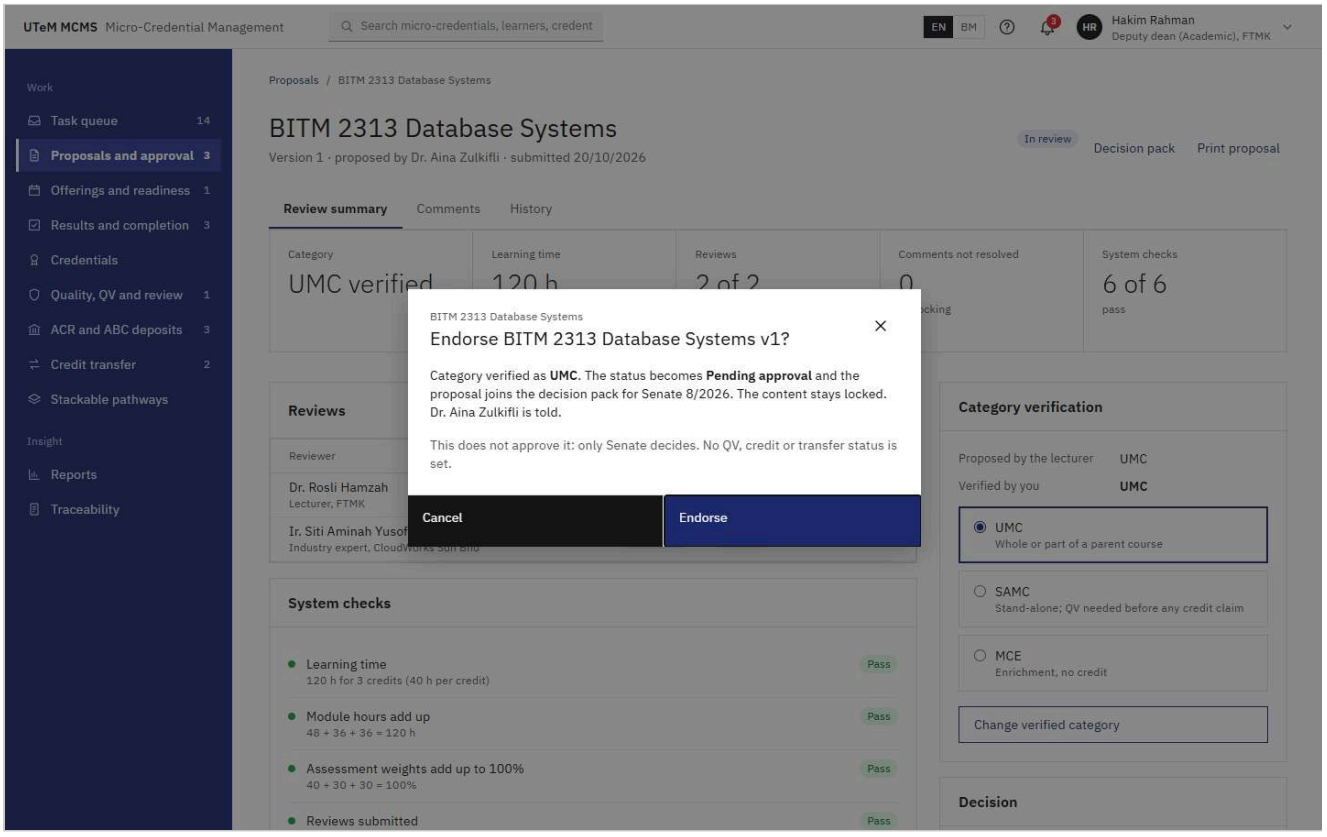


Figure 217: *Confirmation: Endorse BITM 2313 Database Systems v1? (record state: Proposal review (all comments resolved, category verified)).*

Confirm readiness (checklist sent, evidence accepted)

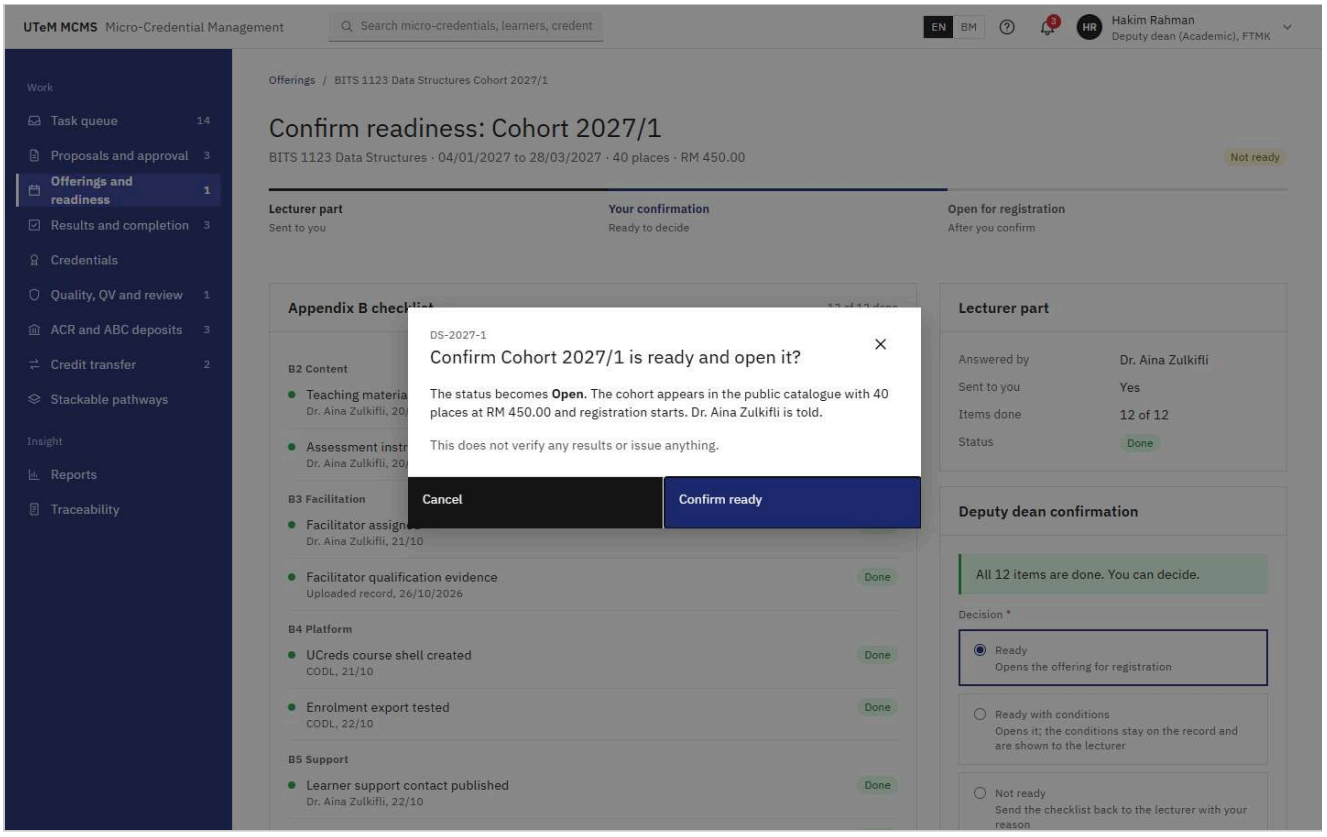


Figure 218: *Confirmation: Confirm Cohort 2027/1 is ready and open it? (record state: Confirm readiness (checklist sent, evidence accepted)).*

Verify results (held learner)

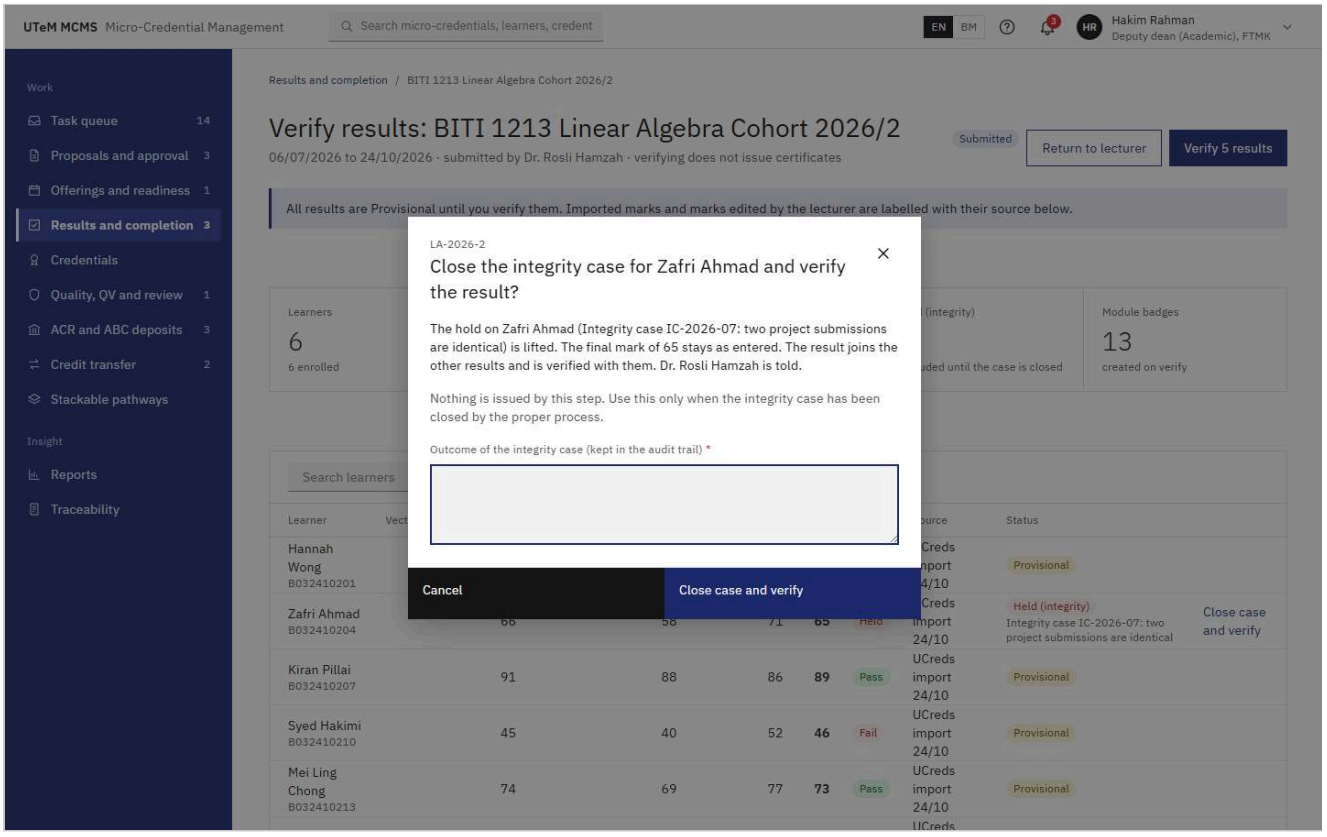


Figure 219: *Confirmation: Close the integrity case for Zafri Ahmad and verify the result? (record state: Verify results (held learner)).*

QV record (preparing)

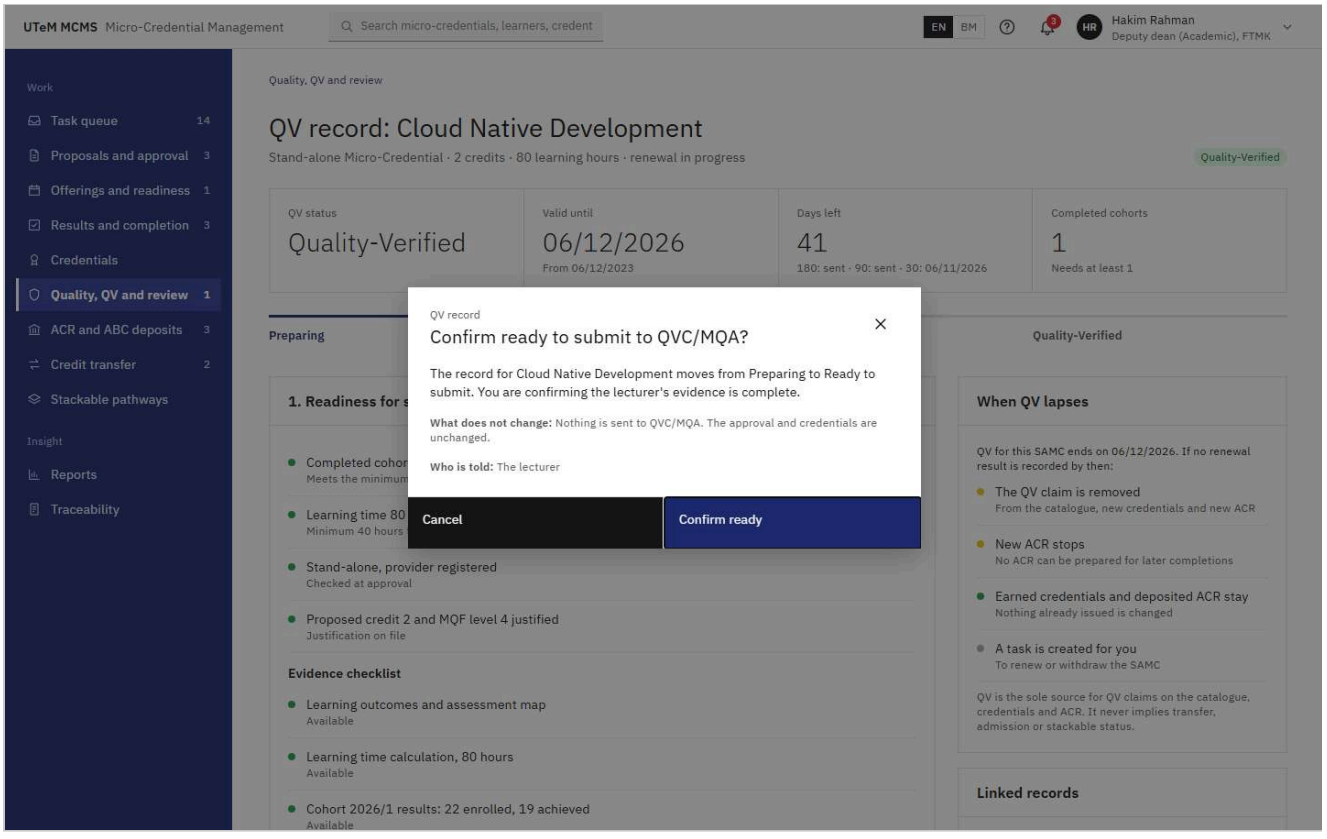


Figure 220: *Confirmation: Confirm ready to submit to QVC/MQA? (record state: QV record (preparing)).*

QV record (submitted, record the result)

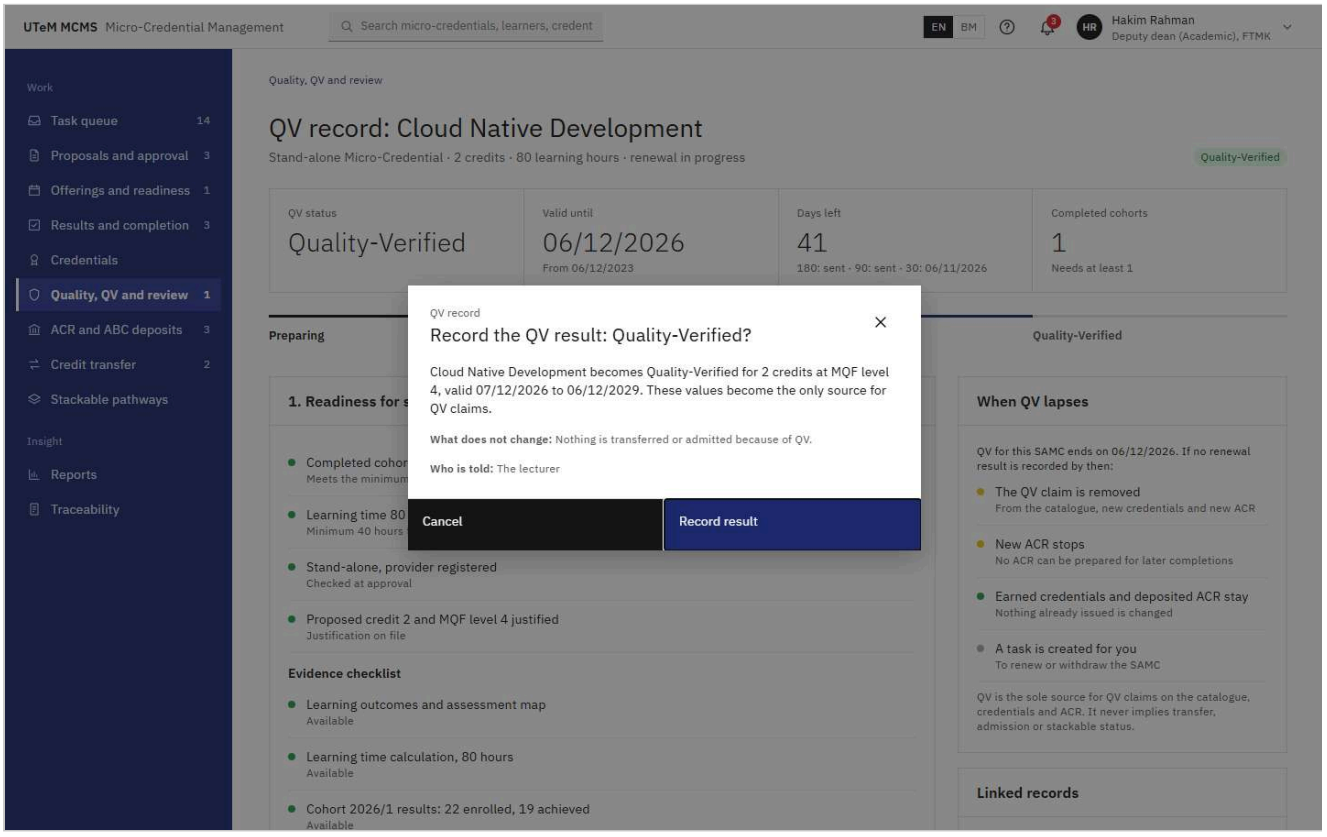


Figure 221: *Confirmation: Record the QV result: Quality-Verified? (record state: QV record (submitted, record the result)).*

Periodic review (outcome chosen)

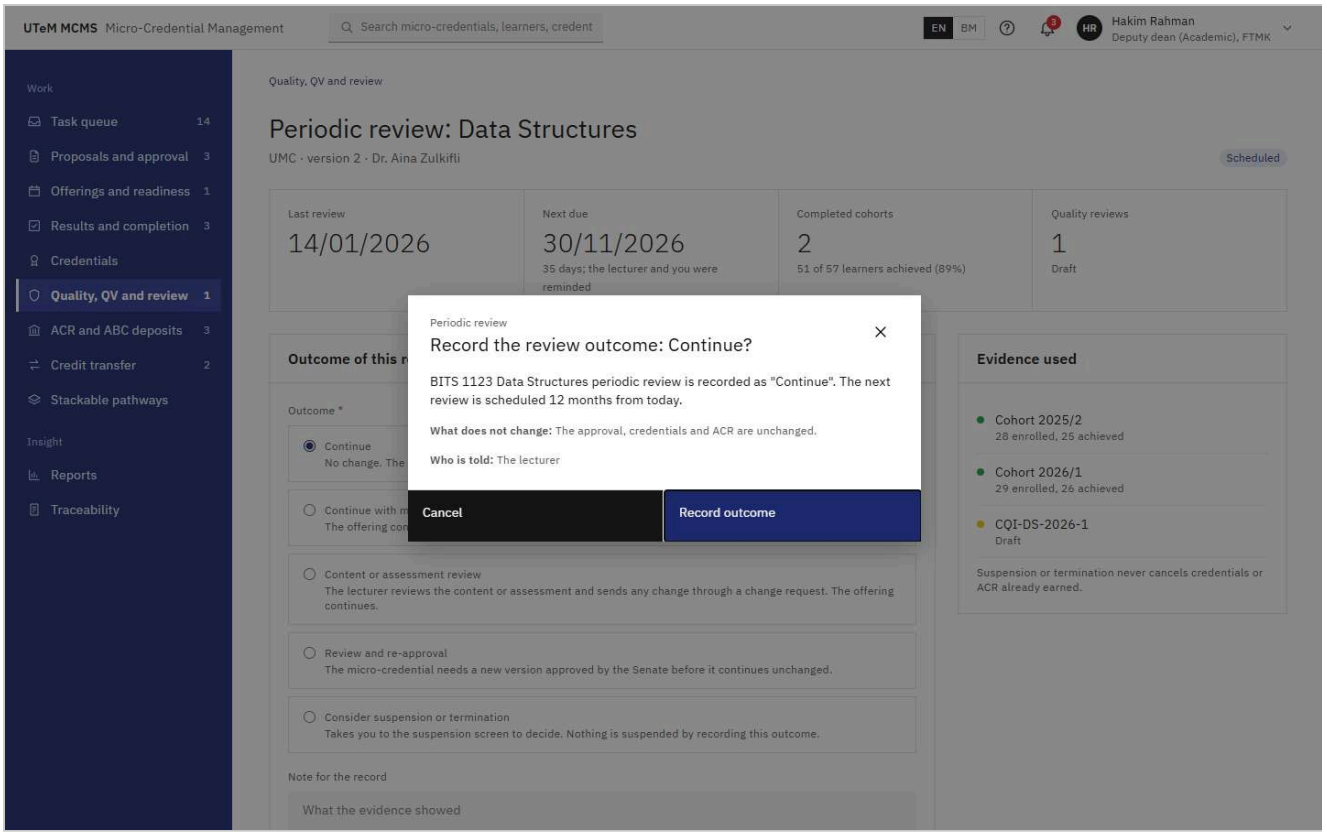


Figure 222: *Confirmation: Record the review outcome: Continue? (record state: Periodic review (outcome chosen)).*

Credit transfer (approved, provisionally redeemed)

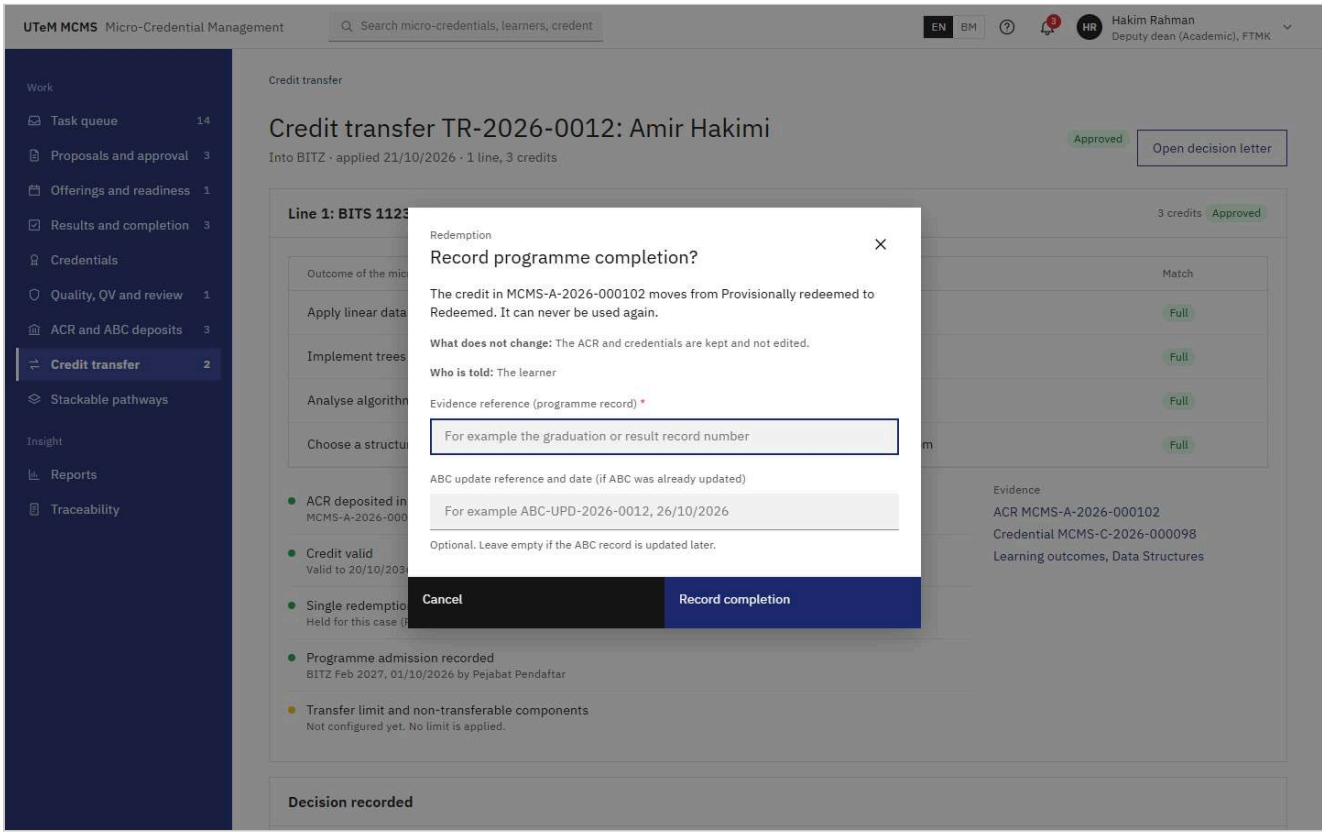


Figure 223: *Confirmation: Record programme completion? (record state: Credit transfer (approved, provisionally redeemed)).*

Credit transfer (approved, provisionally redeemed)

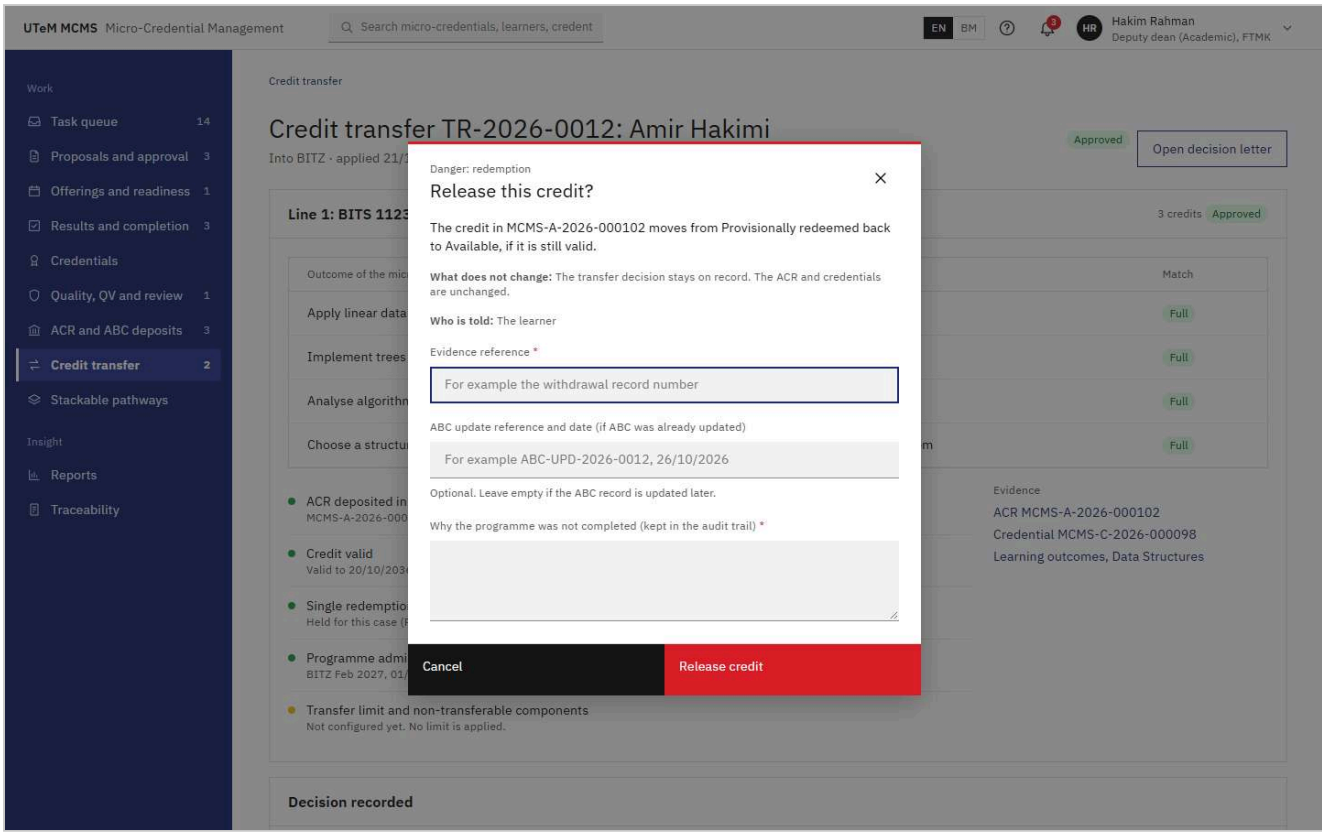


Figure 224: *Warning (destructive): Release this credit? (record state: Credit transfer (approved, provisionally redeemed)).*

Pathway mapping (submitted to the Senate)

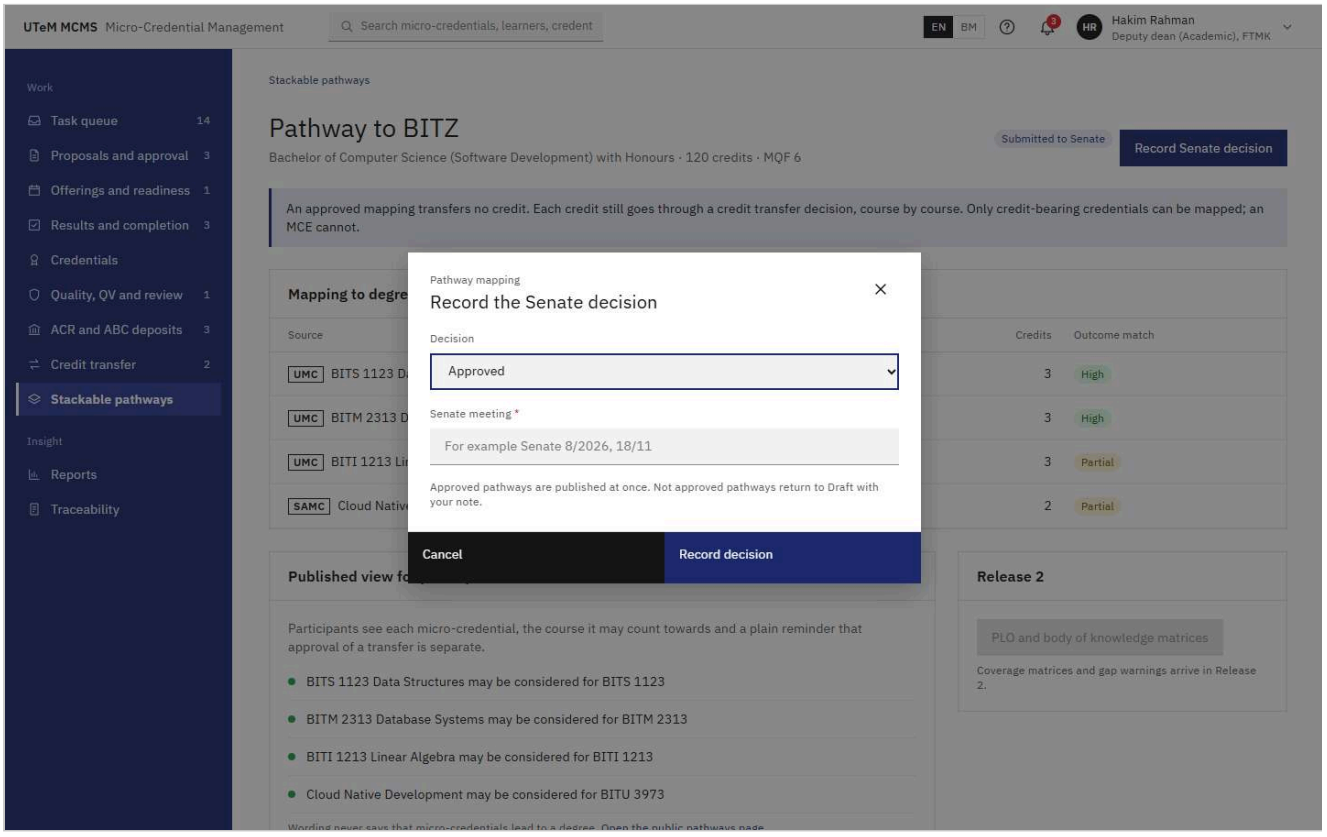


Figure 225: *Dialog: Record the Senate decision (record state: Pathway mapping (submitted to the Senate)).*

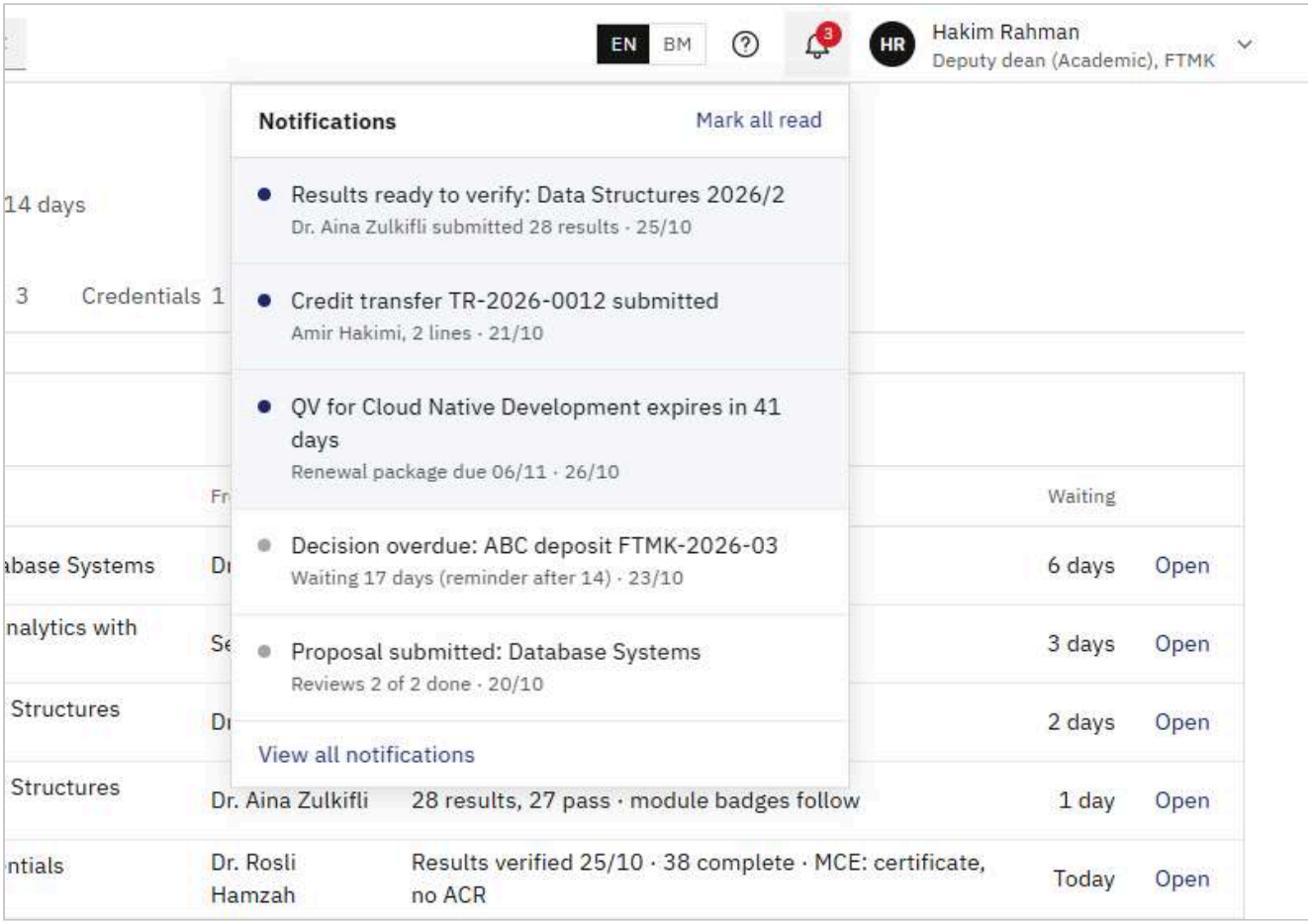


Figure 226: *Deputy dean: Notifications menu.*

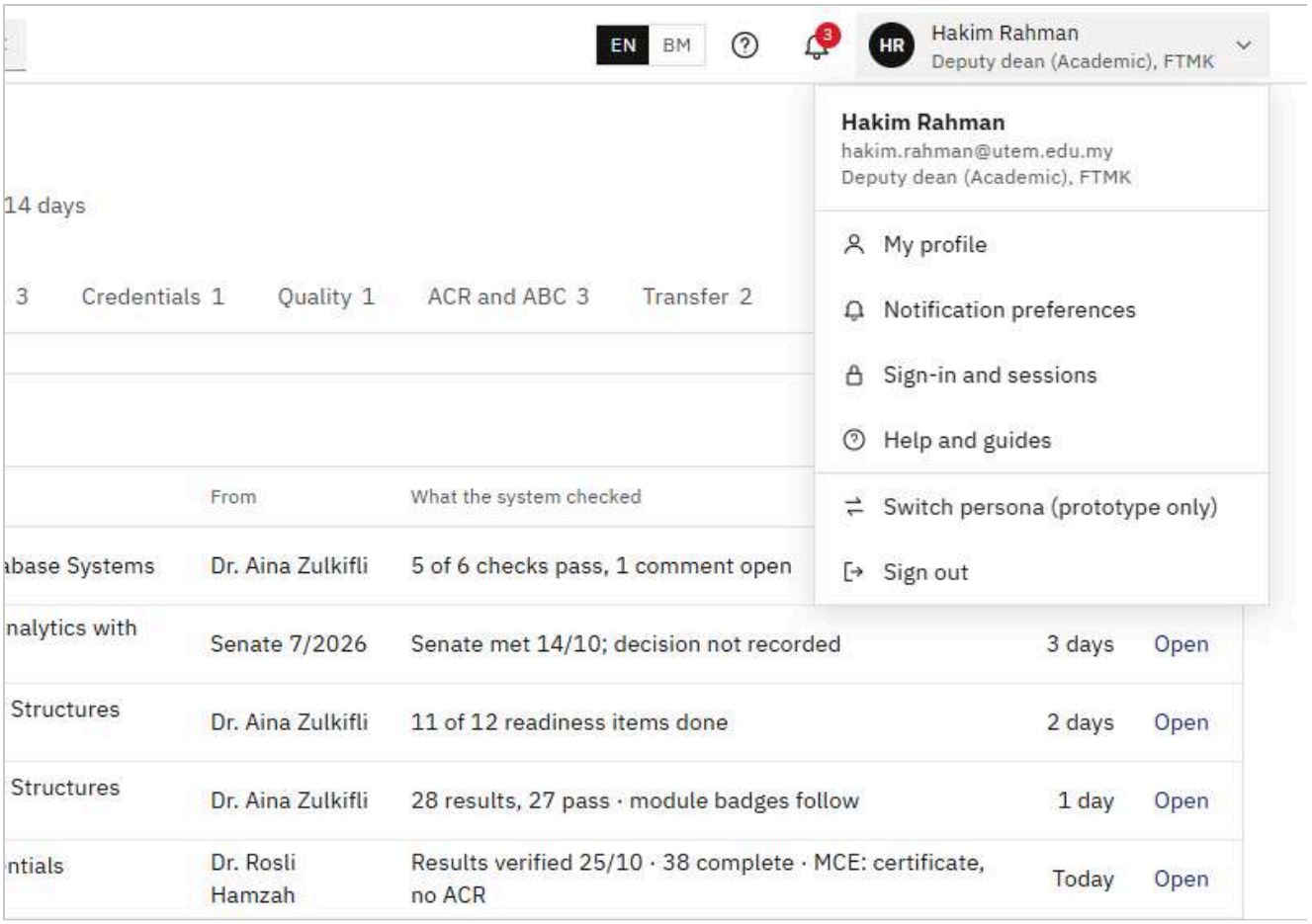


Figure 227: *Deputy dean: Profile menu.*

B.7 CODL system administrator

CODL-01 Users and roles

`#/codl/users`

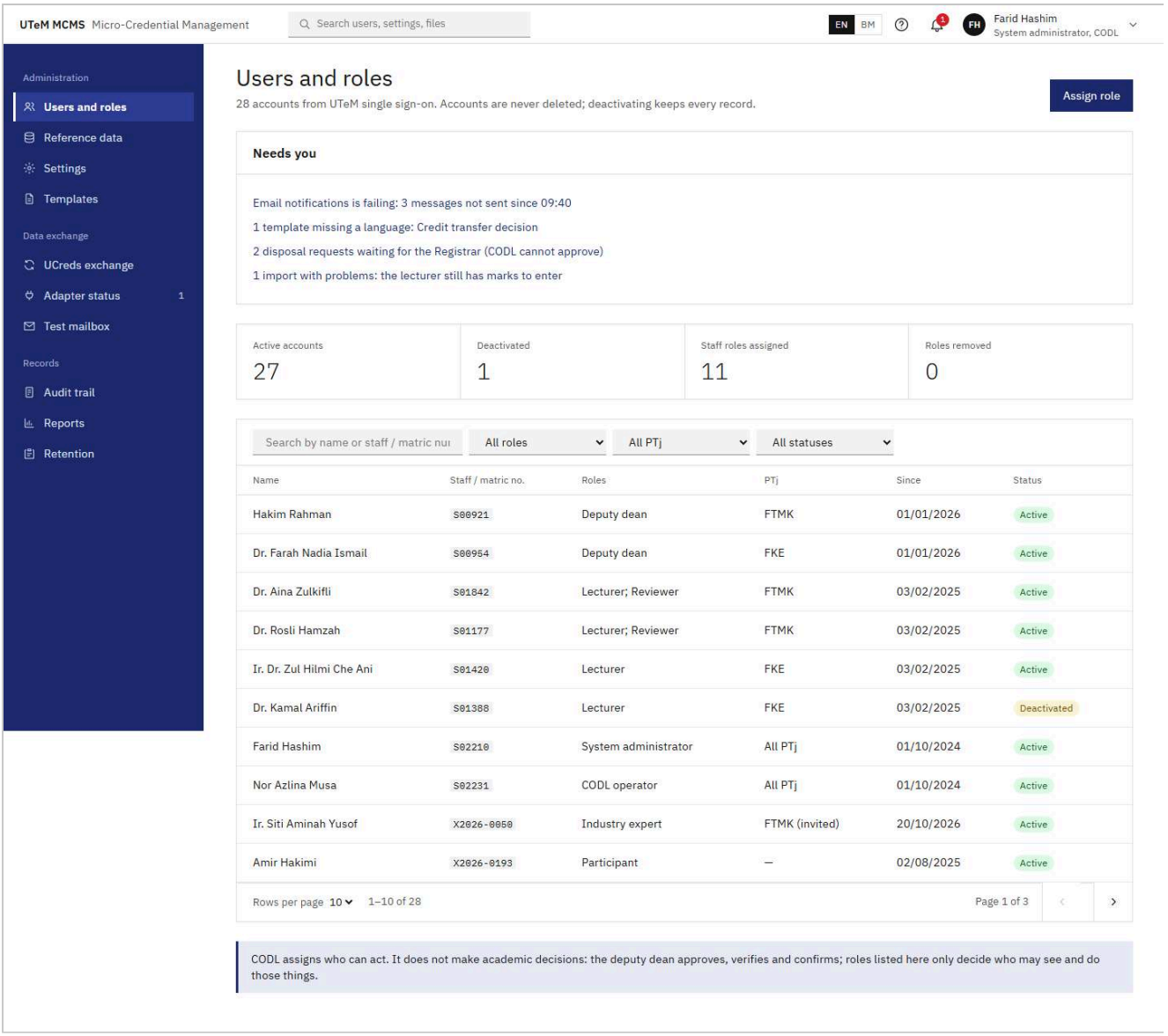


Figure 228: *CODL-01 Users and roles.*

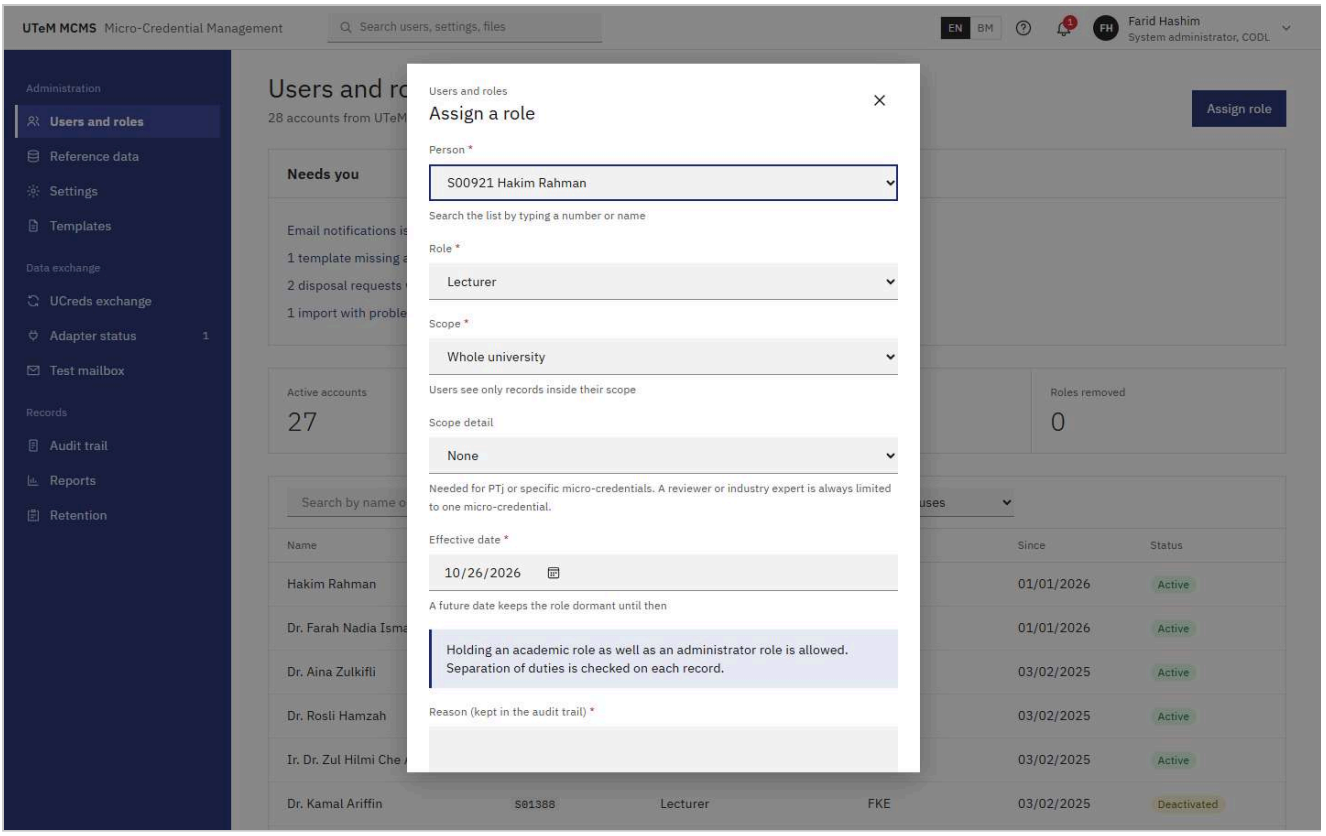


Figure 229: *CODL-01 Dialog: Assign a role.*

CODL-02 User

`#/codl/user/S01842`

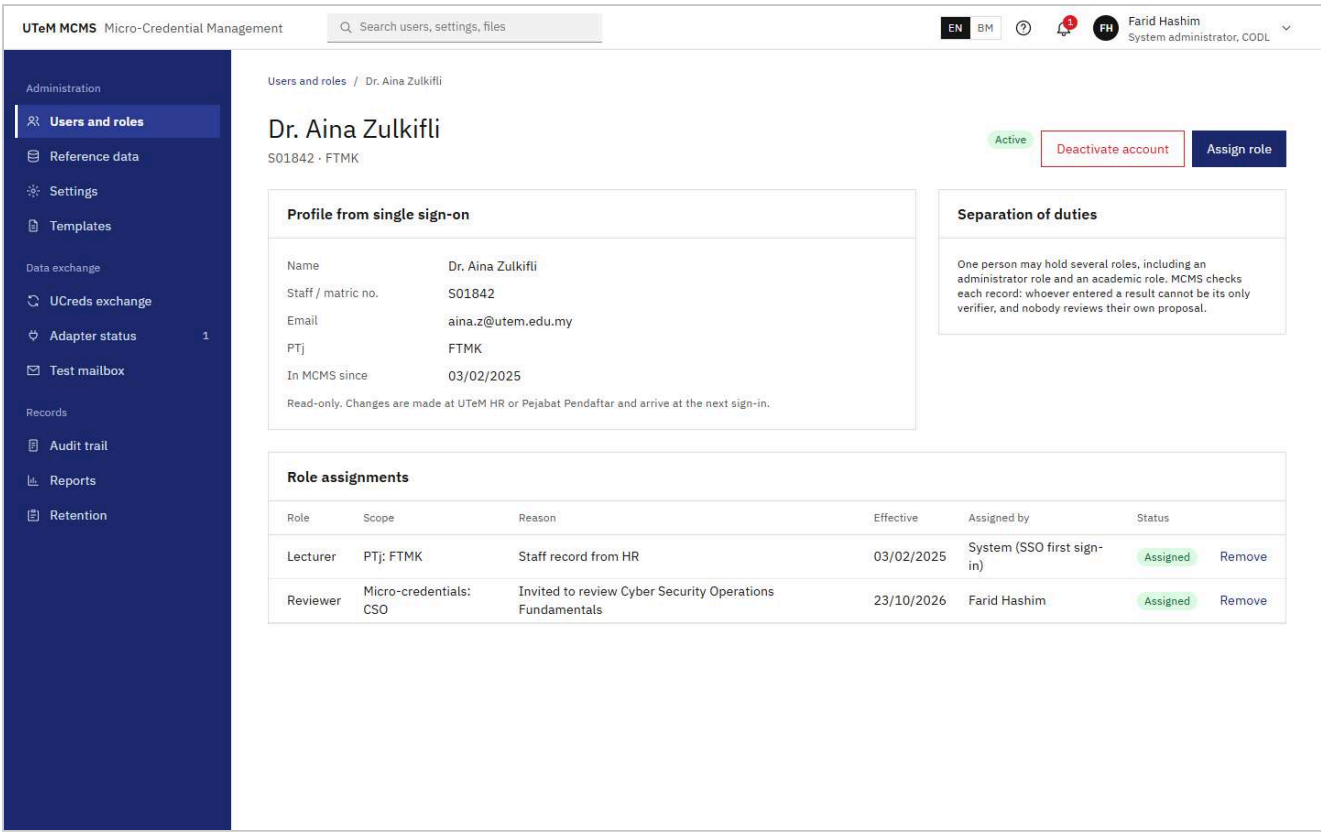


Figure 230: *CODL-02 User.*

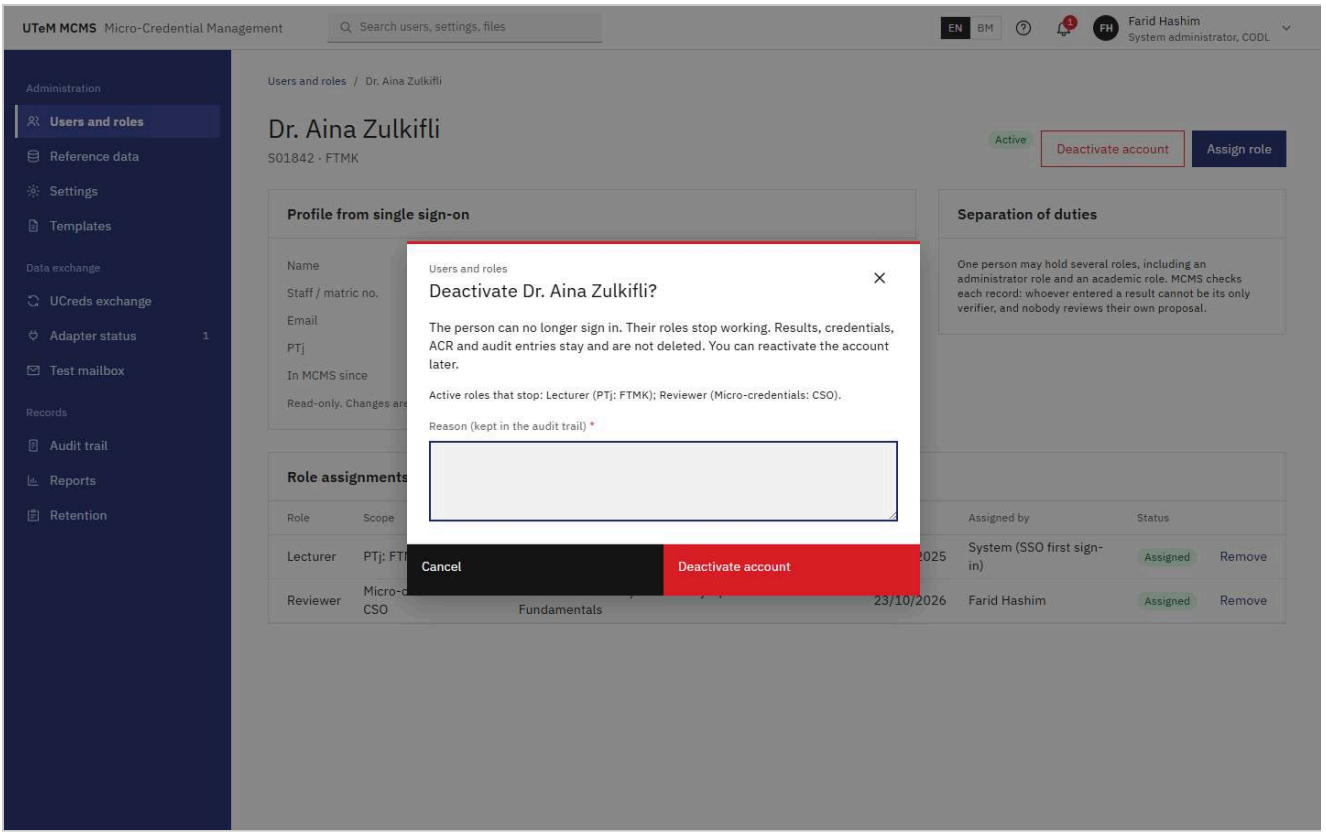


Figure 231: *CODL-02 Warning (destructive): Deactivate Dr. Aina Zulkifli?*

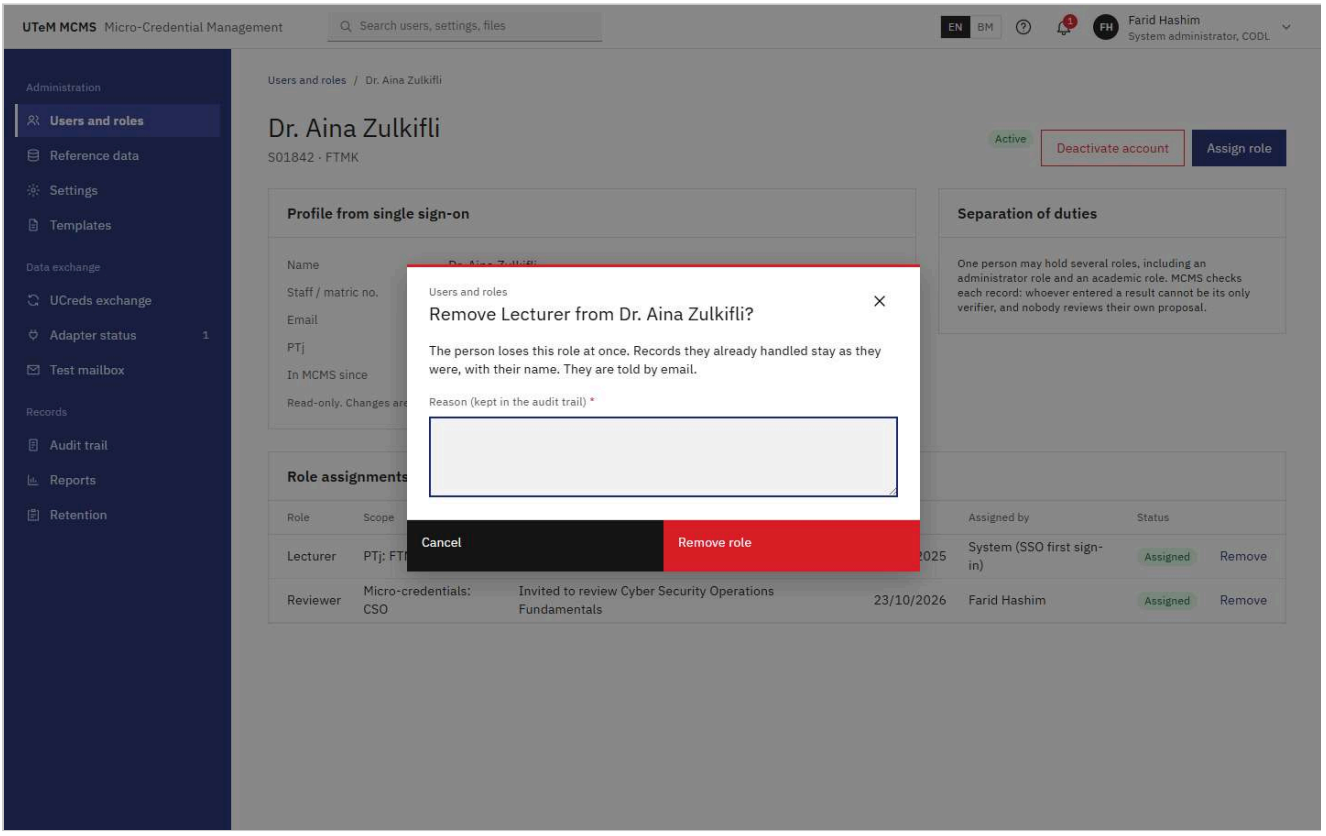


Figure 232: *CODL-02 Warning (destructive): Remove Lecturer from Dr. Aina Zulkifli?*

CODL-03 Reference data

`#/codl/refdata`

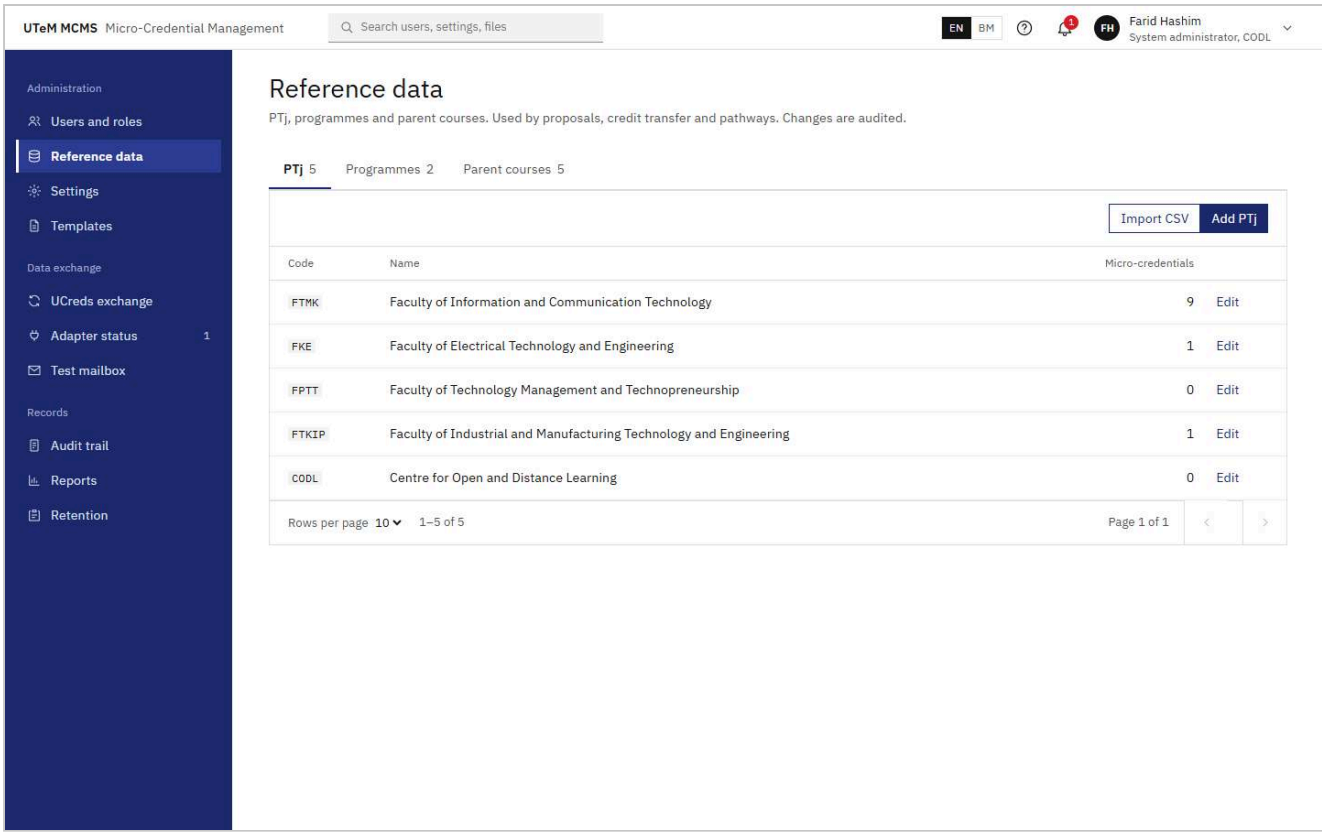


Figure 233: *CODL-03 Reference data.*

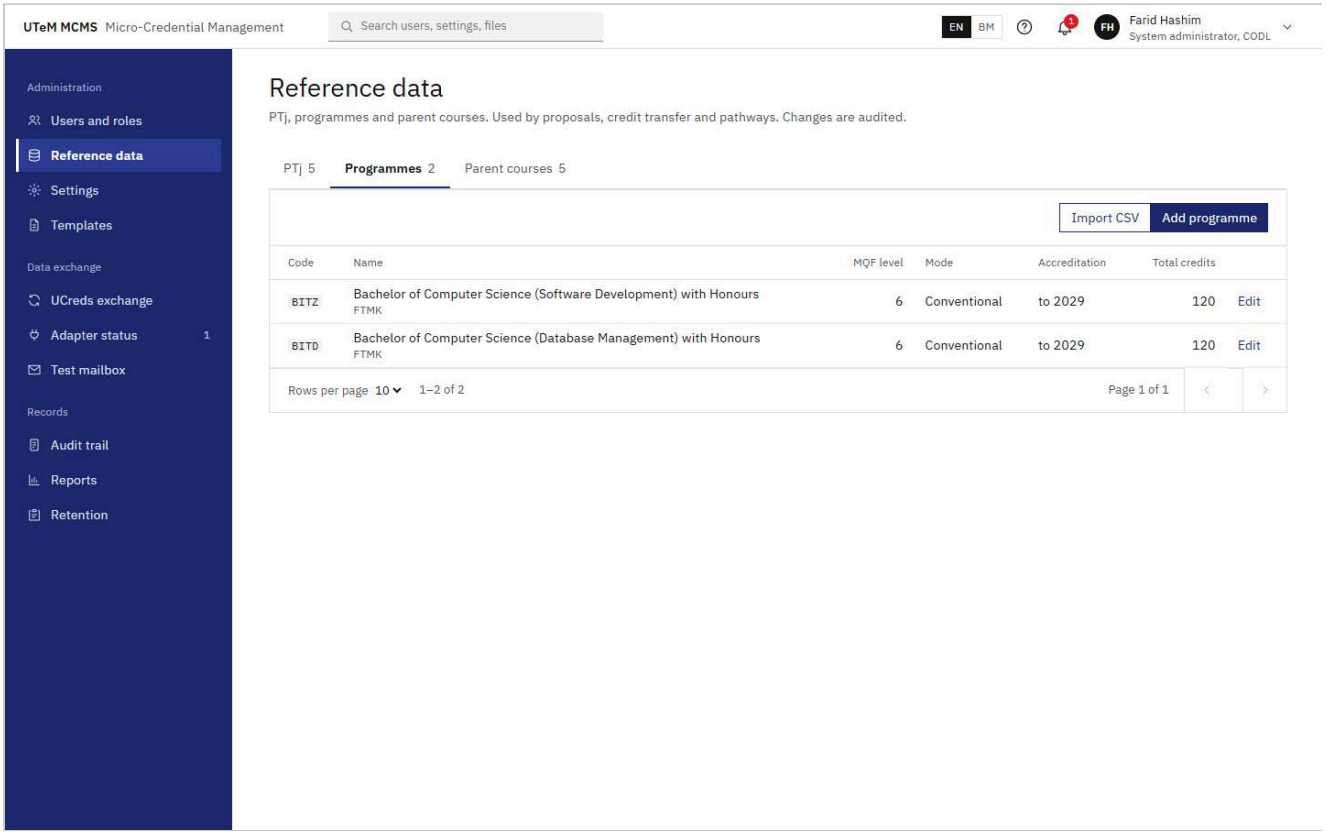


Figure 234: *CODL-03 Reference data: Tab: Programmes.*

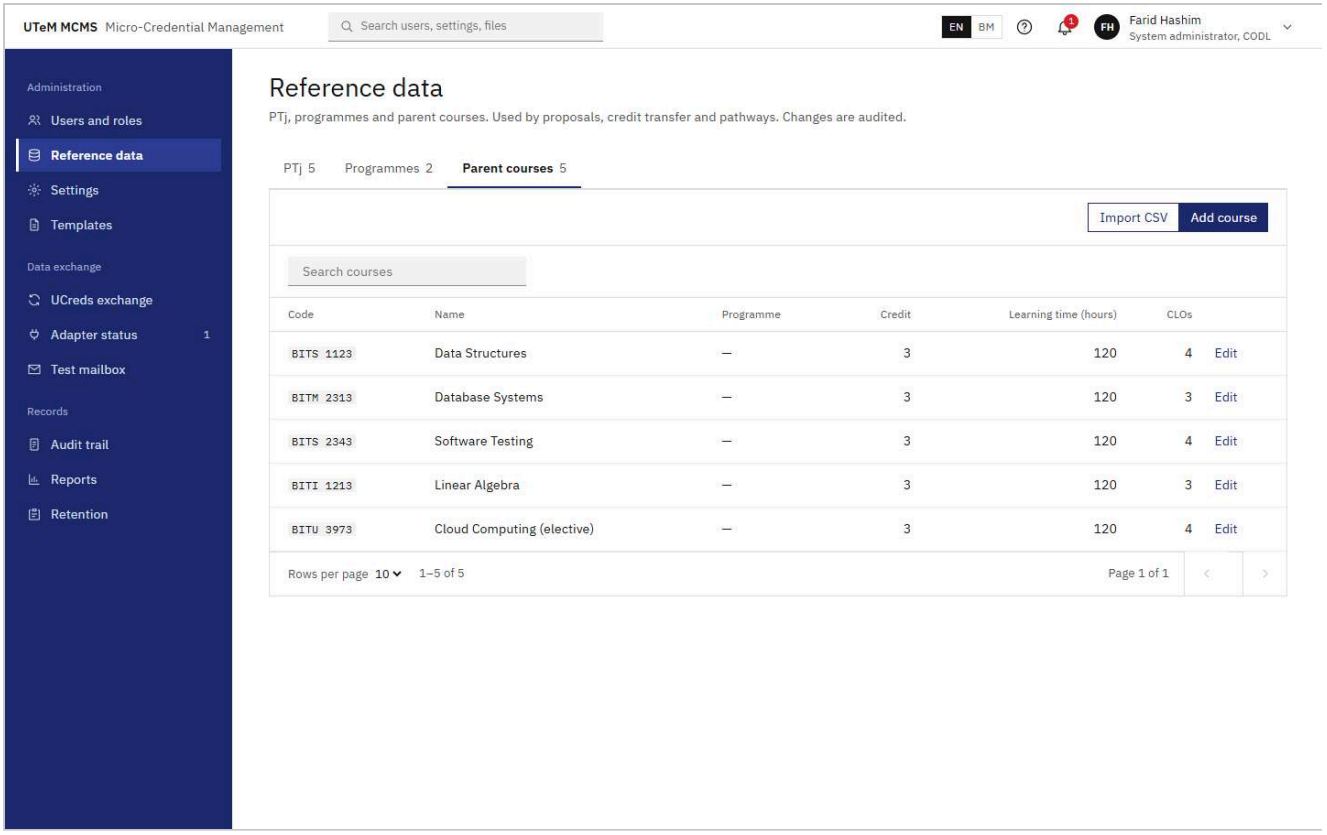


Figure 235: *CODL-03 Reference data: Tab: Parent courses.*

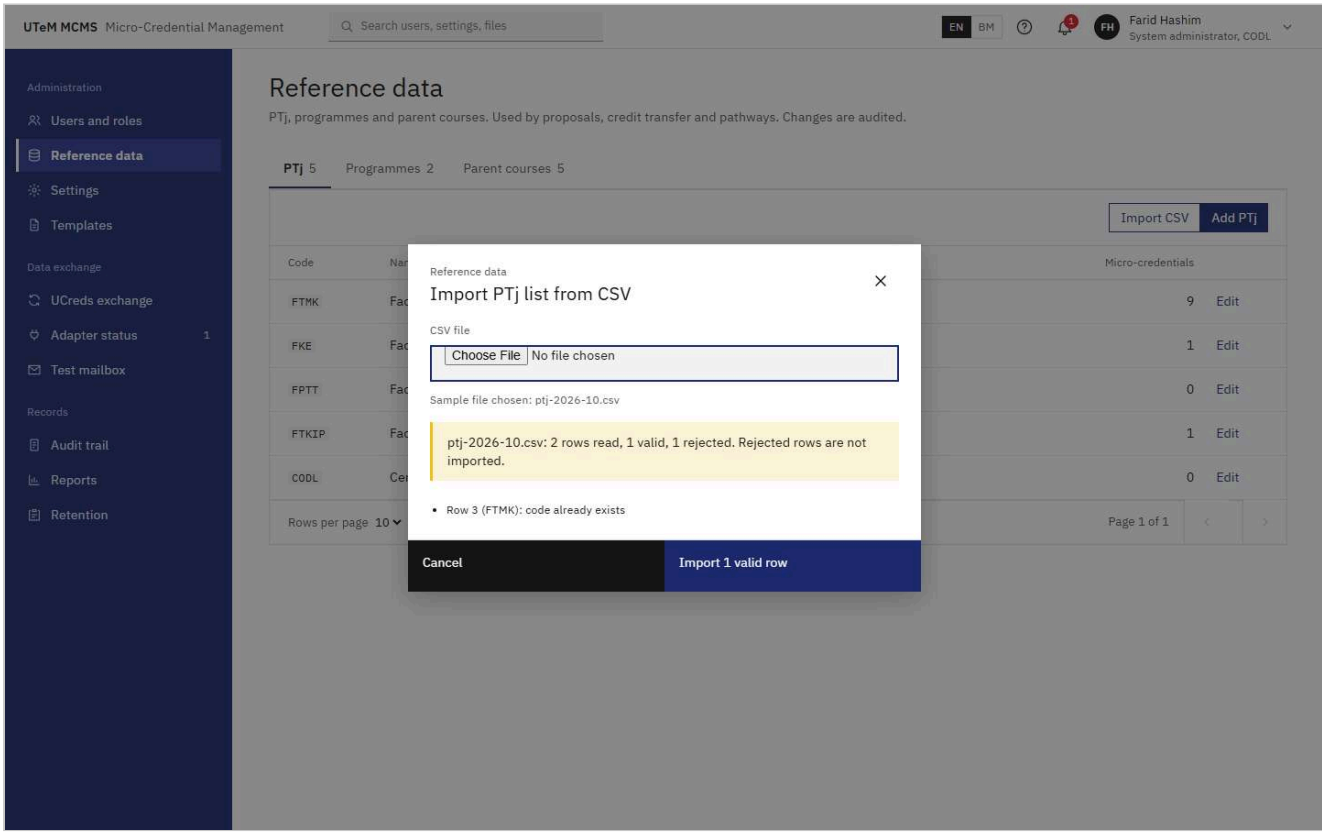


Figure 236: *CODL-03 Dialog: Import PTj list from CSV.*

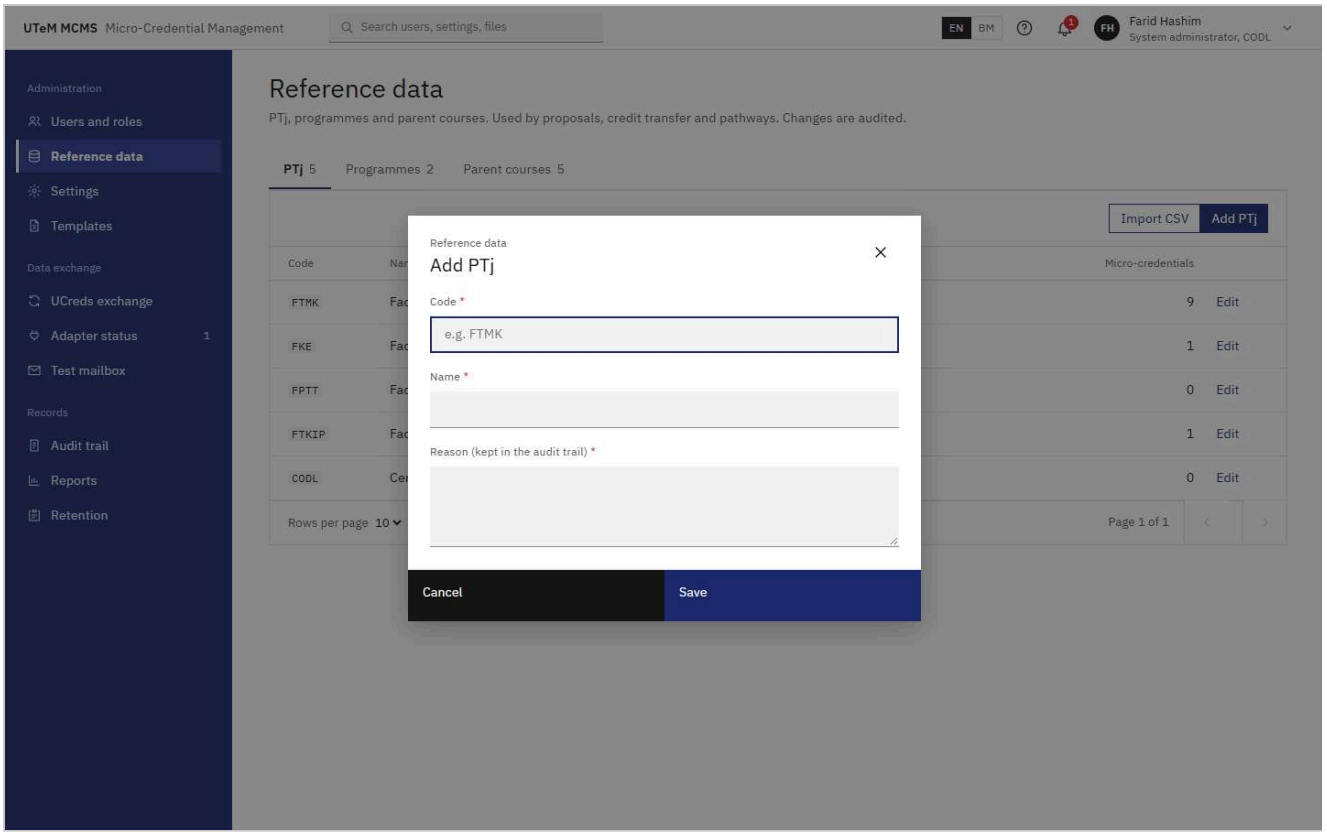


Figure 237: *CODL-03 Dialog: Add PTj.*

CODL-04 Settings

`#/codl/settings`

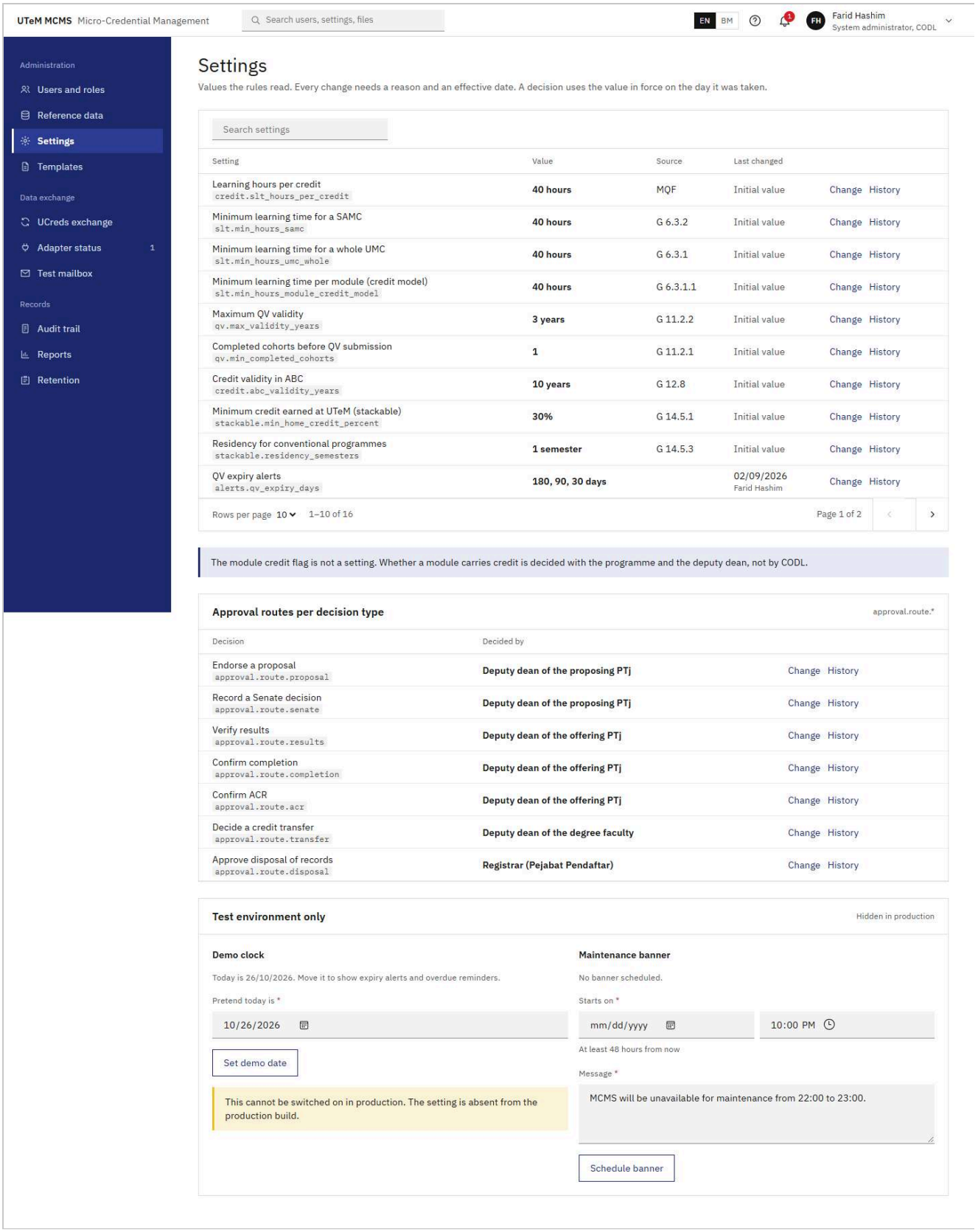


Figure 238: *CODL-04 Settings.*

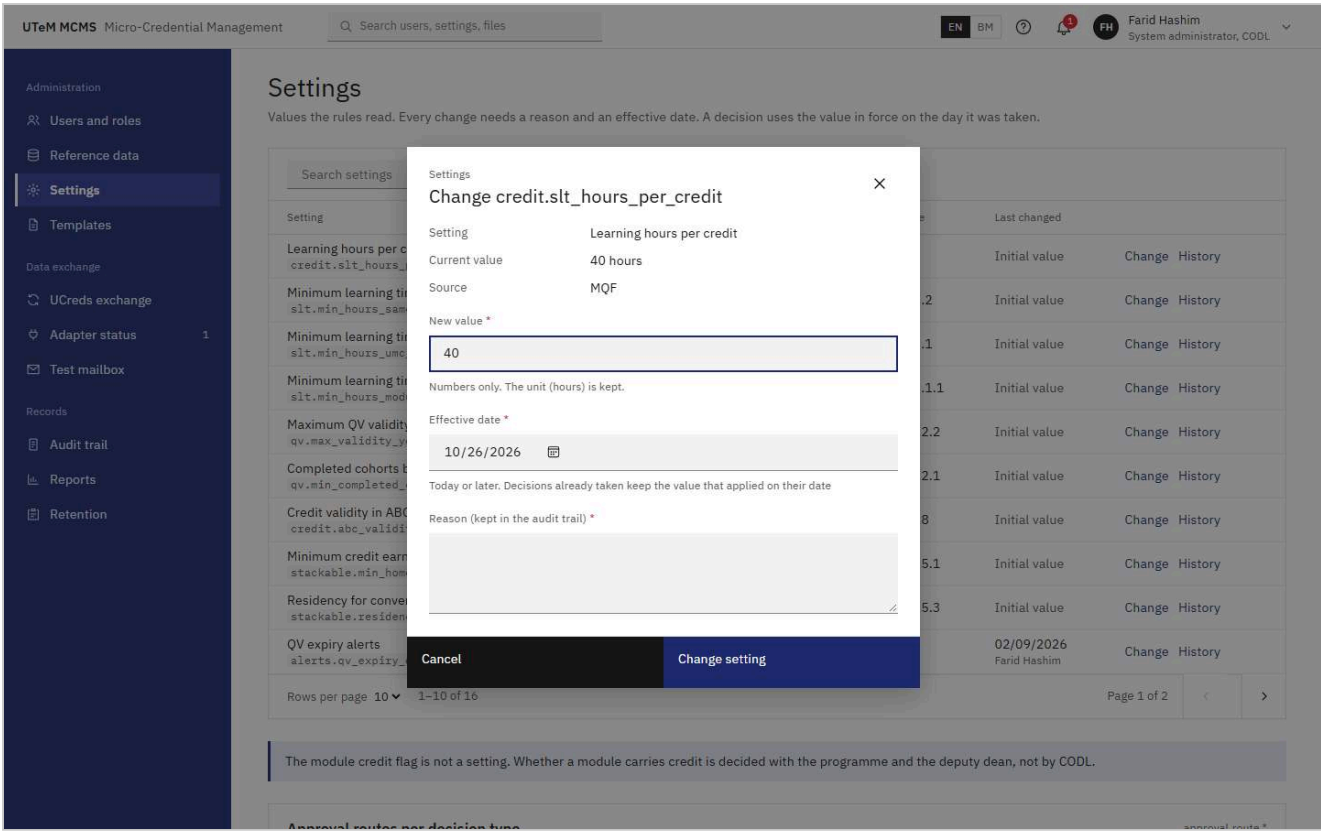


Figure 239: *CODL-04 Dialog: Change credit.slt_hours_per_credit.*

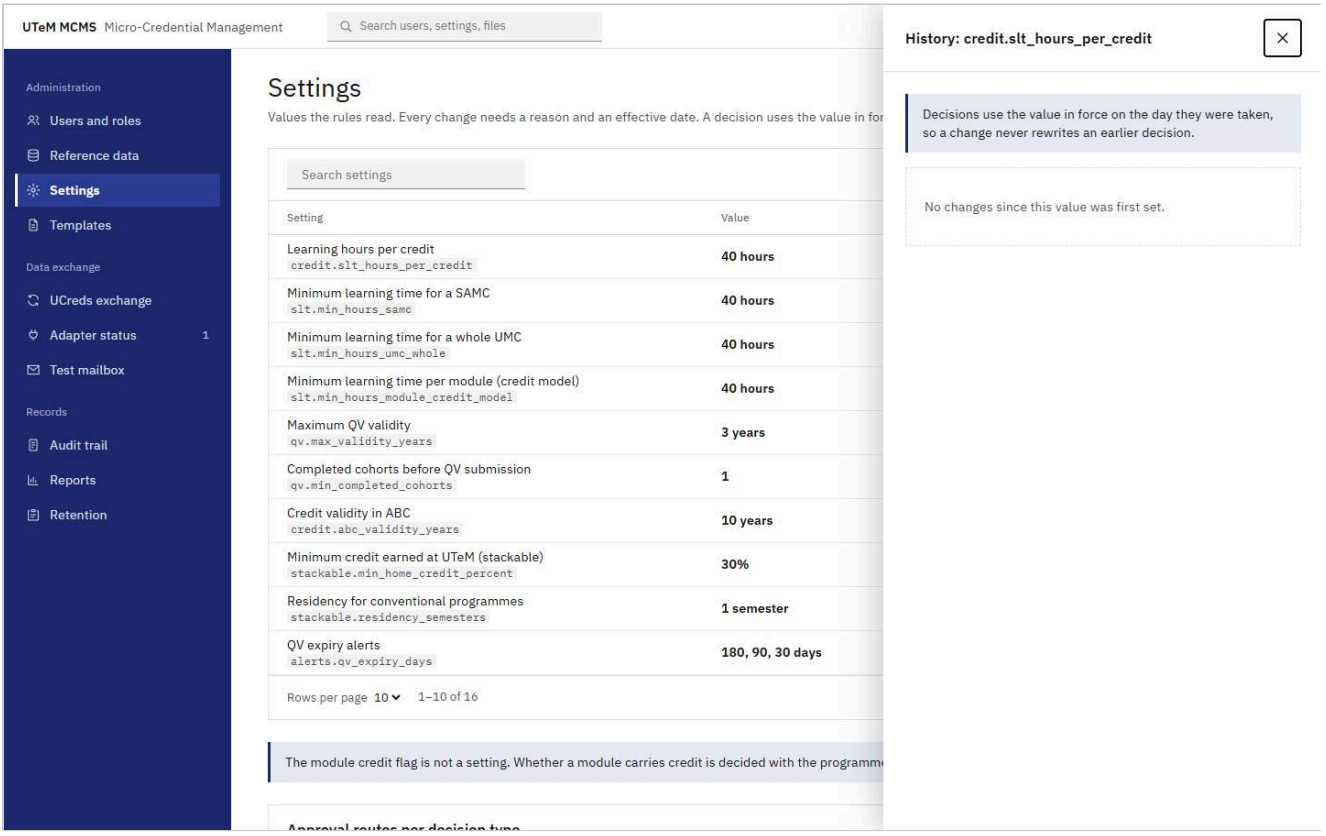


Figure 240: *CODL-04 Side panel: History: credit.slt_hours_per_credit.*

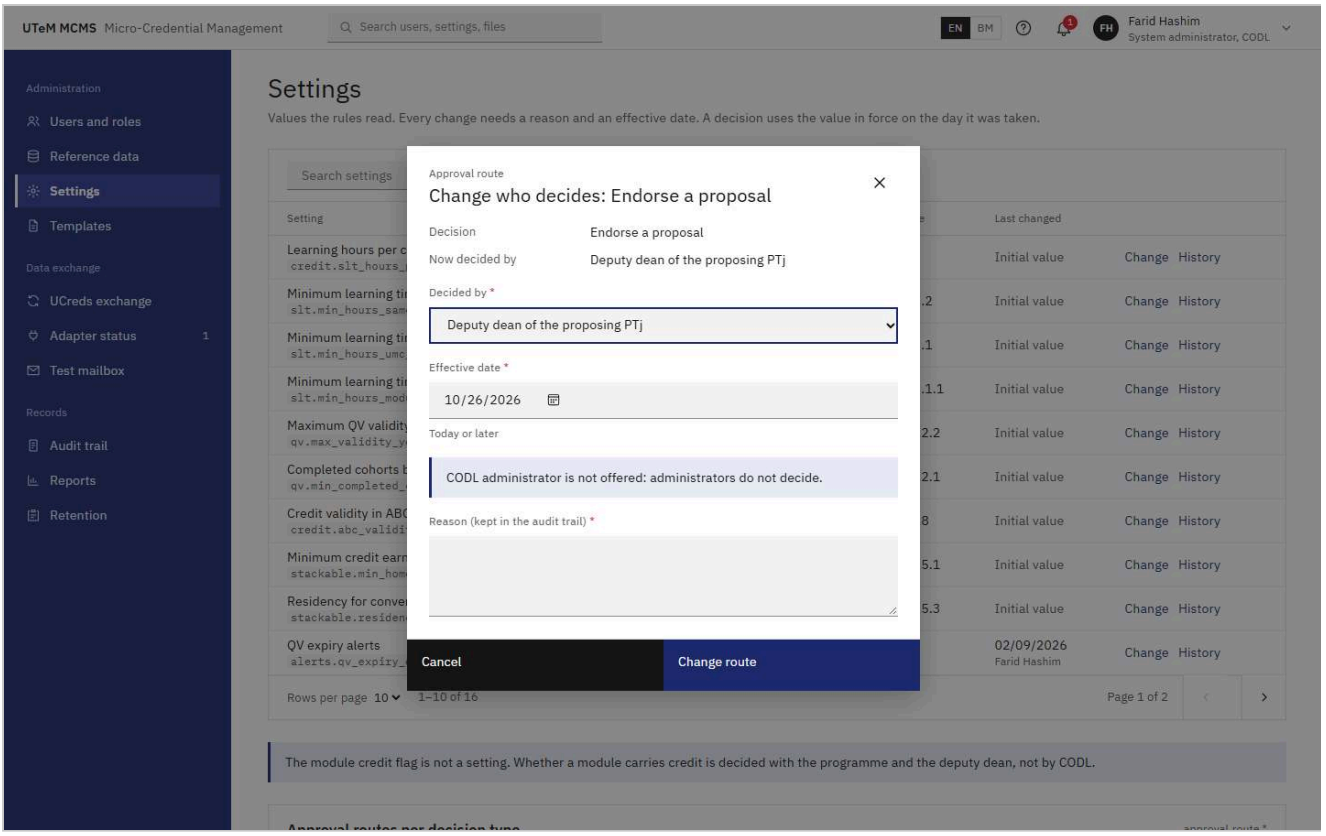


Figure 241: *CODL-04 Dialog: Change who decides: Endorse a proposal.*

CODL-05 Templates

`#/codl/templates`

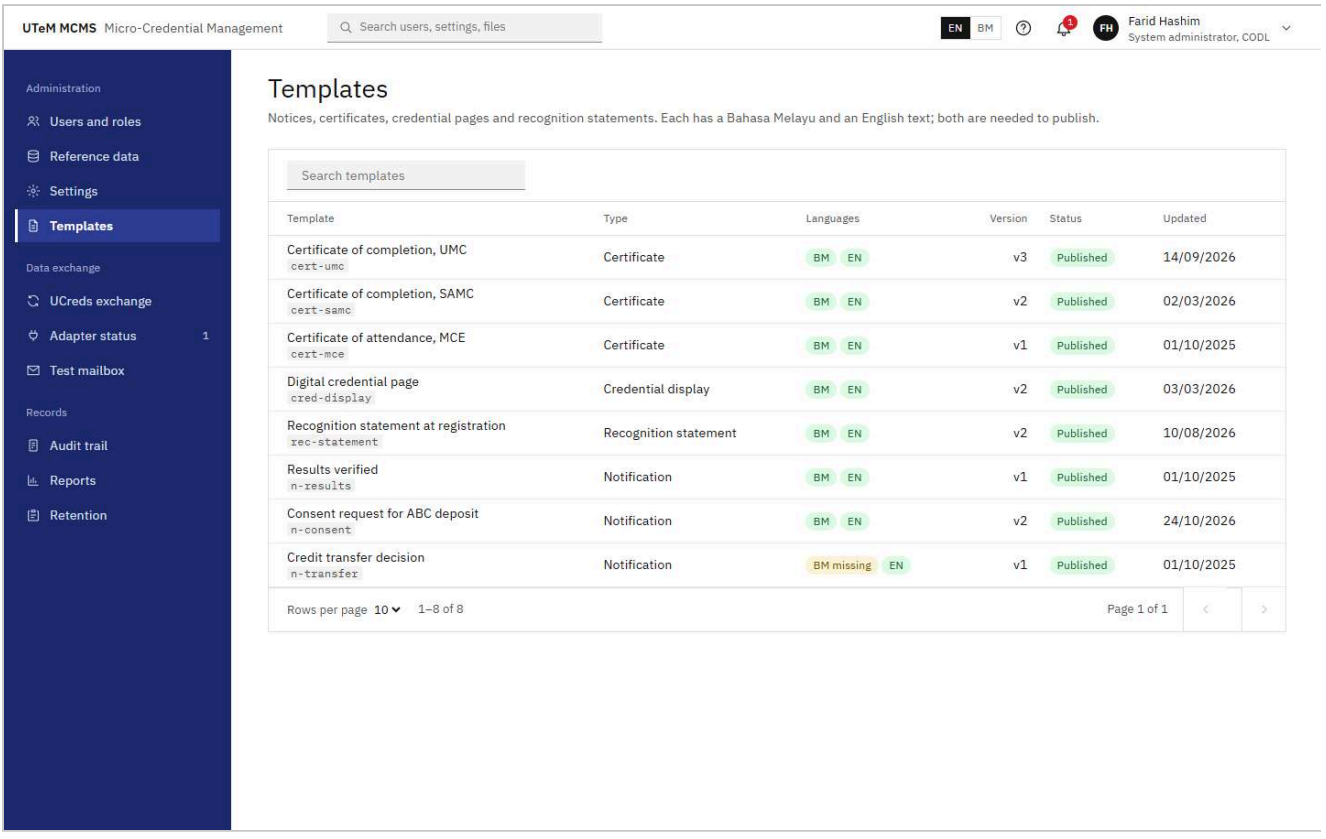


Figure 242: *CODL-05 Templates.*

CODL-06 Template

`#/codl/template/cert-umc`

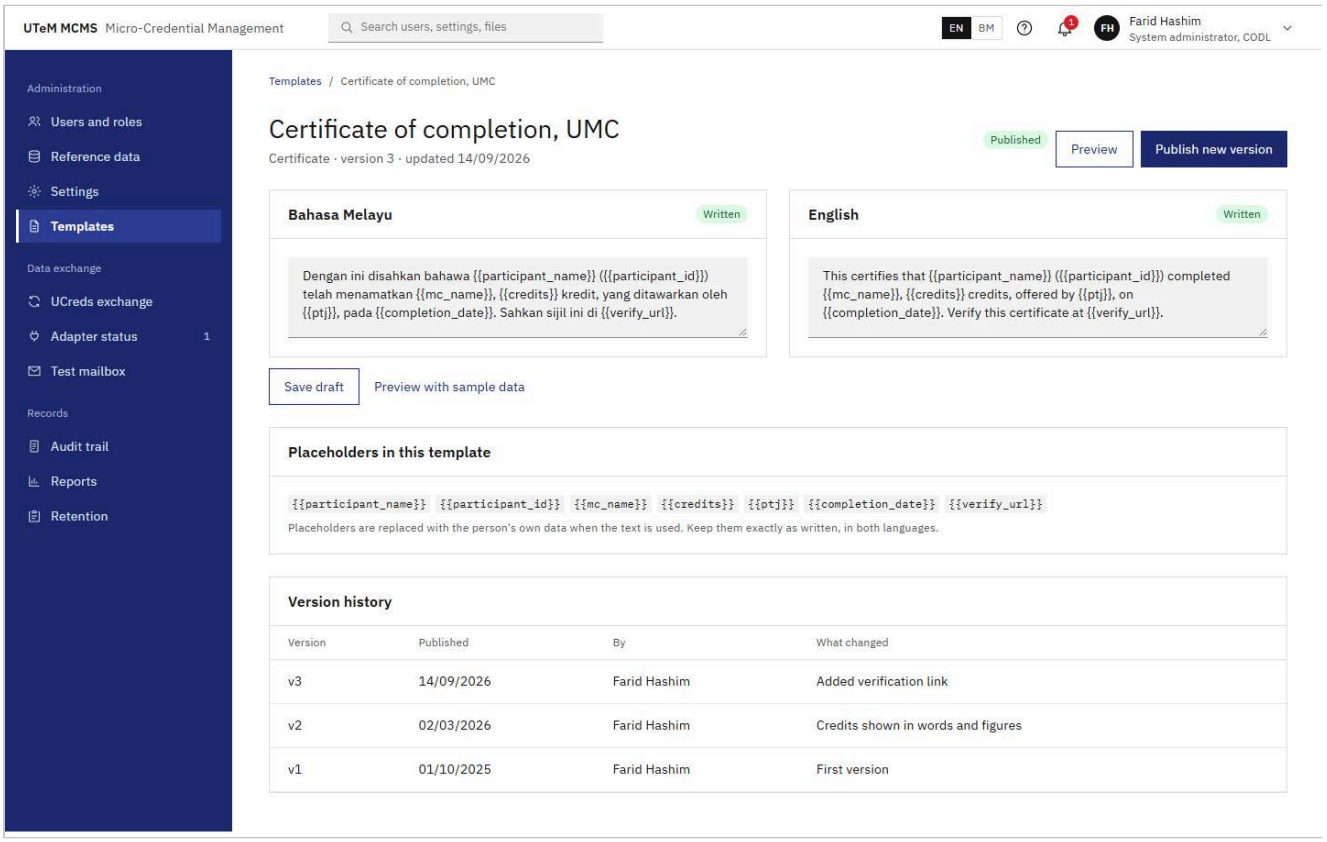


Figure 243: *CODL-06 Template.*

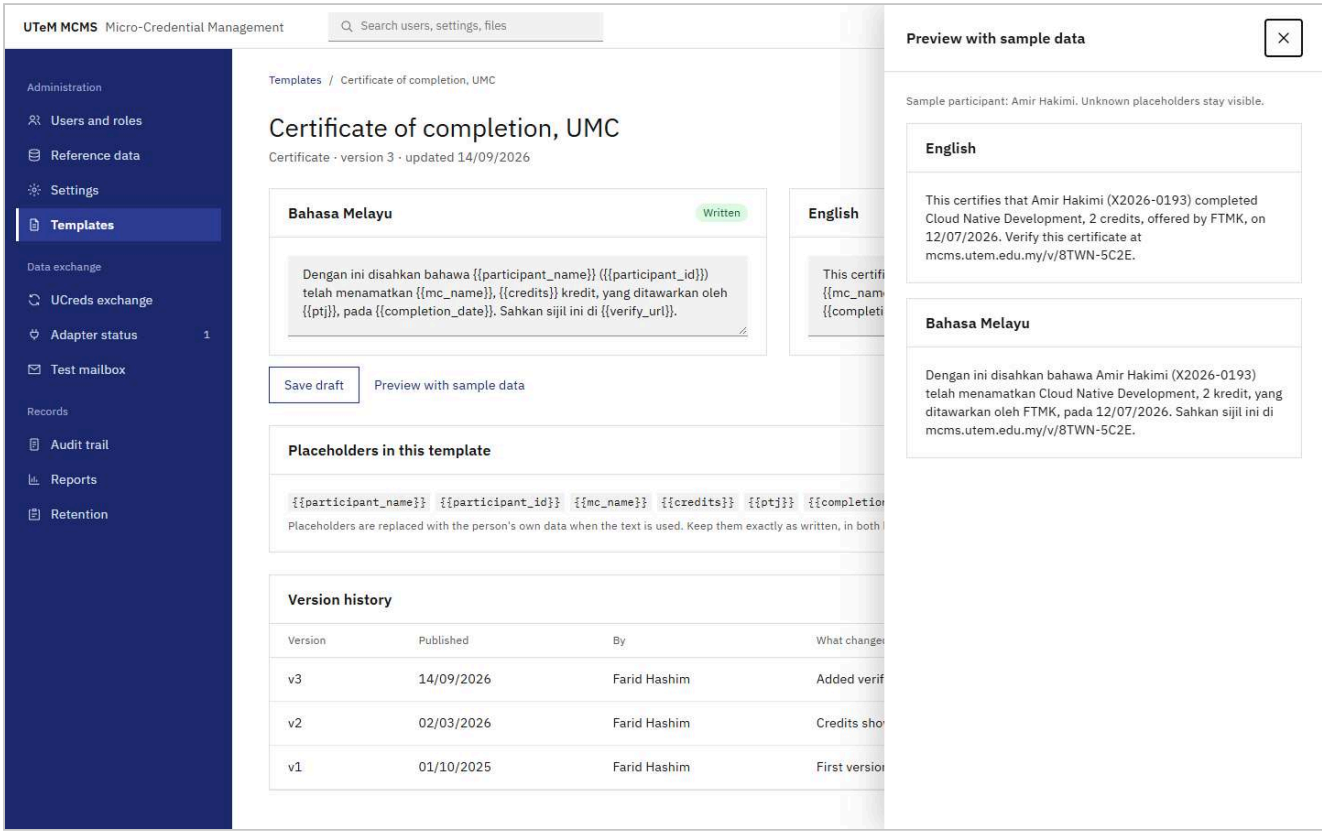


Figure 244: *CODL-06 Side panel: Preview with sample data.*

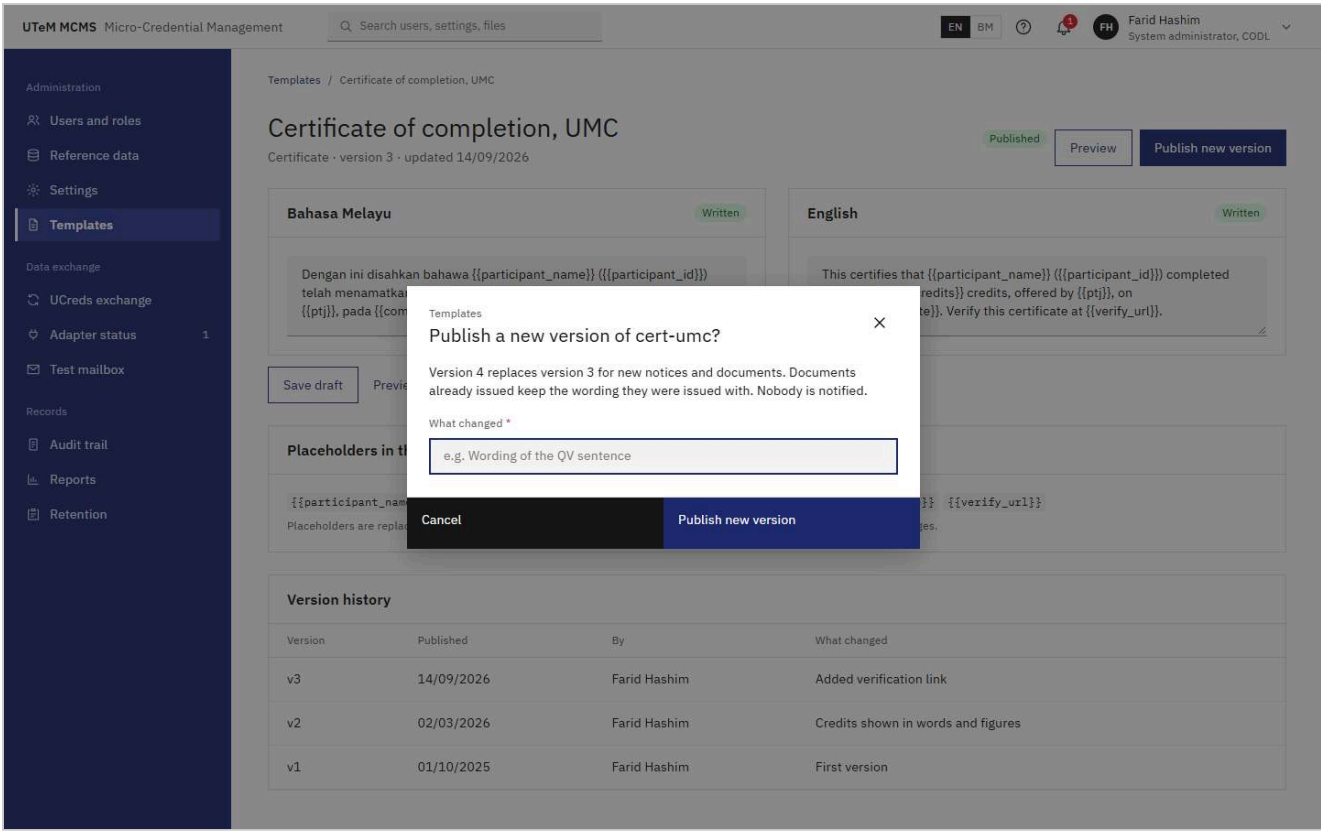


Figure 245: *CODL-06 Confirmation: Publish a new version of cert-umc?*

CODL-07 UCreds exchange

`#/codl/ucreds`

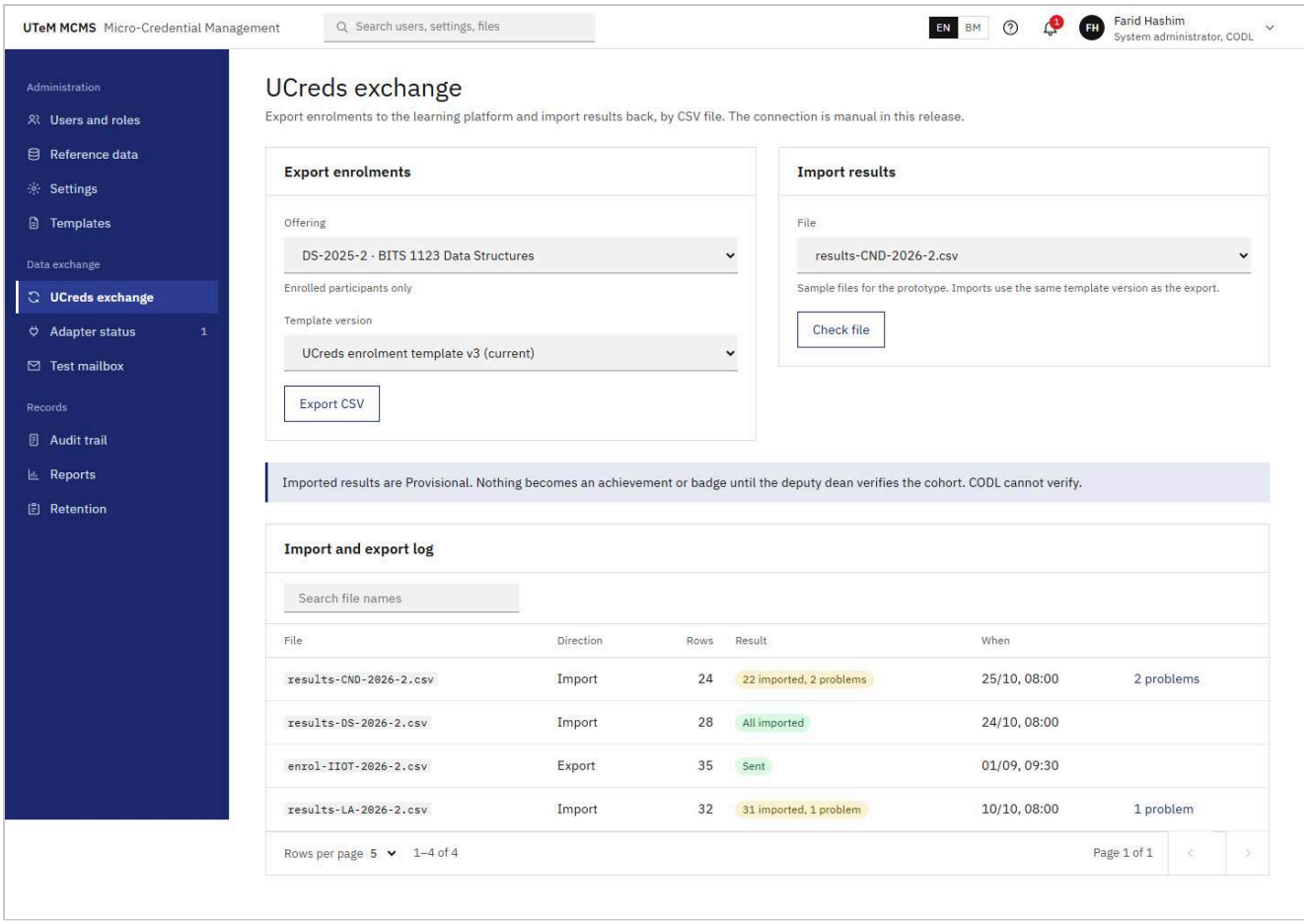


Figure 246: *CODL-07 UCreds exchange.*

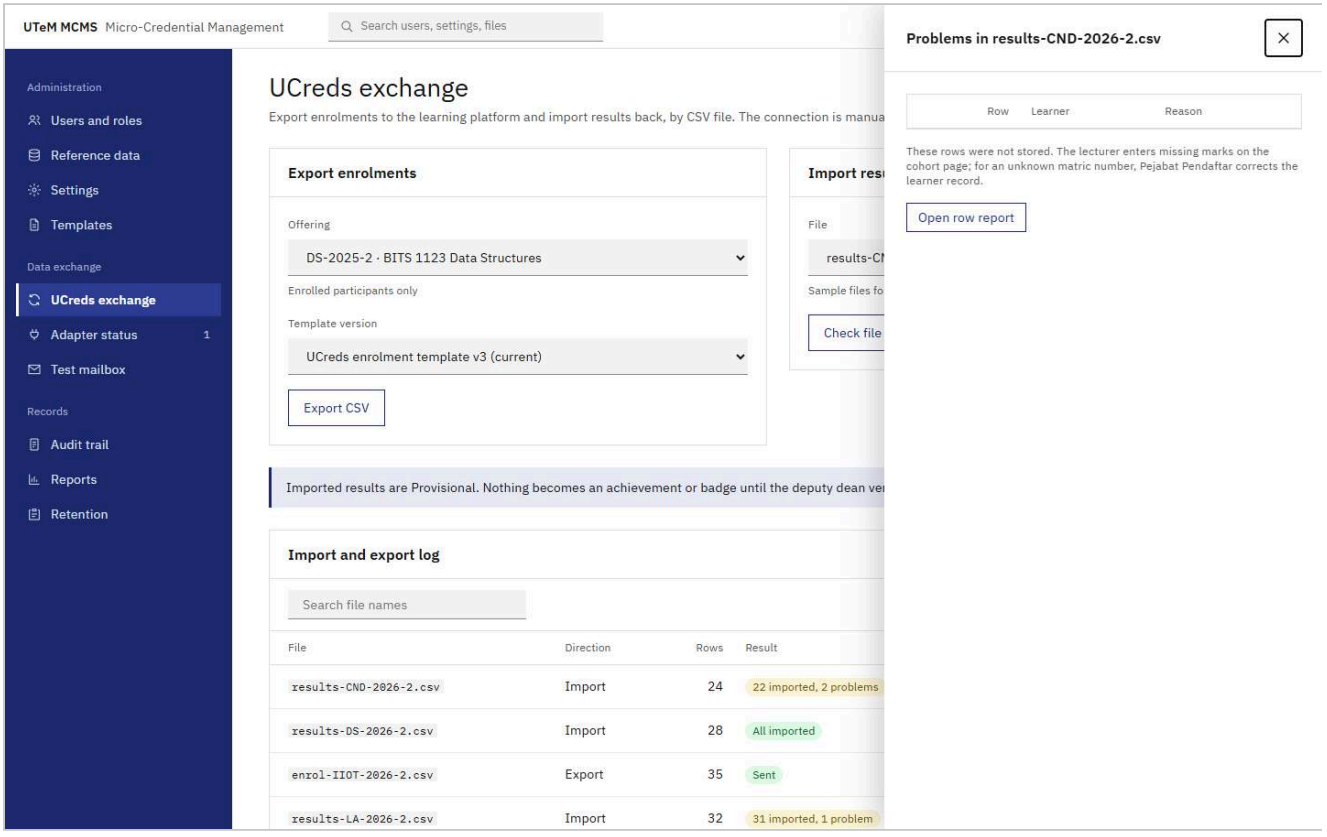


Figure 247: *CODL-07 Side panel: Problems in results-CND-2026-2.csv.*

CODL-08 Adapter status

`#/codl/adapters`

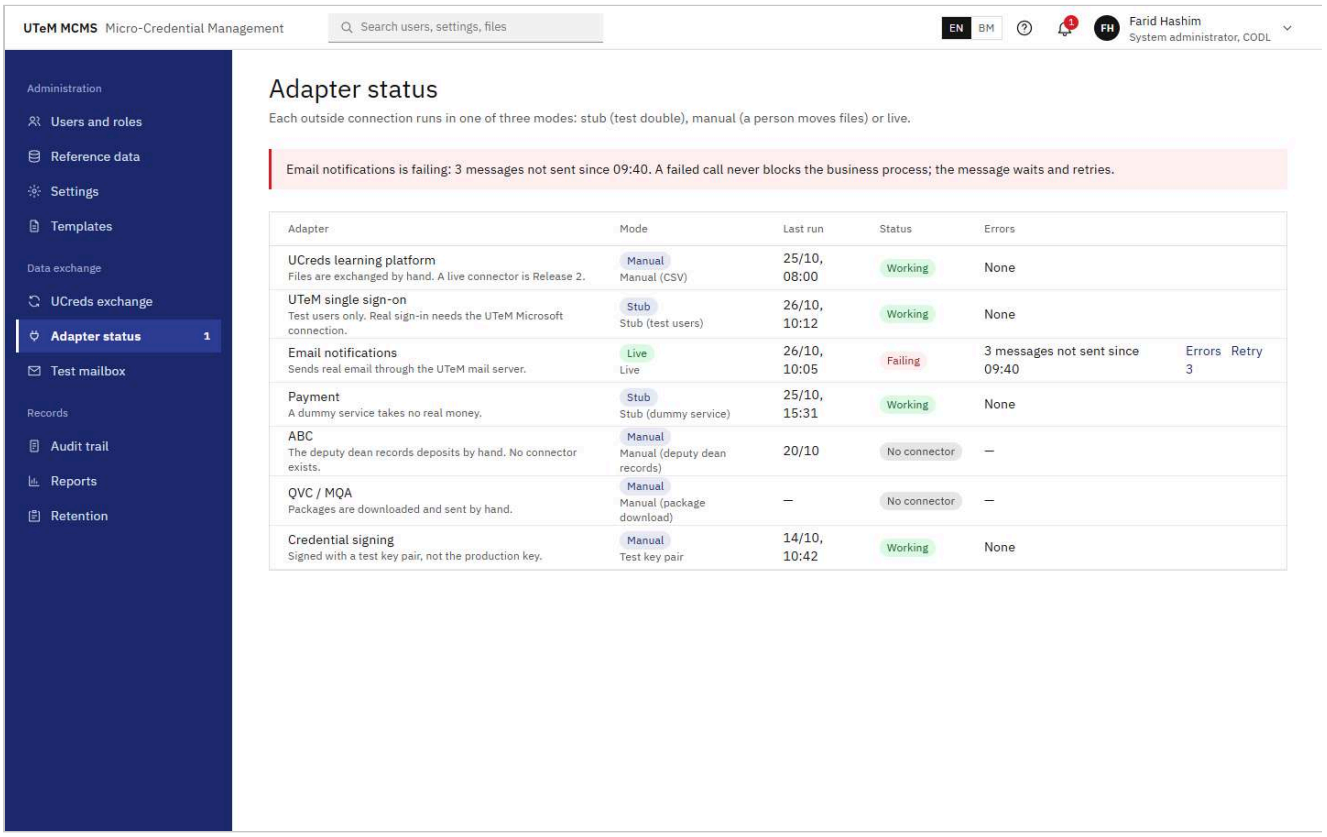


Figure 248: *CODL-08 Adapter status.*

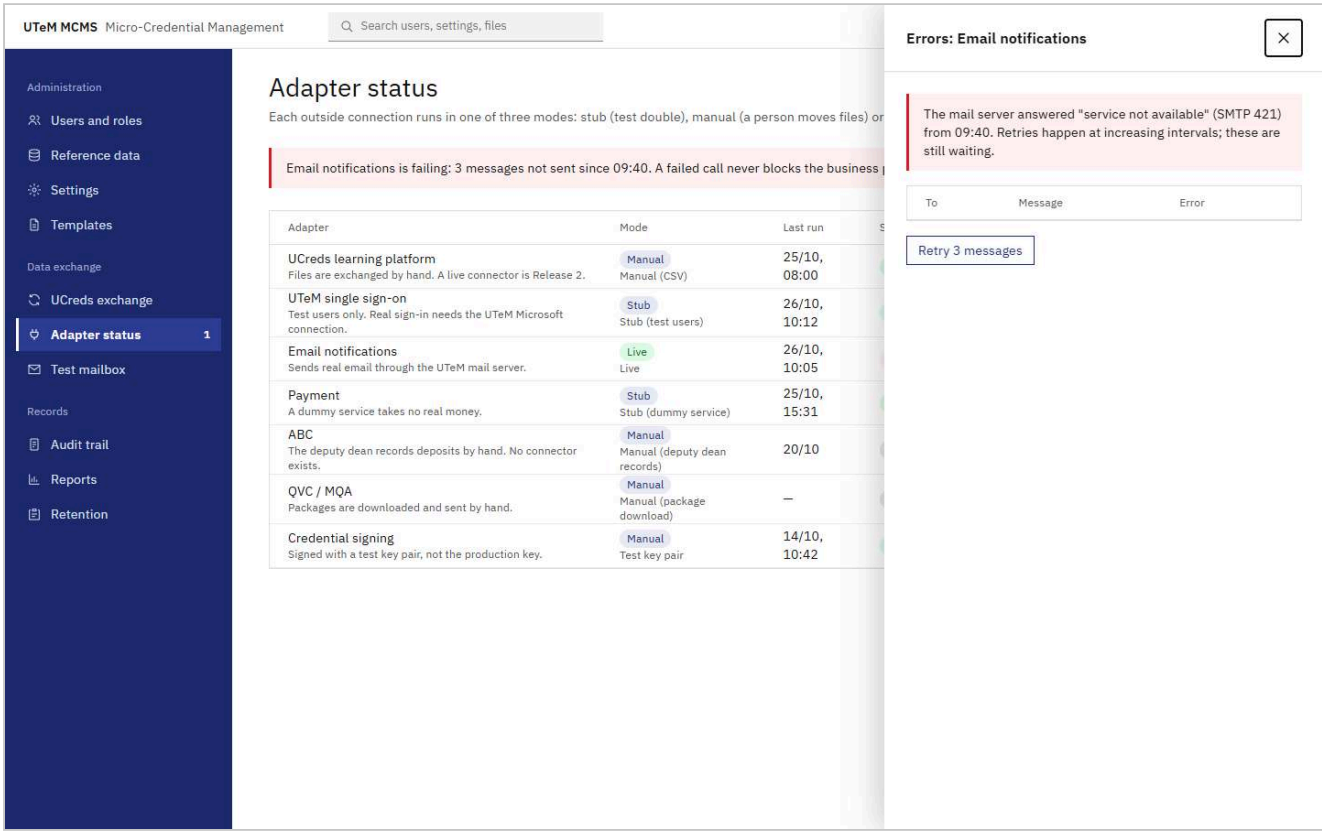


Figure 249: *CODL-08 Side panel: Errors: Email notifications.*

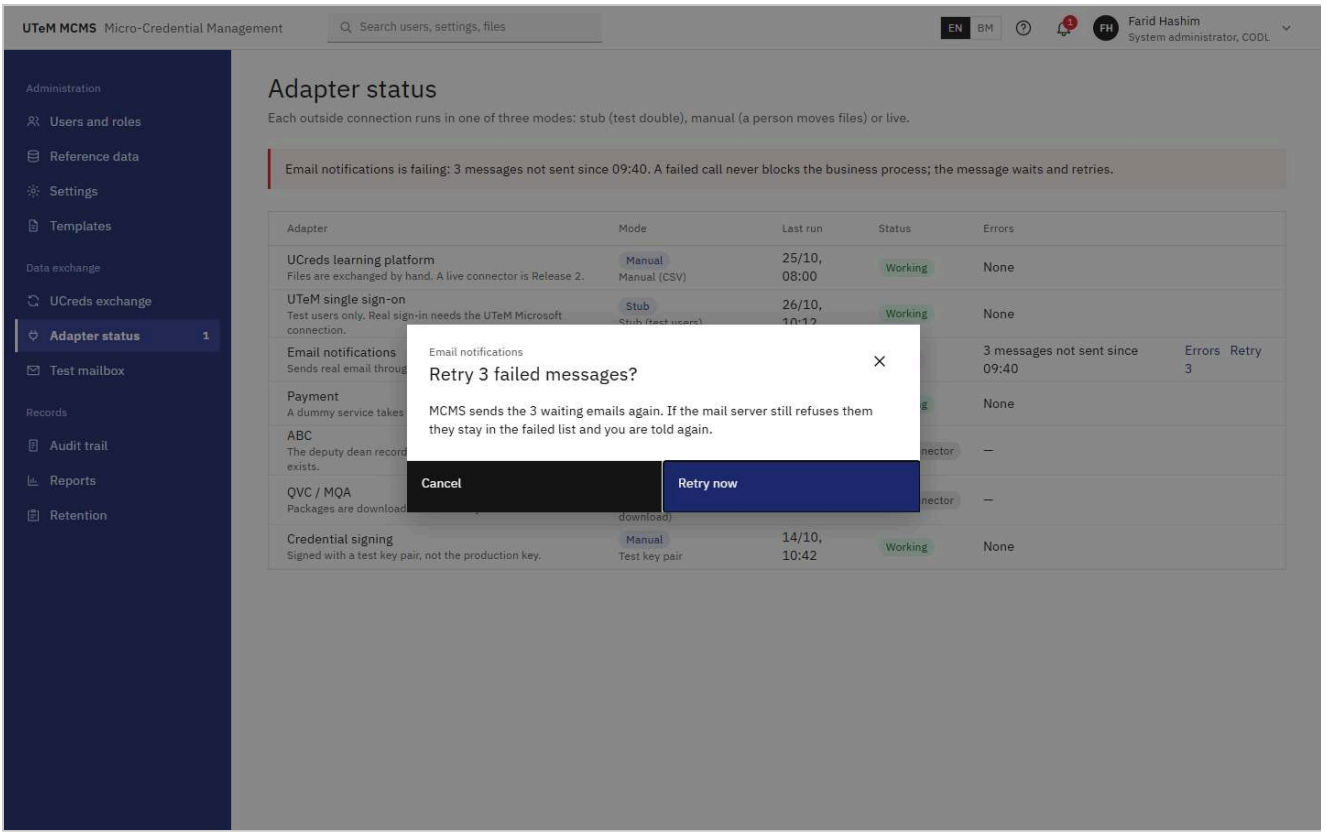


Figure 250: *CODL-08 Confirmation: Retry 3 failed messages?*

CODL-09 Test mailbox

`#/codl/mailbox`

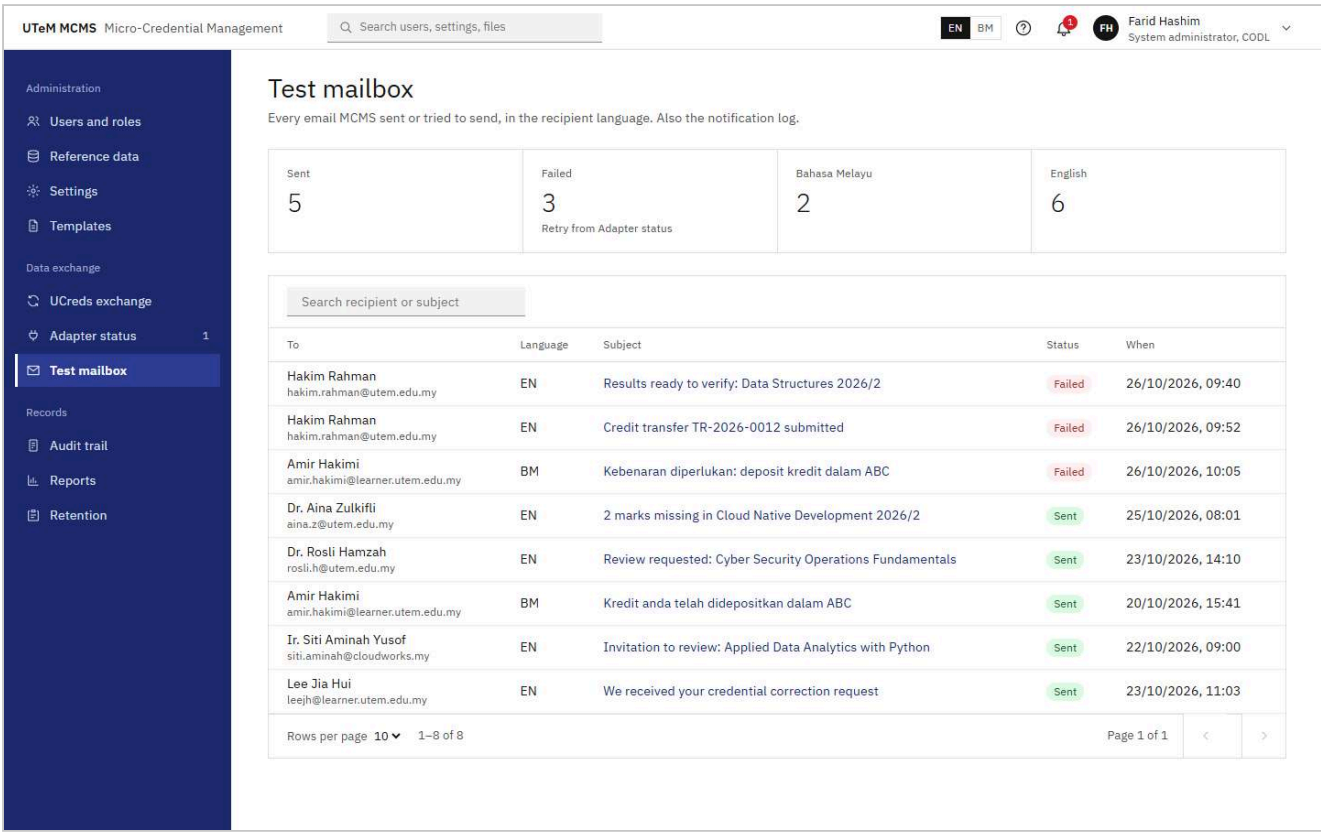


Figure 251: *CODL-09 Test mailbox.*

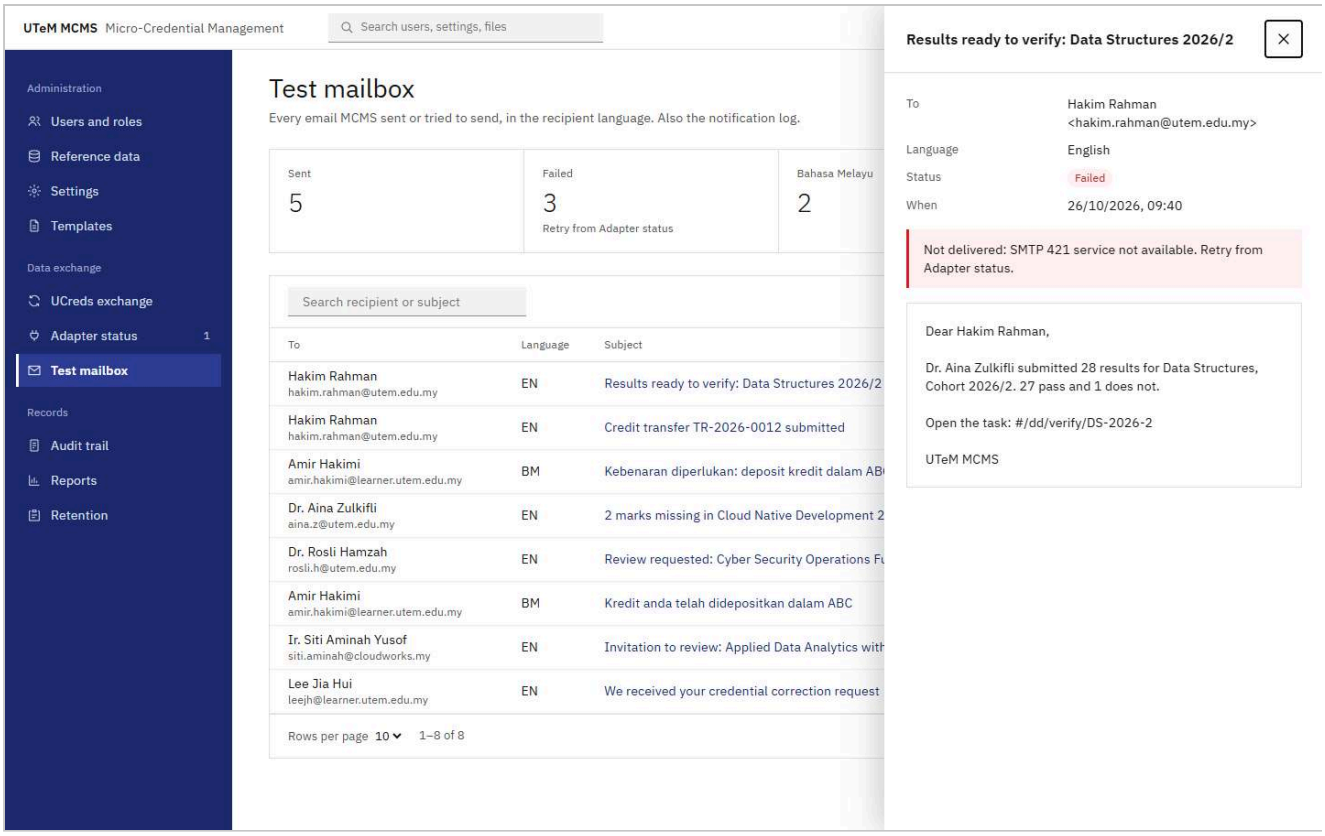


Figure 252: *CODL-09 Side panel: Results ready to verify: Data Structures 2026/2.*

CODL-10 Audit trail

`#/codl/audit`

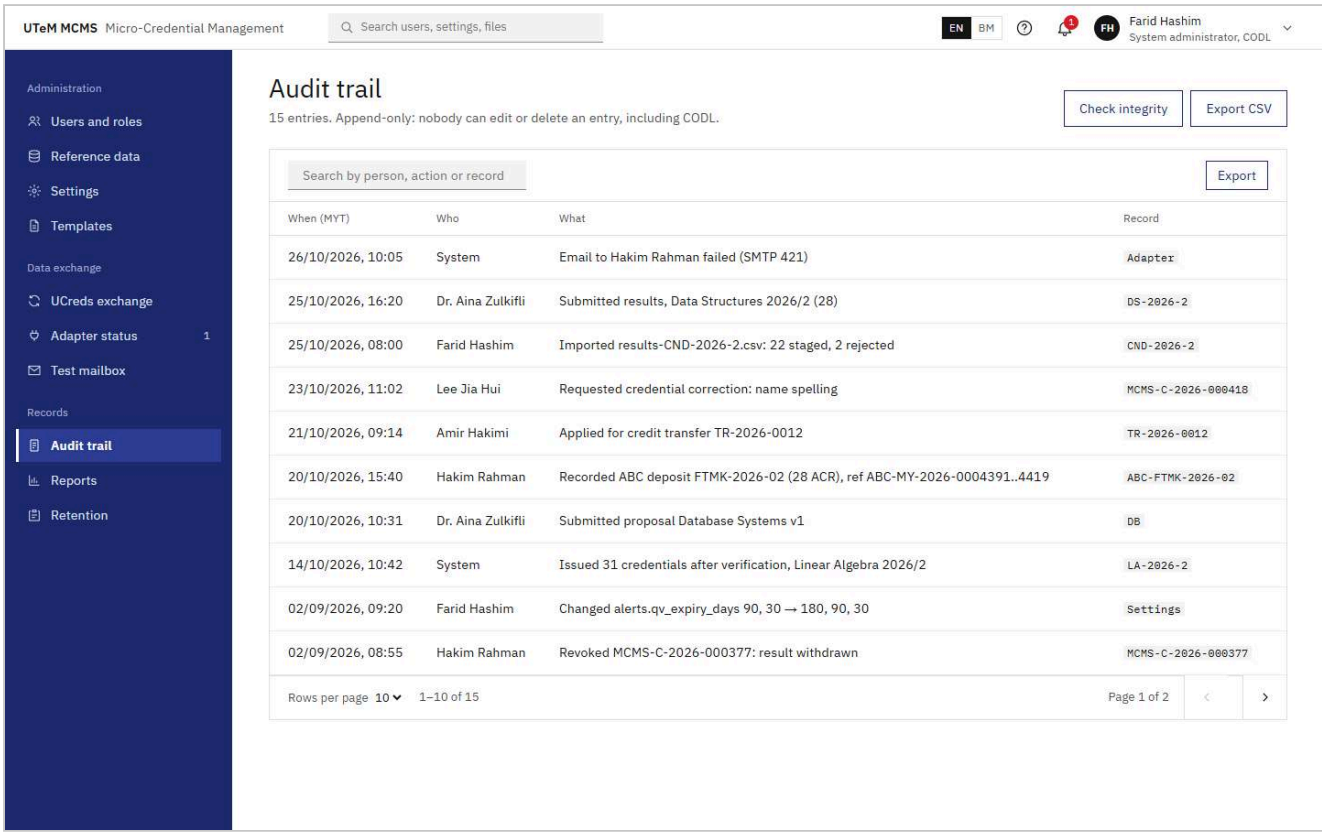


Figure 253: *CODL-10 Audit trail.*

CODL-11 Reports

`#/codl/reports`

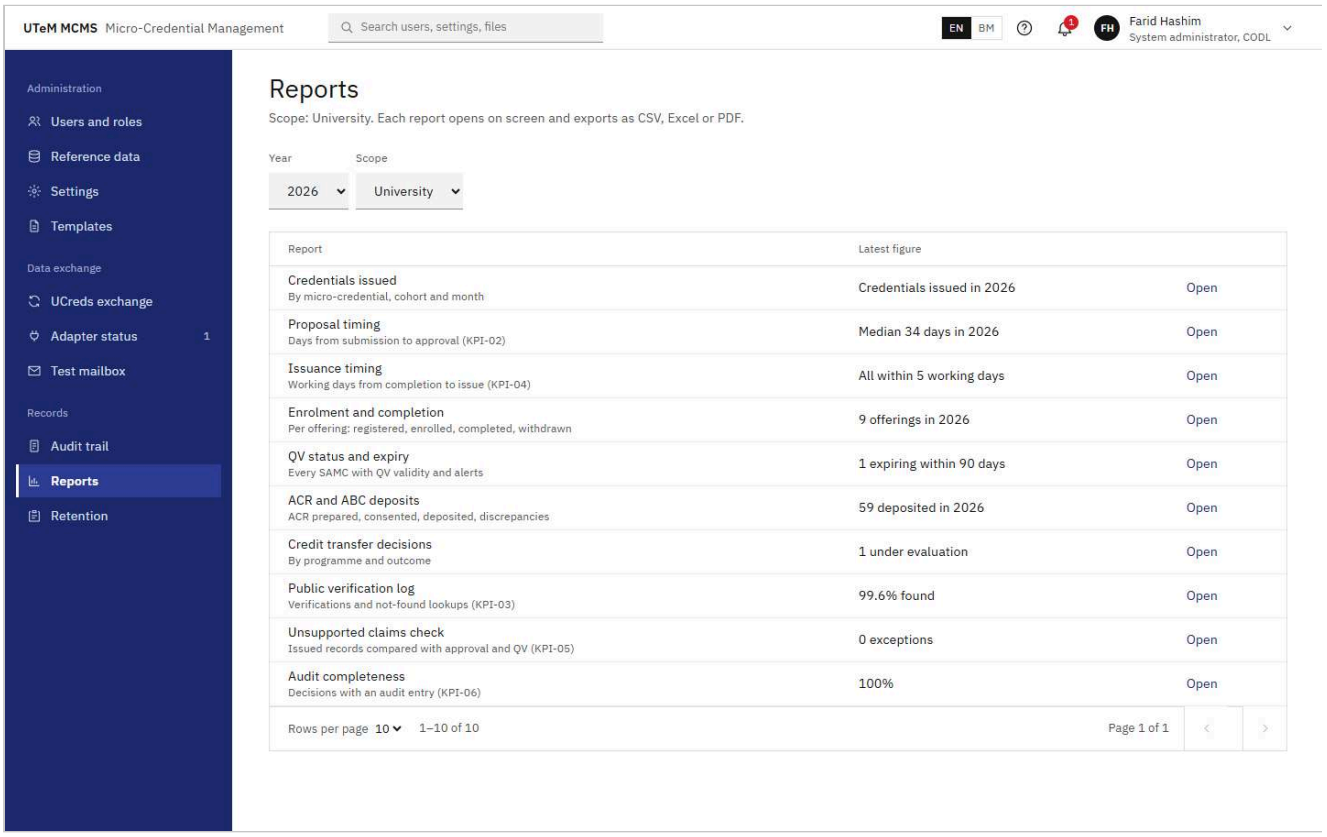


Figure 254: *CODL-11 Reports.*

CODL-12 Report

`#/codl/report/credentials-issued`

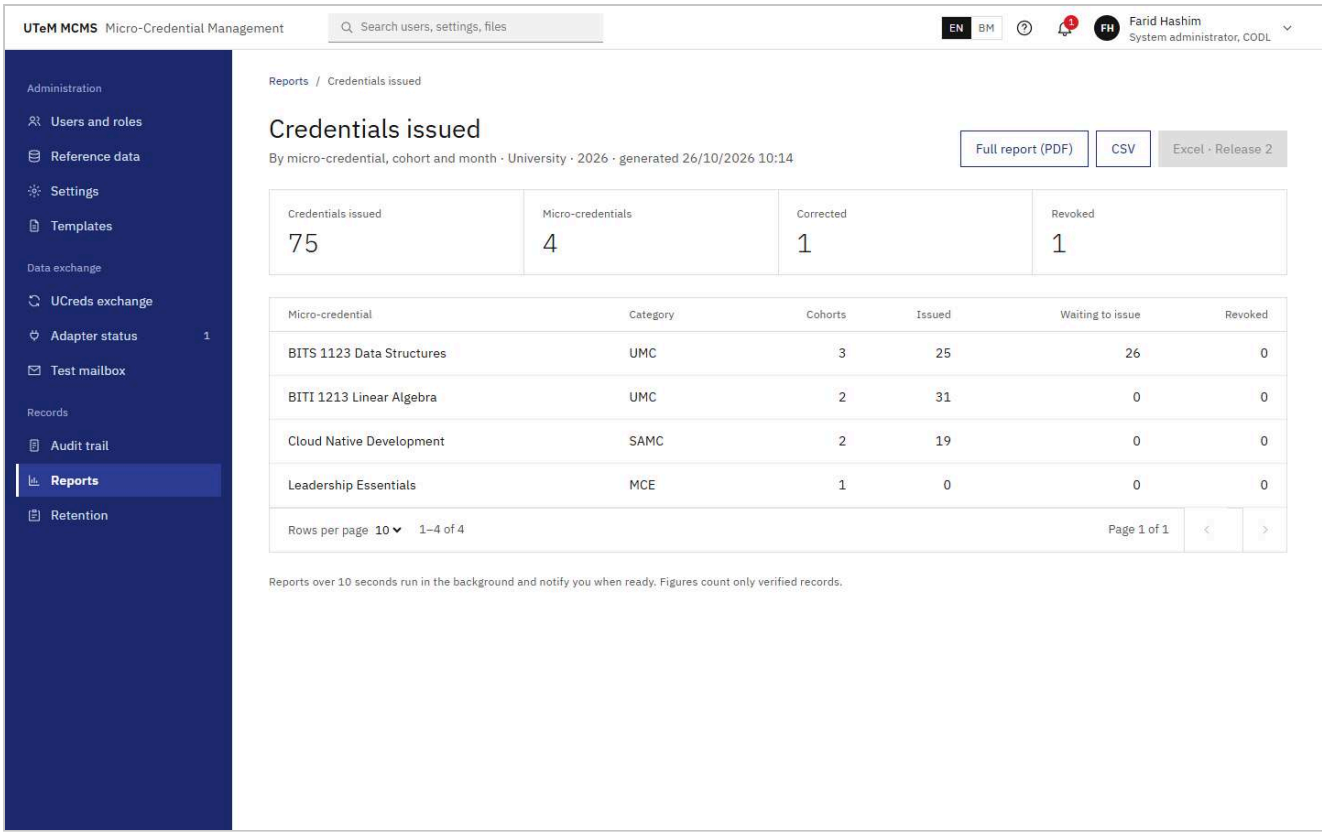


Figure 255: *CODL-12 Report.*

CODL-13 Retention

`#/codl/retention`

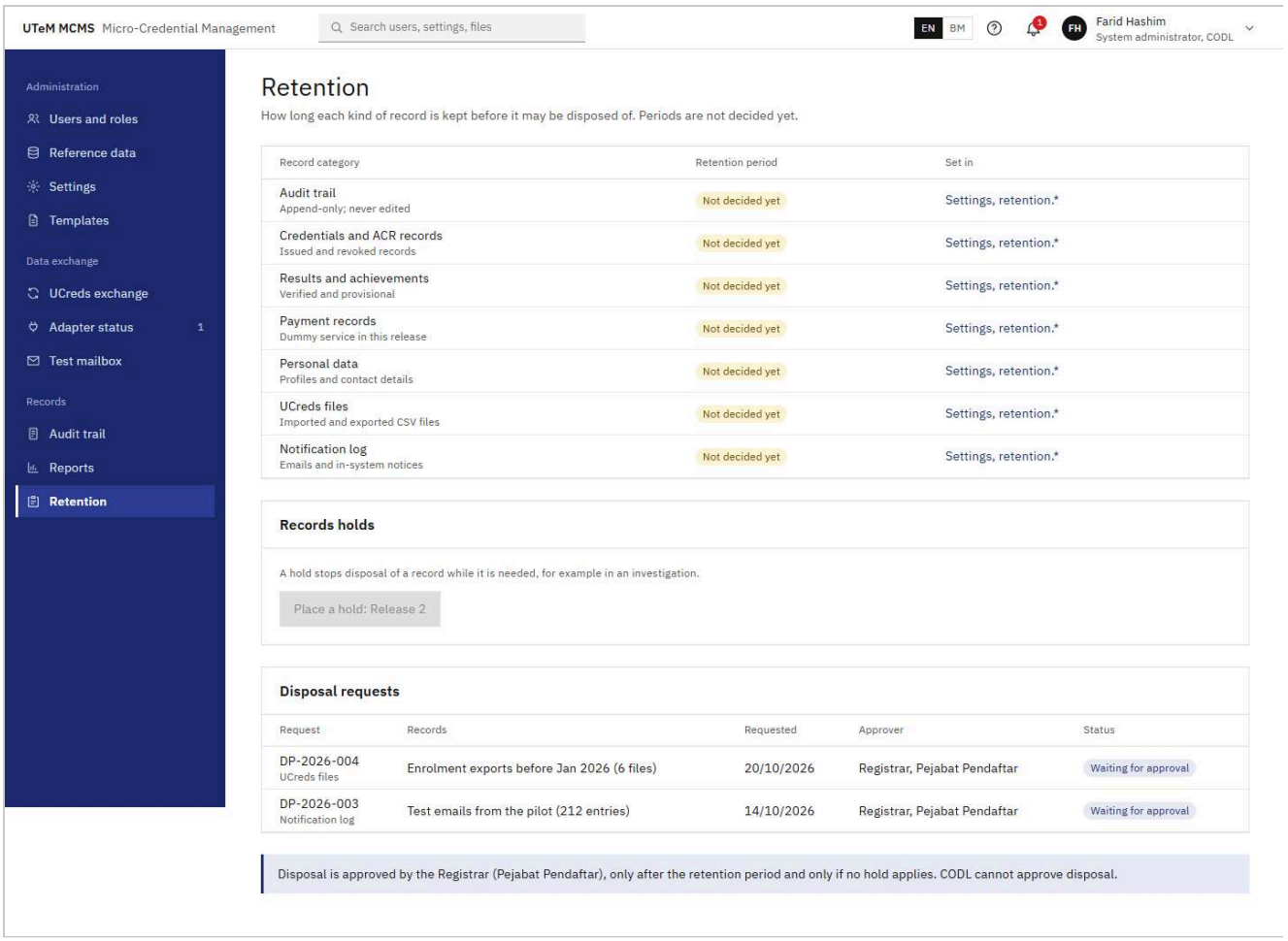


Figure 256: *CODL-13 Retention.*

User (deactivated account)

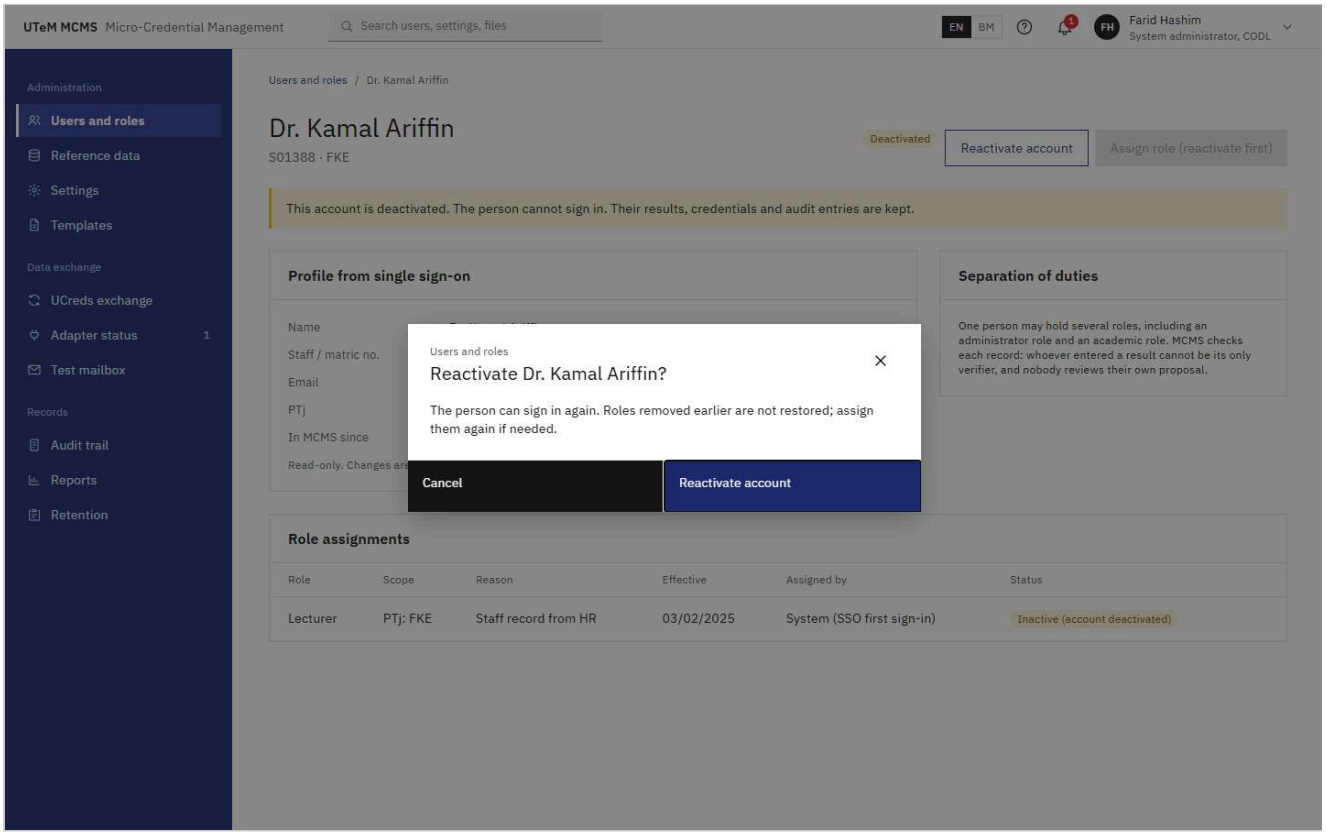


Figure 257: *Confirmation: Reactivate Dr. Kamal Ariffin? (record state: User (deactivated account)).*

UCreds exchange (file validated)

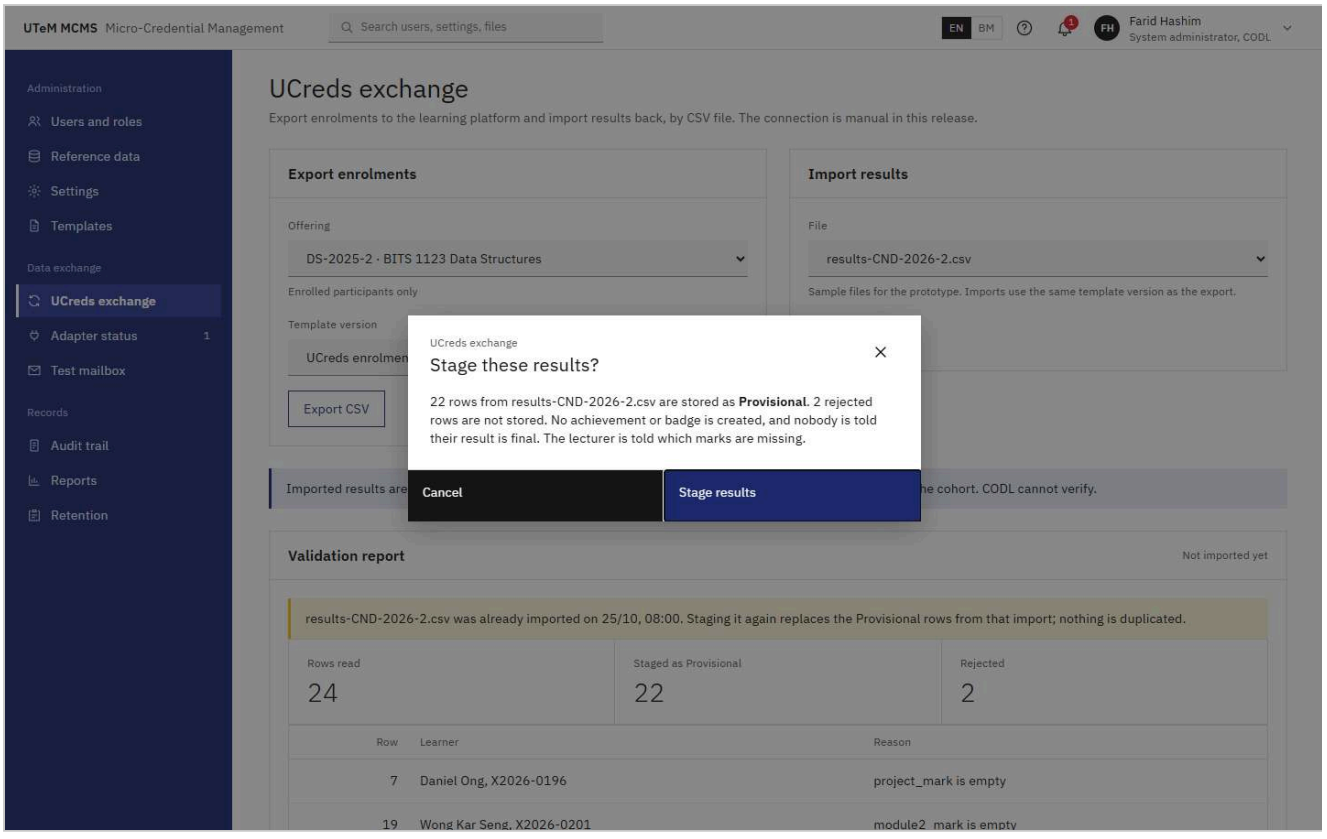


Figure 258: *Confirmation: Stage these results? (record state: UCreds exchange (file validated)).*

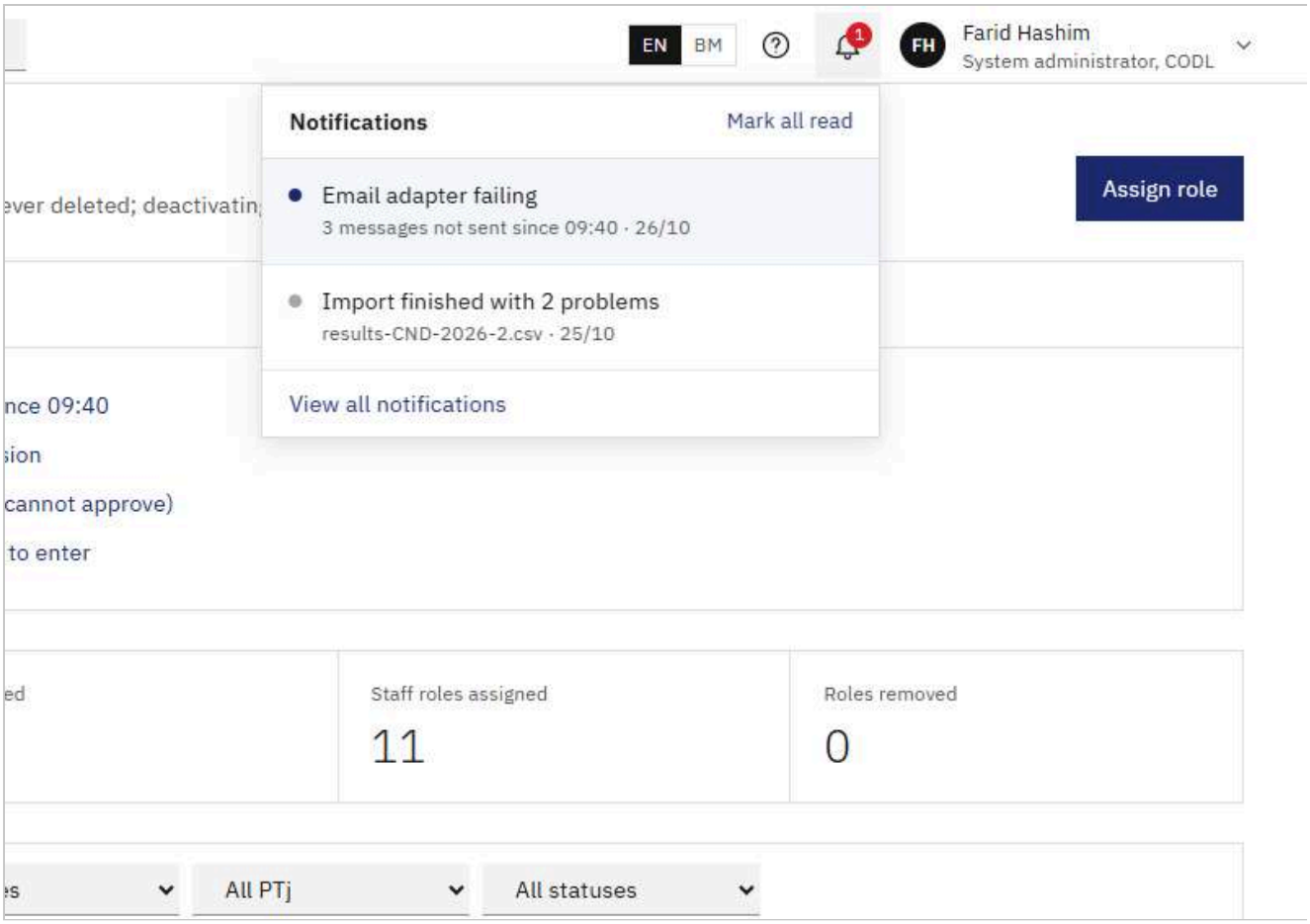


Figure 259: *CODL system administrator: Notifications menu.*

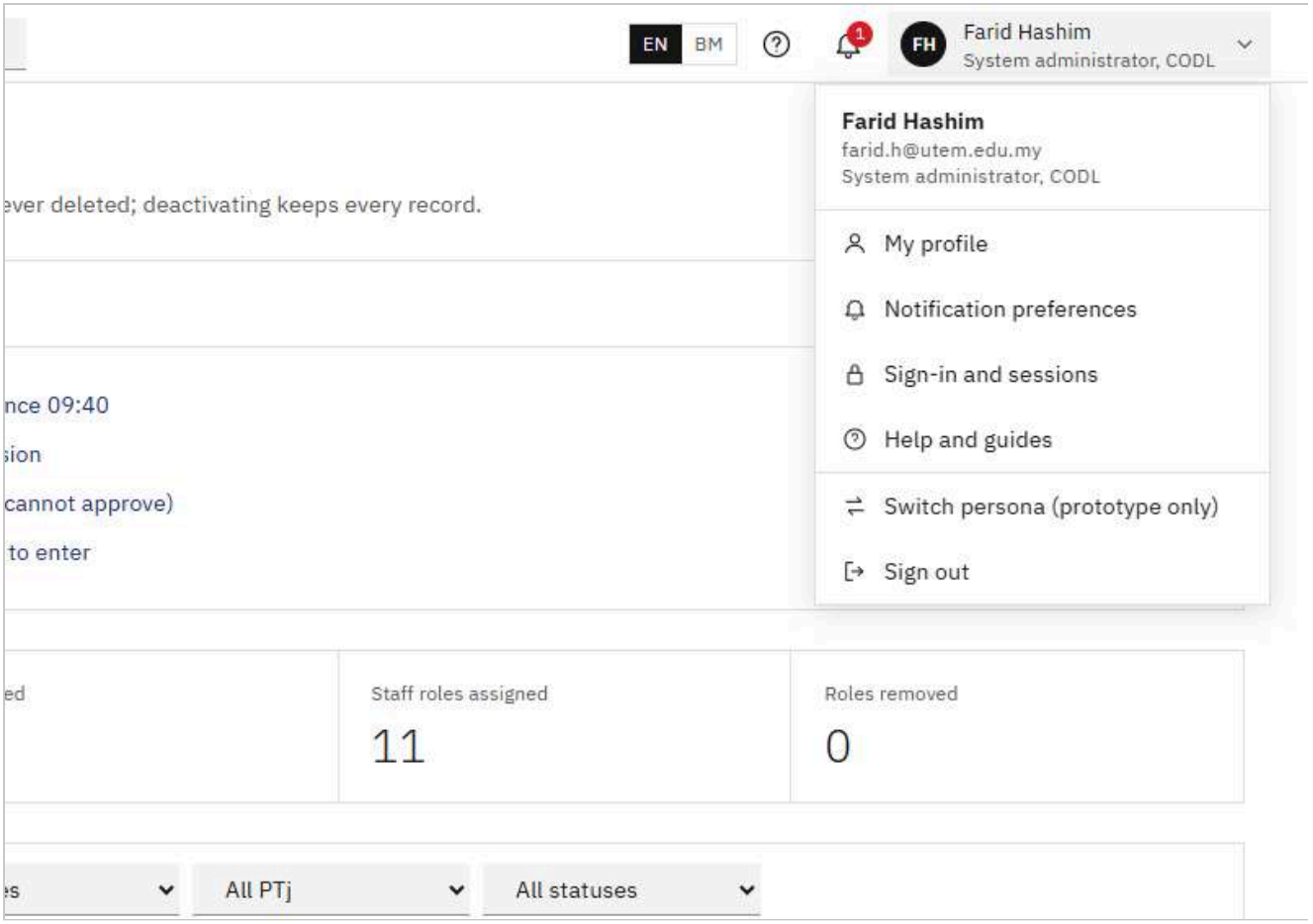


Figure 260: *CODL system administrator: Profile menu.*

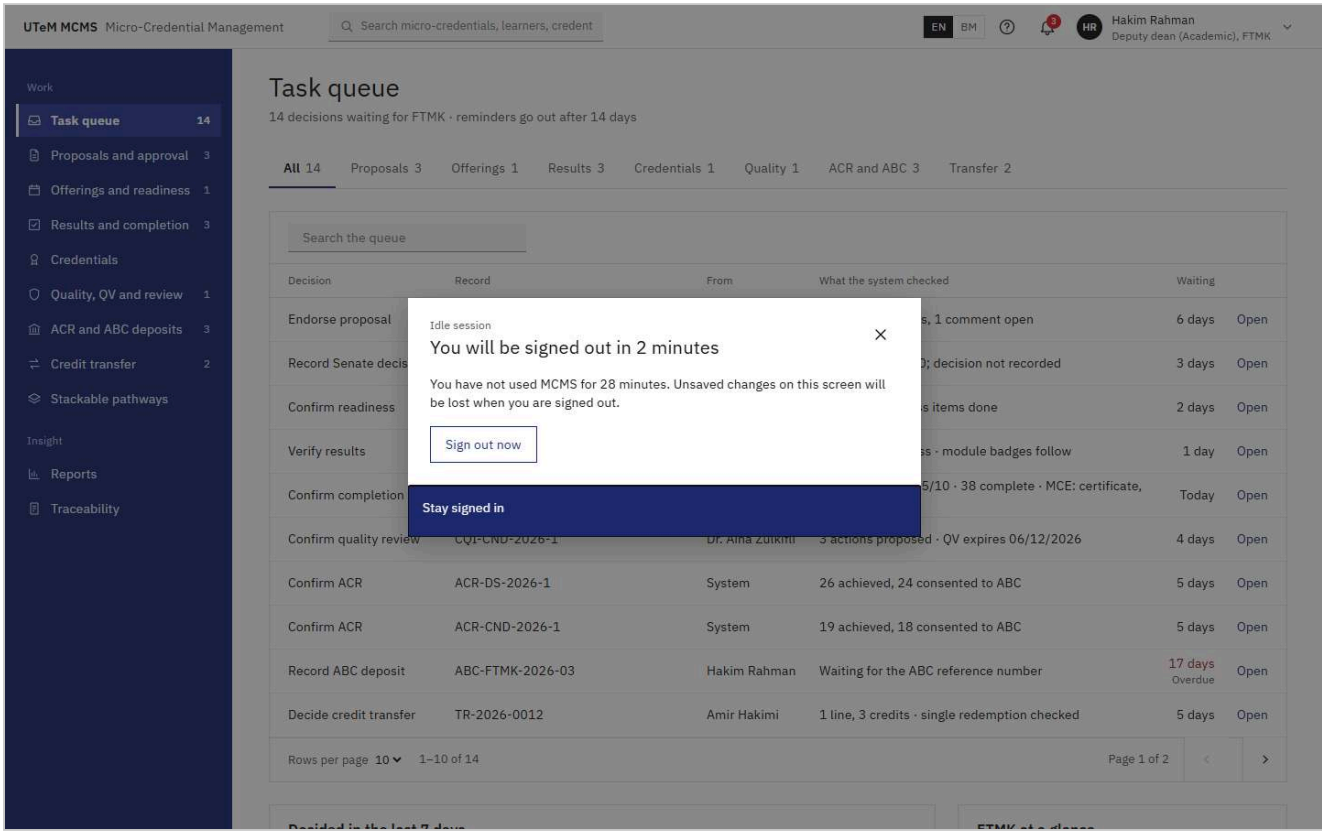


Figure 261: *Shell: Warning: idle sign-out.*

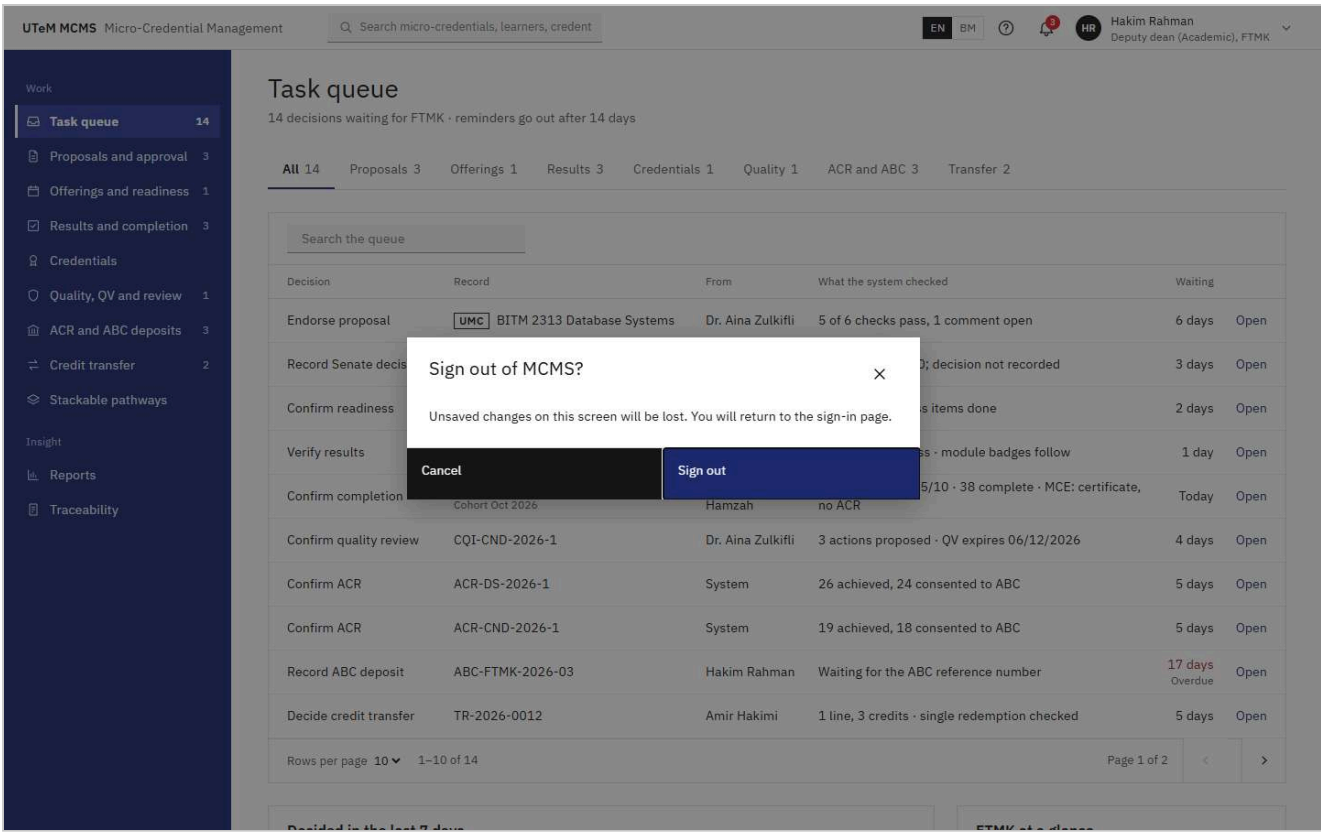


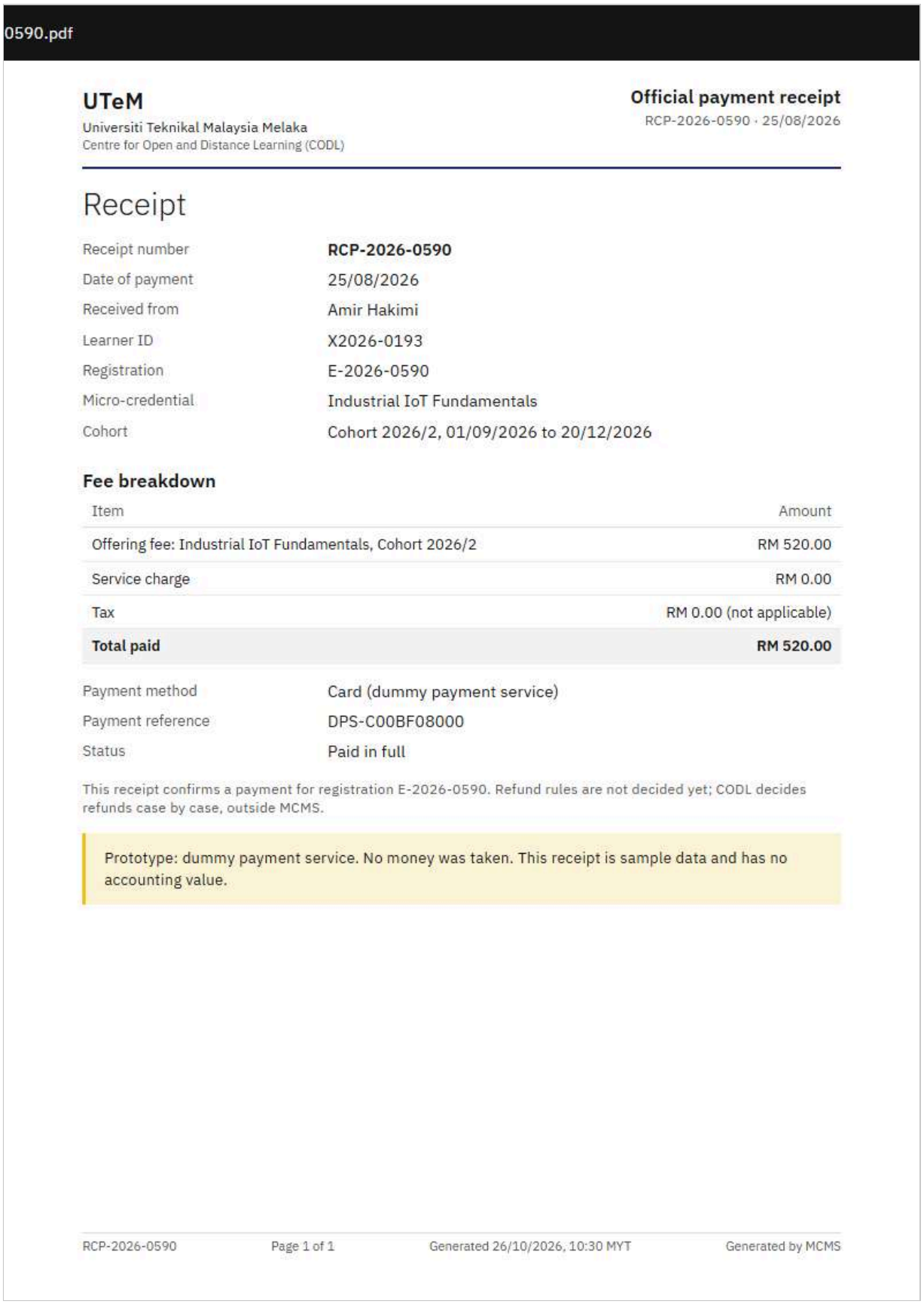
Figure 262: *Shell: Confirmation: sign out.*

Appendix C. Printable Document Gallery

Every page of every printable document and report template of Section 8 (33 pages), as the user prints it or saves it as PDF. Documents use sample data; the report template is shown for the "credentials issued" report and lays out the other nine reports the same way.

DOC-01 Payment receipt

`#/doc/receipt/E-2026-0590`



0590.pdf

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Official payment receipt
RCP-2026-0590 · 25/08/2026

Receipt

Receipt number	**RCP-2026-0590**
Date of payment	25/08/2026
Received from	Amir Hakimi
Learner ID	X2026-0193
Registration	E-2026-0590
Micro-credential	Industrial IoT Fundamentals
Cohort	Cohort 2026/2, 01/09/2026 to 20/12/2026

Fee breakdown

Item	Amount
Offering fee: Industrial IoT Fundamentals, Cohort 2026/2	RM 520.00
Service charge	RM 0.00
Tax	RM 0.00 (not applicable)
Total paid	**RM 520.00**

Payment method	Card (dummy payment service)
Payment reference	DPS-C00BF08000
Status	Paid in full

This receipt confirms a payment for registration E-2026-0590. Refund rules are not decided yet; CODL decides refunds case by case, outside MCMS.

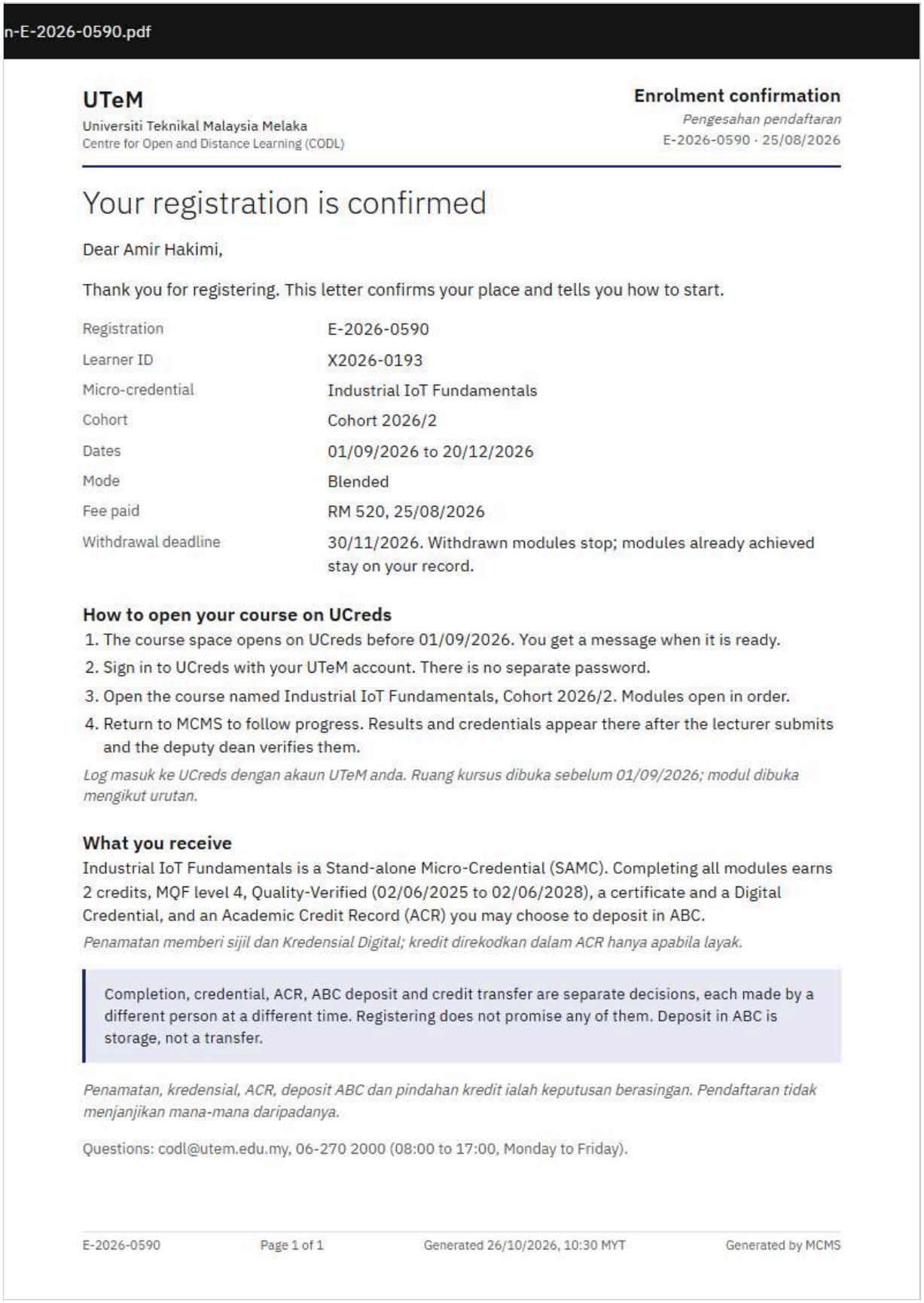
> Prototype: dummy payment service. No money was taken. This receipt is sample data and has no accounting value.

RCP-2026-0590 Page 1 of 1 Generated 26/10/2026, 10:30 MYT Generated by MCMS

Figure 263: *DOC-01 Payment receipt, page 1 of 1.*

DOC-02 Enrolment confirmation

`#/doc/enrolment/E-2026-0590`



n-E-2026-0590.pdf

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Enrolment confirmation
Pengesahan pendaftaran
E-2026-0590 · 25/08/2026

Your registration is confirmed

Dear Amir Hakimi,

Thank you for registering. This letter confirms your place and tells you how to start.

Registration	E-2026-0590
Learner ID	X2026-0193
Micro-credential	Industrial IoT Fundamentals
Cohort	Cohort 2026/2
Dates	01/09/2026 to 20/12/2026
Mode	Blended
Fee paid	RM 520, 25/08/2026
Withdrawal deadline	30/11/2026. Withdrawn modules stop; modules already achieved stay on your record.

How to open your course on UCreds

1. The course space opens on UCreds before 01/09/2026. You get a message when it is ready.
2. Sign in to UCreds with your UTeM account. There is no separate password.
3. Open the course named Industrial IoT Fundamentals, Cohort 2026/2. Modules open in order.
4. Return to MCMS to follow progress. Results and credentials appear there after the lecturer submits and the deputy dean verifies them.

Log masuk ke UCreds dengan akaun UTeM anda. Ruang kursus dibuka sebelum 01/09/2026; modul dibuka mengikut urutan.

What you receive

Industrial IoT Fundamentals is a Stand-alone Micro-Credential (SAMC). Completing all modules earns 2 credits, MQF level 4, Quality-Verified (02/06/2025 to 02/06/2028), a certificate and a Digital Credential, and an Academic Credit Record (ACR) you may choose to deposit in ABC.

Penamatan memberi sijil dan Kredensial Digital; kredit direkodkan dalam ACR hanya apabila layak.

> Completion, credential, ACR, ABC deposit and credit transfer are separate decisions, each made by a different person at a different time. Registering does not promise any of them. Deposit in ABC is storage, not a transfer.

Penamatan, kredensial, ACR, deposit ABC dan pindahan kredit ialah keputusan berasingan. Pendaftaran tidak menjanjikan mana-mana daripadanya.

Questions: codl@utem.edu.my, 06-270 2000 (08:00 to 17:00, Monday to Friday).

E-2026-0590 Page 1 of 1 Generated 26/10/2026, 10:30 MYT Generated by MCMS

Figure 264: *DOC-02 Enrolment confirmation, page 1 of 1.*

DOC-03 Certificate

`#/doc/certificate/MCMS-C-2026-000098`

Figure 265: *DOC-03 Certificate, page 1 of 1.*

DOC-04 Digital credential

`#/doc/credential/MCMS-C-2026-000098`

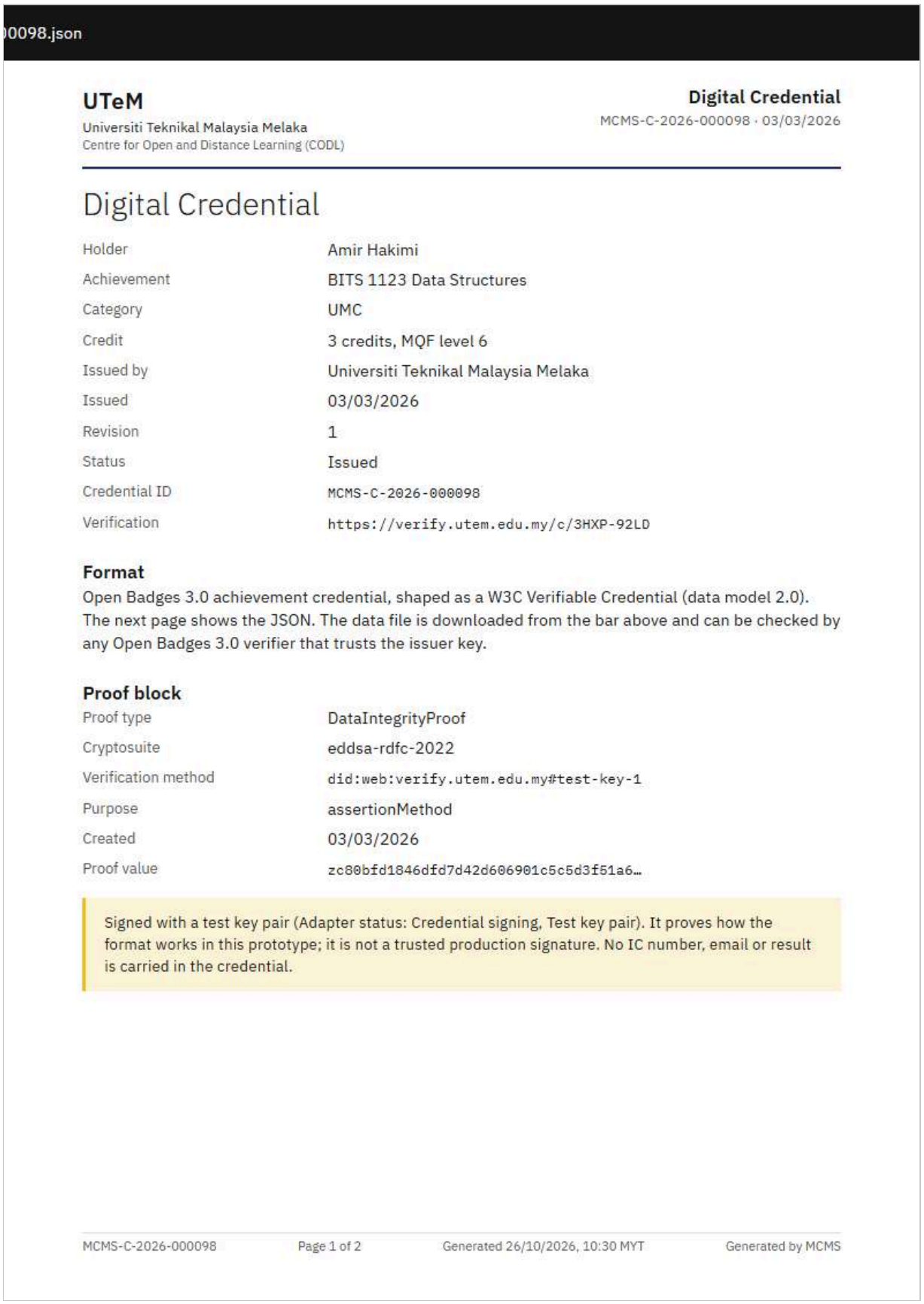
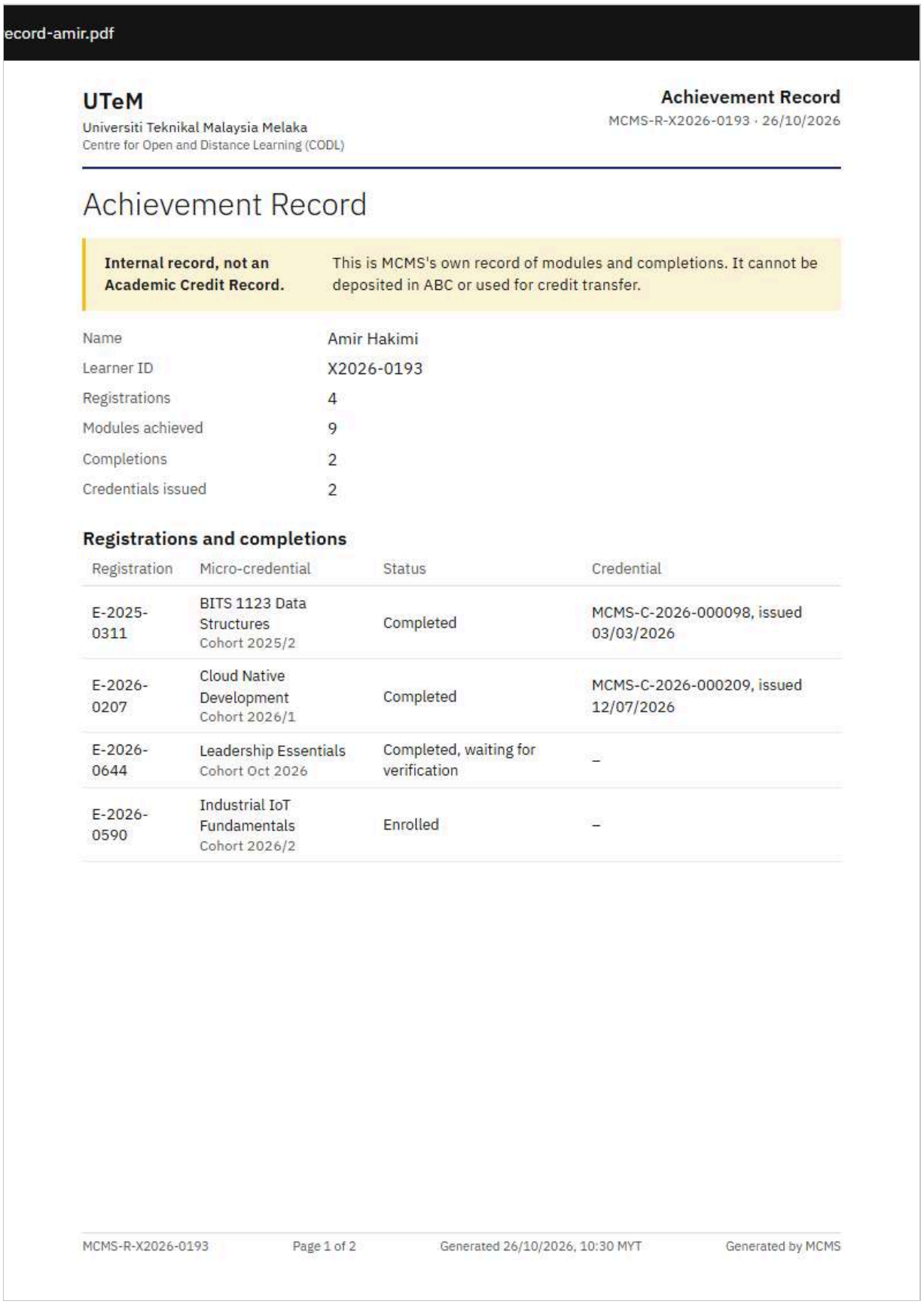


Figure 266: *DOC-04 Digital credential, page 1 of 2.*

Figure 267: *DOC-04 Digital credential, page 2 of 2.*

DOC-05 Achievement record

`#/doc/record/amir`



ecord-amir.pdf

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Achievement Record
MCMS-R-X2026-0193 · 26/10/2026

Achievement Record

Internal record, not an Academic Credit Record. This is MCMS's own record of modules and completions. It cannot be deposited in ABC or used for credit transfer.

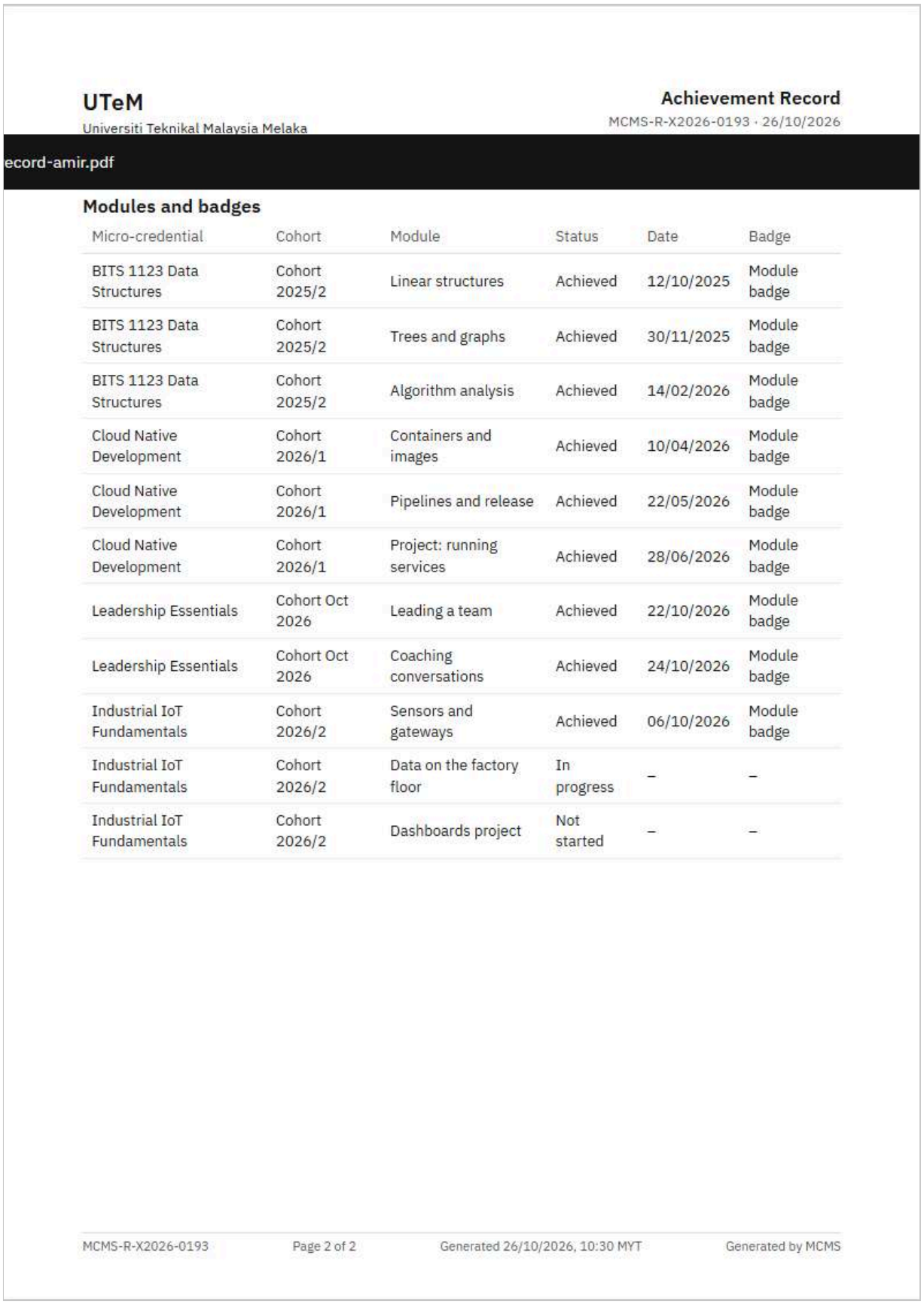
Name	Amir Hakimi
Learner ID	X2026-0193
Registrations	4
Modules achieved	9
Completions	2
Credentials issued	2

Registrations and completions

Registration	Micro-credential	Status	Credential
E-2025-0311	BITS 1123 Data Structures Cohort 2025/2	Completed	MCMS-C-2026-000098, issued 03/03/2026
E-2026-0207	Cloud Native Development Cohort 2026/1	Completed	MCMS-C-2026-000209, issued 12/07/2026
E-2026-0644	Leadership Essentials Cohort Oct 2026	Completed, waiting for verification	–
E-2026-0590	Industrial IoT Fundamentals Cohort 2026/2	Enrolled	–

MCMS-R-X2026-0193 Page 1 of 2 Generated 26/10/2026, 10:30 MYT Generated by MCMS

Figure 268: *DOC-05 Achievement record, page 1 of 2.*



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Achievement Record
MCMS-R-X2026-0193 · 26/10/2026

ecord-amir.pdf

Modules and badges

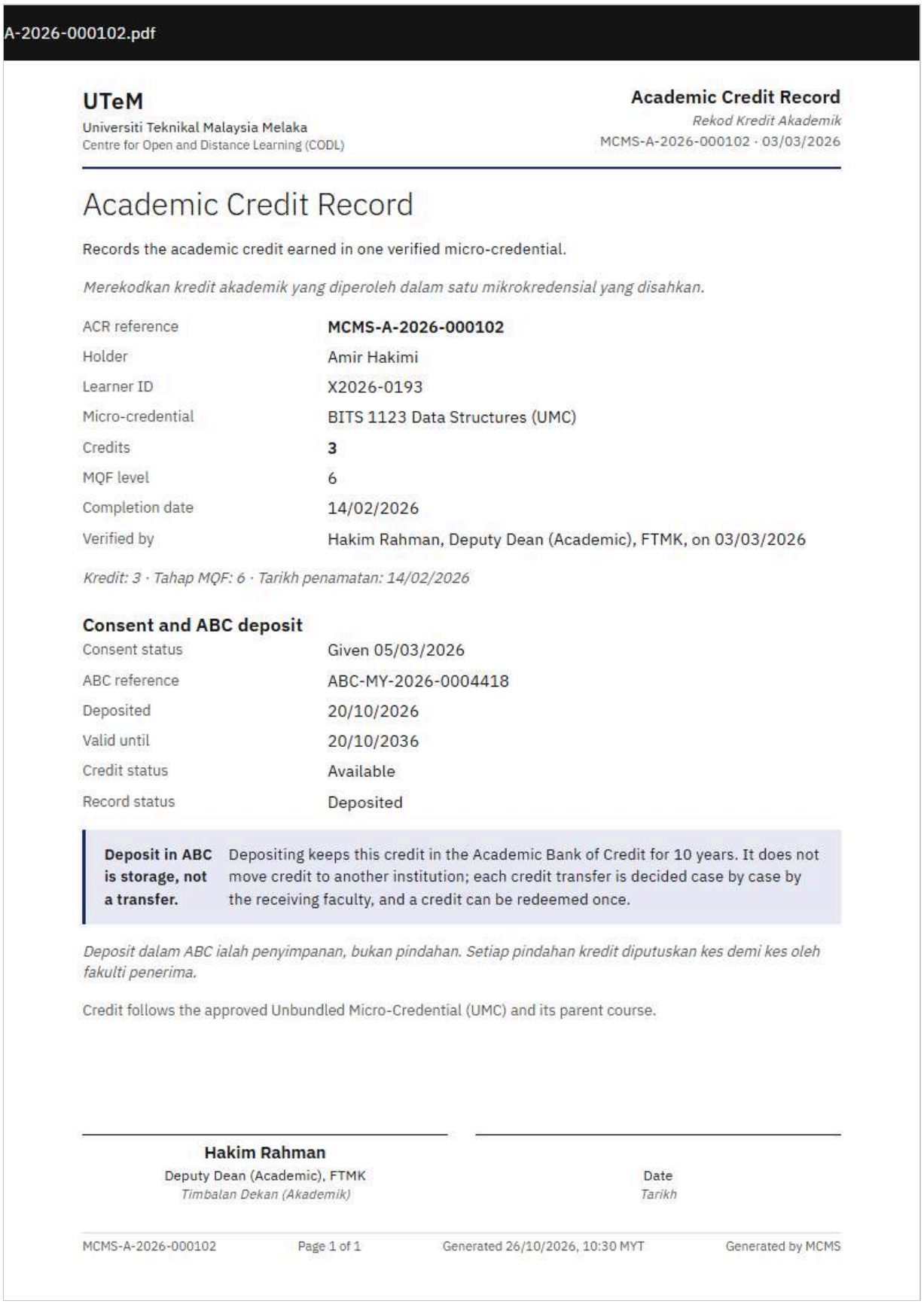
Micro-credential	Cohort	Module	Status	Date	Badge
BITS 1123 Data Structures	Cohort 2025/2	Linear structures	Achieved	12/10/2025	Module badge
BITS 1123 Data Structures	Cohort 2025/2	Trees and graphs	Achieved	30/11/2025	Module badge
BITS 1123 Data Structures	Cohort 2025/2	Algorithm analysis	Achieved	14/02/2026	Module badge
Cloud Native Development	Cohort 2026/1	Containers and images	Achieved	10/04/2026	Module badge
Cloud Native Development	Cohort 2026/1	Pipelines and release	Achieved	22/05/2026	Module badge
Cloud Native Development	Cohort 2026/1	Project: running services	Achieved	28/06/2026	Module badge
Leadership Essentials	Cohort Oct 2026	Leading a team	Achieved	22/10/2026	Module badge
Leadership Essentials	Cohort Oct 2026	Coaching conversations	Achieved	24/10/2026	Module badge
Industrial IoT Fundamentals	Cohort 2026/2	Sensors and gateways	Achieved	06/10/2026	Module badge
Industrial IoT Fundamentals	Cohort 2026/2	Data on the factory floor	In progress	–	–
Industrial IoT Fundamentals	Cohort 2026/2	Dashboards project	Not started	–	–

MCMS-R-X2026-0193 Page 2 of 2 Generated 26/10/2026, 10:30 MYT Generated by MCMS

Figure 269: *DOC-05 Achievement record, page 2 of 2.*

DOC-06 Academic Credit Record

`#/doc/acr/MCMS-A-2026-000102`



A-2026-000102.pdf

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Academic Credit Record
Rekod Kredit Akademik
MCMS-A-2026-000102 · 03/03/2026

Academic Credit Record

Records the academic credit earned in one verified micro-credential.

Merekodkan kredit akademik yang diperoleh dalam satu mikrokredensial yang disahkan.

ACR reference	**MCMS-A-2026-000102**
Holder	Amir Hakimi
Learner ID	X2026-0193
Micro-credential	BITS 1123 Data Structures (UMC)
Credits	**3**
MQF level	6
Completion date	14/02/2026
Verified by	Hakim Rahman, Deputy Dean (Academic), FTMK, on 03/03/2026

Kredit: 3 · Tahap MQF: 6 · Tarikh penamatan: 14/02/2026

Consent and ABC deposit

Consent status	Given 05/03/2026
ABC reference	ABC-MY-2026-0004418
Deposited	20/10/2026
Valid until	20/10/2036
Credit status	Available
Record status	Deposited

Deposit in ABC is storage, not a transfer. Depositing keeps this credit in the Academic Bank of Credit for 10 years. It does not move credit to another institution; each credit transfer is decided case by case by the receiving faculty, and a credit can be redeemed once.

Deposit dalam ABC ialah penyimpanan, bukan pindahan. Setiap pindahan kredit diputuskan kes demi kes oleh fakulti penerima.

Credit follows the approved Unbundled Micro-Credential (UMC) and its parent course.

Hakim Rahman
Deputy Dean (Academic), FTMK
Timbalan Dekan (Akademik)

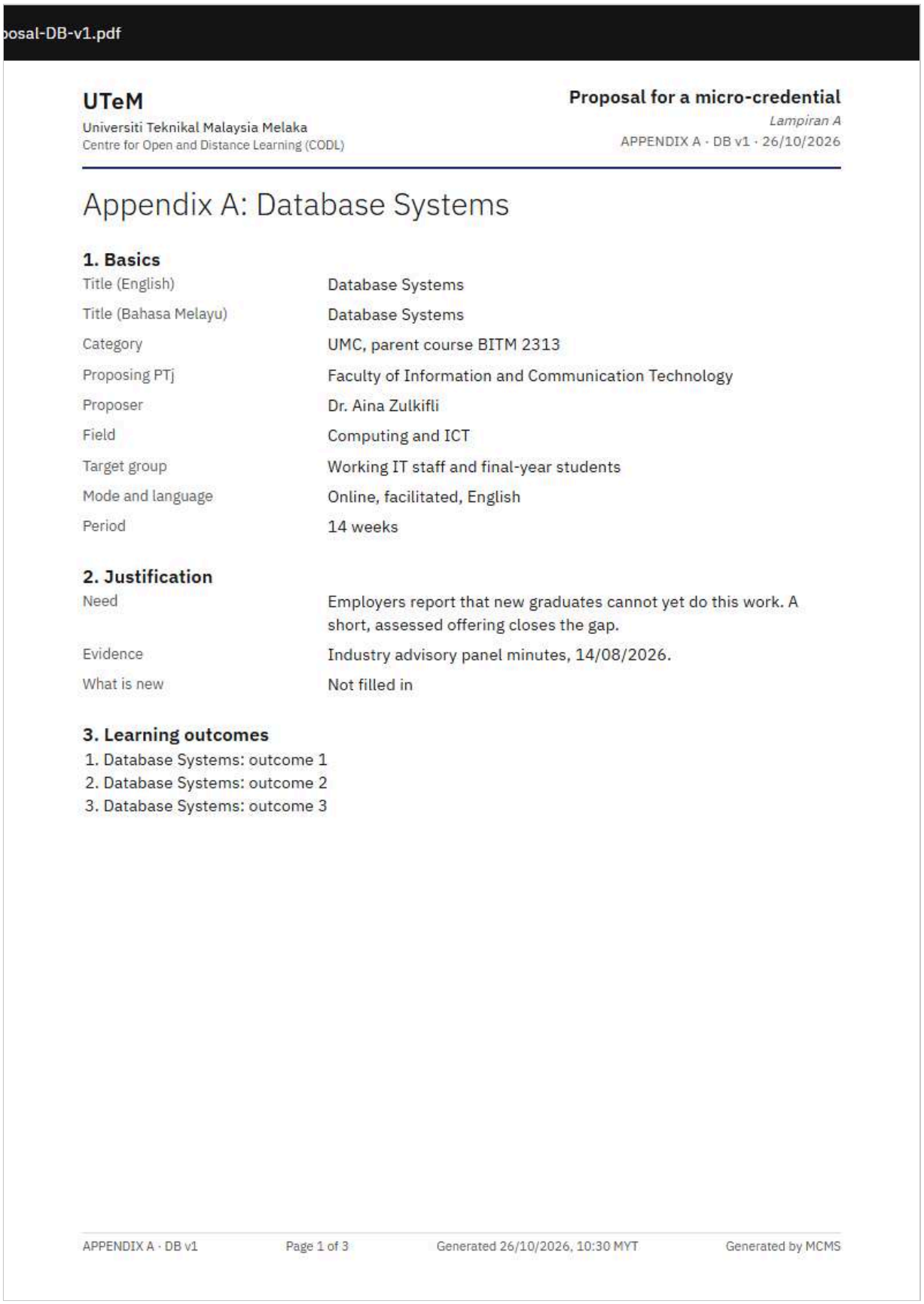
Date
Tarikh

MCMS-A-2026-000102 Page 1 of 1 Generated 26/10/2026, 10:30 MYT Generated by MCMS

Figure 270: *DOC-06 Academic Credit Record, page 1 of 1.*

DOC-07 Proposal print

`#/doc/proposal/DB`



posal-DB-v1.pdf

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Proposal for a micro-credential
Lampiran A
APPENDIX A · DB v1 · 26/10/2026

Appendix A: Database Systems

1. Basics

Title (English)	Database Systems
Title (Bahasa Melayu)	Database Systems
Category	UMC, parent course BITM 2313
Proposing PTj	Faculty of Information and Communication Technology
Proposer	Dr. Aina Zulkifli
Field	Computing and ICT
Target group	Working IT staff and final-year students
Mode and language	Online, facilitated, English
Period	14 weeks

2. Justification

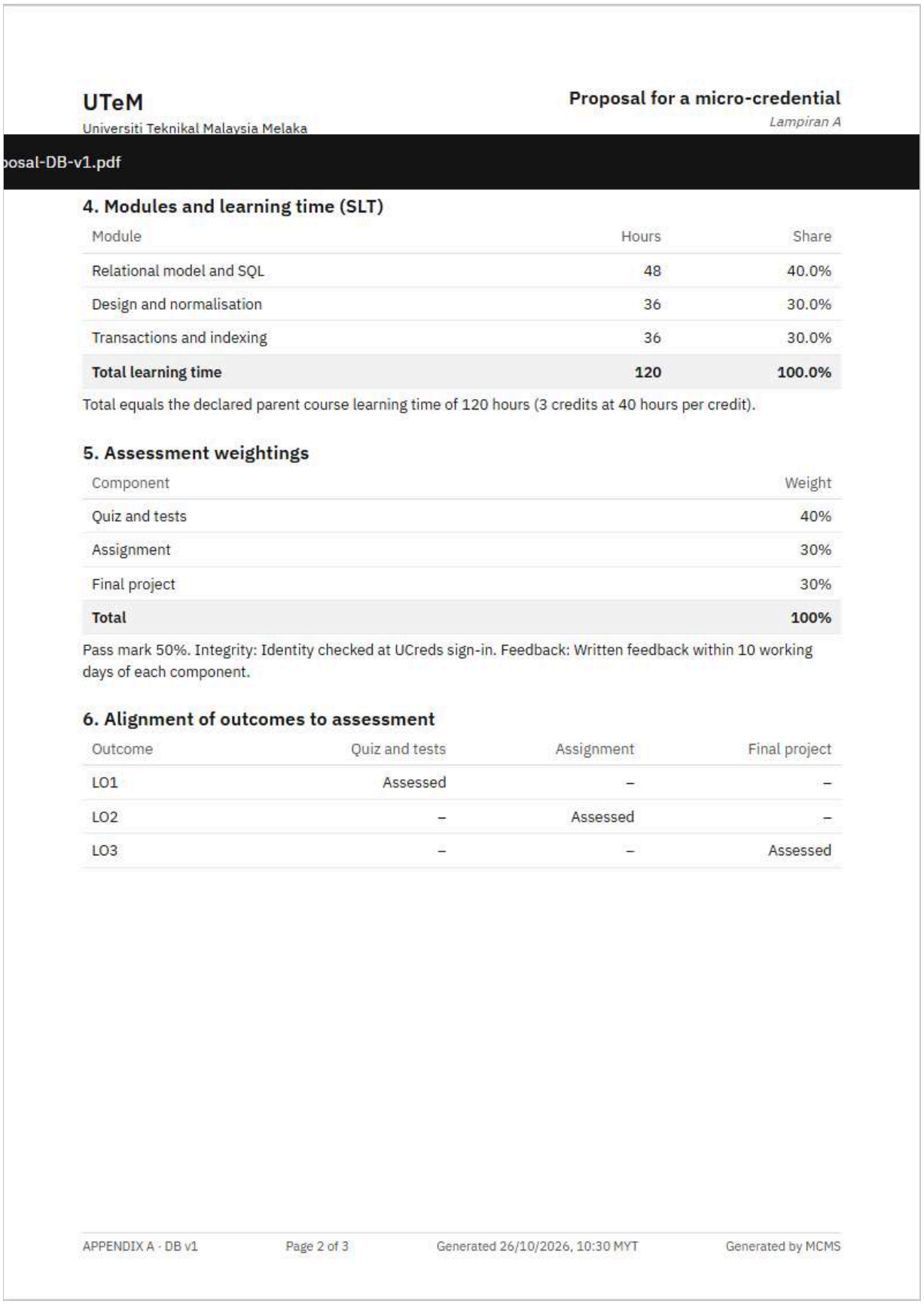
Need	Employers report that new graduates cannot yet do this work. A short, assessed offering closes the gap.
Evidence	Industry advisory panel minutes, 14/08/2026.
What is new	Not filled in

3. Learning outcomes

1. Database Systems: outcome 1
2. Database Systems: outcome 2
3. Database Systems: outcome 3

APPENDIX A · DB v1 Page 1 of 3 Generated 26/10/2026, 10:30 MYT Generated by MCMS

Figure 271: *DOC-07 Proposal print, page 1 of 3.*



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Proposal for a micro-credential
Lampiran A

posal-DB-v1.pdf

4. Modules and learning time (SLT)

Module	Hours	Share
Relational model and SQL	48	40.0%
Design and normalisation	36	30.0%
Transactions and indexing	36	30.0%
Total learning time	**120**	**100.0%**

Total equals the declared parent course learning time of 120 hours (3 credits at 40 hours per credit).

5. Assessment weightings

Component	Weight
Quiz and tests	40%
Assignment	30%
Final project	30%
Total	**100%**

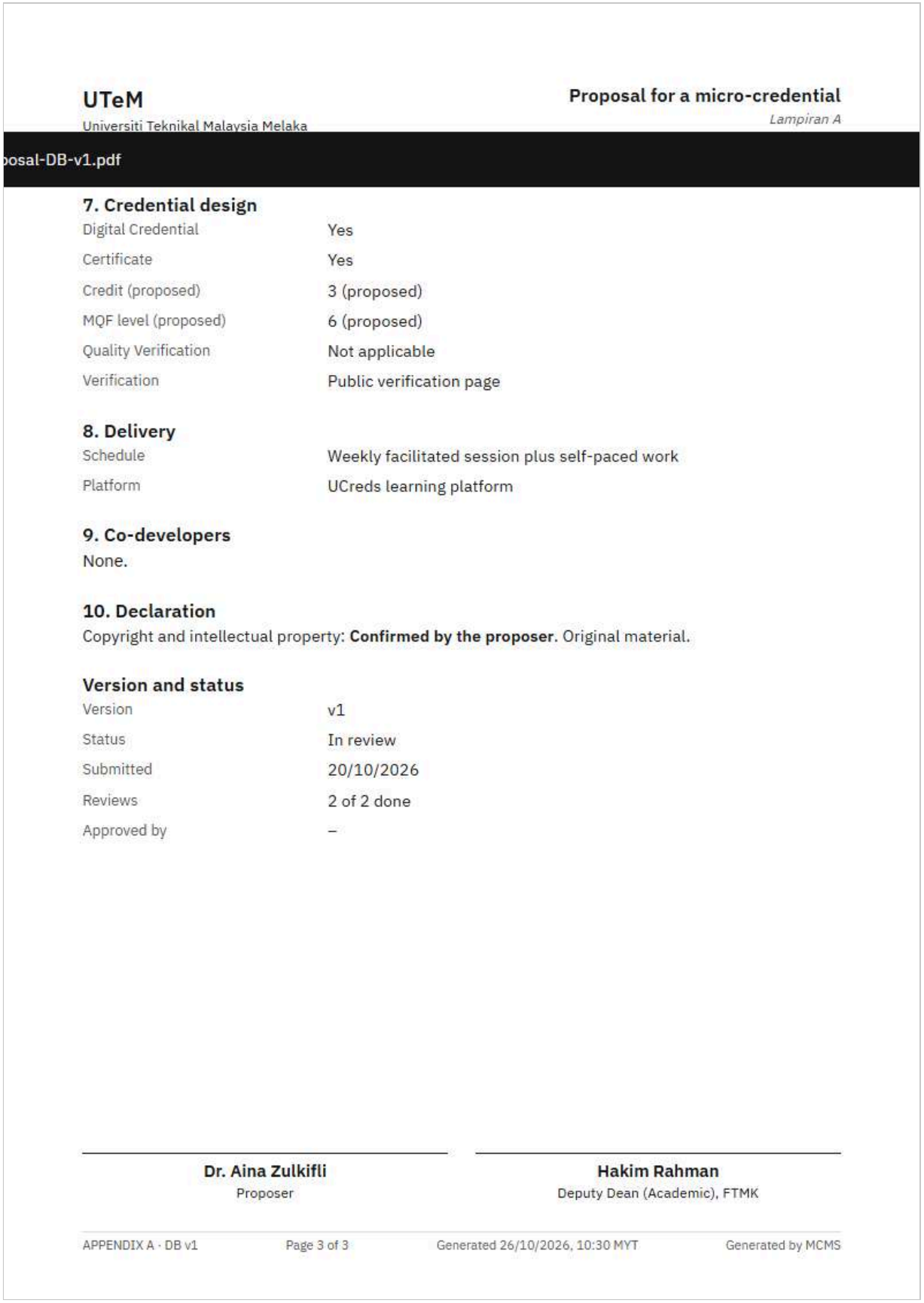
Pass mark 50%. Integrity: Identity checked at UCreds sign-in. Feedback: Written feedback within 10 working days of each component.

6. Alignment of outcomes to assessment

Outcome	Quiz and tests	Assignment	Final project
LO1	Assessed	–	–
LO2	–	Assessed	–
LO3	–	–	Assessed

APPENDIX A · DB v1 Page 2 of 3 Generated 26/10/2026, 10:30 MYT Generated by MCMS

Figure 272: *DOC-07 Proposal print, page 2 of 3.*



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Proposal for a micro-credential
Lampiran A

oposal-DB-v1.pdf

7. Credential design

Digital Credential	Yes
Certificate	Yes
Credit (proposed)	3 (proposed)
MQF level (proposed)	6 (proposed)
Quality Verification	Not applicable
Verification	Public verification page

8. Delivery

Schedule	Weekly facilitated session plus self-paced work
Platform	UCreds learning platform

9. Co-developers

None.

10. Declaration

Copyright and intellectual property: **Confirmed by the proposer**. Original material.

Version and status

Version	v1
Status	In review
Submitted	20/10/2026
Reviews	2 of 2 done
Approved by	–

Dr. Aina Zulkifli
Proposer

Hakim Rahman
Deputy Dean (Academic), FTMK

APPENDIX A · DB v1 Page 3 of 3 Generated 26/10/2026, 10:30 MYT Generated by MCMS

Figure 273: *DOC-07 Proposal print, page 3 of 3.*

DOC-08 Senate decision pack

`#/doc/pack/ADA`

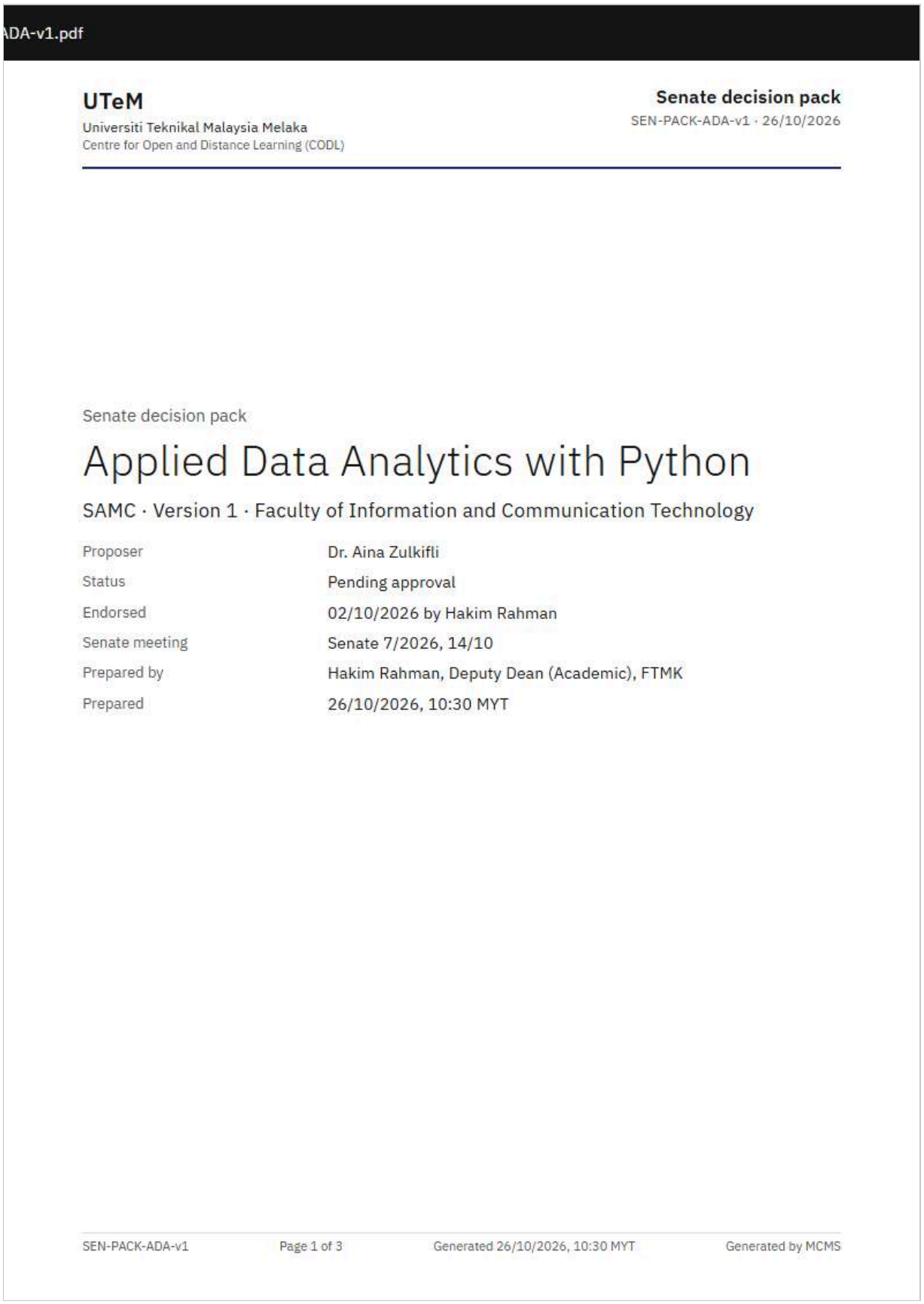
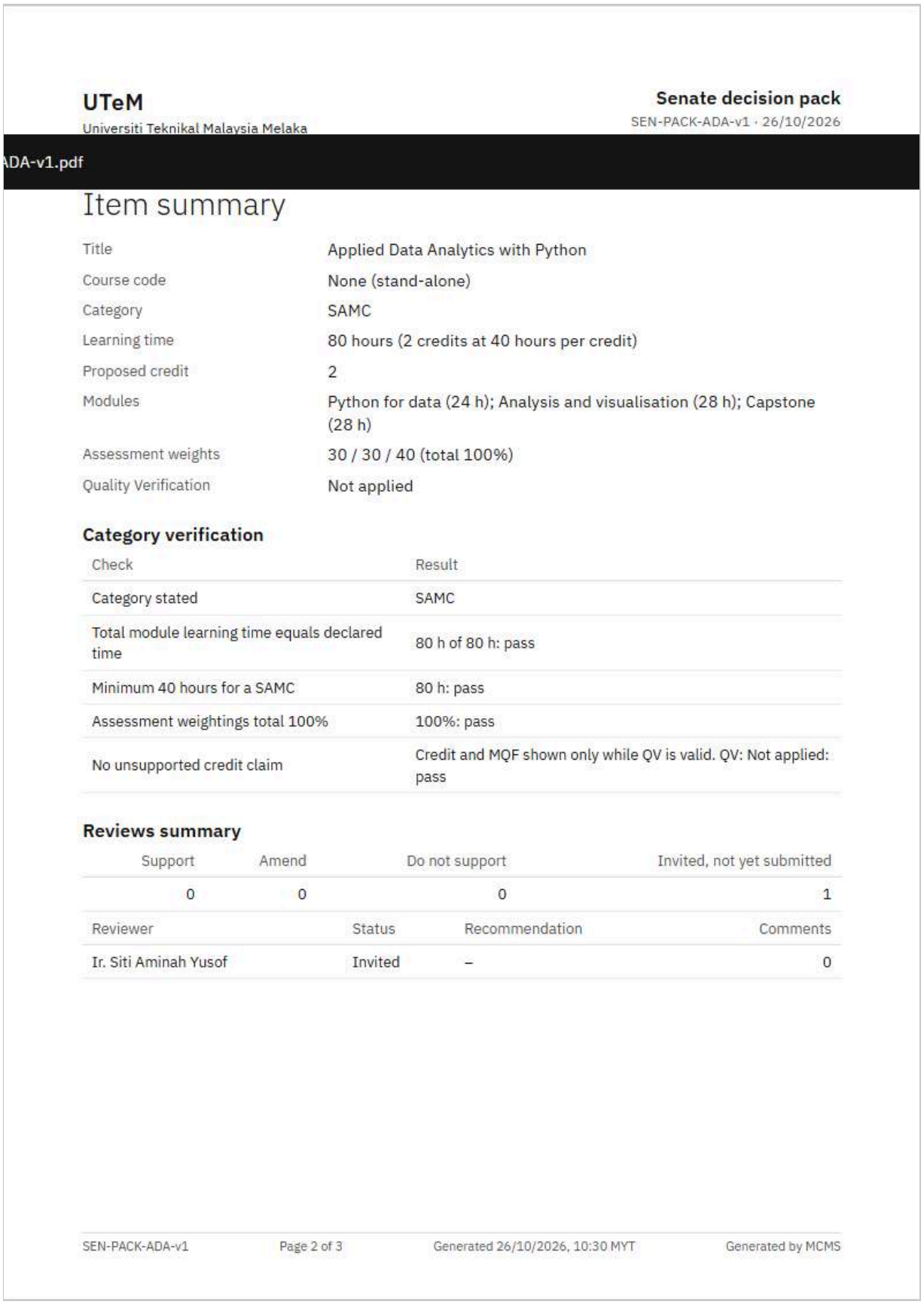


Figure 274: *DOC-08 Senate decision pack, page 1 of 3.*



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Senate decision pack
SEN-PACK-ADA-v1 · 26/10/2026

ADA-v1.pdf

Item summary

Title	Applied Data Analytics with Python
Course code	None (stand-alone)
Category	SAMC
Learning time	80 hours (2 credits at 40 hours per credit)
Proposed credit	2
Modules	Python for data (24 h); Analysis and visualisation (28 h); Capstone (28 h)
Assessment weights	30 / 30 / 40 (total 100%)
Quality Verification	Not applied

Category verification

Check	Result
Category stated	SAMC
Total module learning time equals declared time	80 h of 80 h: pass
Minimum 40 hours for a SAMC	80 h: pass
Assessment weightings total 100%	100%: pass
No unsupported credit claim	Credit and MQF shown only while QV is valid. QV: Not applied: pass

Reviews summary

Support	Amend	Do not support	Invited, not yet submitted
0	0	0	1

Reviewer	Status	Recommendation	Comments
Ir. Siti Aminah Yusof	Invited	–	0

SEN-PACK-ADA-v1 Page 2 of 3 Generated 26/10/2026, 10:30 MYT Generated by MCMS

Figure 275: *DOC-08 Senate decision pack, page 2 of 3.*

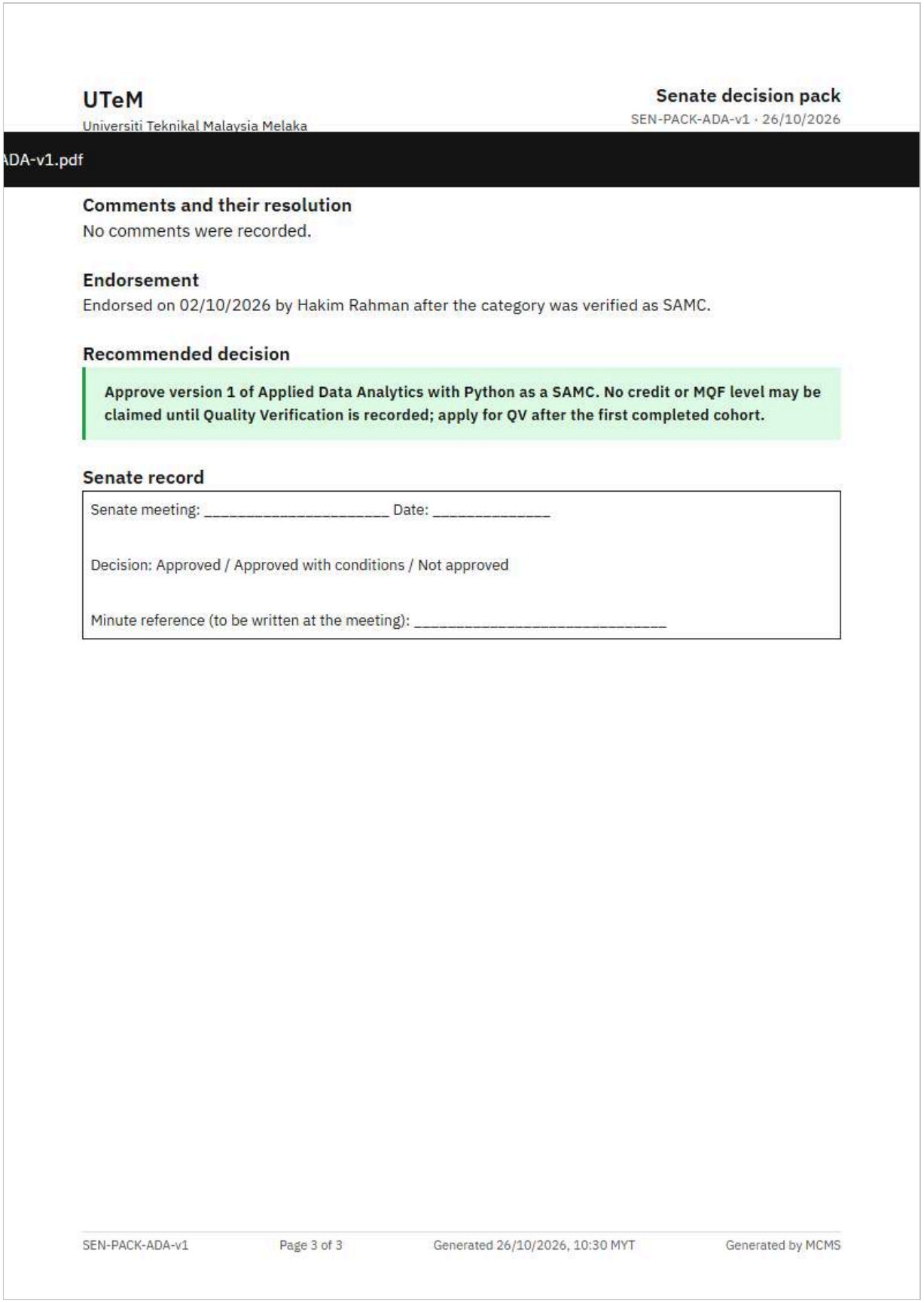
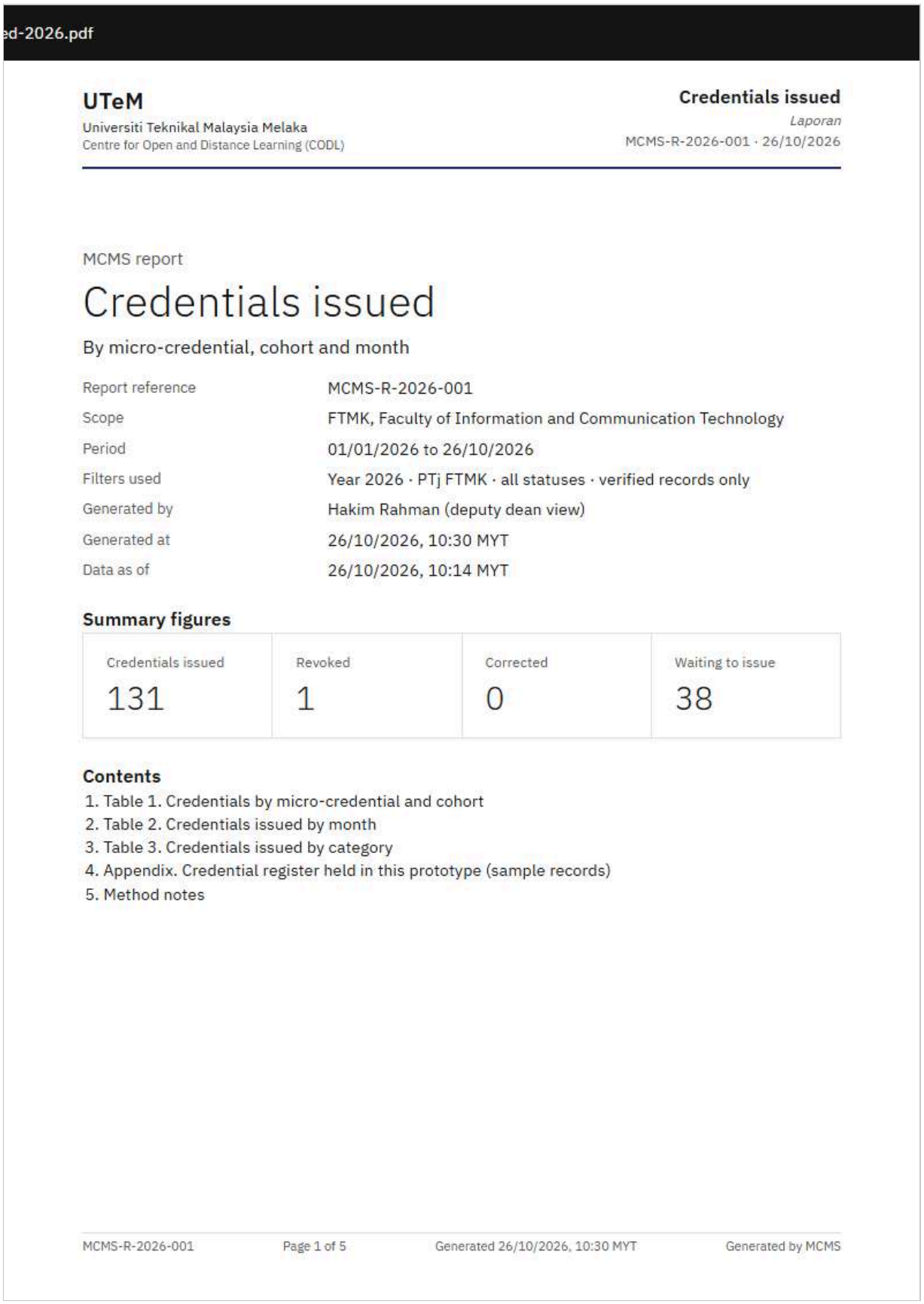


Figure 276: *DOC-08 Senate decision pack, page 3 of 3.*

DOC-09 Report

`#/doc/report/credentials-issued`



ed-2026.pdf

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Centre for Open and Distance Learning (CODL)

Credentials issued
Laporan
MCMS-R-2026-001 · 26/10/2026

MCMS report

Credentials issued

By micro-credential, cohort and month

Report reference	MCMS-R-2026-001
Scope	FTMK, Faculty of Information and Communication Technology
Period	01/01/2026 to 26/10/2026
Filters used	Year 2026 · PTj FTMK · all statuses · verified records only
Generated by	Hakim Rahman (deputy dean view)
Generated at	26/10/2026, 10:30 MYT
Data as of	26/10/2026, 10:14 MYT

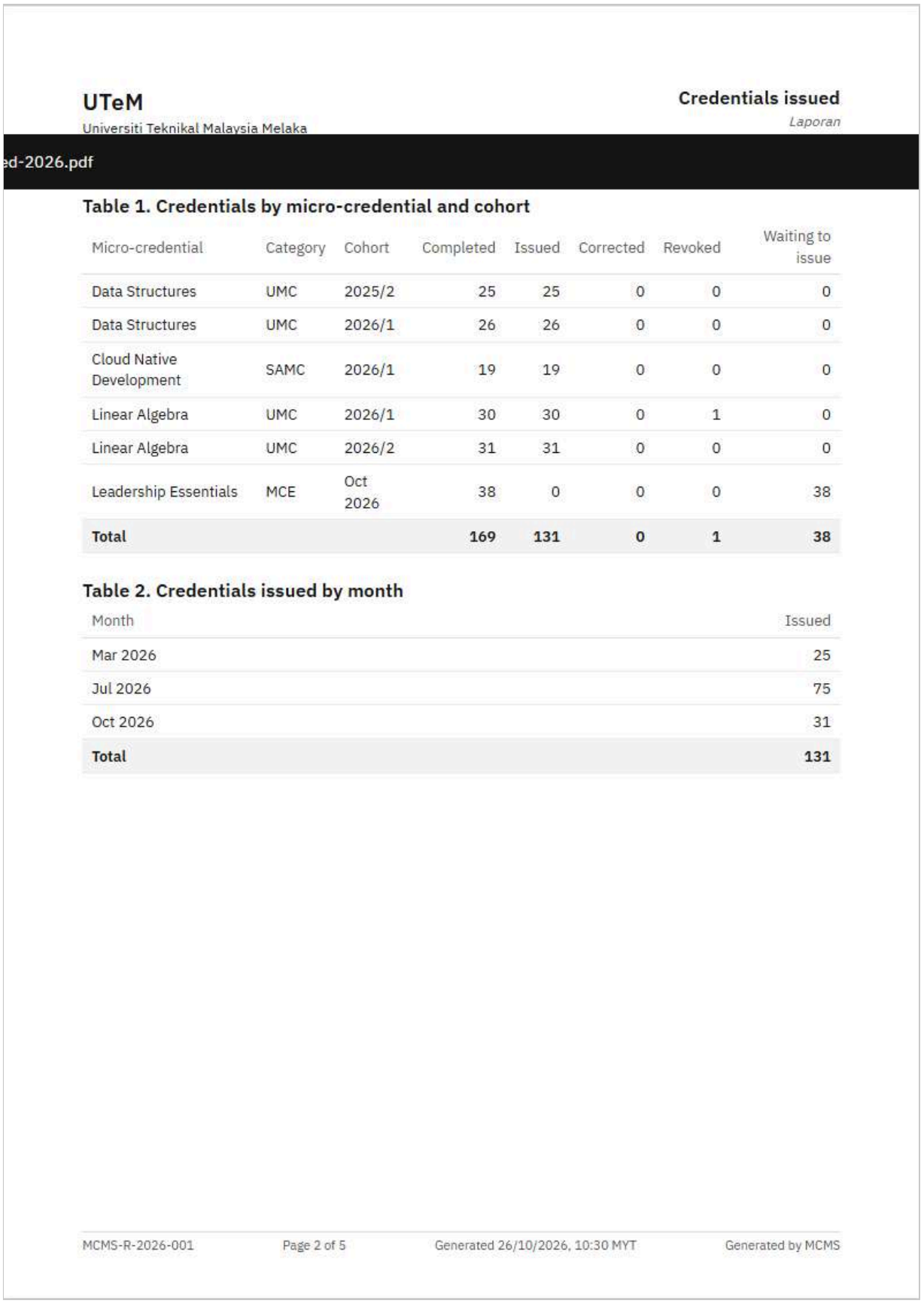
Summary figures

Credentials issued	Revoked	Corrected	Waiting to issue
131	1	0	38

Contents

1. Table 1. Credentials by micro-credential and cohort
2. Table 2. Credentials issued by month
3. Table 3. Credentials issued by category
4. Appendix. Credential register held in this prototype (sample records)
5. Method notes

Figure 277: *DOC-09 Report, page 1 of 5.*



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Credentials issued
Laporan

ed-2026.pdf

Table 1. Credentials by micro-credential and cohort

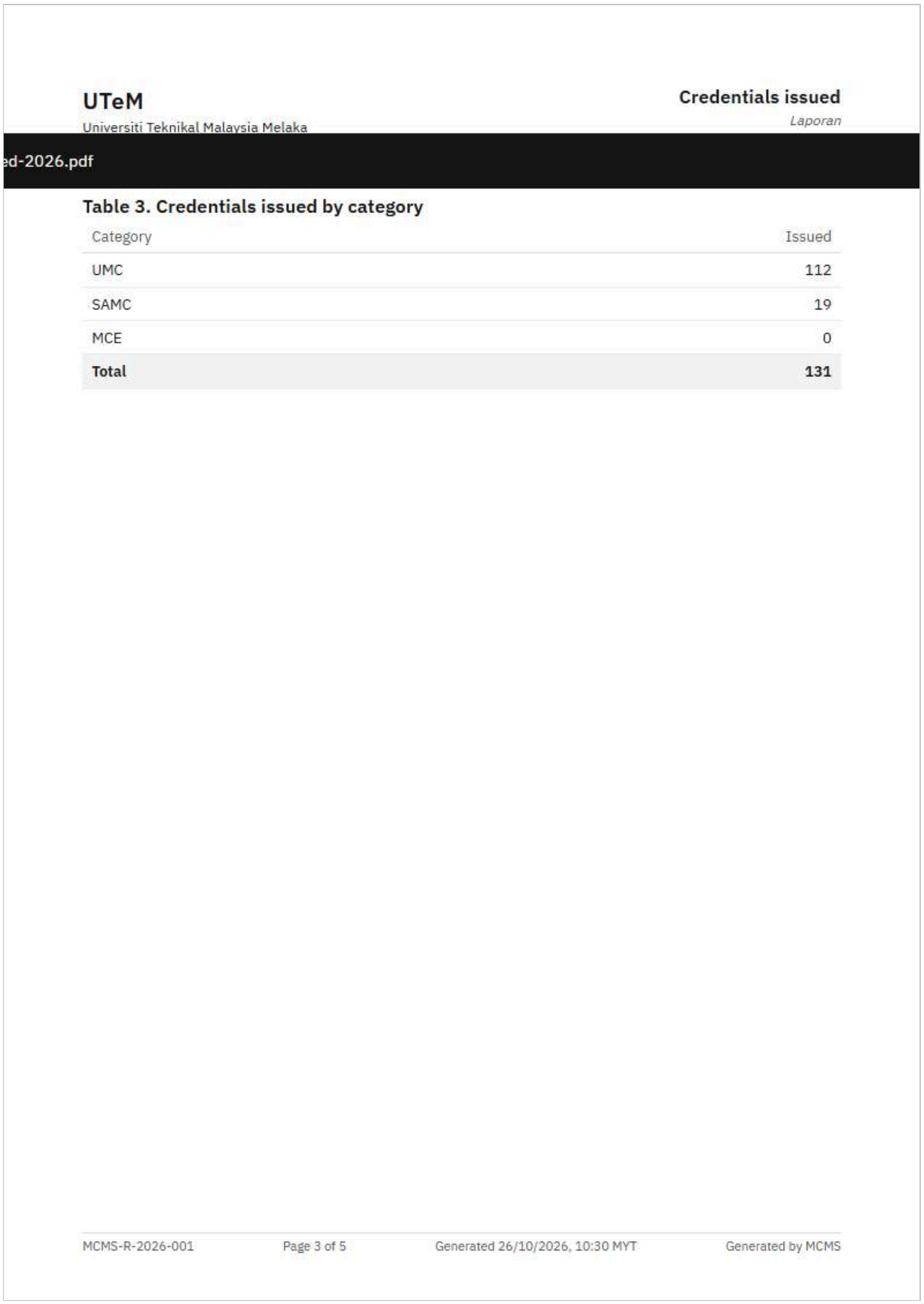
Micro-credential	Category	Cohort	Completed	Issued	Corrected	Revoked	Waiting to issue
Data Structures	UMC	2025/2	25	25	0	0	0
Data Structures	UMC	2026/1	26	26	0	0	0
Cloud Native Development	SAMC	2026/1	19	19	0	0	0
Linear Algebra	UMC	2026/1	30	30	0	1	0
Linear Algebra	UMC	2026/2	31	31	0	0	0
Leadership Essentials	MCE	Oct 2026	38	0	0	0	38
Total			**169**	**131**	**0**	**1**	**38**

Table 2. Credentials issued by month

Month	Issued
Mar 2026	25
Jul 2026	75
Oct 2026	31
Total	**131**

MCMS-R-2026-001 Page 2 of 5 Generated 26/10/2026, 10:30 MYT Generated by MCMS

Figure 278: *DOC-09 Report, page 2 of 5.*



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Credentials issued
Laporan

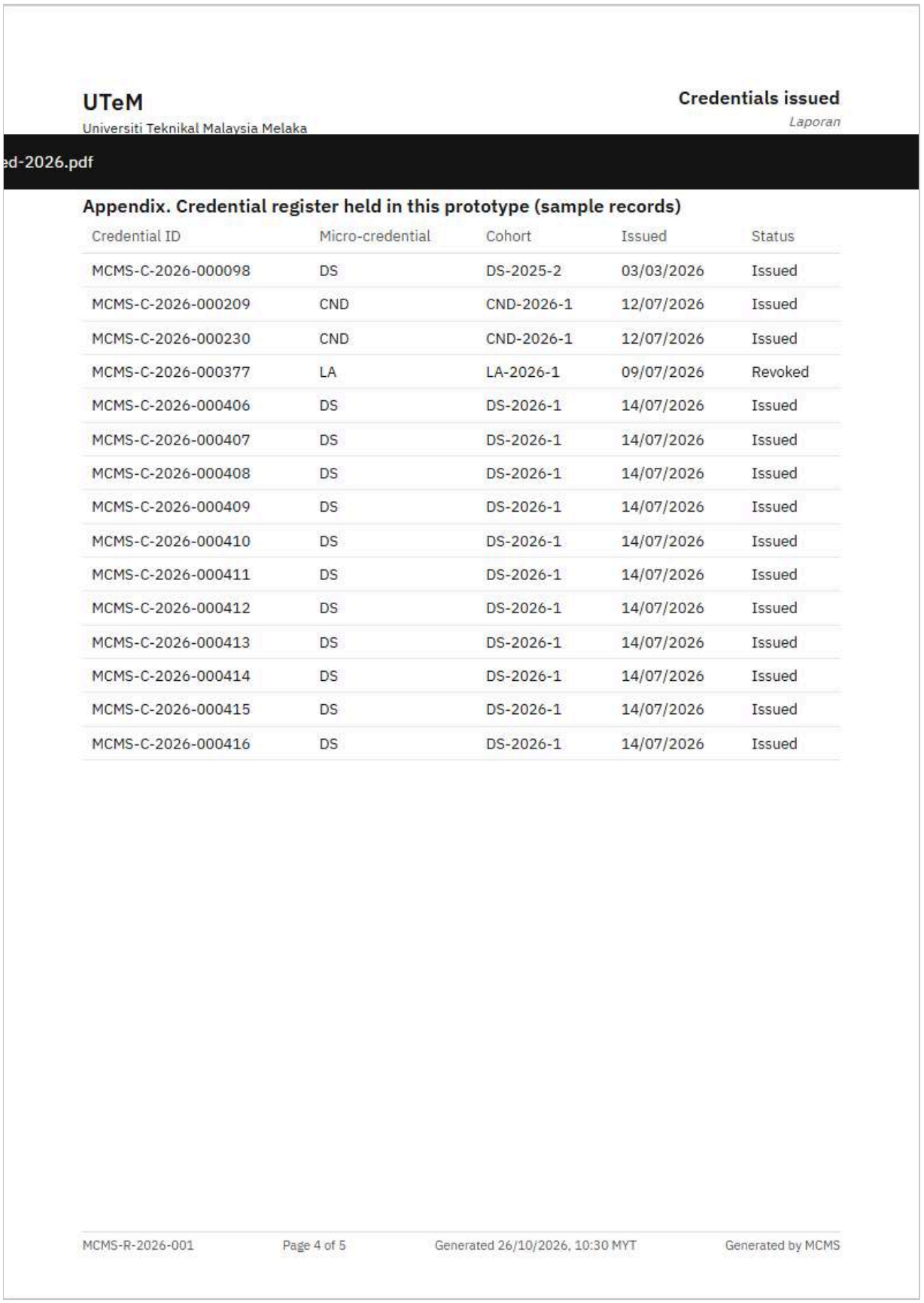
ed-2026.pdf

Table 3. Credentials issued by category

Category	Issued
UMC	112
SAMC	19
MCE	0
Total	**131**

MCMS-R-2026-001 Page 3 of 5 Generated 26/10/2026, 10:30 MYT Generated by MCMS

Figure 279: *DOC-09 Report, page 3 of 5.*



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Credentials issued
Laporan

ed-2026.pdf

Appendix. Credential register held in this prototype (sample records)

Credential ID	Micro-credential	Cohort	Issued	Status
MCMS-C-2026-000098	DS	DS-2025-2	03/03/2026	Issued
MCMS-C-2026-000209	CND	CND-2026-1	12/07/2026	Issued
MCMS-C-2026-000230	CND	CND-2026-1	12/07/2026	Issued
MCMS-C-2026-000377	LA	LA-2026-1	09/07/2026	Revoked
MCMS-C-2026-000406	DS	DS-2026-1	14/07/2026	Issued
MCMS-C-2026-000407	DS	DS-2026-1	14/07/2026	Issued
MCMS-C-2026-000408	DS	DS-2026-1	14/07/2026	Issued
MCMS-C-2026-000409	DS	DS-2026-1	14/07/2026	Issued
MCMS-C-2026-000410	DS	DS-2026-1	14/07/2026	Issued
MCMS-C-2026-000411	DS	DS-2026-1	14/07/2026	Issued
MCMS-C-2026-000412	DS	DS-2026-1	14/07/2026	Issued
MCMS-C-2026-000413	DS	DS-2026-1	14/07/2026	Issued
MCMS-C-2026-000414	DS	DS-2026-1	14/07/2026	Issued
MCMS-C-2026-000415	DS	DS-2026-1	14/07/2026	Issued
MCMS-C-2026-000416	DS	DS-2026-1	14/07/2026	Issued

MCMS-R-2026-001 Page 4 of 5 Generated 26/10/2026, 10:30 MYT Generated by MCMS

Figure 280: *DOC-09 Report, page 4 of 5.*

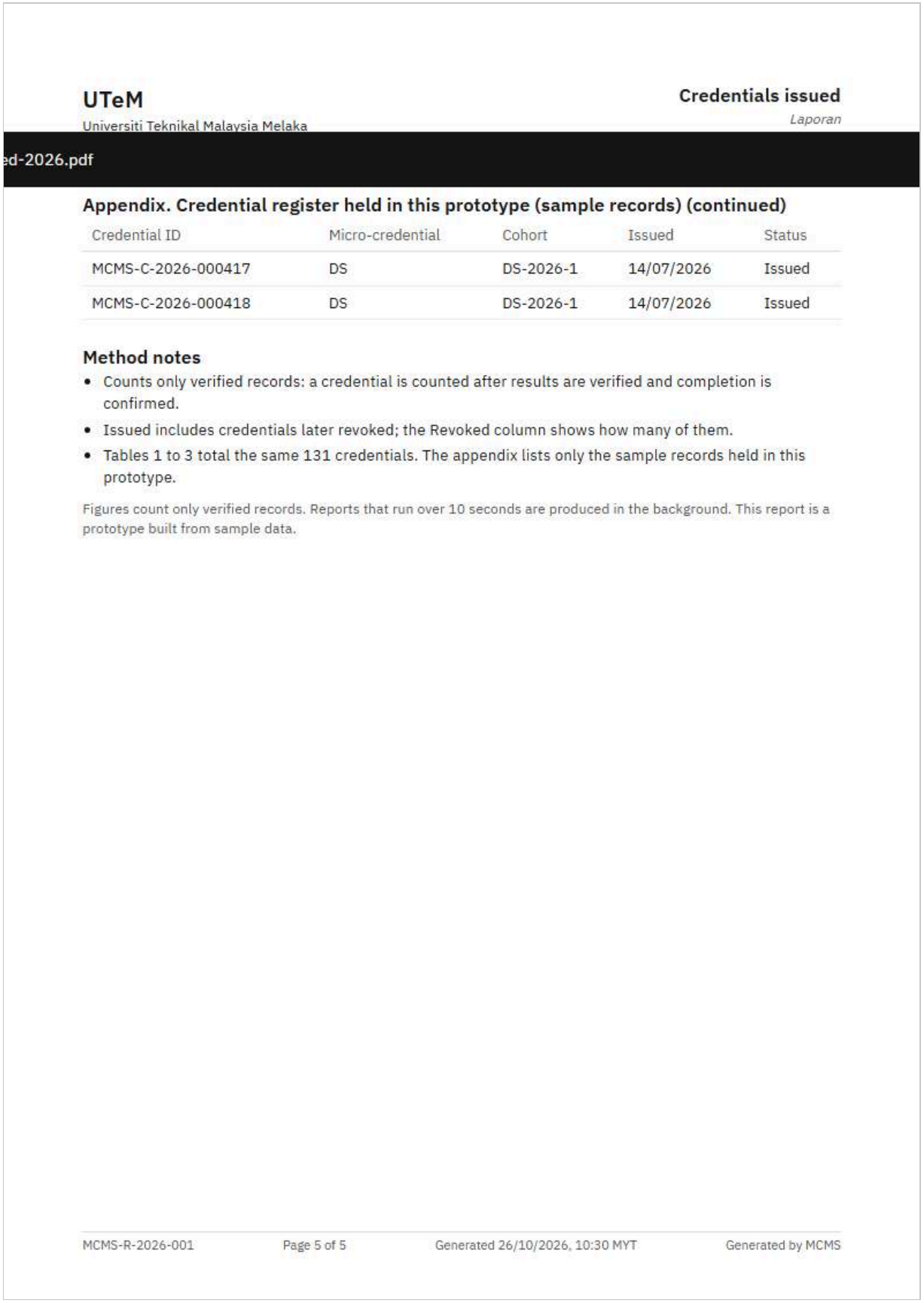
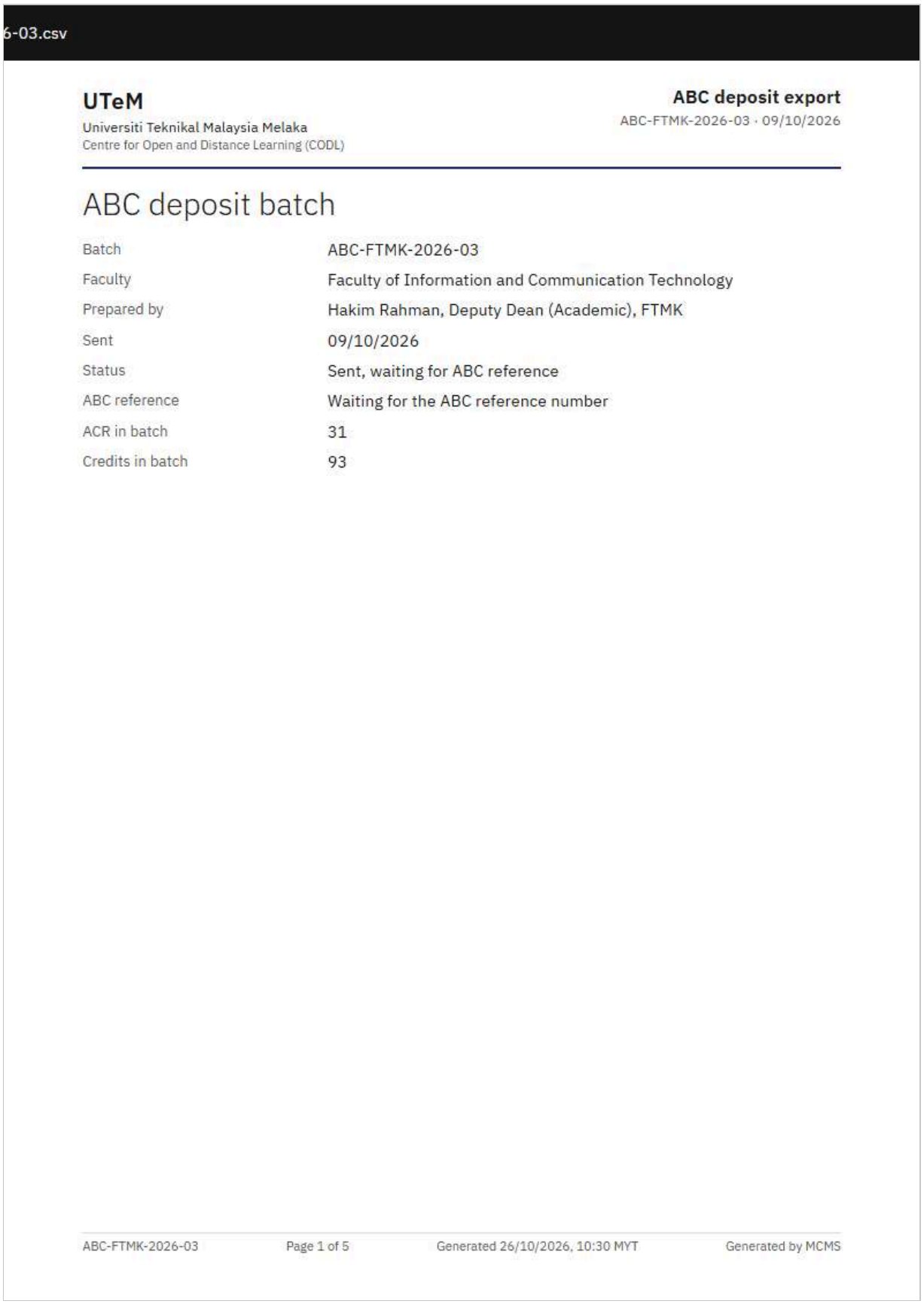


Figure 281: *DOC-09 Report, page 5 of 5.*

DOC-10 ABC deposit export

`#/doc/abc/ABC-FTMK-2026-03`


6-03.csv

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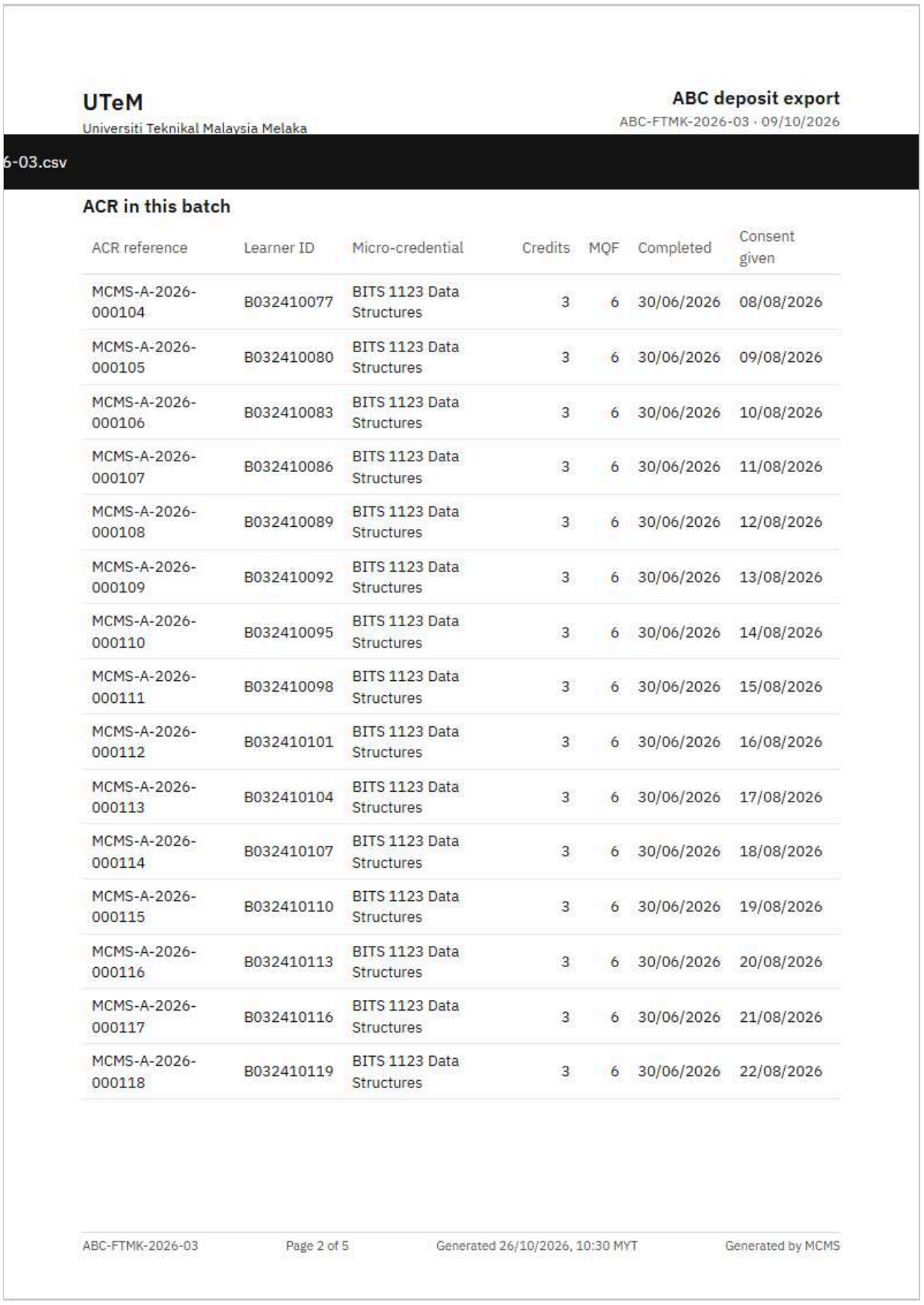
ABC deposit export
ABC-FTMK-2026-03 · 09/10/2026

ABC deposit batch

Batch	ABC-FTMK-2026-03
Faculty	Faculty of Information and Communication Technology
Prepared by	Hakim Rahman, Deputy Dean (Academic), FTMK
Sent	09/10/2026
Status	Sent, waiting for ABC reference
ABC reference	Waiting for the ABC reference number
ACR in batch	31
Credits in batch	93

ABC-FTMK-2026-03 Page 1 of 5 Generated 26/10/2026, 10:30 MYT Generated by MCMS

Figure 282: *DOC-10 ABC deposit export, page 1 of 5.*



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ABC deposit export
ABC-FTMK-2026-03 · 09/10/2026

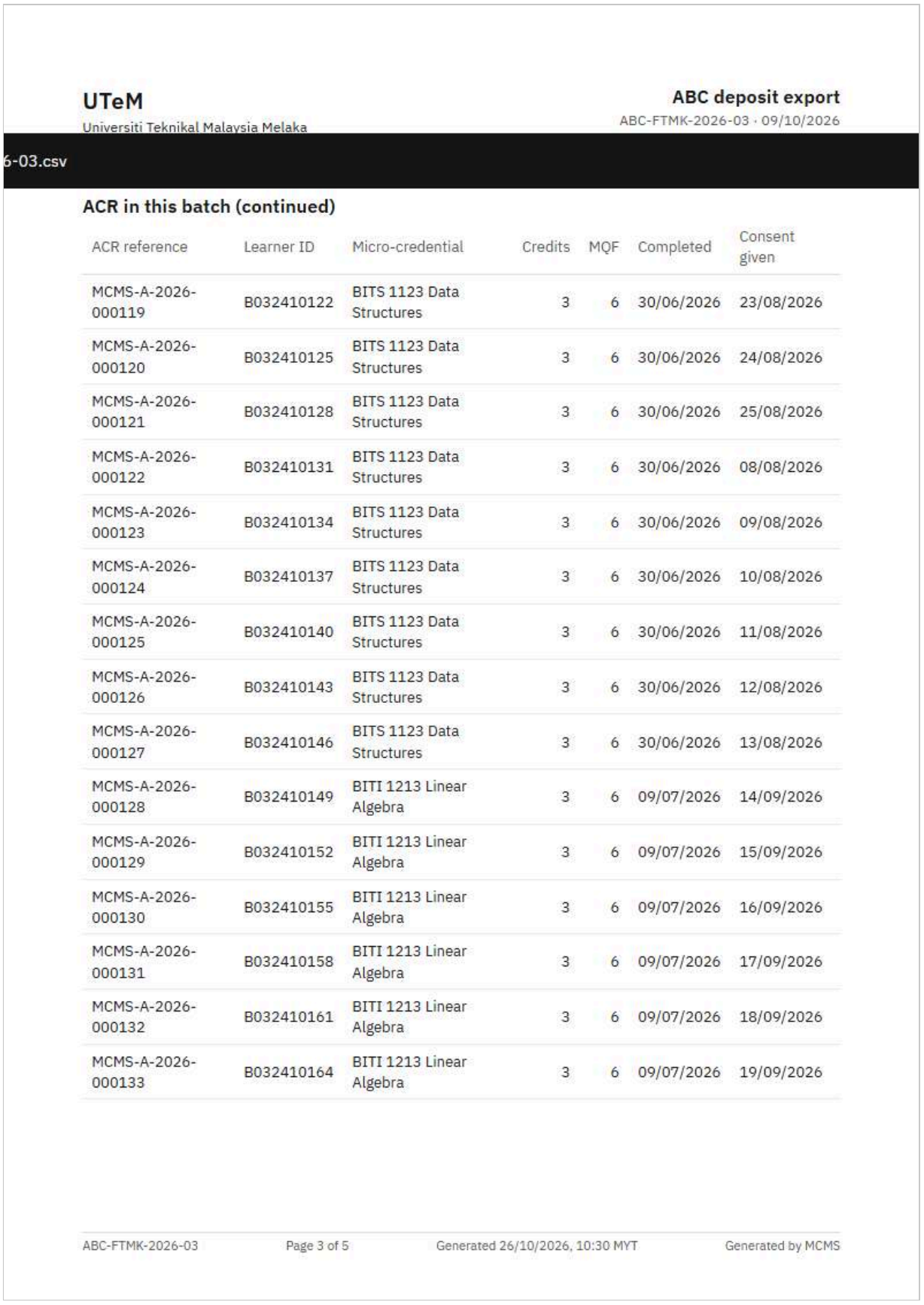
6-03.csv

ACR in this batch

ACR reference	Learner ID	Micro-credential	Credits	MQF	Completed	Consent given
MCMS-A-2026-000104	B032410077	BITS 1123 Data Structures	3	6	30/06/2026	08/08/2026
MCMS-A-2026-000105	B032410080	BITS 1123 Data Structures	3	6	30/06/2026	09/08/2026
MCMS-A-2026-000106	B032410083	BITS 1123 Data Structures	3	6	30/06/2026	10/08/2026
MCMS-A-2026-000107	B032410086	BITS 1123 Data Structures	3	6	30/06/2026	11/08/2026
MCMS-A-2026-000108	B032410089	BITS 1123 Data Structures	3	6	30/06/2026	12/08/2026
MCMS-A-2026-000109	B032410092	BITS 1123 Data Structures	3	6	30/06/2026	13/08/2026
MCMS-A-2026-000110	B032410095	BITS 1123 Data Structures	3	6	30/06/2026	14/08/2026
MCMS-A-2026-000111	B032410098	BITS 1123 Data Structures	3	6	30/06/2026	15/08/2026
MCMS-A-2026-000112	B032410101	BITS 1123 Data Structures	3	6	30/06/2026	16/08/2026
MCMS-A-2026-000113	B032410104	BITS 1123 Data Structures	3	6	30/06/2026	17/08/2026
MCMS-A-2026-000114	B032410107	BITS 1123 Data Structures	3	6	30/06/2026	18/08/2026
MCMS-A-2026-000115	B032410110	BITS 1123 Data Structures	3	6	30/06/2026	19/08/2026
MCMS-A-2026-000116	B032410113	BITS 1123 Data Structures	3	6	30/06/2026	20/08/2026
MCMS-A-2026-000117	B032410116	BITS 1123 Data Structures	3	6	30/06/2026	21/08/2026
MCMS-A-2026-000118	B032410119	BITS 1123 Data Structures	3	6	30/06/2026	22/08/2026

ABC-FTMK-2026-03 Page 2 of 5 Generated 26/10/2026, 10:30 MYT Generated by MCMS

Figure 283: *DOC-10 ABC deposit export, page 2 of 5.*



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ABC deposit export
ABC-FTMK-2026-03 · 09/10/2026

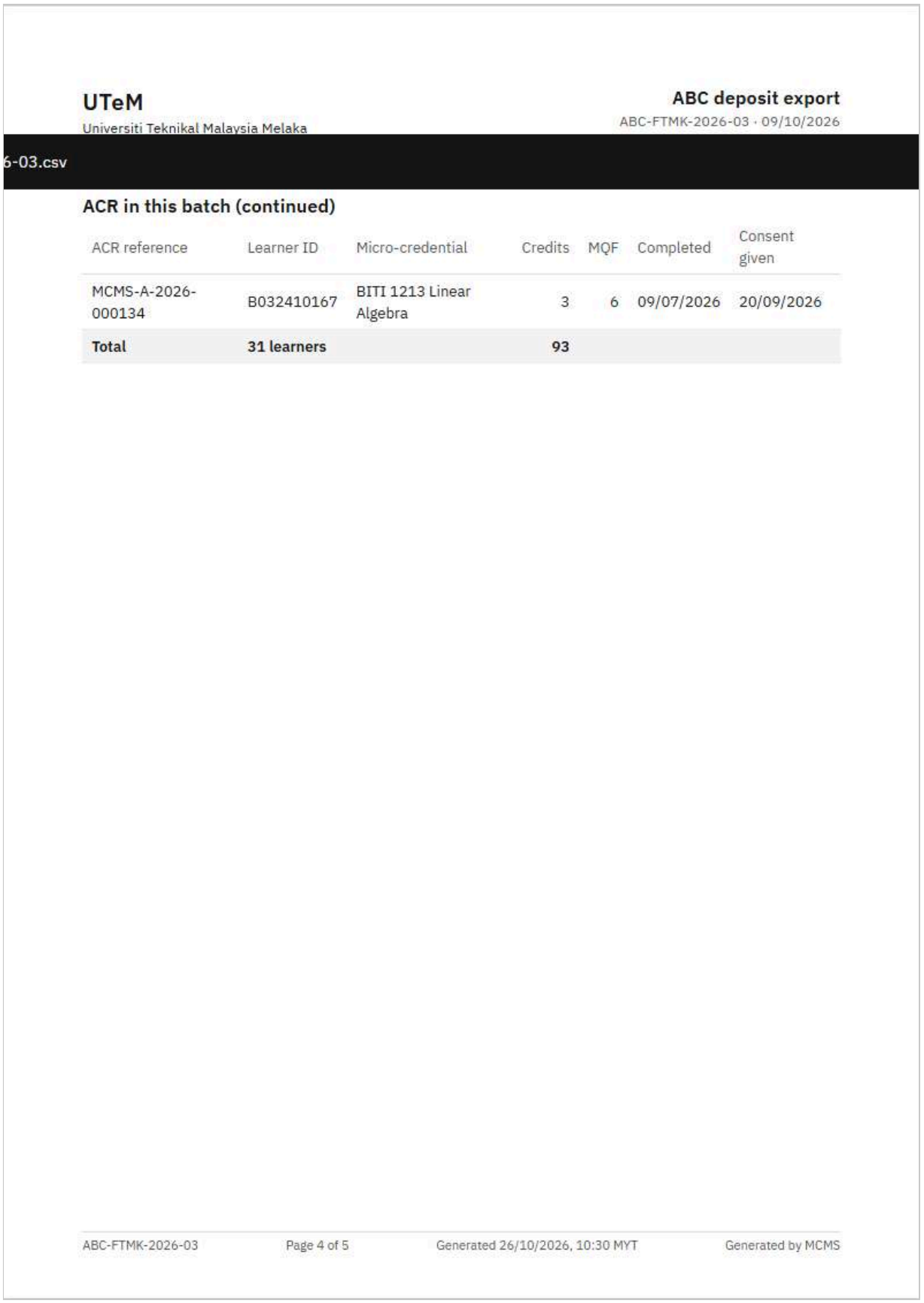
6-03.csv

ACR in this batch (continued)

ACR reference	Learner ID	Micro-credential	Credits	MQF	Completed	Consent given
MCMS-A-2026-000119	B032410122	BITS 1123 Data Structures	3	6	30/06/2026	23/08/2026
MCMS-A-2026-000120	B032410125	BITS 1123 Data Structures	3	6	30/06/2026	24/08/2026
MCMS-A-2026-000121	B032410128	BITS 1123 Data Structures	3	6	30/06/2026	25/08/2026
MCMS-A-2026-000122	B032410131	BITS 1123 Data Structures	3	6	30/06/2026	08/08/2026
MCMS-A-2026-000123	B032410134	BITS 1123 Data Structures	3	6	30/06/2026	09/08/2026
MCMS-A-2026-000124	B032410137	BITS 1123 Data Structures	3	6	30/06/2026	10/08/2026
MCMS-A-2026-000125	B032410140	BITS 1123 Data Structures	3	6	30/06/2026	11/08/2026
MCMS-A-2026-000126	B032410143	BITS 1123 Data Structures	3	6	30/06/2026	12/08/2026
MCMS-A-2026-000127	B032410146	BITS 1123 Data Structures	3	6	30/06/2026	13/08/2026
MCMS-A-2026-000128	B032410149	BITI 1213 Linear Algebra	3	6	09/07/2026	14/09/2026
MCMS-A-2026-000129	B032410152	BITI 1213 Linear Algebra	3	6	09/07/2026	15/09/2026
MCMS-A-2026-000130	B032410155	BITI 1213 Linear Algebra	3	6	09/07/2026	16/09/2026
MCMS-A-2026-000131	B032410158	BITI 1213 Linear Algebra	3	6	09/07/2026	17/09/2026
MCMS-A-2026-000132	B032410161	BITI 1213 Linear Algebra	3	6	09/07/2026	18/09/2026
MCMS-A-2026-000133	B032410164	BITI 1213 Linear Algebra	3	6	09/07/2026	19/09/2026

ABC-FTMK-2026-03 Page 3 of 5 Generated 26/10/2026, 10:30 MYT Generated by MCMS

Figure 284: *DOC-10 ABC deposit export, page 3 of 5.*



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ABC deposit export
ABC-FTMK-2026-03 · 09/10/2026

6-03.csv

ACR in this batch (continued)

ACR reference	Learner ID	Micro-credential	Credits	MQF	Completed	Consent given
MCMS-A-2026-000134	B032410167	BITI 1213 Linear Algebra	3	6	09/07/2026	20/09/2026
Total	**31 learners**		**93**			

ABC-FTMK-2026-03 Page 4 of 5 Generated 26/10/2026, 10:30 MYT Generated by MCMS

Figure 285: *DOC-10 ABC deposit export, page 4 of 5.*

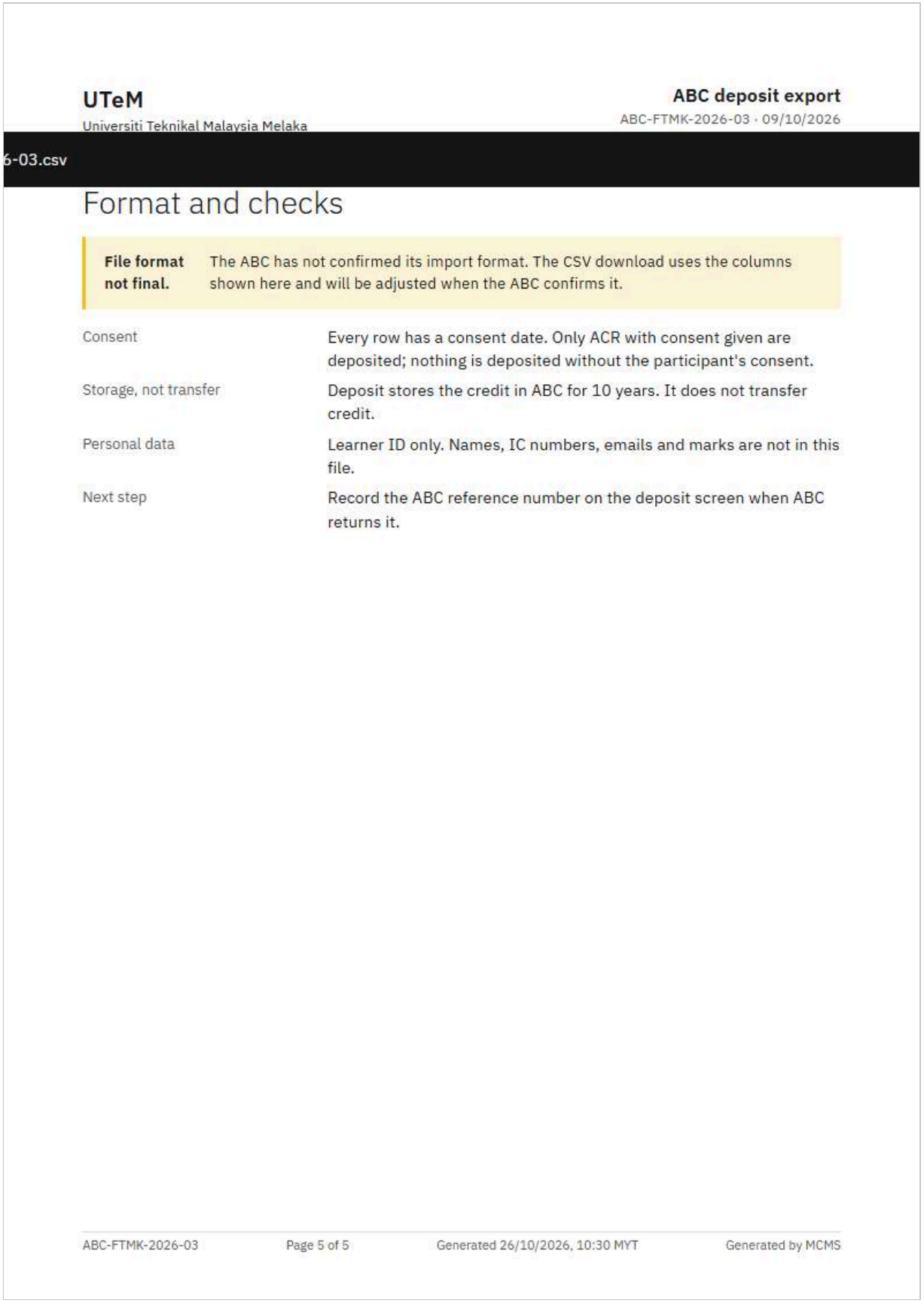
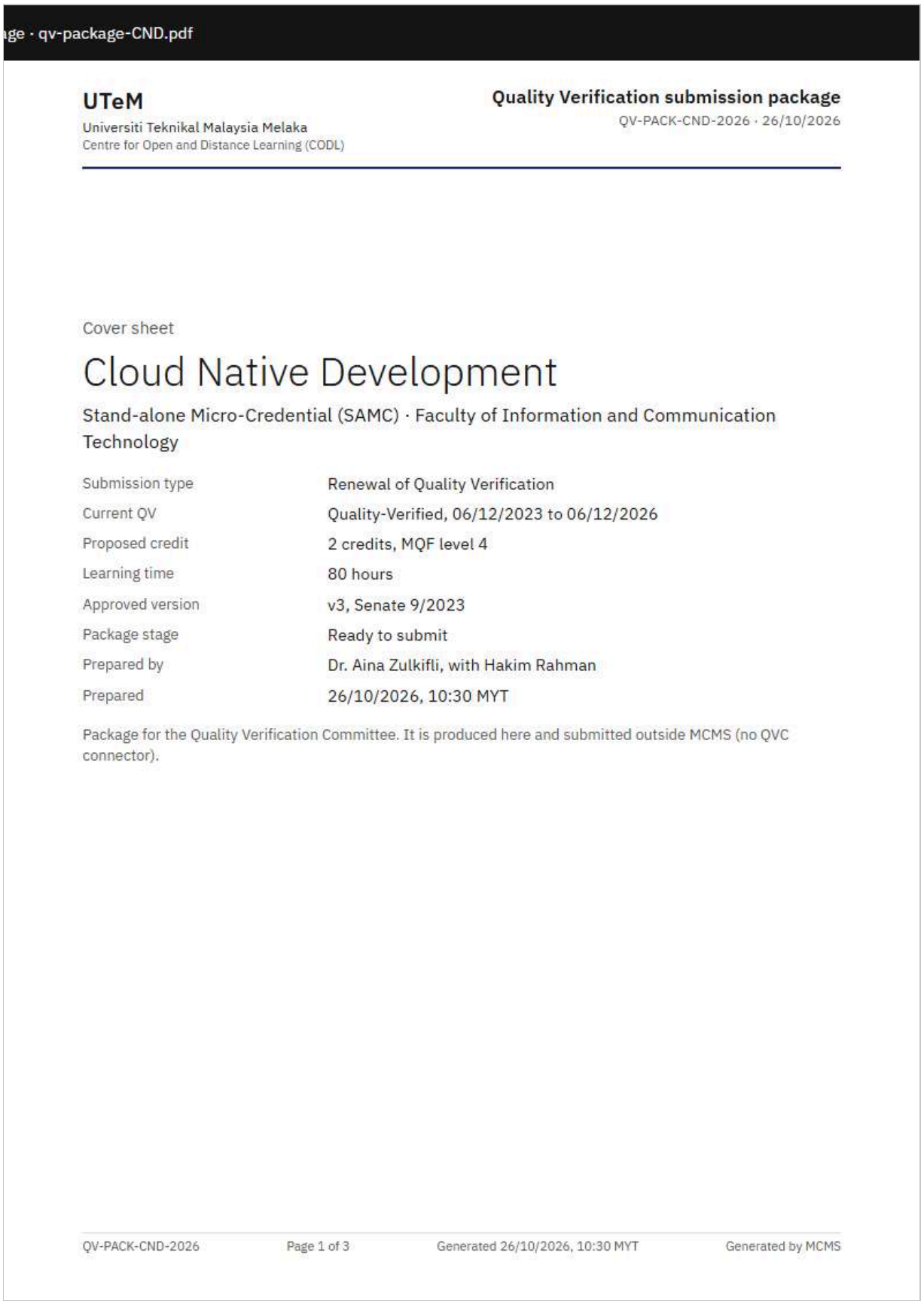


Figure 286: *DOC-10 ABC deposit export, page 5 of 5.*

DOC-11 QV submission package

`#/doc/qvpack/CND`



ge · qv-package-CND.pdf

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Centre for Open and Distance Learning (CODL)

Quality Verification submission package
QV-PACK-CND-2026 · 26/10/2026

Cover sheet

Cloud Native Development

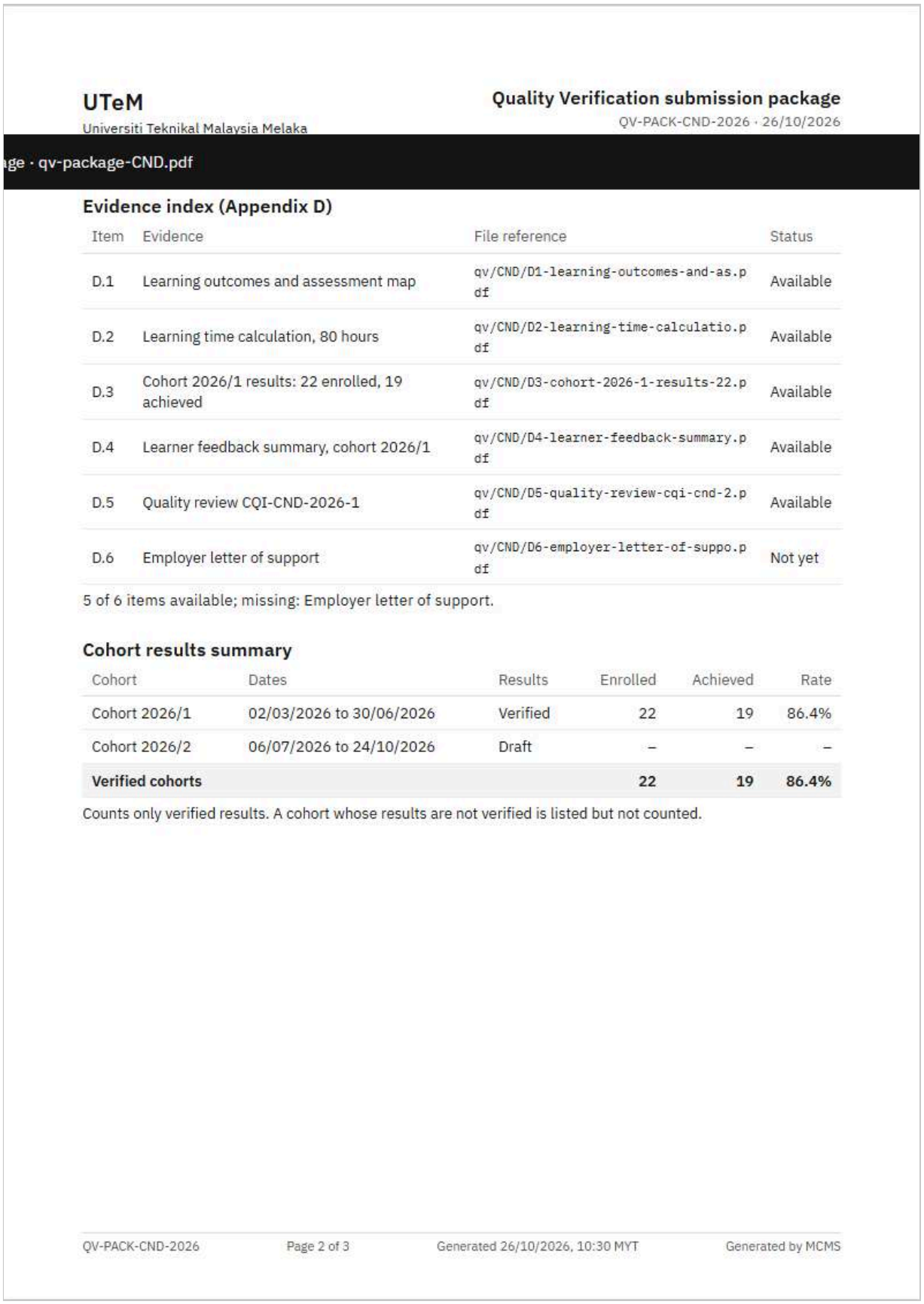
Stand-alone Micro-Credential (SAMC) · Faculty of Information and Communication Technology

Submission type	Renewal of Quality Verification
Current QV	Quality-Verified, 06/12/2023 to 06/12/2026
Proposed credit	2 credits, MQF level 4
Learning time	80 hours
Approved version	v3, Senate 9/2023
Package stage	Ready to submit
Prepared by	Dr. Aina Zulkifli, with Hakim Rahman
Prepared	26/10/2026, 10:30 MYT

Package for the Quality Verification Committee. It is produced here and submitted outside MCMS (no QVC connector).

QV-PACK-CND-2026 Page 1 of 3 Generated 26/10/2026, 10:30 MYT Generated by MCMS

Figure 287: *DOC-11 QV submission package, page 1 of 3.*



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Quality Verification submission package
QV-PACK-CND-2026 · 26/10/2026

age · qv-package-CND.pdf

Evidence index (Appendix D)

Item	Evidence	File reference	Status
D.1	Learning outcomes and assessment map	qv/CND/D1-learning-outcomes-and-as.pdf	Available
D.2	Learning time calculation, 80 hours	qv/CND/D2-learning-time-calculatio.pdf	Available
D.3	Cohort 2026/1 results: 22 enrolled, 19 achieved	qv/CND/D3-cohort-2026-1-results-22.pdf	Available
D.4	Learner feedback summary, cohort 2026/1	qv/CND/D4-learner-feedback-summary.pdf	Available
D.5	Quality review CQI-CND-2026-1	qv/CND/D5-quality-review-cqi-cnd-2.pdf	Available
D.6	Employer letter of support	qv/CND/D6-employer-letter-of-suppo.pdf	Not yet

5 of 6 items available; missing: Employer letter of support.

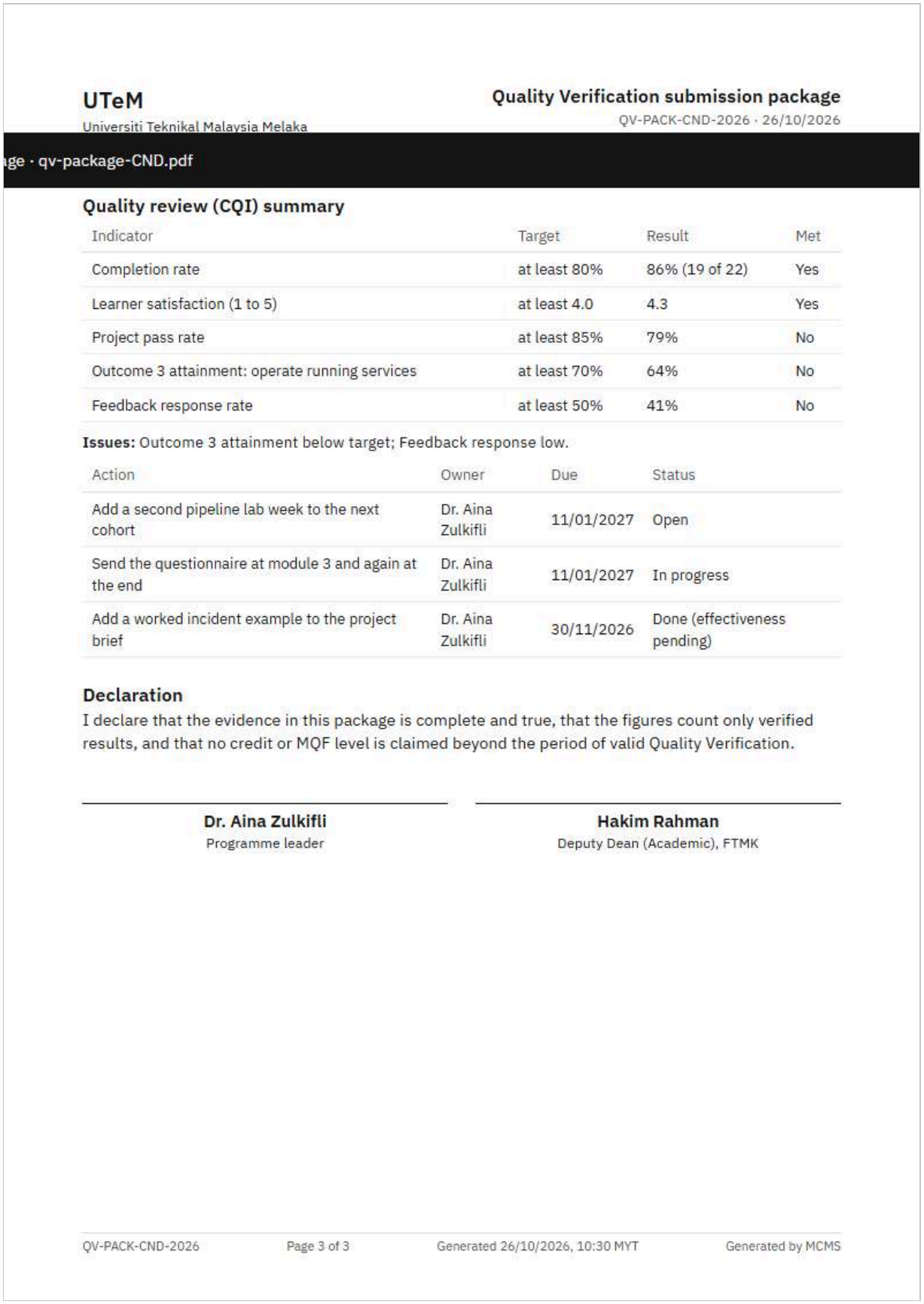
Cohort results summary

Cohort	Dates	Results	Enrolled	Achieved	Rate
Cohort 2026/1	02/03/2026 to 30/06/2026	Verified	22	19	86.4%
Cohort 2026/2	06/07/2026 to 24/10/2026	Draft	–	–	–
Verified cohorts			**22**	**19**	**86.4%**

Counts only verified results. A cohort whose results are not verified is listed but not counted.

QV-PACK-CND-2026 Page 2 of 3 Generated 26/10/2026, 10:30 MYT Generated by MCMS

Figure 288: *DOC-11 QV submission package, page 2 of 3.*



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Quality Verification submission package
QV-PACK-CND-2026 · 26/10/2026

age · qv-package-CND.pdf

Quality review (CQI) summary

Indicator	Target	Result	Met
Completion rate	at least 80%	86% (19 of 22)	Yes
Learner satisfaction (1 to 5)	at least 4.0	4.3	Yes
Project pass rate	at least 85%	79%	No
Outcome 3 attainment: operate running services	at least 70%	64%	No
Feedback response rate	at least 50%	41%	No

Issues: Outcome 3 attainment below target; Feedback response low.

Action	Owner	Due	Status
Add a second pipeline lab week to the next cohort	Dr. Aina Zulkifli	11/01/2027	Open
Send the questionnaire at module 3 and again at the end	Dr. Aina Zulkifli	11/01/2027	In progress
Add a worked incident example to the project brief	Dr. Aina Zulkifli	30/11/2026	Done (effectiveness pending)

Declaration

I declare that the evidence in this package is complete and true, that the figures count only verified results, and that no credit or MQF level is claimed beyond the period of valid Quality Verification.

Dr. Aina Zulkifli
Programme leader

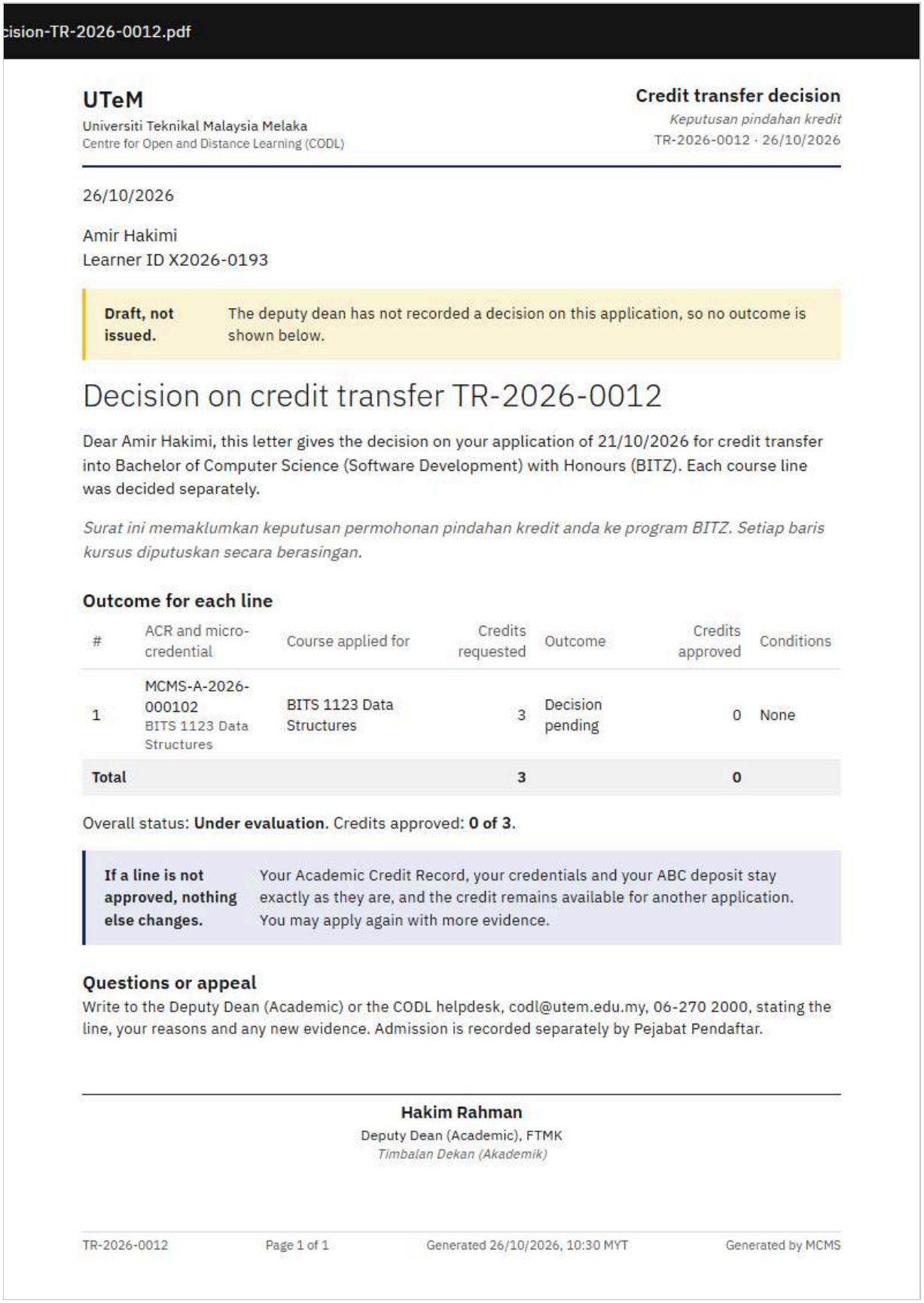
Hakim Rahman
Deputy Dean (Academic), FTMK

QV-PACK-CND-2026 Page 3 of 3 Generated 26/10/2026, 10:30 MYT Generated by MCMS

Figure 289: *DOC-11 QV submission package, page 3 of 3.*

DOC-12 Credit transfer decision letter

`#/doc/transfer/TR-2026-0012`



cision-TR-2026-0012.pdf

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Centre for Open and Distance Learning (CODL)

Credit transfer decision
Keputusan pindahan kredit
TR-2026-0012 · 26/10/2026

26/10/2026

Amir Hakimi
Learner ID X2026-0193

Draft, not issued. The deputy dean has not recorded a decision on this application, so no outcome is shown below.

Decision on credit transfer TR-2026-0012

Dear Amir Hakimi, this letter gives the decision on your application of 21/10/2026 for credit transfer into Bachelor of Computer Science (Software Development) with Honours (BITZ). Each course line was decided separately.

Surat ini memaklumkan keputusan permohonan pindahan kredit anda ke program BITZ. Setiap baris kursus diputuskan secara berasingan.

Outcome for each line

#	ACR and micro-credential	Course applied for	Credits requested	Outcome	Credits approved	Conditions
1	MCMS-A-2026-000102 BITS 1123 Data Structures	BITS 1123 Data Structures	3	Decision pending	0	None
Total			**3**		**0**	

Overall status: **Under evaluation**. Credits approved: **0 of 3**.

If a line is not approved, nothing else changes. Your Academic Credit Record, your credentials and your ABC deposit stay exactly as they are, and the credit remains available for another application. You may apply again with more evidence.

Questions or appeal

Write to the Deputy Dean (Academic) or the CODL helpdesk, codl@utem.edu.my, 06-270 2000, stating the line, your reasons and any new evidence. Admission is recorded separately by Pejabat Pendaftar.

Hakim Rahman
Deputy Dean (Academic), FTMK
Timbalan Dekan (Akademik)

TR-2026-0012 Page 1 of 1 Generated 26/10/2026, 10:30 MYT Generated by MCMS

Figure 290: *DOC-12 Credit transfer decision letter, page 1 of 1.*

DOC-13 Suspension notice

`#/doc/suspend/CND`

notice-CND.pdf

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Centre for Open and Distance Learning (CODL)

Notice of suspension or termination
Notis penggantungan
NOT-CND-2026 · 26/10/2026

Draft for review. No suspension or termination has been recorded for this micro-credential. This is the notice that participants would receive.

Cloud Native Development: suspension notice

To all participants registered in Cloud Native Development.

Kepada semua peserta yang berdaftar dalam Cloud Native Development.

The Faculty of Information and Communication Technology has decided on the suspension of Cloud Native Development, effective 26/10/2026. This notice explains what it means for you.

Fakulti telah memutuskan penggantungan Cloud Native Development berkuat kuasa 26/10/2026.

Who is affected

Cohort	Dates	Status	Participants
Cohort 2027/1	11/01/2027 to 04/04/2027	Open	9
Total			**9**

Transition plan

Participants already registered in an open or running cohort finish it as planned. No new registrations are accepted. Participants who prefer to leave may withdraw, and CODL contacts them about fees. The lecturer stays responsible for teaching and marking until the last cohort ends.

Credentials and credit you have already earned stay valid. Certificates, Digital Credentials, Academic Credit Records and any ABC deposit are not changed by this decision.

Kredensial dan kredit yang telah anda peroleh kekal sah. Sijil, Kredensial Digital, ACR dan deposit ABC tidak berubah.

Questions: codl@utem.edu.my, 06-270 2000 (08:00 to 17:00, Monday to Friday).

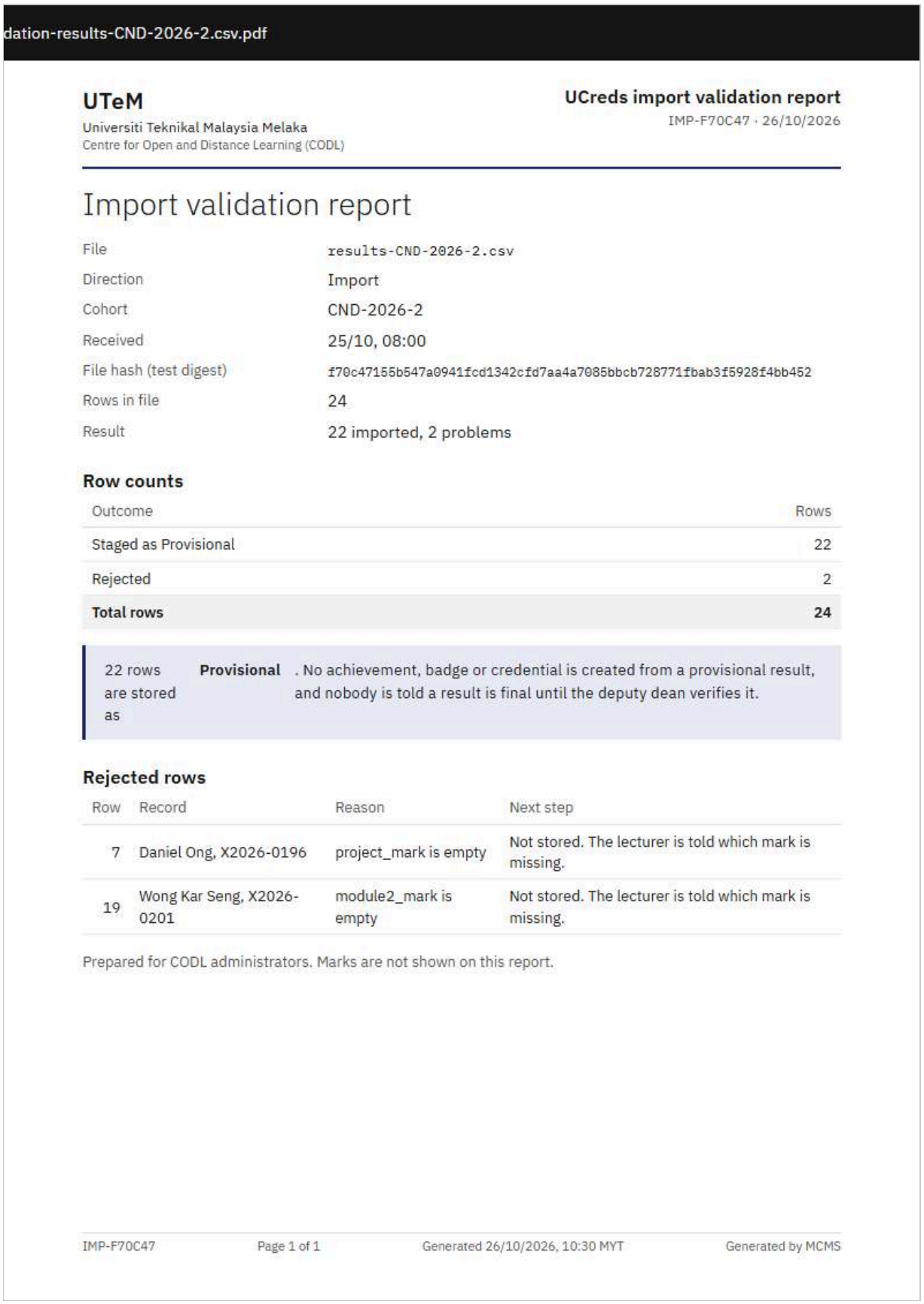
Hakim Rahman
Deputy Dean (Academic), FTMK
Timbalan Dekan (Akademik)

NOT-CND-2026 Page 1 of 1 Generated 26/10/2026, 10:30 MYT Generated by MCMS

Figure 291: *DOC-13 Suspension notice, page 1 of 1.*

DOC-14 Import validation report

`#/doc/import/results-CND-2026-2.csv`



dation-results-CND-2026-2.csv.pdf

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Centre for Open and Distance Learning (CODL)

UCreds import validation report
IMP-F70C47 · 26/10/2026

Import validation report

File	`results-CND-2026-2.csv`
Direction	Import
Cohort	CND-2026-2
Received	25/10, 08:00
File hash (test digest)	`f70c47155b547a0941fcd1342cfd7aa4a7085bbcb728771fbab3f5928f4bb452`
Rows in file	24
Result	22 imported, 2 problems

Row counts

Outcome	Rows
Staged as Provisional	22
Rejected	2
Total rows	**24**

22 rows are stored as **Provisional** . No achievement, badge or credential is created from a provisional result, and nobody is told a result is final until the deputy dean verifies it.

Rejected rows

Row	Record	Reason	Next step
7	Daniel Ong, X2026-0196	project_mark is empty	Not stored. The lecturer is told which mark is missing.
19	Wong Kar Seng, X2026-0201	module2_mark is empty	Not stored. The lecturer is told which mark is missing.

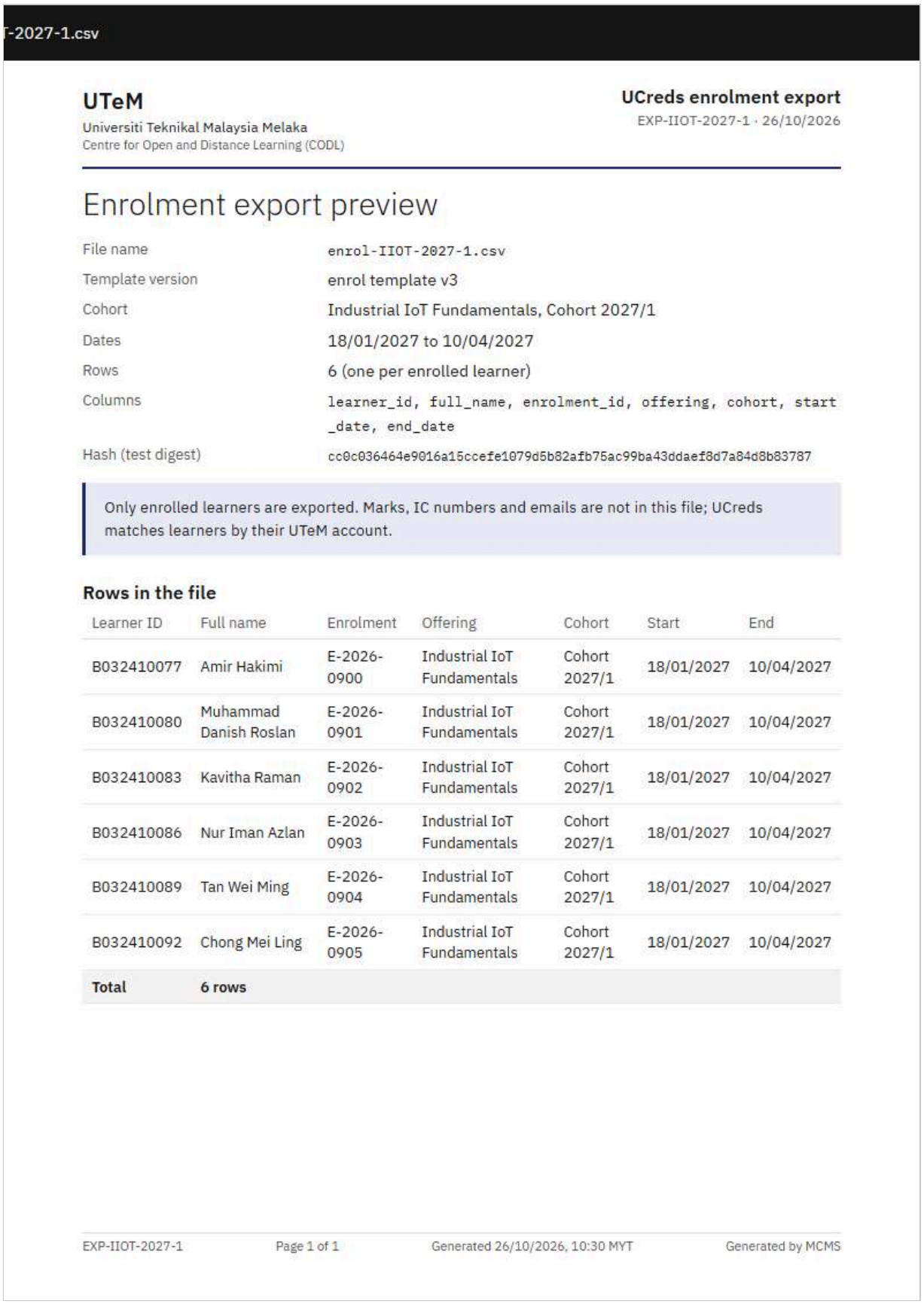
Prepared for CODL administrators. Marks are not shown on this report.

IMP-F70C47 Page 1 of 1 Generated 26/10/2026, 10:30 MYT Generated by MCMS

Figure 292: *DOC-14 Import validation report, page 1 of 1.*

DOC-15 UCreds enrolment export

`#/doc/export/IIOT-2027-1`



-2027-1.csv

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UCreds enrolment export
EXP-IIOT-2027-1 · 26/10/2026

Enrolment export preview

File name	`enrol-IIOT-2027-1.csv`
Template version	enrol template v3
Cohort	Industrial IoT Fundamentals, Cohort 2027/1
Dates	18/01/2027 to 10/04/2027
Rows	6 (one per enrolled learner)
Columns	`learner_id, full_name, enrolment_id, offering, cohort, start_date, end_date`
Hash (test digest)	`cc0c036464e9016a15ccefe1079d5b82afb75ac99ba43ddaef8d7a84d8b83787`

> Only enrolled learners are exported. Marks, IC numbers and emails are not in this file; UCreds matches learners by their UTeM account.

Rows in the file

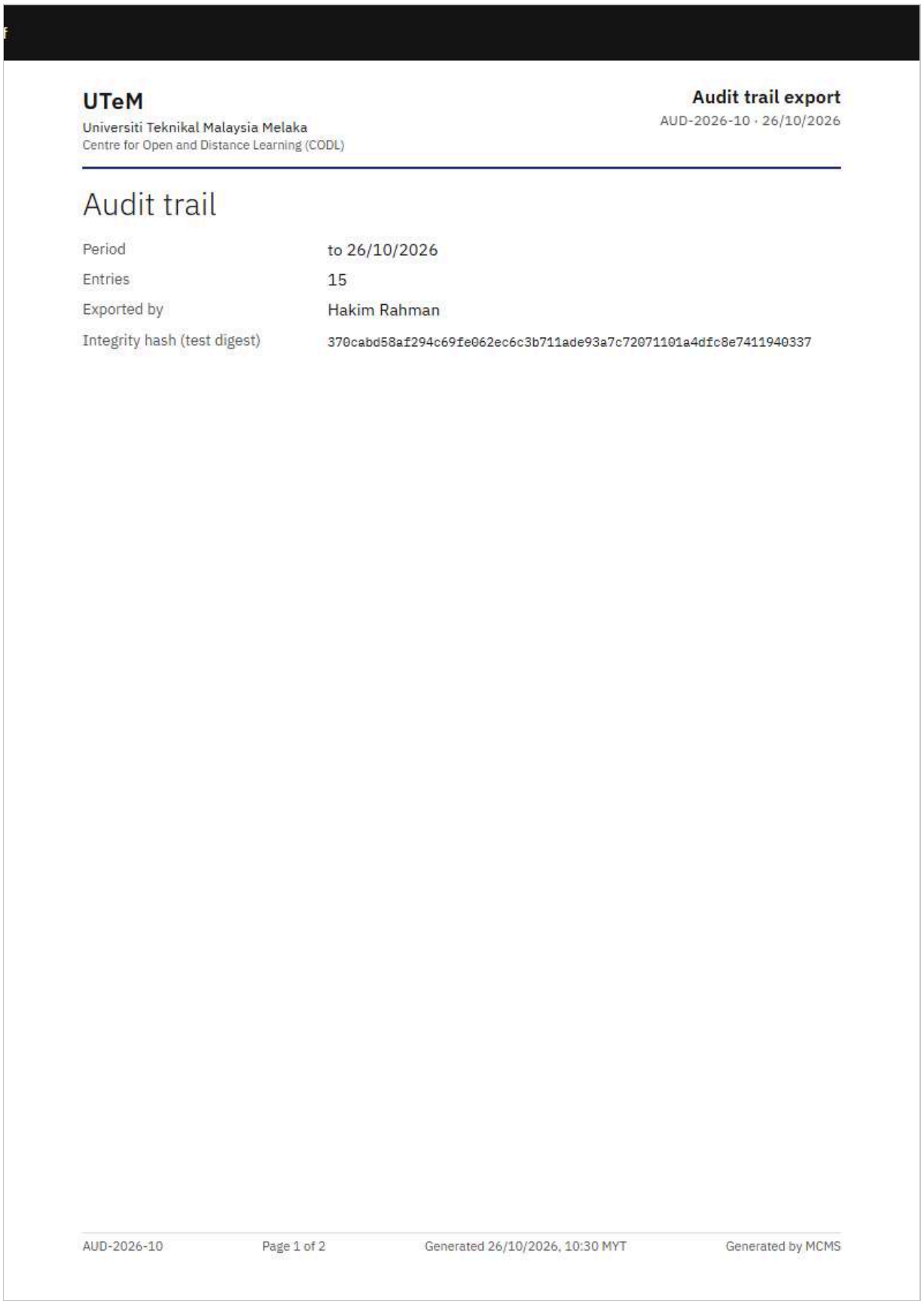
Learner ID	Full name	Enrolment	Offering	Cohort	Start	End
B032410077	Amir Hakimi	E-2026-0900	Industrial IoT Fundamentals	Cohort 2027/1	18/01/2027	10/04/2027
B032410080	Muhammad Danish Roslan	E-2026-0901	Industrial IoT Fundamentals	Cohort 2027/1	18/01/2027	10/04/2027
B032410083	Kavitha Raman	E-2026-0902	Industrial IoT Fundamentals	Cohort 2027/1	18/01/2027	10/04/2027
B032410086	Nur Iman Azlan	E-2026-0903	Industrial IoT Fundamentals	Cohort 2027/1	18/01/2027	10/04/2027
B032410089	Tan Wei Ming	E-2026-0904	Industrial IoT Fundamentals	Cohort 2027/1	18/01/2027	10/04/2027
B032410092	Chong Mei Ling	E-2026-0905	Industrial IoT Fundamentals	Cohort 2027/1	18/01/2027	10/04/2027
Total	**6 rows**					

EXP-IIOT-2027-1 Page 1 of 1 Generated 26/10/2026, 10:30 MYT Generated by MCMS

Figure 293: *DOC-15 UCreds enrolment export, page 1 of 1.*

DOC-16 Audit trail export

`#/doc/audit`



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Audit trail export
AUD-2026-10 · 26/10/2026

Audit trail

Period	to 26/10/2026
Entries	15
Exported by	Hakim Rahman
Integrity hash (test digest)	370cabd58af294c69fe062ec6c3b711ade93a7c72071101a4dfc8e7411940337

AUD-2026-10 Page 1 of 2 Generated 26/10/2026, 10:30 MYT Generated by MCMS

Figure 294: *DOC-16 Audit trail export, page 1 of 2.*

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Universiti Teknikal Malaysia Melaka

Audit trail export
AUD-2026-10 · 26/10/2026

Entries, newest first

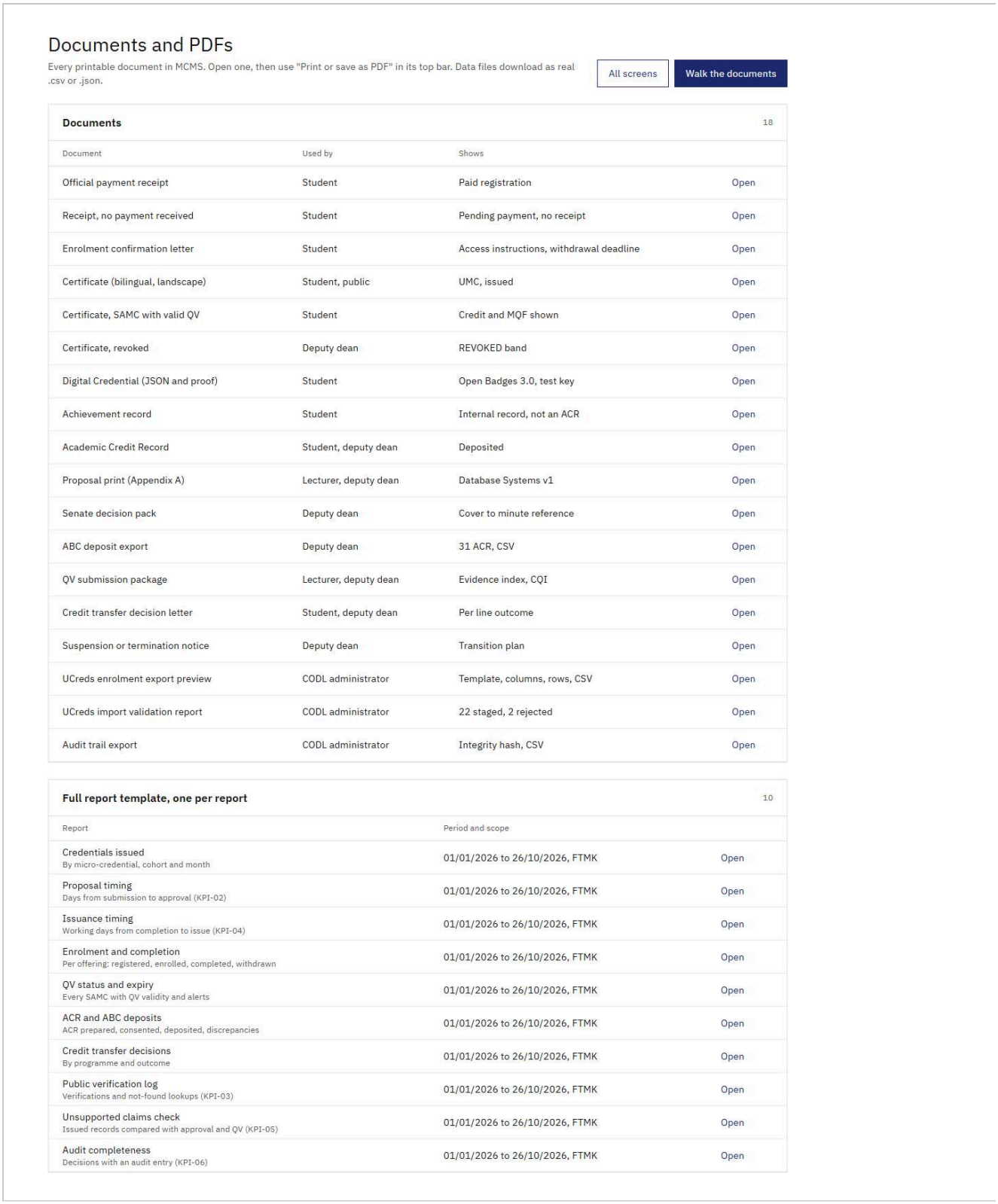
When (MYT)	Who	What	Record
26/10/2026, 10:05	System	Email to Hakim Rahman failed (SMTP 421)	Adapter
25/10/2026, 16:20	Dr. Aina Zulkifli	Submitted results, Data Structures 2026/2 (28)	DS-2026-2
25/10/2026, 08:00	Farid Hashim	Imported results-CND-2026-2.csv: 22 staged, 2 rejected	CND-2026-2
23/10/2026, 11:02	Lee Jia Hui	Requested credential correction: name spelling	MCMS-C-2026-000418
21/10/2026, 09:14	Amir Hakimi	Applied for credit transfer TR-2026-0012	TR-2026-0012
20/10/2026, 15:40	Hakim Rahman	Recorded ABC deposit FTMK-2026-02 (28 ACR), ref ABC-MY-2026-0004391..4419	ABC-FTMK-2026-02
20/10/2026, 10:31	Dr. Aina Zulkifli	Submitted proposal Database Systems v1	DB
14/10/2026, 10:42	System	Issued 31 credentials after verification, Linear Algebra 2026/2	LA-2026-2
02/09/2026, 09:20	Farid Hashim	Changed alerts.qv_expiry_days 90, 30 → 180, 90, 30	Settings
02/09/2026, 08:55	Hakim Rahman	Revoked MCMS-C-2026-000377: result withdrawn	MCMS-C-2026-000377
05/03/2026, 21:12	Amir Hakimi	Gave consent to deposit Data Structures ACR in ABC	MCMS-A-2026-000102
03/03/2026, 10:00	System	Issued credential MCMS-C-2026-000098 to Amir Hakimi	MCMS-C-2026-000098
14/10/2026, 15:00	System	Placed on the Senate 7/2026 agenda	ADA
02/10/2026, 11:20	Hakim Rahman	Endorsed Applied Data Analytics with Python v1; category verified as SAMC	ADA
28/09/2026, 09:05	Dr. Aina Zulkifli	Submitted proposal Applied Data Analytics with Python v1	ADA

AUD-2026-10 Page 2 of 2 Generated 26/10/2026, 10:30 MYT Generated by MCMS

Figure 295: *DOC-16 Audit trail export, page 2 of 2.*

DOC-17 Documents and PDFs

`#/doc/index`



Documents and PDFs

Every printable document in MCMS. Open one, then use "Print or save as PDF" in its top bar. Data files download as real .csv or .json.

All screens | Walk the documents

Documents 18

Document	Used by	Shows	
Official payment receipt	Student	Paid registration	Open
Receipt, no payment received	Student	Pending payment, no receipt	Open
Enrolment confirmation letter	Student	Access instructions, withdrawal deadline	Open
Certificate (bilingual, landscape)	Student, public	UMC, issued	Open
Certificate, SAMC with valid QV	Student	Credit and MQF shown	Open
Certificate, revoked	Deputy dean	REVOKED band	Open
Digital Credential (JSON and proof)	Student	Open Badges 3.0, test key	Open
Achievement record	Student	Internal record, not an ACR	Open
Academic Credit Record	Student, deputy dean	Deposited	Open
Proposal print (Appendix A)	Lecturer, deputy dean	Database Systems v1	Open
Senate decision pack	Deputy dean	Cover to minute reference	Open
ABC deposit export	Deputy dean	31 ACR, CSV	Open
QV submission package	Lecturer, deputy dean	Evidence index, CQI	Open
Credit transfer decision letter	Student, deputy dean	Per line outcome	Open
Suspension or termination notice	Deputy dean	Transition plan	Open
UCreds enrolment export preview	CODL administrator	Template, columns, rows, CSV	Open
UCreds import validation report	CODL administrator	22 staged, 2 rejected	Open
Audit trail export	CODL administrator	Integrity hash, CSV	Open

Full report template, one per report 10

Report	Period and scope	
Credentials issued By micro-credential, cohort and month	01/01/2026 to 26/10/2026, FTMK	Open
Proposal timing Days from submission to approval (KPI-02)	01/01/2026 to 26/10/2026, FTMK	Open
Issuance timing Working days from completion to issue (KPI-04)	01/01/2026 to 26/10/2026, FTMK	Open
Enrolment and completion Per offering: registered, enrolled, completed, withdrawn	01/01/2026 to 26/10/2026, FTMK	Open
QV status and expiry Every SAMC with QV validity and alerts	01/01/2026 to 26/10/2026, FTMK	Open
ACR and ABC deposits ACR prepared, consented, deposited, discrepancies	01/01/2026 to 26/10/2026, FTMK	Open
Credit transfer decisions By programme and outcome	01/01/2026 to 26/10/2026, FTMK	Open
Public verification log Verifications and not-found lookups (KPI-03)	01/01/2026 to 26/10/2026, FTMK	Open
Unsupported claims check Issued records compared with approval and QV (KPI-05)	01/01/2026 to 26/10/2026, FTMK	Open
Audit completeness Decisions with an audit entry (KPI-06)	01/01/2026 to 26/10/2026, FTMK	Open

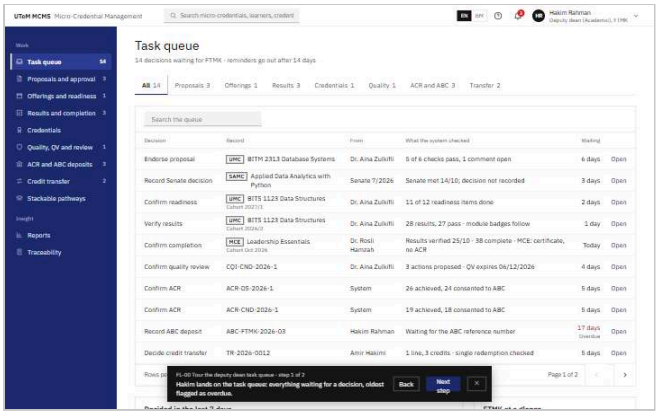
Figure 296: *DOC-17 Documents and PDFs.*

Appendix D. Flow Storyboards

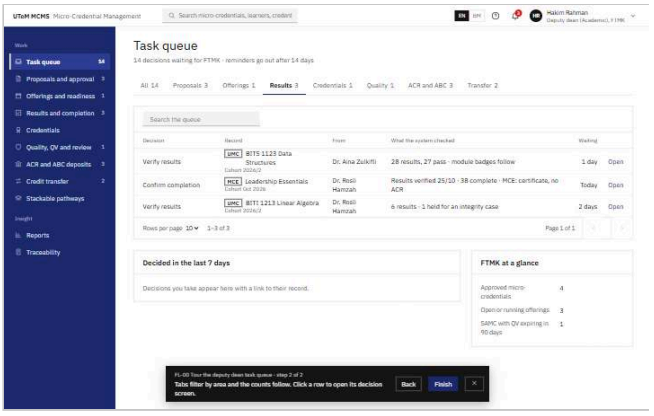
Every step of every flow of Section 7 (26 flows, 121 steps), as the prototype shows it with the step bar at the bottom. Read each flow left to right, top to bottom; the text under each picture is the step bar's explanation.

D.1 FL-00 Tour the deputy dean task queue

Deputy dean · 2 steps



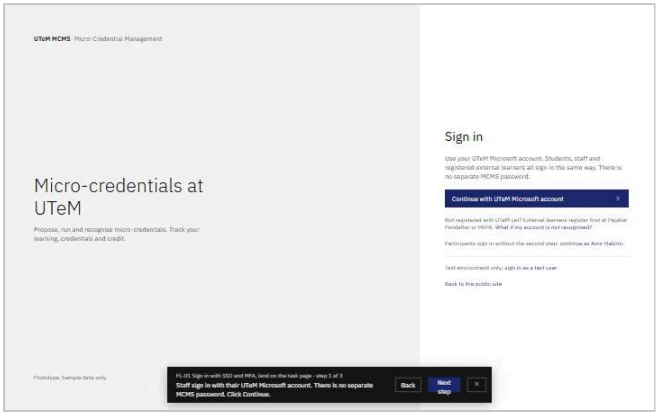
Step 1 of 2. Hakim lands on the task queue: everything waiting for a decision, oldest flagged as overdue. (`#/dd/queue`)



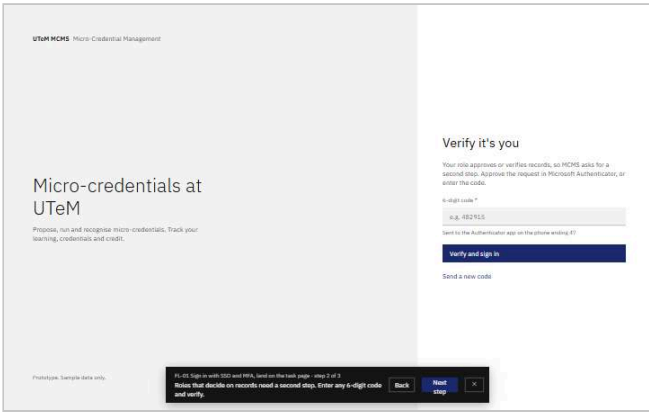
Step 2 of 2. Tabs filter by area and the counts follow. Click a row to open its decision screen. (`#/dd/queue?area=results`)

D.2 FL-01 Sign in with SSO and MFA, land on the task page

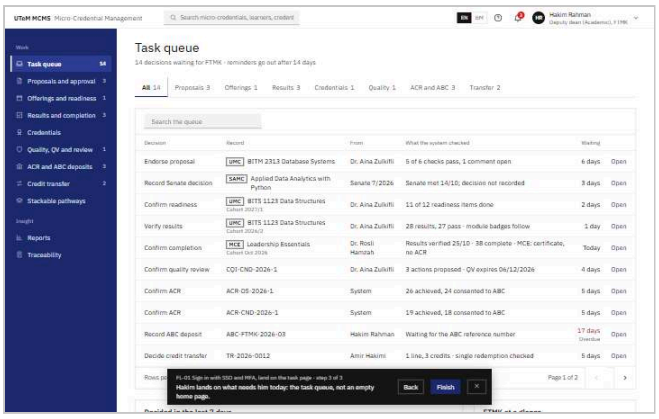
Any staff · 3 steps



Step 1 of 3. Staff sign in with their UTeM Microsoft account. There is no separate MCMS password. Click Continue. (`#/signin`)



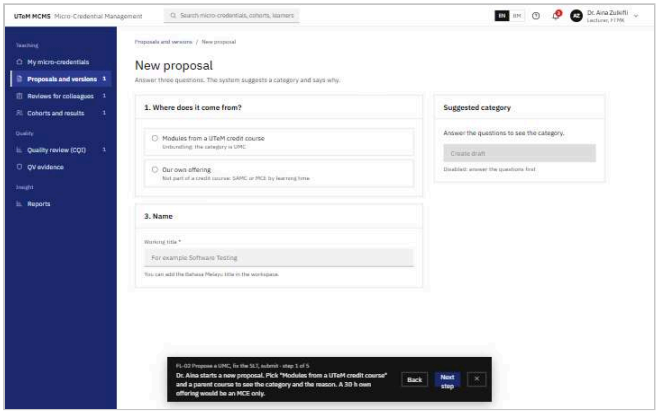
Step 2 of 3. Roles that decide on records need a second step. Enter any 6-digit code and verify. (`#/signin/mfa`)



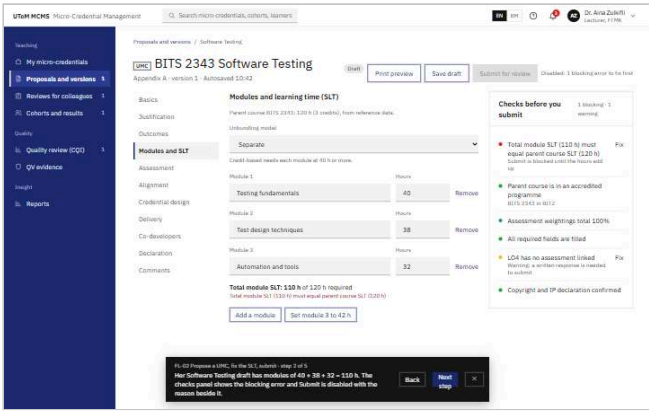
Step 3 of 3. Hakim lands on what needs him today: the task queue, not an empty home page. (`#/dd/queue`)

D.3 FL-02 Propose a UMC, fix the SLT, submit

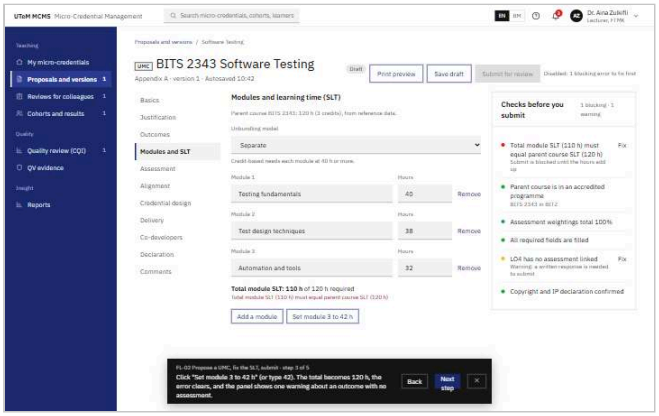
Lecturer · 5 steps



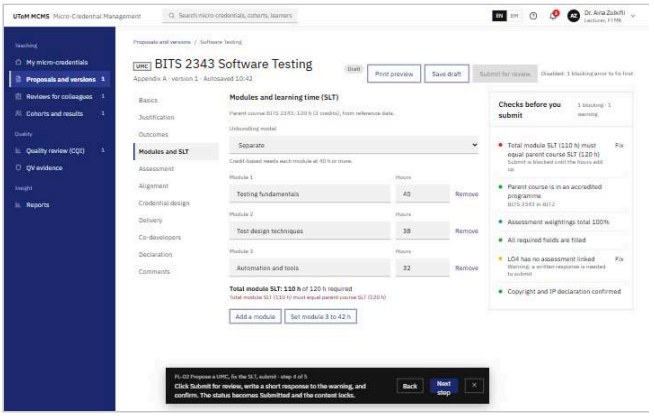
Step 1 of 5. Dr. Aina starts a new proposal. Pick "Modules from a UTeM credit course" and a parent course to see the category and the reason. A 30 h own offering would be an MCE only. (`#/lec/new`)



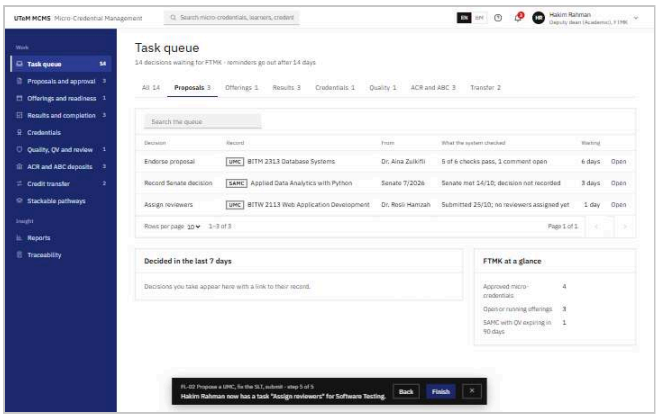
Step 2 of 5. Her Software Testing draft has modules of 40 + 38 + 32 = 110 h. The checks panel shows the blocking error and Submit is disabled with the reason beside it. (`#/lec/proposal/ST?tab=mods`)



Step 3 of 5. Click "Set module 3 to 42 h" (or type 42). The total becomes 120 h, the error clears, and the panel shows one warning about an outcome with no assessment. (`#/lec/proposal/ST?tab=mods`)



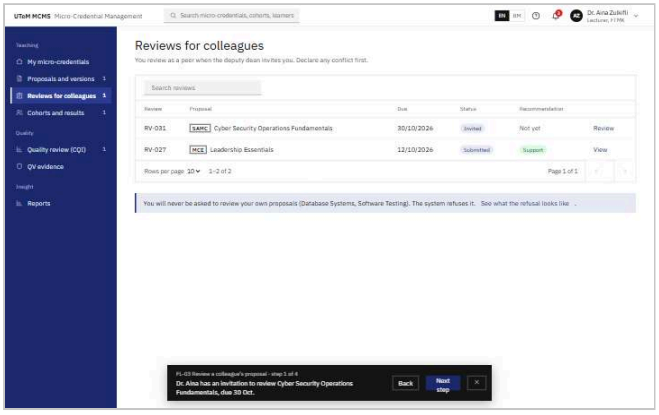
Step 4 of 5. Click Submit for review, write a short response to the warning, and confirm. The status becomes Submitted and the content locks. (`#/lec/proposal/ST?tab=mods`)



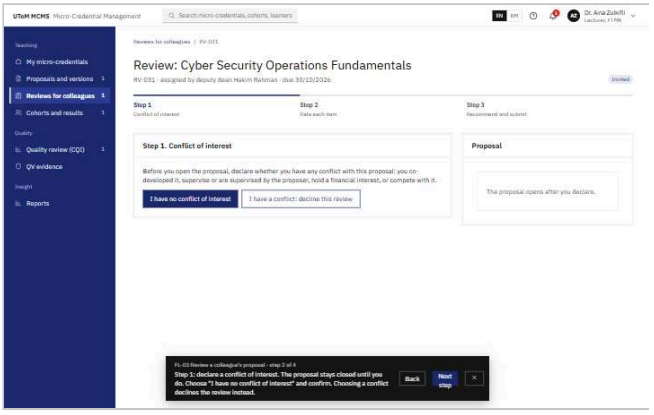
Step 5 of 5. Hakim Rahman now has a task "Assign reviewers" for Software Testing. (`#/dd/queue?area=proposals`)

D.4 FL-03 Review a colleague's proposal

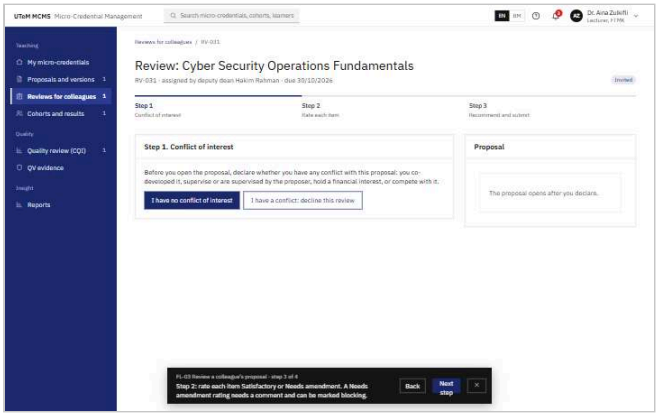
Lecturer as reviewer · 4 steps



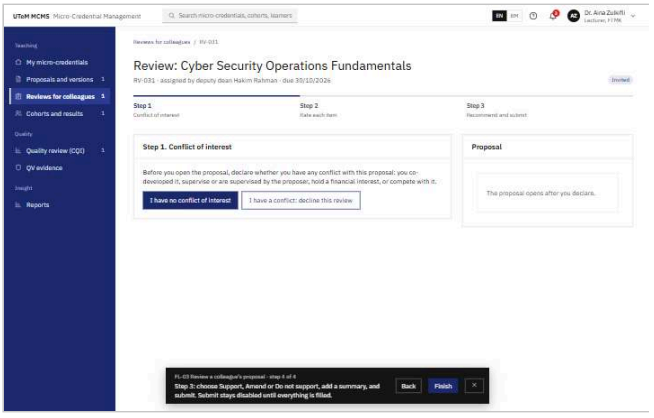
Step 1 of 4. Dr. Aina has an invitation to review Cyber Security Operations Fundamentals, due 30 Oct. (`#/lec/reviews`)



Step 2 of 4. Step 1: declare a conflict of interest. The proposal stays closed until you do. Choose "I have no conflict of interest" and confirm. Choosing a conflict declines the review instead. (`#/lec/review/RV-031`)



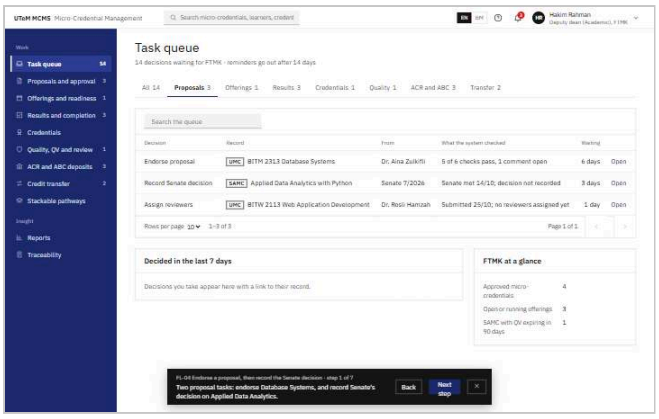
Step 3 of 4. Step 2: rate each item Satisfactory or Needs amendment. A Needs amendment rating needs a comment and can be marked blocking. (`#/lec/review/RV-031`)



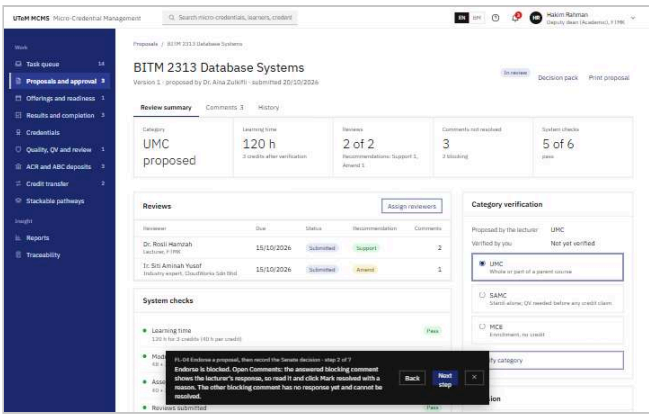
Step 4 of 4. Step 3: choose Support, Amend or Do not support, add a summary, and submit. Submit stays disabled until everything is filled. (`#/lec/review/RV-031`)

D.5 FL-04 Endorse a proposal, then record the Senate decision

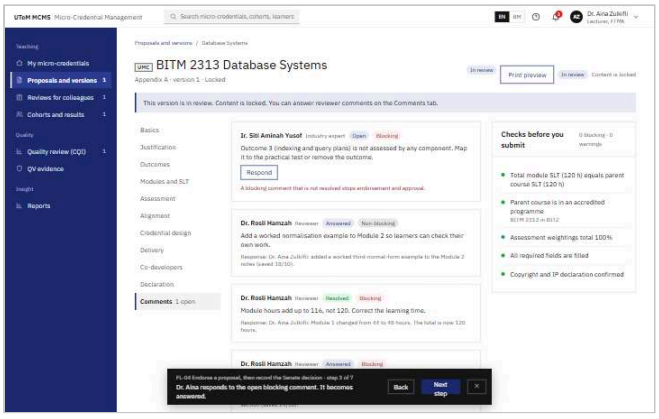
Deputy dean · 7 steps



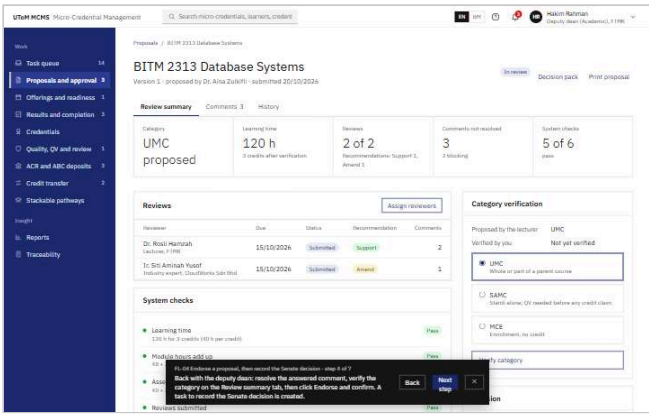
Step 1 of 7. Two proposal tasks: endorse Database Systems, and record Senate's decision on Applied Data Analytics. (`#/dd/queue?area=proposals`)



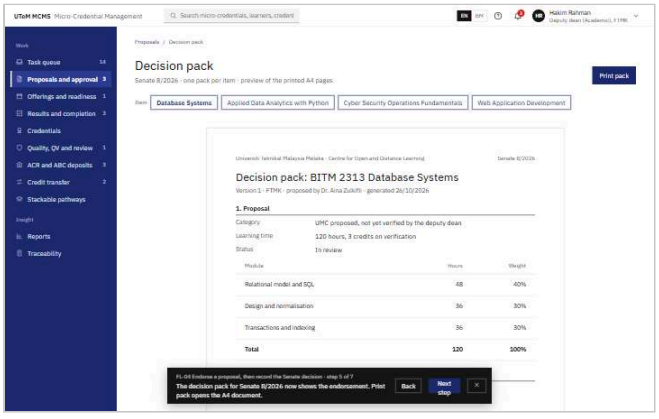
Step 2 of 7. Endorse is blocked. Open Comments: the answered blocking comment shows the lecturer's response, so read it and click Mark resolved with a reason. The other blocking comment has no response yet and cannot be resolved. (`#/dd/proposal/DB`)



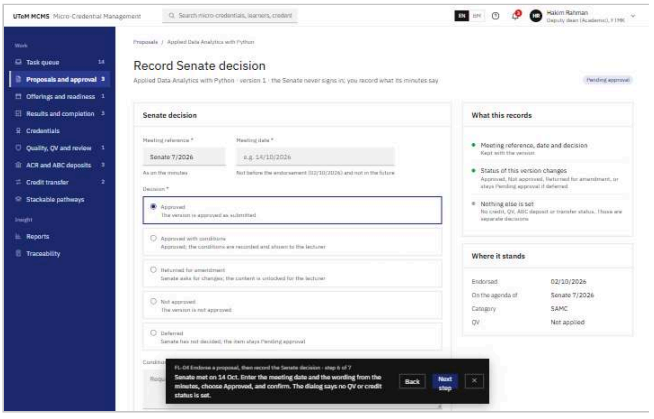
Step 3 of 7. Dr. Aina responds to the open blocking comment. It becomes answered. (`#/lec/proposal/DB?tab=comments`)



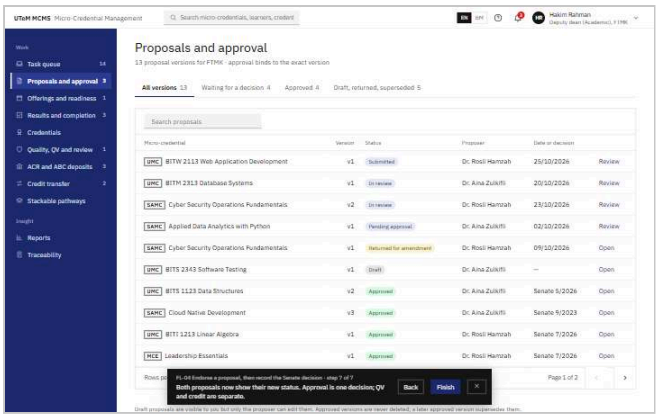
Step 4 of 7. Back with the deputy dean: resolve the answered comment, verify the category on the Review summary tab, then click Endorse and confirm. A task to record the Senate decision is created. (`#/dd/proposal/DB`)



Step 5 of 7. The decision pack for Senate 8/2026 now shows the endorsement. Print pack opens the A4 document. (`#/dd/pack?id=DB`)



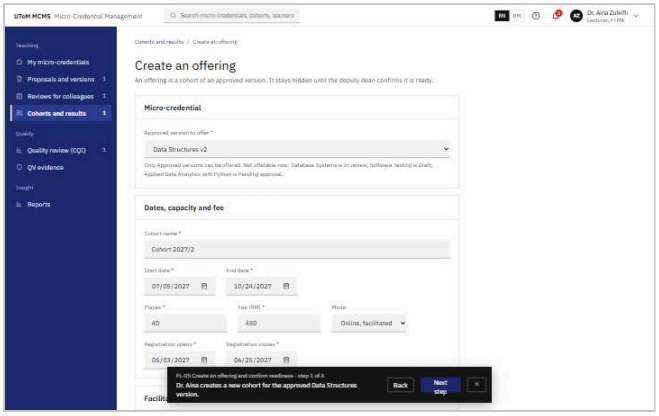
Step 6 of 7. Senate met on 14 Oct. Enter the meeting date and the wording from the minutes, choose Approved, and confirm. The dialog says no QV or credit status is set. (`#/dd/senate/ADA`)



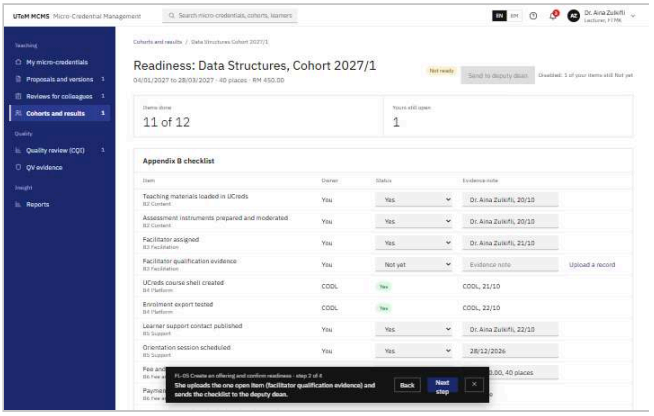
Step 7 of 7. Both proposals now show their new status. Approval is one decision; QV and credit are separate. (`#/dd/proposals`)

D.6 FL-05 Create an offering and confirm readiness

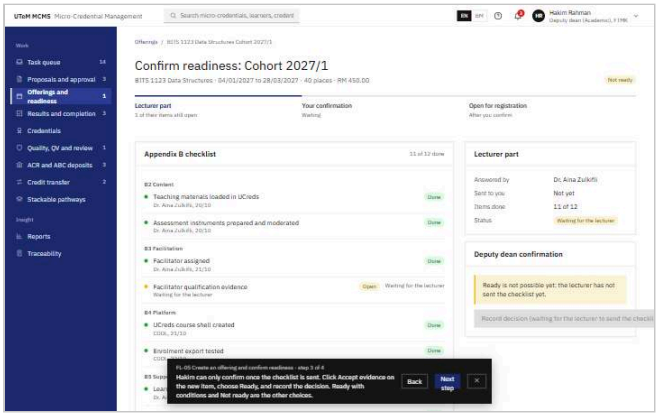
Lecturer, deputy dean · 4 steps



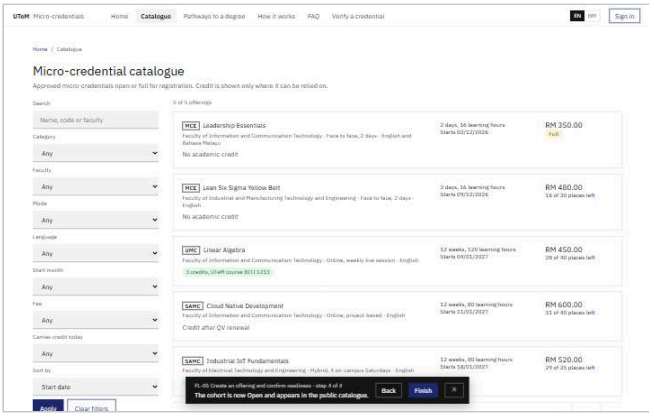
Step 1 of 4. Dr. Aina creates a new cohort for the approved Data Structures version. (`#/lec/offering/new`)



Step 2 of 4. She uploads the one open item (facilitator qualification evidence) and sends the checklist to the deputy dean. (`#/lec/readiness/DS-2027-1`)



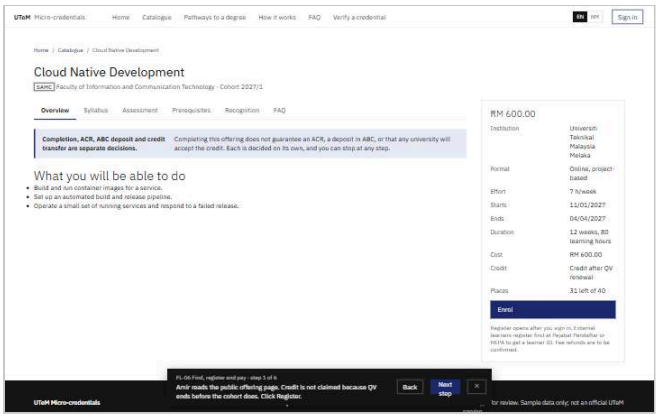
Step 3 of 4. Hakim can only confirm once the checklist is sent. Click Accept evidence on the new item, choose Ready, and record the decision. Ready with conditions and Not ready are the other choices. (`#/dd/readiness/DS-2027-1`)



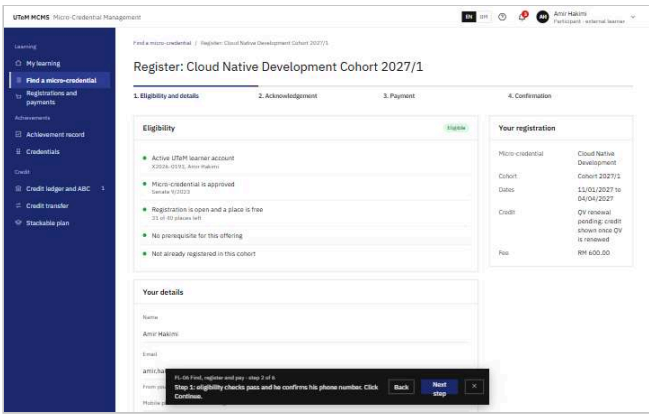
Step 4 of 4. The cohort is now Open and appears in the public catalogue. (`#/public/catalogue`)

D.7 FL-06 Find, register and pay

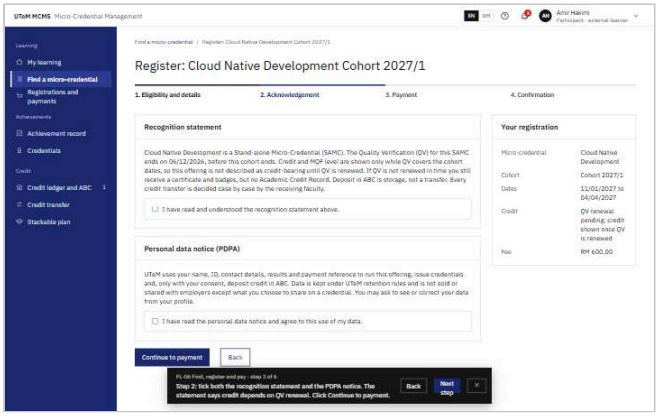
Student · 6 steps



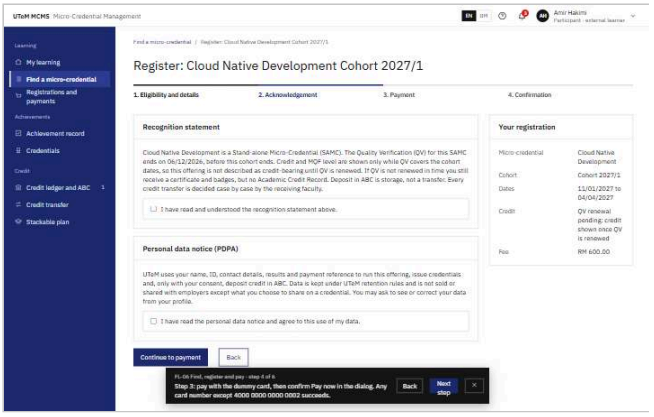
Step 1 of 6. Amir reads the public offering page. Credit is not claimed because QV ends before the cohort does. Click Register. (`#/public/offering/CND-2027-1`)



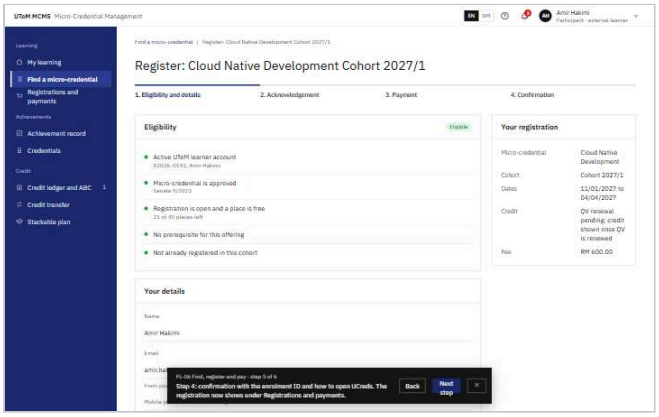
Step 2 of 6. Step 1: eligibility checks pass and he confirms his phone number. Click Continue. (`#/st/register/CND-2027-1?step=1`)



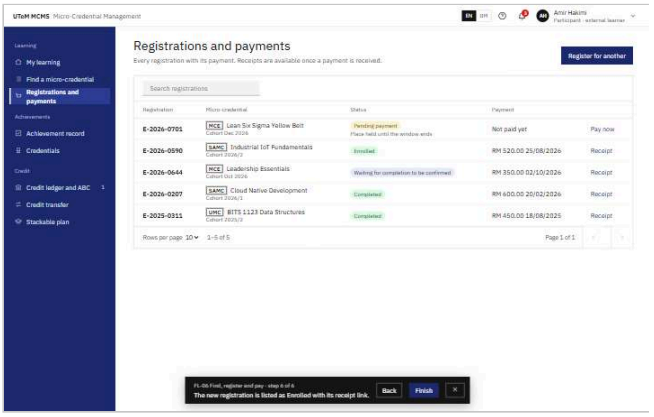
Step 3 of 6. Step 2: tick both the recognition statement and the PDPA notice. The statement says credit depends on QV renewal. Click Continue to payment. (`#/st/register/CND-2027-1?step=2`)



Step 4 of 6. Step 3: pay with the dummy card, then confirm Pay now in the dialog. Any card number except 4000 0000 0000 0002 succeeds. (`#/st/register/CND-2027-1?step=3`)



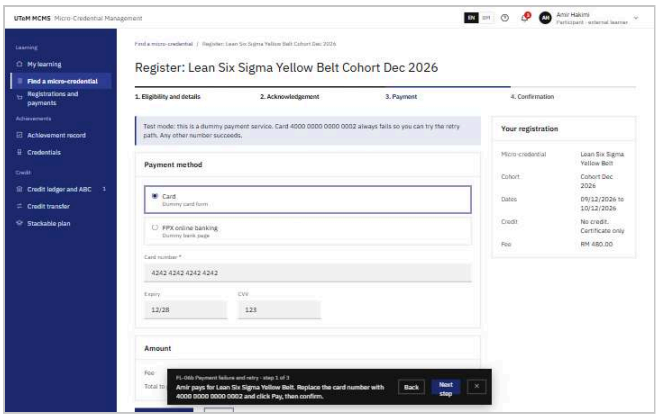
Step 5 of 6. Step 4: confirmation with the enrolment ID and how to open UCreds. The registration now shows under Registrations and payments. (`#/st/register/CND-2027-1?step=4`)



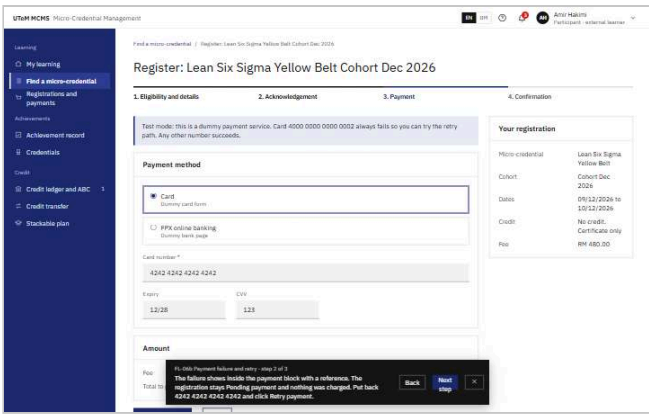
Step 6 of 6. The new registration is listed as Enrolled with its receipt link. (`#/st/enrolments`)

D.8 FL-06b Payment failure and retry

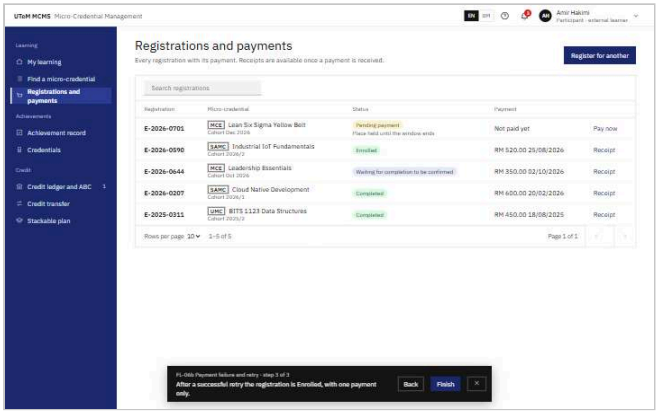
Student · 3 steps



Step 1 of 3. Amir pays for Lean Six Sigma Yellow Belt. Replace the card number with 4000 0000 0000 0002 and click Pay, then confirm. (`#/st/register/LSS-2026-12?step=3`)



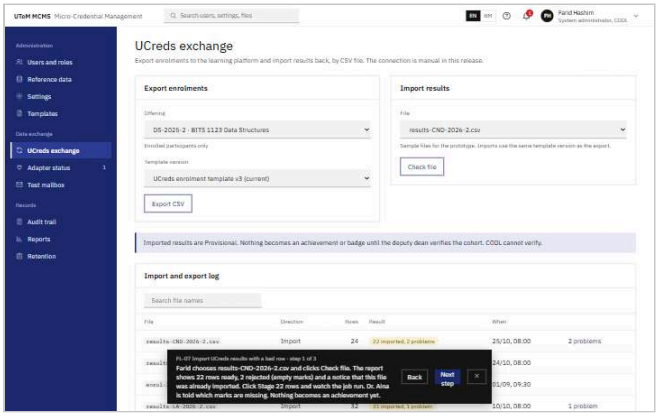
Step 2 of 3. The failure shows inside the payment block with a reference. The registration stays Pending payment and nothing was charged. Put back 4242 4242 4242 4242 and click Retry payment. (`#/st/register/LSS-2026-12?step=3`)



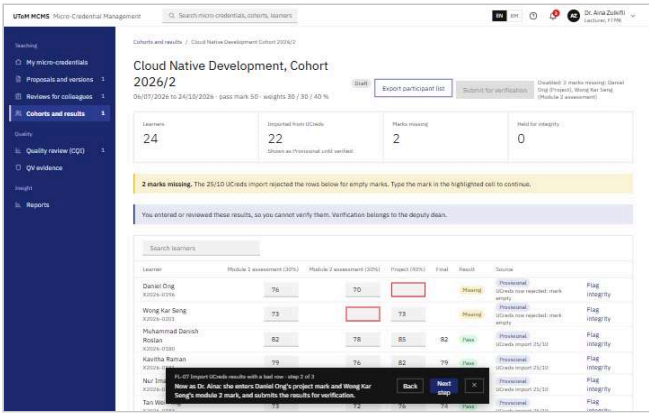
Step 3 of 3. After a successful retry the registration is Enrolled, with one payment only. (`#/st/enrolments`)

D.9 FL-07 Import UCreds results with a bad row

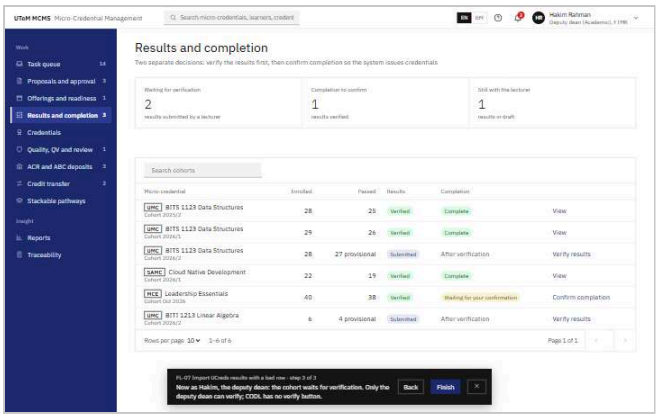
CODL, lecturer, deputy dean · 3 steps



Step 1 of 3. Farid chooses results-CND-2026-2.csv and clicks Check file. The report shows 22 rows ready, 2 rejected (empty marks) and a notice that this file was already imported. Click Stage 22 rows and watch the job run. Dr. Aina is told which marks are missing. Nothing becomes an achievement yet. (`#/codl/ucreds`)



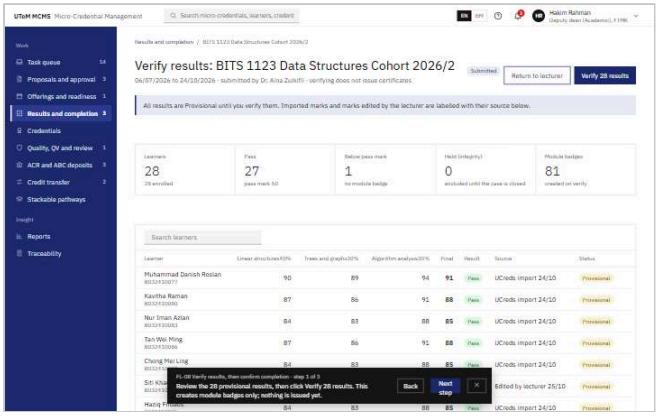
Step 2 of 3. Now as Dr. Aina: she enters Daniel Ong's project mark and Wong Kar Seng's module 2 mark, and submits the results for verification. (`#/lec/cohort/CND-2026-2`)



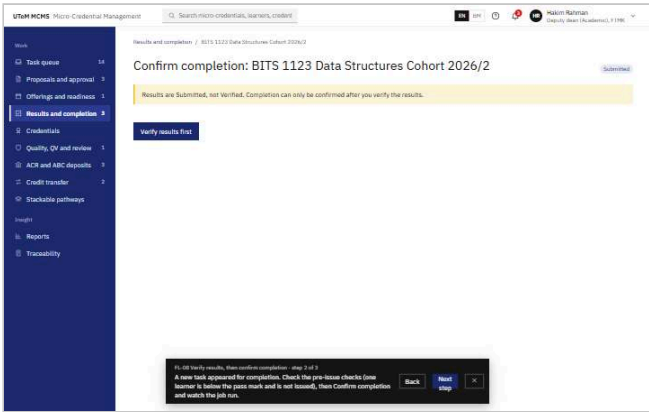
Step 3 of 3. Now as Hakim, the deputy dean: the cohort waits for verification. Only the deputy dean can verify; CODL has no verify button. (`#/dd/results`)

D.10 FL-08 Verify results, then confirm completion

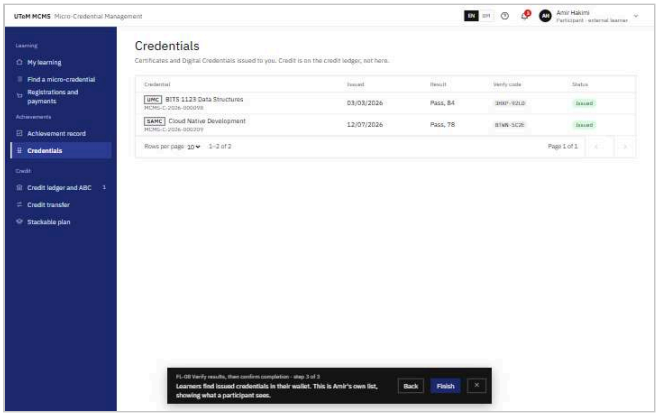
Deputy dean · 3 steps



Step 1 of 3. Review the 28 provisional results, then click Verify 28 results. This creates module badges only; nothing is issued yet. (`#/dd/verify/DS-2026-2`)



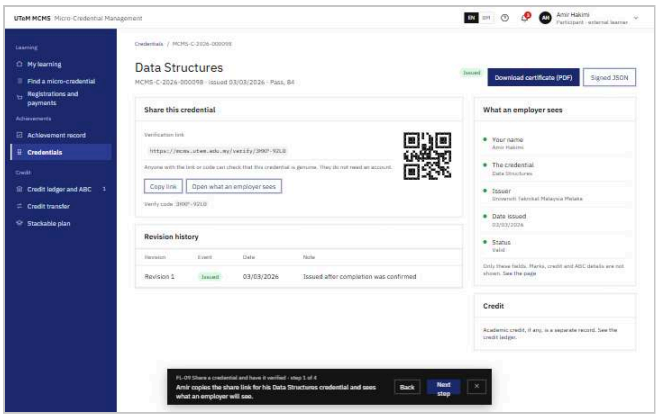
Step 2 of 3. A new task appeared for completion. Check the pre-issue checks (one learner is below the pass mark and is not issued), then Confirm completion and watch the job run. (`#/dd/completion/DS-2026-2`)



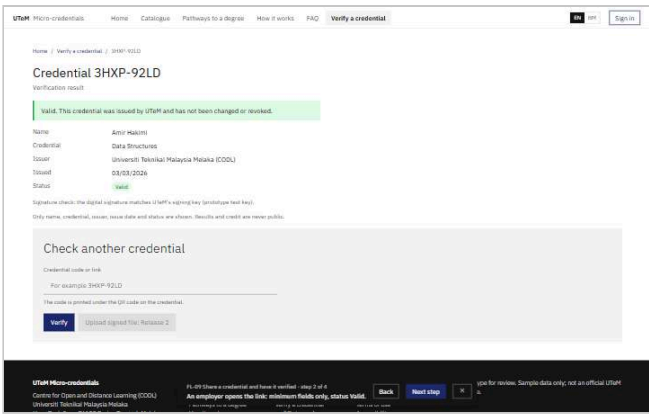
Step 3 of 3. Learners find issued credentials in their wallet. This is Amir's own list, showing what a participant sees. (`#/st/credentials`)

D.11 FL-09 Share a credential and have it verified

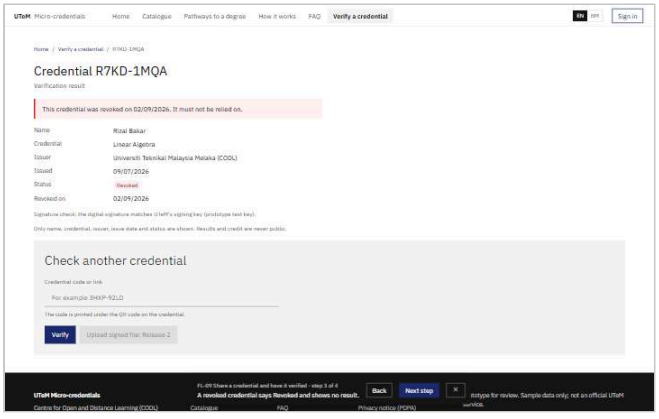
Student, public · 4 steps



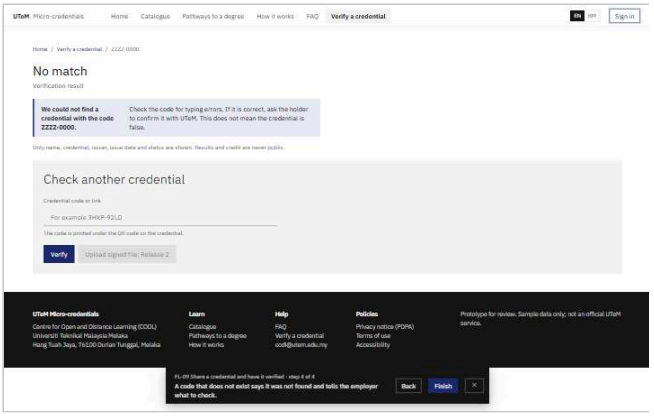
Step 1 of 4. Amir copies the share link for his Data Structures credential and sees what an employer will see. (`#/st/credential/MCMS-C-2026-000098`)



Step 2 of 4. An employer opens the link: minimum fields only, status Valid. (`#/public/verify/3HXP-92LD`)



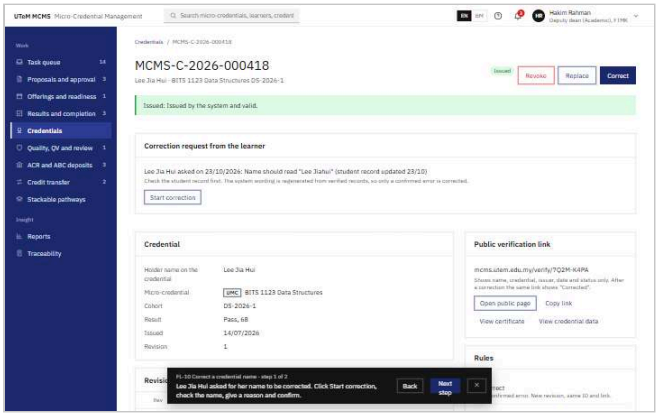
Step 3 of 4. A revoked credential says Revoked and shows no result. (`#/public/verify/R7KD-1MQA`)



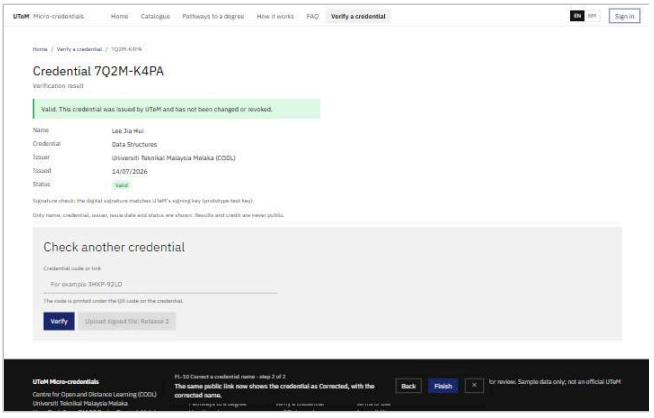
Step 4 of 4. A code that does not exist says it was not found and tells the employer what to check. (`#/public/verify/ZZZZ-0000`)

D.12 FL-10 Correct a credential name

Deputy dean · 2 steps



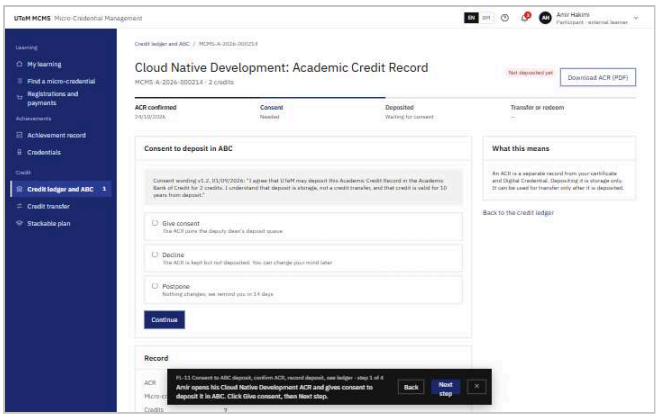
Step 1 of 2. Lee Jia Hui asked for her name to be corrected. Click Start correction, check the name, give a reason and confirm. (`#/dd/cred/MCMS-C-2026-000418`)



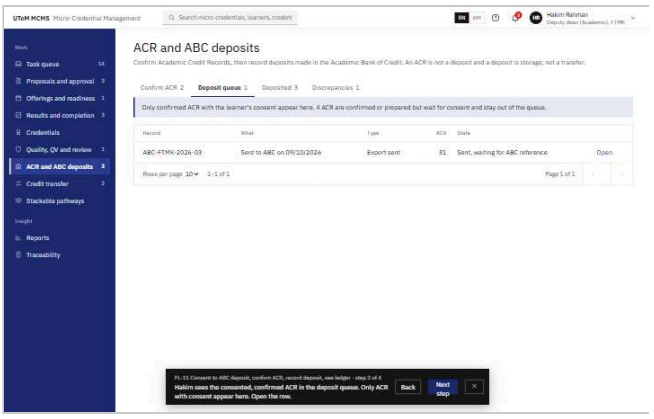
Step 2 of 2. The same public link now shows the credential as Corrected, with the corrected name. (`#/public/verify/7Q2M-K4PA`)

D.13 FL-11 Consent to ABC deposit, confirm ACR, record deposit, see ledger

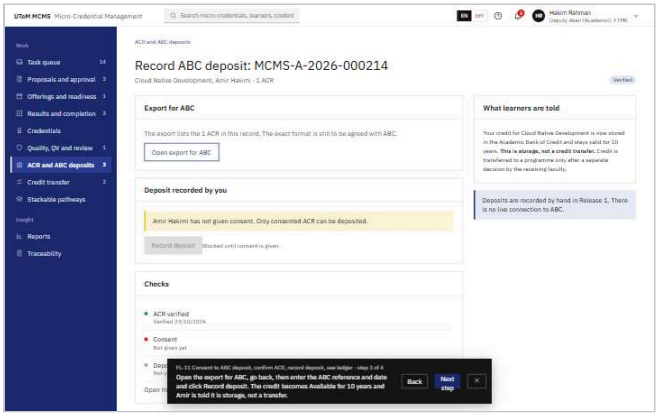
Student, deputy dean · 4 steps



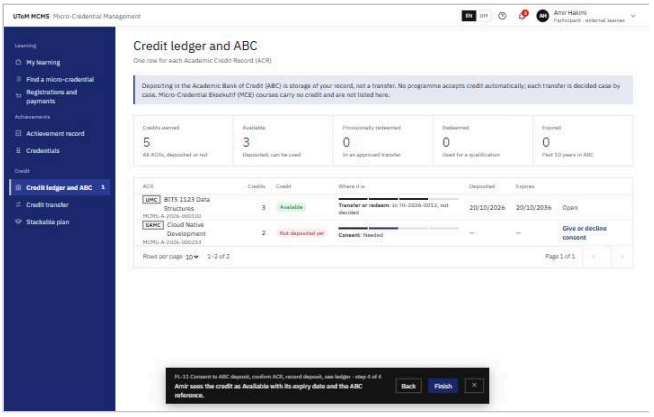
Step 1 of 4. Amir opens his Cloud Native Development ACR and gives consent to deposit it in ABC. Click Give consent, then Next step. (`#/st/acr/MCMS-A-2026-000214`)



Step 2 of 4. Hakim sees the consented, confirmed ACR in the deposit queue. Only ACR with consent appear here. Open the row. (`#/dd/acr?tab=queue`)



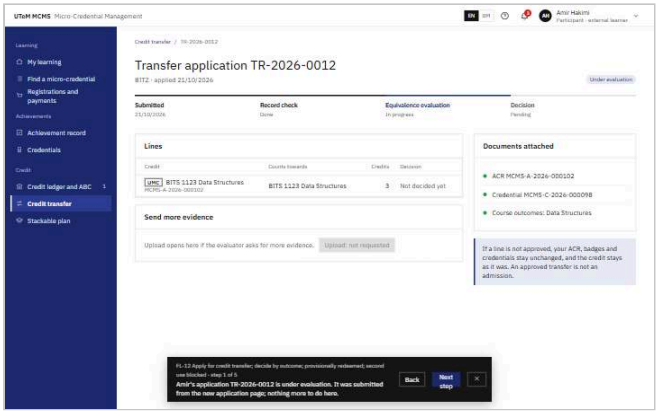
Step 3 of 4. Open the export for ABC, go back, then enter the ABC reference and date and click Record deposit. The credit becomes Available for 10 years and Amir is told it is storage, not a transfer. (`#/dd/deposit/MCMS-A-2026-000214`)



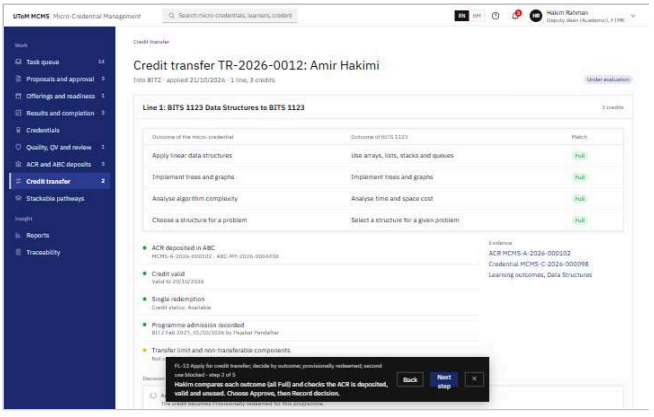
Step 4 of 4. Amir sees the credit as Available with its expiry date and the ABC reference. (`#/st/ledger`)

D.14 FL-12 Apply for credit transfer; decide by outcome; provisionally redeemed; second use blocked

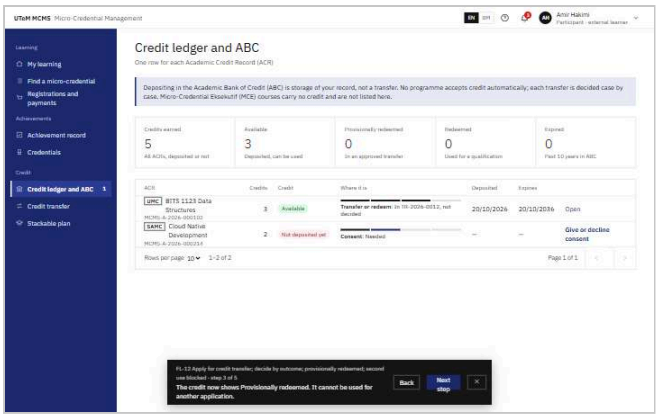
Student, deputy dean · 5 steps



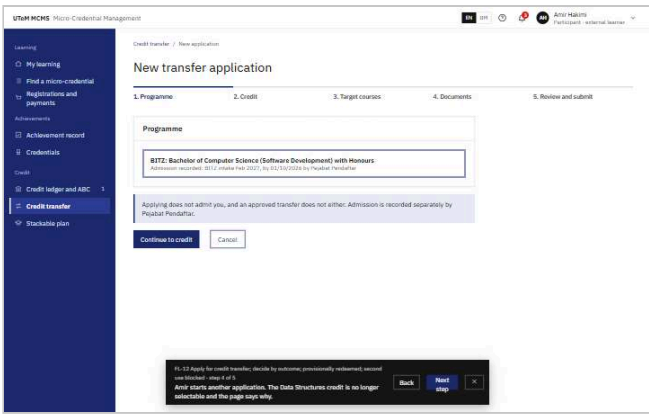
Step 1 of 5. Amir's application TR-2026-0012 is under evaluation. It was submitted from the new application page; nothing more to do here. (`#/st/transfer/TR-2026-0012`)



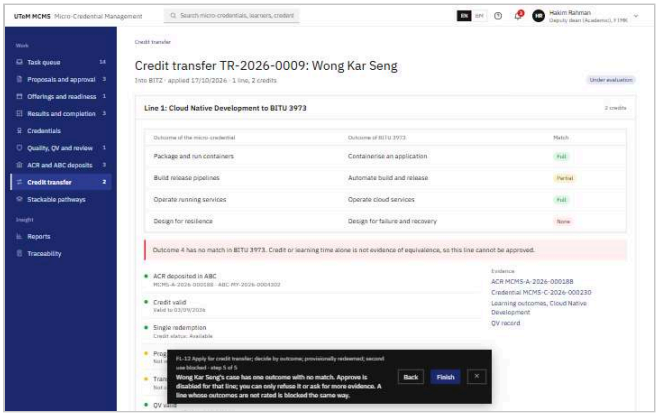
Step 2 of 5. Hakim compares each outcome (all Full) and checks the ACR is deposited, valid and unused. Choose Approve, then Record decision. (`#/dd/transfer/TR-2026-0012`)



Step 3 of 5. The credit now shows Provisionally redeemed. It cannot be used for another application. (`#/st/ledger`)



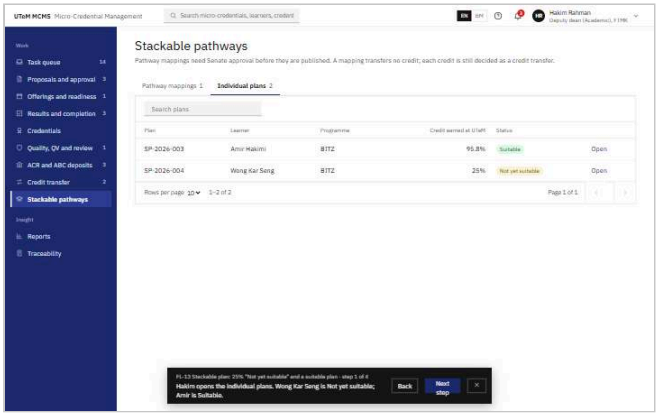
Step 4 of 5. Amir starts another application. The Data Structures credit is no longer selectable and the page says why. (`#/st/transfer/new`)



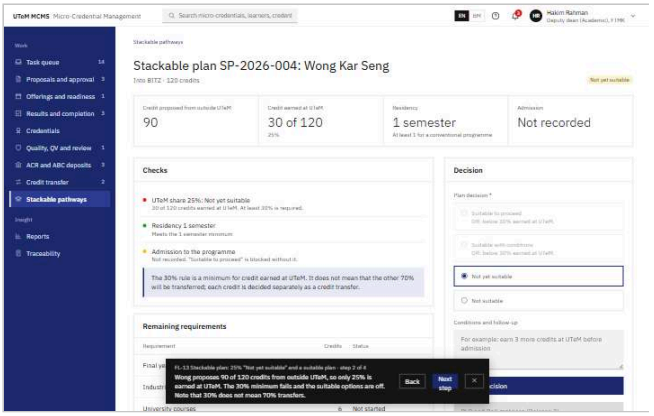
Step 5 of 5. Wong Kar Seng's case has one outcome with no match. Approve is disabled for that line; you can only refuse it or ask for more evidence. A line whose outcomes are not rated is blocked the same way. (`#/dd/transfer/TR-2026-0009`)

D.15 FL-13 Stackable plan: 25% "Not yet suitable" and a suitable plan

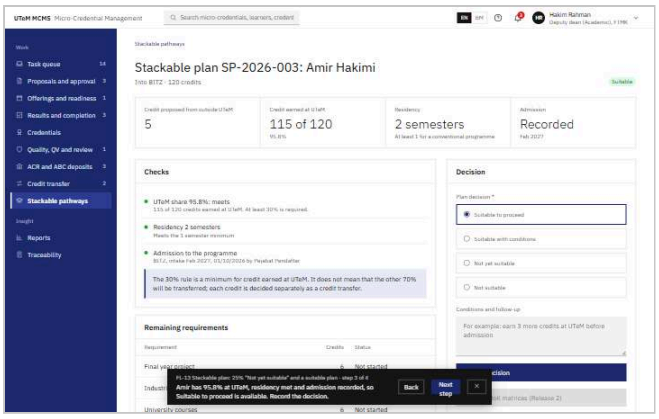
Deputy dean · 4 steps



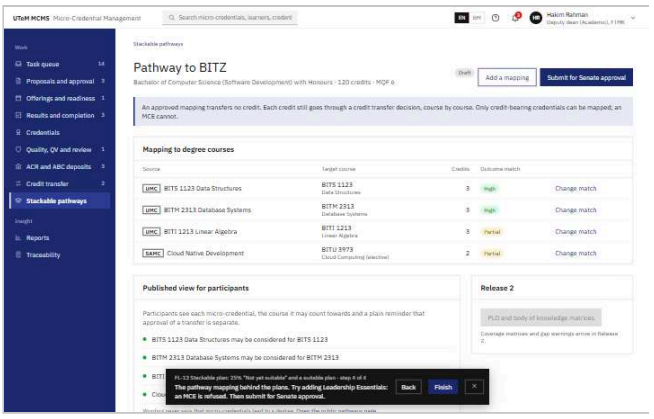
Step 1 of 4. Hakim opens the individual plans. Wong Kar Seng is Not yet suitable; Amir is Suitable. (`#/dd/stack?tab=plans`)



Step 2 of 4. Wong proposes 90 of 120 credits from outside UTeM, so only 25% is earned at UTeM. The 30% minimum fails and the suitable options are off. Note that 30% does not mean 70% transfers. (`#/dd/plan/SP-2026-004`)



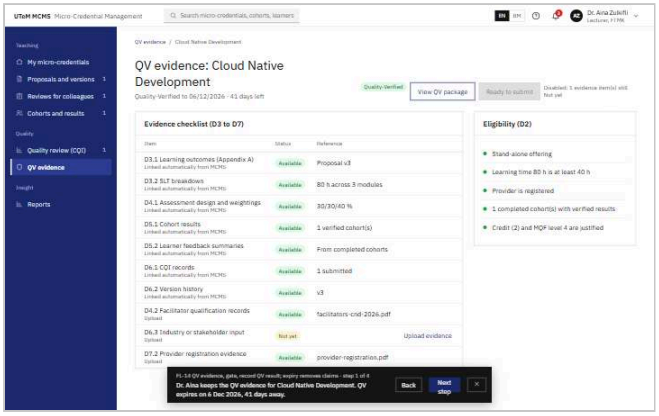
Step 3 of 4. Amir has 95.8% at UTeM, residency met and admission recorded, so Suitable to proceed is available. Record the decision. (`#/dd/plan/SP-2026-003`)



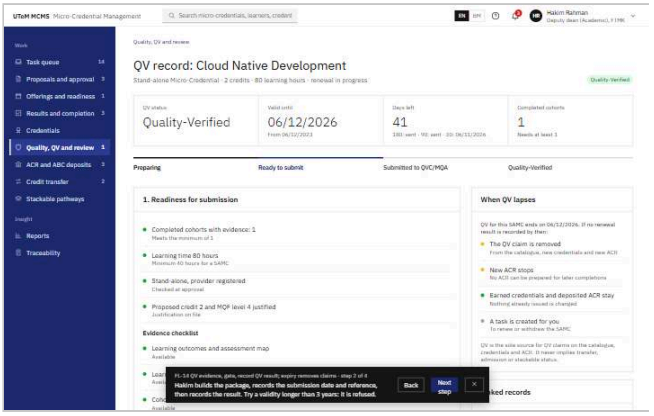
Step 4 of 4. The pathway mapping behind the plans. Try adding Leadership Essentials: an MCE is refused. Then submit for Senate approval. (`#/dd/pathway/PW-BITZ-1`)

D.16 FL-14 QV evidence, gate, record QV result; expiry removes claims

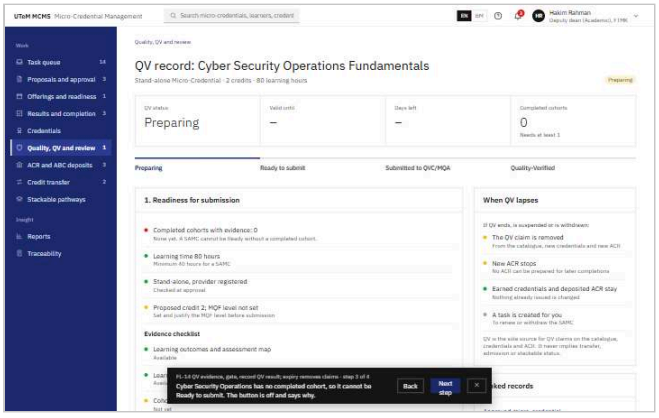
Lecturer, deputy dean · 4 steps



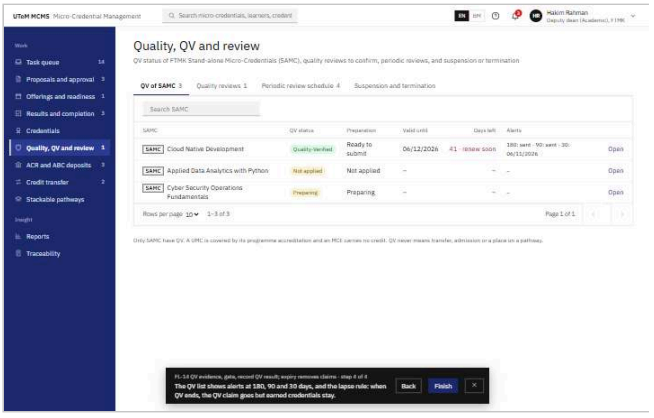
Step 1 of 4. Dr. Aina keeps the QV evidence for Cloud Native Development. QV expires on 6 Dec 2026, 41 days away. (`#/lec/qv/CND`)



Step 2 of 4. Hakim builds the package, records the submission date and reference, then records the result. Try a validity longer than 3 years: it is refused. (`#/dd/qv/CND`)



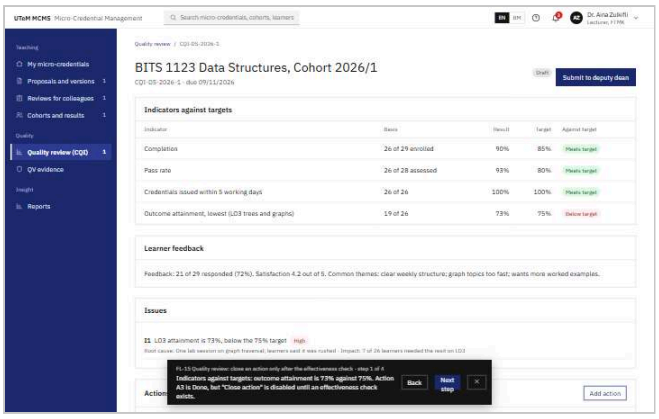
Step 3 of 4. Cyber Security Operations has no completed cohort, so it cannot be Ready to submit. The button is off and says why. (`#/dd/qv/CSO`)



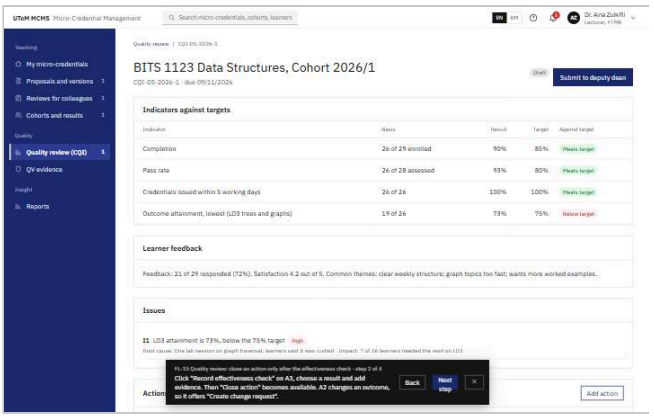
Step 4 of 4. The QV list shows alerts at 180, 90 and 30 days, and the lapse rule: when QV ends, the QV claim goes but earned credentials stay. (`#/dd/quality`)

D.17 FL-15 Quality review: close an action only after the effectiveness check

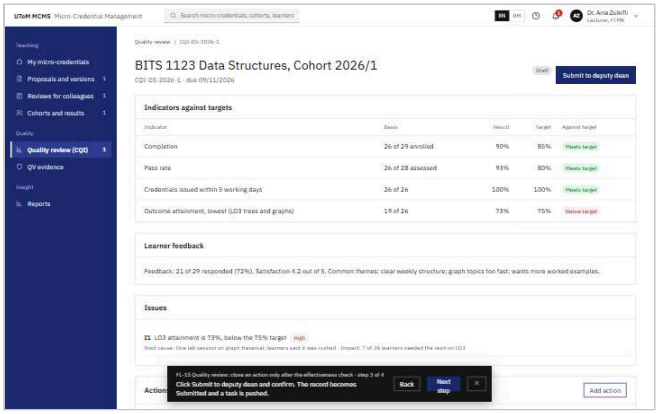
Lecturer, deputy dean · 4 steps



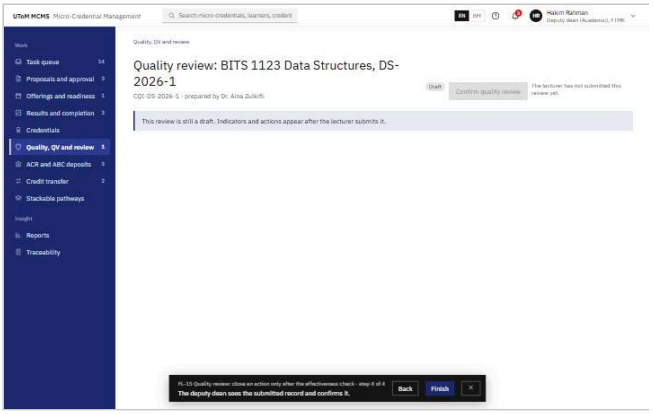
Step 1 of 4. Indicators against targets: outcome attainment is 73% against 75%. Action A3 is Done, but "Close action" is disabled until an effectiveness check exists. (`#/lec/cqi/CQI-DS-2026-1`)



Step 2 of 4. Click "Record effectiveness check" on A3, choose a result and add evidence. Then "Close action" becomes available. A2 changes an outcome, so it offers "Create change request". (`#/lec/cqi/CQI-DS-2026-1`)



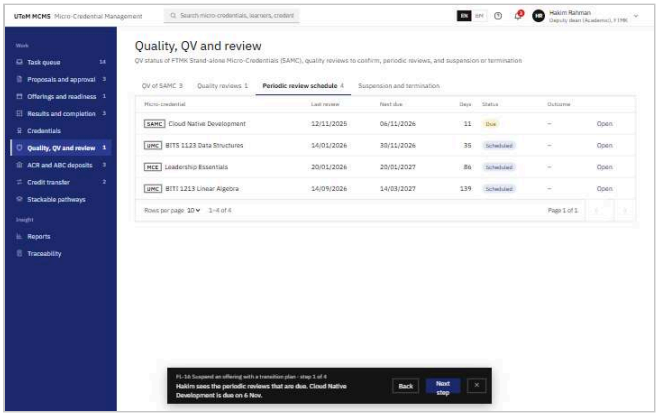
Step 3 of 4. Click Submit to deputy dean and confirm. The record becomes Submitted and a task is pushed. (`#/lec/cqi/CQI-DS-2026-1`)



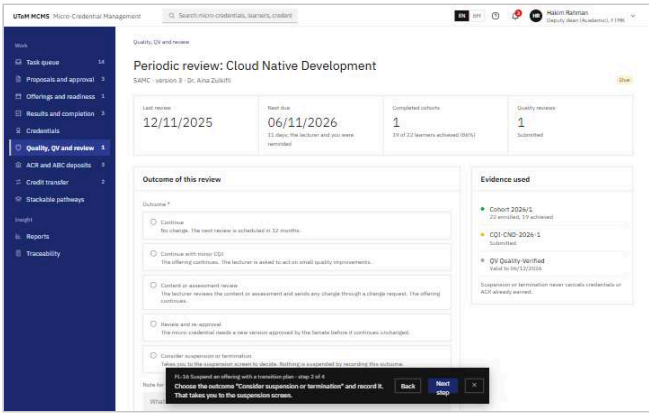
Step 4 of 4. The deputy dean sees the submitted record and confirms it. (`#/dd/cqi/CQI-DS-2026-1`)

D.18 FL-16 Suspend an offering with a transition plan

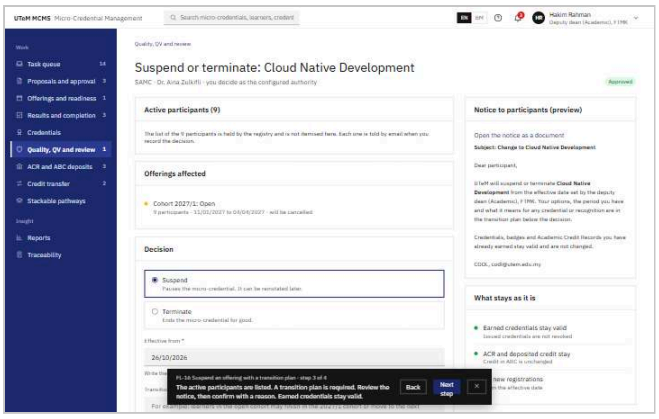
Deputy dean · 4 steps



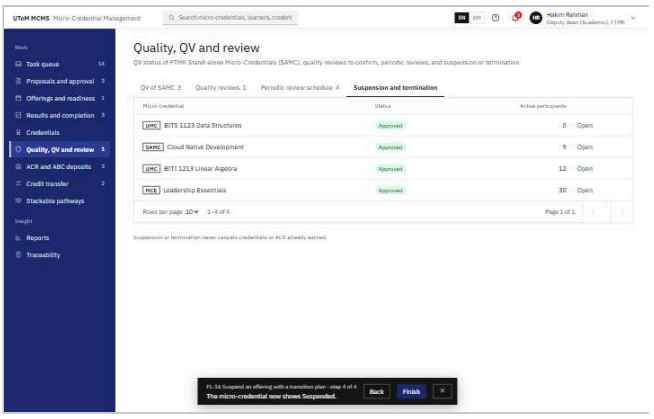
Step 1 of 4. Hakim sees the periodic reviews that are due. Cloud Native Development is due on 6 Nov. (`#/dd/quality?tab=periodic`)



Step 2 of 4. Choose the outcome "Consider suspension or termination" and record it. That takes you to the suspension screen. (`#/dd/review/CND`)



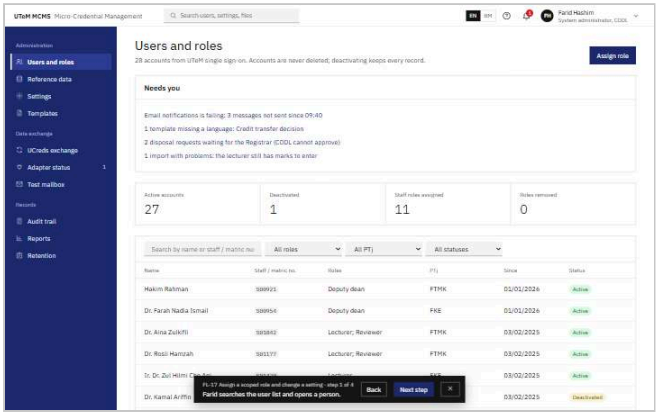
Step 3 of 4. The active participants are listed. A transition plan is required. Review the notice, then confirm with a reason. Earned credentials stay valid. (`#/dd/suspend/CND`)



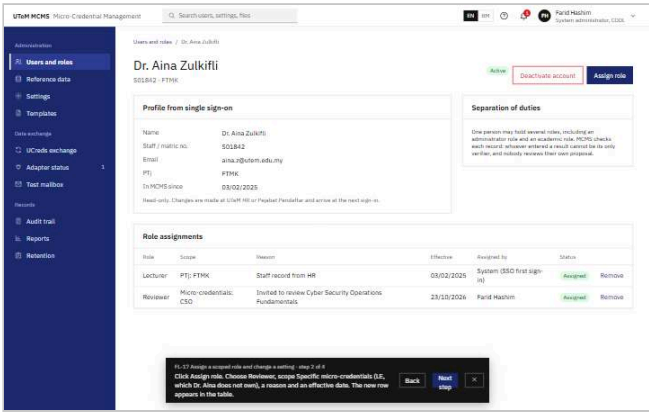
Step 4 of 4. The micro-credential now shows Suspended. (`#/dd/quality?tab=susp`)

D.19 FL-17 Assign a scoped role and change a setting

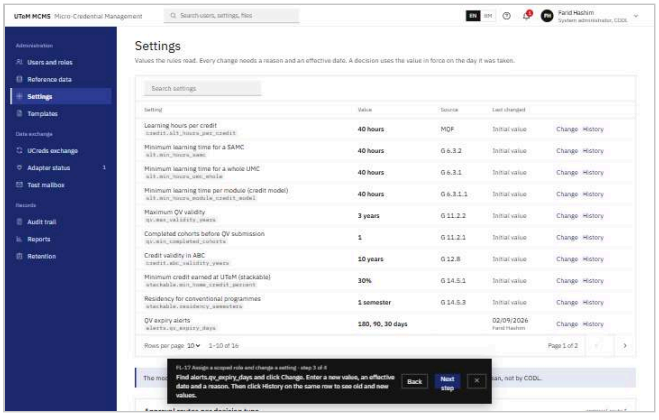
CODL administrator · 4 steps



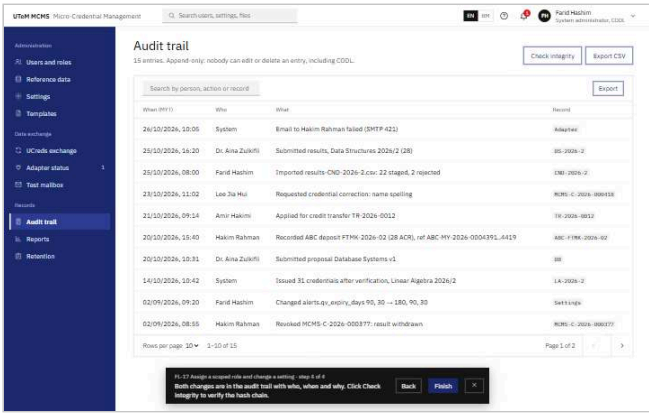
Step 1 of 4. Farid searches the user list and opens a person. (`#/codl/users`)



Step 2 of 4. Click Assign role. Choose Reviewer, scope Specific micro-credentials (LE, which Dr. Aina does not own), a reason and an effective date. The new row appears in the table. (`#/codl/user/S01842`)



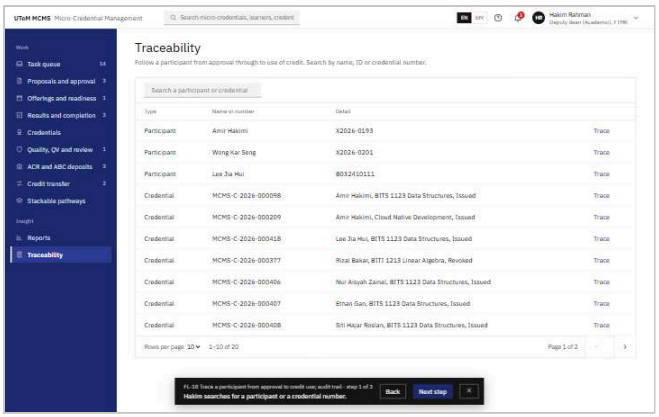
Step 3 of 4. Find alerts.qv_expiry_days and click Change. Enter a new value, an effective date and a reason. Then click History on the same row to see old and new values. (`#/codl/settings`)



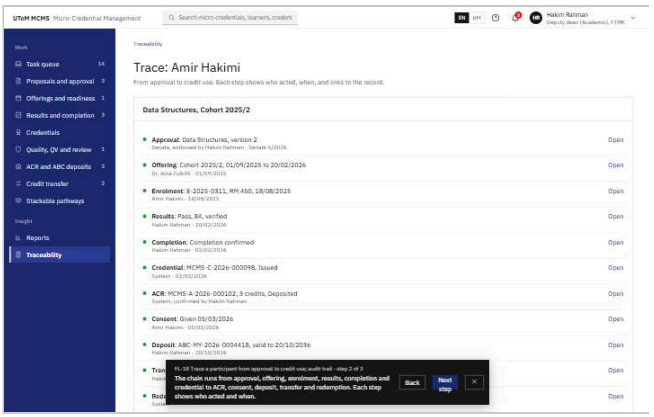
Step 4 of 4. Both changes are in the audit trail with who, when and why. Click Check integrity to verify the hash chain. (`#/codl/audit`)

D.20 FL-18 Trace a participant from approval to credit use; audit trail

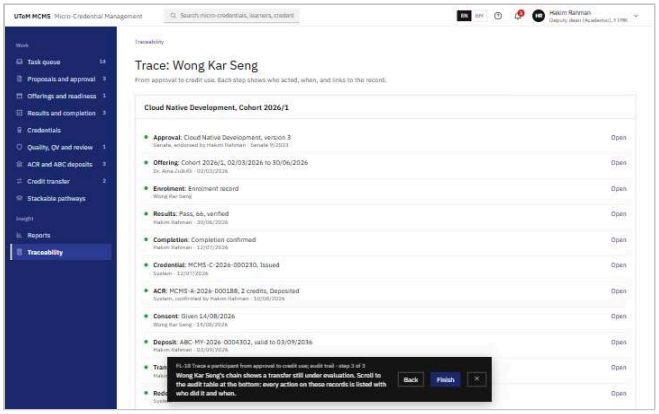
Deputy dean · 3 steps



Step 1 of 3. Hakim searches for a participant or a credential number. (`#/dd/trace`)



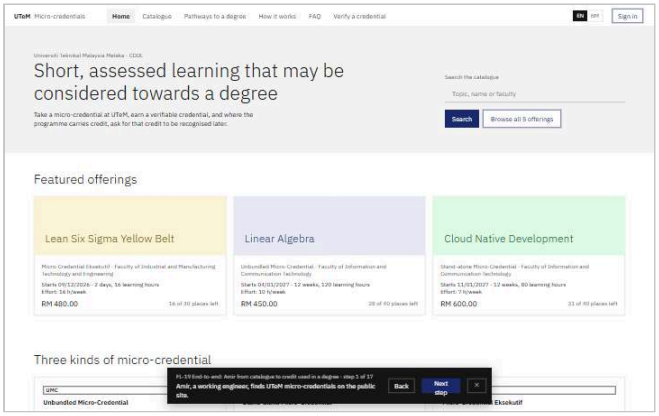
Step 2 of 3. The chain runs from approval, offering, enrolment, results, completion and credential to ACR, consent, deposit, transfer and redemption. Each step shows who acted and when. (`#/dd/trace/amir`)



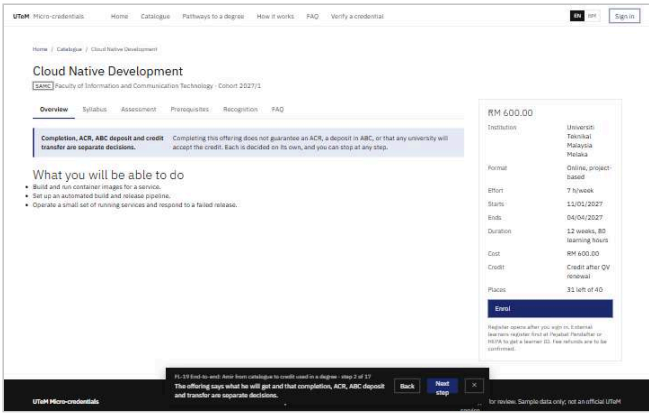
Step 3 of 3. Wong Kar Seng's chain shows a transfer still under evaluation. Scroll to the audit table at the bottom: every action on these records is listed with who did it and when. (`#/dd/trace/wong`)

D.21 FL-19 End-to-end: Amir from catalogue to credit used in a degree

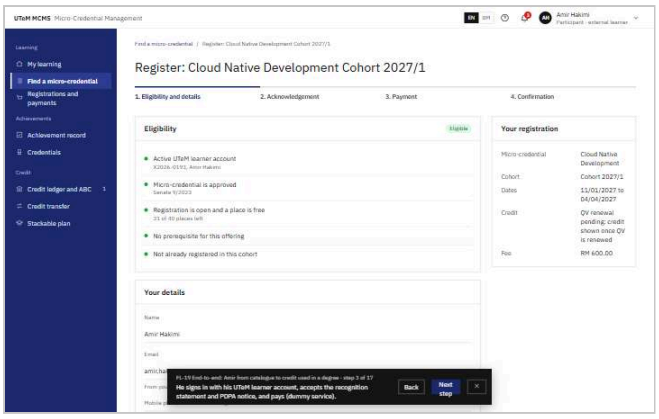
All roles · 17 steps



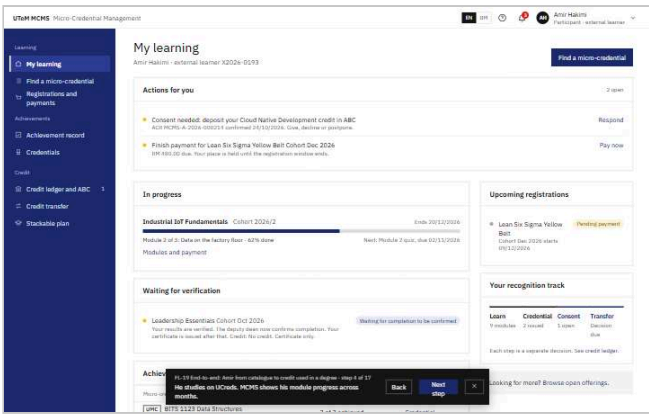
Step 1 of 17. Amir, a working engineer, finds UTeM micro-credentials on the public site. (`#/public/home`)



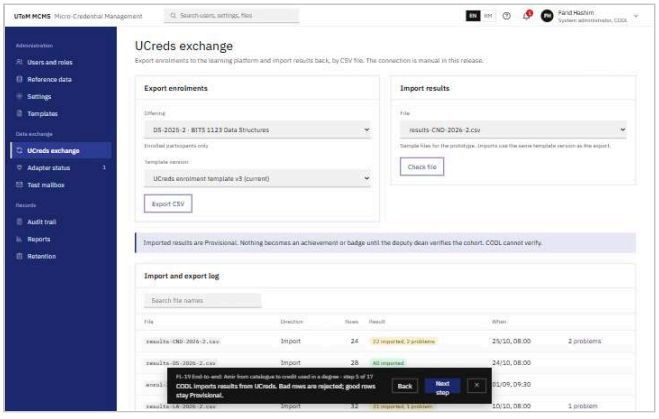
Step 2 of 17. The offering says what he will get and that completion, ACR, ABC deposit and transfer are separate decisions. (`#/public/offering/CND-2027-1`)



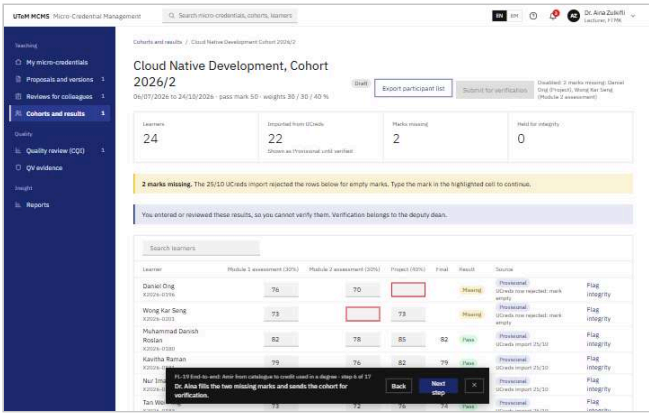
Step 3 of 17. He signs in with his UTeM learner account, accepts the recognition statement and PDPA notice, and pays (dummy service). (`#/st/register/CND-2027-1`)



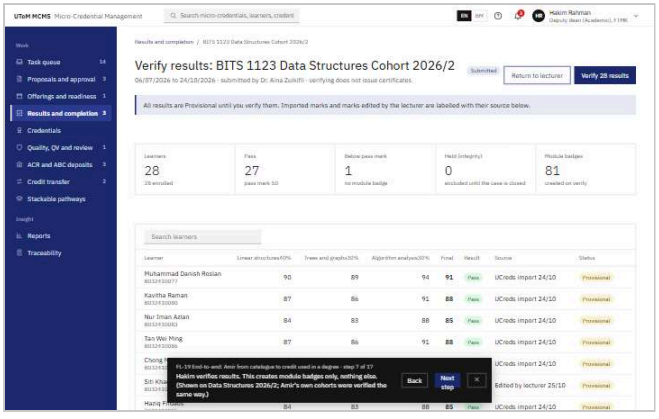
Step 4 of 17. He studies on UCreds. MCMS shows his module progress across months. (`#/st/home`)



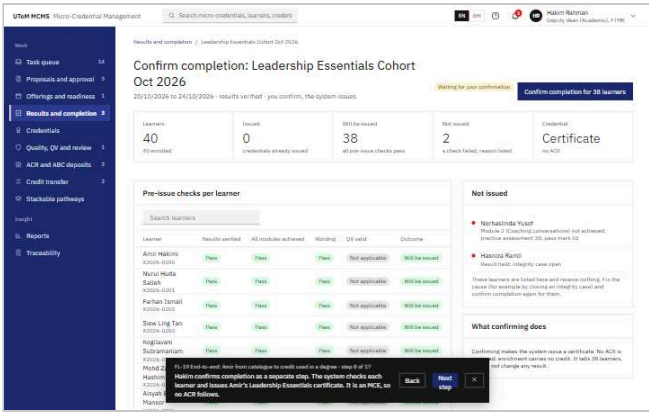
Step 5 of 17. CODL imports results from UCreds. Bad rows are rejected; good rows stay Provisional. (`#/codl/ucreds`)



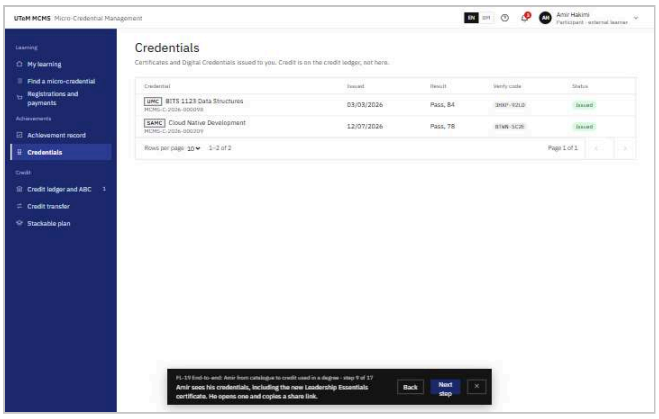
Step 6 of 17. Dr. Aina fills the two missing marks and sends the cohort for verification. (`#/lec/cohort/CND-2026-2`)



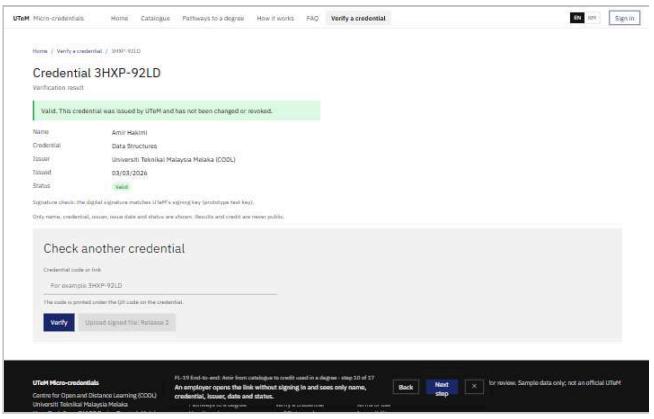
Step 7 of 17. Hakim verifies results. This creates module badges only, nothing else. (Shown on Data Structures 2026/2; Amir's own cohorts were verified the same way.) (`#/dd/verify/DS-2026-2`)



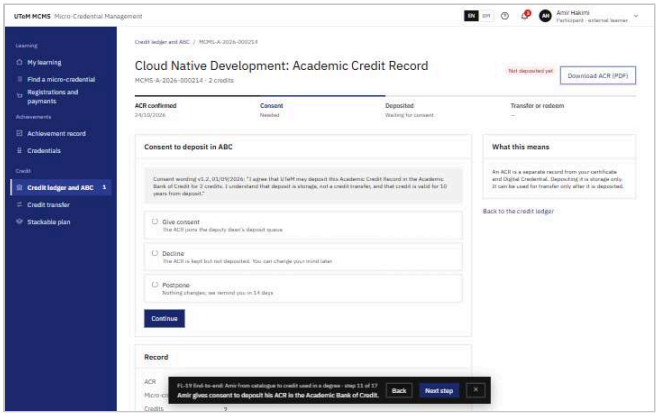
Step 8 of 17. Hakim confirms completion as a separate step. The system checks each learner and issues Amir's Leadership Essentials certificate. It is an MCE, so no ACR follows. (`#/dd/completion/LE-2026-10`)



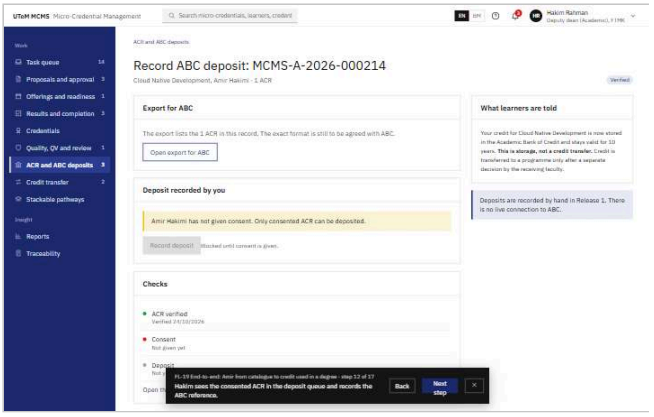
Step 9 of 17. Amir sees his credentials, including the new Leadership Essentials certificate. He opens one and copies a share link. (`#/st/credentials`)



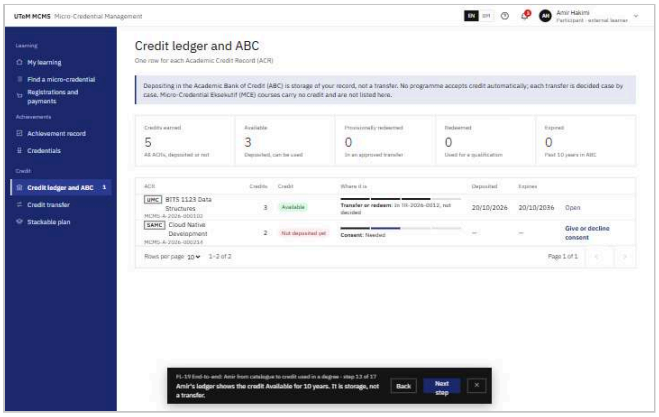
Step 10 of 17. An employer opens the link without signing in and sees only name, credential, issuer, date and status. (`#/public/verify/3HXP-92LD`)



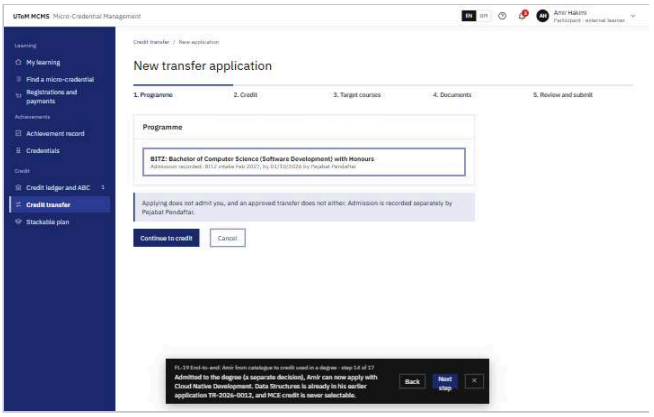
Step 11 of 17. Amir gives consent to deposit his ACR in the Academic Bank of Credit. (`#/st/acr/MCMS-A-2026-000214`)



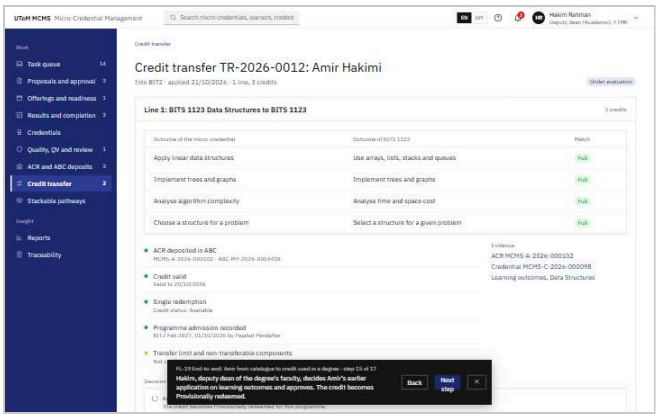
Step 12 of 17. Hakim sees the consented ACR in the deposit queue and records the ABC reference. (`#/dd/deposit/MCMS-A-2026-000214`)



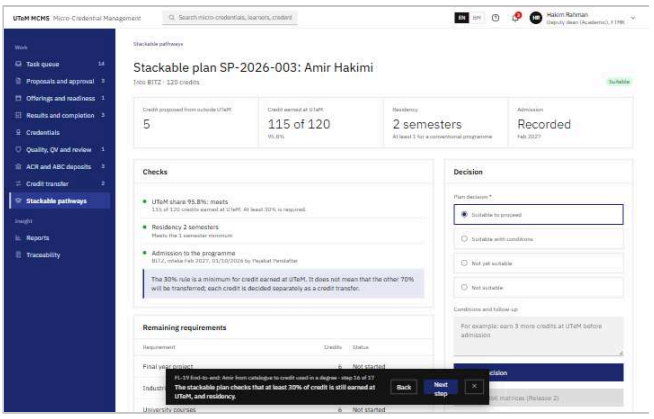
Step 13 of 17. Amir's ledger shows the credit Available for 10 years. It is storage, not a transfer. (`#/st/ledger`)



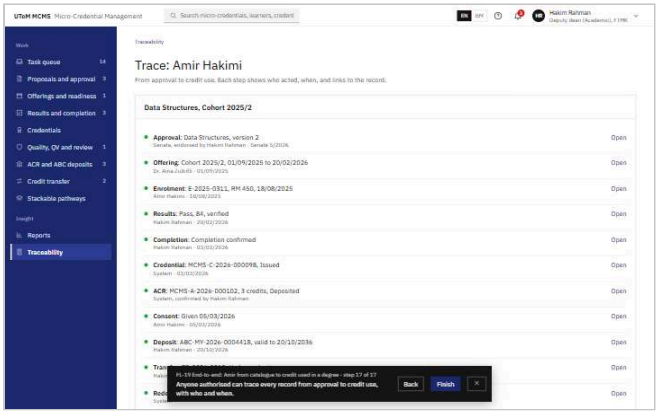
Step 14 of 17. Admitted to the degree (a separate decision), Amir can now apply with Cloud Native Development. Data Structures is already in his earlier application TR-2026-0012, and MCE credit is never selectable. (`#/st/transfer/new`)



Step 15 of 17. Hakim, deputy dean of the degree's faculty, decides Amir's earlier application on learning outcomes and approves. The credit becomes Provisionally redeemed. (`#/dd/transfer/TR-2026-0012`)



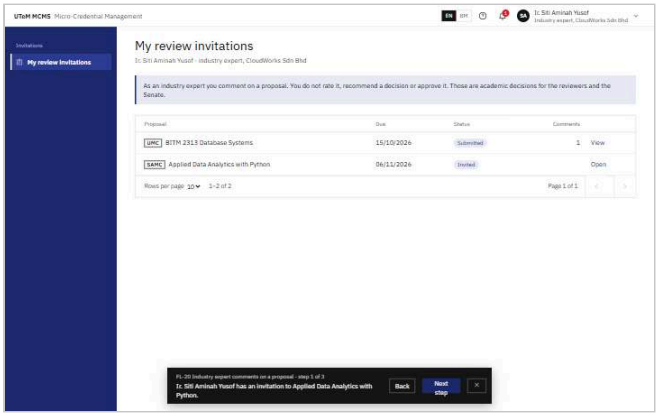
Step 16 of 17. The stackable plan checks that at least 30% of credit is still earned at UTeM, and residency. (`#/dd/plan/SP-2026-003`)



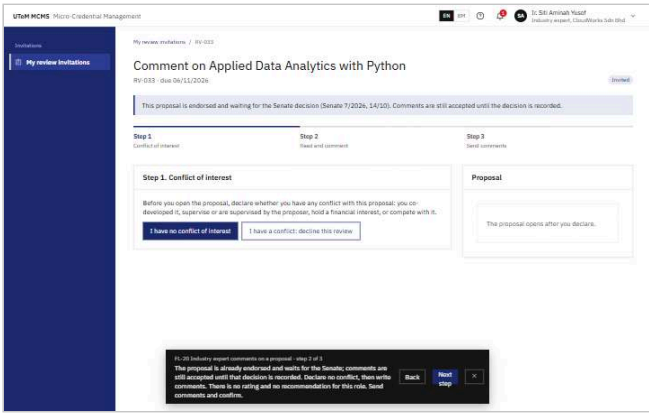
Step 17 of 17. Anyone authorised can trace every record from approval to credit use, with who and when. (`#/dd/trace/amir`)

D.22 FL-20 Industry expert comments on a proposal

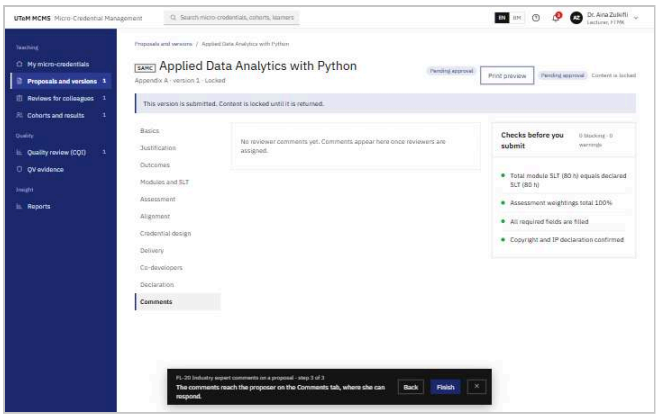
Industry expert · 3 steps



Step 1 of 3. Ir. Siti Aminah Yusof has an invitation to Applied Data Analytics with Python. (`#/ext/home`)



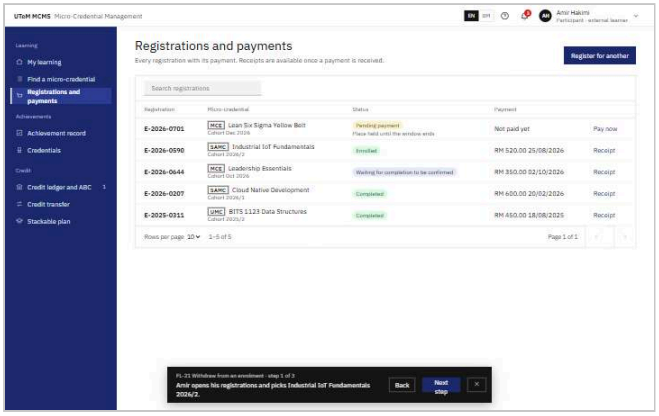
Step 2 of 3. The proposal is already endorsed and waits for the Senate; comments are still accepted until that decision is recorded. Declare no conflict, then write comments. There is no rating and no recommendation for this role. Send comments and confirm. (`#/ext/review/RV-033`)



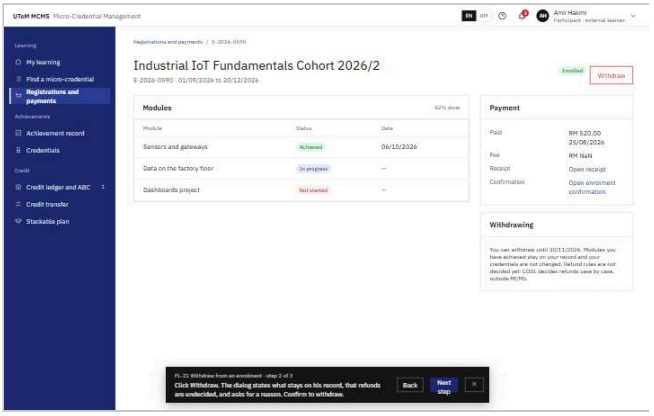
Step 3 of 3. The comments reach the proposer on the Comments tab, where she can respond. (`#/lec/proposal/ADA?tab=comments`)

D.23 FL-21 Withdraw from an enrolment

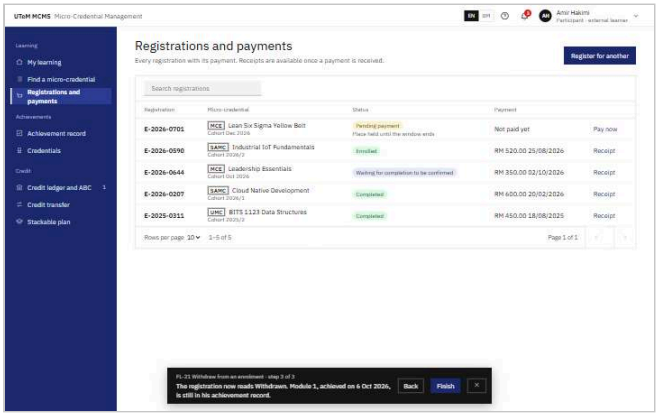
Student · 3 steps



Step 1 of 3. Amir opens his registrations and picks Industrial IoT Fundamentals 2026/2. (`#/st/enrolments`)



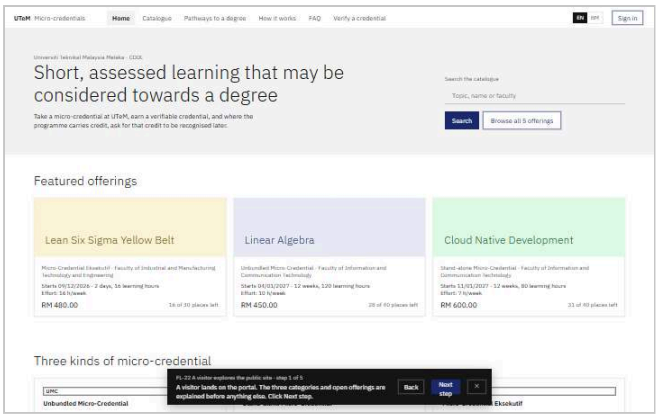
Step 2 of 3. Click Withdraw. The dialog states what stays on his record, that refunds are undecided, and asks for a reason. Confirm to withdraw. (`#/st/enrolment/E-2026-0590`)



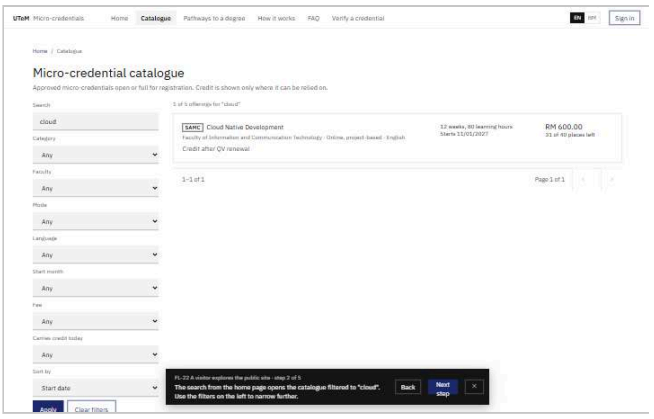
Step 3 of 3. The registration now reads Withdrawn. Module 1, achieved on 6 Oct 2026, is still in his achievement record. (`#/st/enrolments`)

D.24 FL-22 A visitor explores the public site

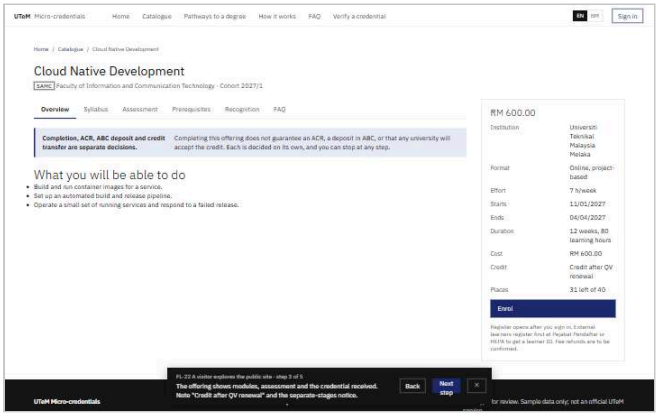
Public visitor · 5 steps



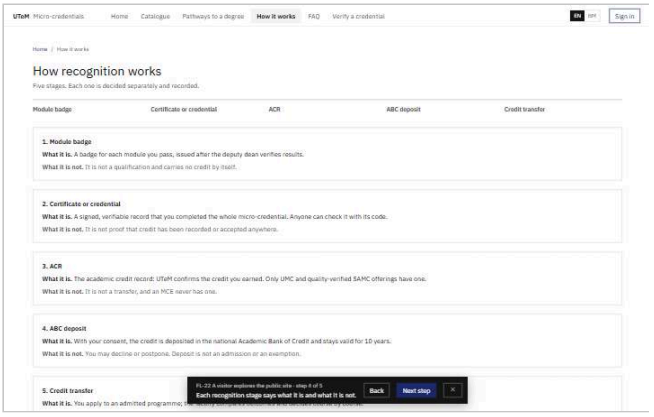
Step 1 of 5. A visitor lands on the portal. The three categories and open offerings are explained before anything else. Click Next step. (`#/public/home`)



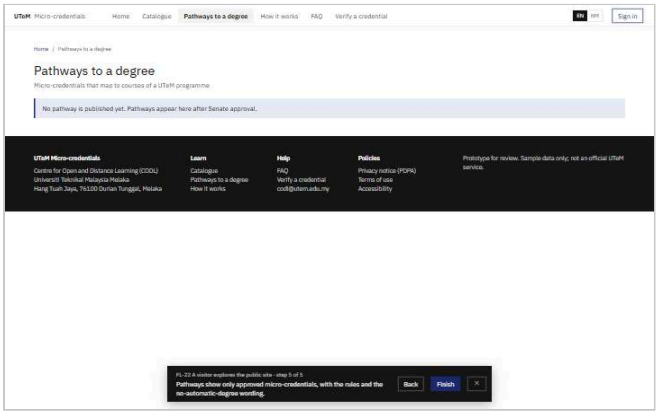
Step 2 of 5. The search from the home page opens the catalogue filtered to "cloud". Use the filters on the left to narrow further. (`#/public/catalogue?q=cloud`)



Step 3 of 5. The offering shows modules, assessment and the credential received. Note "Credit after QV renewal" and the separate-stages notice. (`#/public/offering/CND-2027-1`)



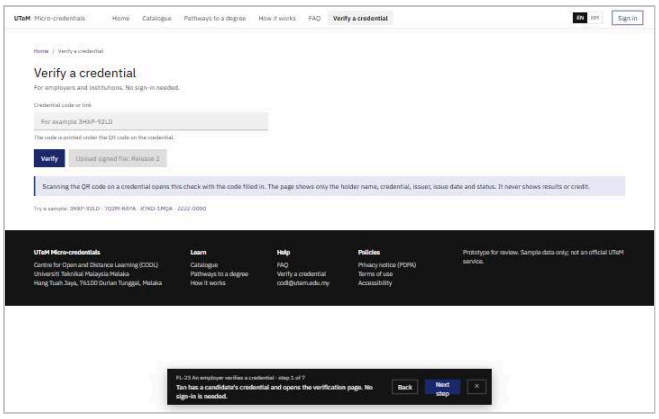
Step 4 of 5. Each recognition stage says what it is and what it is not. (`#/public/how`)



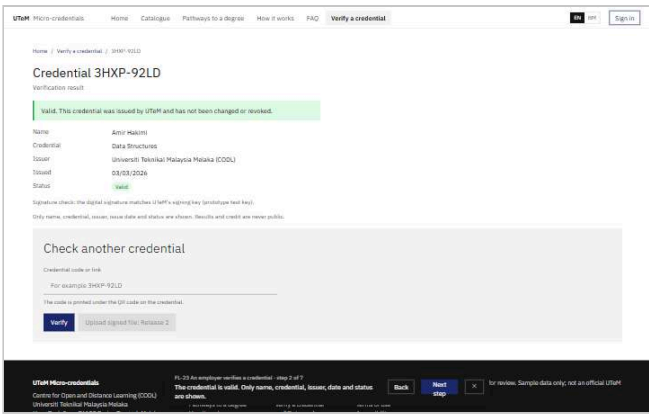
Step 5 of 5. Pathways show only approved micro-credentials, with the rules and the no-automatic-degree wording. (`#/public/pathways`)

D.25 FL-23 An employer verifies a credential

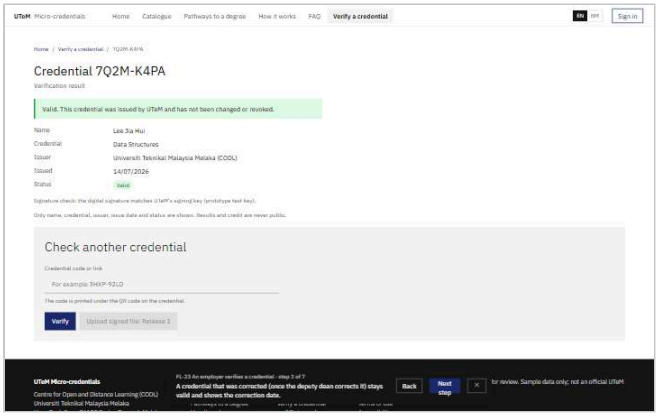
Tan Mei Ling, HR manager (public) · 7 steps



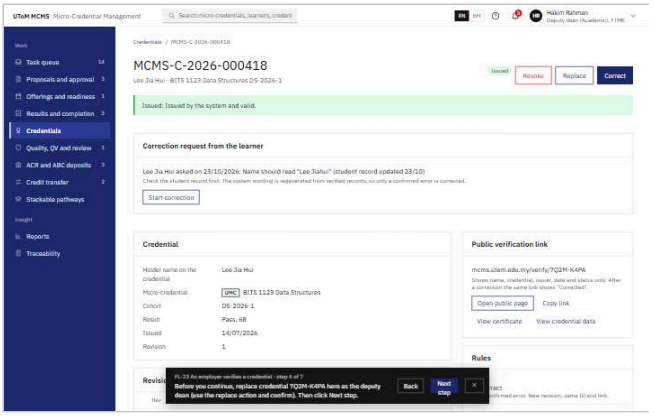
Step 1 of 7. Tan has a candidate's credential and opens the verification page. No sign-in is needed. (`#/public/verify`)



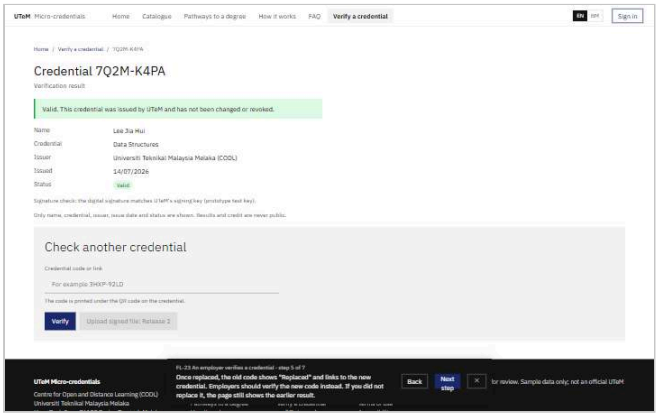
Step 2 of 7. The credential is valid. Only name, credential, issuer, date and status are shown. (`#/public/verify/3HXP-92LD`)



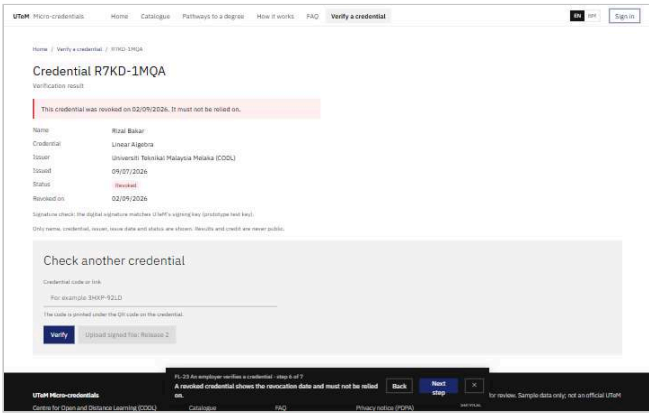
Step 3 of 7. A credential that was corrected (once the deputy dean corrects it) stays valid and shows the correction date. (`#/public/verify/7Q2M-K4PA`)



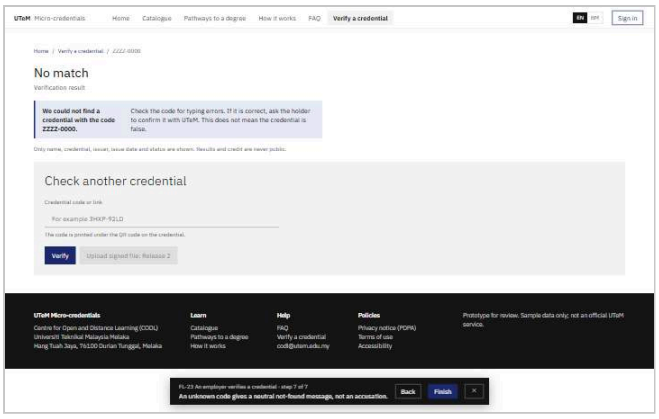
Step 4 of 7. Before you continue, replace credential 7Q2M-K4PA here as the deputy dean (use the replace action and confirm). Then click Next step. (`#/dd/cred/MCMS-C-2026-000418`)



Step 5 of 7. Once replaced, the old code shows "Replaced" and links to the new credential. Employers should verify the new code instead. If you did not replace it, the page still shows the earlier result. (`#/public/verify/7Q2M-K4PA`)



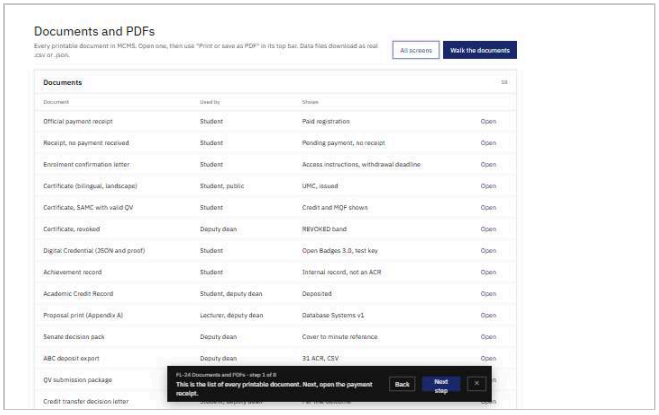
Step 6 of 7. A revoked credential shows the revocation date and must not be relied on. (`#/public/verify/R7KD-1MQA`)



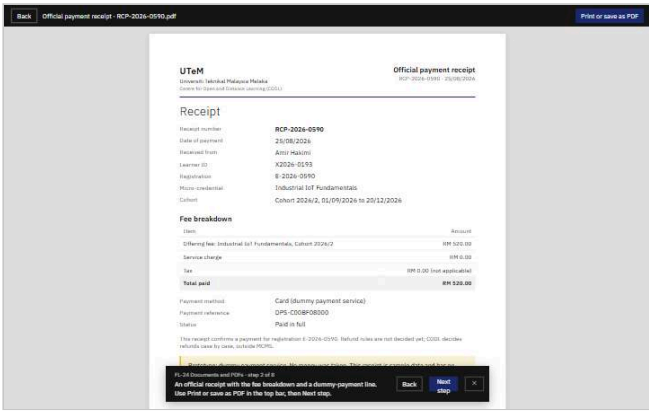
Step 7 of 7. An unknown code gives a neutral not-found message, not an accusation. (`#/public/verify/ZZZZ-0000`)

D.26 FL-24 Documents and PDFs

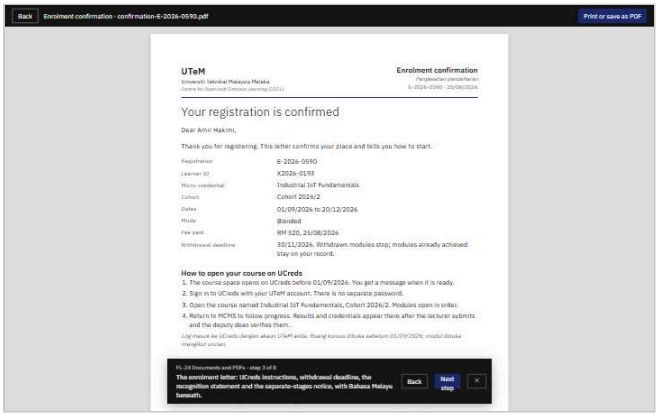
All roles · 8 steps



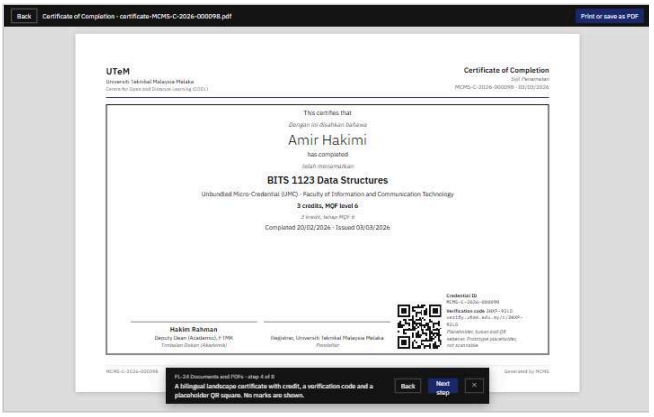
Step 1 of 8. This is the list of every printable document. Next, open the payment receipt. (`#/doc/index`)



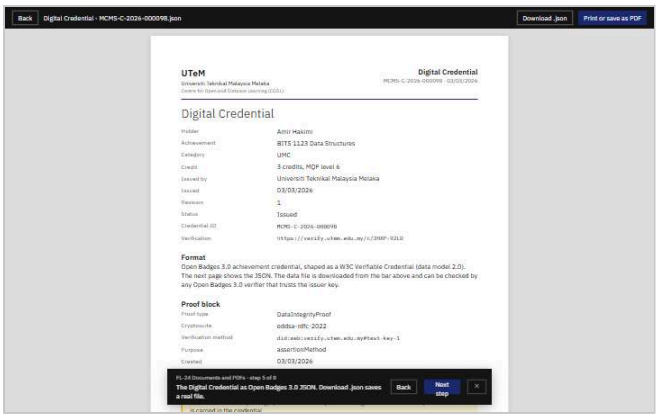
Step 2 of 8. An official receipt with the fee breakdown and a dummy-payment line. Use Print or save as PDF in the top bar, then Next step. (`#/doc/receipt/E-2026-0590`)



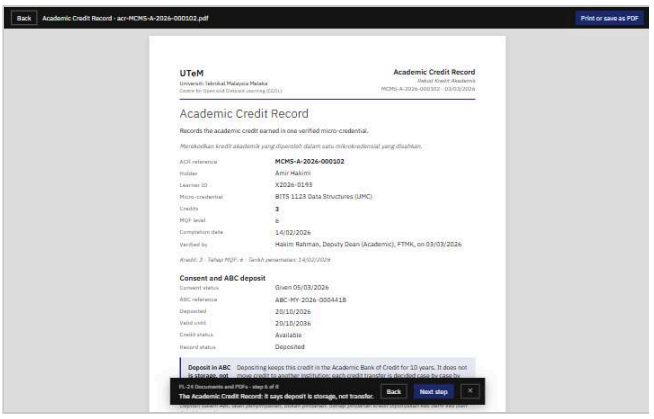
Step 3 of 8. The enrolment letter: UCreds instructions, withdrawal deadline, the recognition statement and the separate-stages notice, with Bahasa Melayu beneath. (`#/doc/enrolment/E-2026-0590`)



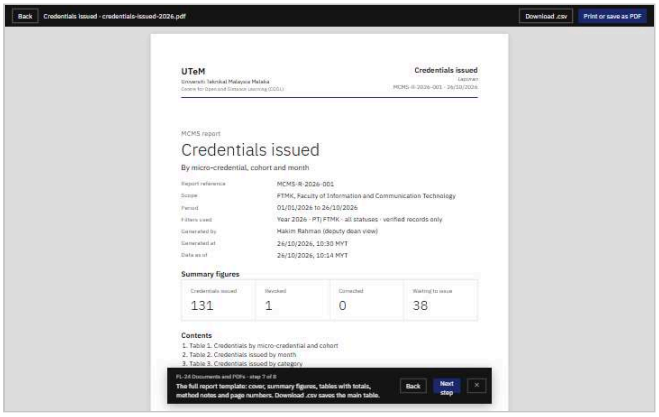
Step 4 of 8. A bilingual landscape certificate with credit, a verification code and a placeholder QR square. No marks are shown. (`#/doc/certificate/MCMS-C-2026-000098`)



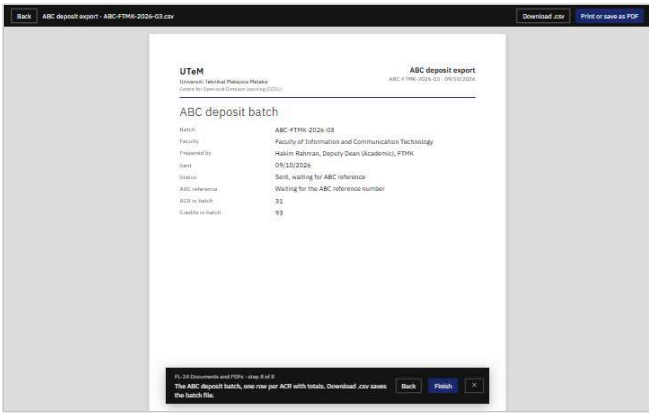
Step 5 of 8. The Digital Credential as Open Badges 3.0 JSON. Download .json saves a real file. (`#/doc/credential/MCMS-C-2026-000098`)



Step 6 of 8. The Academic Credit Record: it says deposit is storage, not transfer. (`#/doc/acr/MCMS-A-2026-000102`)



Step 7 of 8. The full report template: cover, summary figures, tables with totals, method notes and page numbers. Download .csv saves the main table. (`#/doc/report/credentials-issued`)



Step 8 of 8. The ABC deposit batch, one row per ACR with totals. Download .csv saves the batch file. (`#/doc/abc/ABC-FTMK-2026-03`)